SINCE
2016
10 years in service

Software Requirements Specification · ISO/IEC/IEEE 29148

SCHEDULE · Academic Decision System

What the system must provide, for whom, and under which guarantees.

v1.0 | 2026-11-07 | English

627	20	560	291
Requirements	Modules	Server routes	Test files

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Prepared by

Document control

System	SCHEDULE · Academic Decision System
Document	Software Requirements Specification
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The document in numbers

627 Requirements	**20** Modules	**100%** Implemented	**89%** Test-proven
560 Server routes	**165** Screens and components	**291** Test files	**1810** Acceptance criteria

Revision history

Version	Date	Changes
`1.0`	`2026-11-07`	First release: the baseline document with all its modules, requirement-to-code-and-test traceability, and the interactive page.

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1 Introduction

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1.1 Purpose

This document specifies the functional and non-functional requirements of the SCHEDULE system. It is a shared reference for academic administration, the development team and testers during design, implementation and acceptance.

It follows the structure of ISO/IEC/IEEE 29148. Every requirement is testable and linked to its place in the code and, where one exists, to the test that proves it.

1.2 Scope

The scope covers building department and college timetables from terms, courses, instructors and halls; conflict detection; schedule drafts and scenarios; the approval and freeze cycle; import of the official authority file; faculty requests; section planning and study plans; student, advising and seat services; and analytics, reports and notifications.

The scope excludes the official registration system, grades and fees. The authority file is the reference for course numbers and departments; the system compares against it and does not replace it.

1.3 Intended audience

- Academic administration and schedule committees, to confirm the system covers university policy.
- Development and test teams, to turn a requirement into work and a test case.
- Reviewing bodies, to understand the guarantees, permissions and scope.

1.4 Document conventions

- **Identifier:** `FR` functional, `NFR` non-functional, `DR` data, `IR` interface; then a three-letter module code and three digits, e.g. `FR-SCH-012`. A number is never reused after deletion.
- **Priority:** `Must` the system cannot operate without it, `Should` important with a temporary workaround, `Could` an enhancement.
- **Status:** every requirement in this release is `Implemented` and present in the current version, proven by its code reference and its test.
- **Inferred:** a mark placed on a requirement derived from the behaviour of the code that has no test proving it yet, and needs owner confirmation.
- **References:** every requirement carries the file paths that implement it and the tests that prove it, and these are checked automatically on every build.
- **Language:** the Arabic and English texts carry the same requirements under the same numbering.

1.5 References

1. ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes: Requirements engineering.
2. W3C, Web Content Accessibility Guidelines (WCAG) 2.1, level AA.
3. RFC 5545, Internet Calendaring and Scheduling Core Object Specification (iCalendar).
4. RFC 8030, Generic Event Delivery Using HTTP Push.

1.6 Definitions

Appendix C lists 156 terms with their Arabic and English equivalents.

2 Overall description

↑ Contents

2.1 Summary

SCHEDULE is an operational web platform for building, reviewing, approving and publishing academic timetables inside the university. It is driven by a deterministic scheduling engine that detects a conflict at the moment it occurs and refuses to approve a timetable that contains a material one. This document states what the system must provide, for whom, and under which guarantees, using stable numbering that can be traced to code and tests.

Deterministic engine
Scheduling and conflict rules are explicit and tested; the same input always yields the same result.

Governed approval
A cycle of submission, review, return, acceptance and freeze, each step with its own permission and recorded trail.

Safe import
Reading proposes, deterministic validators decide, and nothing is written without explicit confirmation.

Scope and permissions
Every role sees only its own college or department, and the guard lives on the server, not in the interface.

Secret public links
Faculty and students reach their card through a secret link with guess-limited access.

Arabic first
A right-to-left interface that adapts to the phone and prints its reports on A4 with a fixed design.

2.2 Product perspective

The system is a progressive web application (PWA) with a React interface and a Node.js server running on Google Cloud Run. Its database is Firestore, reached by the server alone with administrative credentials, so there is no direct door from the browser.

Beside the application there are public pages built by the server for each secret link: the faculty card, the department card, the room door, the building board, partner pages and the signature verification page.

2.3 History and use



The system is not new: it has been in service since 2016, and every past term is kept for the system to read and analyse. This history is what makes its comparisons and forecasts rest on facts rather than assumptions.

2016 — **Go-live** (2016 – 2026)
The system went live with a large pilot study and began keeping academic terms one after another.

A decade of kept terms (Migration)
Every past term is frozen read-only and stays a reference for comparisons and forecasts.

FR-APR-016

Legacy data under its ids (2026-10-17)
The legacy data moved under its original ids, and its historical tables stay archived as they were.

FR-DAT-021

Live audience measurement (2026-11-07)
From this date measured figures add to the approved baseline, so usage figures refresh on their own.

Rec doc v1.(
A dc tied and refr ever

What the system reads from its history

Each department’s rhythm

Learns a department’s habitual starts and lengths from all its terms and gently notes what departs from them.

FR-SCH-020

Season history and course lineage

Gathers each course’s history from earlier terms of the same season and lets a renumbered course inherit its predecessor’s.

FR-PLN-005

Next-term forecast

Forecasts each course’s sections as a value and a range from previous terms.

FR-INT-018

Visiting history across years

One yearly matrix of visiting instructors that keeps anyone delegated even after leaving the directory.

FR-RPT-011

Back to a previous version

Every version of a schedule is kept and can be restored with explicit confirmation and full checks.

FR-DRF-012

Use in figures

Aggregated, anonymous figures from the system administration as of 2026-10-17; “about” is an estimate and “over” is a lower bound. The interactive page shows them live from the system.

about
80,000
students served

over
5.3 million
visits

about
786
faculty members

over
75,000
faculty card opens

over
90%
of faculty open their card

2.4 Architecture and technologies

Three layers work together: a front end built once for computer and phone, a server that guards the rules and the data, and a cloud operation with automatic checks before every release. Version numbers are read from the project files themselves on every build.

01 Front end

- **Programming languages** — JavaScript · TypeScript `v5.8`
- **UI library** — React `v19`
- **Build and development tool** — Vite `v6`
- **Install and offline use** — PWA · Service Worker
- **Icons and typeface** — Lucide · IBM Plex Sans Arabic

02 Server and data

- **Server runtime** — Node.js `v22`
- **Server framework** — Express `v4`
- **Database** — Google Cloud Firestore `v8`
- **Server-to-data access** — Firebase Admin SDK `v14`
- **Security and performance** — express-rate-limit · compression

03 Operations and quality

- **Hosting and cloud runtime** — Google Cloud Run
- **Version and project management** — Git · GitHub
- **Automated tests, quality and conflict checks** — **291** test files
- **Continuous integration and delivery** — CI/CD
- **Build and run tooling** — esbuild · tsx

Supporting services

- **PDF reading** — PDF.js `v6`
- **Assisted import reading** — Gemini
- **Mobile push notifications** — Web Push
- **Deployment packaging** — Docker
- **Demo and development mode** — Isolated local data
- **Scanned document reading** — Tesseract OCR `v7`
- **Excel import and export** — SheetJS `v0.20`
- **Faculty calendar feed** — iCalendar (ICS)
- **Image processing** — Canvas · HEIC

2.5 How the system computes its decisions

Does the system rely on algorithms or on plain calculation?

Algorithms. Every decision in the system is produced by an explicit, written and tested algorithm: 50 algorithms across 17 modules, each with its steps, thresholds and code references. Most are deterministic: the same input always yields the same result, and every result traces back to its rule. Some search the possible options and rank them, and some are descriptive statistics over the terms kept since 2016. No trained model or probabilistic output enters a decision; where the Gemini service is used (import reading and text queries) it only reads and proposes, while deterministic validators and the user’s confirmation decide.

32 — Deterministic

Explicit rules and comparisons: the same input always yields the same result.

11 — Search and ranking

Explores the possible options within constraints and ranks them by a stated criterion.

5 — Statistics from history

Descriptive statistics over past terms with stated thresholds; deterministic code, not a trained model.

2 — Assisted reading, confirmed

The Gemini service reads and proposes, deterministic validators decide, and the user confirms.

Algorithm index

ID	Algorithm	Module	Kind
ALG-ACC-01	Scope decision	ACC	Deterministic
ALG-ACD-01	Duplicate course and spelling-twin detection	ACD	Deterministic
ALG-ACD-02	Instructor name-identity matching	ACD	Deterministic
ALG-ACD-03	Teaching-load computation and classification	ACD	Deterministic
ALG-LOC-01	Building and hall resolution against the official registry	LOC	Deterministic
ALG-LOC-02	Hall barter windows and their enforcement	LOC	Deterministic
ALG-LOC-03	Ranking candidate halls for an instructor request	LOC	Search and ranking
ALG-SCH-01	Indexed conflict detection	SCH	Deterministic
ALG-SCH-02	Regulation rules: length, start ladder, patterns and hours ceiling	SCH	Deterministic
ALG-SCH-03	Department rhythm and memory from history	SCH	Statistics from history
ALG-SCH-04	Travel time between campuses and buildings	SCH	Deterministic
ALG-SCH-05	Time and hall suggestion engine	SCH	Search and ranking
ALG-SCH-06	Least-impact repair chain (domino)	SCH	Search and ranking
ALG-SCH-07	Cover availability and substitute ranking	SCH	Search and ranking
ALG-SCH-08	Common meeting time	SCH	Search and ranking
ALG-DRF-01	Scenario overlay: apply, drift and write plan	DRF	Deterministic
ALG-DRF-02	Green, yellow and red impact indicator	DRF	Deterministic
ALG-DRF-03	Problem reading and three ranked solutions	DRF	Search and ranking
ALG-DRF-04	Time-collision repair for a start-of-term draft	DRF	Search and ranking
ALG-VIW-01	Two-snapshot diff and instructor schedule fingerprint	VIW	Deterministic
ALG-APR-01	Approval state machine and guards	APR	Deterministic
ALG-APR-02	Effective deadline and wholesale-change detection	APR	Deterministic
ALG-APR-03	Registrar inbox priority and insistence escalation	APR	Deterministic
ALG-IMP-01	Header proof, file department, course and section identity	IMP	Deterministic
ALG-IMP-02	Scan legibility and cell plausibility checks	IMP	Deterministic
ALG-IMP-03	Comparison with the authority source file	IMP	Deterministic
ALG-IMP-04	Gemini-assisted reading with cross-verification	IMP	Assisted reading, confirmed
ALG-REQ-01	Faculty request item verdict and meeting length	REQ	Deterministic
ALG-PLN-01	Section-count suggestion from same-season history	PLN	Statistics from history
ALG-PLN-02	Section gap and surplus review	PLN	Deterministic
ALG-PLN-03	Placement search for new sections	PLN	Search and ranking
ALG-PLN-04	Not-yet-reached course verdict	PLN	Deterministic
ALG-PLN-05	Plan-sheet rule check and reconciliation	PLN	Deterministic
ALG-PLN-06	Study proposal check, alternatives and vacancy ranking	PLN	Search and ranking
ALG-STU-01	Passed-course closure over prerequisites	STU	Deterministic
ALG-STU-02	Graduate plan choice and graduation-sheet proof	STU	Deterministic
ALG-STU-03	Registration rehearsal and semester readiness	STU	Search and ranking
ALG-STU-04	Shortfall cause attribution by scenario comparison	STU	Deterministic

ID	Algorithm	Module	Kind
ALG-STU-05	Advisor seat cap and section-request capacity	STU	Deterministic
ALG-INT-01	Next-term demand forecast with confidence	INT	Statistics from history
ALG-INT-02	Academic decision lab	INT	Search and ranking
ALG-INT-03	Natural-language query layer	INT	Assisted reading, confirmed
ALG-INT-04	Guide question understanding by lexicon and weights	INT	Deterministic
ALG-RPT-01	Load fairness index	RPT	Statistics from history
ALG-RPT-02	Occupancy rate grid	RPT	Deterministic
ALG-NTF-01	Live broadcast coalescing, foreign-write detection and stale-write refusal	NTF	Deterministic
ALG-ENG-01	Ad image quality check from its pixels	ENG	Deterministic
ALG-ENG-02	Aggregated anonymous audience statistics	ENG	Statistics from history
ALG-DSH-01	Term early warning	DSH	Deterministic
ALG-DAT-01	Legacy migration with count parity	DAT	Deterministic

2.6 Module map

Foundation

3.1 · ACC
Identity, accounts, roles and access scope
Who signs in, in which role, what they may read or write, and in which department.

3.2 · ACD
Academic structure and reference data
Terms, colleges, departments, courses, faculty and visiting instructors: the reference every schedule is built on.

3.3 · LOC
Locations
One official registry of buildings and halls, hall barter between departments, door pages and occupancy

3.19 · DAT
Data model and integrity
What is stored, how it relates, and what keeps it sound

Scheduling core

3.4 · SCH
Deterministic scheduling engine
Every appointment passes fixed rules before it is written: conflicts, regulation, teaching day, holds and a rollback version.

3.5 · DRF
Drafts, scenarios and the time machine
Try on a copy, compare the effect, apply in one piece and go back with one click.

3.6 · VIW
Viewing, sharing, copying and transferring schedules
The department schedule to read, a link for whoever needs it, copying from term to term, and an export that keeps the whole picture.

3.7 · APR
Approval cycle
From committee signature to registrar acceptance: rounds, notes, deadlines, term freeze and verifiable stamps.

3.8 · IMP
Import
From authority documents, scans and spreadsheets to a reviewed and confirmed schedule.

Collaboration and services

3.9 · REQ
Faculty requests and public links
A secret personal link lets each faculty member request schedule changes, and an inbox lets the department decide item by item.

3.10 · PLN
Section planning and curricula
From the remaining report and plan sheet to the term's sections and the study proposal

3.11 · STU
Student and advising services
Masari for students, the registration sheet, section requests, seats and the registration rehearsal

3.14 · NTF
Notifications, activity and live channel
A bell derived from state, a live channel that announces change, an immutable trail, and a calendar you subscribe to.

Insight and reporting

3.12 · INT
Analytics Center and Guide
Deterministic schedule readings, next-term forecasts, a natural-language query layer that never executes, and an in-app Guide that runs in the browser.

3.13 · RPT
Reports, queries, printing and export
One question, many lenses; A4 landscape sheets in their approved design; Excel files with seven-digit course numbers.

3.16 · DSH
Dashboards and term awareness
The home dashboard, term story, semester readiness and the welcome tour.

3.15 · ENG
Engagement and public presence
Student survey, partner advertising, audience dashboard and the public pages.

Platform

3.17 · OPS
Operations
From server boot to release: runtime, backups, the demo sandbox and the installable app.

3.18 · NFR
Non-functional requirements
Performance, security, privacy, accessibility, Arabic-first design and reliability, with measured figures.

3.20 · EXT
External interfaces
Everything the system connects to from outside, and under what contract

2.7 User classes

Actor	Requirements	Modules
Root administrator	**94**	**17**
System administrator	**76**	**16**
College dean	**63**	**10**
Vice dean for academic affairs	**18**	**6**
Dean of registration	**11**	**4**
University registrar	**30**	**7**
College registrar	**32**	**8**
Registration officer	**49**	**8**
Department head	**269**	**17**
Department advisor	**18**	**5**
Schedule committee chair	**322**	**17**

Actor	Requirements	Modules
Standard user	**50**	**10**
Faculty member	**50**	**9**
Student	**60**	**9**
Public visitor	**50**	**13**
Partner organisation	**7**	**2**
The system (automatic)	**180**	**19**
External service	**9**	**5**

2.8 Design constraints

- **The engine is deterministic:** no schedule changes without an explicit rule, and the deterministic validators are the final judge before any write.
- **Assisted reading never writes:** every import or text query passes the validators and then the user's confirmation.
- **Course numbers are seven digits:** college (2) + department (2) + course (3), e.g. `0109105`, on every screen, report and export.
- **The header decides the department** in the authority file, even if a course number carries another department's code.
- **Printed reports keep their design:** content and data are fixed, not layout.
- **The server is the guard:** hiding screens in the interface is courtesy to the user's time, not protection.

2.9 Assumptions and dependencies

- Users reach the system with a modern browser on a computer or a phone at least 320 pixels wide.
- The official authority file (SWRSCHA) is the approved source of course numbers and departments.
- Google Cloud and Firestore are available, and the system runs in an isolated demonstration mode when the database is unavailable or for showcasing.
- Every account belongs to one or more colleges or departments within a scope set by the administrator.
- Enabling point-in-time recovery, delete protection and scheduled exports for the production database, and unifying rate limits at the network edge, are decisions for the hosting party on its own project; the documented procedure is ready to run.

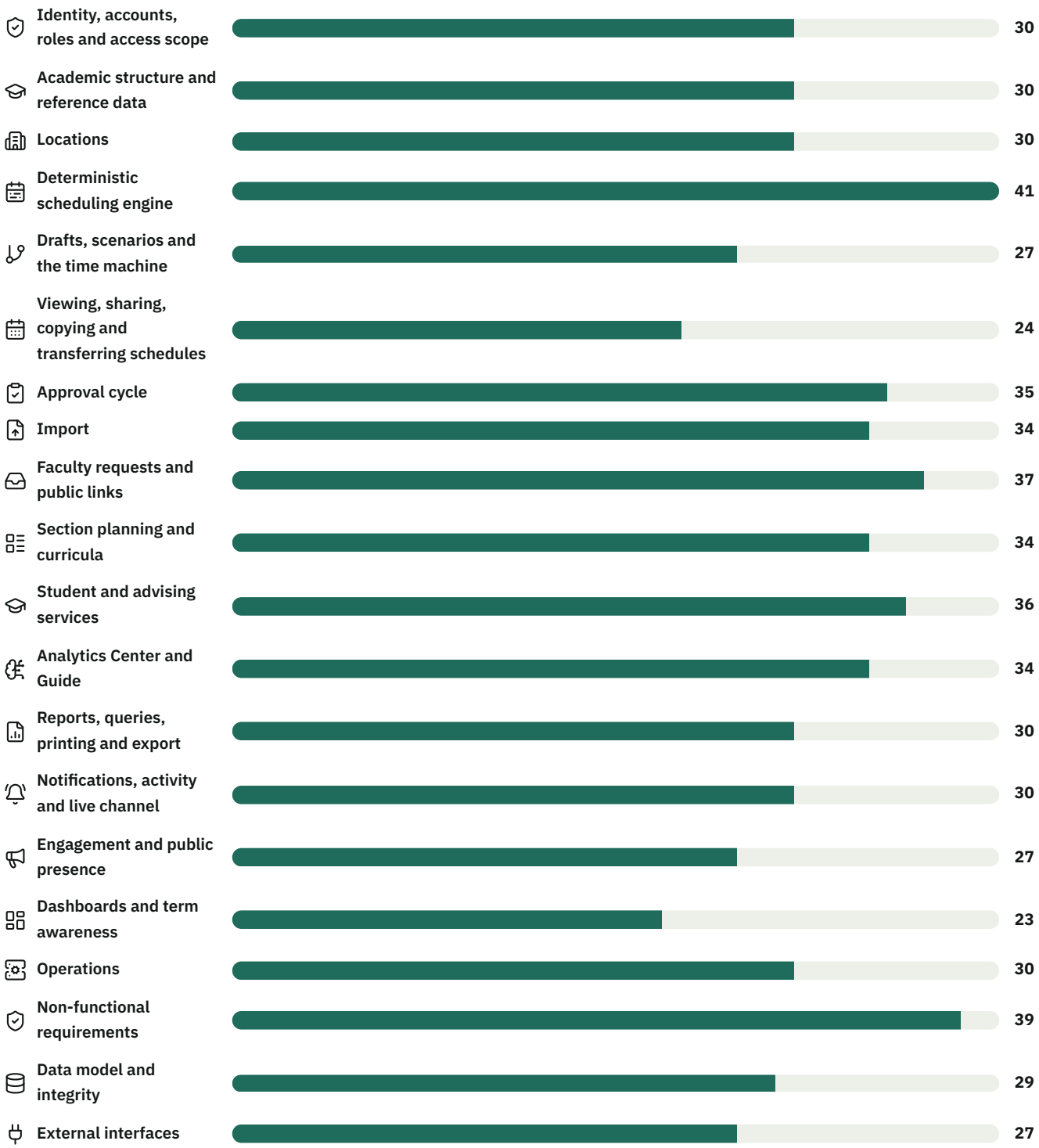
3 Specific requirements

↑ Contents

This part holds 20 modules. Each has an overview, user journeys, diagrams, screens, requirements with acceptance criteria and evidence, and edge cases.

Implemented

Requirements by module



3.1 · ACC

Identity, accounts, roles and access scope

Who signs in, in which role, what they may read or write, and in which department.

30	100%	49	27
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.1.1 Overview

This module describes sign-in, sessions and passwords, and account administration from the Users screen. It defines the ten roles that decide read or write access, screens and scope, and the server guard that enforces them on every write request. It also covers college and department scope and how data is filtered by it, rate limits on anonymous doors, the demo environment and the audit trail. The server is always the authority; hiding buttons in the interface is only a visual convenience.

Root administrator · System administrator · College dean · Vice dean for academic affairs · Dean of registration · University registrar · College registrar · Registration officer · Department head · Department advisor · Schedule committee chair · Standard user · Public visitor · The system (automatic)

3.1.2 User journeys

System administrator — **Creating a read-only dean account**

1. The system administrator opens Users and chooses to add an account.
2. They enter the name, login and password, and pick the role College dean, which reveals the college list.
3. They select the college and save; the system writes screen 14 and a single scope row with department 0 for that college.
4. The account is flagged to change its password at first sign-in, and an access card is issued for the administrator to send to its owner.
5. The dean signs in and sees every department of that college, including ones created later, with no editing controls.

Department head — **First sign-in with the initial password**

1. The department head receives an invitation link carrying the login name and an initial password from the administrator.
2. Opening the link fills in the login name and the # fragment is removed from the browser address.
3. Signing in with the initial password opens a session, but the server allows nothing beyond the sign-in paths.
4. The Choose your password screen appears without a current-password field; they type a new password of at least 8 characters and confirm it.
5. The password is saved, the account's other sessions are closed, the flag is cleared and the workspace opens with department-head permissions.

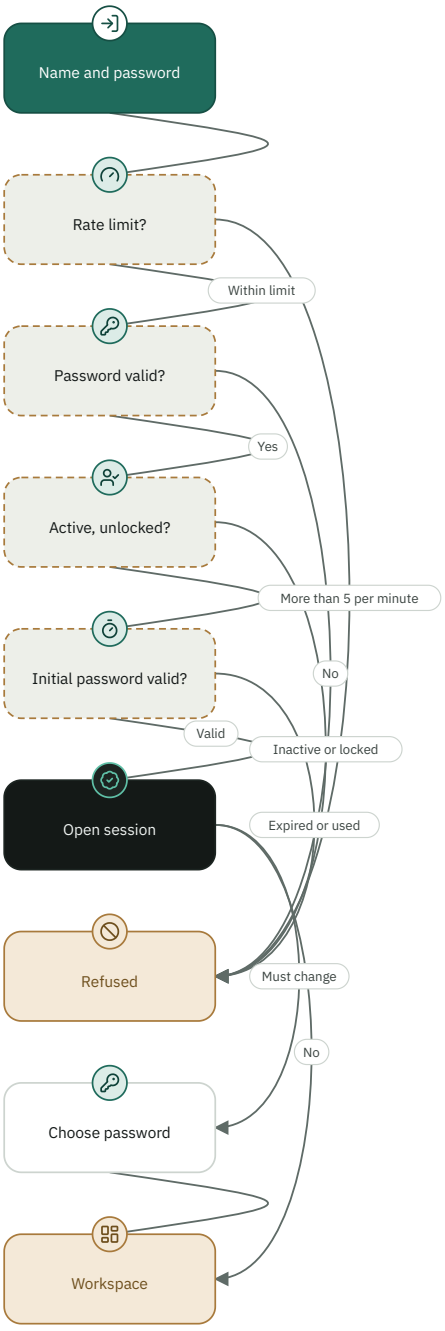
College dean — **A write attempt from a view-only account**

1. The dean opens the department report for their college and reads without obstruction.
2. A modifying request is sent straight to the server, bypassing the interface.
3. The guard decides the role is read-only with no exception path and answers 403.
4. The message names the role and carries a readOnly marker so the interface can show it clearly.
5. The refusal is recorded in the audit log as refused (outside permission).

3.1.3 Diagrams

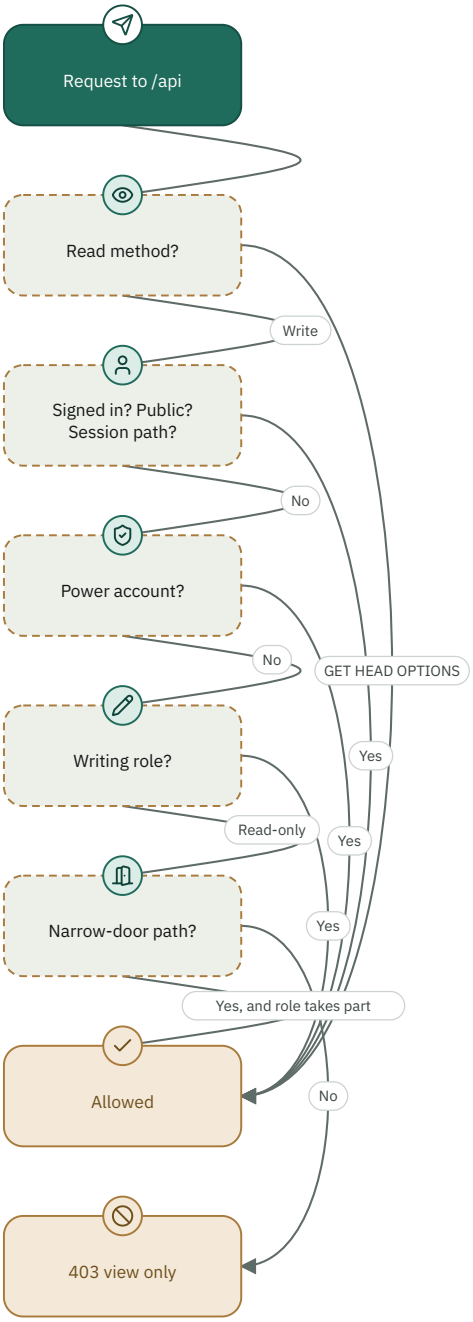
Sign-in flow

Order of checks in POST /api/auth/login; refusals after the password check reveal nothing to someone who does not know it.



Role write guard

The roleWriteDecision order; each step precedes the next for a reason.



Permission matrix: roles × capabilities

	Schedule workspace (screen 7)	Reports and queries	Cell comments	Schedule signature	Inbox: accept and return	Delivery deadline and extension	Users permissi scope
Root administrator	✎	👁	—	—	—	—	✎
System administrator	✎	👁	—	—	—	—	✎
College dean	—	👁	—	—	—	—	—
Vice dean for academic affairs	—	👁	—	—	—	—	—
Dean of registration	—	👁	—	—	👁	—	—
University registrar	—	👁	✎	—	✎	✎	—
College registrar	—	👁	✎	—	✎	✎	—
Registration officer	—	👁	✎	—	✎	—	—
Department head	—	👁	✎	✎	—	—	—
Department advisor	—	👁	—	—	—	—	—
Schedule committee chair	✎	👁	✎	✎	—	—	—
Standard user	✎	—	—	—	—	—	—

✎ Write 👁 Read — No access

3.1.4 Screens

Sign-in
User name and password, the invitation link fragment that pre-fills the name and is never sent to the server, and the demo entry button when enabled.`src/components/Login.tsx`

Choose your password
Blocks the workspace until the account owner picks a password; the current-password field is hidden for someone who has just signed in with the initial password.`src/components/ChangePassword.tsx`

Users, permissions, scopes and audit log
The administration screen in its modes: users (role, template, password), permissions, scopes, audit log, and backup for the root account only.`src/components/AdminUsers.tsx`

Access card
SCHEDULE header, name, role caption and a QR code for the personal link, with copy-message, copy-link and save-card/save-code-as-image actions.`src/components/AccessCard.tsx`

Advising college and department picker
Lists only what the reader's scope allows and pins a sole option as a fixed label instead of a list.`src/components/AdvisingScopePicker.tsx`

Welcome tour
A tour shown once per account and role, with its own variant for seven roles.`src/components/Onboarding.tsx`

3.1.5 Requirements

FR-ACC-001 **Sign-in with user name and password**

The system shall verify the user name and password and open a session whose random identifier is held in an HttpOnly cookie with SameSite=Lax (and Secure in production), and shall answer unknown name, wrong password and deleted account with one identical message.

Implemented · Must · Functional · Standard user, Schedule committee chair, Department head, College dean

RATIONALE

> A single message does not reveal whether an account exists to someone guessing names.

ACCEPTANCE CRITERIA

1. A request without a name or password returns 400; a wrong password for an existing name returns 400 with the same message as an unknown name.
2. An inactive account returns 403, and a locked one returns 403, both only after the password has been verified.
3. The session identifier has the form sess_ followed by 64 hexadecimal characters and is not echoed in the response body.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/components/Login.tsx` `tests/owner-approved-2026-10-29-audit.ts` `tests/password-reset-audit.ts`

FR-ACC-002 **Rate limiting of anonymous requests**

The system shall limit sign-in attempts to five per address per minute, shall limit the public surface under /api/public to 60 requests per address per minute by default, and shall answer 429 when a limit is exceeded.

Implemented · Must · Functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. A sixth request to the sign-in route within 60 seconds returns 429, and a successful sign-in clears the counter for that address.
2. The same counter is shared by the change-password, demo sign-in and short invitation-link routes.
3. A public-surface refusal carries a Retry-After header; the limit is set by PUBLIC_RATE_LIMIT_MAX, the window by PUBLIC_RATE_LIMIT_WINDOW_MS, and 0 disables it.
4. The faculty card has two further limits: 240 requests per link and 1500 per address per minute.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/server/core/rateLimits.ts` `tests/run-tests.ts`

FR-ACC-003 Idle expiry and presence-based renewal

Inferred

The system shall end a session after 15 minutes of inactivity, renew it from user presence rather than a write on every request, and warn in the interface one minute before it ends.

Implemented · Must · Functional · Standard user, Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. The session record is renewed at most once every two minutes however many requests arrive.
2. The interface sends a beat to /api/auth/heartbeat every 4 minutes while active, and the server renews the session immediately, bypassing the throttle.
3. A warning appears 60 seconds before the end and the user is signed out automatically at 15 minutes.
4. A demo session lasts 60 minutes.

EVIDENCE

`src/server/core/auth.ts` `src/server/routes/authRoutes.ts` `src/App.tsx`

FR-ACC-004 Sign-out and session invalidation

On sign-out the system shall drop the cached identity before deleting the session record, clear the cookie, and keep sign-out available to every role including view-only ones.

Implemented · Must · Functional · Standard user, College dean, Dean of registration, Department advisor

ACCEPTANCE CRITERIA

1. No request after sign-out is served from the cached identity of that session.
2. The server sends Set-Cookie with Max-Age=0 for session_id.
3. The role guard allows POST /auth/logout for every role including College dean.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/server/roleGuard.ts` `tests/role-guard-audit.ts`

NFR-ACC-005 Anonymous-door limits: count failures, never block success

The system shall not query the database for a session identifier it did not mint, shall remember sessions that no longer exist, and shall limit lookups of nonexistent sessions or links per address without limiting the successful ones.

Implemented · Must · Non-functional · Public visitor, The system (automatic)

RATIONALE

> A university network is one address for thousands of people, so legitimate users must never meet a wall.

ACCEPTANCE CRITERIA

1. An identifier is accepted only as sess_ or demo_ followed by 64 lowercase hexadecimal characters.
2. A gone session is remembered for 30 minutes with a capacity of 20000 identifiers, evicting the oldest when full.
3. An address inventing identifiers is looked up for 120 of them in 10 minutes and the rest are read as signed out, with an absolute ceiling of 1200.
4. Public link pages count tokens matching nothing: 40, then 429 with Retry-After, and an absolute ceiling of 400 in 10 minutes.
5. Two thousand sessions from one address, two hundred of them expired, produce no refusals.

EVIDENCE

`src/server/anonymousGuard.ts` `src/server/core/auth.ts` `tests/anonymous-guard-audit.ts`

DR-ACC-006 Password storage and non-disclosure

The system shall store a password as a salted scrypt hash, shall never return the hash or the encrypted copy in any response, and shall show an administrator only whether a password exists through the HasPassword field.

Implemented · Must · Data · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. A stored hash begins with scrypt$ and is derived from a salt with a 64-byte derived key.
2. GET /api/users, GET /api/auth/me and the sign-in response contain neither SystemUserPass nor SystemUserPassVault.
3. A temporary reset password is stored as a hash only, with the encrypted-copy field empty.

EVIDENCE

`src/db/repository.ts#hashPassword` `src/server/core/scope.ts#safeSystemUser` `src/server/routes/usersRoutes.ts` `tests/password-reset-audit.ts`

FR-ACC-007 Password-change gate

The system shall deny an account flagged for a password change every /api path except /auth/, /public/, health, version and demo configuration, until the password is changed.

Implemented · Must · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. GET /api/schedules from a flagged account returns 403 with the code password-change-required.
2. /auth/change-password, /auth/me, /auth/logout, /auth/login and /health remain open to it.
3. The gate is mounted after identity resolution and before any data route.

EVIDENCE

`src/server/accountPassword.ts` `server.ts` `tests/owner-approved-2026-10-29-audit.ts`

FR-ACC-008 New-password rules and change

When an account owner changes their password the system shall require a length of 8 to 128 characters, reject the old shared password and the current password itself, and require the current password except in the waived case.

Implemented · Must · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A password of 7 or 129 characters is refused with 400 and a reason message.
2. The legacy shared first password and a value equal to the current password are both refused.
3. A successful change saves the hash, clears the mandatory-change flag and the initial-password deadline, and closes the account's other sessions but not the current one.
4. The service is unavailable in the demo environment and returns 403.

EVIDENCE

`src/server/accountPassword.ts` `src/server/routes/authRoutes.ts` `src/components/ChangePassword.tsx` `tests/first-password-room-loans-audit.ts` `tests/owner-approved-2026-10-29-audit.ts`

FR-ACC-009 Current-password waiver at first sign-in

The system shall not ask for the current password from someone who has just signed in with the initial password, provided three conditions hold together: the account is flagged for change, the session was opened with that same password, and less than 15 minutes have passed since it opened.

Implemented · Must · Functional · Department head, Schedule committee chair, Standard user

ACCEPTANCE CRITERIA

1. The session record carries a mark derived from the stored password hash, neither the password nor its hash, and it is not returned to the browser.
2. An administrator changing the password after the session opened, or more than 900 seconds passing, drops the waiver and a request without a current password returns 400 with the code current-password-required.
3. A team-member session (carrying a member identifier) is never waived.
4. The screen shows the current-password field when /api/auth/me says it is required.

EVIDENCE

`src/server/accountPassword.ts` `src/components/ChangePassword.tsx` `tests/first-login-password-audit.tsx`

FR-ACC-010 Initial password, flagging and lifetime

The system shall flag for a password change every account whose password an administrator sets for someone else, shall expire the default initial password 7 days after issue, and shall flag, give a deadline to, but not lock out, an account that signs in with the old shared password.

Implemented · Must · Functional · System administrator, Root administrator, Department head

ACCEPTANCE CRITERIA

1. Creating an account from Users sets MustChangePassword and sets a deadline only when the password is the default one.
2. Signing in exactly 7 days after issue returns 403 with the code initial-password-expired, and one moment earlier succeeds.
3. An existing account signing in with the shared password and no deadline receives one from that sign-in, and the shared password is refused as a new password.
4. A password an owner changes for themselves from Users loses only the initial-password deadline.

EVIDENCE

`src/server/accountPassword.ts` `src/server/routes/authRoutes.ts` `src/server/routes/usersRoutes.ts` `tests/first-password-room-loans-audit.ts` `tests/password-reset-audit.ts`

FR-ACC-011 Temporary password for reset

On reset the system shall issue the team head a random temporary password of 14 characters valid for 24 hours and a single sign-in, shall sign out their sessions, and shall limit an administrator to 12 resets in 10 minutes.

Implemented · Must · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Characters are drawn from 54 symbols excluding look-alikes (0 O 1 l I); the password is shown once and only its trace is stored.
2. A second sign-in with the same password from another device returns 403 with the code temporary-password-used, and after 24 hours with the code initial-password-expired.
3. A random initial password for a new account is 10 characters.
4. Any change of password after issue invalidates the reset record, and no secret appears in the audit trail.

EVIDENCE

`src/server/accountPassword.ts` `src/server/seatsRoutes.ts` `src/server/routes/authRoutes.ts` `tests/password-reset-audit.ts`

DR-ACC-012 Catalogue of the ten roles

The system shall define ten roles from a single source shared by server and interface, each with an identifier, label, one-line hint, read-only flag, granted screen numbers, scope mode, landing screen and display order.

● Implemented · Must · Data ·

System administrator, College dean, Vice dean for academic affairs, Dean of registration, University registrar, College registrar, Registration officer, Department head, Department advisor, Schedule committee chair, Standard user

RATIONALE

> The server guard is the real protection, so the definition of a role must not diverge from what the interface shows.

ACCEPTANCE CRITERIA

1. The ten roles in order: College dean 1, Vice dean 2, Dean of registration 3, University registrar 4, College registrar 4.5, Registration officer 5, Department head 6, Department advisor 6.5, Schedule committee chair 7, Standard user 8.
2. Screen numbers: Dean [14]; Vice dean, Dean of registration, University registrar and College registrar [14,8,9,16]; Registration officer [14,9]; Department head and Advisor [14]; Committee chair [7,14,8,9,10,16,17,6]; Standard user [7].
3. Only Schedule committee chair and Standard user are not read-only; scope modes are college (Dean, Vice dean, College registrar), all colleges (Dean of registration, University registrar), section (Department head, Advisor, Committee chair) and manual (Registration officer, Standard user).
4. An account with no role or an unknown role reads as Schedule committee chair, and an unknown role submitted on create or edit is refused with 400.
5. GET /api/roles returns the same fields to any signed-in user.

EVIDENCE

`src/utils/academicRoles.ts` `src/server/routes/usersRoutes.ts` `tests/role-guard-audit.ts` `tests/demo-roles-audit.ts`

FR-ACC-013 Server guard for read-only roles

The server shall refuse with 403 every write request from a read-only role regardless of its screens or interface tampering, deciding in a fixed order: read, unauthenticated, public, session, power account, writing role, then the exception paths.

● Implemented · Must · Functional ·

College dean, Vice dean for academic affairs, Dean of registration, University registrar, College registrar, Registration officer, Department head, Department advisor

ACCEPTANCE CRITERIA

1. Every read-only role is refused on every tested combination of write method and path, while the committee chair and standard user write.
2. Power accounts are above roles, and session paths (sign-in, sign-out, heartbeat, presence, mobile notification registration) are open to every role.
3. The refusal carries a message naming the role, the role identifier and readOnly:true.
4. The guard is mounted after identity resolution and before the first write route, and paths under /public/ are governed by the link token rather than the role.

EVIDENCE

`src/server/roleGuard.ts` `server.ts` `tests/role-guard-audit.ts`

FR-ACC-014 Narrow doors for read-only roles

The server shall open to read-only roles that take part in the approval cycle only the /approvals, /schedule-notes and /student-registration paths, shall exclude the Dean, Vice dean and Dean of registration from them entirely, and shall open named paths to the Department head, Advisor and University registrar.

Implemented · Must · Functional · Department head, Department advisor, University registrar, Registration officer, College registrar, College dean

ACCEPTANCE CRITERIA

1. The Department head writes to only four plan-review paths (rule, group, group/delete, course-details) and computes meeting times with POST /schedules/meeting-slots, but cannot write to /courses.
2. The Advisor passes only to /advisor-seats/add and /advisor-seats/exception and does not pass through the approval-cycle paths.
3. Only the University registrar writes PUT /campus-travel-policy; everyone except view-only roles writes /section-requests.
4. POST /report-stamps is open to read-only roles, and the route itself checks sign-in and scope.
5. A similar prefix such as /approvals-export is not opened.

EVIDENCE

`src/server/roleGuard.ts` `src/utils/academicRoles.ts` `tests/role-guard-audit.ts` `tests/demo-roles-audit.ts`

FR-ACC-015 Role template: screens and scope in one operation

When creating an account with a chosen role or changing its role the system shall write the role's screens and scope together, shall not rewrite the template on a save that leaves the role unchanged, and shall remove whole-college rows when the derivation type changes.

Implemented · Must · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. The all-colleges mode writes one row per stored college with department 0 and rewrites it on every save; the college mode writes one row per submitted college with department 0.
2. An empty college list sent explicitly withdraws the scope, while omitting the field leaves it as it is.
3. Moving from University registrar (all colleges) to Dean (college) deletes the university-wide rows before the new template is written.
4. Saving an account with its current role, even one with no stored role that reads as committee chair, does not erase manually granted screens.
5. Creating without naming a role grants screen 7 only; at start-up accounts with no role are migrated to Schedule committee chair and screen 7 is removed from every Department head.

EVIDENCE

`src/server/routes/usersRoutes.ts` `server.ts` `tests/approval-integrity-audit.ts` `tests/role-guard-audit.ts`

FR-ACC-016 User administration

The system shall let a holder of screen 11 list, create, edit and delete accounts, shall reject a duplicate login and missing fields, shall warn the administrator when a role does not match the name, and shall close the sessions of anyone whose password another person sets.

Implemented · Must · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Creating without a name, login or password returns 400; an already registered login returns 400 on both create and edit.
2. Setting a password for another account through PUT /api/users/:id deletes all of its sessions and empties the identity cache.
3. A review-role badge appears for a name containing the registration word with a non-registration role, or beginning with the dean word with a non-dean role; a warning, not a change.
4. Screen 11 is reserved for main system administration, so a committee chair cannot reach it even with a stale grant row.

EVIDENCE

`src/server/routes/usersRoutes.ts` `src/utils/academicRoles.ts` `src/components/AdminUsers.tsx` `tests/approval-integrity-audit.ts`

FR-ACC-017 Main-administration screens and the root account

Inferred

The system shall reserve screens 2, 3, 4, 5, 11, 12 and 15 for the main administration account (IsAdminUser or the root identifier, default 1), shall require a FormSecurity row for the other screens even from an administrator, and shall restrict the vault to the root account.

Implemented · Must · Functional · Root administrator, System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. A non-administrator receives 403 on a reserved screen even if a legacy permission row exists.
2. A missing row for an unreserved screen returns 403, and a request with no session returns 401.
3. The backup routes and the advisor-account consolidation return 403 for anyone but the root account.
4. Advisor-account consolidation touches only an inactive account that has no password.

EVIDENCE

`src/server/core/auth.ts` `src/server/routes/usersRoutes.ts` `src/server/routes/adminRoutes.ts` `tests/colleague-guard-measure-audit.ts`

FR-ACC-018 Permission and scope administration

Inferred

The system shall let screen 12 add, edit and delete permission rows, let screen 15 write scope rows, let a holder of 11 or 15 read scopes, and refuse a scope whose department does not belong to its college.

Implemented · Must · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. A permission row requires non-zero SystemUserId and FormNameId or the answer is 400.
2. Adding a scope with a department not in the college returns 400, and department 0 is rejected from this screen.
3. Duplicates are not removed automatically because the legacy database contains duplicated rows.

EVIDENCE

`src/server/routes/usersRoutes.ts` `src/components/AdminUsers.tsx` `tests/demo-roles-audit.ts`

DR-ACC-019 Screen and scope records

The system shall hold a screen permission as a row (legacy id, account, screen number from 1 to 17) and a scope as a row (legacy id, account, college, department) whose zero department means the whole college for the college and all-colleges roles only.

Implemented · Must · Data · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. The 17 screens are: Home, Colleges, Instructors, Departments, Terms, Courses, Schedules and Analytics Center, Instructor query, Room query, Time query, Users, Permissions, Indicators board, Department report, Scopes, Rooms and times, Advanced search.
2. A row does not require uniqueness of (account, screen), and saving reconciles surplus and missing rows by count.
3. Demo-environment accounts are seeded with permissions and scope derived from the role definition itself.

EVIDENCE

`src/db/repository.ts` `src/utils/academicRoles.ts` `src/server/routes/usersRoutes.ts` `tests/demo-roles-audit.ts`

FR-ACC-020 The scope decision isScopeAllowed

The system shall answer whether an account may see a department from one place: an administration account sees everything, an all-colleges role sees everything even without a row, a row with department 0 means the whole college for college and all-colleges roles only, and anything else matches literally.

● Implemented · Must · Functional ·

College dean, Vice dean for academic affairs, Dean of registration, University registrar, College registrar, Schedule committee chair

ACCEPTANCE CRITERIA

1. A college created after the account was saved is readable by the University registrar without adding a row.
2. A query with department 0 is answered yes if the account has any row in that college.
3. A committee chair with a stale department-0 row is not widened to the college; only a Dean is.
4. A non-administrator with no scope sees nothing.

EVIDENCE

`src/server/core/scope.ts` `tests/dean-notify-audit.ts` `tests/role-scope-audit.ts`

FR-ACC-021 Scope filtering and directed reads

The system shall filter lists with the same decision function and shall not trust a request's stated scope: a named department is checked, a college paired with another college's department is refused, and a whole-college read is allowed only to someone covering all its departments.

● Implemented · Must · Functional · College dean, Dean of registration, Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A data row lacking a department requires a literal match against a department-0 scope row and does not widen to everyone with anything in the college.
2. When nothing is named a department is chosen from inside the account's scope only, busiest first; the whole university is never read.
3. A Dean with one whole college is read at college level, and a Dean of registration with several colleges must choose one.

EVIDENCE

`src/server/core/scope.ts` `src/server/readScope.ts` `tests/dean-notify-audit.ts`

IR-ACC-022 Identity payload sent to the interface

On sign-in and in GET /api/auth/me the system shall send the interface the user without secrets, the role descriptor, the permissions resolved for the session, scope expanded with college and department names, the password-change state and the data mode.

● Implemented · Must · Interface · Standard user, College dean, Department advisor, Schedule committee chair

ACCEPTANCE CRITERIA

1. The role descriptor carries id, label, readOnly, landing, canReview, canManageDeadline, canExtendDeadline, watchesInbox, canAnnotate, signatureStage and canSign.
2. A scope row with department 0 is expanded to one row per department in the college with an AdCollegeWide marker, for display only; authorization reads the raw zero.
3. An account with one department in one college is not asked for college or department, one with a department at several sites is asked for the site, and others see college, department and term pickers.
4. An account with no written role describes itself as Schedule committee chair, but the server requires a written role wherever a prohibition or permission is built on it.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/utils/scopeShape.ts` `src/utils/scopeContext.ts` `src/components/AdvisingScopePicker.tsx` `tests/role-scope-audit.ts` `tests/single-department-scope-audit.ts`

FR-ACC-023 Hiding instructors' personal data from read-only roles

The system shall blank the civil ID and mobile fields of every instructor record delivered to a read-only role, in every route that returns the instructor directory.

Implemented · Must · Functional · College dean, Vice dean for academic affairs, Dean of registration, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. AdInstructorCivil and AdInstructorMobile reach the Dean, Department head and registration roles empty, and reach the Committee chair and system administration complete.
2. The fields are not removed from the shape; they remain with an empty value.

EVIDENCE

`src/server/core/scope.ts#instructorsForReader` `tests/approval-integrity-audit.ts`

FR-ACC-024 Provisioning department-head and advisor accounts from a previewed plan

The system shall show the administrator a plan of department-head and advisor accounts before writing, create only rows marked create, inactive and without a password, with a login derived from the department account prefix, and never create anything twice.

Implemented · Should · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. The department account Techdept proposes Techhead and Techadvisor; when the prefix repeats in another college the college code is appended.
2. A leading 01 prefix is removed from the login, and a second run yields exists for every row and creates nothing.
3. One advisor account per department covers the boys' and girls' colleges, and an account with no scope whose prefix is shared is classed review, not create.
4. Preview and creation are for the main administrator only.

EVIDENCE

`src/utils/departmentHeadAccounts.ts` `src/server/routes/usersRoutes.ts` `tests/department-head-accounts-audit.ts`

NFR-ACC-025 One identity resolution per request with a short cache

Inferred

The system shall resolve session, user, scope and screens once and hold them for two minutes per session, clear them at once on sign-out or any change to an account, scope or permission, and read a locked, inactive or deleted account as anonymous and delete its session.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. The cache lasts 120 seconds and is emptied by an identity-change announcement.
2. Session, user, permission and scope reads happen once per session, not per request.
3. A demo session without its sandbox is read as anonymous and never takes a real administrator identity.

EVIDENCE

`src/server/core/auth.ts`

FR-ACC-026 Demo environment and role switching

The system shall open an isolated demo environment with a demo_ session from POST /api/auth/demo, allow switching role among fictitious accounts carrying their real permissions, and keep administration screens read-only.

Implemented · Should · Functional · Public visitor, Root administrator

ACCEPTANCE CRITERIA

1. SCHEDULE_DEMO_ENABLED=false makes the route answer 404.
2. Writing to /users, /permissions, /user-scopes, /system-backup, /admin/teaching-load and /admin/university-day returns 403 in the demo.
3. POST /api/demo/role outside a demo request returns 404, an unknown role 400, and the same guard applies to the switched role.
4. Signing in with a name and password on an isolated demo service is refused with 403, and role switching is limited to 30 requests per minute.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/server/core/auth.ts` `tests/demo-roles-audit.ts` `tests/demo-every-role-audit.ts`

FR-ACC-027 Audit trail for sign-in and refusals

Inferred

The system shall record every successful write, sign-in, sign-out, refused sign-in attempt, permission refusal 403 and conflict 409, excluding heartbeat and presence, and let the administrator read the log.

Implemented · Should · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. A refused sign-in is recorded under the account tried as a refused sign-in attempt, and the password is never recorded.
2. GET /api/audit-logs returns 250 records by default (the limit parameter) and only an administration account can reach it; the screen asks for 500.
3. A refusal of a view-only role is recorded as refused (outside permission).

EVIDENCE

`server.ts` `src/server/routes/adminRoutes.ts` `src/server/routes/authRoutes.ts` `src/components/AdminUsers.tsx`

FR-ACC-028 Backup vault for the root account

Inferred

The system shall restrict full backup, its import, reset and undo to the root account, and decompress an uploaded archive as a stream with a ceiling of 80 megabytes after decompression.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Every /api/system-backup route returns 403 for anyone but the root account.
2. An archive exceeding 80 megabytes after decompression is abandoned when the ceiling is crossed with a refusal message, without blocking other requests.
3. Live session tokens and the server's own rollback copies are not exported.

EVIDENCE

`src/server/routes/adminRoutes.ts` `src/server/core/auth.ts`

FR-ACC-029 Invitation link and access card without sending the password to the server

The system shall carry the login name and initial password in the # fragment of the link, which the browser never sends to the server, shall limit the short /i/ link to the login name alone, and the interface shall clear the fragment from the address after reading it.

Implemented · Must · Functional · System administrator, Department head, Standard user

ACCEPTANCE CRITERIA

1. GET /i/:username answers 302 to /#invite= with Cache-Control: no-store and Referrer-Policy: no-referrer.
2. An empty name or one longer than 200 characters returns 400, and a link password is truncated at 128 characters.
3. Both #invite= and #invite& forms are read for older messaging links.
4. The short link is not completed with the shared password, so no guessed name becomes a ready invitation.

EVIDENCE

`src/utils/invitationLink.ts` `src/server/routes/authRoutes.ts` `src/components/Login.tsx` `src/components/AccessCard.tsx` `tests/scope-controls-audit.tsx` `tests/password-reset-audit.ts`

FR-ACC-030 Remember-this-device key for the faculty card

The system shall issue the instructor's device a key signed with HMAC and bound to the link token, internal instructor number and expiry, in an HttpOnly, SameSite=Strict cookie scoped to that link path alone, and carrying no civil ID.

Implemented · Should · Functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. The key is not valid for another link, after its expiry, or with a single character of its signature changed.
2. It expires at the earlier of the link end or 180 days.
3. The cookie is named sk_dev with path /api/public/staff/<link token> and is cleared by the forget-this-device action.
4. The secret is derived from the shared secret with the label staff-device-key-v1 so one key is never used for two purposes.

EVIDENCE

`src/server/staffDeviceKey.ts` `tests/staff-device-key-audit.ts`

3.1.6 Algorithms

ALG-ACC-01 · Identity, accounts, roles and access scope

Scope decision

Deterministic

May this account see a given department of a college?

1. An administration account sees everything, and an all-colleges role sees everything even without a scope row.
2. Department zero in the question means any department of that college.
3. A scope row with department zero means the whole college, for college and all-colleges roles only.
4. Otherwise a literal match of college and department.

Inputs

The user, role and scope rows, and the requested college and department.

Output

Yes or no.

One source of the decision; never widens an old row that merely happens to have the same shape.

`src/server/core/scope.ts#isScopeAllowed` `tests/role-scope-audit.ts`

`FR-ACC-020` `FR-ACC-021`

3.1.7 Edge cases

⚠ A college created after the dean's account was saved

Scenario A dean's account has one scope row with department 0, then a new department is added to the college.

Behaviour The dean reads it immediately with no new row, because the zero row means the whole college for college roles; a zero row on a committee chair does not widen scope.

⚠ Moving from an all-colleges role to a one-college role

Scenario A University registrar (one row per college) is changed to a Dean of one college from the Users screen.

Behaviour The derived whole-college rows are deleted before the new template is written, so the dean keeps no university-wide door.

⚠ Saving an account whose role has not changed

Scenario The administrator suspends an account or changes its password, and the interface sends the same role with every save.

Behaviour The server compares the resolved role, not the raw value, so the template is not rewritten and manually granted screens are not erased.

⚠ A stale initial-password session

Scenario An owner signs in with the initial password then leaves the screen for 20 minutes before choosing a password, or the administrator changes it meanwhile.

Behaviour The current-password waiver lapses, the server returns 400 with the code current-password-required and the screen shows the field.

⚠ A temporary password used from two devices at once

Scenario Two devices type the same temporary password at the same moment.

Behaviour The sign-in is counted inside an atomic write, so one succeeds and the other receives 403 with the code temporary-password-used.

⚠ A university network behind one address

Scenario Two thousand real sessions, two hundred of them expired, arrive from one address, while another address invents identifiers.

Behaviour Nobody from the network is refused because each success pays its lookup back, and the inventor stops after 120 failed lookups in 10 minutes.

3.1.8 Standing owner decisions

- The default initial password must be changed at first sign-in and lapses after seven days (owner decision 2026-11-05).
- Reset uses a random temporary password valid for 24 hours and one sign-in, and civil ID digits are never used as a password (owner decision 2026-11-06).

- Someone who has just signed in with the initial password is not asked for it again when choosing their own (owner decision 2026-10-30).
- Accounts are seats, not people: the holder changes while the seat and its scope remain (owner decision 2026-10-29, docs/SEATS.md).
- The Department head reads, comments and signs but does not edit a schedule; the committee builds.
- Printed reports keep their design; permissions change the data they show, never their layout.

↑ Contents

3.2 · ACD

Academic structure and reference data

Terms, colleges, departments, courses, faculty and visiting instructors: the reference every schedule is built on.

30	100%	83	66
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.2.1 Overview

This module holds the reference data without which no schedule can work: the term and its calendar, the college, the department, the course with its seven-digit official number, the faculty member with identity, rank and teaching load, and the offering state of a course in a term. It enforces input limits and delete guards so that no historical record is left without its reference. It also manages visiting instructors, the department visiting directory and the conversion of an instructor between visiting and appointed, and it detects duplicate courses and spelling twins and merges them only after explicit confirmation. System administrators and schedule committee chairs maintain the data; the other roles read it within their scope.

Root administrator · System administrator · Schedule committee chair · Department head · College dean · Registration officer · The system (automatic)

3.2.2 User journeys

System administrator — **Adding a new department to a college**

1. The administrator opens the Departments screen and picks the college.
2. He enters the department code and name; the code is up to 40 characters and the name up to 120.
3. If the code is already registered in the same college the request is refused with a clear message.
4. On success the department is created together with its own faculty-placeholder record, without duplicates.
5. The administrator adds the department's courses; the department can no longer be deleted while it has courses.

Schedule committee chair — **Converting a visiting instructor to appointed**

1. The schedule committee chair opens the faculty card and reads the current visiting state.
2. He chooses the appointed direction and the departments from which the secondment is lifted.
3. The dry-run preview lists every change: removal from the visiting directory and from the rosters of the current and future terms.
4. On confirmation the changes are applied to these records only; no schedule and no past-term roster is touched.

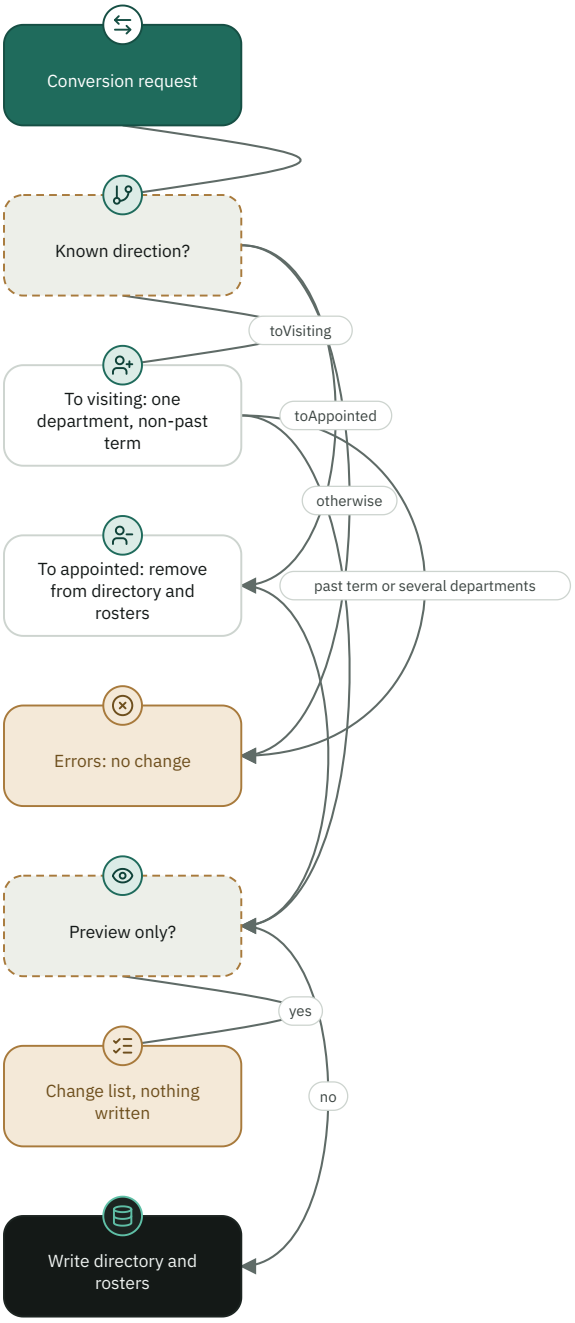
Root administrator — **Merging two spellings of one course**

1. The root administrator runs the spelling-twin scan and the groups appear per department.
2. For each copy he sees the appointment and sheet-link counts and the reason a merge is blocked, if any (student data or sheet rules).
3. He picks the keeper, the duplicate and the canonical name from the course's own spellings.
4. The scan is repeated at merge time and, after explicit confirmation, appointments and links move to the keeper and the duplicate is archived.

3.2.3 Diagrams

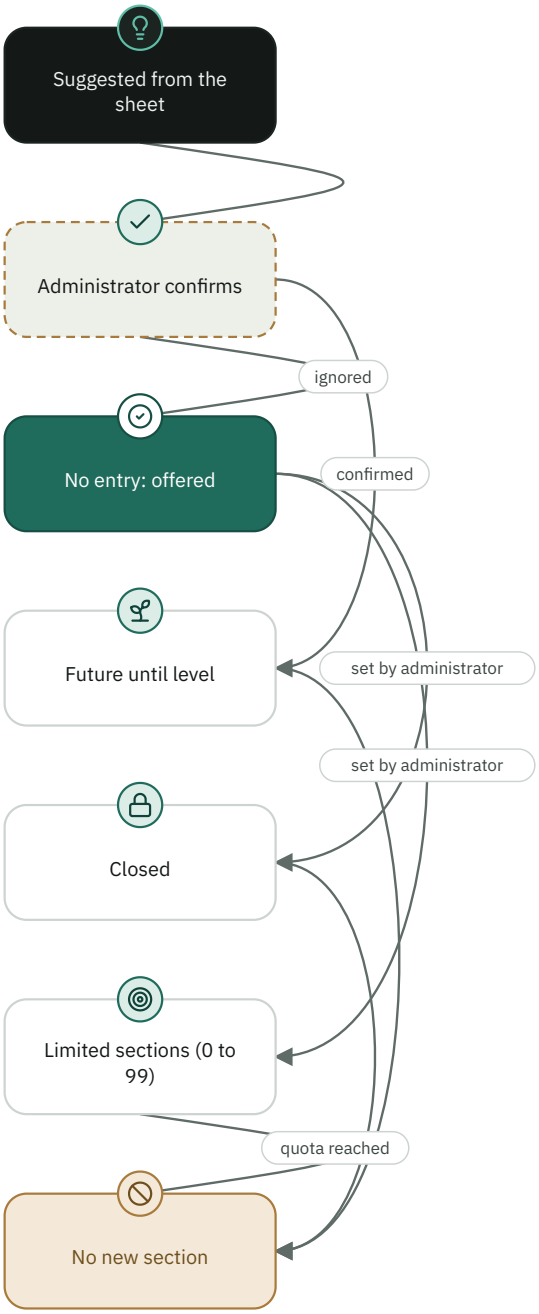
Visiting / appointed conversion

The plan is a pure function used for both preview and execution; execution writes only the visiting directory and the rosters of the current and future terms.



Course offering states

A course with no entry is offered; only a confirmed entry changes what the instructor and the proposal engines see.



Who reads and who writes reference data

	Terms	Colleges and departments	Courses	Faculty members	Visiting and offering state
Root administrator	✎	✎	✎	✎	✎
System administrator	✎	✎	✎	✎	✎
Schedule committee chair	👁	👁	✎	✎	✎
Department head	👁	👁	👁	👁	👁
College dean	👁	👁	👁	👁	👁
Registration officer	👁	👁	👁	—	—

✎ Write 👁 Read — No access

3.2.4 Screens

Terms
Create and edit a term with start date, number of weeks and the ended flag, with a suggested name for the next term.`src/components/Terms.tsx`

Colleges
College and its code, and the travel-time matrix between the college's buildings.`src/components/Colleges.tsx`

Departments
Department, its code and college, and the department's degree rule.`src/components/Sections.tsx`

Courses
Course catalogue with credits, hours, capacity, room type and teaching mode.`src/components/Courses.tsx`

Faculty members
Members in four groups: faculty, TA, on sabbatical, retired; with status, load and visiting conversion.`src/components/Instructors.tsx`

Visiting / appointed conversion
Preview of the changes, then execution, from the faculty card.`src/components/VisitingConversionPanel.tsx`

Course offering state
Offered, closed, future or limited sections for each course in a term and department.`src/components/CourseOfferingPanel.tsx`

Duplicate courses and spelling twins
Scan, then merge or archived delete after administrator confirmation.`src/components/SpellingTwinsMerge.tsx`

3.2.5 Requirements

FR-ACD-001 Term create, update and delete

The system shall let a holder of the terms permission create and edit a term with a name of at most 120 characters, a start date in YYYY-MM-DD form, a whole number of weeks from 1 to 30 and an optional ended flag.

Implemented · Must · Functional · System administrator, Root administrator

RATIONALE

> The shared calendar depends on the start date and week count; the ended flag is stored only as a real boolean so that "not stated" never turns into "open".

ACCEPTANCE CRITERIA

1. A request with no name or a name longer than 120 characters is refused with status 400.
2. A week value outside 1 to 30 or a date in another format is not stored and the term is saved with the remaining fields.
3. Updating a missing term returns 404.
4. Deleting a term with at least one schedule row returns 409 and deletes nothing, and deleting an empty term succeeds.

EVIDENCE

`src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `src/db/repository.ts#termCalendarFields` `src/components/Terms.tsx` `src/server/routes/referenceRoutes.ts` `tests/term-calendar-audit.ts`

FR-ACD-002 Automatic creation of the next term

The system shall create the next term by itself at start-up and every 24 hours: from 1 October the second term Y/Y+1, from 1 January the summer term (Y-1)/Y, from 1 April the first term Y/Y+1; it shall skip a term of the same name that already exists and shall not run in demo mode.

Implemented · Should · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. On 15 November 2026 the term named second term 2026/2027 is created with a Sunday start date and 14 weeks.
2. The summer term lasts 7 weeks and the first and second 14 weeks, anchored at 13 September, 31 January and 13 June to the nearest Sunday.
3. Re-running the job after creation does not duplicate the term.

EVIDENCE

`src/server/autoTerms.ts#runAutoTermJob` `src/utils/termSequence.ts#autoTermDue` `src/utils/termSequence.ts#TERM_CALENDAR` `tests/term-calendar-audit.ts`

FR-ACD-003 Automatic closing of an ended term

The system shall mark a term whose end date has passed as ended (AdTermClosed=true) by itself, unless it was genuinely re-opened after its end.

Implemented · Must · Functional · The system (automatic)

RATIONALE

> Owner decision of 2026-10-01: a stored false alone does not protect, because the edit form writes it on every save.

ACCEPTANCE CRITERIA

1. An unclosed term past its end date is closed on the first run and the event is logged.
2. A term re-opened (AdTermReopenedAt) at or after its end is not closed again.

EVIDENCE

`src/server/autoTerms.ts#closeEndedTerms` `src/utils/termSequence.ts#termHasEnded` `tests/term-calendar-audit.ts`

FR-ACD-004 Scope-aware term list

The system shall show a reader only the terms that have rows in his scope, always keeping the newest created term even when empty, and shall cache the result for 60 seconds per scope.

Implemented · Should · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. A past term with no rows in the reader's scope does not appear in his list.
2. The newest term appears even with no rows, and the current and newest term ids are returned with it.

EVIDENCE

`src/utils/scopedTerms.ts#visibleTermIds` `src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `tests/fixed-scope-terms-audit.ts`

FR-ACD-005 College CRUD and delete guard

The system shall allow creating and editing a college with a code of at most 40 characters and a name of at most 120, shall refuse with status 409 to delete a college that has departments, and shall restrict a non-administrator to the colleges in his scope.

Implemented · Must · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Creating a college without a code or without a name returns 400.
2. Deleting a college with at least one department returns 409.
3. The college list for a non-administrator is limited to his scope and sorted in Arabic.

EVIDENCE

`src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `src/components/Colleges.tsx` `tests/role-scope-audit.ts`

FR-ACD-006 Department, faculty placeholder and delete guard

The system shall allow creating, editing and moving a department between colleges within the user's scope, shall not register a duplicate department code in the same college, and shall create for each new department exactly one faculty-placeholder record with a synthetic 12-digit civil number starting with 29999.

Implemented · Must · Functional · System administrator, Root administrator

RATIONALE

> The placeholder receives sections that have no instructor yet and receives the visiting instructor in an imported schedule; its number cannot be confused with a person's because its month digits are 99.

ACCEPTANCE CRITERIA

1. Creating a department with a code that exists in the college returns 400 with a registered-code message.
2. The department's synthetic civil number is 29999 plus the college id in three digits plus the department id in four digits.
3. Calling the provisioning again for a department that already has its record creates no second record.
4. Deleting a department with at least one course returns 409 and deletes nothing.

EVIDENCE

`src/server/sectionFaculty.ts#ensureSectionFaculty` `src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `src/utils/placeholderInstructor.ts` `tests/section-faculty-audit.ts`

DR-ACD-007 Seven-digit authority course number

The system shall build every course number a reader sees (printed report, query or Excel) in seven digits: college (2), department (2), course (3), such as 0109105, through authorityCourseCode, and shall never display the stored CourseCode, CourseCodeSnapshot or AdCourseId.

● Implemented · Must · Data · The system (automatic)

RATIONALE

> A fixed owner rule: the catalogue usually stores only the last three digits, and the internal id means nothing to the reader.

ACCEPTANCE CRITERIA

1. authorityCourseCode("105", "0109") returns 0109105, ("5", "0109") returns 0109005, and Arabic-Indic digits are read as digits.
2. A code containing letters (ENG101), or given no department key, is returned as stored and no number is invented.
3. A seven-digit course of another department (0102102 under 0101) is not re-linked.
4. npm run test:course-number-seven-digit passes with no failure.

EVIDENCE

`src/utils/authorityAcademicCodes.ts#authorityCourseCode` `src/utils/authorityAcademicCodes.ts#authorityDepartmentCode` `tests/course-number-seven-digit-audit.ts` `tests/authority-course-prefix-audit.ts`

DR-ACD-008 Department key from college and department codes

The system shall derive the four-digit department key from the first two digits of the college code plus the two-digit local department code, so that the boys' and girls' branches share the same key, and shall never match a key against another department's key.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. authorityDepartmentCode("011", "09") and ("012", "09") both return 0109.
2. A department code of four digits that starts with another college's code is re-linked to the report's college: 0101 under college 02 becomes 0201.
3. 0201 does not match 0202.

EVIDENCE

`src/utils/authorityAcademicCodes.ts#authorityDepartmentCode` `src/utils/authorityAcademicCodes.ts#authorityDepartmentMatches` `tests/course-number-seven-digit-audit.ts` `tests/authority-course-prefix-audit.ts`

FR-ACD-009 Linking a course to its printed number

The system shall not link a course to the catalogue by the three-digit tail alone, and shall require the department key or the printed name to agree, so that 0114464 remains a different course from 0120464, and a linked row is kept under its printed number.

● Implemented · Must · Functional · The system (automatic)

RATIONALE

> Standing owner decision of 2026-11-05: the page header decides the department and the printed code does not change a course's identity.

ACCEPTANCE CRITERIA

1. A printed number 0114101 matches the catalogue row "101" of department 0120 only if the printed name agrees, and does not match the full row 0120101.
2. A six-digit number starting with the department key (013099 under 0130) is read as 0130099.
3. The row snapshot keeps the seven-digit printed number when its tail matches the catalogue row.

EVIDENCE

`src/utils/authorityAcademicCodes.ts#authorityInheritedCourseMatches` `src/utils/authorityAcademicCodes.ts#authorityCourseCodeMatches` `src/utils/authorityAcademicCodes.ts#authorityCourseSnapshot` `tests/authority-course-prefix-audit.ts`

FR-ACD-010 Course creation and numeric limits

The system shall accept course creation with a non-empty code and name and a college and department that belong together, with whole credits from 0 to 20, hours from 0 to 40 (fractions allowed) and a whole capacity from 0 to 200, and shall refuse anything else with a message naming the field.

Implemented · Must · Functional · System administrator, Root administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. Capacity 201, credits 21 or hours 41 returns 400 with a message naming the field and its range.
2. A department that does not belong to the chosen college returns 400.
3. A course code is not repeated within a department and a repeated attempt returns 400.
4. A room type longer than 60 characters, or a teaching mode other than theoretical, applied, practical or field_training, is refused.

EVIDENCE

`src/server/routes/coursesRoutes.ts#registerCoursesRoutes` `src/server/core/input.ts` `src/components/Courses.tsx` `tests/department-course-catalog-audit.ts`

FR-ACD-011 Freezing the identity of a course with history

Inferred

The system shall refuse with 409 to change a course's code or name when it has schedule history or belongs to a non-active sheet, shall still allow correcting credits, hours and capacity, and shall prevent moving it to another department while it is linked to a sheet.

Implemented · Must · Functional · System administrator, Root administrator, Schedule committee chair

RATIONALE

> Code and name protect links and the teaching history; the name is compared after normalising presentation forms, so that is not counted as a change.

ACCEPTANCE CRITERIA

1. Editing the name of a course that has one appointment in any term returns 409 and saves nothing.
2. Changing only the capacity of a course in a transition sheet succeeds.
3. Moving a course linked to a sheet to another department returns 409.

EVIDENCE

`src/server/routes/coursesRoutes.ts#registerCoursesRoutes`

FR-ACD-012 Block deleting a linked course

Inferred

The system shall refuse with status 409 to delete a course that is linked to an academic sheet or to a schedule row.

Implemented · Must · Functional · System administrator, Root administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. Deleting a course that is a member of a sheet returns 409 with a message directing to remove it from the sheet.
2. Deleting a course with a schedule row returns 409.

EVIDENCE

`src/server/routes/coursesRoutes.ts#registerCoursesRoutes`

FR-ACD-013 Room type and teaching mode sync across copies

The system shall, when a non-empty room type or teaching mode is saved on a course, apply it to its sibling copies with the same code in the same department family and college family, within the user's scope, and report what it skipped and why.

Implemented · Should · Functional · System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. The course copy at another campus of the same family changes; a course of another department with the same code does not change and is not mentioned.
2. A copy outside the user's scope is skipped and reported as outside his permissions; at most 8 skips are returned.
3. Copies are written before the edited course so that the next save redoes what was missed.

EVIDENCE

`src/utils/courseRoomTypeSync.ts#roomTypeSiblingIds` `src/server/routes/coursesRoutes.ts#registerCoursesRoutes` `tests/course-room-type-sync-audit.ts`

FR-ACD-014 Course tags in a fixed order

The system shall show every applicable course tag in one order on all screens: the sheet, then the requirement type (college, university, department or elective), then not reached yet, and shall show no tag for a mandatory course.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A course in the 2026 sheet that is a college requirement appears with two tags in that order: sheet, then requirement.
2. The last tag, not reached yet, does not print, and the requirement tag prints only for service courses.
3. The coverage guard fails if a course name is drawn on screen without tags and without a justified exception.

EVIDENCE

`src/utils/courseTags.ts#courseTagsOf` `src/components/CourseTags.tsx` `tests/course-tags-coverage-audit.ts`

FR-ACD-015 Duplicate course detection and archived delete

The system shall detect duplicate courses within one department by the same printed number, shall pick the keeper by usage priority (schedule, then students, then sheet rules, then capacity, then sheet link), shall offer for deletion only an unused duplicate, and shall rescan before an archived delete.

Implemented · Should · Functional · Root administrator, System administrator

ACCEPTANCE CRITERIA

1. Weights are 16 for schedule, 8 for students, 4 for rules, 2 for capacity and 1 for link, and the highest score is kept.
2. A duplicate with a schedule is shown with the reason has a schedule and is not removable.
3. Deletion reports the number deleted and then rescans.

EVIDENCE

`src/server/planReviewRoutes.ts` `src/components/DuplicateCoursesPanel.tsx` `tests/department-course-catalog-audit.ts`

FR-ACD-016 Spelling-twin merge with confirmation

The system shall let only the root account preview courses whose names are spelling twins and then merge them with explicit confirmation, so that appointments and sheet links move to the keeper, the canonical name is applied and the duplicate is archived, and shall refuse automatic merging of a course referenced by student data or sheet rules.

Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A merge request without confirm=true returns 428 and without the root account returns 403.
2. The scan is repeated at merge time: a group that changed returns 409 and a canonical name from outside the course's spellings returns 400.
3. A course used by students returns 409 with reason student data, and a course used in rules returns 409 with reason sheet rules.

EVIDENCE

`src/utils/courseSpellingTwins.ts#spellingTwinGroups` `src/utils/courseNameRelation.ts#courseNameRelation` `src/server/planReviewRoutes.ts` `src/components/SpellingTwinsMerge.tsx` `tests/course-spelling-merge-audit.ts` `tests/course-name-relation-audit.ts`

DR-ACD-017 Faculty identity: civil number

Inferred

The system shall store the civil number in one form (Latin digits only, with every non-digit removed), shall accept a 12-digit number that satisfies the check digit, and shall not register a number already held by another member.

Implemented · Must · Data · System administrator, Root administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. A number typed with Arabic digits or spaces is stored as clean Latin digits.
2. A number whose length is not 12 returns the exactly-12-digits message, and one with a wrong check digit is refused.
3. Editing with another member's number returns 400.

EVIDENCE

`src/utils/civilId.ts#validateCivilId` `src/utils/civilId.ts#normalizeCivilId` `src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `tests/role-scope-audit.ts`

FR-ACD-018 Quick instructor add and department enrolment

The system shall allow creating a faculty member by a holder of the instructors or schedule permission, shall enrol him in the department faculty directory when created from a department context within his scope, shall enrol an existing holder of a registered number instead of refusing, and shall not accept the faculty-placeholder name as a new instructor.

Implemented · Must · Functional · System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. A registered number with an in-scope department returns the existing record with existing=true and enrols it in the department.
2. A registered number with no department context returns 400 naming its holder so he can be searched.
3. The name faculty placeholder returns 400.
4. Edit and delete require the instructors permission (3); the schedule permission is not enough.

EVIDENCE

`src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `src/server/sectionFaculty.ts` `src/utils/placeholderInstructor.ts#isFacultyPlaceholder` `tests/section-faculty-audit.ts` `tests/instructor-apply-atomic-audit.ts`

FR-ACD-019 Instructor search by name identity

The system shall match instructor names with the single name-identity rule that folds hamza forms, alef maqsura, ta marbuta, diacritics, titles and spaces, shall find in search everyone the import matching finds, and shall match a civil number only when complete (12 digits).

● Implemented · Must · Functional · Schedule committee chair, System administrator

RATIONALE

> Partial civil-number matching across the university would let anyone complete the number of any employee.

ACCEPTANCE CRITERIA

1. Searching for one spelling of a name finds the other, and the joined and split forms of a compound name find each other.
2. Six digits of a civil number return nobody and the full 12 digits return its holder.
3. The faculty placeholder does not appear in the university search, and results are capped at 60 (default 40).

EVIDENCE

`src/utils/instructorIdentity.ts#foldInstructorText` `src/utils/instructorIdentity.ts#instructorIdentityKey` `src/utils/instructorPickerChoices.ts` `src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `tests/inbox-proposal-pickers-r2-audit.ts`

NFR-ACD-020 Civil number and phone hidden from non-members

The system shall not reveal the full civil number or the phone except to the department circle (those who taught in the reader's scope or are in his department directory or visiting list), to system administration and to the holder of the instructors screen, and shall show others the name and last four digits only, with no phone.

● Implemented · Must · Non-functional · Schedule committee chair, College dean, Department head

ACCEPTANCE CRITERIA

1. An instructor of another college who never taught in the reader's scope appears as a masked number ending in 1234 with no phone.
2. Read-only roles never see the civil number, and a full civil number that exactly matches the search is revealed to the searcher.
3. The faculty placeholder's synthetic number is never shown.

EVIDENCE

`src/server/routes/instructorsRoutes.ts#instructorCircleFor` `src/utils/placeholderInstructor.ts#withoutFacultyPlaceholderCivil` `tests/role-scope-audit.ts`

FR-ACD-021 Instructor status and category

The system shall allow marking an instructor retired or on sabbatical or classifying him as TA, shall not assign new work to a retired or sabbatical instructor, and shall make the TA-to-instructor conversion one-directional so an instructor cannot become a TA from the department tools.

● Implemented · Must · Functional · System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. isAssignableInstructor returns false for retired or sabbatical and true otherwise.
2. A section-roster update with category ta for a non-TA instructor returns 400.
3. The TA conversion plan counts the rows of explicitly tagged people only and never infers the category from a course number.

EVIDENCE

`src/utils/instructorStatus.ts#isAssignableInstructor` `src/utils/taConversion.ts#planTaConversion` `src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `tests/instructor-status-audit.ts` `tests/ta-conversion-audit.ts`

FR-ACD-022 Rank and teaching load limits

The system shall let the root account define up to 30 academic ranks, each with a name of at most 60 characters, minimum and maximum up to 60 hours and overtime up to 30 hours, all whole hours without fractions, and shall allow a per-member exception with a reason (administrative post, task, sabbatical, other) and an optional end term.

● Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A rank whose minimum exceeds its maximum, or has a fraction, or repeats a name is refused with one Arabic message.
2. A member exception is capped at 60 hours with overtime 30, and an unknown reason is refused.
3. Precedence is the member's active exception, then his rank; a member with neither is unset, with no invented number.

EVIDENCE

`src/utils/teachingLoad.ts#teachingLoadProblem` `src/utils/teachingLoad.ts#teachingLoadLimits` `src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `tests/teaching-load-audit.ts`

FR-ACD-023 Load computation and over-load warning

The system shall compute an instructor's term load as the sum of his courses' hours counted once per section (college, department, course, section code) however many appointments it has, shall classify it as unset, under, within, overtime or over, and shall keep the warning read-only so that it never blocks saving an appointment.

● Implemented · Should · Functional · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. A load above the maximum up to the ceiling (maximum plus overtime) is overtime, not an overrun.
2. Anything above the ceiling is over, with excess computed as the difference rounded to two decimals.
3. The feature is off by default: until enabled nothing is computed or sent.

EVIDENCE

`src/utils/teachingLoad.ts#teachingHoursByInstructor` `src/utils/teachingLoad.ts#evaluateTeachingLoad` `src/server/routes/referenceRoutes.ts#registerReferenceRoutes` `tests/teaching-load-audit.ts` `tests/instructor-course-load-audit.ts`

FR-ACD-024 Visiting status through directory and term roster

The system shall treat an instructor as visiting for a department when he is in the department's visiting directory (its family across colleges) and has an instructor record, shall let the term roster decide who teaches in that term, shall show the visiting badge per department and not per person, and shall give all screens one lookup cached for one minute.

● Implemented · Must · Functional · Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. A name in the roster but not in the directory does not appear as visiting, and a directory id with no instructor record is ignored.
2. An instructor visiting in one department is not visiting in another department where he teaches as a member.
3. GET /api/visiting-lookup without termId returns empty scopes and the reader is limited to his departments.

EVIDENCE

`src/server/core/visiting.ts#readVisitingStatus` `src/utils/liveVisiting.ts#liveVisitingIds` `src/utils/visitingLookup.ts#isVisitingIn` `src/components/VisitingBadge.tsx` `tests/visiting-lookup-audit.ts` `tests/visiting-status-permanent-audit.ts` `tests/visiting-badge-group1-audit.ts`

FR-ACD-025 Visiting roster and copy between terms

The system shall let a holder of the schedule permission add a new visiting instructor (valid civil number and a name of 3 to 100 characters) to the department directory and the term roster, and copy the visitors of one term into another while adding them to the directory, within his scope.

Implemented · Should · Functional · Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. A copy with no ids takes all visitors of the source term, and an id not in the source is ignored.
2. Copying into a term that already has visitors merges without duplicates.
3. A department outside the scope returns 403.

EVIDENCE

`src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `src/server/core/visiting.ts#readLiveVisitingRoster` `tests/visiting-lookup-audit.ts`

FR-ACD-026 Converting an instructor between visiting and appointed

The system shall convert an instructor to visiting for one department in a current or future term, and to appointed by removing him from the directory of the chosen departments and from the rosters of the current and future terms, with a preview that writes nothing, and shall never touch a schedule or a past-term roster.

Implemented · Must · Functional · Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. Converting to visiting with more than one department returns the one-department-at-a-time error, and with a past term returns the past-roster error.
2. A request with more than 50 departments is cut to 50, and a direction other than toAppointed or toVisiting returns 400.
3. Lifting from a department whose directory does not contain him returns an error, and adding an already-visiting instructor returns already-visiting.
4. Every department in the request is checked against the reader's scope, otherwise 403.

EVIDENCE

`src/utils/visitingConversion.ts#planVisitingConversion` `src/db/visitingConversion.ts#runVisitingConversion` `src/server/routes/instructorsRoutes.ts#registerInstructorsRoutes` `src/components/VisitingConversionPanel.tsx` `tests/visiting-conversion-audit.ts`

FR-ACD-027 Visiting instructor becomes faculty placeholder on import

The system shall, on schedule import, reassign every row whose instructor is visiting to the department's faculty-placeholder record without touching the visiting status, shall restore to a retired or sabbatical instructor his row by the full printed name when it matches exactly one record and no active instructor shares it, and shall leave rows unchanged when the department has no placeholder record.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A row whose instructor is among the visiting set is switched to the placeholder id and carries replacedVisitingInstructorId.
2. With placeholderId equal to zero or no visitors no row changes and zero is returned.
3. Recovering a retired instructor's name requires two or more tokens and an exact match, and is skipped when an active instructor shares the name.

EVIDENCE

`src/utils/visitingToFaculty.ts#applyVisitingToFaculty` `src/utils/instructorIdentity.ts#recoverRetiredFromPrintedName` `tests/visiting-to-faculty-audit.ts`

FR-ACD-028 Course offering state per term

The system shall store for each course in a term and department one of four offering states: offered (the default with no entry), closed, future with a level of 100 to 400, or limited sections with a maximum of 0 to 99, shall not store a bare offered entry, and shall apply it to the proposal engines, the card and the inbox.

Implemented · Should · Functional · Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. A level written as 3 is stored as 300, and a limited course capped at 5 with 3 existing sections has 2 remaining.
2. A closed or future course gets no proposed new section (allowed count zero).
3. An unconfirmed suggestion changes nothing, and the open-for-offering decision persists unless explicitly sent.
4. Save and read for a department outside the scope return 403.

EVIDENCE

`src/utils/courseOffering.ts#normalizeOfferingEntry` `src/utils/courseOffering.ts#newSectionsAllowed`
`src/server/routes/offeringRoutes.ts#registerOfferingRoutes` `src/server/offeringStates.ts#offeringStatesForTerm`
`src/components/CourseOfferingPanel.tsx` `tests/prepublish-reach-audit.tsx` `tests/owner-features-2026-10-28-audit.ts`

IR-ACD-029 Scoped course reads with sheet tags

The system shall return GET /api/courses sorted in Arabic and limited to the reader's scope, and shall accept sectionId for one department, operational=1 for operational courses and plans=1 to add the sheet tag, the sheet side and the departments that need a service course.

Implemented · Should · Interface · Schedule committee chair, Department head, Registration officer

ACCEPTANCE CRITERIA

1. A response for one department holds only that department's courses (a fraction of about 1,400 courses university-wide).
2. With plans=1 the course carries curriculum, sheetSide and serviceFor where they apply.
3. The accepted permissions are 6, 7, 8, 9, 10, 14, 16 and 17.

EVIDENCE

`src/server/routes/coursesRoutes.ts#registerCoursesRoutes` `src/server/planBadgeShared.ts` `tests/sheet-membership-audit.ts`

NFR-ACD-030 Reference-data input validation

Inferred

The system shall validate every textual or numeric reference-data input with an explicit bound (text up to 120 characters, code up to 40, numbers whole within a range), shall normalise Arabic-Indic digits, shall never store NaN, an object or an array in place of a name, and shall answer with one Arabic message naming the field.

Implemented · Must · Non-functional · System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. A name sent as an array, a boolean or longer than 120 returns 400.
2. The Arabic-Indic string for 45 is read as 45 in integer fields.
3. An empty capacity field means not stated and takes the default 0.

EVIDENCE

`src/server/core/input.ts#cleanText` `src/server/core/input.ts#boundedInt`
`src/server/routes/coursesRoutes.ts#registerCoursesRoutes` `tests/comprehensive-verification.ts`

3.2.6 Algorithms

ALG-ACD-01 · Academic structure and reference data

Duplicate course and spelling-twin detection

Deterministic

Which two courses of a department are one course written twice, and which is kept?

1. Course identity: a short number is expanded with the department code to seven digits (105 → 0109105); same identity with the same name = duplicate.
2. Spelling: the name is normalised (hamza forms, ta marbuta, alef maqsura, the article, spaces) yielding same, spelling, related or different; the first two group twins.
3. The keeper is chosen by usage score: schedule 16, students 8, sheet rules 4, capacity 2, sheet link 1, then shorter code, then id.
4. Only an unused duplicate is offered for deletion, with a rescan before delete; a spelling merge needs explicit confirmation and a course referenced by student data is not merged.

Inputs

Department courses, college and department codes, schedules, student data, sheet rules and links.

Output

Duplicate groups with a keeper and removable or mergeable duplicates with reasons.

Preview first; no silent delete or merge; root account only.

`src/utils/courseNameRelation.ts#courseNameRelation` `src/utils/courseSpellingTwins.ts#spellingTwinGroups` `src/server/planReviewRoutes.ts` `tests/course-spelling-merge-audit.ts` `tests/course-name-relation-audit.ts`

FR-ACD-015 FR-ACD-016

ALG-ACD-02 · Academic structure and reference data

Instructor name-identity matching

Deterministic

Is this typed or printed name this registry instructor, without ambiguity?

1. Text is folded (hamza, alef maqsura, ta marbuta, diacritics, titles) and split into tokens; the fragments 'abd/al/abu/bu' are joined to the next token.
2. Match: equal tokens, equal without spaces, family rotated to the front (3+ names), the registry name as the prefix of the printed one, or first and family with the middle names in order.
3. A link is accepted only when exactly one person matches; otherwise candidates are named (exact, or partial sharing two names).

Inputs

The written name and the instructor registry.

Output

One instructor, or 'unregistered', or 'ambiguous' with candidates.

No containment in the middle or end (a father is not his son) and no choice between two people; a civil number matches only when complete.

`src/utils/instructorIdentity.ts#instructorIdentityTokens` `src/utils/instructorIdentity.ts#sameInstructorIdentity` `src/utils/instructorIdentity.ts#uniqueExactIdentityMatch` `src/utils/instructorIdentity.ts#registryCandidatesFor` `tests/pdf-import-hardening.ts` `tests/pdf-import-regression.ts`

FR-ACD-019 FR-IMP-013

ALG-ACD-03 · Academic structure and reference data

Teaching-load computation and classification

Deterministic

How many hours does a member teach this term, and where does that fall against his load?

1. Each section (college, department, course, section code) is counted once for its instructor at its course hours, however many appointments it has.
2. Limits: the member's active exception first, then his academic rank; ceiling = maximum + overtime.
3. Classification: above the ceiling = over, between maximum and ceiling = overtime, below minimum = under, otherwise within, and no limits = unset.
4. An exception expires when the computed term is chronologically after its end term.

Inputs

Term rows, course hours, rank and exception configuration, and the term list.

Output

Hours, state, excess and overtime used.

A read-only warning that never blocks saving; no invented figure when no load is set.

Up to 30 ranks, maximum up to 60 hours, overtime up to 30.

`src/utils/teachingLoad.ts#teachingHoursByInstructor` `src/utils/teachingLoad.ts#teachingLoadLimits`
`src/utils/teachingLoad.ts#evaluateTeachingLoad` `tests/teaching-load-audit.ts`

FR-ACD-022 FR-ACD-023

3.2.7 Edge cases

⚠ Three-digit course tail across similar departments

Scenario The authority report prints 0114464 while another department's catalogue holds course 464 under department code 0120.

Behaviour It is not linked by the tail alone: the printed name must agree and the file must not print the department's own number with the same tail, so 0114464 stays different from 0120464, and when no link is proven the course is created, after a single confirmation, under its printed number.

⚠ Editing the name of a course with history

Scenario An administrator corrects the spelling of a course name that has appointments in past terms.

Behaviour It is refused with 409 if the code or name changes after presentation-form normalisation, and the administrator is directed to create the new course and link it from the sheets screen; a spelling twin is handled by the confirmed merge, not by editing.

⚠ Visiting in one department, member in another

Scenario An instructor is seconded to a department in one college and teaches as a member in a different department.

Behaviour The visiting badge appears beside his name only in the first department's rows, because a visitor is known by department and term; the department is a family across colleges, so its sister's directory in the other college is read.

⚠ Visitor removed from the directory but left in an old roster

Scenario A name is removed from the department visiting directory but remains in a past term's roster.

Behaviour The old roster stays as historical evidence and is never rewritten, but it does not keep a live visiting badge in today's schedule because the directory is the authority for live membership.

⚠ New department without a faculty-placeholder record

Scenario A newly created department with no history, and a coordinator opens its instructor list.

Behaviour Its record is created when the department is created; if that fails the department is not lost and the record is retried at first import. While it has none, other departments' records are offered for selection, its own college's first and then the university's, and only once.

⚠ Member with an explicit zero load

Scenario A member with full exemption: maximum 0 and overtime 0.

Behaviour An explicit zero is a real limit with ceiling 0, not no limit, so any teaching hour for him is over; the absence of both rank and exception means unset with no warning.

3.2.8 Standing owner decisions

- Course numbers are seven digits only: college (2), department (2), course (3), such as 0109105; the stored code and AdCourseId are never shown, the number is built with authorityCourseCode and guarded by test:course-number-seven-digit.
- The page header decides the department in the authority report (SWRSCHA) even when a course number carries another department's code; a missing department or course is created only after one confirmation under its printed number and name with the usual permission, a three-digit tail alone never links, and scans never create.
- Printed reports keep their design: values and data are fixed, not the layout, and any unavoidable layout change reuses the existing print classes and is checked on a Chrome-printed A4 landscape PDF before merging.
- A term whose end date has passed is declared ended automatically (decision of 2026-10-01), and only a genuine re-opening after its end protects it.

↑ Contents

3.3 · LOC

Locations

One official registry of buildings and halls, hall barter between departments, door pages and occupancy

30	100%	31	46
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.3.1 Overview

The module keeps the official registry of buildings and halls on which every schedule appointment is built, and decides who may book which hall. It lets a department borrow a free window of another department's hall on the same campus, with the host's approval and for one term only. It publishes a public door page for each hall showing its state now (free, busy or starting soon) and a corridor board for each building, with a printable QR sticker. It also covers occupancy figures, the companion facility of a hall and the travel policy between campuses.

Root administrator · System administrator · Department head · Schedule committee chair · University registrar · Public visitor · Student · Faculty member · The system (automatic)

3.3.2 User journeys

Schedule committee chair — **Borrowing a hall window from another department**

1. The committee chair opens Hall barter for the department and term; free windows in the college halls are listed per day.
2. She narrows the list by host department, day or building and reads the fit warning if the hall is specialised or smaller than the department's students.
3. She picks a window and requests it; the request goes digitally to the host department and no appointment changes.
4. The host department approves and the window becomes digitally reserved for the requesting department, visible to both.
5. She adds the lecture from the appointments screen inside the approved window only; anything outside is refused with a message naming her windows.

Department head — **Answering a barter request and taking the hall back**

1. The host department head sees the number of waiting requests and the age of the oldest in the side bar.
2. He opens incoming requests and approves or rejects; hall usage is re-read inside the lock before approval.
3. He uses his hall freely outside the lent window, but not inside it until the loan is ended.
4. He ends the loan and the window returns to his department, unless the borrower teaches in it, in which case ending is refused until that appointment is moved.

Root administrator — **Adding a building and its halls and printing its door stickers**

1. The root administrator picks the campus and enters the building number; the official code is the site prefix plus two digits (for example 012B07).
2. He adds the building's halls by code in one go and links them to one or more departments; a hall linked to several departments is shared.
3. He opens the building links and gets the building board, the corridor screen and a link for each hall.
4. He prints an A4 sheet with four QR stickers and fixes each sticker on its hall door.

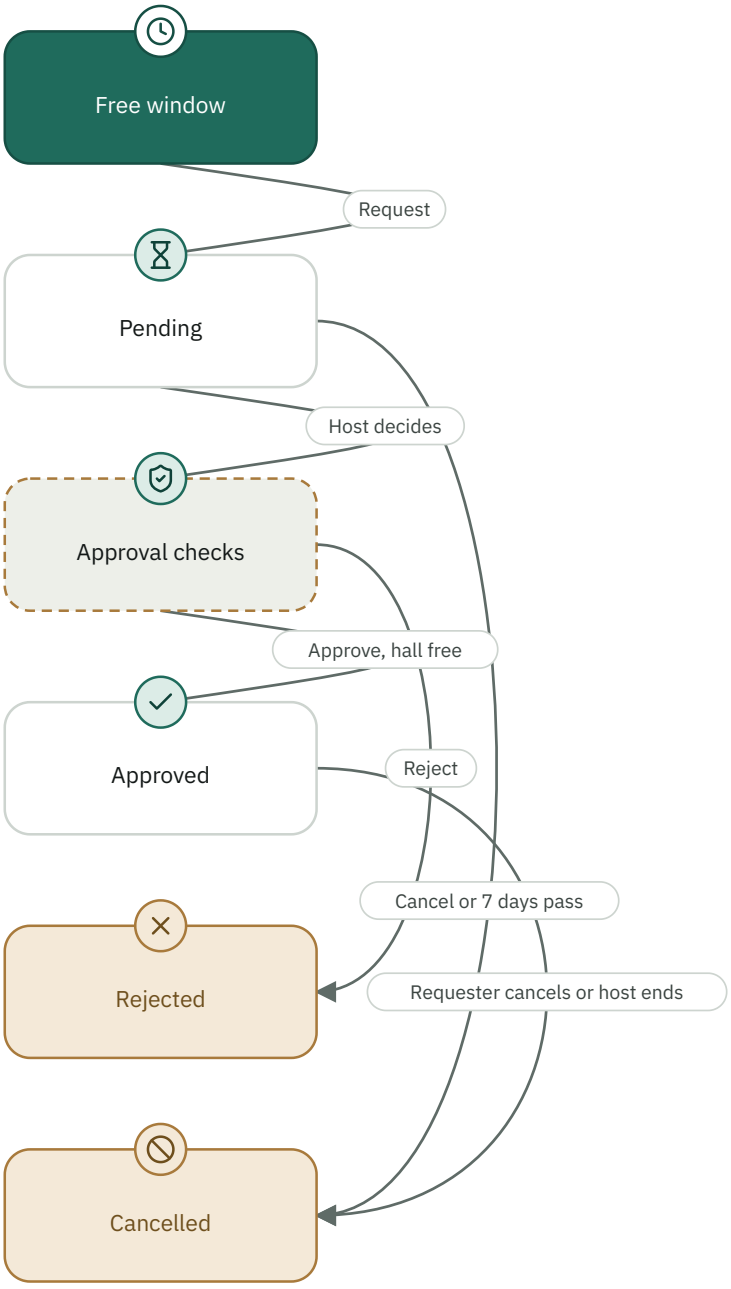
Public visitor — **Scanning a hall door sticker**

1. A student or visitor scans the QR code on a hall door between lectures.
2. The hall page opens without sign-in and says free until when, busy until when, or starting in a few minutes.
3. He follows the building link to see the board of the other halls and finds a free one nearby.

3.3.3 Diagrams

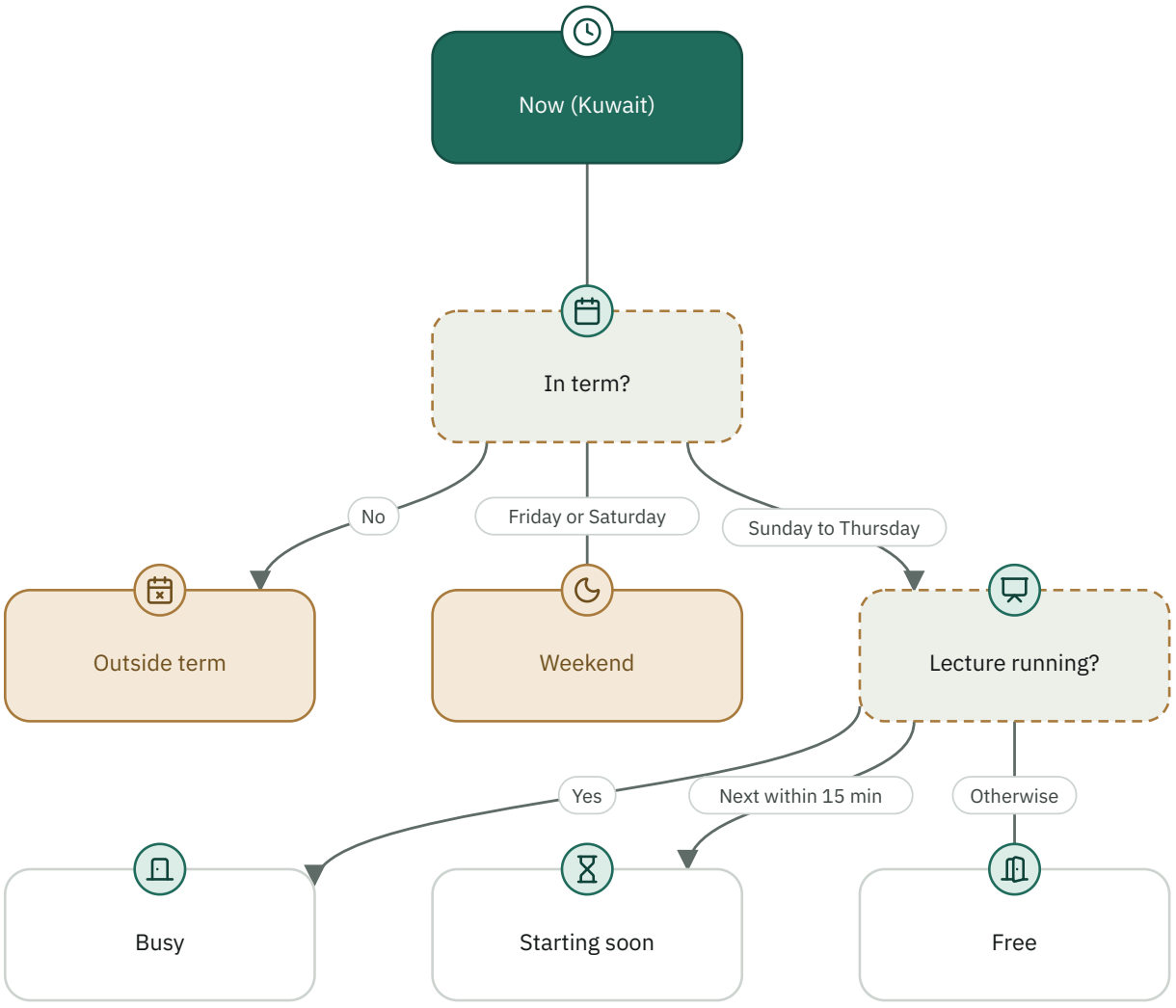
Barter request lifecycle

Request states as stored, and the guard on each transition.



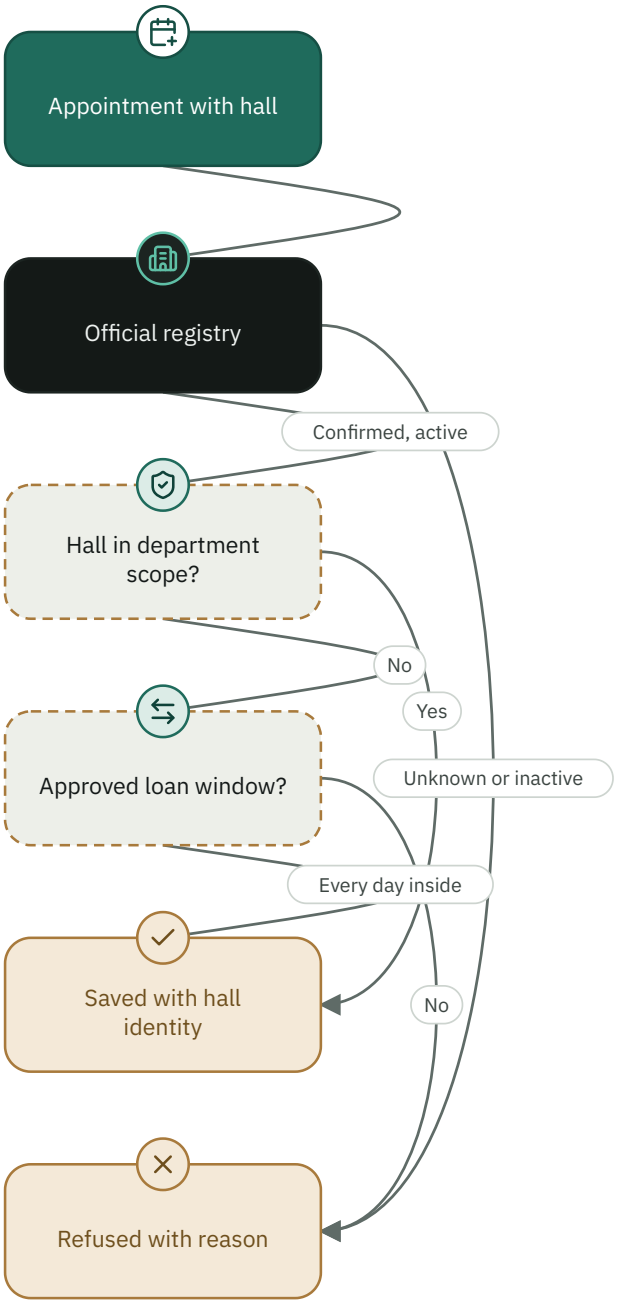
Hall state on the door

What the door page shows at a given instant in Kuwait time.



Location check on save

What an appointment passes before it is saved with a hall.



Permissions in Locations

	Read registry	Edit halls and buildings	Add and deactivate (root only)	Request barter	Host response and ending	Door links and stickers	Door pages	Travel policy
Root administrator	✎	✎	✎	✎	✎	✎	👁	✎
System administrator	✎	✎	—	✎	✎	✎	👁	✎
University registrar	👁	—	—	—	—	—	👁	✎
Department head	👁	—	—	✎	✎	—	👁	—
Schedule committee chair	👁	—	—	✎	✎	—	👁	—
Faculty member	👁	—	—	—	—	—	👁	—
Student	—	—	—	—	—	—	👁	—
Public visitor	—	—	—	—	—	—	👁	—

✎ Write 👁 Read — No access

3.3.4 Screens

Add buildings and halls
The single registry screen of the root administrator; the old Buildings and halls link opens it. It also carries the door links and stickers.`src/components/LocationQuickSetup.tsx`

Location picker
Building and hall choice in the appointment form, naming the sharing departments and the owner of a borrowed hall.`src/components/LocationPicker.tsx`

Hall barter
Board of free windows and the department's outgoing and incoming requests, refreshed automatically every minute.`src/components/HallBarterBoard.tsx`

Loan hours strip
One weekly drawing of what the department lent and borrowed.`src/components/RoomLoanStrip.tsx`

Hall types
Manage hall types and assign a type to departments.`src/components/LocationRoomTypeManager.tsx`

Door tools
Door stickers, the corridor screen and a QR code for each hall.`src/components/RoomDoorTools.tsx`

Occupancy

Companion facility

Day by hour grid with lecture counts and hall occupancy; selecting a cell opens its panel.`src/components/OccupancyHeatGrid.tsx`

Pick a studio or lab booked alongside the main hall, with availability for the appointment time.`src/components/CompanionFacilityPicker.tsx`

Travel between campuses

Registrar card inside the submission deadlines.`src/components/CampusTravelPolicyCard.tsx`

3.3.5 Requirements

DR-LOC-001 **Official building identity**

Every building shall carry an official code made of the four-character site prefix followed by a two-digit building number, and the prefix shall be derived from the college name through a fixed official table of twelve sites.

Implemented · Must · Data · Root administrator

RATIONALE

> The official code is what imported schedules and past terms match on, so it is never guessed or re-based.

ACCEPTANCE CRITERIA

1. The girls' Basic Education college with building number 7 yields 012B07, and number 0 or more than three digits is rejected.
2. A college with no official site prefix cannot receive a building and the reply says the reference code is missing.

EVIDENCE

`src/utils/locationCollegePrefixes.ts` `src/server/routes/locationsRoutes.ts` `tests/location-registry-audit.py`

DR-LOC-002 **Hall record**

A hall code (1 to 12 Latin letters or digits including at least one digit, never PENDING_ROOM or a placeholder) shall identify the hall inside its building without duplicates, and the hall may carry an integer capacity from 1 to 5000, a type up to 60 characters and one of three booking policies: normal, companion only or blocked.

Implemented · Must · Data · Root administrator, System administrator

ACCEPTANCE CRITERIA

1. A code such as ZX901 is accepted; a code with no digit, over 12 characters or a placeholder such as TBA is rejected.
2. A capacity of 0, 5001 or a fraction is rejected with the range message, and a policy outside the three is rejected.
3. A code already in the building returns 409.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/utils/locationRegistry.ts` `tests/rooms-one-screen-audit.ts` `tests/companion-rooms-audit.ts`

DR-LOC-003 Registry confidence and seed

Only active and confirmed buildings and halls shall appear in any picker, and the registry seed bundled with the software shall be merged with the stored registry by adding missing rows only, never replacing stored ones, and not merged into the demo sandbox.

Implemented · Must · Data · The system (automatic), Root administrator

RATIONALE

> The seed comes from the real university history and holds 31 buildings (28 confirmed, 3 probable), 355 halls (265 confirmed, 90 probable) and 444 review cases; a probable row stays inactive until a person confirms it.

ACCEPTANCE CRITERIA

1. Every building and hall with probable confidence in the seed is inactive.
2. A demo request reads its own sandbox registry without seed rows; the registry cache expires after 60 seconds and is cleared on every edit.

EVIDENCE

`src/server/locationRegistryEngine.ts` `src/server/core/locations.ts` `src/generated/locationRegistrySeed.ts` `tests/location-registry-audit.py`

FR-LOC-004 Add a building

The system shall let only the root administrator add a building by choosing the site, the building number and an optional department, and shall reject a duplicate code or one that does not follow the site prefix.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Adding an existing building returns 409 stating the building already exists.
2. A department that does not belong to the chosen site returns 400.
3. A user other than the root account gets 403.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/components/LocationQuickSetup.tsx` `tests/rooms-one-screen-audit.ts`

FR-LOC-005 Add halls in bulk

The system shall let the root administrator add 1 to 100 halls to an active building in one request, normalising codes (NFKC, upper case, no spaces) and linking them to at least one department of the same site.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Three codes with a department return 201 and created equals 3.
2. A hall linked to more than one department is saved as shared, and any code already in the building fails the whole request with 409.
3. More than 100 codes or a department from another site is rejected with 400.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `tests/rooms-one-screen-audit.ts`

FR-LOC-006 Edit a building

The system shall allow editing a building's site name, branch name, description, active state and departments from one form, an omitted field leaving its value unchanged, and departments outside the building's official site shall not be linked.

Implemented · Must · Functional · Root administrator, System administrator

ACCEPTANCE CRITERIA

1. Saving active=false with one department returns the building inactive with one department; a later save of the description alone keeps it inactive.
2. Emptying the departments means all departments.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/components/LocationQuickSetup.tsx` `tests/rooms-one-screen-audit.ts`

FR-LOC-007 Deletion is deactivation

The system shall never remove a building or hall from the registry: deleting a hall marks it inactive, deleting a building archives it and deactivates every active hall in it, with an entry in the row's audit history and past appointments still pointing at them.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Deleting a hall returns active=false and the row stays with its historical forms.
2. Deleting a building returns the count of deactivated halls and the building row remains.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `tests/rooms-one-screen-audit.ts`

FR-LOC-008 Rename and move a hall

The system shall let the administrator change a hall's code or move it to another building, silently recording the old code once as a historical form with no clash with another hall, updating the hall's appointments (current and future terms on rename, all terms on move), and reverting both hall and appointments if the write fails.

Implemented · Must · Functional · System administrator, Root administrator

RATIONALE

> A full system backup used to precede the move and exceeded the request timeout so nothing moved; the undo is now row by row from the values read before the write.

ACCEPTANCE CRITERIA

1. Changing ZX901 to ZX911 records ZX901 as a confirmed form and a second save without a code change does not duplicate it.
2. Returning to the original code drops it from the forms, and a code owned by another hall in the building is not recorded as a form.
3. The reply returns updatedScheduleCount.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/utils/locationAliasHistory.ts` `tests/rooms-one-screen-audit.ts`

FR-LOC-009 The list is the permission

The system shall list for each department only halls registered to it, shared with it, borrowed under an approved window this term, pinned in its directory, or ownerless in its branch sites, shall not allow typing a hall by hand, and shall name the sharing departments on a shared hall and the owner on a borrowed one.

Implemented · Must · Functional · Department head, Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. A hall registered to another department does not appear for mine until a loan is approved, then borrowedFrom shows its owner.
2. If the department is left with no building at all, its college's buildings open so it can choose one and then Pending hall confirmation.
3. An administrator without a department reads the raw registry with the type scopes.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/components/LocationPicker.tsx` `src/utils/locationRegistry.ts` `tests/room-types-by-department-audit.ts` `tests/instructor-request-room-universe-audit.ts`

FR-LOC-010 Location check on every save

On saving any appointment the system shall check that the building and hall are known, active and confirmed, that the hall belongs to the building, that the building is within the appointment's college or its branch (three-digit code root), and that the hall is within the department's scope unless shared or inside an approved loan window, and shall accept Pending hall confirmation with a warning.

Implemented · Must · Functional · Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. A Jahra hall on an appointment opened from the main campus is not rejected for a different college, whereas another branch (boys versus girls) is rejected.
2. A Tuesday appointment outside a Sunday loan window is refused with a message naming the department's approved windows, up to four, with days and times.
3. An appointment with a pending hall is saved with a pending_room warning.

EVIDENCE

`src/server/locationRegistryEngine.ts` `src/server/core/locations.ts` `tests/location-registry-audit.py` `tests/instructor-request-availability-audit.ts`

FR-LOC-011 Registry initialisation, migration and health

The system shall let the administrator preview the conversion of historical appointments to hall identities and then apply it with explicit confirmation, with a per-row journal, automatic rollback on failure and a later manual rollback, and shall show registry health counts and open review cases.

Implemented · Should · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Applying without the initialize-location-registry confirmation header returns 409, and rolling back without rollback-location-registry returns 409.
2. If a completed run exists for this version the reply is alreadyInitialized with no repetition.
3. A probable hall is never promoted to verified by usage alone and goes to review.
4. Closing a review case as resolved with no resolution text is rejected.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/server/locationRegistryEngine.ts` `src/components/LocationRegistryAdmin.tsx` `tests/location-registry-audit.py`

FR-LOC-012 Hall type differs by department

The system shall keep for each hall a general type and, where needed, a different type per department, shall restrict a type to chosen departments so that others read it as General hall, and shall treat hall suitability for a course as ranking and warning, never as a ban.

Implemented · Should · Functional · Root administrator, Department head

ACCEPTANCE CRITERIA

1. A hall whose general type is a sound studio restricted to a technology department reads as General hall for another department.
2. A hall smaller than the course lowers the rank by 24 points without blocking the save; a matching type raises it by 12.
3. Renaming a type updates the linked halls and courses.

EVIDENCE

`src/utils/roomTypeForSection.ts` `src/utils/roomSuitability.ts` `src/components/LocationRoomTypeManager.tsx` `src/server/routes/coursesRoutes.ts` `tests/room-types-by-department-audit.ts` `tests/course-room-type-sync-audit.ts`

FR-LOC-013 Free-window board for barter

The system shall compute for the requesting department the free hall windows in its college for this term in 30-minute steps within the working-day window, after subtracting the term's appointments, instructors' request holds and approved windows, shall not show a window shorter than 50 minutes, and shall cap the display at 400 windows shared equally among host departments.

Implemented · Must · Functional · Department head, Schedule committee chair

RATIONALE

> A blind cut at 400 windows used to remove whole departments from the filter, so selection now alternates between them.

ACCEPTANCE CRITERIA

1. A hall free from 08:00 to 10:00 gives one window of 120 minutes and a 40-minute gap is not shown.
2. No inactive, unconfirmed or non-normal-policy hall appears, and neither does a hall of the requesting department.
3. When windows exceed 400 the reply returns the full total, truncated=true and the complete filter lists.
4. The owner is the registered department, otherwise the one booking it most this term and then in college history.

EVIDENCE

`src/server/routes/hallBarterRoutes.ts` `src/utils/hallBarterFairness.ts` `src/utils/scheduleTime.ts` `tests/first-password-room-loans-audit.ts` `tests/pdf-import-regression.ts`

FR-LOC-014 Boys and girls campuses stay separate

Inferred

The system shall neither list nor approve a barter between a boys' college and a girls' college, and shall fail closed when a college's campus cannot be told from its name.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Approving a request across the two campuses returns 409 stating the halls are fully separate.
2. A college whose name does not indicate its campus neither borrows nor lends.

EVIDENCE

`src/server/routes/hallBarterRoutes.ts`

FR-LOC-015 Requesting and approving a barter

The system shall create a barter request from an offered window without changing any teaching appointment, shall let only the host department approve or reject, and shall serialise approvals per term and re-read occupancy inside the lock so two overlapping requests are never both approved.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Requesting a window no longer offered returns 409 asking for a refresh, and a department requesting from itself returns 409.
2. Approving while the hall is busy in the window or approved to another returns 409 with the state unchanged.
3. A response from a non-host department returns 403 and an already answered request returns 409.

EVIDENCE

`src/server/routes/hallBarterRoutes.ts` `src/server/core/hallBarter.ts` `tests/instructor-request-availability-audit.ts` `tests/first-password-room-loans-audit.ts`

FR-LOC-016 Cancelling and ending a barter

The system shall let the requesting department cancel its pending or approved request and the host department end a running loan whenever it chooses, and shall refuse either while the borrowing department's schedule uses the hall in that window.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Ending a loan carrying a borrower lecture returns 409 asking to move the appointment first.
2. Cancelling an approved request that the requester's schedule uses returns 409; afterwards the request is cancelled and the window is released.
3. Ending writes an audit line naming the building and hall.

EVIDENCE

`src/server/routes/hallBarterRoutes.ts` `tests/first-password-room-loans-audit.ts` `tests/instructor-request-availability-audit.ts`

FR-LOC-017 Request expiry and inbox counter

Inferred

The system shall automatically cancel any pending request older than seven days with the reason that the one-week limit passed without a reply, and shall show in the side bar the count of requests waiting for the user's departments and the age of the oldest, excluding a request the user sent and excluding an administrator with no department.

Implemented · Should · Functional · Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. An 8-day-old request shows cancelled with its reason when the board is built or the counter is read.
2. The counter returns pending and oldestDays and does not count the user's own request.

EVIDENCE

`src/server/routes/hallBarterRoutes.ts` `src/components/HallBarterBoard.tsx`

FR-LOC-018 Enforcing loan windows on save

When writing any official appointment the system shall prevent the borrower from using the hall outside its approved windows, prevent the lender from using a lent window, keep a tombstone of a loan cancelled after approval so an in-flight write cannot reuse it, and check the companion facility as a separate hall.

Implemented · Must · Functional · Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. The lender inside the loan is blocked, outside it allowed, and after it ends allowed.
2. An existing appointment occupying a requested window shows as a clash for both sides on the request.
3. No window is offered for a time occupied by an appointment or a held instructor request.

EVIDENCE

`src/utils/hallBarterWindow.ts` `src/server/core/hallBarter.ts` `src/server/routes/hallBarterRoutes.ts` `tests/first-password-room-loans-audit.ts` `tests/instructor-request-availability-audit.ts`

FR-LOC-019 Loan hours drawing

The system shall draw for each department a Sunday-to-Thursday weekly strip of its approved windows only, lent by it or borrowed by it, with the minute total of each kind.

Implemented · Could · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A 60-minute lent and a 90-minute borrowed window give lentMinutes=60, borrowedMinutes=90 and five day strips.
2. A department unrelated to the window sees zero.

EVIDENCE

`src/utils/hallBarterWindow.ts` `src/components/RoomLoanStrip.tsx` `tests/first-password-room-loans-audit.ts`

FR-LOC-020 Hall door page

The system shall show for each active confirmed hall a public read-only page at /d/ with its token that states in Kuwait time (UTC+3) whether it is free until when, busy until when, or starting in a few minutes, with its Sunday-to-Thursday week.

Implemented · Must · Functional · Public visitor, Student, Faculty member

ACCEPTANCE CRITERIA

1. Sunday 10:20 with a lecture ending 10:45 reads busy and ends 10:45; Sunday 09:20 with the next at 09:30 reads starting in 10 minutes.
2. The threshold is 15 minutes, Friday and Saturday read weekend, and outside the term it reads outside term with the next term's start date.
3. Two sections of the same course at the same time show one lecture 01 · 02, and a lecture cancelled today does not occupy the hall and is shown with its mark.
4. The page reads the operationally current term, not the upcoming registration term.

EVIDENCE

`src/utils/roomDoor.ts` `src/server/roomDoorData.ts` `src/server/roomDoorPage.ts` `src/server/roomDoorRoutes.ts` `tests/room-door-audit.ts`

FR-LOC-021 Building board and corridor screen

The system shall show for each active building a public board at /b/ with one tile per hall ordered by floor then number, and a screen mode (?screen=1) that fills a corridor TV without scrolling; pages shall refresh about every minute (60 seconds plus up to 7 seconds of jitter) and reload when the system build changes.

● Implemented · Should · Functional · Public visitor, Student, Faculty member

ACCEPTANCE CRITERIA

1. G01, G02, F01, F10, S01, L01 are ordered ground, first, second then other floors.
2. After two failed pulls the page says not connected with the last update time and recovers automatically.
3. The board carries only time and code for each lecture.

EVIDENCE

`src/server/roomDoorRoutes.ts` `src/server/roomDoorPage.ts` `src/server/roomDoorData.ts` `src/components/room-door.css` `tests/room-door-audit.ts`

NFR-LOC-022 Sticker token is permanent and unguessable

A hall sticker token shall be sixteen base64url characters (96 bits) produced by an HMAC over the hall's permanent registry id with a key derived from the stored shared secret under a door-specific label; it shall not change with a new deployment or with rotation of the calendar secret, and a building token shall differ from a hall token.

● Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> A sticker stays on a door for years so its token cannot die. The token rests on the registry id, not the printed hall code, so it survives a move or a code correction.

ACCEPTANCE CRITERIA

1. The token matches ^[A-Za-z0-9_-]{16}$ and is identical after a restart and after rotating CALENDAR_SECRET.
2. A hall token opens no building or other hall; a demo token carries a ten-character sandbox tag and opens only its owner's sandbox.

EVIDENCE

`src/server/roomDoorSecret.ts` `tests/room-door-audit.ts`

NFR-LOC-023 Public page privacy and limits

Door pages shall expose only the course code, name, section and time, shall be neither indexed by search engines nor cached by browsers, shall give their scripts a per-page nonce (CSP nonce), shall be rate limited to 600 requests per minute per address by default (ROOM_DOOR_PER_IP_PER_MINUTE), and shall answer a wrong token or an inactive hall with 404 and a quiet page.

● Implemented · Must · Non-functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. No instructor name, student count or row or course id appears in the data or the page.
2. The reply carries X-Robots-Tag noindex and Cache-Control no-store, and a request over the limit returns 429.
3. Each hall's and building's data is cached for 60 seconds in the real world only, and the service worker never stores /d/ or /b/ pages.

EVIDENCE

`src/server/roomDoorRoutes.ts` `src/server/roomDoorData.ts` `src/server/roomDoorPage.ts` `tests/room-door-audit.ts`

FR-LOC-024 Door links and stickers

The system shall give only registry administrators the links of the building board, its screen and every hall, and shall print an A4 portrait sheet with four QR stickers drawn in the browser by a local generator with no external service, admitting no unsafe text or SVG into the sheet.

● Implemented · Should · Functional · Root administrator, System administrator

ACCEPTANCE CRITERIA

1. Requesting /api/room-door/links without sign-in returns 401 and an inactive or unknown building returns 404.
2. Each sticker says scan to know the hall now, stickers are laid four to a sheet, and the short link gives a QR of at most 29 modules per side.
3. In the demo the links carry the visitor's sandbox tag and are forgotten after two hours.

EVIDENCE

`src/server/roomDoorRoutes.ts` `src/utils/roomDoorStickers.ts` `src/components/RoomDoorTools.tsx` `tests/room-door-audit.ts`

FR-LOC-025 Seven-digit course number on the door

The door shall show each course number in seven digits (college 2, department 2, course 3) through authorityCourseCode, not as stored in the catalogue.

● Implemented · Must · Functional · Public visitor, Student

RATIONALE

> The owner’s standing decision: the official number has seven digits. The catalogue stores most courses with three, so the displayed number is built from the department and course codes at display time only, without changing what is stored.

ACCEPTANCE CRITERIA

1. A course stored as 105 in department 0109 appears on the door as 0109105.
2. Neither CourseCode nor AdCourseId is shown raw.
3. The building board shows the same seven-digit number, and sections of one course are still merged by the stored code.

EVIDENCE

`src/server/roomDoorData.ts#doorWeek` `src/server/roomDoorRoutes.ts`
`src/utils/authorityAcademicCodes.ts#authorityDepartmentKeys` `tests/room-door-audit.ts`
`tests/course-number-seven-digit-audit.ts`

FR-LOC-026 Occupancy rate

The system shall compute for each day and hour the number of lectures occupying it, the number of busy halls and their share of the scope's halls, plus the day and week rates, count a hall once per hour, and let a cell open a panel of its lectures grouped under their campuses with short names and a search appearing from nine rows.

● Implemented · Should · Functional · Department head, Schedule committee chair, University registrar

ACCEPTANCE CRITERIA

1. A lecture from 9:30 to 10:45 falls in both the nine and ten o'clock hours and two lectures in one hall count as one hall.
2. The colour level runs 1 to 5 for non-empty cells and a lecture with no hall counts as a lecture and occupies no hall.
3. The college name is not repeated on every row and course and instructor names are never cut with ellipses.

EVIDENCE

`src/utils/occupancyHeat.ts` `src/utils/occupancySlot.ts` `src/components/OccupancyHeatGrid.tsx`
`src/components/OccupancySlotPanel.tsx` `tests/occupancy-heat-audit.ts` `tests/occupancy-slot-panel-audit.ts`
`tests/room-occupancy-audit.ts`

FR-LOC-027 Companion facility of a hall

The system shall list for a course that requires a companion facility the facilities registered with the companion-only policy of the same type on the same site (first four characters of the building code) with their availability for the appointment time, link the first available one automatically, and occupy both the facility and the main hall for the whole appointment.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A facility booked by another appointment returns available=false and the edited appointment does not count itself (excludeId).
2. Main and companion use of the same studio clash.
3. The badge shows the registered type of the actually booked facility and none is shown for an appointment without a booking.

EVIDENCE

`src/server/routes/locationsRoutes.ts` `src/components/CompanionFacilityPicker.tsx` `src/components/CompanionFacilityBadge.tsx` `src/utils/companionRoomDisplay.ts` `src/utils/locationRegistry.ts` `tests/companion-rooms-audit.ts`

FR-LOC-028 Travel policy between campuses

The registrar general or an administrator shall set the travel time for each pair among Jahra, Fahaheel and the main campus from 0 to 180 minutes (zero disables it) with an exemption list of up to 500 instructors; defaults are 60 minutes between Jahra and Fahaheel and 30 for the others; the time applies as a warning in the department appointment and as an obligation in the faculty card.

Implemented · Must · Functional · University registrar, System administrator, Root administrator

ACCEPTANCE CRITERIA

1. A pair saved at 90 minutes returns a shortfall with need=90 for a 60-minute gap and a 105-minute gap passes.
2. An exempt instructor and zero stop the rule, and the legacy required field is stored but read by no rule.
3. An unauthorised user gets 403 and the policy is cached for 30 seconds.

EVIDENCE

`src/utils/campusTravelPolicy.ts` `src/utils/campusTravel.ts` `src/server/core/travelPolicy.ts` `src/server/routes/referenceRoutes.ts` `src/server/core/scheduleConflicts.ts` `src/components/CampusTravelPolicyCard.tsx` `tests/undo-travel-rule-audit.ts`

FR-LOC-029 Walking time between buildings on a campus

Inferred

The system shall keep for each college a default time between two unconfigured buildings (5 minutes), a time inside one building (3 minutes) and configured pairs between specific buildings of 1 to 120 minutes, editable only by a user with the colleges permission within their scope.

Implemented · Should · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. A pair value of 0 or 500 is clamped to 1 or 120 and the same-building time is limited to 0 to 30.
2. A pair from a building to itself is ignored.

EVIDENCE

`src/utils/campusTravel.ts` `src/server/routes/referenceRoutes.ts`

NFR-LOC-030 **One identity for a hall**

A hall shall be identified in every comparison (occupancy, barter, door page, suggestion) by one key: its registry id, otherwise the normalised building and hall text (NFKC, Latin digits, upper case, no spaces), so that no hall looks free in one place and booked in another.

Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> The barter key used lower case while the registry uses upper case, so a reservation never matched any legacy hall and the reserved-digitally warning never fired.

ACCEPTANCE CRITERIA

1. A row carrying the building and hall as text occupies the hall just like one linked by id.
2. legacy:012B08|G20 matches legacy:012b08|g20 after normalisation.

EVIDENCE

`src/utils/locationRegistry.ts` `src/server/core/hallBarter.ts` `src/server/roomDoorData.ts` `tests/room-occupancy-audit.ts` `tests/companion-rooms-audit.ts`

3.3.6 Algorithms

ALG-LOC-01 · Locations

Building and hall resolution against the official registry

Deterministic

Which official building and hall does this typed or scanned text refer to, or does it need review?

1. The code is normalised and matched against the official code and aliases within the college/department context; a single candidate is accepted, several go to review.
2. A short code (8 or B08) is recovered only when one confirmed building remains in the college context.
3. The hall is searched only inside the resolved building: name match, then repair of reading confusions (O/I/L to 0/1, 6 to G, a phantom digit) only when one confirmed hall results.
4. A bare number is accepted if it matches one hall, or the department's footprint narrows it to one.

Inputs

Building and hall text, the location registry, college, department and branch.

Output

A status (confirmed, review required, invalid) with the entity and written evidence.

Never invents a hall or borrows one from another building; every ambiguity stays for human review.

`src/utils/locationRegistry.ts#resolveBuilding` `src/utils/locationRegistry.ts#resolveRoom` `src/utils/locationRegistry.ts#resolveBuildingFromUniqueRoom` `src/server/core/locations.ts#canonicalizeHistoricalLocationForRuntime` `tests/pdf-import-regression.ts` `tests/location-registry-audit.py`

DR-LOC-003 FR-LOC-010 FR-LOC-011 NFR-LOC-030

ALG-LOC-02 · Locations

Deterministic

Hall barter windows and their enforcement

Where are free hall windows to borrow, and does an appointment respect the approved barter window?

1. Opportunity board: for each hall and day, 30-minute steps are walked and free slots (after subtracting appointments, holds and approved windows) are merged into windows.
2. Above the cap, windows are chosen round-robin between owning departments so no department disappears, then restored to reading order.
3. On write: the lender is blocked from a lent window, and the borrower outside its approved window (which must fully contain the appointment on that day).
4. A loan cancelled after approval leaves a tombstone so an in-flight write cannot reuse it; the companion facility is checked as a separate hall.

Inputs

Barter requests and their states, term rows, the hall, day and time.

Output

A list of windows to borrow, or a block/allow verdict for a write.

The same write guard for every official mutation; a loan is requested and approved, never assumed.

Shortest window 50 minutes, display cap 400 windows.

`src/utils/hallBarterWindow.ts#scheduleViolatesApprovedBarterWindow`
`src/utils/hallBarterWindow.ts#borrowerOutsideApprovedWindow` `src/utils/hallBarterFairness.ts#fairShareByOwner`
`src/server/routes/hallBarterRoutes.ts` `tests/first-password-room-loans-audit.ts`
`tests/instructor-request-review-regressions.ts` `tests/pdf-import-regression.ts`

`FR-LOC-013` `FR-LOC-018` `FR-SCH-012`

ALG-LOC-03 · Locations

Search and ranking

Ranking candidate halls for an instructor request

Which valid halls best suit this requested appointment?

1. A hall whose recorded capacity is below the course capacity is dropped (an unrecorded capacity does not drop it).
2. Score: type suitability × 10, hall of the course's department +400, building near the instructor's halls that day +300, a hall where the course was taught +200.
3. Capacity closeness up to +100, least-used in the term +20, minus the hall's meeting count.
4. Sorted descending, ties broken by hall key.

Inputs

Free valid halls, the course department, the instructor's buildings, previous halls, course capacity and hall usage.

Output

Ranked halls with reason codes (department, near, previous, capacity, fair, facility).

Mandatory conditions (free, eligible, facility) are decided by the verdict engine; this only ranks, stably between two reads.

`src/utils/requestRoomRanking.ts#rankRequestRooms` `src/utils/roomSuitability.ts` `tests/requests-engine-owner-feedback-audit.ts`

`FR-LOC-012`

3.3.7 Edge cases

⚠ Barter reservation on a legacy hall held as text

Scenario A hall with no registry id holds building and hall as text in different letter case between the request and the appointment.

Behaviour Both sides are normalised by the registry's own rule so they match, and the reserved-digitally message and the window-exceeded warning both fire.

⚠ Ending a loan the borrower teaches in

Scenario The host department presses End loan while the borrowing department has a lecture in the window.

Behaviour Ending is refused with 409 asking to move the appointment first, and the repository guards the same rule when two writes race.

⚠ Renaming a hall then returning to its code

Scenario A hall code changes from ZX901 to ZX911, then ZX912, then back to ZX901.

Behaviour Each old code is recorded once as a form; on returning, ZX901 drops from the forms while ZX912 and ZX911 remain, and no code owned by another hall is recorded.

⚠ A lecture cancelled today on the door page

Scenario A Thursday lecture is cancelled for today's date only, at its current time.

Behaviour The hall shows free because of it and it is not counted as next; the following Thursday is busy again, and a merged lecture is cancelled only if all its sections are cancelled.

⚠ A deactivated hall with a sticker on its door

Scenario A hall is deactivated in the registry while its sticker is still on the door.

Behaviour The token returns 404 with a quiet page revealing nothing; reactivating the hall brings the same sticker back because the token rests on the registry id.

⚠ Two overlapping requests approved at once

Scenario Two hosts approve, at the same moment, two different requests for overlapping windows of one hall.

Behaviour A per-term lock serialises the two decisions and re-reads approved loans inside it, so the first passes and the second returns 409 saying another overlapping loan is approved.

3.3.8 Standing owner decisions

- Course numbers are seven digits only in everything a reader sees, including the door page and building board.
- The boys' and girls' campuses are fully separate, so there is no hall barter between them.
- A loan reserves the hall window only and for one term: no fake lecture and no course change; the lender is free outside the window and barred inside it until the loan is ended.
- Travel time between campuses is an obligation in the faculty card and a warning that does not block in the department appointment (decision of 2026-11-05).
- No typing a hall by hand: what is not in the list cannot be booked, and deleting only deactivates so history is kept.
- Printed reports keep their design; hall values and data are fixed, not the layout.

↑ Contents

3.4 · SCH

Deterministic scheduling engine

Every appointment passes fixed rules before it is written: conflicts, regulation, teaching day, holds and a rollback version.

41	100%	42	79
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.4.1 Overview

This module governs appointments (schedule rows) from creation to deletion: field validation, instructor, hall and duplicate-section conflict detection, regulation rules for days, lengths, the pattern library and the weekly-hours ceiling, the university teaching day and the department window, and request holds and colleague holds. Every write is preceded by a rollback snapshot and is pushed live to colleagues. The schedule committee chair and the department head use it daily; registration and the dean read its results. There is one verdict for every path: manual save, proposal approval, batch move and draft checks all run through the same functions.

Schedule committee chair · Department head · Root administrator · System administrator · College registrar · Registration officer · College dean · The system (automatic)

3.4.2 User journeys

Schedule committee chair — **Adding a new appointment without conflicts**

1. Opens «New appointment» and chooses course, section and instructor; the teacher mini-week shows the instructor's week and free slots.
2. Picks a day and start; the days and end time complete from the course's approved pattern, and an advisory hold is announced to colleagues for two minutes.
3. The live check (POST /api/schedules/check-conflicts) lists blockers in red and soft notes separately.
4. On a hall conflict he uses «suggest time and hall»; up to three options that passed the same gate are returned.
5. On save the schedule is snapshotted, the row is written and the change reaches every screen open on the same board.

Department head — **Dragging a lecture to a new slot and undoing it**

1. Drags the lecture card to another cell; the drag preview classifies the move before the drop.
2. On drop the request goes to batch move with each row's revision, and every row is checked against the whole term.
3. If refused the card returns, the first message appears with the number of other blockers, and nothing is written.
4. If accepted an undo bar shows for fifteen seconds and the log keeps the last sixty changes; Ctrl+Z restores the appointment.

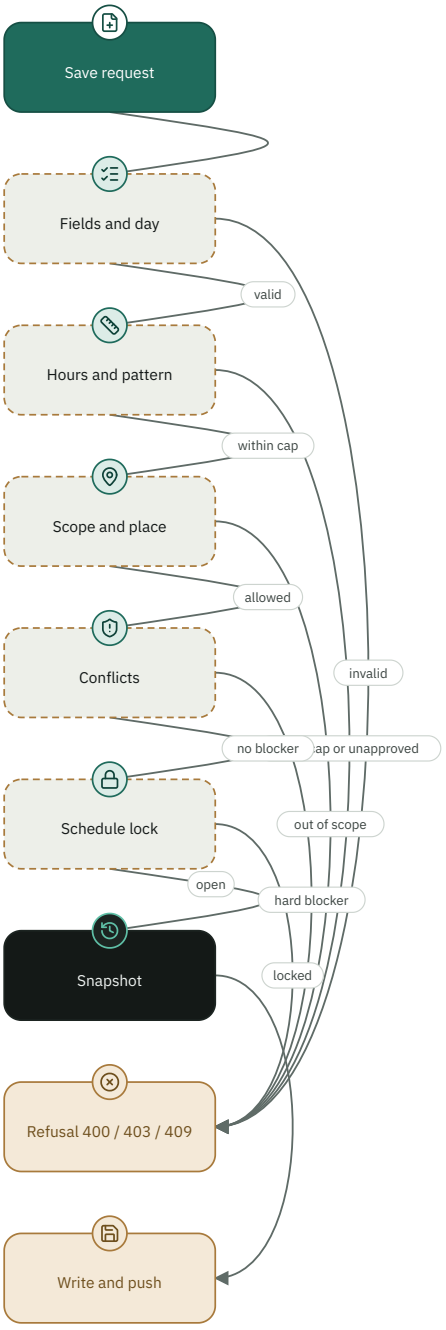
Schedule committee chair — **Assigning a faculty-placeholder section to a real instructor**

1. Searches a time and hall for a real instructor and the results list placeholder sections that match exactly as «replaceable».
2. Chooses an action: assign, swap hall, or replace.
3. The server dry-runs every step on the week as it will be, then writes all steps in one transaction.
4. The reply carries the undo steps in order: the new row is removed first, then the section is restored.

3.4.3 Diagrams

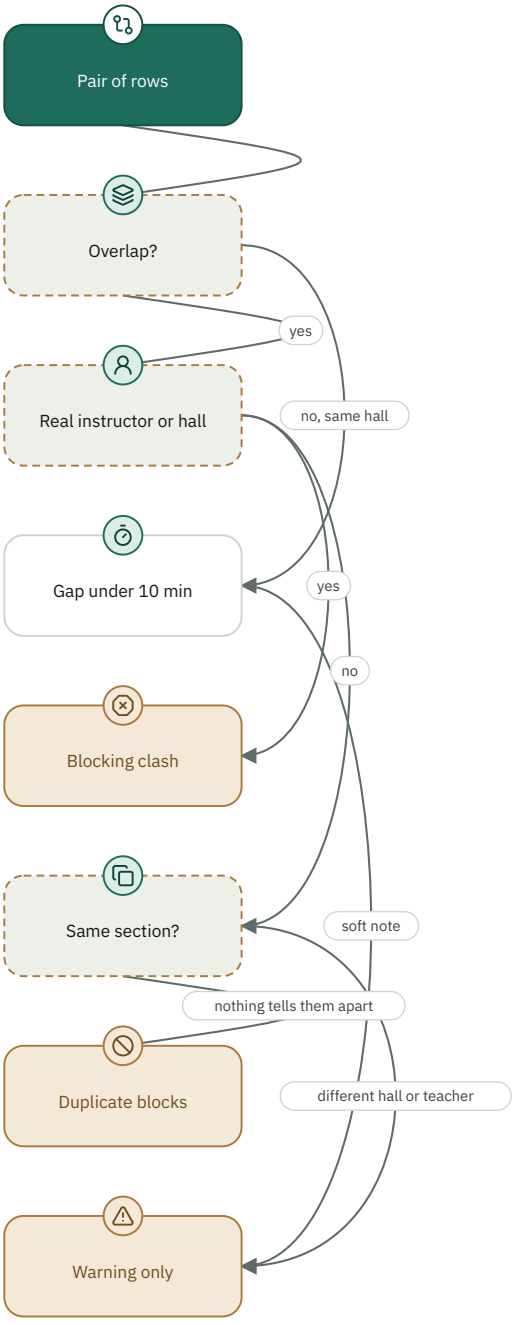
Appointment save gate

Order of checks on create; edit starts by reading the existing row and asking the lock.



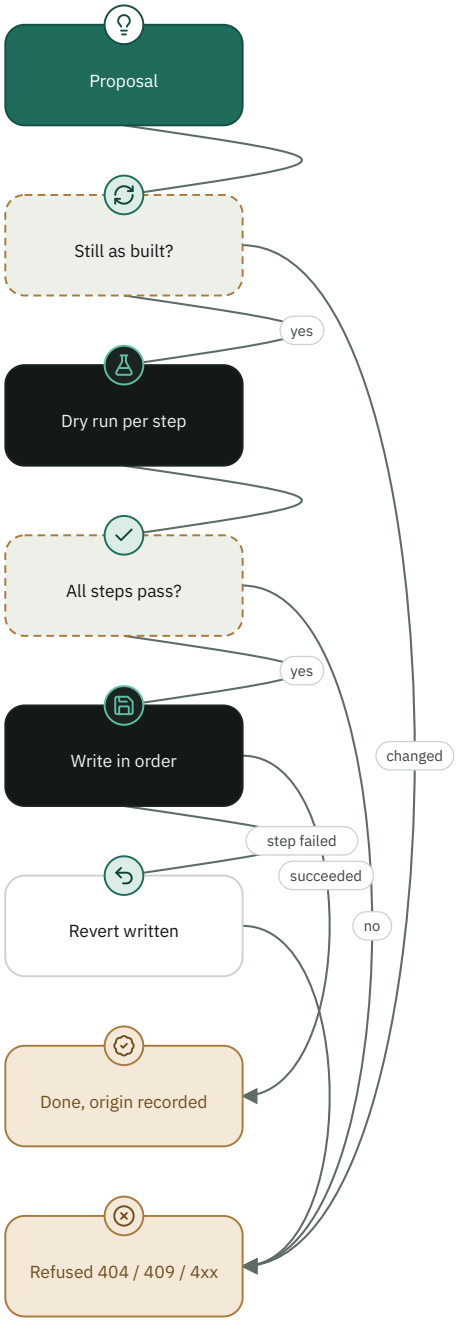
Conflict verdict

Each pair of appointments yields one verdict, named after its most serious reason.



Proposal approval

A move, swap or section opening is dry-run, then written; written steps are reverted if a step fails.



Who can do what in the schedule

	Read appointments	Edit appointments	Cell notes	Wipe schedule	University day
Root administrator	✎	✎	✎	✎	✎
Schedule committee chair	✎	✎	✎	✎	—
Department head	✎	✎	✎	—	—
College registrar	✎	—	✎	—	—
Registration officer	✎	—	✎	—	—
College dean	👁	—	—	—	—

✎ Write 👁 Read — No access

3.4.4 Screens

Schedule workshop
Three views: week, list and rooms; list is the default on phones. Appointment form, live checks, command palette Ctrl+K and undo Ctrl+Z.`src/components/Schedules.tsx`

Teacher mini-week
The instructor's week in five columns with free slots, before/after placement, a load ring and drag-to-draw.`src/components/TeacherMiniWeek.tsx`

When can we meet?
Common free time for chosen instructors across the whole term.`src/components/MeetingSlots.tsx`

Department teaching hours
The university teaching day (root administrator) and narrower department exceptions.`src/components/SectionScheduleHours.tsx`

Wipe schedule
Double confirmation: typed department name and appointment count.`src/components/WipeScheduleButton.tsx`

Saved views
Appears after the first saved view; a dot shows the board differs from the saved state.`src/components/ScheduleViewsMenu.tsx`

Drag preview
Classifies a move before the drop: excellent, good, neutral, suboptimal, impossible.`src/components/SchedulePhysics/SchedulePhysicsLayer.tsx`

3.4.5 Requirements

FR-SCH-001 Create an appointment

The system shall accept creation of an appointment under the schedule permission (screen 7) when college, department, term, course, section number, instructor, start, end, building and room (or «room pending») are present, the section number is Latin digits only, and at least one day is chosen.

Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> A missing field or a lettered section number produces rows that cannot be matched in reports or imports.

ACCEPTANCE CRITERIA

1. A request without building or room (and not in the pending-room state) is refused with 400 «please fill the fields marked red» and nothing is written.
2. Section number «5a1» is refused with 400 «write digits in English»; «501» is accepted.
3. An accepted row returns 201 with revision 1 and a source order of at least 1,000,000 (so it is never coloured as an imported row).

EVIDENCE

`src/server/core/scheduleWrites.ts#createScheduleHandler` `src/server/routes/schedulesRoutes.ts` `tests/schedule-apply-proposal-audit.ts` `tests/appointment-forms-audit.ts`

FR-SCH-002 Edit and delete with revision check

The system shall compare the submitted revision (rev) with the stored one when editing an appointment, return 409 with both the current row and the user's version when they differ, and delete a row only after a snapshot and the scope and lock checks, removing its entry from the instructor request if the department had added it.

Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> So one colleague's save never silently erases another's work.

ACCEPTANCE CRITERIA

1. An edit with rev=3 when the stored value is 4 is refused with 409 and body {conflict: revision, current, yours}; the row is unchanged.
2. An edit with no rev is written without comparison and increments rev by one.
3. Deleting a row in a locked schedule returns 409 with code schedule-locked; deleting outside the user's scope returns 403.

EVIDENCE

`src/server/core/scheduleWrites.ts#updateScheduleHandler` `src/server/core/scheduleWrites.ts#deleteScheduleHandler` `src/db/repository.ts#ScheduleRevisionConflict` `tests/proposal-undo-emulator-audit.ts`

FR-SCH-003 Schedule lock after submission and term end

The system shall refuse every create, edit, delete, move or replace on a schedule that is with registration awaiting a decision, or in a term declared ended for anyone but the schedule committee chair, with 409 and code schedule-locked, and on edit shall ask about the existing row's lock before validating fields.

Implemented · Must · Functional · Schedule committee chair, Department head, College registrar

RATIONALE

> Editing in a frozen term used to return a field error first and the real reason only afterwards.

ACCEPTANCE CRITERIA

1. Editing a row in a schedule in «submitted» state with no amendment round open returns 409 «the schedule is with registration» before any field message.
2. A term with AdTermClosed=true refuses writes for every role except an explicitly written schedule committee chair.
3. A week exception (cancel or cover) ignores the registration lock and obeys only term end.

EVIDENCE

`src/server/core/approvals.ts#scheduleLockRefusal` `src/utils/approvalWorkflow.ts#approvalLockReason` `src/server/core/scheduleWrites.ts` `tests/term-freeze-audit.ts` `tests/approval-integrity-audit.ts`

FR-SCH-004 Scope and course integrity

The system shall limit writes to departments within the user's scope (403 outside, except for administrators), reject a course that does not belong to the stated college and department (400) or is academically archived (409), and block swapping the course of an imported approved-schedule row (source order below 1,000,000) with code COURSE_NAME_LOCKED.

Implemented · Must · Functional · Schedule committee chair, Department head, System administrator

ACCEPTANCE CRITERIA

1. Adding a row for an archived course is refused with 409 «this course is academically archived» and nothing is written.
2. Changing AdCourseId on a row whose source order is 500,000 returns 409 with COURSE_NAME_LOCKED; deleting it and adding a new course remains allowed.
3. A user scoped to another department receives 403 «outside your allowed departments» on create, edit and delete.

EVIDENCE

`src/server/core/scheduleWrites.ts` `src/server/core/scope.ts#isScopeAllowed` `tests/role-scope-audit.ts`

FR-SCH-005 Official place and companion facility

The system shall resolve the building and room to their official registry identities before saving, reject a room that does not accept independent booking, require a course with a companion-facility type to carry a facility of that type at the same site (first four characters of the building code), and reject one for a course that needs none.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A course requiring a «studio» sent without a facility is refused with 400 «the course requires a companion facility of type studio for the whole appointment».
2. A facility at another site is refused with 400 «studio and primary hall must be at the same site».
3. A course with no facility type that arrives with one is refused with 400 «this course needs no companion facility».

EVIDENCE

`src/server/core/scheduleWrites.ts#companionBookingFor` `src/server/core/locations.ts` `tests/companion-rooms-audit.ts`

FR-SCH-006 Instructor, hall and duplicate-section conflict detection

The system shall detect, across the whole term (all colleges), real-instructor conflicts, physical-hall conflicts and exact twins of the same section, count each pair of appointments as one conflict carrying all its reasons and named after the most serious, and treat two rows as overlapping when they share a day and each starts before the other ends.

Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

RATIONALE

> Forty-six real collisions were once reported as 138 because each reason was counted separately.

ACCEPTANCE CRITERIA

1. 08:00–09:00 and 09:00–10:00 for one instructor on one day do not overlap; 08:00–09:00 and 08:30–09:30 do.
2. Sections 1 and 2 of one course with the same instructor, hall and hour yield one high-severity conflict with reasons «instructor» and «hall» and are not called a duplicate.
3. Hall written as «G28», «G/G28» or «G-28» counts as one hall and its clash blocks.
4. Two rows with the same course, section, days and time and nothing distinguishing them (no hall, no real instructor) yield «identical appointments for the same section» and block the save.

EVIDENCE

`src/utils/scheduleIntelligence.ts#findConflicts` `src/server/core/scheduleConflicts.ts#scheduleConflicts`
`src/utils/locationRegistry.ts` `tests/run-tests.ts` `tests/conflict-parity.ts` `tests/companion-rooms-audit.ts`

FR-SCH-007 Same section in two places: warning, not a block

The system shall classify two rows with the same course, section, days and time as a non-blocking «sectionTwice» warning for save, signature and approval when the normalised hall or building differs or the two real instructors differ, and as a blocking duplicate otherwise.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> Owner rule 2026-09-27: only a physical conflict may block signing.

ACCEPTANCE CRITERIA

1. Rows in G28 and G31 with different instructors yield a medium soft warning «same section in two places at the same time» and both are saved.
2. A row with no hall (or room pending) does not distinguish itself from its twin and stays a blocking duplicate.
3. Another section of a specialisation course at the same time yields a low regulation note «Art. 7/5» that does not block.

EVIDENCE

`src/utils/scheduleIntelligence.ts#twinKind` `src/server/core/scheduleConflicts.ts`
`src/utils/scheduleRegulations.ts#sameCourseSectionClash` `tests/blocker-oracle-audit.ts` `tests/run-tests.ts`

FR-SCH-008 A faculty placeholder is not a person who can be double-booked

The system shall not count an instructor conflict for the instructor record that stands for «faculty staff» (the faculty placeholder, identified by the first word of its name), shall not treat it as a person who distinguishes twin rows, and shall apply the same rule on the server, in the browser check and in the exhaustive engine.

● Implemented · Must · Functional · Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. Two different sections of the placeholder at the same hour in different halls yield no instructor conflict.
2. A hall double-booking still blocks even when the instructor is the placeholder.
3. A real instructor assigned to such a section is checked in full.

EVIDENCE

`src/utils/instructorIdentity.ts#placeholderInstructorIds` `src/utils/scheduleIntelligence.ts`
`src/utils/placeholderInstructor.ts` `tests/client-placeholder-conflicts-audit.ts` `tests/one-blocker-number-audit.ts`

FR-SCH-009 Turnaround gap and student clash: two soft notes

The system shall state but not block when the gap between consecutive lectures in the same hall is below the requirement (ten minutes on every day unless the department declares its own figure from 1 to 60) and when two courses needed together by surveyed students overlap, and shall measure the gap from the end of the first to the start of the second.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. 08:00–09:00 then 09:05 in the same hall yields low-severity soft «gap between lectures is insufficient»; 09:10 yields nothing.
2. Monday 08:00–09:20 then 09:30 in the same hall yields nothing.
3. Two different halls need no gap.
4. A student clash is raised only for two different overlapping courses within the survey pairs, medium severity soft:true, naming the number of shared students.

EVIDENCE

`src/utils/scheduleIntelligence.ts` `src/utils/scheduleRegulations.ts#requiredGapForDay`
`src/server/core/rhythm.ts#departmentStyle` `tests/run-tests.ts` `tests/room-door-audit.ts`

FR-SCH-010 What blocks a save and what does not

The system shall refuse a save with 409 and the list of blockers when any non-soft conflict remains, pass soft notes through to the dry run and the reply without blocking, and in the other paths treat as blocking what is not soft and has high severity or type «duplicate» and is not a memory, advice or regulation note.

Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. A 409 carries error with the first message and issues with every blocker; soft notes appear under warnings in a dry run.
2. Gap, student clash, rhythm, travel, memory and same-course section are all soft and never block.
3. The blocker count in the approval bar, in review and in the check panel is read from one list, so the numbers agree.

EVIDENCE

`src/utils/scheduleIntelligence.ts#isBlockingConflict` `src/utils/scheduleBlockers.ts` `src/server/core/scheduleWrites.ts` `tests/appointment-entry-parity-audit.ts` `tests/sound-schedule-zero-blockers-audit.ts` `tests/one-blocker-number-audit.ts`

FR-SCH-011 What is disclosed about another department's appointment

The system shall, on a clash with another department's appointment, name only the department, college, day and hour, never the other course, section or instructor, return rowId=0 for rows outside the user's scope, and let the board learn which rows clash with another department without loading the whole term.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A cross-department clash detail reads «Arabic department — college · Sunday and Tuesday · 11:00 – 11:50» with no course name.
2. GET /api/schedules/outside-clashes?collegeId&termId returns ids, pairs and notes for in-scope rows clashing with out-of-scope rows only, never a pair inside the same scope.
3. A missing collegeId or termId returns 400 «specify college and term»; an unauthorised scope returns 403.

EVIDENCE

`src/server/core/scheduleConflicts.ts` `src/server/routes/scheduleReviewRoutes.ts` `src/utils/scheduleIntelligence.ts#outsideScopeClashes` `tests/schedule-blockers-audit.ts` `tests/blocker-oracle-audit.ts` `tests/run-tests.ts`

FR-SCH-012 Other departments' halls and hall barter

Inferred

The system shall block a save in a hall registered to another department in the official registry unless approved barter windows cover all of the appointment's days and time, block a save over another department's approved barter window, and downgrade the scope notice to a non-blocking note when windows fully cover the appointment.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Saving in another department's hall without barter is refused with 409 «hall scope notice» naming its owner.
2. An appointment one minute beyond the barter window is refused with «appointment exceeds the approved barter window».
3. An appointment fully inside the approved window saves and shows «borrowed hall with approved window» at low severity, soft:true.

EVIDENCE

`src/server/core/scheduleConflicts.ts#scheduleConflicts` `src/server/core/scheduleRules.ts#roomOwnership` `src/utils/hallBarterWindow.ts`

FR-SCH-013 Instructor-request holds awaiting the department

The system shall treat a slot chosen by an instructor in a request awaiting the department as held, returning 409 with code request-hold carrying the holder and item instead of a silent block, and accept a resend with overrideRequestHolds (up to ten requests visible to the user) so the change proceeds past those requests only, is logged on their timeline and the item is marked fixed.

Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> The department stays the decision-maker; a coordinator must not be trapped by a hold with no way out.

ACCEPTANCE CRITERIA

1. If every blocker is a hold the answer is request-hold; if a real blocker is mixed in it is an ordinary conflict.
2. A temporary hold with heldUntil does not block the department.
3. A request in state superseded or settled, or outside the user's scope, cannot be overridden.
4. Deleting a department-added row removes its item from the instructor's request; deleting one fixed by his request reopens the item.

EVIDENCE

`src/server/core/scheduleConflicts.ts#applyRequestHoldOverride` `src/server/core/scheduleConflicts.ts#requestHoldRefusal` `src/server/core/scheduleConflicts.ts#noteRequestHoldOverride` `tests/holds-patterns-r4-audit.ts` `tests/undo-travel-rule-audit.ts`

FR-SCH-014 Two-minute advisory colleague hold

The system shall keep one hold per seat announcing what is being built (days Sunday to Thursday, HH:MM start and end, hall, instructor, course, section) that lives 120 seconds from its last renewal, and show colleagues within scope the live holds for the term without account numbers, without ever blocking a save.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. An end at or before the start, a day outside Sunday–Thursday, or text longer than 40 characters is refused with 400.
2. GET /api/schedule-holds?termId= excludes the caller's own hold and any outside scope, and sweeps up to 25 expired holds per read.
3. The client debounces the announcement by 800 ms, renews every 45 s and polls colleagues every 20 s.

EVIDENCE

`src/server/scheduleHolds.ts` `src/server/scheduleHoldsRoutes.ts` `src/utils/scheduleHolds.ts` `tests/colleague-guard-measure-audit.ts`

FR-SCH-015 University teaching day and department window

The system shall apply one university day to browser and server (default 08:00–20:00, five-minute step, from 00:00 to 24:00) that blocks saves outside it, treat a narrower department exception (itself clipped to the university day) as a note that does not block in the appointment form, and not refuse an edit whose time is unchanged because of this limit.

● Implemented · Must · Functional · Root administrator, Schedule committee chair, Department head

RATIONALE

> Owner decision 2026-11-05: the university day blocks; the department exception only warns in the appointment form.

ACCEPTANCE CRITERIA

1. Creating 07:50–08:40 is refused with 400 «lecture time must be within the university day (08:00–20:00)»; start 08:00 and end 20:00 are accepted.
2. Time «10:76» or «25:00» is refused with 400 «hour 0–23, minute 0–59»; an end before the start is refused with «end must be after start».
3. A corrupt stored value or start after end falls back to 08:00–20:00; a department window wider than the university day is clipped to it.
4. Only the root administrator changes the university day (PUT /api/admin/university-day); the count of appointments left outside is shown before saving and does not block.
5. Editing a row with an unchanged time is not refused by the day limit (keptTime); changing its time is refused if it falls outside.

EVIDENCE

`src/utils/scheduleTime.ts#universityDayIssue` `src/utils/scheduleTime.ts#scheduleDayWindowMinutes` `src/server/core/scheduleRules.ts#schedulePayloadIssues` `src/server/core/universityDay.ts` `src/server/routes/referenceRoutes.ts` `src/components/SectionScheduleHours.tsx` `tests/schedule-time-window-audit.ts` `tests/owner-hours-rule-audit.ts` `tests/full-day-setting-audit.ts` `tests/section-schedule-hours-audit.ts`

FR-SCH-016 Regulation days, lengths and start ladder

The system shall apply the regulation: Sunday, Tuesday and Thursday lectures are 50 minutes and start on the hour from 08:00; Monday and Wednesday lectures are 80 minutes and start at 08:00 then every 90 minutes (09:30, 11:00…); a 10-minute gap separates consecutive lectures; and shall only warn, without blocking, about a start off the ladder and a non-conforming length.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. 08:00–08:40 on Sun/Tue/Thu for a 3-hour course yields a length note; 08:00–08:50 and 08:00–09:50 (50+10+50) pass.
2. Mon/Wed 08:00–09:20 passes and 08:00–09:00 is noted; the 08:00–10:50 block (five-minute tolerance) is accepted.
3. A Sunday start of 08:55 yields «outside the regulation start ladder» at low severity soft:true; mixed-family days have no ladder.
4. A length the course has used habitually (±5 minutes) is accepted without a length note.

EVIDENCE

`src/utils/scheduleRegulations.ts#lectureLengthProblem` `src/utils/scheduleRegulations.ts#startLadderNote` `src/utils/scheduleRegulations.ts#requiredGapForDays` `src/server/core/scheduleConflicts.ts` `tests/lecture-length-audit.ts` `tests/approved-patterns-audit.ts`

FR-SCH-017 Approved pattern library

The system shall refuse on create, and on edit when the row's shape (days or time) changes, a distribution of the section's days (combined with the course's other meetings in the same department and term) that does not match a pattern in the library of 17 patterns for courses of one to five hours, and list the approved patterns in the message.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> Owner decision 2026-10-30: the pattern must be approved; history is a suggestion that never overrides the regulation.

ACCEPTANCE CRITERIA

1. A 2-hour course: Sun+Tue, Tue+Thu, Mon+Wed and Tuesday 110 min are accepted; Sun+Mon is refused with 400 listing the approved ones.
2. Single-day blocks are approved with their lengths: 170 for three periods, 230 for four, 290 for five.
3. Re-saving an existing row whose shape is unchanged is not refused by the pattern rule, nor is an exact restore from the time machine.
4. A course with no hours or credits is not pattern-checked.

EVIDENCE

`src/utils/scheduleRegulations.ts#PATTERN_LIBRARY` `src/utils/scheduleRegulations.ts#daysFitLibrary` `src/server/core/scheduleRules.ts#unapprovedPatternIssue` `tests/approved-patterns-audit.ts` `tests/pattern-library-audit.ts` `tests/holds-patterns-r4-audit.ts`

FR-SCH-018 Weekly-hours ceiling per section

The system shall refuse an appointment that raises the section's weekly teaching minutes (this row plus its other distinct meetings) above the ceiling round(hours × 50 × 1.1), not double-count a copy identical to the same row, and allow editing a row already over the ceiling when the total does not increase.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A 3-hour course: ceiling 165 minutes; six 50-minute lectures are refused with 400 «what exceeds the approved hours is not saved».
2. A Wednesday 170-minute block of three periods counts 150 teaching minutes; Mon+Wed 80/80 counts 160.
3. An identical copy (same days and time) does not turn 150 into 300 minutes.
4. Hours come from CourseHours, else CourseCredit; a course with neither is not checked.

EVIDENCE

`src/utils/scheduleRegulations.ts#maxWeeklyMinutesFor` `src/utils/scheduleRegulations.ts#sectionWeeklyHours` `src/server/core/scheduleRules.ts#scheduleLengthIssue` `tests/lecture-length-audit.ts` `tests/appointment-forms-r2-audit.ts`

FR-SCH-019 Inter-campus travel time

The system shall warn, not block, when an instructor's appointment leaves less than the configured travel time to an appointment at another campus (default 30 minutes between main and a branch and 60 between Jahra and Fahaheel; registration may set 0 to 180 and exempt up to 500 instructors), with at most three warnings and no repeat per campus pair and day.

● Implemented · Should · Functional · Schedule committee chair, Department head

RATIONALE

> Owner decision 2026-11-05: the duration is mandatory in the teacher card and a warning in the appointment form.

ACCEPTANCE CRITERIA

1. A 20-minute gap between Jahra and Fahaheel yields a soft «tight transfer» warning stating need and available; a 60-minute gap yields nothing.
2. An exempt instructor or a pair set to 0 yields no warning.
3. Campus is derived from the college name (Jahra, Fahaheel, otherwise main).

EVIDENCE

`src/utils/campusTravelPolicy.ts#travelShortfalls` `src/utils/campusTravel.ts` `src/server/core/travelPolicy.ts` `src/server/core/scheduleConflicts.ts` `tests/undo-travel-rule-audit.ts` `tests/admin-rules-r4-audit.ts`

FR-SCH-020 Department rhythm and memory

The system shall read the department's habit from its whole history (every term) and state in one soft note that a start is more than zero and under 20 minutes from a habitual start of the department, or that its length is outside the library, never rank habit above the regulation, and cache the reading per department, answering at once from a reading up to 10 minutes old and refreshing it in the background.

● Implemented · Should · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. A start of 08:55 where the department habitually starts at 08:50 yields soft «outside the department's usual rhythm»; a start 30 minutes away is a decision and is not noted.
2. A break habit needs at least 6 pairs at 55% agreement, a length habit at least 8 lectures at 35% agreement.
3. A declared room_doorway constraint outranks the learned figure and is capped at 60 minutes.
4. Memory notes: one sentence about the slot and one about the hall, computed only at check time.

EVIDENCE

`src/server/core/rhythm.ts#departmentStyle` `src/utils/departmentRhythm.ts#offRhythm` `src/utils/departmentRhythm.ts#learnRhythm` `tests/run-tests.ts` `tests/approved-patterns-audit.ts`

FR-SCH-021 Live appointment check before saving

The system shall provide POST /api/schedules/check-conflicts running in parallel the same check steps that govern the save gate, returning conflicts, validationIssues (fields, length, pattern, place and companion) and pattern, length and hours notes, and term-start drafts shall use the same functions row by row.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. What the check refuses is refused on save with the same message, and a saved row whose shape is unchanged is not refused.
2. The reply adds patternNotice, lectureNotice and departmentHoursNotice when present.
3. A draft row failing pattern, ceiling or hall appears in rowIssues, a soft note in rowWarnings, and a check that could not run reads as a blocker, not as clean.

EVIDENCE

`src/server/routes/scheduleReviewRoutes.ts` `src/server/core/draftChecks.ts#draftAppointmentChecks` `tests/appointment-entry-parity-audit.ts` `tests/import-term-tools-audit.ts`

FR-SCH-022 Atomic batch move

Inferred

The system shall accept one to sixty appointments in POST /api/schedules/move-batch, change only days, time and place on them, check each row against the term treating the travelling rows as already moved, check the moved rows against each other in their new places, and write all in one batch or nothing.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A request with none or more than 60 is refused with 400 «select one to sixty appointments per move».
2. A refused move returns 409 «not moved: <first message> (+N more)» with conflicts and writes nothing.
3. Two rows moved to the same hall and hour together are refused with «between two appointments moved together».
4. A stale revision on any row returns 409 conflict=revision and nothing moves; a locked schedule returns schedule-locked.

EVIDENCE

`src/server/routes/scheduleReviewRoutes.ts` `src/db/repository.ts#moveSchedulesBatch`

FR-SCH-023 Approving a proposal into the live schedule

The system shall execute a proposal (open a section, move a row, swap two rows) from a known source (registration radar, student case, instructor request, Analytics Center, section planning) through the very same create and edit paths, after confirming it is not stale, with a dry run of each step on the week as it will be, and reverting what was written if a later step fails.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A source outside the closed list is refused with 400 code bad-origin; a kind other than move/swap/opening is refused with bad-kind.
2. A row whose day or time changed since the proposal was built is refused with 409 code stale; a deleted row with 404 stale.
3. A swap is checked on a week with both rows moved; failure of the second step restores the first and returns applied=[] and reverted=true.
4. Opening a section with no free instructor is refused with 422 needs-instructor; a proposed section number already used is replaced by highest + 1 (or 501 if none exist).
5. The written row carries origin (kind, reference, note up to 160 characters) and the audit records it as a proposal approval.

EVIDENCE

`src/server/core/scheduleWrites.ts#applyProposalCore` `src/server/core/scheduleWrites.ts#runScheduleHandler` `src/server/routes/schedulesRoutes.ts` `tests/schedule-apply-proposal-audit.ts` `tests/proposal-undo-emulator-audit.ts`

FR-SCH-024 Assign, swap and replace a faculty-placeholder section

The system shall execute in POST /api/schedules/placeholder-action an «assign» (a real instructor to a section of the same course), a «swap» (move the section to a free hall and add the instructor's row in its hall) or a «replace» (delete the section and add the row) in one transaction after a dry run of each step, and return the undo steps.

Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Assign changes the instructor of the same row, creates no row, and requires the same course (400 otherwise).
2. Replace takes the section's days, time and hall as they are, deletes it and adds the new row in the same transaction.
3. An instructor conflict or stale revision returns 409, the section stays and no row is created; a storage failure returns 503.
4. A section no longer assigned to the placeholder is refused with 409, and an instructor who is himself a placeholder with 400.

EVIDENCE

`src/server/routes/scheduleToolsRoutes.ts` `src/utils/placeholderMatch.ts#placeholderMatch` `src/db/repository.ts` `tests/placeholder-action-audit.ts` `tests/appointment-forms-r2-audit.ts`

FR-SCH-025 Replace an instructor across a term

Inferred

The system shall show in a preview the effect of handing all of an instructor's appointments in a department and term to a replacement or to «no instructor», and execute it on commit=true when no conflict arises with the replacement's schedule, after a snapshot labelled from ← to, without deleting any teaching.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. from equal to to is refused with 400 «choose a different replacement»; a clash with the replacement returns 409 with up to 20 conflicts.
2. The preview returns affected, compatible and reasons with no write; the commit returns moved and undoVersionId.
3. GET /api/schedules/replace-instructor/history lists up to 20 snapshots whose label starts «before replacing the instructor».

EVIDENCE

`src/server/routes/scheduleToolsRoutes.ts`

FR-SCH-026 Convert TA staff to the faculty placeholder

The system shall convert the appointments of explicitly classified TA staff to the department's single «faculty» placeholder record, with a preview carrying a fingerprint (previewKey) of the rows and their revisions, an explicit confirmation in the x-schedule-confirm header, a limit of 400 appointments, within one department and term, in an atomic batch under a serial lock.

Implemented · Could · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A run without the token ta-conversion is refused with 409 «conversion needs explicit confirmation».
2. A schedule or classification change after the preview is refused with 409 «reopen the preview»; more than 400 appointments or more than one placeholder record blocks.
3. A frozen term or locked schedule blocks the conversion.

EVIDENCE

`src/server/routes/scheduleToolsRoutes.ts` `src/utils/taConversion.ts` `tests/ta-conversion-audit.ts`

FR-SCH-027 Time and hall suggestions

The system shall suggest in POST /api/schedules/suggest-slots times and halls on a 30-minute step within the department window, from official, pinned and approved-barter halls, pass each ranked candidate through the same conflict gate (up to 48 candidates) and return three (six for «suggest a hall»), and write nothing.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The requested duration is clamped between 30 and 300 minutes (60 by default); with no days an empty slots list is returned with reason days.
2. What the gate rejects is counted in rejectedByScheduleGate and not shown; if nothing remains a named reason is returned.
3. Up to four exactly matching faculty-placeholder sections are returned as replaceable; a course needing a companion facility gets no slot without a free one.
4. If the approved-hours ceiling is exceeded the result is empty with reason hours.

EVIDENCE

`src/server/routes/scheduleToolsRoutes.ts` `src/utils/suggestionEngine.ts` `src/utils/slotSuggestionScope.ts` `tests/slot-suggestion-recovery-audit.ts` `tests/appointment-forms-audit.ts` `tests/placeholder-action-audit.ts`

FR-SCH-028 Common meeting time

The system shall compute common free time for two or more instructors (duration 30 to 180 minutes) from the whole term's schedule across all colleges on a 30-minute step, drop on the server the faculty placeholder, visiting instructors and anyone with a non-empty status, and return only busy/free per chosen name.

Implemented · Could · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Fewer than two instructors is refused with 400 «choose at least two instructors».
2. A department head (permission 14) may choose only instructors of his own department (403 otherwise).
3. The reply carries busyScope=allColleges, and the best window is ordered by fewest busy, then longest run, then nearest to 10:00.
4. Inspecting a specific time outside the university day is refused with 400.

EVIDENCE

`src/utils/meetingSlots.ts` `src/server/routes/scheduleToolsRoutes.ts` `src/components/MeetingSlots.tsx` `tests/meeting-slots-audit.ts`

FR-SCH-029 Cancel or cover one date, and rank substitutes

The system shall record against an appointment a dated exception YYYY-MM-DD (cancel, or cover by another instructor) that never touches the row or its revision, refuse a second exception for the same date, and rank up to 12 substitutes free in that hour.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A date on a day the appointment does not meet is refused with 400 «this appointment does not meet on the chosen day»; a repeated date returns 409.
2. Covering with the same instructor is refused with 400; with an instructor busy at that hour or already covering returns 409 (cover-busy or cover-double-booked).
3. The note is cut to 240 characters. Substitutes score: taught the course 50 + min(12, 2×terms), same department 20, same building that day 10 or same day 6, plus 10 minus load hours.
4. A mobile number is withheld from anyone outside the circle of departments and the reply says «contact via his department».

EVIDENCE

`src/server/routes/schedulesRoutes.ts` `src/server/core/scheduleRules.ts#weekExceptionDayKey` `src/utils/coverAvailability.ts` `tests/doctor-journey-audit.ts` `tests/term-freeze-audit.ts`

FR-SCH-030 Cell notes, department rebuttal and registration verdict

The system shall let registration and the department attach a note to a specific cell (time, days, hall, instructor, section number, course, or whole row), compute its state (open, changed, answered, resolved, removed) from the cell's present value rather than a flag, and require a rebuttal reason of at least three characters.

Implemented · Should · Functional · College registrar, Registration officer, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Note text is cut to 600 characters and optional, defaulting to «review <field>»; a second click on the same cell corrects the text and clears the rebuttal and verdict.
2. A registration note belongs to the current approval round (at least 1); each department author keeps their own note on the same cell.
3. A rebuttal under 3 characters is refused with 400; verdict «accepted» resolves the note, «insisted» reopens it and archives the argument, escalating on the third insistence.
4. Writing is refused in an ended term; deletion is limited to the author or the administrator.

EVIDENCE

`src/server/routes/scheduleNotesRoutes.ts` `src/server/core/notes.ts#noteState` `src/server/core/notes.ts#noteFieldValue` `tests/schedule-changes-audit.ts` `tests/approval-integrity-audit.ts` `tests/relay-fixes-audit.ts`

FR-SCH-031 Live push and colleague presence

The system shall push every schedule change to each open screen within about 350 ms (and across all instances), show the presence of colleagues on the same board (cell under pointer, card in the air, row open) scoped so that nobody learns who works in a college outside their own, and replace the open stream with a short pulse when the user is alone.

● Implemented · Should · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. The stream (GET /api/schedules/events) closes once 400 connections are open, sends a ping every 25 s and retries after 4000 ms.
2. A presence mark counts as stale after 45 seconds and a silent connection leaves the roster after 90 seconds.
3. A stream cannot claim a board outside its holder's scope and registers with an empty scope; the pulse returns company, changedAt and notifiedAt from memory with no database read.
4. A pulse from a demo sandbox wakes only that sandbox's screens.

EVIDENCE

`src/server/routes/scheduleLiveRoutes.ts` `src/server/core/liveChannel.ts` `src/server/livePulse.ts` `src/components/schedulePresence.ts` `tests/live-pulse-audit.ts` `tests/live-pulse-http-audit.ts`

FR-SCH-032 Snapshot before every write and no silent overwrite

The system shall capture a snapshot of the stored schedule (not the cache) before every add, edit, delete, move, copy, wipe and replace, record the after-state from the operation's own rows, and refuse to restore a version over a schedule that changed after its base unless «overwrite-newer» is confirmed, listing what would be lost.

● Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. Each snapshot is labelled by what precedes it: «before adding an appointment», «before editing an appointment», «before deleting an appointment».
2. Moving a row between scopes captures two snapshots: «before editing…» for the source and «before moving an appointment into this schedule» for the destination.
3. A restore over a changed schedule returns 409 code scope-changed-since-base with up to 40 changes and the total.
4. In Firestore the version list carries no rows; whoever needs the snapshot fetches it by id.

EVIDENCE

`src/server/core/versions.ts#captureScopeVersion` `src/server/core/versions.ts#scopeOverwriteRefusal` `src/server/core/versions.ts#hydrateVersionRows` `tests/schedule-versions-list-audit.ts` `tests/schedule-wipe-audit.ts`

FR-SCH-033 Wipe and restore a department schedule

The system shall allow wiping a department's schedule only in a term being prepared, only for the system administrator and the schedule committee chair, after explicit confirmation, a typed department name, a matching appointment count, a limit of 450 appointments and a full backup, and allow restoring the last wipe into an empty schedule.

● Implemented · Must · Functional · Root administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. The current, ended or non-planning term is refused with its reason; a signed schedule can be wiped only by the administrator.
2. A mismatched department name returns 409 wipe-name, a changed count wipe-count, an empty schedule wipe-empty, over 450 wipe-too-large.
3. If the backup cannot be saved the answer is 500 and nothing is wiped; after the wipe old instructor requests are closed and the schedule generation advances.
4. Restore requires an empty schedule (wipe-not-empty), a backup with all rows and the same version id, and validates the rows before writing.

EVIDENCE

`src/utils/scheduleWipePolicy.ts#wipeDecision` `src/server/routes/schedulesRoutes.ts` `src/components/WipeScheduleButton.tsx` `tests/schedule-wipe-audit.ts`

FR-SCH-034 Copy a term's schedule into another term

The system shall allow copying a department's schedule from one term to another for the main administrator (SystemUserId = 1) only, provided the target term is empty, the rows hold no archived courses and the copied rows pass the import validation, with a snapshot before it.

Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Anyone but the main administrator is refused with 403; a target term with appointments is refused with 409 «a schedule already exists in the target term».
2. Rows with archived courses are refused with 409 stating their number; rows failing validation are refused with 400 with issues.
3. The preview GET /api/schedules/copy-preview returns canCopy only when a source exists, the target is empty and there are no issues.

EVIDENCE

`src/server/routes/schedulesRoutes.ts` `tests/approval-integrity-audit.ts` `tests/term-freeze-audit.ts`

FR-SCH-035 Read boundaries by role and scope

The system shall read an appointment reader's rows only from departments within their scope, restrict the dean and vice dean to each department's final schedule (live if approved, from the last-accepted version if returned to registration, nothing if never accepted), and, with no explicit term, read the newest term rather than all terms.

Implemented · Must · Functional · College dean, Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. GET /api/schedules with no termId reads the newest term; it filters by instructorId, building, hall, a start/end window and days sun..thr combined with OR.
2. The X-Schedule-Finality response header carries «accepted» or «historical» per department for those who see the final view only.
3. GET /api/schedules/workspace returns colleges, departments, terms, appointments, instructors and courses in one request, and is confined to the account's scope for non-administrators.

EVIDENCE

`src/server/core/scheduleReads.ts` `src/server/routes/schedulesRoutes.ts` `tests/role-scope-audit.ts` `tests/dean-notify-audit.ts`

FR-SCH-036 Arabic ordering of terms, names and courses

The system shall order terms newest to oldest by the key start-year × 10 + term order (first 1, second 2, summer 3), falling back to the term id, order reference lists alphabetically in Arabic (numeric comparison, ignoring case and punctuation), and order appointments by course code, then section number, then course name, then id.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. «Summer term 2025/2026» precedes «Second term 2025/2026», which precedes «First term 2025/2026» in a newest-first list.
2. A term name without two years is ordered by AdTermId.
3. Two sections of one course order by section number, then by id on a tie.

EVIDENCE

`src/server/core/ordering.ts#termChronologyServer` `src/server/core/ordering.ts#arabicUiCollator` `src/utils/scheduleOrder.ts#compareCourseSection` `tests/dashboard-home-term-audit.ts`

FR-SCH-037 Teacher mini-week and week window

The system shall show in the appointment form the chosen instructor's week in five columns on a compact axis (at least six hours), free slots fitting the course pattern with the required gap, «before/after» options adjacent to each lecture's edges on start-ladder steps (0, 30 and 60 minutes where no ladder applies), and a header with load (now → after), attendance days and gaps.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Free slots stop at the 10-minute gap from each lecture and snap to a 60- or 90-minute ladder from 08:00 for a single family.
2. «Before/after» skips the anchor that is the appointment's own course and section, and returns the anchor's hall if free, otherwise its building.
3. A drawn appointment starts on the five-minute grid and lasts at least half an hour if no regulation length is known; an edited row is not counted as an addition to the load.

EVIDENCE

`src/utils/teacherMiniWeek.ts` `src/utils/lectureStick.ts` `src/components/TeacherMiniWeek.tsx` `src/components/LectureStickChips.tsx` `tests/teacher-mini-week-audit.ts` `tests/week-window-audit.ts` `tests/lecture-stick-own-audit.ts`

DR-SCH-001 Appointment (schedule row) data model

The system shall store an appointment with a numeric id, a revision rev starting at 1, college, department, term, course, section number, instructor, five day flags (Sunday to Thursday), start and end in normalised HH:MM, building and hall with their official identities, a snapshot of the course name and code, the legacy fdetail day encoding, a source order sourceOrder, and a change origin when approved from a proposal.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. «9:00» is stored as «09:00»; anything that is not a valid clock is not stored.
2. A manually added row carries sourceOrder ≥ 1,000,000; an imported row is below it.
3. origin.kind comes from the closed list (registration radar, student case, instructor request, Analytics Center, section planning), refId up to 64 characters and note up to 160.

EVIDENCE

`src/types.ts` `src/utils/scheduleTime.ts#normalizeClock` `src/server/core/scheduleWrites.ts` `tests/schedule-time-window-audit.ts`

NFR-SCH-001 Indexed conflict detection with identical results

The system shall detect conflicts by lookup in indexes (day + instructor, day + hall, day + paired course, course + section) instead of comparing every appointment with every other, and return exactly the result of the original exhaustive sweep kept as reference — pairs, reasons, severity, wording and order.

● Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> A single reading held the production server for 2.5 seconds; a missed conflict is worse than a slow schedule.

ACCEPTANCE CRITERIA

1. The parity test compares both paths on random whole-term schedules and fails on the first difference in pair, reason or order.
2. One cached current-term read is used, so a full decade is never scanned during interactive use.
3. Heavy reads are cached: department habit 10 minutes, department window 30 seconds, university day 30 seconds with a maximum 2-second wait.

EVIDENCE

`src/utils/scheduleIntelligence.ts#findConflicts` `src/utils/scheduleIntelligence.ts#findConflictsExhaustive` `src/server/core/rhythm.ts` `src/server/core/universityDay.ts` `tests/conflict-parity.ts`

IR-SCH-001 Drag preview before the drop

Inferred

The schedule interface shall classify the effect of moving a lecture before the drop into five classes (excellent, good, neutral, suboptimal, impossible), never count soft notes as a blocker that sends the card home, and reserve «impossible» for a move with a blocking conflict.

Implemented · Should · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A move creating a high-severity blocker is «impossible»; one that adds conflicts, 90 or more minutes of gap, or 20% or more day pressure is «suboptimal».
2. A one-day row shorter than 100 minutes takes the target day's length (50 or 80) when moved.
3. The reduced-motion preference is honoured and the animated preview stops when requested.

EVIDENCE

`src/components/SchedulePhysics/physics.ts#classifyDecision` `src/components/SchedulePhysics/physics.ts#buildMoveCandidate` `src/components/SchedulePhysics/useReducedMotion.ts`

IR-SCH-002 Saved views, undo and offline work

Inferred

The schedule interface shall keep named saved views (at most three favourites) with a dot signalling that the state differs from them, offer undo (Ctrl+Z) with a log of sixty changes and a fifteen-second bar, and defer row edit, delete and batch move when the connection drops (up to 120 operations), showing later server refusals for review instead of discarding them.

Implemented · Should · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Saving a fourth favourite is refused; the views menu does not appear until the first view is saved.
2. Creating a new appointment offline is not deferred (a POST other than batch move); repeated PUTs of the same row are coalesced into the last one.
3. What the server refuses after reconnecting moves to the review shelf and can be retried or discarded.
4. The default view on phones is «list»; on desktop it is the saved choice of week, list or rooms.

EVIDENCE

`src/utils/scheduleViews.ts` `src/components/ScheduleViewsMenu.tsx` `src/utils/offlineScheduleQueue.ts` `src/components/Schedules.tsx` `tests/run-tests.ts`

3.4.6 Algorithms

ALG-SCH-01 · Deterministic scheduling engine

Indexed conflict detection

Deterministic

Does this appointment collide with any other appointment in the whole term, and what is the most serious reason?

1. Four indexes are built from the term's rows across all colleges: day+instructor, day+hall, day+student-paired course, and course+section.
2. For each checked row only the rows found in those indexes become candidates, walked in ascending original order.
3. Two rows overlap when they share a day and each starts before the other ends; the faculty placeholder is never treated as a person who can be double-booked.
4. All reasons for the pair (instructor, hall, duplicate, same section in two places, student clash, short turnaround) are collected and the pair is reported once, named after the most serious.
5. The result is classified: a double booking or an indistinguishable duplicate blocks; same section in two places, short turnaround and student clash are non-blocking notes.

Inputs

The candidate row, all rows of the term, the department's declared turnaround, course pairs from the student survey, and the placeholder instructor ids.

Output

A list of conflicting pairs, each with a type, severity and written reason.

Fully deterministic: the indexed path returns exactly the same pairs, reasons, order and wording as the exhaustive reference sweep kept beside it, and this equality is tested automatically.

Required turnaround in the same hall: 10 minutes on every day unless the department declares 1 to 60.

`src/server/core/scheduleConflicts.ts#scheduleConflicts` `src/utils/scheduleRegulations.ts#requiredGapForDays` `src/utils/locationRegistry.ts#sharePhysicalRoom` `src/utils/scheduleBlockers.ts` `tests/conflict-parity.ts` `tests/client-placeholder-conflicts-audit.ts` `tests/blocker-oracle-audit.ts`

FR-SCH-006 FR-SCH-007 FR-SCH-008 FR-SCH-009 FR-SCH-010 NFR-SCH-001

ALG-SCH-02 · Deterministic scheduling engine

Regulation rules: length, start ladder, patterns and hours ceiling

Deterministic

Does the appointment's shape comply with the regulation and the approved pattern library without exceeding the section's hours?

1. A length is accepted when it equals k lectures of the day's length with 10-minute gaps, k not exceeding the course hours, or equals a library block length.
2. The start is compared with the regulation ladder: 60-minute steps on Sun/Tue/Thu and 90-minute steps on Mon/Wed from 08:00; a start off the ladder is a non-blocking note.
3. The section's days must match a library pattern for the course hours: exactly, as a one-day block, or with the same lengths on non-adjacent days.
4. The section's weekly minutes (this row plus its other distinct meetings, never counting an identical copy twice) are summed and compared with the ceiling.

Inputs

The appointment's days and times, the course hours, and the section's other meetings.

Output

A refusal message for length or ceiling, a note for ladder or pattern, or nothing.

Fixed rules using the regulation's figures; department history never overrides them.

50 minutes on Sun/Tue/Thu, 80 minutes on Mon/Wed, 10-minute gap. Weekly-minutes ceiling = round(hours × 50 × 1.1). The Mon/Wed double block is accepted 5 minutes short (08:00–10:50).

`src/utils/scheduleRegulations.ts#lectureLengthProblem` `src/utils/scheduleRegulations.ts#startLadderNote` `src/utils/scheduleRegulations.ts#meetingsMatchApprovedPattern` `src/utils/scheduleRegulations.ts#sectionWeeklyHours` `src/server/core/scheduleRules.ts#scheduleLengthIssue` `tests/lecture-length-audit.ts` `tests/approved-patterns-audit.ts` `tests/section-schedule-hours-audit.ts` `tests/pattern-library-audit.ts`

FR-SCH-016 FR-SCH-017 FR-SCH-018

ALG-SCH-03 · Deterministic scheduling engine

Statistics from history

Department rhythm and memory from history

What do the department's past terms say about this time and hall, and does the appointment break its habit?

1. All the department's terms are read and lectures are split into two day families (Sun/Tue/Thu, Mon/Wed); a lecture spanning both is left out.
2. Gaps between consecutive lectures are measured within one term and one day; the mode and its agreement share are computed, and likewise the length mode with its 5th–95th percentile range.
3. The department's start ladder = starts used at least twice, top 12.
4. The memory counts, per day, hour and hall, how many terms used it, and speaks one sentence only when it is notable (never used, rare, or abandoned).
5. On save: the regulation ladder first; then a start more than 0 and up to 20 minutes from a habitual start (for mixed-family days), then a length outside the library — as a soft note.

Inputs

All historical rows of the department, courses and instructors, and the candidate row.

Output

A rhythm reading per day family, and at most one note about habit and one about the hall.

Deterministic descriptive statistics with no trained model; silent when the sample is small or agreement weak, never blocks a save and never outranks the regulation. The learned gap does not set the enforced turnaround rule.

Gap habit: at least 6 pairs at 55% agreement, gaps capped at 45 minutes.

Length habit: at least 8 lectures at 35% agreement.

Memory: at least 3 terms; a habit at 70%, rare at 12% or less.

The reading is cached per department and refreshed in the background after 10 minutes.

`src/utils/departmentRhythm.ts#learnRhythm` `src/utils/departmentRhythm.ts#offRhythm` `src/utils/departmentMemory.ts#readDepartmentMemory` `src/server/core/rhythm.ts#departmentStyle` `tests/run-tests.ts` `tests/approved-patterns-audit.ts`

`FR-SCH-020`

ALG-SCH-04 · Deterministic scheduling engine

Deterministic

Travel time between campuses and buildings

Does the appointment leave the instructor enough time to travel from an adjacent appointment at another campus?

1. Each college's campus (main, Jahra, Fahaheel) is identified and the time for each pair is read from the registrar's policy with its exemption list.
2. For each day of the appointment and each other appointment of the instructor at a different campus, the gap between the adjacent end and start is measured; overlaps are ignored (they have their own rule).
3. If the gap is below the configured time a shortfall is recorded; duplicates by day and pair are removed.
4. Within one campus the walking time between two buildings is read from the college's settings.

Inputs

The candidate appointment, the instructor's appointments across the term, the travel policy and college campuses.

Output

Up to three 'tight transfer' warnings stating the minutes needed and available.

A non-blocking warning in the department form; zero disables a pair and exempt instructors are not checked.

60 minutes Jahra–Fahaheel and 30 for other pairs by default; range 0–180 with up to 500 exemptions.

3 minutes inside a building, 5 between unconfigured buildings, configured pairs 1–120.

`src/utils/campusTravelPolicy.ts#travelShortfalls` `src/utils/campusTravel.ts#interCampusMinutes` `src/server/core/travelPolicy.ts#readTravelPolicy` `src/server/core/scheduleRules.ts#travelMinutesFor` `tests/undo-travel-rule-audit.ts`

`FR-SCH-019` `FR-LOC-028` `FR-LOC-029`

ALG-SCH-05 · Deterministic scheduling engine

Search and ranking

Time and hall suggestion engine

Which available times and halls are best for this appointment, and why is there no option when none remains?

1. Every start on a 30-minute step within the department window is combined with every official, pinned or approved-barter hall.
2. A candidate is dropped if the instructor is busy, the hall is taken, it is outside the barter window, travel time is insufficient, or no companion facility is free; each rejection reason is counted.
3. Score = 100 − instructor idle/3 − walking/2 − spread of the department day/6 (per day), plus the hall-suitability adjustment, clamped to 0–100.
4. Candidates are sorted by score, then start and hall, keeping one option per time in 'time and hall' mode.
5. The ranked list is passed through the same save gate until the requested count is reached.

Inputs

Days and duration, eligible halls, hall/instructor/department rows, travel times, barter windows, suitability and companion facility.

Output

Three suggestions (six for 'suggest a hall') with reasons, or the dominant reason for emptiness.

Writes nothing; every suggestion passes the same conflict gate that governs saving; ordering is stable on ties.

A day with no lecture for the instructor costs 90 idle minutes; up to 48 candidates are sent through the gate.

`src/utils/suggestionEngine.ts#rankSuggestionCandidates` `src/utils/suggestionEngine.ts#orderSuggestionCandidates` `src/utils/suggestionEngine.ts#emptySuggestionReason` `src/server/routes/scheduleToolsRoutes.ts#registerScheduleToolsRoutes` `tests/slot-suggestion-recovery-audit.ts` `tests/appointment-forms-audit.ts`

FR-SCH-027

ALG-SCH-06 · Deterministic scheduling engine

Search and ranking

Least-impact repair chain (domino)

What is the shortest chain of moves that removes a conflict without creating another?

1. For each row alternative placements are generated (day × 30-minute start × department hall) and ranked by acceptability: keep the hall, the instructor's days, the same day, closeness to his lectures, morning.
2. Depth-first search with iterative deepening: depth 1 is exhausted before depth 2 opens, so the first chain found is the shortest.
3. A placement that displaces exactly one row makes that row the next card; a row already moved in the chain is never displaced again, and never more than one at a step.
4. A chain is accepted when the conflict count drops by at least one; for a closed hall the hall is forbidden and the board must be no worse.

Inputs

The target row, the term's rows, forbidden halls and placeholder ids.

Output

A list of moves with conflict counts before and after and the number of instructors and halls affected, or nothing.

Proposes, never writes; never returns a chain that leaves more conflicts than it found; the chain is committed whole through the atomic batch move or not at all; with no honest plan it returns none rather than a partial one.

Maximum depth 5 moves (3 for a hall closure), branching 6 out of the best 24 placements.

Ranking weights: same hall +30, a day the instructor teaches +24, same day +18, proximity up to +20, start 08:00–14:00 +8.

`src/utils/repairChain.ts#findRepairChain` `src/utils/repairChain.ts#planDisruption` `src/components/Schedules.tsx` `tests/run-tests.ts`

FR-SCH-022 FR-DRF-020

ALG-SCH-07 · Deterministic scheduling engine

Cover availability and substitute ranking

Search and ranking

Who can cover this lecture on this date, and who fits best?

1. An instructor is busy if a weekly lecture overlaps the hour on that day, unless it is cancelled on that date or covered by someone else.
2. He is also busy if another cover is recorded for him on the same date overlapping the hour.
3. The lecture's own instructor, anyone with a non-empty status, and anyone neither teaching this term nor with history on the course are excluded.
4. Score: taught the course 50 + min(12, 2 × terms), same department +20, same building that day +10 (or lectures that day +6), + max(0, 10 − load hours).

Inputs

The appointment and date, all term rows, the college's history of the course, and week exceptions.

Output

Up to 12 ranked substitutes with reasons.

The same rule guards recording a cover and the ranking; nothing is sent automatically; an exception never touches the row or its revision.

`src/utils/coverAvailability.ts#coverConflict` `src/server/routes/schedulesRoutes.ts` `tests/doctor-journey-audit.ts`

`FR-SCH-029`

ALG-SCH-08 · Deterministic scheduling engine

Common meeting time

Search and ranking

When are selected members free together for a meeting of a given length?

1. For every day and every start on a 30-minute step, the set of busy members is computed from the whole term across colleges.
2. Consecutive starts with the same busy set are merged into one window.
3. Windows are ranked by fewest busy, then longest, then closest to 10:00.
4. At most two windows per day enter the ranked list.

Inputs

Term rows, selected members and duration.

Output

Free windows per day, near misses, a ranked list of up to 6, and the best time when everyone is free.

Deterministic; the server drops the placeholder, visiting instructors and anyone with a status, and reveals only free/busy per name.

Duration 30 to 180 minutes.

`src/utils/meetingSlots.ts#computeMeetingSlots` `src/server/routes/scheduleToolsRoutes.ts` `tests/meeting-slots-audit.ts`

`FR-SCH-028`

3.4.7 Edge cases

⚠ Two combined sections with one instructor, hall and hour

Scenario Sections 501 and 502 of one course with the same instructor in the same hall at 11:00.

Behaviour One high-severity conflict with instructor and hall reasons is raised and blocks the save; combined delivery is no exemption, because an unreported double booking is a decision nobody made.

⚠ The same section in two halls with two instructors

Scenario Two rows with the same course, section, days and time: one in G28 and one in G31 with different instructors.

Behaviour A medium «same section in two places» warning blocks no save, signature or approval; if either row has no hall it returns to a blocking duplicate.

⚠ Two simultaneous edits of one appointment

Scenario Two colleagues open a row at revision 3; the first saves making it 4, then the second saves.

Behaviour The second is refused with 409 carrying the current row and their own version to choose from, and nothing is written; without rev it is written without comparison.

⚠ An existing row left outside a narrowed university day

Scenario The administrator narrows the day to 08:00–18:00 while the schedule holds a 18:30–19:20 row.

Behaviour The row stays and an edit that does not touch its time is not blocked; changing its time or creating a similar one is refused with 400. The administrator is told how many rows fall outside before saving.

⚠ A slot held by an instructor request awaiting the department

Scenario An instructor chose Sunday 10:00 in his request and a coordinator tries to place another lecture for him at the same hour.

Behaviour A 409 request-hold is returned with the holder and item; the coordinator confirms or cancels; confirming resends with overrideRequestHolds and passes only that request while other instructors' holds keep guarding.

⚠ Second step of a two-row swap fails

Scenario A swap of two rows passes its dry run, then one row changes at the same instant and the second write is refused.

Behaviour What was written is restored and applied=[] with reverted=true is returned; if the first step cannot be restored a message names the course for manual review.

⚠ Appointment count changes while confirming a wipe

Scenario A user saw a warning to wipe 120 appointments, then a colleague added one before «wipe» was pressed.

Behaviour A 409 wipe-count with the new number is returned and nothing is wiped; even if the count changes inside the lock the repository throws ScopeChangedError and nothing is wiped.

⚠ A term declared ended

Scenario A department tries to edit an appointment in a term flagged AdTermClosed.

Behaviour 409 schedule-locked is returned before any field message, and only the schedule committee chair with an explicitly written role can work in it; to reopen, the flag is lifted in the terms screen.

3.4.8 Standing owner decisions

- Only a physical conflict blocks signing and approval: a real instructor in two places, a hall booked twice, or an indistinguishable duplicate (rule 2026-09-27).
- The pattern must be approved from the library; history is a suggestion that never overrides the regulation, and department habit does not lift the hours ceiling (2026-10-30).
- The university day blocks everywhere; a narrower department exception warns but does not block in the appointment form (2026-11-05).
- Inter-campus travel time is mandatory in the teacher card and a non-blocking warning in the department's appointment form (2026-11-05).
- Two sections of a specialisation course at the same time: a warning, not a block, in the appointment form, and a block in the teacher card for specialisation courses only (owner note 8).
- Every course number a reader sees is seven digits; printed reports keep their design.

↑ Contents

3.5 · DRF

Drafts, scenarios and the time machine

Try on a copy, compare the effect, apply in one piece and go back with one click.

27	100%	29	18
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.5.1 Overview

This module holds everything that is tried before the live schedule is touched: schedule drafts and start-of-term drafts with publishing behind a safety point, the version log with restore and term comparison, the time machine with its scenarios, solutions, atomic apply and restore point, what-if simulation on the registration radar, and ending the previous plan sheet. It serves the schedule committee chair who writes, and the department head and dean who read and simulate. Nothing is written to the live schedule without explicit confirmation, and nothing changed after a draft's base is erased without being named.

Schedule committee chair · Department head · College dean · System administrator · Root administrator · The system (automatic)

3.5.2 User journeys

Schedule committee chair — **Starting a new term from the previous one**

1. Opens the living layer and picks the previous term as the source for the department.
2. Reads the summary of what will be copied (read only), then builds the «بداية الفصل» draft.
3. Sections appear numbered from 501 with each row's issues and soft warnings beneath it.
4. Edits the instructor, time or room of a flagged row and the check reruns, showing N issues remaining.
5. When the draft is ready, publishes with explicit confirmation and sees the number of appointments and how many had their time adjusted.
6. If it was a mistake, presses undo and the version captured before publishing comes back.

Schedule committee chair — **Emergency: a room is down on Sunday**

1. Opens the time machine and writes «قاعة G23 تعطلت الأحد», or picks an example taken from the department's own schedule.
2. Sees up to three solutions of different philosophies, the best one marked, and a comparison table.
3. Checks the chosen solution's effect on instructors, rooms and students (from the radar) and its indicator.
4. Saves the scenario under a name and presses «اعتمد الكل»; the sheet shows what changes and who is notified.
5. After applying, sends the affected instructors their cards by WhatsApp himself.
6. If something is wrong, presses restore and the schedule goes back as it was, unless it was edited after.

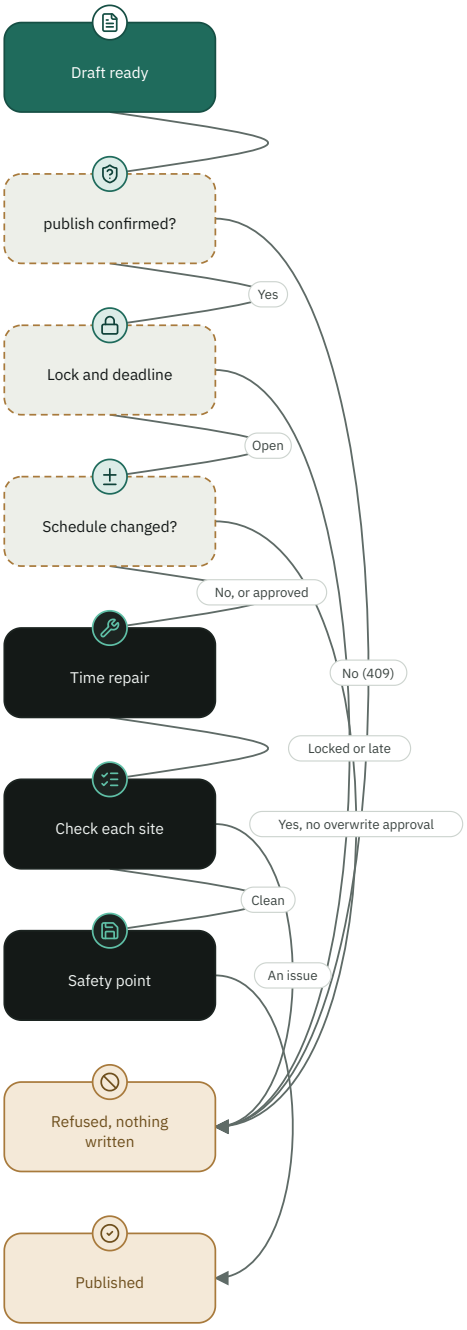
College dean **The dean reviews a shared scenario**

1. Opens the time machine for a department in the college and picks it from the selector.
2. Sees only the scenarios the coordinator shared for viewing.
3. Simulates and reads the before/after effect, and sees «عرض ومحاكاة — الاعتماد لمن يكتب الجدول».
4. Any attempt to save or apply is refused by the server with 403.

3.5.3 Diagrams

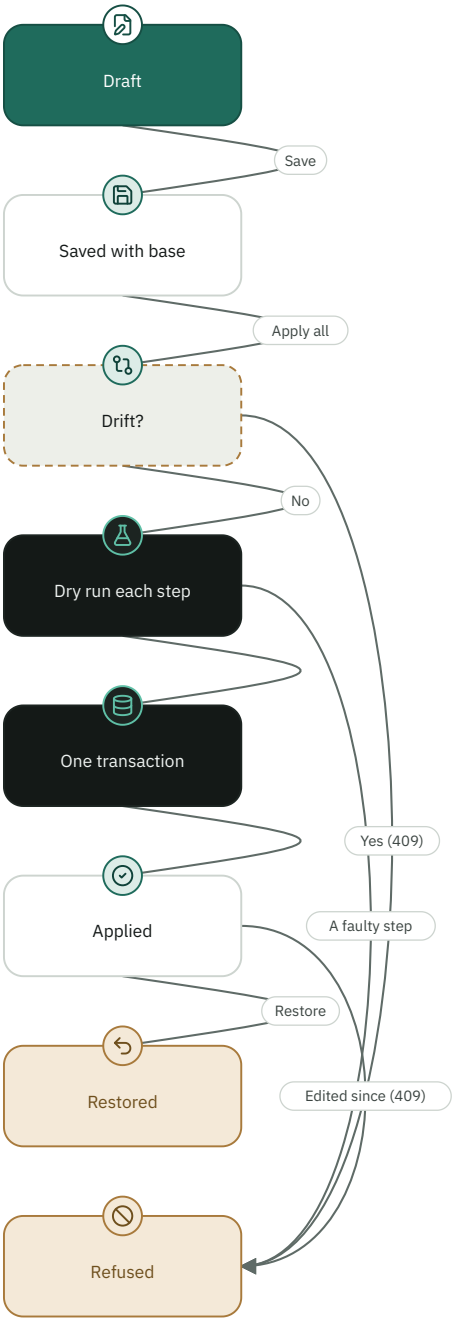
Publishing a draft

Every gate is checked before the first write; the publish happens entirely or not at all.



Scenario lifecycle

Applying is one transaction and the restore point is written in it.



Who can do what in the time machine

	Read and simulate	Save and share	Apply and restore
Schedule committee chair	✎	✎	✎
Department head	👁	—	—
College dean	👁	—	—

✎ Write 👁 Read — No access

3.5.4 Screens

Time machine
Three modes: emergency (sentence → three solutions), lab (free editing), goal; steps: write, compare, apply; college and department pickers in the header.`src/components/TimeMachine.tsx`

Scenario workbench
Impact indicator, before/after, time slider, domino chain, student impact, save, share, copy, apply, restore and instructor notice cards.`src/components/ScenarioWorkbench.tsx`

Section simulator
Section cards with remove, add, merge and split in a basket measured together.`src/components/SectionSimulator.tsx`

Move decision preview
«لم يُحفظ شيء بعد»: from and to, effect on blockers, gap, quality and pressure, then confirm or cancel.`src/components/SchedulePhysics/ScheduleDecisionPreview.tsx`

End the previous plan sheet
For the main account: pick departments and press end or restore.`src/components/PreviousPlansPanel.tsx`

3.5.5 Requirements

FR-DRF-001 **Create a schedule draft**

The system shall create a schedule draft for a given department and term from rows supplied by the user, and store its base (the fingerprint of the live schedule at creation time) without changing the live schedule.

● Implemented · Must · Functional · Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. A create request for a department outside the user's scope is refused with 403 and no draft is created.
2. An empty name is stored as «سيناريو جديد», and a name longer than 100 characters is cut to 100.
3. A draft source outside the list (what-if, auto, import, manual) is stored as what-if.
4. After a successful create (201) no appointment of the live schedule has changed.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `tests/role-guard-audit.ts` `tests/intelligence-scope-api.mjs` `tests/blockers-stream-audit.ts`

FR-DRF-002 Validate a draft before saving and cap its size

Inferred

The system shall refuse to save a draft while preview issues remain, required fields are missing, or the stored size of its rows exceeds 960,000 bytes, and shall state the reason before any write.

Implemented · Must · Functional · Schedule committee chair

RATIONALE

> A draft is a single database document limited to 1 MiB; the case is reported with the row count before writing instead of a generic failure.

ACCEPTANCE CRITERIA

1. A non-empty preview-issue list (cut at 80 distinct items) yields 400 and no draft is saved.
2. A row missing required fields yields 400 with the list of issues.
3. Rows exceeding 960,000 bytes of JSON yield 413 with code DRAFT_TOO_LARGE and nothing is written.

EVIDENCE

`src/server/routes/draftsRoutes.ts`

FR-DRF-003 Edit and delete draft rows

The system shall allow editing only specific fields of a draft row (instructor, time, days, location), deleting one row or all rows, and shall re-check the draft after each change and return issues per row.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Editing a row's course or section number through the edit route changes nothing (only 15 fields are allowed).
2. The reply carries issues, issueRowIds, rowIssues, rowWarnings and ready, and ready is true only when there are no issues.
3. Deleting all rows without the header x-schedule-confirm: delete-all-draft-rows yields 409.
4. A row that is not in the draft yields 404.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `tests/import-term-tools-audit.ts`

FR-DRF-004 Start-of-term draft from the previous term

The system shall build a draft for a department in a new term from the schedule of a different, earlier term, without writing any live appointment, renumbering the sections of each course from 501 upwards and skipping appointments of archived courses while stating their number.

Implemented · Must · Functional · Schedule committee chair

RATIONALE

> The result is a draft that is reviewed and edited before publishing; the live schedule does not change.

ACCEPTANCE CRITERIA

1. A source term equal to the target term yields 400, and a source with no schedule for this department yields 400.
2. Sections of a copied course numbered 1, 10 and 11 become 501, 502 and 503.
3. A missing instructor is replaced by the instructor of a sibling section of the same course when one exists, and an empty room is taken from a sibling section.
4. The reply is 201 with coverage (sourceRows, archivedSkipped, copiedRows, skippedRows, adjustedRows) and the count of items needing review.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts` `tests/import-term-tools-audit.ts` `tests/blockers-stream-audit.ts`

FR-DRF-005 Appointment gate for start-of-term drafts

The system shall ask every row of a start-of-term draft the same questions a new appointment is asked (fields and location, approved hours, approved pattern, conflicts) against the term as it will become, and a soft warning shall not block.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. The check calls schedulePayloadIssues, scheduleLengthIssue, unapprovedPatternIssue and scheduleConflicts for every row.
2. A soft conflict appears in rowWarnings, not in issues, and does not block publishing.
3. A check that fails internally adds «تعذّر فحص هذا الموعد. أعد المحاولة قبل النشر.» and counts as an issue.
4. The check runs on create, edit, delete and publish, and rows are checked in batches of 8.

EVIDENCE

`src/server/core/draftChecks.ts` `src/server/routes/draftsRoutes.ts` `tests/import-term-tools-audit.ts`

FR-DRF-006 Publish a draft with explicit confirmation, lock and deadline gates

The system shall publish a draft only with explicit confirmation in the header, and shall refuse to publish when the site schedule is locked or a wholesale replacement falls outside its allowed deadline, the deadline being measured at publish time and not at save time.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A publish request without the token publish in x-schedule-confirm yields 409.
2. A locked schedule yields 409 with code schedule-locked and nothing is written.
3. An imported or start-of-term draft outside the wholesale-replacement deadline yields 409 with code deadline-wholesale.
4. A draft outside the user's scope yields 403, and a missing draft yields 404.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/server/core/versions.ts` `tests/approval-integrity-audit.ts` `tests/blockers-stream-audit.ts`

FR-DRF-007 Never erase changes made after the draft's base

The system shall, when publishing a draft that has not been published yet, compare the live schedule with its base and refuse once with a list of what would be erased if the schedule changed, proceeding only with the overwrite-newer approval.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A draft saved on Sunday and published on Tuesday after a colleague's Monday edit yields 409 with code scope-changed-since-base.
2. The list is cut at 40 lines and the full count is sent in total.
3. Repeating the request with the header «publish, overwrite-newer» publishes after the user agrees.
4. A difference only in identifiers of the very same appointments, with nothing erased, does not cause a refusal.

EVIDENCE

`src/server/core/versions.ts` `src/utils/scopeFingerprint.ts` `src/utils/scopeOverwrite.ts` `src/server/routes/draftsRoutes.ts` `tests/blockers-stream-audit.ts`

FR-DRF-008 Repair time collisions when publishing a start-of-term draft

Inferred

The system shall, when time collisions in the new term block a start-of-term draft, run the optimiser up to 4 passes moving time only (never course, instructor, day or room), then re-check and report how many rows had their time changed.

Implemented · Should · Functional · Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. A successful reply carries adjusted = the number of rows whose time differs from the stored one.
2. Issues remaining after repair yield 400 with issues, issueRowIds and rowIssues and nothing is written.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/server/routes/intelligenceInsightRoutes.ts`

FR-DRF-009 Publish every site or nothing

The system shall, when publishing a draft that carries branch sites, place every row in the department of its site according to its building, and check permission, lock, deadline and validation for every site before the first write, so that the publish either succeeds on all sites or writes nothing.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A row whose site cannot be resolved yields 400 with code BRANCH_SCOPE_UNRESOLVED and no row is published.
2. A site outside the user's scope yields 403 with code BRANCH_SCOPE_FORBIDDEN.
3. A course with no twin in the other site's department is copied from the home catalogue with the same code and listed in plantedCourses; an unknown course yields 400 with code BRANCH_COURSE_MISSING.
4. A sibling site's lock or deadline yields 409 naming the site and no row is published.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `tests/approval-integrity-audit.ts`

FR-DRF-010 Safety point before publish and approval-cycle notice

The system shall capture a version of every site before replacing it, mark the draft as published with its time, notify the approval cycle for each site separately, and return the version identifiers so the publish can be undone.

Implemented · Must · Functional · Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. The version is labelled «قبل نشر: <draft name>» with source publish.
2. The reply carries count, scopes (siteLabel and count), undoVersions, adjusted and plantedCourses.
3. An accepted schedule that is replaced returns to a new approval round at its own site, not at the draft's site.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/server/core/versions.ts` `tests/blockers-stream-audit.ts` `tests/approval-integrity-audit.ts`

FR-DRF-011 Version list and two-version comparison

Inferred

The system shall list the latest 80 versions of a department and term scope without row snapshots, and compare two versions of the same scope, counting what was added, removed and unchanged.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A failure reading the log returns an empty list with the header X-Reading-Partial: versions instead of a 500 error.
2. Two versions from different scopes yield 403, and a missing version yields 404.
3. The comparison key is course, section, instructor, days, start, end and room.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/utils/scheduleVersionList.ts` `tests/intelligence-scope-api.mjs`

FR-DRF-012 Restore a previous version

The system shall restore a version only with explicit confirmation, after checking lock, deadline, overwrite and validation, and shall capture a version of the current state before replacing so the restore can itself be undone.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Without the token restore in x-schedule-confirm the result is 409.
2. A version holding invalid times or conflicts yields 400 with issues (historical values are tolerated through resolveHistorical).
3. A change to the schedule after the version's base yields 409 with code scope-changed-since-base before any write.
4. The version captured before restoring is labelled «قبل استرجاع: <version label>».

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/server/core/versions.ts` `tests/blockers-stream-audit.ts`

FR-DRF-013 Compare two terms of a department

The system shall compare a department's schedule in two terms, listing appeared, disappeared and moved appointments (up to 80 per group) with each term's score and the visiting-instructor mark according to each side's own term.

Implemented · Should · Functional · Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. The reply carries appeared, disappeared, moved, fromTermName, toTermName, fromScore and toScore.
2. Each moved row carries fields, before and after; rooms are shown with a slash, not a pipe.
3. A missing one of the four identifiers or a scope that is not allowed yields 403.
4. Comparing a term with itself at department scope reads a single college in perCollege.

EVIDENCE

`src/server/routes/draftsRoutes.ts` `src/utils/previousTermRequest.ts` `tests/intelligence-scope-api.mjs` `tests/visiting-badge-group3-audit.ts`

DR-DRF-014 Scenario record

The system shall store a scenario as one record holding its name (up to 80 characters), mode (emergency, lab, goal, sections), problem (up to 200 characters), list of changes (up to 80), the revision of every row it touches, status (draft, applied, restored), share flag, revision number and the last 40 history entries.

Implemented · Must · Data · Schedule committee chair

ACCEPTANCE CRITERIA

1. Saving stamps the revision of every row a change touches (base[rowId] = the row's rev).
2. Saving with a stale revision number yields 409 with code revision, and saving an applied scenario yields 409 with code applied.
3. A change pointing to an appointment outside the scope or no longer present yields 409 with code stale.
4. There are four change types: move, assign, remove and add; remove and assign are cut at 12 rows, add at 4 rows, and an unknown mode becomes lab.

EVIDENCE

`src/server/timeMachineRoutes.ts` `src/utils/timeMachine.ts` `tests/time-machine-audit.ts`

FR-DRF-015 Who may read and simulate, and who may save and apply

The system shall let everyone who reads the schedule within their scope read and simulate the time-machine context, restrict save, share, copy, apply, restore and delete to those who write the schedule (permission 7 and a non-read-only role), and show a non-writer only what has been shared with them.

Implemented · Must · Functional · Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. The dean reads the context (canWrite=false) and sees the shared scenario.
2. A save or apply by the dean or department head yields 403 from the server, not from hiding the button.
3. Reading a frozen term returns frozen and canWrite=false.

EVIDENCE

`src/server/timeMachineRoutes.ts` `tests/time-machine-audit.ts`

FR-DRF-016 Mask the rest of the term outside the scope

The system shall send the appointments of other departments to the simulator masked: time and room without course or department name, keeping the instructor only when they belong to the scope's instructors, so double booking can be detected without revealing another department's data.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Every row outside the scope carries the course name «موعد قسمٍ آخر», department 0, college 0 and a section code x1 onwards.
2. An instructor from outside the scope's instructors is replaced by a negative number.

EVIDENCE

`src/server/timeMachineRoutes.ts` `tests/time-machine-audit.ts`

FR-DRF-017 Apply a scenario atomically

The system shall apply a scenario in one piece or not at all: refuse a frozen term, take the term lock, refuse if the live schedule has drifted from the scenario's base, dry-run every step through the same two save handlers against the final week, then write the rows and the scenario record in one transaction.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A frozen term yields 409 with code term-frozen, and a concurrent apply in the same term yields 409 with code apply-busy.
2. A row edited or deleted after the scenario was built yields 409 with code drift and the list of rows.
3. One faulty step voids everything: the reply carries applied=[] and reverted=true and no row or revision number changes.
4. An applied scenario cannot be applied twice (409).
5. A row at a locked site is not deleted; the result is 409 with code schedule-locked and the step number.

EVIDENCE

`src/server/timeMachineRoutes.ts` `tests/time-machine-audit.ts`

FR-DRF-018 One-click restore point

The system shall store at apply time a restore point (images of the rows before the change, ids of created rows and post-apply revisions) in the same transaction, put the schedule back with one click, and refuse the restore if any row was edited after applying.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Restoring a scenario that is not applied yields 409 with code not-applied.
2. A manual edit of a row after applying makes the restore yield 409 with code revision and nothing is written.
3. A successful restore sets the status to restored, clears the restore point and deletes the rows the apply created.
4. An applied scenario cannot be deleted (409), and copying it builds its base on the live schedule now and yields an unshared draft.

EVIDENCE

`src/server/timeMachineRoutes.ts` `tests/time-machine-audit.ts`

FR-DRF-019 Green, yellow and red impact indicator

The system shall compare the schedule before and after a scenario using the schedule's own measures and give an indicator: red if approval blockers rise, instructors over their load rise or students without a seat rise; yellow if not red and a time or instructor changed, gaps grew by more than 30 minutes or regulation notes rose; otherwise green.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Changing two rows of an instructor gives a non-green indicator and names the affected instructor.
2. Affected instructors do not include placeholder names such as «هيئة تدريسية».
3. The «اعتمد الكل» button is disabled when there are no changes, or when the indicator is red and blockers increased.

EVIDENCE

`src/utils/timeMachine.ts` `src/components/ScenarioWorkbench.tsx` `tests/time-machine-audit.ts`

FR-DRF-020 Problem reading and three ranked solutions

The system shall read a written problem sentence (room down, instructor unavailable, demand up on a course, clear a day, reduce gaps, free rooms) and propose up to 3 solutions of different philosophies (minimal change, students first, instructor comfort, fewer rooms, placeholder-instructor flexibility), none of which raises approval blockers, with the best one marked.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A sentence that is not understood returns nothing, not an invented solution.
2. Every solution in every mode has approval blockers after applying that are less than or equal to those before.
3. Every solution in the answer has a different philosophy, and only one carries the best mark.
4. A link sentence is cut at 200 characters, run once and then removed from the address; a long course number is matched in full, not by its last digits.

EVIDENCE

`src/utils/timeMachine.ts` `src/utils/timeMachineLink.ts` `tests/time-machine-audit.ts` `tests/time-machine-links-audit.ts`

FR-DRF-021 Student impact of a scenario from the radar only

The system shall measure a scenario's impact on students as counts, not names, from the registration radar alone and for the radar term only, and shall leave the figure empty when there is no radar instead of inventing zero.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A term other than the radar term returns available=false with the reason, and an internal error returns available=false too.
2. Changes pointing to rows outside the scenario's scope are dropped before measuring.
3. With no radar: shortfall and studentsMoved are null, not 0.
4. The impact verdict is apply, caution, avoid or neutral: harm to any student makes it avoid unless those helped outnumber those harmed, in which case it is caution.

EVIDENCE

`src/server/timeMachineRoutes.ts` `src/server/simulationData.ts` `tests/time-machine-audit.ts`

FR-DRF-022 Notify only the affected instructors

The system shall, after applying, show every instructor whose schedule changed a before → after card with their personal link (an existing link is reused) and a ready WhatsApp message that is sent only when the coordinator presses the button.

Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Cards cover the affected instructors only, and each card has at least one before or after line.
2. The instructor's message does not mention the room name.
3. When no staff card is published the screen shows «انشر بطاقات الأساتذة أولاً من أداة النشر».
4. The server sends nothing automatically.

EVIDENCE

`src/components/ScenarioWorkbench.tsx` `src/server/routes/shareRoutes.ts` `tests/time-machine-audit.ts` `tests/import-term-tools-audit.ts`

FR-DRF-023 What-if on a proposal's impact before approval

The system shall measure the effect of moving an appointment, swapping two appointments, opening a section or a batch of sections on the radar term's students without writing anything, and shall refuse any term other than the radar term.

Implemented · Should · Functional · Schedule committee chair, Department head, System administrator

ACCEPTANCE CRITERIA

1. kind accepts move, swap, opening and openings; swap needs exactly two moves, a batch holds up to 60 items, and a move holds up to two appointments.
2. A frozen term yields 409 with code term-frozen, and a term other than the radar term yields 409 with code other-term.
3. An appointment no longer present yields 409 with code stale, and one outside the scope for a non-administrator yields 403.
4. The reply carries impact, basis and stale and does not write to the schedule.

EVIDENCE

`src/server/simulationRoutes.ts` `src/server/simulationData.ts` `tests/radar-insight-audit.ts`

FR-DRF-024 Refresh and read the registration radar within scope

The system shall let the radar summary be read within the reader's scope, restrict running to the system administrator with one run at a time, and refresh it automatically once every 24 hours while student wishes are open and once more after they close.

Implemented · Should · Functional · System administrator, Root administrator, The system (automatic), Department head

ACCEPTANCE CRITERIA

1. A run by a non-administrator yields 403, and a second run while one is running yields 409 with code busy.
2. RADAR_AUTO_HOURS is 1 or more, otherwise 24, and RADAR_AUTO=false stops the automatic refresh.
3. After a schedule change the radar refreshes after at least 120 seconds of quiet and at least 10 minutes between two runs.
4. A frozen earlier term is readable for reference and not recalculated, and running it by name yields 409 with code term-frozen.

EVIDENCE

`src/server/simulationRoutes.ts` `src/server/simulationData.ts` `tests/registration-flow-audit.ts` `tests/radar-insight-audit.ts`

FR-DRF-025 End the previous plan sheet for departments

The system shall let the main account alone end the previous plan sheet for chosen departments (up to 400) so only the new sheet's courses appear in the schedule, and bring it back with a counter command.

Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Reading the list or executing as anyone but the main account yields 403.
2. Ending archives transition sheets only when their track has an active new sheet containing courses, otherwise the department is skipped and appears in skipped.
3. Restoring brings back only what this action archived.
4. The screen shows each department's state: running, ended, or no new sheet yet.

EVIDENCE

`src/components/PreviousPlansPanel.tsx` `src/server/planReviewRoutes.ts` `tests/plan-sheet-import-route-audit.ts`

IR-DRF-026 The «before apply» sheet

The interface shall not write when «Apply all» is pressed, but open a sheet showing the number of affected appointments, instructors and students, the first 6 changes by name, who is notified and how to go back, and write only on «Apply now».

Implemented · Should · Interface · Schedule committee chair

ACCEPTANCE CRITERIA

1. The back button and Escape close the sheet without writing, and inside the «data tools» dialog Escape closes only the topmost sheet.
2. More than 6 changes show 6 and say «و N أخرى».
3. After switching college or department the previous department's context is not shown and «اعتمد» is not active on it.

EVIDENCE

`src/components/ScenarioWorkbench.tsx` `src/components/TimeMachine.tsx` `src/components/timeMachineShared.tsx` `src/utils/timeMachineScope.ts` `tests/time-machine-scope-escape-audit.ts`

NFR-DRF-027 **Seven-digit course number on time-machine screens**

The time-machine and section-simulator screens shall display the course number in seven digits (college, department, course) in examples, labels and section cards, not the three digits stored in the catalogue.

Implemented · Must · Non-functional · Schedule committee chair

RATIONALE

> The official number has seven digits wherever the user reads it; the stored value stays as is, and the parser accepts both the long and the short form.

ACCEPTANCE CRITERIA

1. The demand example reads «زاد الطلب على 0109468», not «زاد الطلب على 468».
2. The section card, the add label and the course option in the section simulator show the number built with authorityCourseCode.
3. Typing «demand rose for 0109468» resolves to the stored course 468, and the stored code itself is still accepted.

EVIDENCE

`src/utils/timeMachine.ts#tmCourseNumber` `src/components/TimeMachine.tsx` `src/components/SectionSimulator.tsx` `src/components/AllCourses.tsx` `src/server/timeMachineRoutes.ts` `tests/time-machine-links-audit.ts` `tests/course-number-seven-digit-audit.ts`

3.5.6 Algorithms

ALG-DRF-01 · Drafts, scenarios and the time machine

Deterministic

Scenario overlay: apply, drift and write plan

What does the schedule look like after these changes, and has the live schedule moved since the scenario was built?

1. Changes (move, assign, remove, add) are applied in order to a copy of the schedule without touching the original.
2. The revision of every live row the scenario touches is stored at build time.
3. Drift: a missing row is 'gone', a row whose revision changed is 'edited'.
4. Write plan: update only changed fields with the expected revision, delete, or create; all are dry-run before one atomic write.

Inputs

The live schedule, the change list, and base revisions.

Output

The resulting schedule, the drift list and write operations.

Pure functions; never writes over a colleague's edit; applies all or nothing.

`src/utils/timeMachine.ts#applyChanges` `src/utils/timeMachine.ts#driftOf` `src/utils/timeMachine.ts#planOperations` `src/server/timeMachineRoutes.ts` `tests/time-machine-audit.ts`

`FR-DRF-017`

ALG-DRF-02 · Drafts, scenarios and the time machine

Deterministic

Green, yellow and red impact indicator

Does this scenario make the schedule worse, or only change someone's week?

1. Both boards are measured with the same measures: blocking conflicts, over-load instructors, students without a seat, gap minutes and regulation notes.
2. Changed rows, affected instructors and halls, and time and hall changes are counted.
3. Red if blockers, over-load or seatless students increase.
4. Yellow if not red and a time or instructor changed, gaps grew by more than 30 minutes or regulation notes rose; otherwise green.

Inputs

Schedule before and after, courses, instructors and load limits, registration radar data when present.

Output

A level (green/yellow/red), short reasons and impact counts.

Student counts come only from the radar, otherwise left empty; no invented number.

Gap threshold 30 minutes.

`src/utils/timeMachine.ts#compareBoards` `src/utils/timeMachine.ts#measure` `tests/time-machine-audit.ts`

`FR-DRF-019` `FR-DRF-021`

ALG-DRF-03 · Drafts, scenarios and the time machine

Search and ranking

Problem reading and three ranked solutions

What are the three best solutions of different philosophies for a written problem (hall down, instructor absent, demand up, clear a day, gaps, halls)?

1. The sentence is read with fixed word rules into a problem kind and its parties.
2. Drafts are built under different philosophies: minimal change (domino chain), students first, instructor comfort, fewer halls, placeholder flexibility.
3. Duplicate drafts are removed, as is any draft that raises blocking conflicts, leaves the patterns or introduces a clash.
4. Cost = 5000 if unsolved + 1000×added blockers + 400×over-load + 60×seatless students + 10×changes + 6×instructors + 5×time changes + 2×hall changes + gaps/30 + 8×regulation.
5. Sorted ascending by cost; the best three are returned and the first is marked.

Inputs

The problem text, live schedule, courses, instructors, halls and radar.

Output

Up to three solutions, each with its changes, impact indicator and cost.

No solution raises approval blockers; proposes without writing, application goes through the atomic scenario path.

`src/utils/timeMachine.ts#parseProblem` `src/utils/timeMachine.ts#solveProblem` `src/utils/timeMachine.ts#compareScenarios` `tests/time-machine-audit.ts` `tests/review-1105-audit.ts`

`FR-DRF-020`

ALG-DRF-04 · Drafts, scenarios and the time machine

Time-collision repair for a start-of-term draft

Search and ranking

Can a schedule copied from an earlier term be published by moving times only?

1. Rows are ordered by instructor then id, and each is tried at every 30-minute start in the university day with its own duration.
2. Any alternative that raises the row's conflicts above their current number is rejected.
3. Cost = 10000×conflicts + 0.38×distance from the original time (−18 for staying) + a penalty after 16:00 + 0.16×gaps to the instructor's lectures (capped at 180 minutes each).
4. Repeated up to 4 passes or until nothing changes, then re-checked and the number of rows whose time changed is reported.

Inputs

Draft rows and the rest of the new term's rows.

Output

Rows with adjusted times and the changed count, or the remaining blockers.

Safety-first greedy: never changes course, instructor, day or hall, and never trades a clean row for a new conflict.

At most 4 passes, 30-minute step.

`src/server/routes/draftsRoutes.ts` `src/server/core/draftChecks.ts#draftAppointmentChecks`

`FR-DRF-008`

3.5.7 Edge cases

⚠ An old draft published after a colleague's edit

Scenario A draft was saved on Sunday, a colleague edited the live schedule on Monday, then «publish» was pressed on Tuesday.

Behaviour The server refuses once with 409 scope-changed-since-base and names what would be erased (up to 40 lines plus the full count); if the user agrees the request is repeated with «publish, overwrite-newer». A draft that was already published skips this comparison.

⚠ A start-of-term draft collides with other departments

Scenario Copied appointments clash on room or instructor with other departments' schedules in the new term.

Behaviour The optimiser moves time only, up to 4 passes at creation and at publish; if issues remain they are returned per row and nothing is written. A soft warning is shown under the row and does not block.

⚠ The live schedule moved after the scenario was built

Scenario A colleague edited an appointment the scenario touches before «اعتمد الكل» was pressed.

Behaviour The apply is refused with 409 drift naming the changed row (gone or edited), and the workbench shows the notice that the live schedule changed. No row is written.

⚠ Restore after a manual edit

Scenario A scenario was applied, someone then edited by hand a row it changed, and a restore was requested.

Behaviour The restore is refused with 409 revision because it would overwrite the edit; nothing is written and the scenario stays applied.

⚠ A file carrying branch sites where one site fails

Scenario A draft with rows for the main site, Jahra and Fahaheel, where one site is locked or a row has no known site.

Behaviour Every site is checked before the first write; the server returns 409 or 400 naming the site and no row is published at any site.

3.5.8 Standing owner decisions

- A course number is always seven digits (college 2 + department 2 + course 3, such as 0109105); the stored three-digit code is never displayed.
- Printed reports keep their design; fixes go in values and data, not layout.

↑ Contents

3.6 · VIW

Viewing, sharing, copying and transferring schedules

The department schedule to read, a link for whoever needs it, copying from term to term, and an export that keeps the whole picture.

24	100%	7	16
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.6.1 Overview

This module covers everything about looking at a schedule and handing it on: the read-only department schedule screen, saved views, the living-schedule layer with its six scenes including start of term and copying from the previous term, the previous-term ghost and the decision panel, the transfer dialog (export, instructor replacement, publish), share links with faculty cards, personal links and their lifetimes, and the journey screen. It serves the schedule committee chair, department head, instructors, students and the public visitor. What reaches people who are not signed in is aggregate and without names, and the link alone is the permission.

Schedule committee chair · Department head · College dean · Faculty member · Student · Public visitor · The system (automatic)

3.6.2 User journeys

Schedule committee chair — Publishing faculty cards, then notifying whoever's schedule changed

1. Opens «data tools» → «publish» and issues the department's faculty card with a last date for requests.
2. Picks the instructors, a personal link is minted for each and WhatsApp opens with a ready message.
3. Each link records the send time and the fingerprint of its owner's schedule.
4. The schedule is later edited and the panel marks «changed since sent» next to whoever the edit touched.
5. Resends to them alone, and if a new department card is issued it replaces the old one while personal links stay alive.

Department head — Reading and printing the department schedule

1. Opens «department schedule» from an account bound to one department.
2. Sees the current term's appointments sorted by days then time, with the visiting-instructor mark.
3. Reads the notice that the schedule may be a draft and checks approval status in «schedule changes».
4. Prints the page with the print button.

Schedule committee chair — **Retiring an instructor with all their appointments**

1. Opens «replacement» and chooses the instructor, then a replacement or «leave without an instructor».
2. Sees the number of appointments and the compatibility verdict before any change.
3. If there is a conflict, reads its reasons and chooses another replacement.
4. Executes; a version «قبل استبدال الأستاذ» is captured and the decision appears in the replacement log.
5. If it was a mistake, undoes it from the log.

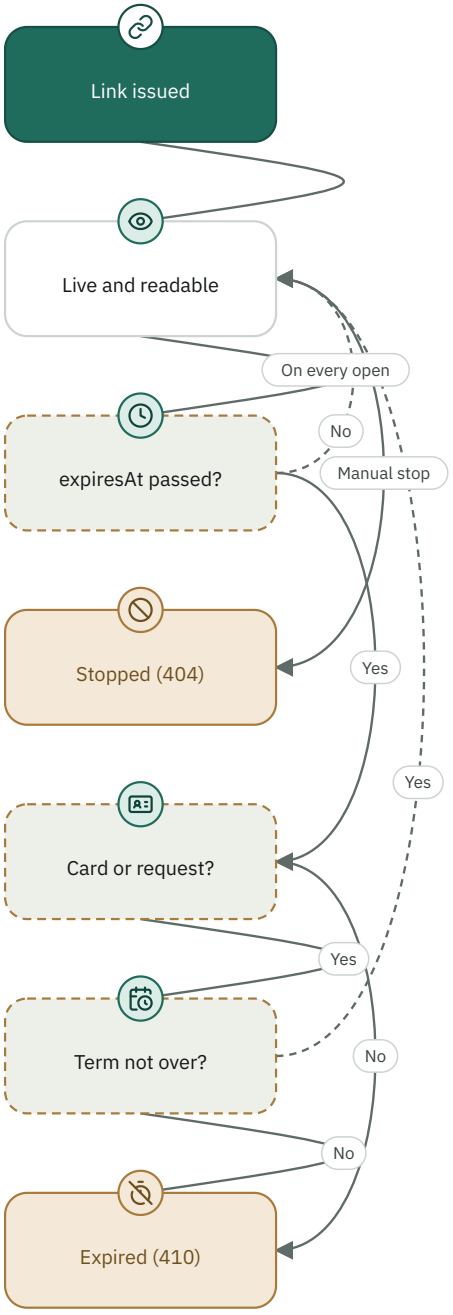
Public visitor — **A visitor opens the SCHEDULE journey before login**

1. Presses the journey button on the login page.
2. A full-screen sheet opens and the memory is read from the server as aggregate counts without names.
3. Browses the impact, the scale, the path and the philosophy; numbers move once, or show at once when reduced motion is preferred.
4. Closes it with Escape and focus returns to the button.

3.6.3 Diagrams

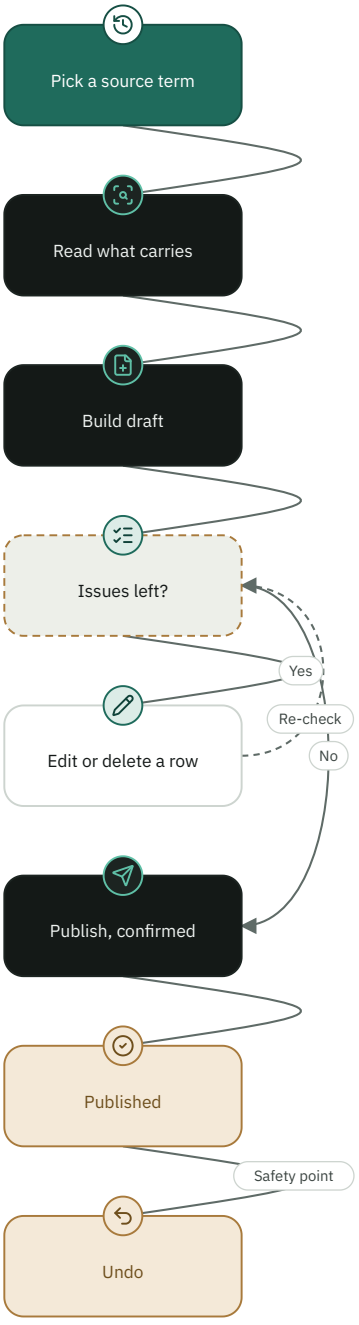
Life of a share link

Reading follows the link's kind; the written requests deadline governs writing only.



Copying the previous term

The reading and the draft never touch the live schedule; only publishing writes, and it has an undo point.



Who reaches what

	Department schedule	Issue links	Export and replace	Own card by personal link
Schedule committee chair	✎	✎	✎	—
Department head	👁	—	—	—
Faculty member	—	—	—	👁

✎ Write 👁 Read — No access

3.6.4 Screens

Department schedule
Read only: table and cards, print, and the visiting-instructor mark.`src/components/DepartmentSchedule.tsx`

Data tools (schedule transfer)
The export, publish and replacement tabs are covered here; import belongs to its own module.`src/components/ScheduleTransfer.tsx`

Publish the schedule
The department schedule for whoever holds the link, or a card for every instructor with a personal link, and the send log.`src/components/SchedulePublish.tsx`

Living schedule
Overview, pressure map, health and fairness, one-minute brief, start of term, safety net.`src/components/LivingScheduleLayer.tsx`

Previous-term ghost and decision panel
A comparison with the previous term over the schedule, and the decision-now panel.`src/components/ScheduleExperienceLayer.tsx`

Saved views
A menu that appears after the first saved view, and a naming dialog with one field.`src/components/ScheduleViewsMenu.tsx`

The SCHEDULE journey
Four narrative acts on a dark canvas with a fixed palette.`src/components/ScheduleJourney.tsx`

3.6.5 Requirements

FR-VIW-001 Read-only department schedule

The system shall show an account bound to a single department that department's schedule for the current term, read only, as a table on wide screens and cards on phones, sorted by days then time, with the visiting-instructor mark and course tags, a print button, and a notice that the schedule may be a draft.

Implemented · Must · Functional · Department head, Schedule committee chair, Standard user

ACCEPTANCE CRITERIA

1. An account without a single-department scope sees «هذا الحساب لا يملك نطاق قسم واحد لعرض جدوله.» and no rows.
2. The displayed term is the current term, or the first term in the list when it cannot be determined.
3. Columns: course, section, instructor, days, time and room, and every appointment appears both in the table and in the card.
4. The screen has no edit button, and an empty term shows «لا توجد شعب لهذا الفصل».
5. A visiting instructor carries the mark according to the row's own college and department.

EVIDENCE

`src/components/DepartmentSchedule.tsx` `tests/visiting-badge-group2-audit.ts`

FR-VIW-002 Export the term to a file

The system shall export a department's schedule for a given term to an Excel or JSON file carrying the appointments, courses, staff, halls and the term's visiting roster, and refuse the export outside the user's scope.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. A request without college or term yields 400 «حدد الكلية والفصل قبل التصدير.», and a scope that is not allowed yields 403.
2. The Excel file has a sheet «الجدول الدراسي» with nine columns (course code, course, section, instructor, civil number, days, time, building, hall) and a sheet «معلومات الملف» carrying the term name.
3. The file name is «جدول_<department>_<term>.xlsx» with forbidden characters replaced by an underscore.
4. An export by college with no department from a department account does not carry the courses of the college's other departments.

EVIDENCE

`src/components/ScheduleTransfer.tsx` `src/server/routes/scheduleImportRoutes.ts` `tests/role-scope-audit.ts`

FR-VIW-003 An ended term is read and export only

Inferred

The schedule-transfer dialog shall limit an ended term to the export and publish tabs only, hiding import, section planning, replacement and visiting instructors, and shall return any other open tab to export, so nothing is written into an archive.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Opening an ended term on a tab other than export or publish returns it to export.
2. The publish control does not issue a link for an archived term.

EVIDENCE

`src/components/ScheduleTransfer.tsx` `src/utils/termSequence.ts`

FR-VIW-004 Replace an instructor in a term in two steps

The system shall hand all of an instructor's appointments in a term to a named replacement or leave them without an instructor, in two steps: a preview showing the number of appointments and the replacement's compatibility, then execution with an explicit commit request, and shall never delete any appointment.

● Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. The preview returns preview=true, affected, compatible and reasons and writes nothing.
2. A replacement whose own appointments conflict yields 409 at execution with the reasons (up to 20 conflicts) and nothing changes.
3. A replacement equal to the instructor handed from yields 400, and execution on a locked schedule yields 409 with code schedule-locked.
4. Before executing a version «قبل استبدال الأستاذ: A ← B» is captured and undoVersionId is returned, and the approval log is notified of an edit.
5. The replacement log shows the last 20 decisions drawn from the last 80 versions, and they are reversed through restore.

EVIDENCE

`src/server/routes/scheduleToolsRoutes.ts` `src/components/ScheduleTransfer.tsx` `tests/term-freeze-audit.ts` `tests/approval-integrity-audit.ts`

FR-VIW-005 Read what will carry over before copying a term

The system shall, before a start-of-term draft is built, read the source term read-only and say what carries over with confidence and what carries a decision: a course no longer in the catalogue, a course whose hours or credits changed, an unavailable instructor, a room no longer used, listing up to 40 concerning appointments and up to 6 notes on the department's pattern.

● Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. The reply carries sentence, concerns (up to 40), concernCount, style and guardrail «قراءة فقط. لم يُنشأ شيء ولم يتغيّر شيء.»
2. Each concern carries one or more flags: course-gone, course-changed, instructor-away and room-retired.
3. An empty source term says so plainly («لا جدول»).
4. A missing department or source term, or a scope not allowed, yields 403.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts` `src/utils/termRollover.ts` `tests/run-tests.ts`

FR-VIW-006 Start-of-term screen: build, review, publish and undo

The screen shall build a start-of-term draft from the chosen previous term, show its rows with each row's issues and warnings, allow editing a row, deleting it or deleting all with confirmation, publish with explicit confirmation that asks before erasing anything that changed, and then allow undoing the publish.

● Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Rows with issues are marked and scrolled to automatically, and after every edit or delete the issues are refreshed and the screen says «ملاحظة N بقيت» or «المسودة جاهزة للنشر».
2. «حذف جميع المواعيد» asks first and sends x-schedule-confirm: delete-all-draft-rows, and its message is «لم يُنشر أي تغيير على الجدول الرسمي».
3. Publishing goes through the overwrite question (publish then overwrite-newer), and after success the number of appointments and the number whose time was adjusted are shown.
4. Undo goes through the safety point with the token decision-undo and captures a version of the current state before restoring.
5. When reduced motion is preferred the visual choreography is skipped and only the plain result remains.

EVIDENCE

`src/components/LivingScheduleLayer.tsx` `src/components/GenesisChoreography.tsx` `tests/safety-net-restore-audit.ts` `tests/blockers-stream-audit.ts` `tests/import-term-tools-audit.ts`

FR-VIW-007 The six living-schedule scenes

The living layer shall present six scenes: overview, pressure map, health and fairness, one-minute brief, start of term and safety net, from a single reading carrying the schedule pulse and the quality, resilience, fairness, fragility, rooms, topology and brief figures, within the reader's scope.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The reading returns pulse, health, fairness, fragility, roomIntelligence, topology, brief, memory, constraints.count and capabilities.
2. A missing department or term yields 400, and a scope that is not allowed yields 403.
3. analysisScope=department restricts the analysis to the department, otherwise it covers the college.
4. Every scene carries the visiting-instructor mark wherever an instructor is named.

EVIDENCE

`src/components/LivingScheduleLayer.tsx` `src/components/LivingScheduleScenes.tsx` `src/server/routes/intelligenceInsightRoutes.ts` `tests/blockers-stream-audit.ts` `tests/visiting-badge-group3-audit.ts`

FR-VIW-008 One approval-blocker number everywhere

The living-schedule reading shall count approval blockers with the approval bar's own options (placeholder exemption and the same room identity), so the number never differs between the two screens.

Implemented · Must · Functional · Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. The living route calls approvalBlockerOptions once and passes the same object to every memoised reading.
2. «مانع اعتماد N — حالة الجدول» equals the approval bar's number.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts` `tests/one-blocker-number-audit.ts`

NFR-VIW-009 The living reading is computed once, not once per asker

Inferred

The server shall cache the living-reading reply per scope for 120 seconds and clear it on any schedule write, and the interface shall share the in-flight request between components within an 8-second window and at most 8 scopes, without one component's abort cancelling another's request.

Implemented · Should · Non-functional · The system (automatic)

RATIONALE

> A cold reading was measured on production at up to 27 seconds on the single-threaded server.

ACCEPTANCE CRITERIA

1. Two consecutive requests for one scope within 120 seconds are served from memory, and demo sessions are never cached.
2. In the interface the oldest reading is dropped when more than 8 scopes are held.
3. Changing scope does not cancel a reading in flight; its result is only discarded.

EVIDENCE

`src/utils/livingCache.ts` `src/server/routes/intelligenceInsightRoutes.ts` `src/components/ScheduleExperienceLayer.tsx`

FR-VIW-010 Last reading of the three indicators

Inferred

The interface shall keep in the browser the last quality, resilience and fairness values per scope and show them dimmed and labelled as the last reading as soon as the layer opens, until the real reading arrives and replaces them.

Implemented · Could · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. The key schedule.lastScores.v1 in browser storage.
2. Blocked storage or an old value makes the rings wait as before, and decides nothing.

EVIDENCE

`src/utils/livingCache.ts` `src/components/LivingScheduleLayer.tsx`

FR-VIW-011 Previous-term ghost over the schedule

Inferred

The interface shall let a user turn on a previous-term ghost over the current schedule for one department or for all, marking each earlier appointment as unchanged, changed or removed, and summarising how many are unchanged, how many changed instructor, room or day, and how many are missing.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The previous term is the first term in the genome history, otherwise the term next after the current one from newest to oldest.
2. With no previous term the screen shows «لا يوجد فصل سابق محفوظ لهذا القسم بعد.» and requests nothing.
3. An appointment is compared by identity (course and section) and by placement (days, time, instructor and room).
4. Turning the ghost on needs no college or department: with none chosen the whole term is read, and changing scope switches the ghost off.

EVIDENCE

`src/components/ScheduleExperienceLayer.tsx`

FR-VIW-012 The «decision now» panel

Inferred

The interface shall open a decision panel that asks the server for the decision room of the department and term scope, shall end every reading within a 35-second ceiling with a clear Arabic reason for any failure, and shall show an empty term as guidance, not as a panel.

Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. No reply within 35 seconds gives «استغرقت القراءة وقتاً أطول من المتوقع. حاول مرة أخرى.»
2. The statuses 429, 401, 403 and 500 and above each give a specific Arabic message, and no English server text reaches the screen.
3. An empty room shows its message as guidance.

EVIDENCE

`src/components/ScheduleExperienceLayer.tsx` `src/server/routes/intelligenceRoutes.ts`

NFR-VIW-013 Analysis reads are deferred and quiet on failure

Inferred

Reads of genome, constraints and the living reading in the experience layer shall be deferred to browser idle time (at most 6 seconds), and a soft failure (timeout or interruption) shall be ignored without an error banner while the last reading is kept, because the schedule works fully without them.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Nothing is requested on every click in the schedule, only once after idle time.
2. An error marked soft raises no banner, and a non-soft one shows its Arabic text.

EVIDENCE

`src/components/ScheduleExperienceLayer.tsx`

FR-VIW-014 Issue a share link

The system shall issue a share link for a department and term of four kinds (department schedule, faculty cards, course survey, students' schedule), with a lifetime computed on the server, and restrict issuing to those who own the department.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Without college, department or term the result is 400, and a scope that is not allowed yields 403.
2. The lifetime of a department or survey link is 1 to 180 days (30 by default), and a students' link lasts 365 days.
3. A live students' link is returned and no second one is issued, so the printed QR sticker does not change.
4. A survey for an ended term yields 409 with code term-ended.
5. Hiding instructor names is stored in showInstructors (on unless false is sent).

EVIDENCE

`src/server/routes/shareRoutes.ts` `src/server/core/shareLinks.ts` `tests/teacher-card-flow-audit.ts` `tests/publish-pick-teachers-audit.ts` `tests/student-journey-audit.ts`

FR-VIW-015 One live department card per term

The system shall make a new faculty card for a department (and its family of sites) replace the earlier one: old department cards are stopped, personal links stay alive and take the new requests deadline, and issuing is protected by a serial lock.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. The issue reply carries replaced = the number of old cards stopped.
2. The link list shows only the single newest live card of the family.
3. The requests deadline requestsCloseAt is stored separately and carried to personal links.

EVIDENCE

`src/server/routes/shareRoutes.ts` `src/server/core/staffCard.ts` `tests/teacher-card-flow-audit.ts` `tests/publish-pick-teachers-audit.ts`

FR-VIW-016 Link lifetime: reading until the end of the term

Faculty cards (department link and personal link) and the request link shall stay readable until the end of their term, not until the requests deadline, with the end of life computed from the term's dates, else from its name, else after 150 days; the requests deadline governs writing only.

Implemented · Must · Functional · Schedule committee chair, Faculty member

ACCEPTANCE CRITERIA

1. A link for a term ending on 31 December ends with the declared end of the term, and a link for a past term is not born expired.
2. A term of unknown date gives +150 days (TERM_LINK_FALLBACK_DAYS).
3. A department-schedule or survey link is readable only until expiresAt.
4. The requests deadline is written at the end of the Kuwait day (23:59:59+03:00), the same instant as the approval deadline.

EVIDENCE

`src/utils/shareLinkLifetime.ts` `src/server/core/shareLinks.ts` `tests/doctor-journey-audit.ts`

FR-VIW-017 Opening and stopping a link

Inferred

The system shall, when a link token is opened, answer 404 if it is missing or stopped and 410 if it expired, and build the public payload without civil number, phone or account, sorted by start time, and without instructor names when they are hidden.

Implemented · Must · Functional · Public visitor, Student, Faculty member

ACCEPTANCE CRITERIA

1. A stopped token: «(404) «الرابط غير موجود أو تم إيقافه»; an expired one: «(410) «انتهت صلاحية هذا الرابط».
2. The public row carries code, name, section, start, end, days, room, the revision rev, instructor and the visiting flag.
3. showInstructors=false empties the instructor name and drops the visiting mark.

EVIDENCE

`src/server/core/shareLinks.ts` `tests/doctor-journey-audit.ts`

FR-VIW-018 Revoke a link

The system shall stop a link on a delete request, and stopping a department card shall stop the whole department family together with unsent request links, leaving sent or signed requests and their reading links untouched.

Implemented · Must · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. Stopping a department card by a user who owns none of the family's sites yields 403.
2. Stopping a personal link is allowed to anyone who owns the department at any of its sites.
3. A missing link yields 404.

EVIDENCE

`src/server/routes/shareRoutes.ts` `tests/teacher-card-flow-audit.ts`

FR-VIW-019 Personal instructor links and send tracking

The system shall mint a personal link for every eligible instructor from the department card (up to 300 per request), reused when a live one exists, living until the end of the term and without names, and shall record for each link the send time and the fingerprint of the instructor's schedule, so it later knows whose schedule changed since they were notified.

Implemented · Must · Functional · Schedule committee chair, Faculty member

ACCEPTANCE CRITERIA

1. Eligible are those who teach in the department family this term, delegates or faculty of the department; others are skipped silently.
2. The reply links carries instructorId, id, expiresAt and reused.
3. GET /api/share-personal returns, for each link, changedSinceSent true when the schedule fingerprint differs from the fingerprint at sending.
4. The delivery list returns only the identifier, name and mobile, never the civil number.

EVIDENCE

`src/server/routes/shareRoutes.ts` `tests/doctor-journey-audit.ts` `tests/publish-requests-audit.ts`

FR-VIW-020 The calendar subscription lives while a card link lives

A fixed calendar subscription of an instructor shall keep showing their lectures as long as they hold at least one live card link (personal, their request link, or a department card of a department they teach in) readable now by the same lifetime rule, and when none is left it shall answer a valid empty calendar.

Implemented · Should · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. A stopped link, or an expired one whose term has ended, does not keep the calendar alive.
2. Department-schedule, survey and students' links do not count as card links.
3. An instructor with no identifier (0) gets no subscription.

EVIDENCE

`src/utils/shareLinkLifetime.ts` `tests/calendar-ics-audit.ts`

FR-VIW-021 Saved views

Inferred

The schedule screen shall let a user save a way of looking at the schedule (department and term, view mode, what colour means, legend folds, quick search and the rooms matrix) under a name of up to 40 characters, with up to 3 favourites, in the user's own browser, and opening a view changes nothing in the schedule.

Implemented · Could · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The storage key is schedule-views-<user id>, and a corrupted or old record is ignored and does not take the screen down.
2. A fourth favourite arrives unfavourited, and toggling a fourth is not carried out.
3. The menu is not in the toolbar before a first view is saved; the first save comes from the «حفظ العرض الحالي» command.
4. A dot after the name means the board no longer matches what was saved, with «تحديث العرض الحالي» and «العودة إلى المحفوظ».

EVIDENCE

`src/utils/scheduleViews.ts` `src/components/ScheduleViewsMenu.tsx`

FR-VIW-022 Warn when a saved view lost its scope

Inferred

The interface shall, when opening a saved view that names a college, department or term that is no longer available, say which one was lost and open what is possible instead of an empty board with no explanation.

Implemented · Could · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A view naming a merged department shows «القسم في هذا العرض لم تعد متاحة — فُتح ما أمكن منه.»
2. A view with an intact scope shows no message.

EVIDENCE

`src/utils/scheduleViews.ts`

FR-VIW-023 The journey screen

The SCHEDULE journey screen shall show a live reading of aggregate counts, not names (the number of appointments as the headline figure, then courses, instructors, departments, colleges and terms), in four narrative acts, to anyone who opens it before or after login.

Implemented · Could · Functional · Public visitor, Standard user, System administrator

ACCEPTANCE CRITERIA

1. The route /api/journey is public on purpose and returns scoped=false and counts without names or civil numbers.
2. Zero terms shows «الرحلة تبدأ من هنا.», and a failed reading shows «تعذّرت قراءة الذاكرة الآن».
3. Escape closes the sheet, tabbing is trapped inside it, and focus returns to the button that opened it.
4. The sheet is rendered into body so nothing in the sidebar paints over it, and when reduced motion is preferred the content shows at once.

EVIDENCE

`src/components/ScheduleJourney.tsx` `src/server/routes/dashboardRoutes.ts` `src/styles/10-journey.css` `tests/role-scope-audit.ts` `tests/presentation-coverage-audit.ts`

NFR-VIW-024 **Seven-digit course number on viewing and transfer screens**

The department schedule, the start-of-term preview, the appointment picker in the scenes and the export file shall show the course number in seven digits (college, department, course), not the code stored in the catalogue.

● Implemented · Must · Non-functional · Schedule committee chair, Department head

RATIONALE

> The official number has seven digits in everything the user reads or exports; the stored value stays as is.

ACCEPTANCE CRITERIA

1. The course column of the department schedule shows a seven-digit number built with authorityCourseCode.
2. The «رمز المقرر» column of the exported Excel file and the preview line in start of term show the same number.
3. The exported Excel file re-imports as is: the seven-digit number matches the course without replacing the stored code.

EVIDENCE

`src/components/DepartmentSchedule.tsx` `src/components/LivingScheduleLayer.tsx`
`src/components/LivingScheduleScenes.tsx#ContextPicker` `src/components/ScheduleTransfer.tsx`
`src/server/routes/scheduleImportRoutes.ts` `src/server/routes/smartImportRoutes.ts` `tests/course-number-seven-digit-audit.ts`

3.6.6 Algorithms

ALG-VIW-01 · Viewing, sharing, copying and transferring schedules

Two-snapshot diff and instructor schedule fingerprint

Deterministic

What changed since the last review, and has this instructor's schedule changed since it was sent?

1. Rows are matched by their stable id: absent before = added, absent after = removed, otherwise six fields are compared (time, days, hall, instructor, section, course).
2. The report is ordered removed first, then added, then changed, then by course and section.
3. Fingerprint: one line per row (id, course, section, days, time, hall and facility), sorted and hashed with 32-bit FNV-1a, prefixed with the row count.
4. The fingerprint is stored at send time and compared with the current one.

Inputs

Two saved schedule snapshots, or an instructor's rows.

Output

Added/removed/changed entries with counts, or a text fingerprint.

Deterministic and independent of row order; the fingerprint carries nothing readable from the schedule.

`src/utils/scheduleDiff.ts#diffSchedules` `src/utils/scheduleFingerprint.ts#instructorScheduleFingerprint`
`src/server/routes/shareRoutes.ts` `tests/schedule-changes-audit.ts` `tests/doctor-journey-audit.ts`

FR-VIW-019 FR-APR-026 FR-DRF-022

3.6.7 Edge cases

⚠ An faculty card after the requests deadline

Scenario The department closed the request window on a date, and an instructor opens their card after that date.

Behaviour The card stays readable until the end of the term (schedule and calendar) and only the request door closes at the requests deadline; the card does not say «permanent subscription».

⚠ A survey for an ended term

Scenario The coordinator asks for a survey link for a term that ended or closed.

Behaviour The server refuses with 409 term-ended and says to choose the next term, because answers nobody can act on are of no use. Other link kinds are unaffected.

⚠ A conflicting replacement instructor

Scenario The coordinator picks a replacement for a retired instructor who has an appointment overlapping one of the moved appointments.

Behaviour The preview says «يوجد تعارض» and names the reason, execution yields 409 and no appointment changes; the «no instructor» option is not checked because there is no replacement.

⚠ The previous-term ghost on «All»

Scenario The reader chose «All», so no college or department is set, and presses the previous-term ghost.

Behaviour The whole previous term is read without scope restriction and the ghost appears; the press used to return silently with no effect or message, and was fixed so the only condition is the term.

⚠ A saved view for a merged department

Scenario A view saved a year ago names a department that was merged or a term that was archived.

Behaviour What can be opened is opened and the message «فُتح ما أمكن منه — القسم في هذا العرض لم تعد متاحة.» appears, instead of an empty board with no explanation.

3.6.8 Standing owner decisions

- A course number is always seven digits (college 2 + department 2 + course 3, such as 0109105); the stored code and AdCourseId are never displayed.
- Printed reports keep their design; fix values and data, not layout, and any unavoidable change reuses the existing print classes and is checked on A4 landscape.

↑ Contents

3.7 · APR

Approval cycle

From committee signature to registrar acceptance: rounds, notes, deadlines, term freeze and verifiable stamps.

35	100%	18	31
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.7.1 Overview

The module governs how each department's schedule for a given term moves through six statuses: in preparation, signed by the committee, signed by the department head, with the registrar, returned with notes, and accepted. The schedule committee chair signs, then the department head, and the schedule reaches the registrar as a new round; the registrar writes notes on cells and either returns or accepts, and no schedule with a material conflict can be accepted. After acceptance every edit opens an amendment round automatically. The module also covers the term submission deadline, department exceptions and extension requests, freezing of past terms, the registrar inbox, the department balance and the council bulletin for the dean, the approval dossier, and printed-report stamps with public verification pages.

Schedule committee chair · Department head · University registrar · College registrar · Registration officer · Dean of registration · College dean · Vice dean for academic affairs · System administrator · Public visitor · Faculty member · The system (automatic)

3.7.2 User journeys

Schedule committee chair — **From signature to acceptance**

1. The committee chair opens the approval bar and signs once the schedule has no material conflict and every course has a registered capacity.
2. The department head signs next, and the schedule is sent to the registrar automatically as round 1.
3. A registration officer clicks the offending room or time to write a cell note, then returns the schedule.
4. The committee edits the flagged cell so the note becomes 'changed', or replies with a reason for keeping it.
5. The schedule is resent, opening a new round without a new signature; the registrar then accepts and any remaining notes are closed.

University registrar — Setting the deadline and granting an extension

1. The university registrar sets the term's last submission date in the Submission deadlines panel.
2. A department nearing its deadline sends an extension request with a reason and a number of days from its approval bar.
3. The request reaches the inbox with a suggested date; the registrar grants it or rejects it with a reason the department can read.
4. The exception stays specific to that department and its new date is counted from the later of the term deadline and its exception.

Department head — Amending an accepted schedule

1. The committee finds a room error after acceptance and edits the appointment directly.
2. The schedule returns automatically to 'with the registrar' in an amendment round whose baseline is the last accepted version.
3. Further edits accumulate in the same round as long as the registrar has not written a note in it.
4. If a section is added after the department head signed and nothing was ever accepted, sending waits for his one-click acknowledgement.

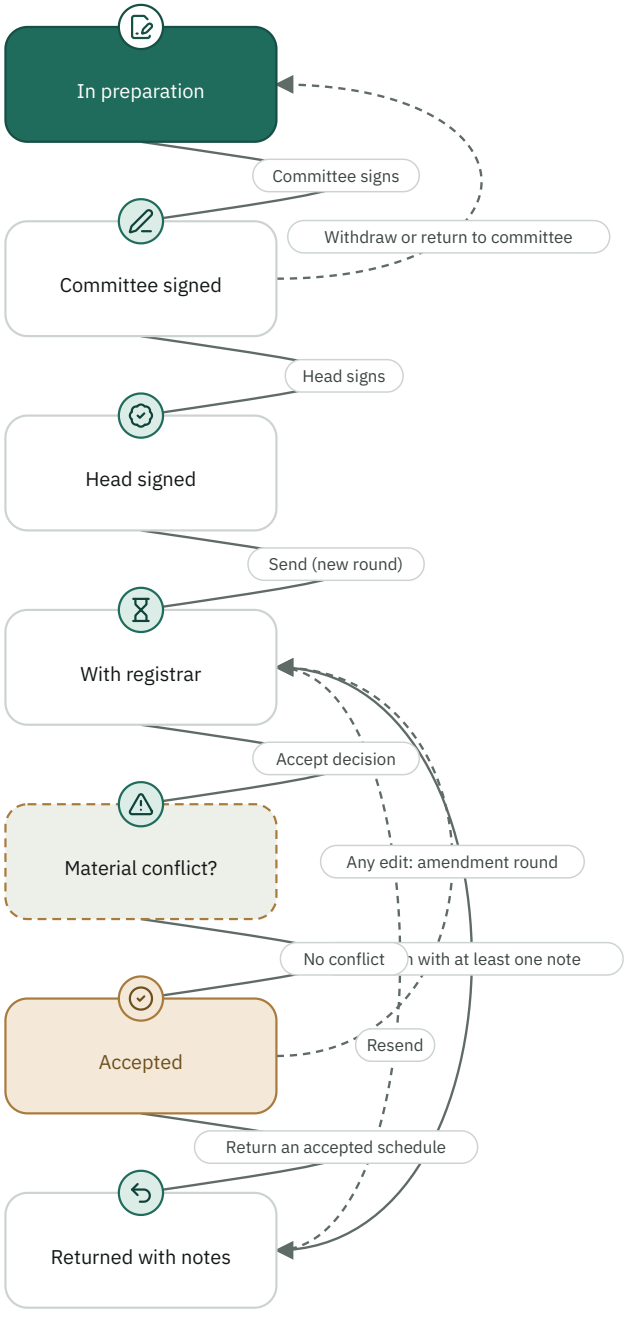
Public visitor — Verifying a printed sheet

1. Someone holding a comprehensive or change report scans the QR code in its footer or types the eight-character code.
2. The page compares the sheet's rows with the live schedule field by field, without sign-in.
3. It states either 'the sheet matches the schedule now' or 'the schedule changed after this sheet was printed', listing the affected course lines.

3.7.3 Diagrams

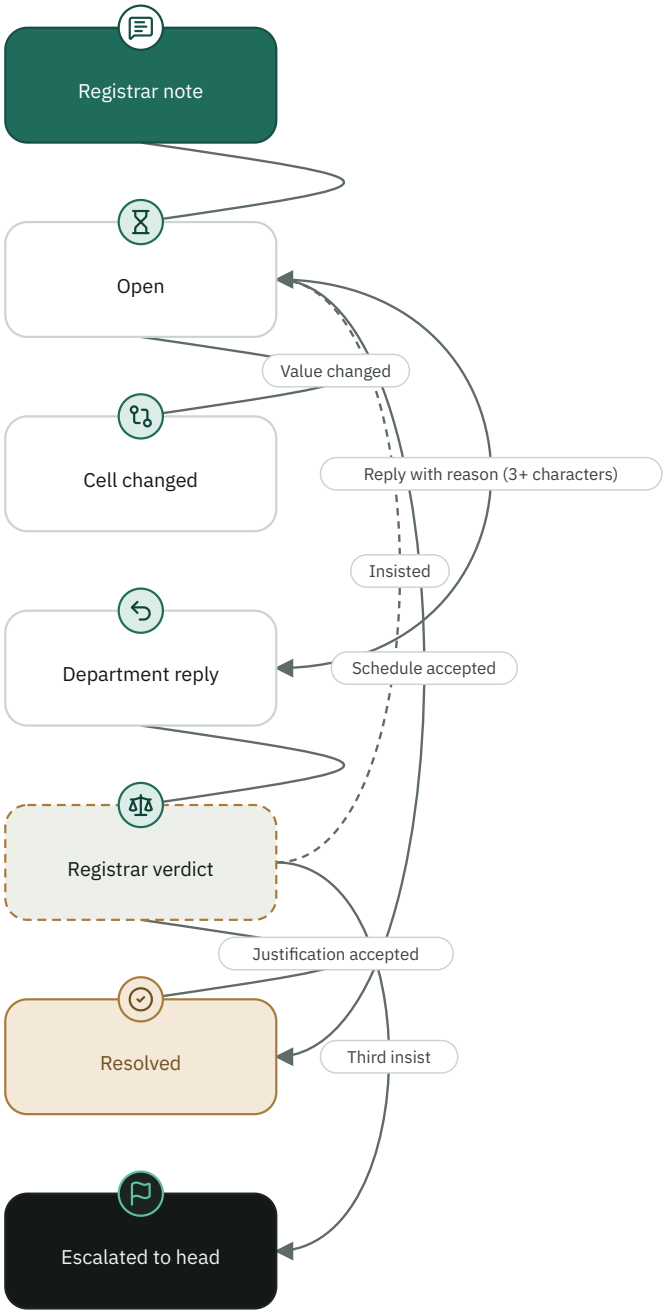
Approval states

The six statuses of APPROVAL_STATUS_LABEL and the transitions the server enforces.



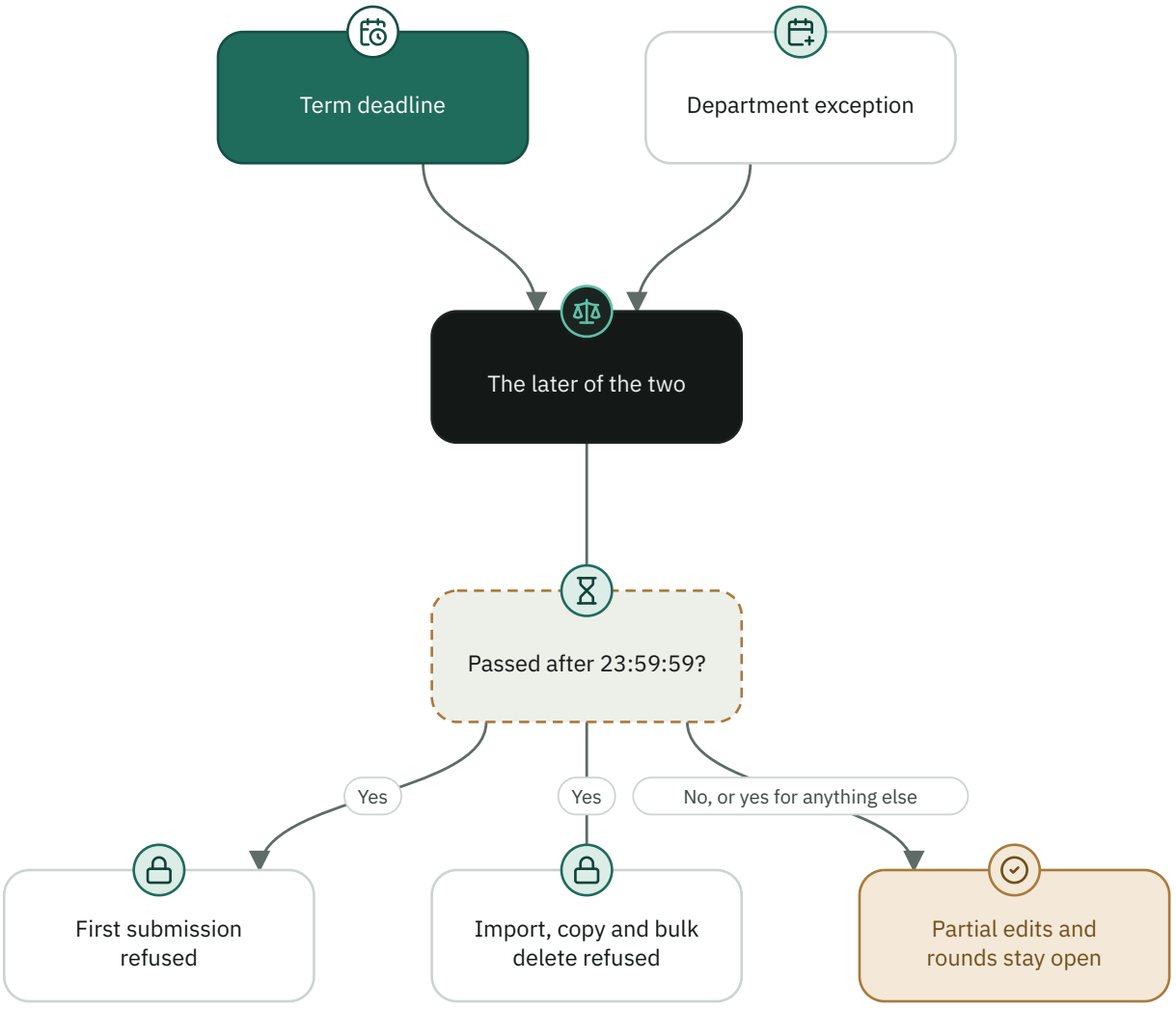
Cell note lifecycle

A note's state is computed from the cell's current value, never stored.



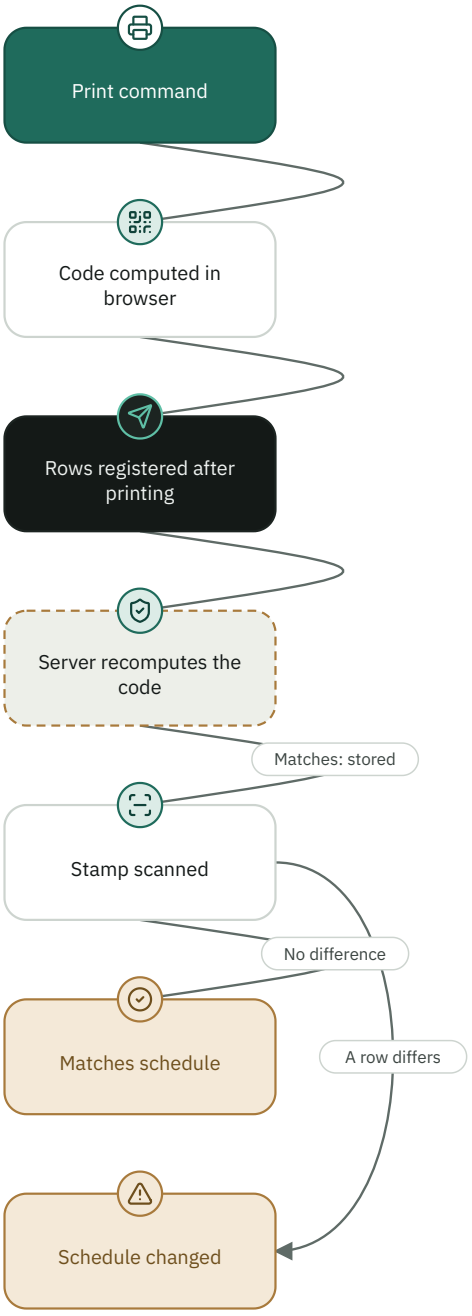
Deadline calculation and effect

The date is a full day in Kuwait time, and an exception never shortens a deadline.



Printed sheet stamp

The code is derived from the sheet's rows, registered after the print command, and compared with the live schedule when scanned.



Who can do what in the approval cycle

	Sign and send	Cell notes	Return and accept	Request extension	Grant extension	Set term deadline	Balance and inbox
Schedule committee chair	✎	✎	—	✎	—	—	—
Department head	✎	✎	—	✎	—	—	—
Registration officer	—	✎	✎	—	—	—	👁
College registrar	—	✎	✎	—	✎	—	👁
University registrar	—	✎	✎	—	✎	✎	👁
Dean of registration	—	—	—	—	—	—	👁
College dean	—	—	—	—	—	—	👁
Vice dean for academic affairs	—	—	—	—	—	—	👁

✎ Write 👁 Read — No access

3.7.4 Screens

Approval bar
Sits above the department schedule and shows only what the viewer must do now: sign, send, acknowledge added sections, return to the committee, request an extension.`src/components/ApprovalBar.tsx`

Schedule changes (registrar inbox)
The registrar's workplace: one row per department, cell notes, return and accept; the department sees the registrar's notes in place.`src/components/ScheduleChanges.tsx`

Submission deadlines
Panel for the term deadline, exceptions, department requests and the submission progress bar.`src/components/SubmissionDeadlines.tsx`

Approval dossier
A preview-and-print A4 sheet: what changed, blockers, signatures and department status.`src/components/ApprovalDossier.tsx`

Approval review
The last reading before approval: blockers, regulation findings and the approval score.`src/components/ScheduleReview.tsx`

Past term, read-only ribbon
Announces a frozen term and disables write buttons on the three screens that use it.`src/components/TermFrozenRibbon.tsx`

Department balance and council bulletin

A table with one row per department including approval status and deadline; printable as a council bulletin.`src/components/Reports.tsx`

3.7.5 Requirements

DR-APR-001 **Approval record and its event log**

The system shall keep exactly one approval record per college, department and term combination, keyed 'college:section:term', in one of six statuses (in preparation, committee signed, head signed, with the registrar, returned with notes, accepted), and shall not write the record before the first action occurs. It shall also record every action (sign, withdraw, send, return, accept, extension, extension request, amendment round opened) in the approval document as an event carrying actor, role, round and time, keeping only the latest 200 events.

● Implemented · Must · Data · The system (automatic)

RATIONALE

> A department that has not started is shown with an empty record built at read time, so storage is not filled with 'not started' records.

ACCEPTANCE CRITERIA

1. Reading a scope with no record returns status 'in preparation', round 0 and no signatures.
2. The record key for scope (1, 5, 20) is '1:5:20', and exactly six named statuses exist.
3. Appending 250 events leaves only the 200 most recent.
4. Each event carries the actor's name, role and the round number.

EVIDENCE

`src/utils/approvalWorkflow.ts#emptyApproval` `src/server/core/approvals.ts#readApproval` `src/utils/approvalWorkflow.ts#appendApprovalEvent` `src/server/core/approvals.ts#withEvent` `tests/approval-workflow-audit.ts` `tests/relay-fixes-audit.ts`

DR-APR-002 **Signature and captured version**

The system shall capture a version of the scope's schedule at each signature and store on the signature its stage, the signer's role label, the appointment count, the regulation-notice count, the version id and a six-character uppercase verification code derived from the version id, the signer and the signing instant.

● Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The code matches [0-9A-Z]{6}, is stable for the same version, signer and instant, and differs when any of them changes.
2. A code computed from empty input is still 6 characters long.

EVIDENCE

`src/utils/approvalWorkflow.ts#verificationCode` `src/server/routes/approvalsRoutes.ts` `tests/approval-workflow-audit.ts`

FR-APR-001 Signing order and who may sign

The system shall let the schedule committee chair sign first and the department head second, and shall refuse signing on an empty schedule, on a schedule with the registrar, or when the same stage is already signed, and shall refuse the head's signature before the committee's. It shall also restrict signing and withdrawing to accounts of the schedule committee chair (committee stage) and the department head (head stage) that are not seat team members, and shall refuse with code same-person-both-stages when the same person, identified by the instructor id on the seat, signs both stages of the same schedule.

● Implemented · Must · Functional · Schedule committee chair, Department head, System administrator

ACCEPTANCE CRITERIA

1. A head signature before the committee returns code wrong-order, and a repeated signature returns already-signed.
2. A schedule with no appointments returns empty, and one with status 'with the registrar' returns locked.
3. An administrator or registrar account attempting to sign receives 403.
4. A person occupying both seats who signs the second stage receives 409 naming the capacity of the first signature.

EVIDENCE

`src/utils/approvalWorkflow.ts#canSign` `src/server/routes/approvalsRoutes.ts` `src/utils/academicRoles.ts#signatureStage` `src/utils/seats.ts#actorMaySign` `tests/approval-workflow-audit.ts` `tests/role-scope-audit.ts`

FR-APR-002 Material conflict blocks signing, sending and acceptance

The system shall refuse signing, sending and acceptance while the scope's schedule contains a material conflict, meaning an instructor or room double booking in overlapping appointments or an indistinguishable duplicate entry, evaluated against the whole term and counted once per pair, and shall not let the 'same section recorded twice' warning block them.

● Implemented · Must · Functional · Schedule committee chair, Department head, University registrar, College registrar, Registration officer

RATIONALE

> Owner rule 2026-09-27: only a material conflict prevents signing; and one blocker number is used on every screen.

ACCEPTANCE CRITERIA

1. Accepting a schedule with a material conflict returns 409 with code blocking-conflicts and tells the registrar to return it to the department.
2. A clash with another department's appointment counts in both departments and is described generically, without naming that department or instructor.

EVIDENCE

`src/utils/scheduleBlockers.ts#readScopeConflicts` `src/server/routes/approvalsRoutes.ts#blockingSummaryFor` `tests/blocker-oracle-audit.ts` `tests/one-blocker-number-audit.ts` `tests/approval-workflow-audit.ts`

FR-APR-003 Other signing gates: capacity, requirements, regulation

The system shall refuse signing if the schedule contains a course with no registered capacity (code missing-capacity) or a course lacking its suitable room type or teaching nature (code course-requirements-missing), and shall name the incomplete courses in the message. It shall also refuse signing and sending with code regulation-blockers whenever the regulation review reports a decision 1912/2016 finding whose approval effect is 'block' within the scope's rows (theoretical-course pattern art. 7/8, fixed section time art. 7/9, consecutive teaching limits), and shall not require this check when the registrar accepts.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> This behaviour is narrower than the code comments' statement that the regulation is displayed and does not block; it is documented as implemented.

ACCEPTANCE CRITERIA

1. A course with no capacity blocks signing with a message to register the capacity first.
2. Other data gaps are shown as notices and do not block signing.
3. A schedule containing a section whose meetings start at two different times is refused signing with a 'resolve before signing' message.
4. Regulation findings with effect 'note' or 'review' do not block.

EVIDENCE

`src/utils/approvalWorkflow.ts#canSign` `src/server/routes/approvalsRoutes.ts` `src/utils/scheduleDataReadiness.ts` `src/server/core/approvals.ts#regulationNoticesForScope` `src/utils/scheduleRegulations.ts` `tests/approval-workflow-audit.ts` `tests/decision-1913-audit.ts`

FR-APR-004 Head signature, sending and its conditions

The system shall send the schedule to the registrar automatically in the same save that records the department head's signature, and if a blocker prevents sending (passed deadline, pending note, conflict) the signature shall stay and the response shall carry submitted=false with the blocking reason. It shall also refuse sending unless both the committee and department head signatures exist, and while sections added after the head's signature await his acknowledgement, a material conflict exists or a registrar note is open, shall refuse the first submission after the deadline has passed (code deadline) while allowing later rounds, and shall refuse sending a schedule that is already sent or accepted and unchanged.

● Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A successful signature returns submitted=true and the round number, and the status becomes 'with the registrar'.
2. A signature with an open registrar note keeps the signature and returns submitBlocked with the reason.
3. A single signature returns not-signed, a repeated send returns already-submitted, and an unchanged accepted schedule returns already-accepted.
4. The first submission after the deadline is refused while a second-round submission after it passes.

EVIDENCE

`src/server/routes/approvalsRoutes.ts#submitToRegistrar` `src/utils/approvalWorkflow.ts#canSubmit` `tests/relay-fixes-audit.ts` `tests/approval-workflow-audit.ts`

FR-APR-005 A new round for every send

The system shall open a sequentially numbered round at every send carrying the send time, the sender and the captured version id, shall write on the closing round the number of rows that actually moved since the registrar last saw it, and shall clear any pending extension request on sending.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. The current round is the one numbered with the current number, and the comparison baseline is the last version the registrar saw.
2. The first review has no earlier baseline: the whole schedule counts as new.

EVIDENCE

`src/utils/approvalWorkflow.ts#openRound` `src/utils/approvalWorkflow.ts#roundBaselineVersionId`
`src/server/routes/approvalsRoutes.ts#submitToRegistrar` `tests/approval-workflow-audit.ts` `tests/relay-fixes-audit.ts`

FR-APR-006 Signature withdrawal and head return to committee

The system shall let only the signer withdraw a signature, shall drop the department head's signature and its pending additions when the committee's signature is withdrawn, shall refuse withdrawal from a schedule with the registrar (locked), accepted (accepted) or with no signature by the requester (not-signed), and shall keep a returned schedule 'returned' after a withdrawal. It shall also let the department head return a schedule the committee signed and he has not signed, with a written reason of at least three characters (up to 400), dropping signatures, returning status to 'in preparation' and storing the reason in headReturn, which is cleared when the committee signs again.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The committee withdrawing its signature leaves the schedule with no signatures and status 'in preparation'.
2. The department head cannot withdraw the committee's signature.
3. A reason shorter than three characters returns 400, and a return after the head has signed returns 409 wrong-status.
4. The reason is shown in the committee's bar until it signs again.

EVIDENCE

`src/utils/approvalWorkflow.ts#canWithdraw` `src/utils/approvalWorkflow.ts#withRemainingSignatures`
`src/utils/approvalWorkflow.ts#canHeadReturn` `src/server/routes/approvalsRoutes.ts` `tests/relay-fixes-audit.ts`

FR-APR-007 Sections added after the head's signature

Unless the registrar has already accepted once, the system shall record a section added (or whose course or section number was swapped) after the department head signed in a queue awaiting his acknowledgement, naming at most 60 and counting the overflow, and shall block sending until the head acknowledges with one click limited to the ids shown to him.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Importing 300 rows after signing names 60, counts 240 overflow and blocks the whole send.
2. Whatever was added between opening the screen and clicking approve stays awaiting acknowledgement.

EVIDENCE

`src/utils/approvalWorkflow.ts#mergePendingAdditions` `src/utils/approvalWorkflow.ts#acknowledgeAdditions`
`src/server/core/approvals.ts#noteScheduleMutation` `src/server/routes/approvalsRoutes.ts` `tests/relay-fixes-audit.ts`
`tests/approval-workflow-audit.ts`

FR-APR-008 Cell notes and their computed state

The system shall let the registrar and the department write a note with one click on any of seven cells (time, days, room, instructor, section number, course, whole appointment) with optional text up to 600 characters, shall derive the origin from the author's role (registrar or department), shall tag the note with the current round and the cell value at writing time, and shall make a second click on the same cell replace the first note and clear its reply and verdict. It shall also derive a note's state from the cell's current value: 'resolved' if closed or its justification was accepted, 'answered' if a reply exists, 'changed' if the value differs from writing time, 'removed' if the row was deleted, otherwise 'open', and shall count as open only registrar notes whose state is 'open' in any round.

● Implemented · Must · Functional ·

Registration officer, College registrar, University registrar, Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. An empty text defaults to a 'review the time' style message naming the cell.
2. The department head and the committee chair write two independent notes on the same cell while the registrar speaks with one voice.
3. A note from an earlier round that the registrar insisted on is counted and the schedule is returned with it.
4. The department head's internal note does not inflate the registrar counter and does not block sending.

EVIDENCE

`src/server/routes/scheduleNotesRoutes.ts` `src/server/core/notes.ts#noteFieldValue` `src/server/core/notes.ts#noteState` `src/utils/approvalWorkflow.ts#countOpenRegistrarNotes` `tests/schedule-changes-audit.ts` `tests/approval-integrity-audit.ts` `tests/relay-fixes-audit.ts`

FR-APR-009 Department reply, registrar verdict and escalation

The system shall let the department reply to a registrar note with a reason of at least three characters (up to 600), shall let the registrar either accept the justification, ending the note, or insist, returning it to 'open' and moving the reply to the reply history, and shall record at the third insist the moment the dispute is escalated so the department head sees it in his bar.

● Implemented · Should · Functional · Schedule committee chair, Department head, Registration officer, College registrar, University registrar

ACCEPTANCE CRITERIA

1. A reply shorter than three characters returns 400.
2. The third insist sets escalatedAt once and is counted in escalatedNotes in the department bar.

EVIDENCE

`src/server/routes/scheduleNotesRoutes.ts` `src/utils/approvalWorkflow.ts#insistOutcome` `tests/relay-fixes-audit.ts` `tests/approval-integrity-audit.ts`

FR-APR-010 Returning a schedule with notes

The system shall let a registration officer, college registrar or university registrar return a schedule that is 'with the registrar' or 'accepted' to the department provided at least one registrar note is open, shall record on the round the return time, the returner and the note count, shall keep both signatures in place, and shall open a new round for a returned accepted schedule based on what the registrar sees now.

● Implemented · Must · Functional · Registration officer, College registrar, University registrar

ACCEPTANCE CRITERIA

1. A return with no open note returns 409 with code no-notes, and an unsuitable status returns not-reviewable.
2. The dean of registration and the deans cannot return (403).

EVIDENCE

`src/utils/approvalWorkflow.ts#canReturn` `src/server/routes/approvalsRoutes.ts` `tests/relay-fixes-audit.ts` `tests/term-freeze-readonly-audit.ts`

FR-APR-011 Final acceptance by the registrar

The system shall let the registrar accept only a schedule whose status is 'with the registrar' and which has no material conflict, capturing an accepted version, writing acceptedAt, acceptedBy and acceptedVersionId on the current round, emptying the additions queue, and, after the save succeeds, closing remaining open and answered registrar notes as 'closed by acceptance' while keeping their text.

● Implemented · Must · Functional · Registration officer, College registrar, University registrar

ACCEPTANCE CRITERIA

1. Accepting a schedule that is not with the registrar returns 409, and an acceptance that loses a revision race closes no note.
2. The log's accept event states the number of notes closed 'closed by acceptance'.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/utils/approvalWorkflow.ts#CLOSED_BY_ACCEPTANCE_LABEL` `tests/relay-fixes-audit.ts` `tests/approval-workflow-audit.ts`

FR-APR-012 Decision serialisation and binding to what was seen

The system shall accept from the client the round, appointment count and status as the decider saw them, shall refuse the decision with code stale-view and 409 if any changed, and shall not check what the client did not send. It shall also run every decision that writes the approval document on a scope lock queue 'approval:college:section:term' with a revision number, shall retry the decision once if another decision reached the document first, and if the conflict repeats shall answer 409 with code approval-revision, shall not repeat what must not repeat (version capture runs once across both attempts), and shall write no side effect before the save succeeds.

● Implemented · Must · Functional ·

Schedule committee chair, Department head, Registration officer, College registrar, University registrar, The system (automatic)

ACCEPTANCE CRITERIA

1. A signature after the appointment count rose since the screen was opened is refused with the 'the schedule changed since you opened the screen — refresh' message.
2. A request without expected* fields passes unchecked for backward compatibility.
3. Two attempts by default, and both failing returns 'conflict', not 'done'.
4. Bulk exceptions write each department on its own queue without a single response for all.

EVIDENCE

`src/utils/approvalWorkflow.ts#staleViewRefusal` `src/server/routes/approvalsRoutes.ts#refuseIfStale` `src/server/approvalAttempts.ts#runApprovalAttempts` `src/server/routes/approvalsRoutes.ts#approvalTransaction` `tests/relay-fixes-audit.ts`

FR-APR-013 Amendment round after acceptance

The system shall, on the first edit (add, edit or delete) to an accepted schedule, open an amendment round based on the last accepted version and return the status to 'with the registrar', shall keep editing open in that round unless the registrar writes a note in it, and shall mark the document amendmentPending and send the X-Approval-Amendment header when the record cannot be written, so the round opens on the next read or decision.

● Implemented · Must · Functional · Schedule committee chair, The system (automatic)

RATIONALE

> So a department does not have to ask for manual unlocking each time and end up fixing errors outside the system.

ACCEPTANCE CRITERIA

1. Two consecutive edits after acceptance fall in one round, then the schedule locks at the first registrar note.
2. An addition through any write door (entry, import, term copy, batch move, study proposal) passes through the same function.

EVIDENCE

`src/server/core/approvals.ts#noteScheduleMutation` `src/server/core/approvals.ts#amendmentAfterEdit` `src/utils/approvalWorkflow.ts#amendmentStillOpen` `tests/relay-fixes-audit.ts` `tests/approval-integrity-audit.ts`

FR-APR-014 Schedule lock after sending

The system shall refuse, at every schedule write door (add, edit, delete, import, term copy, batch move, study proposal), edits to a schedule whose status is 'with the registrar' unless an amendment round is open, and shall exempt week exceptions, instructor-request decisions and issuing their links from the registrar lock, leaving only the ended-term rule on them.

Implemented · Must · Functional · Schedule committee chair, Standard user

ACCEPTANCE CRITERIA

1. Trying to edit an appointment in a schedule with the registrar returns 409 with the 'the schedule is with the registrar and awaits his decision' message.
2. The same message appears on screen before the user acts, through lockReason.

EVIDENCE

`src/server/core/approvals.ts#scheduleLockRefusal` `src/utils/approvalWorkflow.ts#approvalLockReason` `tests/approval-workflow-audit.ts` `tests/term-freeze-audit.ts` `tests/approval-integrity-audit.ts`

FR-APR-015 Ended term: the cycle is closed

The system shall, when a term carries an explicit 'ended' flag, refuse signing, withdrawing, sending, returning, accepting, notes and extensions with code term-closed and 409, shall allow its schedule to be edited only by an account whose written role is schedule committee chair, and shall record a committee edit of an accepted schedule as a closed-term-edit event without opening a round.

Implemented · Must · Functional · Schedule committee chair, System administrator

RATIONALE

> Only the explicit flag is read, not one derived from term order, so ten years of terms are not frozen at once.

ACCEPTANCE CRITERIA

1. An account without a written role (such as the root administrator) does not bypass the lock.
2. Clearing the 'ended' flag on the terms screen is the only visible way to reopen.

EVIDENCE

`src/server/core/approvals.ts#scheduleLockRefusal` `src/server/core/approvals.ts#refuseIfTermClosed` `src/utils/approvalWorkflow.ts#TERM_CLOSED_APPROVAL_MESSAGE` `tests/term-freeze-audit.ts`

FR-APR-016 Past terms frozen read-only

The system shall treat a term as frozen if it was explicitly closed, or its window has passed and it is not the current term, or it is chronologically older than the current term, shall not freeze the next term or an unknown term, shall refuse with code term-frozen and the 'past term: view only' message every write door in registration status, instructor wishes, section requests, schedule changes and proposals, and the screen shall show the 'Past term · view only' ribbon and disable write buttons.

Implemented · Must · Functional · Registration officer, Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. With current term 12 and next term 13: term 11 is frozen, 12 and 13 are open, and id 0 or an unknown id is not frozen.
2. The server door and the client read the same termIsFrozen function.

EVIDENCE

`src/utils/termSequence.ts#termIsFrozen` `src/server/core/approvals.ts#refuseIfTermFrozen` `src/components/TermFrozenRibbon.tsx` `tests/term-freeze-readonly-audit.ts` `tests/term-freeze-audit.ts`

FR-APR-017 Term deadline and its effect after it passes

The system shall let only the university registrar set the schedule submission date as YYYY-MM-DD or clear it with an empty value, shall refuse setting a date for an ended term, shall treat a deadline as passed after 23:59:59 of its day in Kuwait time (UTC+3), shall compute a department's effective deadline as the later of the term deadline and its exception, and shall call it 'near' when seven days or fewer remain. It shall also The system shall, after the department's effective deadline has passed, refuse importing a schedule from a file, copying a schedule from another term, and bulk deletion exceeding 30% of the scope's appointments, while keeping partial edits (adding a section, changing a room, instructor or time) and rounds after the first open.

● Implemented · Must · Functional · University registrar, System administrator, Schedule committee chair

ACCEPTANCE CRITERIA

1. A malformed date returns 400 and an ended term returns 409 term-closed.
2. A deadline 4 days ago has passed with daysLeft = -4; one in 4 days has tone = near.
3. Deleting 40 of 100 is wholesale, deleting 30 of 100 is exactly the limit and is not, and an empty schedule has no ratio.
4. The refusal names the passed date and says an extension from the university registrar is the remedy.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/utils/approvalWorkflow.ts#readDeadline` `src/utils/approvalWorkflow.ts#effectiveSubmissionDate` `src/utils/approvalWorkflow.ts#isWholesaleChange` `src/server/core/approvals.ts#wholesaleRefusal` `src/server/routes/scheduleImportRoutes.ts` `src/server/routes/schedulesRoutes.ts` `tests/approval-workflow-audit.ts` `tests/deadlines-panel-audit.ts`

FR-APR-018 Bulk exception grants

The system shall let the university registrar and the college registrar grant or lift a submission exception, or reject an extension request, for all departments, one college or named departments (up to 500) within their scope, using 1 to 60 days or an explicit date, with a reason of at least three characters (240 maximum), never before the term deadline and never in a term with no deadline or an ended term, shall count days from the later of the term deadline and the department's existing exception, and shall write each department on its own queue and revision and report to each what happened to it.

● Implemented · Must · Functional · University registrar, College registrar

ACCEPTANCE CRITERIA

1. A registration officer receives 403 on the route; a department outside the grantor's scope is reported as outside the allowed departments.
2. Lifting an exception returns the department to the term deadline, granting folds away the department's request, and rejecting a request needs a reason the department sees.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/utils/submissionDeadlines.ts#decideException` `src/utils/submissionDeadlines.ts#resolveExceptionTargets` `tests/deadlines-panel-audit.ts`

FR-APR-019 Extension request from a department

The system shall let the schedule committee or department head request an extension with a reason of at least three characters (240 maximum) and a number of days from 1 to 60 (default 7), shall accept it only for a schedule that has not reached the registrar and is not accepted (in preparation, signed or returned), with an existing effective deadline that is three days or fewer away or has passed, and with no earlier request pending, and the same rule shall govern the request button in the bar.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A request for an accepted schedule or one with the registrar returns 409 with code extension-request-refused and its reason.
2. The request reaches the inbox with a suggested date equal to the later of the effective deadline and today plus the days.

EVIDENCE

`src/utils/approvalWorkflow.ts#extensionRequestRefusal` `src/server/routes/approvalsRoutes.ts` `tests/relay-fixes-audit.ts` `tests/deadlines-panel-audit.ts`

FR-APR-020 Term student dates

The system shall let only the university registrar save the term's student-journey dates (wishes, registration rounds with a maximum of 8 rounds, and request closing) after validating them with the same rule the client shows, shall refuse inconsistent ordering or an ended term, and shall clear all dates when an empty value is sent.

● Implemented · Could · Functional · University registrar, System administrator

ACCEPTANCE CRITERIA

1. Wishes that start before the schedule submission deadline are refused with 400 and a message per issue.
2. An empty value clears the student dates and returns phases = null.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/utils/termStudentPhases.ts#checkStudentPhases` `src/components/StudentPhasesBlock.tsx` `tests/term-student-phases-audit.ts`

FR-APR-021 Registrar inbox, lateness and counter

The system shall show, for each department in the reader's scope, one row with status, round, appointment count, conflicts, open and answered notes, pending sections, deadline, lateness and extension request, shall order rows by priority: new arrival (0), returned after a return (1), has blockers (2), returned (3), other (4), accepted (5), then by blocker count and name, and shall include departments that have not started. It shall also treat a department as late if its effective deadline has passed in Kuwait time and it never sent a round and is not with the registrar, returned or accepted, shall compute lateness at read time without storing it, and shall make /api/approvals/badge show the registrar the number of schedules 'with the registrar', the department head what awaits his signature, additions acknowledgement and open registrar notes, and the committee the open registrar notes, summed over the current term and the planning term.

● Implemented · Must · Functional ·

Registration officer, College registrar, University registrar, Dean of registration, Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. New before returning, returned before accepted, and accepted at the bottom.
2. The whole term is read with three fixed reads (appointments, notes, options) however many departments exist.
3. A sent, returned or accepted department is not late even if the deadline passed.
4. Notes of a schedule with the registrar are not counted in the department's counter because it cannot act on them.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/utils/approvalWorkflow.ts#inboxPriority` `src/utils/lateness.ts#isLate` `src/server/routes/approvalsRoutes.ts#approvalBadgeForTerm` `tests/approval-workflow-audit.ts` `tests/approval-integrity-audit.ts` `tests/deadlines-panel-audit.ts` `tests/dean-notify-audit.ts`

FR-APR-022 Term-wide status and department balance

The system shall return from /api/approvals/term each department's status in the term within the reader's scope together with departments that have not started, shall distinguish a department with appointments but no record ('in preparation') from one with no appointments ('not started'), and the department balance shall show one row per department with an approval column ordered by priority: late, not started, in preparation, committee signed, head signed, returned, with the registrar, accepted.

● Implemented · Should · Functional · College dean, Vice dean for academic affairs, Dean of registration, University registrar

ACCEPTANCE CRITERIA

1. A department that wrote six appointments and did not sign appears as 'in preparation', not 'not started'.
2. A department in the reader's scope with no data appears with explicit zeros rather than a silent absence.

EVIDENCE

`src/server/routes/approvalsRoutes.ts` `src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx#BalancePanel` `src/utils/submissionDeadlines.ts#progressSegment` `tests/deadlines-panel-audit.ts` `tests/role-scope-audit.ts`

FR-APR-023 Dean sees the final schedule only; council bulletin

The system shall read for the dean and vice dean each department as the registrar last accepted it (live if accepted, from the last accepted version if it returned to the registrar, and never shown if never accepted except in an ended term), and shall allow printing the 'Council bulletin — department balance' at any time even before the first acceptance, in pages of 14 departments, with approval and deadline columns, in the approved printed-report design.

Implemented · Should · Functional · College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. Draft appointments and the 'heaviest instructor' of a department not yet accepted are excluded from the dean's balance.
2. Main administration reads drafts as they are, the dean sees his college and the dean of registration sees all.

EVIDENCE

`src/server/core/scheduleReads.ts#readsFinalSchedulesOnly` `src/utils/finality.ts#finalSourceFor` `src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx` `tests/dean-notify-audit.ts` `tests/role-scope-audit.ts` `tests/report-approval-audit.ts`

FR-APR-024 Review readiness and detailed blockers

The system shall return from /api/schedules/review-readiness the blockers with the department's appointment details and the warnings (the same section recorded twice in two halls or with two instructors) separate from the count, the two numbers for blockers and the appointments they touch, shall gather in family=1 mode the rows of the department family allowed to the reader, shall describe a clash with another department generically without names, and shall add blockers from approved hall barter windows.

Implemented · Must · Functional · Schedule committee chair, Department head, Registration officer

ACCEPTANCE CRITERIA

1. The blockers list length equals blockingConflicts, and blockingRows is the number of distinct appointments in its pairs.
2. A counterpart appointment outside the reader's scope is described as an appointment outside the current view.

EVIDENCE

`src/server/routes/scheduleReviewRoutes.ts` `src/utils/scheduleBlockers.ts#approvalBlockers` `src/utils/scheduleBlockers.ts#approvalWarnings` `tests/one-blocker-number-audit.ts` `tests/blockers-stream-audit.ts`

FR-APR-025 Approval score as an explicit formula

The system shall compute the approval score = sound appointments ÷ all scope appointments × 100 rounded to a whole number, shall count each appointment once at the first category touching it in severity order (blockers, then regulation), shall show per category the projected score if it alone is fixed, and shall display the formula as a sentence the reader can check by hand.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The ring's slices plus the sound appointments equal all appointments, and a scope with no appointments shows 'no appointments in scope' and a score of 100.
2. 'If you fix this → N' counts the appointments touched only by that category.

EVIDENCE

`src/utils/approvalScore.ts#approvalScoreBreakdown` `src/components/ScheduleReview.tsx` `tests/approval-review-audit.ts`

FR-APR-026 Approval dossier

The system shall build an approval dossier from the approval record, the change report and the review readiness, showing counts of added, modified and removed appointments as percentages summing to 100, the changed courses (up to 12 lines then 'and N more'), blockers (up to 4), and the timeline of rounds and signatures with their codes, shall drop anything received for another scope, shall not present an unavailable value (it says 'not available'), and shall print on A4 as previewed.

● Implemented · Should · Functional · Registration officer, College registrar, University registrar, Department head

ACCEPTANCE CRITERIA

1. A sibling department's approval record is not shown even if it arrives under this scope's key (scopeMismatch).
2. No reasons for changes are shown: the record carries what changed, not why.

EVIDENCE

`src/utils/approvalDossier.ts#buildApprovalDossier` `src/components/ApprovalDossier.tsx` `src/utils/approvalScope.ts` `tests/approval-dossier-audit.ts` `tests/approval-scope-audit.ts`

FR-APR-027 Printed sheet stamp and its verification page

The system shall draw in the footer of the comprehensive department report (both kinds) and the change report a QR code and an eight-character code (Crockford alphabet) derived from the sheet kind, term, sites and its rows' fingerprints (up to 3000 rows and 12 sites), shall register it after the print command rather than before, and the server shall recompute the code from the sent rows and refuse with 409 any that does not match, count a new print for an existing stamp, and flag rows that differ from the system (a stale screen). It shall also provide a public page with no sign-in and no script at /v/:code that accepts the code in lower case or with a dash (and O for 0, I or L for 1) and redirects it to its single canonical link, compares the sheet's rows with the live schedule row by row and states 'matches the schedule now' or 'the schedule changed after this sheet was printed' with the changed, removed and added rows (up to 60 lines per list), and the approval status for a single-department sheet, without any name or civil id.

● Implemented · Should · Functional · Registration officer, Department head, Schedule committee chair, The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. The same rows give the same code and any edit before printing gives a different code.
2. A registration that failed on a network drop is queued and retried when the query screens open (up to 12 queued).
3. An unknown code returns 404 with state 'unknown', a malformed one 404 with state 'invalid', and a read failure 503.
4. The page is limited to 40 requests per minute per address and is not indexed (noindex).
5. Course numbers shown are seven digits.

EVIDENCE

`src/utils/reportStamp.ts#reportStampCode` `src/utils/reportStampClient.ts` `src/components/PrintStamp.tsx` `src/server/routes/reportsRoutes.ts` `src/server/reportStampPage.ts#reportStampPage` `src/utils/reportStamp.ts#compareReportStamp` `tests/report-stamp-audit.ts`

FR-APR-028 Instructor request signature check by civil id

The system shall provide at /verify and /verify/:code a public page where an instructor enters his civil id (12 valid digits) and optionally the signature code (6 characters) and sees his signatures (up to 20 requests) and what was signed at the time, without storing the number, answering identically for a number with no signature and a code that does not match, and limiting requests to 30 per minute plus a per-address attempt quota.

● Implemented · Should · Functional · Faculty member, Public visitor

RATIONALE

> This check is specific to instructor request signatures; the codes of the committee and head signatures appear in the approval dossier and have no public page.

ACCEPTANCE CRITERIA

1. A malformed civil id returns 400 and is not counted as an attempt, and a number with no signature returns 404 with one message.
2. A code that was typed and did not match does not close the door: the signatures are shown and the mismatch is stated.

EVIDENCE

`src/server/routes/verifyRoutes.ts` `src/server/signatureVerifyPage.ts#signatureVerifyPage` `tests/request-signature-audit.ts`

NFR-APR-001 Scope on every route and inbox performance

The system shall verify on every approval route that the requested college and department are within the user's scope (403 'outside the departments allowed to you'), shall apply the same scope to inbox rows, the counter and the balance, and shall require college, department and term together (400 with one Arabic sentence). It shall also build the term inbox with a fixed number of reads (term appointments, term notes, blocker options) distributed to departments in memory, shall not read the term schedule once per department, and shall measure a note's state against the department's own rows rather than the term's rows.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A request for a department outside the user's scope returns 403 and nothing is read from it.
2. A request without a term or department returns 400 asking to choose the term, college and department first.
3. The inbox route contains no per-department read loop.
4. A row that moved between departments does not make its note appear 'open' in the new department.

EVIDENCE

`src/server/routes/approvalsRoutes.ts#approvalScopeFromBody` `src/server/routes/scheduleReviewRoutes.ts` `src/server/routes/approvalsRoutes.ts` `src/server/core/approvals.ts#countBlockingConflicts` `tests/role-scope-audit.ts` `tests/approval-scope-audit.ts` `tests/approval-integrity-audit.ts`

NFR-APR-002 Kuwait-time dates with a single rule

The system shall read 'today' in every deadline, extension and lateness calculation in Kuwait time (UTC+3) rather than universal time, shall have the client and server use the same functions (readDeadline, kuwaitDateISO, isLate), and shall count days by calendar rather than hours.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Between midnight and 03:00 Kuwait time a deadline's day counts as passed, not still running.
2. The difference between 2024-02-28 and 2024-03-01 is two days (leap year).

EVIDENCE

`src/utils/approvalWorkflow.ts#kuwaitDateISO` `src/utils/approvalWorkflow.ts#deadlineEndsAt` `src/utils/lateness.ts` `tests/approval-workflow-audit.ts` `tests/relay-fixes-audit.ts`

NFR-APR-003 Privacy and attempt limits on public pages

The public verification pages shall not store a civil id nor show a name or civil id on the stamp page, shall send Cache-Control: no-store and a noindex marker, shall limit requests (stamp page 40 per minute, signature check 30 per minute, stamp registration 60 per minute), and shall not distinguish in the answer between a number with no signature and a number that belongs to no one.

Implemented · Must · Non-functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. The civil id is sent once for matching, cleared from the field and not kept in the browser.
2. Both pages send no-store and contain robots noindex.

EVIDENCE

`src/server/routes/verifyRoutes.ts` `src/server/reportStampPage.ts` `src/server/signatureVerifyPage.ts` `tests/request-signature-audit.ts` `tests/report-stamp-audit.ts`

IR-APR-001 Approval bar shows only what awaits the viewer

The approval bar shall order what it shows in the order of action: returned with notes, then sections awaiting the head's acknowledgement, then signing, then sending, then 'with the registrar' as quiet news, shall show nothing to someone with nothing waiting, and shall phrase the conflict sentence in correct Arabic number forms (one, dual, few, many) together with the number of appointments it touches.

Implemented · Should · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A department at the start of preparation with nothing waiting sees no bar.
2. The phrase for 1, 2, 5 and 11 follows the counted noun: singular, dual, plural for few, accusative for many.

EVIDENCE

`src/components/ApprovalBar.tsx` `src/utils/approvalBarPresence.ts` `src/utils/approvalWorkflow.ts#blockingConflictPhrase` `tests/approval-bar-visual-audit.ts` `tests/approval-workflow-audit.ts`

IR-APR-002 Submission deadlines panel

The interface shall show in the header of 'Schedule changes' one panel that is the only place the term deadline and its exceptions are written: the date with the days left, three ready dates (in one, two and three weeks) for a term with no deadline, day chips of 1, 2, 3, 5 and 7, a preview 'N departments will become: ...' before applying, the term submission progress bar, and department requests with grant and reject buttons, collapsing to a strip once the deadline is set and nothing waits.

Implemented · Should · Interface · University registrar, College registrar, Registration officer

ACCEPTANCE CRITERIA

1. The university registrar sees the deadline field and extension actions, the college registrar sees the deadline read-only plus extension actions, and a registration officer sees the panel read-only.
2. 'Late' in the panel is the single isLate rule, not a second calculation.

EVIDENCE

`src/components/SubmissionDeadlines.tsx` `src/utils/inboxAudience.ts` `src/utils/submissionDeadlines.ts` `tests/deadlines-panel-audit.ts`

3.7.6 Algorithms

ALG-APR-01 · Approval cycle

Approval state machine and guards

Deterministic

Is this action (sign, withdraw, submit, return, accept) allowed now, and what is the state after it?

1. States: drafting → committee-signed → head-signed → submitted → accepted, or returned with notes.
2. Signing is refused for a submitted or empty schedule, a material conflict, a course without capacity, a repeated signature, or head before committee.
3. Submission requires both signatures, acknowledgement of sections added after the head signed, zero conflicts, zero open notes, and the deadline not passed in the first round.
4. A return requires at least one open note; every submission opens a new round.
5. Every decision carries the round, appointment count and status as seen, and is refused if any changed.

Inputs

The approval record, blocking conflict count, open notes, deadline state, and what the decider saw.

Output

An allowed/refused verdict with a reason code, and the next state.

Pure functions read by both server and interface; decisions are serialised on a scope lock so two cannot race.

Pending additions list names up to 60 and counts the rest; event log up to 200.

`src/utils/approvalWorkflow.ts#canSign` `src/utils/approvalWorkflow.ts#canSubmit` `src/utils/approvalWorkflow.ts#canWithdraw` `src/utils/approvalWorkflow.ts#canReturn` `src/utils/approvalWorkflow.ts#staleViewRefusal` `src/utils/approvalWorkflow.ts#openRound` `tests/approval-workflow-audit.ts` `tests/relay-fixes-audit.ts`

FR-APR-001 FR-APR-002 FR-APR-004 FR-APR-005 FR-APR-006 FR-APR-007 FR-APR-010 FR-APR-012

ALG-APR-02 · Approval cycle

Effective deadline and wholesale-change detection

Deterministic

Which deadline is the department held to, and is this action a wholesale change refused after it passes?

1. Effective deadline = the later of the term deadline and the department's extension.
2. Today is read in Kuwait time and a deadline passes after 23:59:59 of its day; days left are counted in days, not hours.
3. Tone: past, near when 7 days or fewer remain, or ok.
4. Import and term copy are always wholesale; a bulk delete is wholesale when it exceeds 30% of the department's appointments.
5. An extension request is offered at 3 days or fewer, or after the deadline, for a schedule not at the registrar and not accepted.

Inputs

Term deadline, department extension, current instant, action kind and how many of the total are deleted.

Output

Deadline state (effective, days left, tone) and the wholesale refusal verdict.

One rule for server and interface; partial edits stay open after the deadline; a term without a deadline is never locked retroactively.

Wholesale delete ratio 0.3, near window 7 days, extension-request window 3 days, UTC+3.

`src/utils/approvalWorkflow.ts#readDeadline` `src/utils/approvalWorkflow.ts#effectiveSubmissionDate` `src/utils/approvalWorkflow.ts#isWholesaleChange` `src/utils/approvalWorkflow.ts#extensionRequestRefusal` `tests/approval-workflow-audit.ts` `tests/deadlines-panel-audit.ts`

FR-APR-017 FR-APR-019 NFR-APR-002

ALG-APR-03 · Approval cycle

Registrar inbox priority and insistence escalation

Deterministic

Which department should the registrar start with, and when is a note escalated?

1. Submitted in round 1 = 0, submitted after a return = 1, has blockers = 2, returned = 3, other = 4, accepted = 5.
2. Rows are ordered ascending by this priority.
3. Each registrar insistence on a note increments a counter, and the escalation instant is recorded once at the third insistence.

Inputs

Approval status, round, blocking conflict count and insistence counter.

Output

A priority number per department and an escalation stamp for the note.

A fixed explicit rule independent of alphabet or timing.

Escalation after 3 insistences.

`src/utils/approvalWorkflow.ts#inboxPriority` `src/utils/approvalWorkflow.ts#insistOutcome` `src/server/routes/approvalsRoutes.ts` `tests/approval-workflow-audit.ts` `tests/relay-fixes-audit.ts`

FR-APR-021 FR-APR-009

3.7.7 Edge cases

⚠ Head signature with a sending blocker

Scenario The department head signs while a registrar note is open or the deadline passed before the first send.

Behaviour The signature is kept and the schedule is not sent; the response carries submitted=false with submitBlocked giving the reason, and the status remains 'head signed' until the cause is resolved and the schedule is sent.

⚠ Approval record write fails after an accepted edit

Scenario An appointment edit is saved on an accepted schedule and then writing the amendment round fails.

Behaviour The appointment save is not undone; an amendmentPending marker is written over the last revision and X-Approval-Amendment: pending is sent, and the first later reader or decision opens the round attributed to the original actor, so no amended schedule stays accepted without a round.

⚠ Term deadline moved after an exception was granted

Scenario A department was granted an exception until a day, then the term deadline was moved later than that day.

Behaviour The department is held to the later of the two so it is not penalised by an earlier grant, and a new exception counts its days from the later date as well.

⚠ Two simultaneous decisions on one schedule

Scenario The head clicks approve on added sections while someone else imports or the registrar accepts at the same moment.

Behaviour The losing decision is retried once on the document's new state; if it loses again it is refused 409 approval-revision with a request to refresh, and no notes are closed for an acceptance that was not saved.

⚠ Accepting a schedule with outstanding notes

Scenario The registrar accepts a schedule that still has registrar notes open or answered.

Behaviour After the acceptance save succeeds those notes are closed as 'closed by acceptance' and their text stays, so they do not block the first later resend; a failure to close a note is logged and does not undo the acceptance.

⚠ Swapping the course or section number after signing

Scenario The committee edits a row after the head signed, changing its course or section number, or moving it to another department.

Behaviour The swap is treated as an addition rather than an edit and enters the acknowledgement queue, unlike a change of room or time; a row deleted before being acknowledged is removed from the queue.

⚠ Editing an accepted schedule in an ended term

Scenario The schedule committee applies a correction to an accepted schedule in an explicitly closed term.

Behaviour The edit passes because only the committee works in an ended term, and a closed-term-edit event (add, edit or delete) is recorded instead of opening a round that would send the schedule to an inbox that is not expecting it.

3.7.8 Standing owner decisions

- Owner rule (2026-09-27): only a material conflict prevents signing; the same section recorded twice at the same time is stated and is not counted as a blocker.
- After the deadline wholesale hand-over (import, term copy, deletion above 30%) is refused while partial edits remain open.
- System owner's rule: an ended term has no signing, sending or surveys, and editing the schedule is the schedule committee's matter alone.
- A past term is view-only: reading stays open and writing is closed, with the owner's two texts 'Past term · view only' and 'This is a past term: shown for reference only and not edited'.
- An unapproved pattern blocks (owner decision 2026-10-30): history does not exempt a non-standard shape from the hour pattern.
- Printed reports (comprehensive, change report and council bulletin) keep their approved design, and course numbers in them and on the verification page are seven digits.

↑ Contents

3.8 · IMP

Import

From authority documents, scans and spreadsheets to a reviewed and confirmed schedule.

34	100%	8	30
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.8.1 Overview

The module imports the official authority timetable (SWRSCHA) as text or as a scan read by Tesseract in Arabic and English, decides the department, branch and term from the header before any row, and shows the result in a preview with evidence for each cell and a pre-publish check against the whole term. The Gemini service optionally reads pages that still have unread cells and only proposes; nothing is written except through the deterministic validators and with explicit confirmation by the user. The module also covers comparing the live schedule with the document, spreadsheet and JSON import, creating a missing department and courses after one confirmation, and reconciling courses with plan sheets.

Schedule committee chair · Department head · System administrator · Root administrator · College registrar · The system (automatic) · External service

3.8.2 User journeys

Schedule committee chair — **Copying the authority timetable into an empty term**

1. Chooses college, department and term and uploads the authority timetable PDF.
2. Follows the four-station reading progress while the server proves the header before reading rows.
3. The preview shows cell evidence and row notes; he edits an instructor or room or deletes a row.
4. Confirms scanned pages with missing lines using the page-reviewed button.
5. The bar says the check is complete after the whole-term conflict check.
6. Presses apply, the draft is saved with its baseline and published, and the comparison report can then be opened.

Department head — A scanned file with an assisted reading

1. Uploads a black-and-white scan which is read by optical recognition.
2. The preview shows cells the scan could not read on two pages.
3. The button offers the assisted reading for those two pages only and he presses it by choice.
4. Proposed cells for empty cells only are shown and he approves those he accepts.
5. The validators run again and he completes applying like any import.

Department head — Importing a spreadsheet

1. Downloads the template and fills its rows under the Arabic headers.
2. Uploads the file; the first sheet is read and codes and instructors are matched.
3. Sees notes on unmatched lines and fixes them in the preview.
4. The bar checks conflicts, then he applies, or runs a trial if the term already has a schedule.

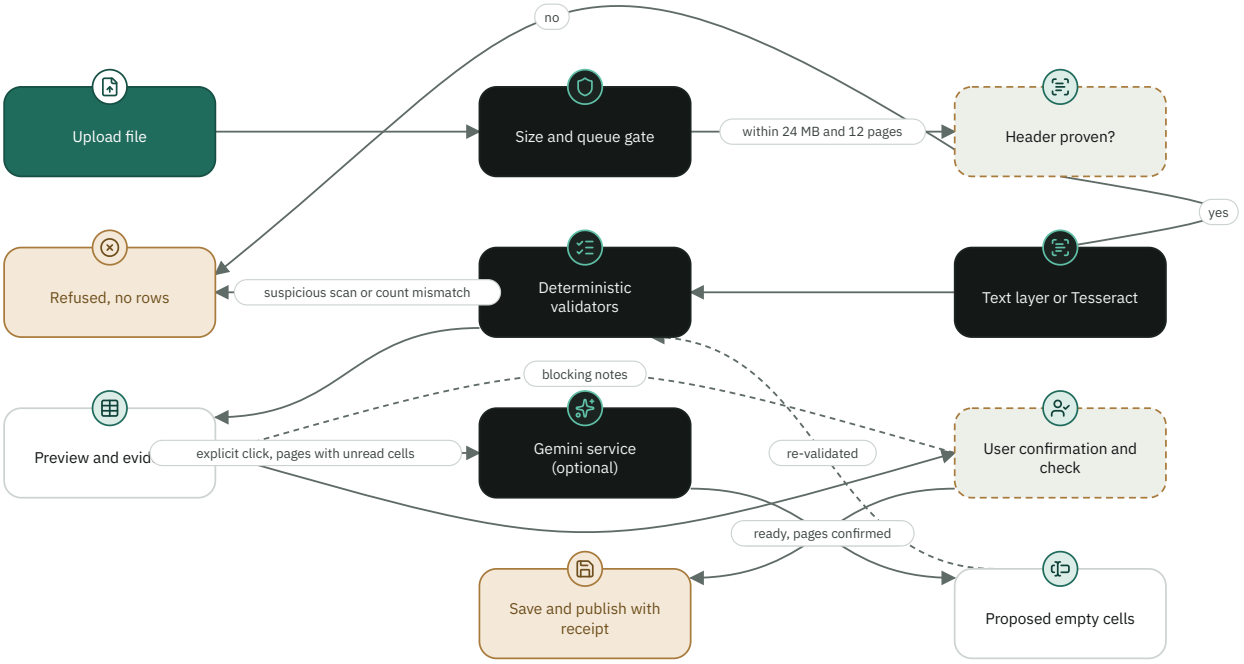
System administrator — A new department from the file header

1. Uploads a file of a department absent from the college and gets PDF_DEPARTMENT_MISMATCH with canBootstrapSection.
2. Confirms creation once; the server re-reads the header and creates the department under its printed number and name.
3. The file is read again into the new department and the usual path continues.

3.8.3 Diagrams

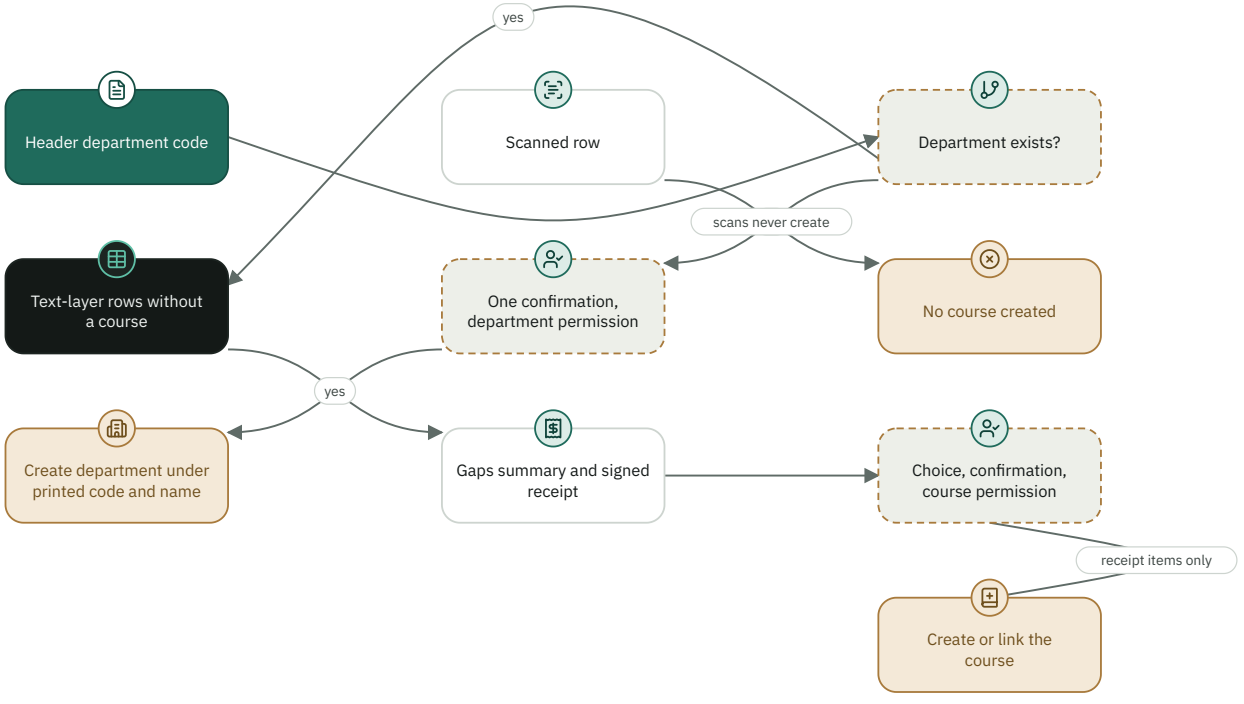
Import pipeline and its gates

The Gemini service is optional and outside the main path: it only reads and proposes, and nothing reaches the schedule except through the deterministic validators and the user's confirmation.



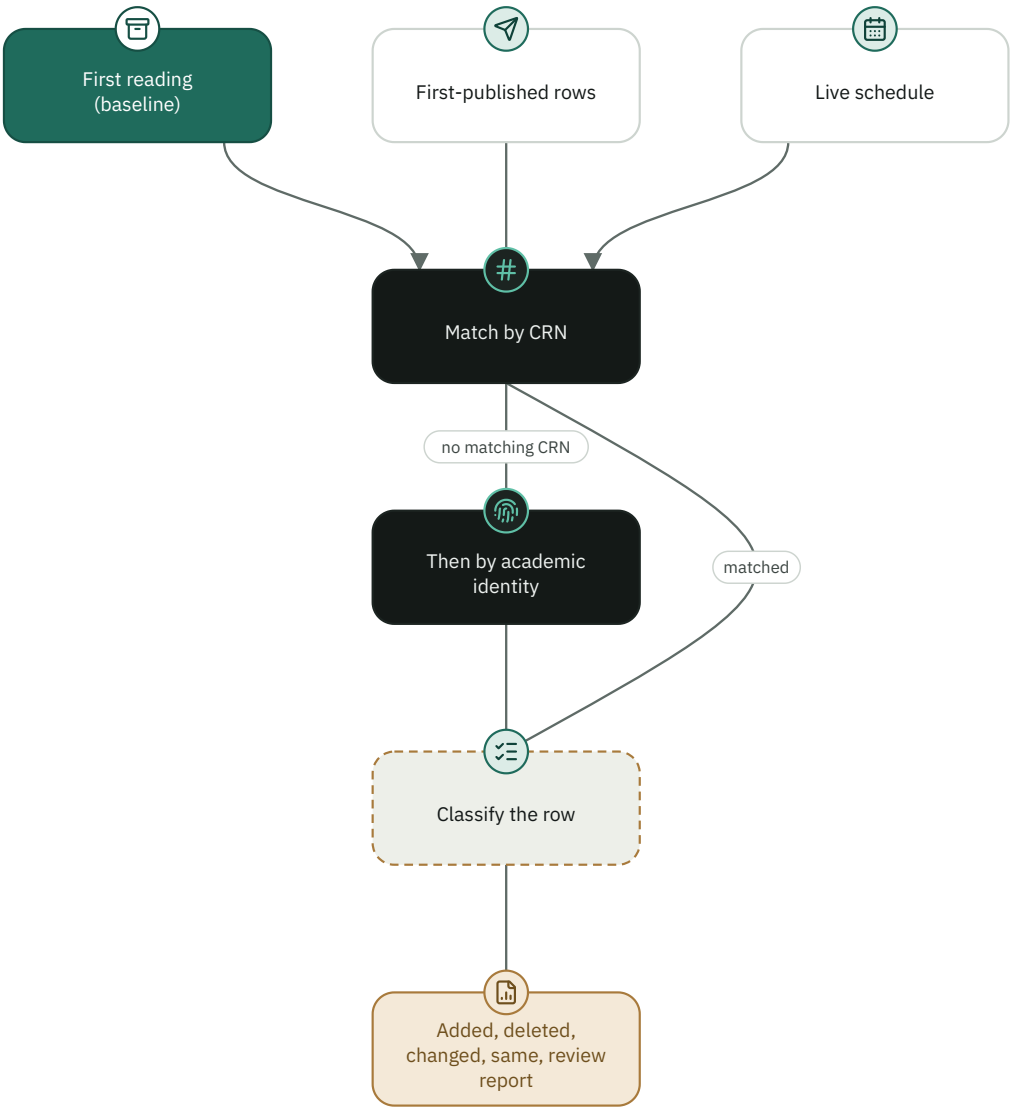
Creating missing departments and courses

Nothing is created by the reading itself: the server re-reads or signs a receipt and creates only after one confirmation with the usual permission, and scans never create.



Comparing the schedule with the authority file

The baseline never changes, the comparison is against the live schedule and not the draft, and the report is read-only.



3.8.4 Screens

Schedule transfer: import

Upload of PDF, spreadsheet or JSON, progress reading, preview and status bar.`src/components/ScheduleTransfer.tsx`

Import preview

Editable table with per-cell evidence and row notes.`src/components/ImportPreviewTable.tsx`

Paged preview

Page by page, with the page-reviewed button for scans.`src/components/PagedImportPreview.tsx`

Import status bar

One state and one apply button: checking, blocking notes, ready.`src/components/ImportStatusBar.tsx`

Trial run
The difference card against the current schedule.`src/components/ImportDryRun.tsx`

Gaps summary
Missing courses and sites, confirmed in one step.`src/components/ImportCatalogueGaps.tsx`

Instructor name fill
Fill from the current schedule or a second PDF on exact match.`src/components/ImportNameFillPanel.tsx`

3.8.5 Requirements

FR-IMP-001 **Authority PDF intake**

The system shall accept the official authority timetable PDF (SWRSCHA) of at most 24 MB and at most 12 pages, within a college, department and term the user is permitted to work in, and shall refuse a larger or longer file entirely instead of reading part of it.

Implemented · Must · Functional · Schedule committee chair, Department head, System administrator

ACCEPTANCE CRITERIA

1. A file over 24 MB is refused with a size error and no row is imported.
2. A file of 13 or more pages is refused whole with a message asking to split it; the first 12 pages are not read alone.
3. A scope outside the user's permissions returns 403.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/documentOcr.ts` `src/server/core/http.ts` `tests/pdf-import-hardening.ts` `tests/pdf-import-regression.ts`

FR-IMP-002 **Text layer or scanned reading**

The system shall read generated SWRSCHA files from their embedded text layer using cell coordinates, shall read scanned files with the Tesseract optical character reader in Arabic and English (ara+eng), and shall treat a file with fewer than 120 embedded characters and fewer than 12 text items as an image.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A text file is read from its text layer and each row is tagged sourceReadMode=pdf-text.
2. A scanned file is read by optical recognition and never creates a course (see FR-IMP-019).

EVIDENCE

`src/utils/documentOcr.ts` `src/server/routes/pdfImportRoutes.ts` `tests/pdf-import-golden.ts` `tests/pdf-import-regression.ts`

FR-IMP-003 Header proof before table reading

The system shall prove the term, college/branch and department from the page-1 header before any row matching, retry with a high-resolution reading when a field is missing, and shall not continue with any unproven field. The high-resolution reading fills only missing fields and never overrides a field proven earlier.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A header still incomplete after the precise reading returns 422 PDF_HEADER_UNRESOLVED naming the missing fields; no row is imported.
2. A complete header that contradicts the scope in the quick probe is refused immediately, before the full reading.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/documentOcr.ts` `tests/header-department-rule-audit.ts` `tests/pdf-import-hardening.ts`

FR-IMP-004 Header decides the department

The system shall decide the file's department and branch from the page header and not from course numbers, shall let the proven numeric department code outrank the printed name, and shall refuse the file with PDF_DEPARTMENT_MISMATCH when the header department differs from the selected one, or with PDF_MIXED_DEPARTMENTS when text-layer pages carry different departments.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A file of department 0121 uploaded to another department returns 409 PDF_DEPARTMENT_MISMATCH with a suggested department and the canBootstrapSection flag.
2. A course carrying another department's code inside a header department's page is accepted by the page header, not refused.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/authorityAcademicCodes.ts` `tests/header-department-rule-audit.ts`

FR-IMP-005 Branch match by code

The system shall match the file's branch by its three-digit code (for example 012) and not by name, shall not treat a different site letter inside one branch (012B, 012J, 012F) as a mismatch, and shall refuse the file with PDF_BRANCH_MISMATCH when the branch code differs.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A branch 011 file into a college of branch 012 returns 409 PDF_BRANCH_MISMATCH.
2. Rows with 012J buildings inside a branch 012 file stay legitimate and carry a site badge.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/branchScope.ts` `tests/pdf-import-golden.ts` `tests/pdf-import-hardening.ts`

FR-IMP-006 Accepted source term rule

The system shall accept a file only when its header term is one of three terms: the selected term itself, the same season of the previous academic year, or the term immediately before it (second accepts the first of its own year, first accepts the previous second, summer accepts itself and last year's summer); otherwise it shall refuse with PDF_TERM_MISMATCH naming the accepted terms.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. With target Second 2026/2027 the files Second 2025/2026, Second 2026/2027 and First 2026/2027 are accepted; First 2025/2026 and Summer 2025/2026 are refused.
2. The refusal names the file term, the selected term and the accepted terms, and imports no row.

EVIDENCE

`src/utils/pdfTermRule.ts` `src/server/routes/pdfImportRoutes.ts` `tests/pdf-term-rule-audit.ts`

FR-IMP-007 Copy into an empty term and trial run

The system shall copy an authority PDF only into a term with no appointments (PDF_TARGET_TERM_OCCUPIED), shall offer a term that already has a schedule a trial run that reads the file with all checks, writes nothing and shows the difference from the current schedule, and shall replace that schedule afterwards only when three conditions hold together: a signed trial-run receipt, an explicit request from the screen, and the replace-filled confirmation.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. dryRun=1 on a filled term reads the file and returns a receipt flagged replaceFilled with no row written.
2. Saving after a trial without replace-filled returns 409 PDF_TARGET_TERM_OCCUPIED; a preview with every row deleted returns 400 PDF_REPLACE_EMPTY.
3. An appointment added to the term between save and publish makes publishing refuse and deletes nothing.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/server/routes/draftsRoutes.ts` `src/utils/importDiff.ts` `src/components/ImportDryRun.tsx` `tests/import-courses-r3-audit.ts` `tests/import-term-tools-audit.ts`

FR-IMP-008 Refusing unusable scans up front

The system shall refuse before the full reading any pages that need turning to landscape (PDF_SCAN_REQUIRES_LANDSCAPE) and pale grey photos (PDF_SCAN_GREY_PHOTO), and afterwards refuse a file whose legibility is not proven or whose extraction is judged suspicious (SUSPICIOUS_EXTRACTION), without producing a table of blank cells.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A grey photo returns 422 PDF_SCAN_GREY_PHOTO with page numbers and no reading.
2. A later page needing rotation returns 422 PDF_SCAN_REQUIRES_LANDSCAPE before row matching.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/documentOcr.ts` `tests/pdf-import-hardening.ts`

FR-IMP-009 Scanned page verdict

The system shall accept a scanned page with a warning when no more than two lines or 15% of the printed lines were missed, and shall stop the file when the shortfall is larger, when the rows read are below 55% of the visible table bounds (3 rows or more), or when more than half the rows (and more than two) have neither time nor building. It shall also stop the import when the table geometry counts 3 rows or more and matching returns fewer than 70% of them (COUNT_RECONCILIATION_FAILED).

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A page with 28 printed lines of which 4 were read stops the file with a message naming both numbers.
2. A page with 20 printed lines of which 18 were read is accepted with a warning and missedLines=2.

EVIDENCE

`src/utils/documentOcr.ts` `src/server/routes/pdfImportRoutes.ts` `tests/pdf-import-hardening.ts`

NFR-IMP-010 One scan read at a time

The system shall not read two scanned files at once: a file is read in its turn, a different file waits up to 45 seconds and is then refused as SCAN_READING_BUSY (503), the same file sent again shares the running reading, and the recognition workers are terminated after each full reading. The reading routes shall admit only two concurrent requests with a queue of 8, and refuse the rest with 503.

● Implemented · Must · Non-functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. A different second file during a scan reading gets 503 SCAN_READING_BUSY after 45 seconds with no row imported.
2. A request beyond two active and eight queued gets 503 immediately.

EVIDENCE

`src/utils/documentOcr.ts` `src/server/core/http.ts` `src/server/routes/pdfImportRoutes.ts` `tests/pdf-import-hardening.ts` `tests/section-planning-audit.ts`

FR-IMP-011 Reading progress and interruption

The system shall stream reading progress line by line as NDJSON with the phases render, orient, read, rescue and match, ending with a done or error line; the screen shall show the four stations (prepare pages, read table, check pages, match registry) without ever going back, and shall name an interrupted reading by the response status (413, 429/502/503/504, 0 and 2xx) stating that no row was imported.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Status 413 shows a message citing the 24 MB limit and asking to split the file.
2. A station whose phase never arrived (clean read) is shown as skipped, not done.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/importStages.ts` `src/utils/importStreamFailure.ts` `tests/pdf-import-hardening.ts` `tests/journey-reveal-audit.ts`

FR-IMP-012 Course identity and section number

The system shall bind each row to a department catalogue course by its printed number alone, take the course name from the system and not from optical reading, generate the section number of each course in the sequence 501, 502, 503… in the order its rows appear, and refuse at save any section outside 501–999.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A row whose course number is unproven has an empty course name and the read text is never shown as a name.
2. A course with three sections is numbered 501, 502, 503 regardless of what the section cell read.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/authorityAcademicCodes.ts` `src/server/core/smartImport.ts` `tests/pdf-import-golden.ts` `tests/authority-course-prefix-audit.ts`

FR-IMP-013 Instructor identity without guessing

The system shall never create an instructor from a PDF, shall bind an instructor cell to a single proven registry record only after removing titles, shall compare name parts position by position so a different family or father name is another person, and shall leave the cell blank when ambiguous while naming the reason (registered more than once, close candidate, unregistered, or unreadable name).

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A family name that contradicts the registry gives no binding and the cell stays blank.
2. Two records with the same folded name give DUPLICATE_REGISTRATION and both candidates are named.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/instructorIdentity.ts` `src/server/core/smartImport.ts` `tests/pdf-import-hardening.ts`

FR-IMP-014 Building then room from the registry

The system shall resolve each row's location with one resolver that matches the building by an explicit official code and then the room inside that building only, shall never show an unconfirmed optical value as a location, shall mark a failing row LOCATION_REVIEW_REQUIRED, and shall never read seat-count welds such as 345045 as a building code. A building of another branch is marked CROSS_BRANCH and not added to the selected college before review.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Room F10 in an unconfirmed building leaves the room cell empty and keeps the raw text for review.
2. Building 011B18 in a branch 012 import is CROSS_BRANCH with a notice blocking publishing until reviewed.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/locationRegistry.ts` `src/utils/branchScope.ts` `tests/pdf-import-golden.ts` `tests/pdf-import-hardening.ts` `tests/location-registry-audit.py`

FR-IMP-015 Registering branch sites the document proves

The system shall register the building and room of another site in the same branch when proven by a full-shape official code (three branch digits, a site letter and a two-digit building) for a site with a registered college, shall not extend the open home site, another branch or accept a shortened shape, and shall not fail the whole import when registration fails.

● Implemented · Should · Functional · The system (automatic), System administrator

ACCEPTANCE CRITERIA

1. Code 012F15 in a branch 012 document registers the building with source AUTHORITY_DOCUMENT, so no manual choice is needed.
2. Code 011B09 in a branch 012 document is not registered.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `tests/location-registry-audit.py`

FR-IMP-016 Plausibility checks on scanned rows

The system shall check scanned rows before calling them confirmed: days × lecture length must equal the course hours × 50 minutes within 40 minutes a week or the days are cleared, a reference number whose length differs from the rest of the page is cleared, and an instructor bound by a shortened name who teaches in another college at the same hour loses that identity. In a multi-page scan only empty cells are recovered, from a unique historical fingerprint, and values the scan read are never overwritten.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A 3-hour course on two days of 80 minutes totals 160 and stays within 150±40; on three days the days are cleared.
2. An ambiguous historical fingerprint invents no course (AdCourseId stays 0).

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/authorityScanRecovery.ts` `src/utils/documentOcr.ts` `tests/authority-scan-recovery.ts` `tests/pdf-import-hardening.ts`

DR-IMP-017 Per-cell evidence on every row

The system shall attach to every imported row evidence for seven cells (course, section, days, time, instructor, building, room) holding the raw text, canonical value, source, method, score and confidence (CONFIRMED, REVIEW_REQUIRED or UNRESOLVED), and shall return a summary of confirmed, derived and review cells with the number of ready rows, plus a summary per page.

● Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A value recovered from a historical fingerprint is REVIEW_REQUIRED with derived=true and is not counted confirmed.
2. A row with one unconfirmed cell is not counted in readyRows.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/server/core/smartImport.ts` `tests/pdf-import-golden.ts`

FR-IMP-018 Creating a missing department after one confirmation

When the header's department code is proven but absent from the college, the system shall allow creating the department with one confirmation under its printed number and name, after re-reading the file on the server rather than trusting the browser, with the permission to create departments (system administrator), provided the code has four digits starting with the user's college code and the branch matches, and shall refuse with PDF_DEPARTMENT_AMBIGUOUS when more than one department carries the code.

● Implemented · Must · Functional · System administrator, Root administrator

RATIONALE

> Owner decision: a missing department is created after one confirmation.

ACCEPTANCE CRITERIA

1. Department 0125 absent ← created as code 25 under its printed name and never linked to a similarly named department (11 Home economics).
2. A user without the create-departments permission gets 403 PDF_DEPARTMENT_CREATE_FORBIDDEN.
3. A repeated request creates no second department and returns the existing one.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/authorityAcademicCodes.ts` `tests/header-department-rule-audit.ts`

FR-IMP-019 Creating missing courses after one confirmation

The system shall summarise courses not registered in the header department from text-layer rows only, shall create only those the user chose and that appear in a signed receipt valid for 6 hours carrying the number, name and units from the file and catalogue and not from the browser, with the permission to create courses inside the department scope and within the committee review window, and shall not duplicate a course on a repeated request. A course is never created from a scanned reading or from a trial run.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> Owner decision: a missing item is created after one confirmation, and scans never create.

ACCEPTANCE CRITERIA

1. A receipt older than 6 hours returns 409 PDF_GAPS_RECEIPT_INVALID and no course is created.
2. A code not in the receipt is ignored.
3. Units and hours are taken from a course with the same code and a name matching at 0.7 or more in any college, otherwise zero and visible in the summary.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/utils/importCatalogueGaps.ts` `src/components/ImportCatalogueGaps.tsx` `tests/header-department-rule-audit.ts`

FR-IMP-020 A three-digit tail never links

The system shall never link a printed number to a catalogue course by its three-digit tail alone: a number matching a short catalogue row by its tail is offered for choice (link, or create a new course under its printed number), and 0114464 stays a different course from 0120464 when the file shows the department's own number with that tail.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> Owner decision: a three-digit tail alone never links.

ACCEPTANCE CRITERIA

1. Number 0114324 against catalogue course 324 appears as a choice and is not linked automatically.
2. An unlinked row loses the invalid-course verdict only after the user chooses.

EVIDENCE

`src/utils/importCatalogueGaps.ts` `src/utils/authorityAcademicCodes.ts` `tests/header-department-rule-audit.ts` `tests/authority-course-prefix-audit.ts`

DR-IMP-021 Signed header-check receipt

Inferred

After each successful reading the system shall sign a receipt (HMAC-SHA256) carrying college, department, term, source header values and review pages, and shall accept saving or publishing an authority-pdf draft only with a receipt valid for that scope, so the term, college and department check cannot be bypassed from the screen (PDF_IMPORT_RECEIPT_REQUIRED and PDF_IMPORT_RECEIPT_INVALID).

● Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Saving a draft with no receipt or one for another department returns 409 PDF_IMPORT_RECEIPT_REQUIRED.
2. Publishing a draft whose receipt is invalid returns 409 PDF_IMPORT_RECEIPT_INVALID.

EVIDENCE

`src/server/routes/pdfImportRoutes.ts` `src/server/routes/draftsRoutes.ts` `src/server/core/smartImport.ts`

FR-IMP-022 Scanned page review confirmation

The system shall not save or publish a draft until the reviewer confirms each scanned page accepted with printed lines that have no row (the page-reviewed button), and the server shall enforce this from the signed receipt and not from a disabled button, returning 400 PDF_PAGE_REVIEW_REQUIRED with the page numbers.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Page 2 with two missing lines and no confirmation returns 400 PDF_PAGE_REVIEW_REQUIRED with pages=[2].
2. An outdated window sending no reviewedPages list gets a message asking to close and re-upload, and nothing is saved.

EVIDENCE

`src/utils/importPageReview.ts` `src/server/routes/draftsRoutes.ts` `src/components/PagedImportPreview.tsx` `tests/pdf-import-hardening.ts`

FR-IMP-023 Pre-publish check against the whole term

The system shall provide a read-only check (POST /api/schedules/import-preflight) applying the same conflict rule the save gate applies to the whole term except the department sites about to be replaced, adding the row-level save checks and sites without a matching department, and returning up to 60 conflicts with their preview row positions and the other appointment named (college, department, course, section, days, time) or only its college and time when outside the reader's permission.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A room booked by another department in the same term shows as a conflict in the preview before pressing, naming the other department.
2. Deleting a preview row makes conflict positions read from current order and not from the negative id.

EVIDENCE

`src/server/routes/scheduleImportRoutes.ts` `src/server/core/smartImport.ts` `src/hooks/useImportPreflight.ts` `tests/import-status-bar-audit.ts`

FR-IMP-024 Single import gate state

The screen shall compute the import state in one place in a fixed order (checking, then current-schedule read failed, then blocking notes, then check failed, then ready), shall neither count notes nor draw a success mark while a check is running, the hook shall re-check 300 ms after the last edit, and a failed check shall be treated as unverified and not as success while the server gate remains the last guard.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A running check with earlier notes gives the state checking, not blocking notes.
2. The first file is checked without delay; consecutive edits wait 300 ms.

EVIDENCE

`src/utils/importGate.ts` `src/hooks/useImportPreflight.ts` `src/components/ImportStatusBar.tsx` `tests/import-status-bar-audit.ts`

FR-IMP-025 Editable preview and name fill

The system shall show rows in a preview where cells are edited and rows deleted with the same vocabularies as adding an appointment, and shall allow filling empty instructor names from the current schedule, the previous term or a second PDF provided the whole line matches in course, section, days, time and room, so only the instructor cell is filled, a partial match is shown with its exact difference for the user to decide, any ambiguity fills nothing, and the fill can be undone exactly.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A line already holding an instructor is never touched.
2. Two matching lines with different instructors fill nothing.
3. Deleting the last row with the comparison open leaves no decision pointing at a missing row.

EVIDENCE

`src/utils/importNameFill.ts` `src/components/ImportNameFillPanel.tsx` `src/components/ImportPreviewTable.tsx` `src/components/PagedImportPreview.tsx` `tests/import-term-tools-audit.ts` `tests/import-review-rows-audit.ts`

FR-IMP-026 Comparing the schedule with the authority file

The system shall keep the first reading of the file as an unchanging baseline and compare it with the live schedule, not the saved draft, matching rows by the authority reference number (CRN) first and then by unique academic identity, classifying each row as added, deleted, changed, unchanged or needing source review (an unlinked imported record is not a proven addition), marking changed cells among twelve fields, comparing a course by its official code and not its internal number, and treating two different registered instructors as a change.

● Implemented · Must · Functional · Schedule committee chair, Department head, College registrar

ACCEPTANCE CRITERIA

1. A row deleted from the preview before first publish appears as deleted in the report because the baseline keeps the first reading.
2. Editing a cell then restoring its original value clears the change mark.
3. An empty live table classifies every original row as deleted and does not return 404.

EVIDENCE

`src/server/core/authority.ts` `src/server/routes/reportsRoutes.ts` `src/utils/authorityRowMatching.ts` `src/utils/authorityNetChange.ts` `src/utils/authorityReportOrdering.ts` `tests/authority-net-diff-audit.ts` `tests/authority-identity-reconciliation-audit.ts` `tests/authority-diff-cell-audit.ts` `tests/schedule-changes-audit.ts`

FR-IMP-027 Assisted reading through Gemini on explicit choice

The system shall not send any file to the Gemini service without an explicit click by the user, shall not offer the assisted reading automatically after an approved reading fails, shall offer it after a successful reading only for pages with an unread course, section, days or time cell, and shall send only those pages as images (longest edge 1700 pixels) in one request per page; the Gemini service only reads and proposes.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A file read clean the first time shows no assisted-reading offer and nothing leaves for Gemini.
2. A request with pages=2,5 sends only pages 2 and 5, read in parallel.

EVIDENCE

`src/server/routes/smartImportRoutes.ts` `src/server/core/gemini.ts` `src/utils/importPageReview.ts` `src/components/IntelligenceWorkspace.tsx` `tests/gemini-schedule-layer.ts`

FR-IMP-028 Personal-data redaction and proposal identity

The system shall never send a civil number or any similar key to Gemini, shall supply it a catalogue of at most 260 courses and 360 instructors carrying only id and name, shall ask for literal transcription of what is printed and for an empty cell instead of a guess, and shall discard any course or instructor identifier the service returns and bind locally from the printed name and code.

● Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. A catalogue carrying a civil field has it stripped before sending.
2. A row returned with an instructor id has the id ignored and the name bound locally or left empty.

EVIDENCE

`src/utils/geminiScheduleLayer.ts` `src/server/routes/smartImportRoutes.ts` `tests/gemini-schedule-layer.ts`

FR-IMP-029 Assisted proposals never write on their own

The system shall ensure the Gemini service writes to no schedule or draft: every row it returns passes the existing deterministic validators (course, instructor, location, time, conflict) before being shown, only empty cells are proposed for filling and no read row or cell is replaced, every proposed cell is shown before anything changes, only the cells the reviewer approved are applied, and saving and publishing remain behind the checks, the receipt and explicit confirmation like any import.

● Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> Design decision: Gemini only reads and explains; no write without a deterministic validator and explicit confirmation.

ACCEPTANCE CRITERIA

1. A sharper reading adding nothing shows the message that the schedule did not change and edits nothing.
2. An approved cell turns its evidence to CONFIRMED with method SMART_FILL and the row counts as ready.
3. A returned row failing the validators appears in blockingIssues and blocks saving.

EVIDENCE

`src/utils/geminiScheduleLayer.ts` `src/server/core/smartImport.ts` `src/server/routes/smartImportRoutes.ts` `tests/gemini-schedule-layer.ts`

IR-IMP-030 Gemini call interface and limits

Inferred

The system shall call Gemini with a JSON request at low thinking level and temperature 0.1 and with system instructions forbidding it to decide the schedule or ask to save, shall allow a document reading 180 seconds and function-call requests 45 seconds, shall not call the service when its key is not configured, and shall return a readable reason for a failed or timed-out request.

● Implemented · Should · Interface · The system (automatic), External service

ACCEPTANCE CRITERIA

1. No configured key makes no call and reports the assisted reading as not configured on the server.
2. Exceeding 180 seconds aborts the request and returns a timeout reason.

EVIDENCE

`src/server/core/gemini.ts` `tests/gemini-schedule-layer.ts`

FR-IMP-031 Spreadsheet import

The system shall read the first sheet of a spreadsheet file with Arabic headers (room, building, time, days, civil number, course instructor, section, course, course code) or the accepted field names, shall match the course by code within the department's operational courses and the instructor by civil number or name, shall accept at most 450 rows and reject more, shall note each line with an unknown course, unknown instructor or duplicate course/section, and shall write nothing at this step. It shall provide a downloadable template with two sample rows and a guide sheet.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. 451 rows return 400 saying the file exceeds the safe import limit.
2. An unknown course code gives a line note and the row stays in the preview.
3. A repeated course and section gives a duplicate note.

EVIDENCE

`src/server/routes/smartImportRoutes.ts` `src/components/ScheduleTransfer.tsx` `src/server/core/smartImport.ts` `tests/import-term-tools-audit.ts`

FR-IMP-032 JSON schedule export and import

Inferred

The system shall export a scope's schedule as schedule-export/2 carrying rows, courses, instructors, halls and visiting staff keyed by code and not by internal id, shall import the file by code and civil number, shall treat an import as a preview unless commit is explicitly requested, shall not replace an existing section (rejecting it as already present), shall reject a row whose building or room is unresolved in the official registry, shall check conflicts against the whole term and refuse the entire import on any problem, shall respect the schedule lock and the wholesale-import deadline, and shall report the change once per batch.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. commit on a locked schedule returns 409 schedule-locked with no write.
2. A preview returns ready, rejected and willAdd with no write.
3. A file carrying an unknown course with commit creates it and counts it in planted.courses.

EVIDENCE

`src/server/routes/scheduleImportRoutes.ts` `src/components/ScheduleTransfer.tsx`

FR-IMP-033 Reading a study-plan sheet

The system shall read a study-plan sheet PDF from its text layer only (not an image), at most 2 MB and 6 pages, with no external fonts or code, shall refuse an oversize file or a non-PDF with an explicit message, and shall restrict upload to the root account and refuse a sheet that combines several sheets or belongs to another department without writing anything.

● Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A 3 MB file gives the file-too-large message.
2. A non-root account gets 403, upload is for the root account only.
3. A sheet of another department returns 422 kind=department and the courses are unchanged.

EVIDENCE

`src/server/planPdfExtract.ts` `src/utils/planPdfParser.ts` `src/server/planReviewRoutes.ts` `tests/plan-sheet-import-route-audit.ts` `tests/plan-sheet-import-audit.ts`

FR-IMP-034 **Reconciling courses with their sheet**

The system shall make the sheet the source in reconciling a department's courses: a course follows the department its code points to and is created there if absent, the record the department actually taught is preferred over one without history, name, units and hours come from the sheet (spelling alone does not rename), another taught course holding the number is never renamed but a separate course is created under the full code, a course in an active sheet is not corrected from an older sheet, and a service department with no department in the system stays pending.

● Implemented · Should · Functional · Root administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. Two courses sharing a tail stay separate (0114464 and 0120464).
2. An older sheet does not correct a course in an active sheet.

EVIDENCE

`src/server/sheetReconcile.ts` `src/utils/planSheetImport.ts` `scripts/reconcile-plan-sheets.ts` `tests/plan-sheet-import-route-audit.ts` `tests/sheet-membership-audit.ts`

3.8.6 Algorithms

ALG-IMP-01 · Import

Header proof, file department, course and section identity

Deterministic

Which term, branch and department does this file belong to, which catalogue course does each row bind to, and what is its section number?

1. Term, college/branch and department are proven from the page-1 header before any row; a missing field triggers a high-resolution re-read of the header band that fills only missing fields; any unproven field stops the import.
2. Department key = 2 college digits + 2 department digits (01 + 09 → 0109), and the header code must equal the selected department's composite key, otherwise PDF_DEPARTMENT_MISMATCH; course numbers never decide the department.
3. A printed row binds by its number alone: equal digits, or the seven-digit key = department key + the course's three digits (0109105); a six-digit print under the department is expanded with a zero; another department's key never links by its tail alone and is offered for choice instead.
4. The course name comes from the catalogue, not from optical reading.
5. Section numbers are generated per course as 501, 502, … in order of appearance; two rows of the same course with the same reference number (CRN) are one section; save refuses anything outside 501–999.

Inputs

The page-1 header text, the rows read from the table, the selected college and department, and the department's course catalogue.

Output

A proven scope or a coded refusal, rows bound to catalogue courses with generated section numbers, and unbound numbers offered for a decision.

No college, department or course identity is invented; the high-resolution reading never overrides an already proven field; no row is imported from another department's file.

Displayed course number is always seven digits: college (2) + department (2) + course (3) Section number range 501–999

`src/server/routes/pdfImportRoutes.ts#registerPdfImportRoutes` `src/utils/authorityAcademicCodes.ts#authorityDepartmentMatches` `src/utils/authorityAcademicCodes.ts#authorityCourseCodeMatches` `src/utils/authorityAcademicCodes.ts#assignAuthoritySections` `src/utils/authorityAcademicCodes.ts#authorityCourseCode` `tests/header-department-rule-audit.ts` `tests/authority-course-prefix-audit.ts` `tests/pdf-import-golden.ts`

FR-IMP-003 FR-IMP-004 FR-IMP-012 FR-IMP-020

ALG-IMP-02 · Import

Deterministic

Scan legibility and cell plausibility checks

Is the scan worth reading, is the page reading complete, and is each read cell plausible before it counts as confirmed?

1. Before reading: the file is refused when at least half its pages are pale grey photos (median grey < 185 and 55% or more of pixels mid-grey between 110 and 179).
2. After reading: it is refused when meaningful characters fall below 140 per page or engine confidence below 42.
3. Page verdict: stop when missed printed lines exceed max(2, ⌈15%⌉), when rows read are below 55% of visible table rows (3 or more), or when more than half the rows (and more than two) have neither time nor building; smaller misses pass with a warning.
4. A time cell must hold two clocks between 07:00 and 20:59, 30 to 240 minutes apart; a days cell must be an ascending 1–5 sequence without repeats (a mirrored reading is accepted); days × lecture length must equal course hours × 50 minutes within 40 minutes a week, otherwise the days are cleared.
5. An empty cell is recovered from history only with two or more independent facts and an evidence score of 9 or more, when all matching historical rows agree; an optical reading is never overwritten.

Inputs

Grey page pixels, the read text and its confidence, printed and read line counts, raw cells, course hours, and the historical timetable.

Output

An early refusal with its reason, or a page verdict (stopped or warned), cells cleared for review with their raw text, and recovered cells tagged with their evidence.

A table of blank cells is never handed over as a reading; a cleared value remains visible evidence for the reviewer; no timetable data is manufactured from one weak fact.

Background grey < 185 and mid-grey share ≥ 55% · Legibility: 140 characters per page and confidence 42 · Hours tolerance 40 minutes a week on a 50-minute hour · Recovery: at least two evidence categories and a score of 9

`src/utils/documentOcr.ts#scanPageTone` `src/utils/documentOcr.ts#greyPhotoVerdict` `src/utils/documentOcr.ts#judgeLegibility` `src/utils/documentOcr.ts#scanPageVerdict` `src/utils/documentOcr.ts#authorityTimeCellLooksPlausible` `src/utils/documentOcr.ts#clearImplausibleScanDays` `src/utils/authorityScanRecovery.ts#recoverAuthorityScanRowsFromHistory` `src/server/routes/pdfImportRoutes.ts#registerPdfImportRoutes` `tests/pdf-import-hardening.ts` `tests/authority-scan-recovery.ts`

FR-IMP-008 FR-IMP-009 FR-IMP-016

ALG-IMP-03 · Import

Deterministic

Comparison with the authority source file

What has actually changed in the live timetable relative to the authority file: added, deleted, changed or unchanged?

1. The first reading stays an unchanging baseline; its section numbers are generated and instructors are overlaid from the first published snapshot (the source often prints none).
2. Baseline rows are matched to live rows in stages, accepting a pair only when its key is unique on both sides: the reference number (CRN) first, then the full academic placement (course, site, building, room, days, time), then relaxed keys dropping room, time, day, building, and building plus room in turn.
3. A course match requires equal seven-digit keys when both are known (0109105 is not 0118105); another department's course never matches.
4. Unmatched baseline → deleted; unmatched imported live rows → added; matched rows are compared by meaning: the same verified room or the same minute in another notation is not a change.

Inputs

The baseline read from the file, the first published snapshot, the live timetable of the scope, course codes and instructor names.

Output

Report rows: added, deleted, changed (with fields), unchanged, or needing source review.

A net difference, not an edit log; neither the baseline nor the timetable is written; an ambiguous pair is never guessed.

`src/server/core/authority.ts#buildAuthorityPdfDiff` `src/utils/authorityRowMatching.ts#matchAuthoritySourceRows` `src/utils/authorityNetChange.ts#authorityFieldMatchesOriginal` `tests/authority-net-diff-audit.ts` `tests/authority-identity-reconciliation-audit.ts`

FR-IMP-026 FR-RPT-023

ALG-IMP-04 · Import

Gemini-assisted reading with cross-verification

Assisted reading, confirmed

Which empty cells of a scanned page can be filled from a second reading, without replacing any value already read?

1. Offered only after a successful approved reading, only for pages with an unread course, section, days or time cell, and sent only on an explicit click.
2. A bounded catalogue (up to 260 courses and 360 instructors) is sent after removing any civil-identity field.
3. Returned rows are bound to the catalogue and pass the deterministic validators (course, instructor, location, time, conflict).
4. Cross-verification per row: the two readings must agree on at least two identity cells (section, start, end, building, room); one contradiction voids the whole candidate unless reference number and section both agree; a matching reference number counts double.
5. Only empty cells are proposed for filling; a disagreement on a read cell is reported (up to 20) and never applied.
6. The reviewer sees every proposed cell before anything changes; only the approved fills are applied and marked confirmed.

Inputs

Images of the selected pages, the rows read by the approved engine, and the bounded catalogue.

Output

A list of proposed cell fills with page and section, and a list of disagreements and notes.

The Gemini service writes to no timetable or draft; the deterministic validators and the reviewer's explicit confirmation decide.

Document reading budget 180 seconds | At least two agreements to accept a candidate

`src/utils/geminiScheduleLayer.ts` `src/utils/geminiScheduleLayer.ts#bindGeminiRowsToCatalogue`
`src/server/core/gemini.ts#requestGeminiScheduleLayer` `src/utils/importPageReview.ts` `tests/gemini-schedule-layer.ts`

`FR-IMP-027` `FR-IMP-028` `FR-IMP-029` `IR-IMP-030`

3.8.7 Edge cases

⚠ A file mixing departments

Scenario A text-layer file carries a third page with a department code different from page 1.

Behaviour The whole file is refused with PDF_MIXED_DEPARTMENTS naming the page and code, asking to upload each department's pages in its own file. A scan proves only the first page header.

⚠ A course tail matching another department

Scenario The file prints 0114464 while the department catalogue holds a short course 464.

Behaviour It is not linked by tail; it is offered as a choice between linking 464 or creating 0114464 as a separate course after one confirmation, and if the file shows the department's own number with that tail it is another course that is created.

⚠ A scanned page with missing lines

Scenario A page prints 20 lines and 18 are read.

Behaviour It is accepted with a warning and nothing is saved or published until the reviewer presses the page-reviewed button and adds the missing lines in the schedule after import; with 28 printed and 4 read the file stops.

⚠ A second scan during the first

Scenario Two users upload two different scanned files together.

Behaviour The first is read and the second is refused after 45 seconds with 503 SCAN_READING_BUSY and no row imported; if it is the same file it shares the running reading.

⚠ Gemini unavailable or timed out

Scenario Every page fails because of load or the 180-second timeout.

Behaviour 503 GEMINI_SMART_IMPORT_UNAVAILABLE with the statement that the schedule did not change; the approved preview stays as it is and the provider is not called again automatically.

⚠ A term that already has a schedule

Scenario A PDF is uploaded to a term that already holds appointments.

Behaviour Saving returns PDF_TARGET_TERM_OCCUPIED, and the trial run shows the difference (added/removed/changed/same); applying afterwards takes two steps and is preceded by a rollback copy.

⚠ A shortened instructor name that collides

Scenario A full printed name binds to a shortened record of a person teaching in another college at the same hour.

Behaviour The identity of every row with that name is cleared and the reason and the other college are stated so the reviewer picks or adds the instructor.

3.8.8 Standing owner decisions

- Header decides the department (authority PDF): department and branch come from the page header even when a course number carries another department's code; a missing department or course is created after one confirmation under its printed number and name with the usual permission; a three-digit tail alone never links (0114464 is not 0120464); scans never create.
- Course numbers are 7 digits only in everything a reader sees (college 2 + department 2 + course 3, for example 0109105); the internal AdCourseId is never shown.
- Printed reports keep their design: fix values and data, not layout, including the report comparing the schedule with the authority document.

↑ Contents

3.9 · REQ

Faculty requests and public links

A secret personal link lets each faculty member request schedule changes, and an inbox lets the department decide item by item.

37	100%	37	52
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.9.1 Overview

The module issues a secret personal link (/r/) to each faculty member. The link opens a request card with a time window set in Kuwait time, where the member signs with the civil ID the changes, deletions and additions requested. One server engine judges every item against the regulations and clashes before submission, holds the chosen time temporarily and settles races between colleagues in a single transaction. The request reaches the Instructor inbox, where the department fixes the item in the schedule through the normal save path, rejects it with a reason and alternatives, or converses with the owner. It also covers the public faculty card (/s/), guess limits on links and numbers, the not-yet-reached course state, and the appointment helpers (save-blocker checks, mini week, lecture stick).

Faculty member · Department head · Schedule committee chair · Standard user · Public visitor · The system (automatic) · Root administrator

3.9.2 User journeys

Schedule committee chair — **Opening requests and the department decision**

1. In Publish, Faculty card, the chair picks the mode (view only or request edit), sets the closing time in Kuwait time and chooses all or selected faculty.
2. If the sheet holds locked courses, the not-yet-reached sheet appears; the chair sends as is or opens courses for offering first.
3. The system issues a personal link and a request in status sent for each member, and logs a line in the send log.
4. As members submit, requests appear in the Instructor inbox ordered: ready first, then needing an exception, then blocked.
5. The department fixes the item, which is written to the schedule through the normal save path and then recorded, or rejects it with one of seven reasons and up to three alternatives.
6. The decision reaches the member's live card within seconds and the member can accept an alternative or reply.

Faculty member — Requesting a time change from the personal link

1. The member opens the link (/r/), sees the week and items, and picks a lecture to change or a course to add.
2. Pressing a day completes the regulation pattern for the course hours; choosing the start computes the end.
3. Start times are coloured: green passes, yellow needs an exception reason, red is blocked; the chosen time is held for ten minutes.
4. On send the member enters the civil ID, which is matched against the record; the request is accepted or refused with the reason of the blocking item.
5. The member follows each item (pending, fixed, rejected with alternatives) and converses with the department or accepts an alternative with a new signature.

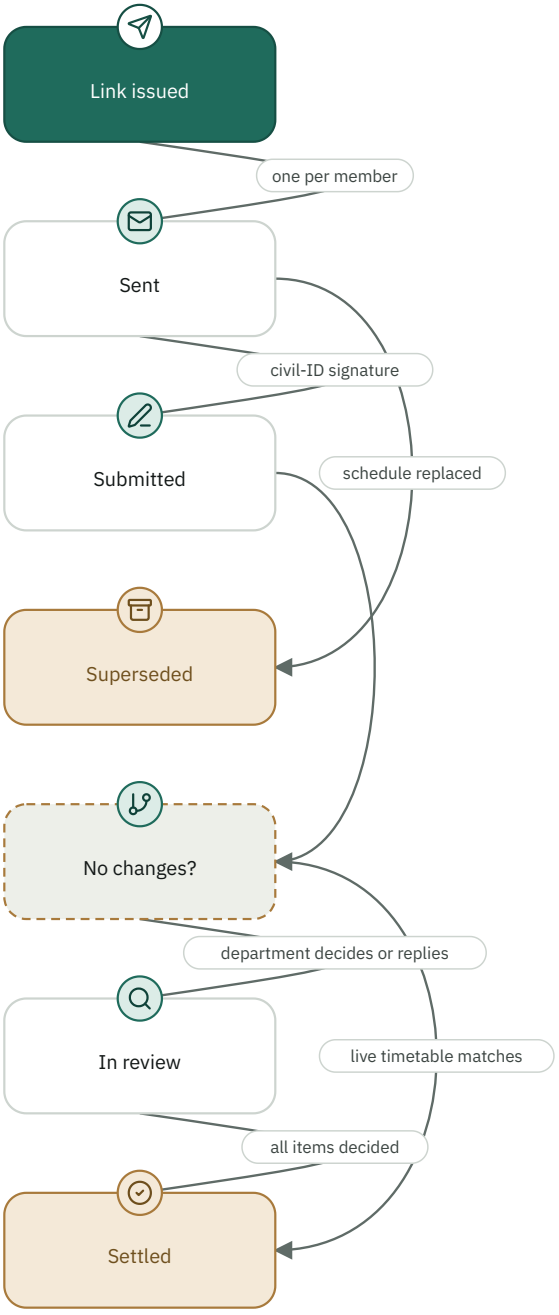
Faculty member — Opening the public faculty card and notifying the department

1. The member opens the card link (/s/) and types the civil ID, optionally choosing remember me.
2. If the number matches no record or the member teaches nothing in the term, the same no-card answer is returned without distinction.
3. The member sees schedule, movement, the next two weeks and each department's approval state, and opens the request links available to them.
4. The member leaves a note (change request or apology) beside a lecture of their own, which reaches the department, capped at eight notes per day.

3.9.3 Diagrams

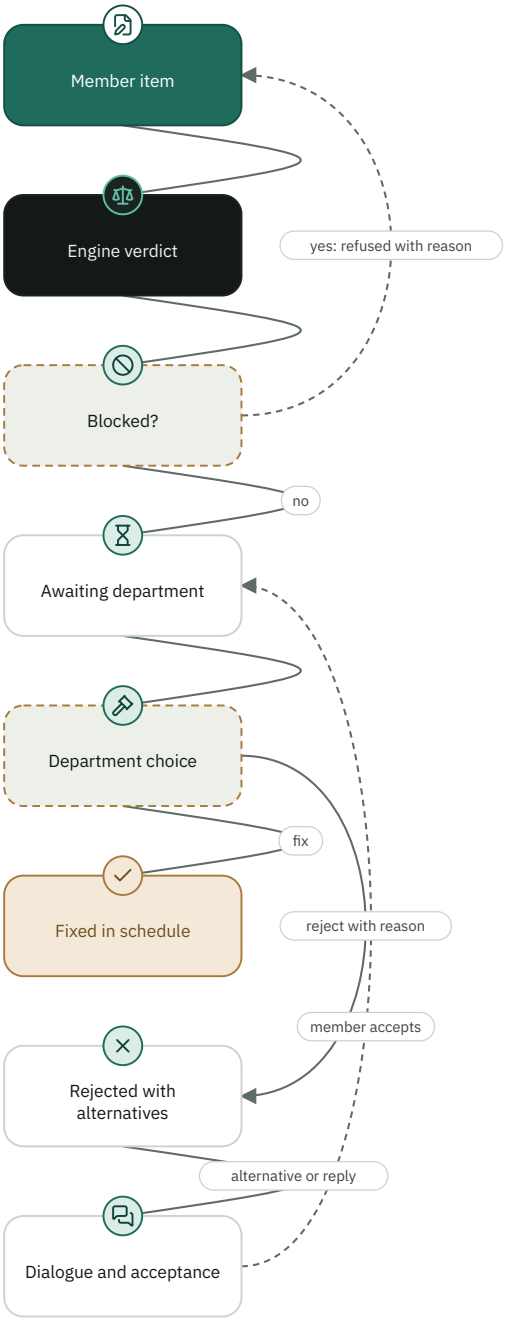
Request lifecycle

Request statuses as the server stores them, and who moves them.



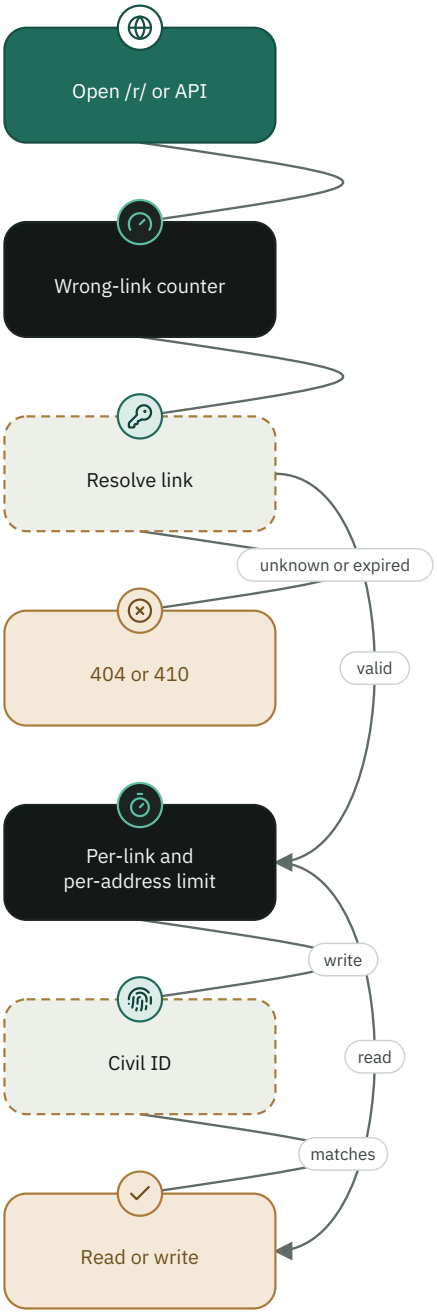
Item decision

The path of one item from the member's submission to the department outcome.



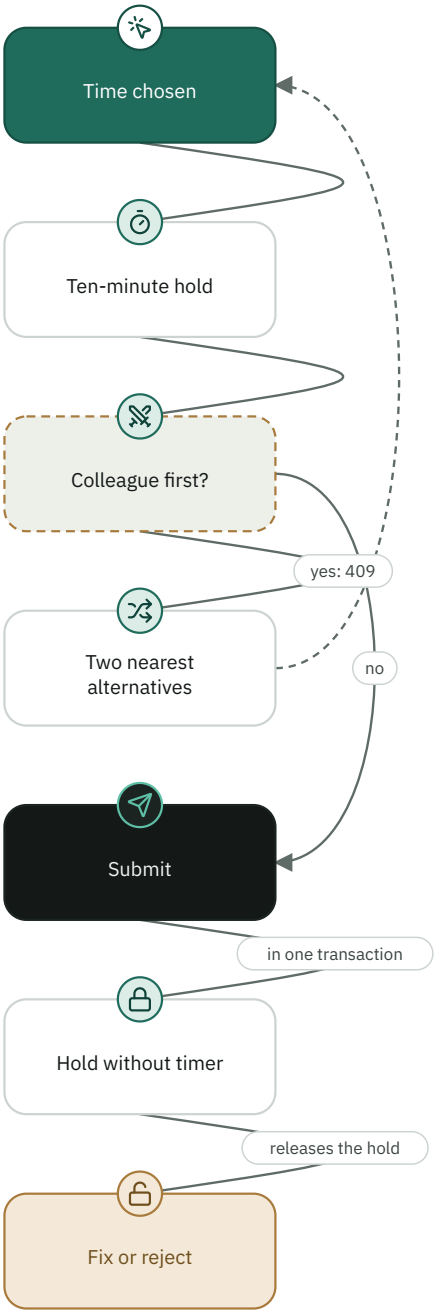
Public link gates

What an anonymous request passes before anything is read.



Hold and race

How two colleagues competing for the same time and room are settled.



Who does what

	Issue links	Submit signed request	Decide on item	Open faculty card
Faculty member	—	✎	—	👁
Schedule committee chair	✎	—	✎	—
Department head	✎	—	✎	—
Standard user	✎	—	✎	—
Public visitor	—	—	—	—

✎ Write 👁 Read — No access

3.9.4 Screens

Instructor request card (/r/)
Public page opened with the personal link: the member's week, item editing, signing, dialogue with the department and term memory.`src/server/pages/instructorRequestPage.ts`

Faculty card (/s/)
Public card opened with the civil ID: schedule, movement, notes and request links.`src/server/pages/staffCardPage.ts`

Instructor inbox
Department decision queue: fix, reject with a reason, dialogue, fix all ready, location filter and congestion heat map.`src/components/InstructorInbox.tsx`

Publish, Faculty card, For whom?
Choose all or selected faculty before sending.`src/components/PublishRecipientPicker.tsx`

Not-yet-reached sheet before sending
Warning listing locked courses with an open-for-offering action.`src/components/PrepublishReachSheet.tsx`

Save-blocker checks
Six status cards for the appointment form and a summary ring.`src/components/AppointmentChecksPanel.tsx`

Faculty mini week
The member's week, free slots and before/after targets inside appointment forms.`src/components/TeacherMiniWeek.tsx`

Course reach state
Reached / not-yet-reached chips and the prerequisite chain.`src/components/CourseReach.tsx`

3.9.5 Requirements

FR-REQ-001 Personal request link per faculty member

The system shall issue to each faculty member in the department family's schedule for the term a secret personal link and a request in status sent holding all of the member's schedule rows, shall never issue a link to the placeholder faculty identity, and shall refuse issuing outside the user's scope or on a frozen term or locked schedule.

● Implemented · Must · Functional · Schedule committee chair, Department head, Standard user

RATIONALE

> The personal link is the only access permission, so it must belong to a real person and sit within the decision-maker's scope.

ACCEPTANCE CRITERIA

1. Issuing for a department with two members returns two issued entries; for a schedule with no rows it returns emptyTerm with reason no-rows.
2. Issuing on a frozen term or locked schedule is refused with 409 (schedule-locked), and outside scope with 403.
3. No link is issued to the placeholder faculty identifier even when it appears in the schedule.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts#registerInstructorRequestsRoutes`
`src/server/core/staffCard.ts#staffFamilyScopes` `tests/instructor-request-server-audit.ts`
`tests/instructor-request-cross-college-audit.ts` `tests/publish-requests-audit.ts`

FR-REQ-002 Request window and link lifetime

The system shall accept the request closing time in Kuwait time (+03:00) only as a valid date or date and time and shall reject it in the past, and shall set the request link's expiry to the greatest of the closing time plus one day, the issue time plus 21 days, and the end of the term.

● Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

RATIONALE

> Links used to expire before the announced deadline, and department decisions and alternatives arrive after closing, so the link must remain readable.

ACCEPTANCE CRITERIA

1. The date 2026-02-31 alone or the time 25:99 is refused with 400, and a past date is refused with 400 closing date in the past.
2. A date without a time closes at the end of that day in Kuwait time.
3. A request with an elapsed window refuses submission with 403 while dialogue and reading remain available.

EVIDENCE

`src/utils/instructorRequestWindows.ts#normalizeInstructorDeadline` `src/utils/instructorRequestWindows.ts#requestWindowOpen`
`src/server/core/instructorRequests.ts#REQUEST_LINK_DAYS` `src/utils/shareLinkLifetime.ts` `tests/staff-request-windows-audit.ts`
`tests/instructor-request-server-audit.ts`

FR-REQ-003 Link mode: view, edit or training

The system shall store the card's send mode on each member's request as one of three values (view only, request edit, training phase) and enforce it on the server, so a view-only link writes nothing and the training phase cannot be opened through the normal issue path.

Implemented · Must · Functional · Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. Submitting a request, a hold or a stick from a view link is refused with 403 view-only or returns held false.
2. Issuing with mode ta is refused with 400 and directed to the training phase panel.
3. In the training phase an item that is not a vacant section is refused, and an absent field means request edit.

EVIDENCE

`src/utils/linkMode.ts` `src/server/routes/instructorRequestsRoutes.ts` `src/server/routes/publicRequestRoutes.ts` `tests/publish-pick-teachers-audit.ts` `tests/ta-phase-flow-audit.ts` `tests/staff-request-windows-audit.ts`

FR-REQ-004 Publishing to selected faculty and the send log

The system shall allow sending to a list of selected faculty (up to 300) or to owners of appointments carrying an open registration note, shall never turn an empty list or an identifier outside the family into all, and shall record every send as a log line holding mode, audience, count and sender name, listing the latest 200 lines.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Empty or more than 300 instructorIds are refused with 400, and an identifier not in the family's schedule is refused with 404 with nothing written.
2. A selected send does not change the shared deadline or the requests of unselected members, and a notes-only send requires edit mode and is refused with 409 when there are no notes.
3. GET send-log returns audience all, picked, exception or ta and never more than 200 lines.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts` `src/components/PublishRecipientPicker.tsx` `src/utils/requestNotedScope.ts` `tests/publish-pick-teachers-audit.ts` `tests/publish-requests-audit.ts`

FR-REQ-005 Reopening follows the new deadline

When reissuing to a member who already has a request, the system shall leave a submitted or in-review request untouched, give an unsubmitted request the new deadline (with a new link if its link is dead), and create a new request only when the previous one is settled or superseded.

Implemented · Must · Functional · Schedule committee chair, Department head

RATIONALE

> The mere existence of any earlier request used to make the system treat it as the member's request, so the member never found the request tab on the card.

ACCEPTANCE CRITERIA

1. reissueMove returns keep for submitted and in-review, extend for sent, and fresh for settled and superseded.
2. A revoked link, or one expiring before the new deadline, is replaced by a new link and the old one is revoked.

EVIDENCE

`src/utils/requestReissue.ts` `src/server/routes/instructorRequestsRoutes.ts` `tests/request-reissue-audit.ts` `tests/publish-requests-audit.ts`

FR-REQ-006 Individual exception after closing

The system shall allow the department to open an exceptional edit window for one member with a written reason (up to 400 characters) and a dedicated closing time, provided an active published card exists, shall allow revoking the exception immediately without cancelling the card, and shall not extend a settled or superseded request.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. An individual issue without a reason is refused with 400, and without a published card with 409.
2. exception/revoke records revokedAt and returns 409 when no exception is active, and works even when the term is frozen.
3. requestWindowOpen returns true inside the exception and false after revokedAt or for a settled request.

EVIDENCE

`src/utils/instructorRequestWindows.ts#individualExceptionOpen` `src/server/routes/instructorRequestsRoutes.ts` `tests/staff-request-windows-audit.ts`

FR-REQ-007 Not-yet-reached course state

The system shall judge every course in every branch (the department in its college) and every sheet as reached, locked or unknown, from the sheet's prerequisites, corequisites, level and that branch's own section history, shall let an open-for-offering decision override the rule, and shall never lock a course without evidence.

Implemented · Must · Functional · Schedule committee chair, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. A course with no prerequisites whose level exceeds the highest level offered in the branch by more than one is locked, and the first term of a new sheet opens only 100-level courses with no prerequisites.
2. The boys' branch history does not open a course for the girls' branch, and a branch with no history stays unknown with no tag.
3. A locked course shows greyed with its reason on the faculty card and cannot be chosen, and the server refuses choosing it.

EVIDENCE

`src/utils/courseReach.ts` `src/server/courseReachShared.ts` `src/components/CourseReach.tsx` `src/server/routes/publicRequestRoutes.ts#lockUnreached` `tests/course-reach-audit.ts` `tests/sheet-membership-audit.ts`

FR-REQ-008 Not-yet-reached sheet before sending

Before sending a card in request-edit or training mode, the system shall show a confirmation sheet listing locked courses by code, name and short reason with an open-for-offering action for each, shall not show it for view only or when no course is locked, and shall not let a failure to read the verdict block sending.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The sheet does not appear for view-only mode, and what the administrator already closed (not offered or future) is not counted.
2. Open-for-offering flips the row to open in place and reduces the count including its corequisite.

EVIDENCE

`src/utils/prepublishReach.ts` `src/components/PrepublishReachSheet.tsx` `tests/prepublish-reach-audit.tsx` `tests/course-reach-audit.ts`

FR-REQ-009 Link resolution and error replies

The system shall resolve the faculty request token, returning 404 for an unknown, revoked or other-kind link and 410 for an expired link or a round closed by schedule replacement, shall make the /r/ page show an invalid-link page with the same status, and shall never return a civil ID or phone number in any public reply.

● Implemented · Must · Functional · Faculty member, Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. A department schedule link token on the request path returns 404 not a schedule request.
2. A link expired after its term end returns 410, and a manually revoked link returns 404 before any other consideration.
3. Opening an /s/ link of kind request is redirected to its own door and never shows the department schedule.

EVIDENCE

`src/server/core/instructorRequests.ts#resolveRequestLink` `src/server/routes/publicRequestRoutes.ts` `src/server/routes/publicStaffRoutes.ts` `tests/instructor-request-server-audit.ts` `tests/instructor-request-followup-audit.ts`

FR-REQ-010 Guess limit on wrong links

The system shall count per address only the /r/ page and proposal opens that match nothing (404), shall block the address after 40 misses in ten minutes with a 429 page and a Retry-After header of 600 seconds, and shall credit the address for each page served successfully, with an absolute ceiling of 400 misses.

● Implemented · Must · Functional · Public visitor, The system (automatic)

RATIONALE

> The university network is one address for thousands of members, so guessing deserves the limit, not entry.

ACCEPTANCE CRITERIA

1. The forty-first miss within ten minutes from the same address returns 429, and an expired link (410) is not counted as a miss.
2. Sub-requests under /r/ other than the page and proposals pass without counting.

EVIDENCE

`src/server/anonymousGuard.ts#createLinkMissGuard` `src/server/publicAttemptLimiter.ts#createAttemptLimiter` `server.ts#PUBLIC_LINK_PAGE_PREFIXES` `tests/anonymous-guard-audit.ts`

NFR-REQ-011 Rate limits on public request APIs

The system shall apply per-minute limits counted per link (token fingerprint) rather than per address: 30 for submit, reply and interest, 90 for time checks, hold and stick, and 120 for opening the card stream, on top of a general ceiling of 240 per link and 1500 per address per minute, configurable from the environment.

● Implemented · Must · Non-functional · The system (automatic), Faculty member

ACCEPTANCE CRITERIA

1. The 31st request in one minute on one token to the submit path returns 429 with an Arabic message.
2. The key is a sha256 fingerprint truncated to 32 characters, not the token itself, so no token stays in memory.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts#rateLimitPublicRequest` `src/server/core/rateLimits.ts#requestTokenKey` `src/server/core/rateLimits.ts#requestLinkLimit` `tests/teacher-card-flow-audit.ts`

FR-REQ-012 Civil-ID signature on requests

The system shall require on submit, reply and acceptance of department alternatives a valid civil ID matching the link owner's record, shall return one answer for a wrong number and for a record with no number, shall store not the number but its fingerprint and a verification code, and shall limit guessing to ten failures in ten minutes per request and address.

Implemented · Must · Functional · Faculty member

RATIONALE

> Links are forwarded over messaging apps, and pressing send alone does not prove the owner wrote the request.

ACCEPTANCE CRITERIA

1. An incomplete number is refused with 400, and a number not matching the record with 403 civil ID does not match the owner of this link.
2. Arabic and Persian digits are normalised before matching, and the eleventh wrong attempt returns 429.
3. The public reply omits the fingerprint and returns only the date and verification code.

EVIDENCE

`src/server/core/instructorRequests.ts#verifyRequestSigner` `src/server/core/shareLinks.ts#publicAttemptReserve` `src/server/publicAttemptLimiter.ts` `tests/request-signature-audit.ts` `tests/instructor-request-server-audit.ts`

FR-REQ-013 Accepting a request and validating its items

On submission the system shall rebuild the items from stored data rather than from what was sent (only day, start, course and action), and shall refuse more than 200 items, a blocking item (conflict), an exception without a reason, a closed term, a closed window, an edit outside appointments carrying a registration note, and a course that is not offered or outside the department catalogue.

Implemented · Must · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. Sending 201 items is refused with 400, and an item with verdict conflict is refused with 400 carrying itemIndex and the first blocking reason.
2. An exception without excuse is refused with 400 asking for the reason, and the excuse is truncated at 400 characters.
3. The general note is truncated at 400 characters and does not enter the verdict, and a course outside the catalogue is refused with 400.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts#instructorRequestItemsFromPayload` `src/server/routes/publicRequestRoutes.ts` `src/utils/requestNotedScope.ts#notedScopeBreach` `tests/instructor-request-server-audit.ts` `tests/instructor-request-followup-audit.ts` `tests/instructor-request-availability-audit.ts`

FR-REQ-014 Engine verdict on a member's item

The system shall judge every item with one engine against the live schedule after applying the whole bundle, with one of three verdicts (clear, needs an exception, not allowed), shall make the regulation binding on the faculty card, shall not block a row the member did not touch, and shall recompute at every read and submit rather than store the verdict.

Implemented · Must · Functional · Faculty member, Department head, The system (automatic)

ACCEPTANCE CRITERIA

1. Moving two lectures to the same hour does not clash each with its old place but with its sibling in the bundle.
2. A review verdict in the regulation passes tagged needs exception, a block verdict prevents submission, and a physical instructor or room overlap blocks.
3. The request window limits the member's submission but not the department's decision: the department read (forDepartment) ignores the window condition.

EVIDENCE

`src/utils/instructorRequestVerdict.ts#judgeRequest` `src/server/core/instructorRequests.ts#judgeRequestItems` `src/utils/instructorRequestWeek.ts#instructorWeekAfter` `tests/instructor-request-availability-audit.ts` `tests/instructor-request-review-regressions.ts` `tests/teacher-card-hours-patterns-audit.ts`

FR-REQ-015 Meeting length and pattern day completion

The system shall compute the meeting end from days and start with a single engine shared by server and page (the appointment's own length when the days match, otherwise the regulation pattern for the course hours, otherwise 50 minutes, and 80 for Monday and Wednesday), and shall complete the regulation pattern's days on the first day pressed, after which each press adds or removes that day alone.

Implemented · Must · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. Monday alone for a three-hour course becomes Monday and Wednesday, and the second press adds or removes one day only.
2. The browser copy of the length engine gives the same result as the server on a table of cases, so the member never reads 16:50 and saves 17:15.
3. Nothing is inferred from the course history when a day is chosen.

EVIDENCE

`src/utils/requestMeetingLength.ts` `src/server/requestLengthScript.ts` `src/server/requestDayAutofillScript.ts` `tests/teacher-card-logic-audit.ts` `tests/request-day-autofill-audit.ts` `tests/teacher-card-hours-patterns-audit.ts`

FR-REQ-016 Temporary hold and race settlement

The system shall temporarily hold the chosen time for ten minutes inside the same ledger transaction that submission uses, so the time disappears from colleagues' maps, shall answer a member beaten by a colleague with 409 and two nearest alternatives, and shall keep a submitted request's hold without a timer until the department decides.

Implemented · Must · Functional · Faculty member, The system (automatic)

RATIONALE

> Forty concurrent submissions for one room must produce exactly one winner.

ACCEPTANCE CRITERIA

1. A second hold for the same time and room returns 409 with code race-lost and an alternatives field of at most two entries.
2. A temporary hold older than ten minutes holds nothing, and cancelling an addition releases only its own hold by identity, not by index.
3. Browsing options and checking times hold nothing, and fixing or rejecting the item releases the hold.

EVIDENCE

`src/utils/instructorRequestReservations.ts` `src/server/routes/publicRequestRoutes.ts#REQUEST_HOLD_TTL_MS` `src/components/RequestHoldSheet.tsx` `tests/instructor-request-availability-audit.ts` `tests/teacher-card-logic-audit.ts` `tests/teacher-card-flow-audit.ts`

FR-REQ-017 Time map and draft check

The system shall measure against the real schedule up to 14 start times for one item, returning ok, warn or bad for each with the same single verdict, with one reason code for a blocked time, shall return the days problem (shape) with approved patterns instead of colouring every time red, shall check a draft of up to 80 items in one batch, and shall fail explicitly rather than claim there is no conflict.

Implemented · Should · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. More than 14 starts are truncated to 14, a non-HH:MM format is dropped, and a batch above 80 is refused with 400.
2. When the days do not cover the course hours the reply is problem shape with up to three patterns and no times.
3. The blocking reason code is one of instructor, room, facility, cohort, load, regulation and travel.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts` `src/server/core/instructorRequests.ts#buildSharedRequestContext` `tests/teacher-card-logic-audit.ts` `tests/instructor-request-followup-audit.ts`

FR-REQ-018 Stick to my lecture

The system shall generate for each of the member's existing lectures two candidate times (before and after), counted from its end or start plus the regulation gap of 10 minutes onto the first start on the regulation ladder, shall judge each candidate with the single engine, and shall order available first without ever sending the member a room or building name.

Implemented · Could · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. A lecture whose move or deletion was requested is not a fixed anchor, and anchors never exceed 12.
2. The reply passes through stickForTeacher so it carries no sameRoom or sameBuilding, and the nearest available candidate stops the check of later ones.

EVIDENCE

`src/utils/lectureStick.ts` `src/server/routes/publicRequestRoutes.ts` `src/components/LectureStickChips.tsx` `tests/lecture-stick-own-audit.ts` `tests/teacher-card-r4-audit.ts` `tests/appointment-forms-r2-audit.ts`

FR-REQ-019 Choosing a vacant section

The system shall offer the member only sections assigned to the placeholder faculty, with all their meetings, within the allowed courses, after excluding what another member reserved or the member already chose, what has no real room identifier, what exceeds the member's load, and what clashes with the member or in a room, and shall never expose another member's timetable.

Implemented · Must · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. A section with a real instructor on any row does not appear, and a section with no room identifier does not appear.
2. Approval assigns the section's own rows without changing times, rooms or code, rejection releases them, and a submitted hold has no timer.
3. A section of a course locked by not-yet-reached is refused by the server.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts#availableInstructorOfferings` `src/server/core/instructorRequests.ts` `tests/instructor-offering-flow-audit.ts` `tests/instructor-request-availability-audit.ts`

FR-REQ-020 Confirming the schedule unchanged

The system shall settle the request at signing (status settled with an unchanged-confirmed event) only when every item is keep and each item matches the member's live schedule exactly (course, section, days and times), with no free-text note, decision or dialogue, and shall store the schedule fingerprint at that moment.

Implemented · Must · Functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. A request with any general note stays submitted and reaches the department, and no department notification is sent on direct confirmation.
2. A settled request with no changes returns 409 on resubmission saying the schedule was already confirmed.
3. A copy carried over from a previous term is not confirmed, since the decision concerns the live schedule alone.

EVIDENCE

`src/utils/instructorRequestConfirmation.ts` `src/server/routes/publicRequestRoutes.ts` `tests/unchanged-request-confirmation-audit.ts` `tests/unchanged-request-confirmation-http-audit.ts`

FR-REQ-021 Replaying the previous term's request

On request the system shall show the member what they submitted in the nearest earlier term chronologically (not by identifier order) and how it was decided, and shall replay it as a draft by one rule that does not replay what was rejected, does not guess a section for a course with more than one section, and does not touch an item the member edited, the department decided, or that is already the current time.

Implemented · Could · Functional · Faculty member

ACCEPTANCE CRITERIA

1. Each skipped item is shown with its reason (rejected that term, more than one section, already the current time).
2. Replay is a draft not a submission, goes through live checking as if just written, and the reply is cached for five minutes and carries nothing meant for the department alone.

EVIDENCE

`src/server/requestReplayScript.ts` `src/utils/previousTermRequest.ts#pickPreviousRequest` `src/server/routes/publicRequestRoutes.ts` `tests/teacher-card-flow-audit.ts` `tests/request-page-browser-syntax-audit.mjs` `tests/doctor-journey-audit.ts`

NFR-REQ-022 What never reaches the member: rooms and internal history

The system shall strip from every public reply of a member's request the rooms and buildings (candidates, options, reserved, switched and the room field), the signature fingerprint, signature history, archived items and revision stamps, and shall scrub room codes from what the committee writes (rejection note, replies, timeline).

Implemented · Must · Non-functional · Faculty member, The system (automatic)

RATIONALE

> Owner decision (2026-10-29): rooms are the department's decision; the member sees only that a time is available or not.

ACCEPTANCE CRITERIA

1. The reply of GET /api/public/request/:token contains no roomCandidates, roomOptions, offeringRows, reservedRoom, roomSwitch or signatureHistory.
2. A rejection note written by the coordinator that contains a room code reaches the member with the generic word instead of the code.
3. Schedule movement on the request page carries no room name, and an edit that touched only the room does not arrive as an event.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts#stripForInstructor` `src/server/core/staffCard.ts#scheduleMovementEntries` `tests/instructor-request-server-audit.ts` `tests/teacher-card-logic-audit.ts` `tests/teacher-card-r4-audit.ts`

IR-REQ-023 Live card stream

The system shall open for the member's page an event stream (SSE) using the same personal link that carries from the request only the event kind and time, capped at 4 connections per request and 600 server-wide with a heartbeat every 25 seconds, and the browser shall fall back to ordinary reading after the cap.

Implemented · Should · Interface · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. The fifth connection for one request returns 429 too many connections on this link, and the registration is cleaned when the connection drops.
2. A department decision reaches an open card within two seconds without reloading.

EVIDENCE

`src/server/routes/publicRequestRoutes.ts#CARD_STREAMS_PER_REQUEST` `src/server/routes/publicRequestRoutes.ts` `tests/teacher-card-flow-audit.ts`

FR-REQ-024 Instructor inbox: decision list and its scope

The system shall return to the department inbox the current-round requests for a term and college (and optionally a department) with the department family across sites, shall show of an item outside the user's scope only a hidden shell with no course, time or room, and shall return totals (sent, answered, unchanged, changed, unopened, settled) without listing the placeholder faculty.

● Implemented · Must · Functional · Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. An item in another department arrives with hidden true and empty slots, keeping action, index and decision state.
2. totals.unopened does not count view-only links, and requests from old rounds (after schedule replacement) do not appear.
3. A request issued by another college appears in this college when it carries an item on a lecture in it.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts` `src/server/core/instructorRequests.ts#requestItemScope` `src/utils/inboxAudience.ts` `tests/instructor-request-cross-college-audit.ts` `tests/inbox-campus-audit.ts` `tests/inbox-live-audit.ts`

FR-REQ-025 Department decision on an item: fix or reject

The system shall record the department decision on an item as one of two states (fixed, rejected) within the item's scope rather than the link's scope, shall require for rejection a reason from seven (room, instructor, regulation, cohort, load, department, other), a note of up to 400 characters and up to three alternatives, and shall not record fixed before the schedule itself confirms it happened.

● Implemented · Must · Functional · Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. A rejection with no valid reasonCode is refused with 400 asking for the reason, and fixing an already-fixed item is refused with 409 already-fixed.
2. Fixing an addition with no matching saved row (member, course, term, college, department) is refused with 409, and fixing a deletion while the row remains is refused with 409.
3. An item with different times across days is not fixed with one time: refused with 409 mixed-times, and alternatives are cut at three.
4. A rejection with alternatives does not close the request, which stays in review until the member chooses.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts` `src/utils/instructorRequestVerdict.ts#requestFullySettled` `src/utils/instructorRequestThread.ts#parseOfferedSlots` `tests/masari-approval-integrity-audit.ts` `tests/instructor-request-server-audit.ts` `tests/instructor-request-followup-audit.ts`

FR-REQ-026 Atomic approval of an item

The system shall fix an item (with its pending linked items) in a single call that dry-runs the bundle through the normal save path, then writes the appointments, the decision and the hold release together or writes nothing, under a per-term apply lease, requiring the request revision stamp and each row's stamp, and shall offer a preview that writes nothing.

● Implemented · Must · Functional · Schedule committee chair, Department head, Standard user

RATIONALE

> Fixing used to be two steps with a gap that could leave a row without a decision or a chain half moved.

ACCEPTANCE CRITERIA

1. A missing request stamp is refused with 428, a stale one with 409 request-revision, a missing row stamp with 428 and a changed row with 409.
2. A fix already running in the same term returns 409 apply-busy, and the lease is renewed every 40 seconds and released at the end.
3. A rejected or fixed item is not approved (409 rejected-item or already-fixed), and linked items are accepted only when pending.
4. preview true returns the plan and warnings with no write.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts` `src/utils/linkedRequestItems.ts#awaitedItemIndexes` `src/utils/instructorRequestItems.ts` `tests/instructor-request-review-regressions.ts` `tests/instructor-request-availability-audit.ts` `tests/linked-requests-audit.ts`

FR-REQ-027 Item dialogue and accepting an alternative

The system shall let the department reply on an item with text of up to 400 characters and up to three proposed times without deciding or writing to the schedule, and shall let the member, with a signature, accept a proposal or the rejection alternatives (singly or in a batch of up to 100) or reply with another proposal, in a thread of at most 40 messages per item whose first message is kept and the overflow archived, remaining open after the window closes unless the term is frozen.

Implemented · Must · Functional · Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. A reply with no text and no time is refused with 400, and a fixed and agreed item is refused with 409.
2. Acceptance with an outdated proposal identifier is refused with 409 request-proposal, and a request not yet sent is refused with 409 send your request first.
3. Beyond 40 messages the first and the last 39 are kept and the older ones move to threadHistory.
4. A department reply with proposed times on a chosen section whose time is fixed is refused with 400.

EVIDENCE

`src/utils/instructorRequestThread.ts` `src/utils/instructorRequestProposalIdentity.ts` `src/utils/requestAlternatives.ts` `src/server/routes/publicRequestRoutes.ts` `src/server/routes/instructorRequestsRoutes.ts` `tests/public-kit-audit.ts` `tests/instructor-request-followup-audit.ts` `tests/instructor-request-review-regressions.ts`

FR-REQ-028 Cancelling a department-placed appointment and undoing a fix

Inferred

The system shall allow cancelling an item the department placed from an appointment by deleting its row through the normal delete path with a line to the member of up to 160 characters (rejected with reason department), and shall allow undoing a fix from the schedule by restoring the item to its earlier state without a notification, provided the row no longer sits on the requested time, with repetition having no effect.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Cancelling with no line is refused with 400, and cancelling an item that is not a department-placed appointment is refused with 409.
2. reopen-fixed while the row is still on the requested time is refused with 409, a superseded request with 409, and repeating it returns reopened 0.

EVIDENCE

`src/server/routes/instructorRequestsRoutes.ts` `src/server/core/scheduleConflicts.ts#reopenedRequest` `tests/instructor-request-followup-audit.ts` `tests/masari-approval-integrity-audit.ts`

NFR-REQ-029 Request revision stamp against overwriting a colleague

The system shall require the request revision number (requestRevision) on every decision, reply, cancellation and public submission, returning 428 when absent and 409 when it differs from storage, so nothing is written over a change that happened after the screen opened, and the member's draft shall stay preserved in a 409 reply.

Implemented · Must · Non-functional · Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. decide, reply, cancel-scheduled and the public POST without requestRevision return 428 with code request-revision-required.
2. An old number returns 409 with code request-revision, and a changed item order returns 409 with code request-item.

EVIDENCE

`src/server/core/instructorRequests.ts#requireInstructorRequestRevision` `src/server/core/instructorRequests.ts#instructorRequestItemIdentity` `tests/instructor-request-review-regressions.ts` `tests/instructor-request-server-audit.ts`

FR-REQ-030 Faculty card identity gate

The system shall open the faculty card (/s/) with the civil ID or a signed device key, shall return one 404 answer for a wrong number and for a member who teaches nothing in the link's term, shall open a personal link only for its owner, shall limit guessing to ten failures in ten minutes per link and address, and shall not count an attempt with neither number nor key.

Implemented · Must · Functional · Faculty member, Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. A civil ID with fewer than 8 digits after normalisation opens no card, and the eleventh wrong attempt returns 429.
2. A personal link with a colleague's number returns the same 404, and the general department link carries no requests, decisions or rejection notes.
3. A member whose every lecture was removed after having a schedule gets an empty card with the movement, not 404.

EVIDENCE

`src/server/routes/publicStaffRoutes.ts` `src/server/core/shareLinks.ts#publicAttempts` `src/server/core/staffCard.ts` `tests/staff-card-global-scope-audit.ts` `tests/staff-device-key-audit.ts` `tests/doctor-journey-audit.ts`

FR-REQ-031 Remember this device

After verification by civil ID and on explicit request, the system shall give the device a signed key (HMAC) bound to the link token and the member's internal identifier in an HttpOnly, SameSite=Strict cookie scoped to that link only, expiring at the link's end or after 180 days, whichever is sooner, and carrying no civil ID.

Implemented · Should · Functional · Faculty member

ACCEPTANCE CRITERIA

1. The key is not valid for another link's token and falls when the link is revoked or renewed or by the forget-this-device button.
2. A key no longer valid is cleared and the number is asked as usual, indistinguishable from no card.

EVIDENCE

`src/server/staffDeviceKey.ts` `src/server/routes/publicStaffRoutes.ts` `tests/staff-device-key-audit.ts`

FR-REQ-032 Member notes from the faculty card

After identity is verified the system shall let the member leave a note (change request or apology) of up to 400 characters and a date or date range beside a lecture of their own schedule only, in the live term, stored in the lecture's actual scope, capped at eight notes per day per member, with no path to the schedule row.

Implemented · Should · Functional · Faculty member, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The ninth note in a day is refused with 429, a lecture not on the member's schedule with 404, and a non-live term with 409.
2. A change note with no text is refused with 400, an apology is accepted with no text, and a date not in YYYY-MM-DD format is dropped.

EVIDENCE

`src/server/routes/publicStaffRoutes.ts#STAFF_NOTES_PER_DAY` `tests/doctor-journey-audit.ts` `tests/teacher-card-polish-audit.ts`

DR-REQ-033 Family card publication and schedule movement

The system shall publish the faculty card to the whole academic department family rather than one site, shall show the schedule-change request tab only with an active publication (not revoked, not expired, not personal), and shall build both the faculty card and the request page's schedule movement from one source: the member's requests and the department decision on them, and what was actually added, moved and deleted between schedule versions.

● Implemented · Must · Data · Faculty member, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. Revoking card publication hides the request tab on the member's card, and request movement starts from the first submission and is cut at 100 lines.
2. Each line carries its source (request or schedule) and its decision (fixed, rejected, pending), and a course removed entirely stays visible in movement.

EVIDENCE

`src/server/core/staffCard.ts#liveStaffPublication` `src/server/core/staffCard.ts#scheduleMovementEntries`
`src/server/routes/publicStaffRoutes.ts` `src/utils/instructorRequestWindows.ts#cardRequestsCloseAt`
`tests/staff-card-global-scope-audit.ts` `tests/instructor-request-cross-college-audit.ts`

IR-REQ-034 Location filter, congestion map and inbox question

The Instructor inbox shall tag each item with its location (an addition where the member chose, an edit or deletion where the row sits), shall offer a location filter that hides items outside it and a request left with no item in it, a congestion map that counts requested appointments per cell (day by start) and warns of room shortage when requests exceed known rooms, and an Arabic-sentence question field that reads and never decides.

● Implemented · Could · Interface · Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. Three requests for two known rooms in the same cell is a shortage, two for two is none, and a request with no known rooms creates no shortage.
2. Keep and delete items and sent-but-unanswered requests are not counted in the map.
3. The filter choice is remembered per account on its device, and what the question field understood is shown on a line beneath it to be cleared.

EVIDENCE

`src/utils/inboxCampus.ts` `src/utils/inboxHeat.ts` `src/utils/inboxAsk.ts` `src/components/InstructorInbox.tsx`
`tests/inbox-campus-audit.ts` `tests/inbox-heat-audit.ts` `tests/inbox-ask-audit.ts` `tests/inbox-studio-audit.ts`

IR-REQ-035 Save-blocker checks for the appointment form

The appointment form shall show one verdict read together by the fix-before-saving strip, the checks panel and the save button, with six cards (instructor, room, section, facility, time, college) each ok, warning, blocker or unchecked, shall not check a field that was not chosen, and shall show a single blocker number.

● Implemented · Should · Interface · Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. An instructor conflict before an instructor is chosen, or a room scope before a room is chosen, stays an idle card, not a blocker.
2. The number in the middle of the ring equals the blocker count the save strip states, and each blocker carries an alternative button.

EVIDENCE

`src/utils/appointmentVerdict.ts` `src/components/AppointmentChecksPanel.tsx` `src/utils/appointmentRegulation.ts`
`tests/appointment-forms-audit.ts` `tests/appointment-forms-r2-audit.ts` `tests/appointment-forms-r3-audit.ts`

IR-REQ-036 Faculty mini week in appointment forms

When a member is chosen in any appointment form the system shall show their week in five columns, the current appointment as a ghost (green when sound, outlined red when clashing), free slots that fit the course pattern, before/after targets by the same stick formula, and the load ring, attendance days and gaps, read-only with no new judgement rule.

● Implemented · Could · Interface · Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. The verdict on the appointment stays with appointmentVerdict alone, and a free slot fits the course pattern and the regulation gap.
2. The window collapses and remembers its state, floats by dragging its header and docks back with a double click, and on a narrow screen becomes a bottom strip.

EVIDENCE

`src/utils/teacherMiniWeek.ts` `src/components/TeacherMiniWeek.tsx` `tests/appointment-forms-r2-audit.ts` `tests/appointment-forms-r3-audit.ts` `tests/appointment-forms-audit.ts`

DR-REQ-037 Anonymous appointment-entry measure

Inferred

The system shall fold the entry measure of each appointment (time, blocker hits, helper used, device, create or edit) into a monthly document of counts and sums only with no account number, name, course or room, shall not count time while the page was hidden, and shall let only the root administrator account read the document.

● Implemented · Could · Data · Root administrator, Schedule committee chair, Department head, Standard user

ACCEPTANCE CRITERIA

1. GET on the measure returns the last three months to the root administrator account only, and any other account is refused.
2. Allowed helpers form a closed whitelist that silently drops anything else, and folding is atomic so two simultaneous measures count as two.

EVIDENCE

`src/utils/appointmentMeasure.ts` `src/server/appointmentMetrics.ts` `src/components/AppointmentMeasureCard.tsx` `tests/appointment-entry-parity-audit.ts`

3.9.6 Algorithms

ALG-REQ-01 · Faculty requests and public links

Faculty request item verdict and meeting length

Deterministic

Does what the instructor asks for on the card go through, need an exception, or is it not allowed — and what are the nearest free times when it is not?

1. Meeting length: the appointment's own length when its days are kept (or one day moves to one day), else the regulation pattern for the course hours, else 50 minutes on Sunday, Tuesday and Thursday and 80 on Monday and Wednesday; the first day pressed completes the regulation pattern containing it.
2. The week is built after applying the whole bundle to this instructor's rows; everyone else's rows stay as they are.
3. A closed window blocks; delete, keep, and an edit that touches nothing are not judged.
4. Checks in order: clock format, campus and department day bounds, weekly minutes against course hours, approved pattern, same-course section clash, start ladder, regulation, instructor clash, travel, cohort overlap (advice only), load, and hall availability.
5. Any blocking reason → not allowed; non-blocking reasons other than cohort and travel → needs an exception with a written excuse; otherwise clear.
6. When blocked, each start on the department ladder ∪ the regulation ladder is tried, the moved bundle is re-judged, and the three nearest sendable starts are returned.

Inputs

The request item (action, days, start), the live term timetable, the department's courses and hours, the start ladder, the request window, the instructor's load allowance, and cohort pairs.

Output

One of three verdicts (clear, exception, not allowed) with reasons and their source, a computed end time, and up to three alternative times.

Writes nothing; a row the department placed and the member did not touch is never blocked; no specific hall is revealed to the instructor; the verdict is recomputed at every read and submit.

50 minutes (Sunday, Tuesday, Thursday) and 80 minutes (Monday, Wednesday) · Default campus day 08:00–20:00 (configurable) · Up to 3 alternative times

`src/utils/instructorRequestVerdict.ts#judgeRequest` `src/utils/requestMeetingLength.ts#meetingMinutes` `src/utils/requestMeetingLength.ts#meetingSlots` `src/server/requestDayAutofillScript.ts#REQUEST_DAY_AUTOFILL_SCRIPT` `tests/instructor-request-availability-audit.ts` `tests/request-day-autofill-audit.ts` `tests/teacher-card-hours-patterns-audit.ts`

FR-REQ-014 FR-REQ-015

3.9.7 Edge cases

⚠ A colleague took the same time first

Scenario Two members choose the same time and the only available room, and the first submits before the second.

Behaviour The first wins in the ledger transaction. The second receives 409 with code race-lost and two nearest alternatives, the draft stays preserved on the page, and no colleague or room name is mentioned.

⚠ Item with different times across days

Scenario A member requests a course with two different times on two days and the coordinator presses fix.

Behaviour The server refuses with 409 code mixed-times because one row has one time, so the coordinator fixes from the schedule with a row per time; fixed is never recorded without verification.

⚠ Schedule replaced after links were issued

Scenario The department schedule is replaced entirely after members opened their links and some submitted.

Behaviour The old round closes: the link returns 410 round closed, its requests leave the inbox and stay in history, and the department issues a link for the current round.

⚠ Term frozen after issue

Scenario The request window is still open by date but the term has become frozen or ended.

Behaviour The server answers the member with 409 in wording of their own (check with your department) without mentioning the schedule committee, and refuses decision, approval and reply from the department; revoking an individual exception stays allowed because it does not touch the schedule.

⚠ Guessed civil ID or shared network

Scenario Someone tries numbers on a card, or thousands of members open cards from the one university address.

Behaviour The limit counts the failure, not the entry: ten failures in ten minutes per link and address block, a successful attempt returns its credit, and opening with neither number nor key is not counted. The same answer is returned for a wrong number and for a member who teaches nothing.

⚠ General department link with a colleague's number

Scenario Someone enters a colleague's civil ID on the general department link or on another member's personal link.

Behaviour A personal link opens only for its owner and returns 404 for anyone else, the general department link shows the schedule alone with no requests, decisions or rejection notes, and request editing remains conditioned on the number matching the member's record.

⚠ Two linked items across two departments

Scenario A member teaching in two colleges requests deleting a lecture in the first and adding another in the second at the same time.

Behaviour Each department sees only its own item, the item waiting on another is marked awaitedElsewhere without that department's details, and approval accepts linked items only when pending and in one transaction.

3.9.8 Standing owner decisions

- Rooms and buildings are the department's decision alone: they never reach the member in any reply, in what the committee writes, or in schedule movement on the request page (owner decision 2026-10-29).
- Pressing send is a civil-ID signature, a rejection is never without a reason, and fixed is said only when it actually happened in the schedule.
- Any course number shown is always seven digits (college + department + course), and printed reports keep their design.
- The regulation binds on the faculty card and advises on department screens: the member asks the department so cannot ask for what is not allowed, while the department is an authority that can grant exceptions.

↑ Contents

3.10 · PLN

Section planning and curricula

From the remaining report and plan sheet to the term's sections and the study proposal

34	100%	69	70
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.10.1 Overview

The module builds the term's class sections from the study plans. The department uploads the remaining report, the system suggests section counts from comparable past terms, and the missing sections are applied to the schedule through one engine with explicit confirmation and undo. It also manages plan sheet rules (levels, prerequisites, electives) from draft to approval and public publishing, course offering states, and the study proposal the department prepares for an instructor and commits after approval. It serves department heads, the schedule committee chair and system administration, while students and visitors read the published sheets.

Department head · Schedule committee chair · Root administrator · Department advisor · Standard user · Faculty member · Public visitor · Student · The system (automatic)

3.10.2 User journeys

Department head — **From the remaining report to sections in the schedule**

1. Uploads the remaining report as PDF or images from section planning and reviews the preview.
2. Reads each course's suggested range and reason, and writes a «chosen» count where he disagrees.
3. Opens «apply sections» and sees required, existing and remaining per course.
4. Previews a week built with halls, instructors and conflict-free times, plus reasons for what could not be built.
5. Confirms the apply, writing all sections or none, and may undo with one click.

Schedule committee chair — From plan sheet draft to the public page

1. The main administrator imports the sheets, creating «under review» drafts.
2. The department corrects its courses' levels, prerequisites and elective groups.
3. The chair reviews the blockers and approves the version he saw.
4. Publishes the approved sheet from «sheet publishing» and obtains a link, QR and PDF.
5. Students read the page at /plans/ without an account, and the snapshot refreshes on each later approval.

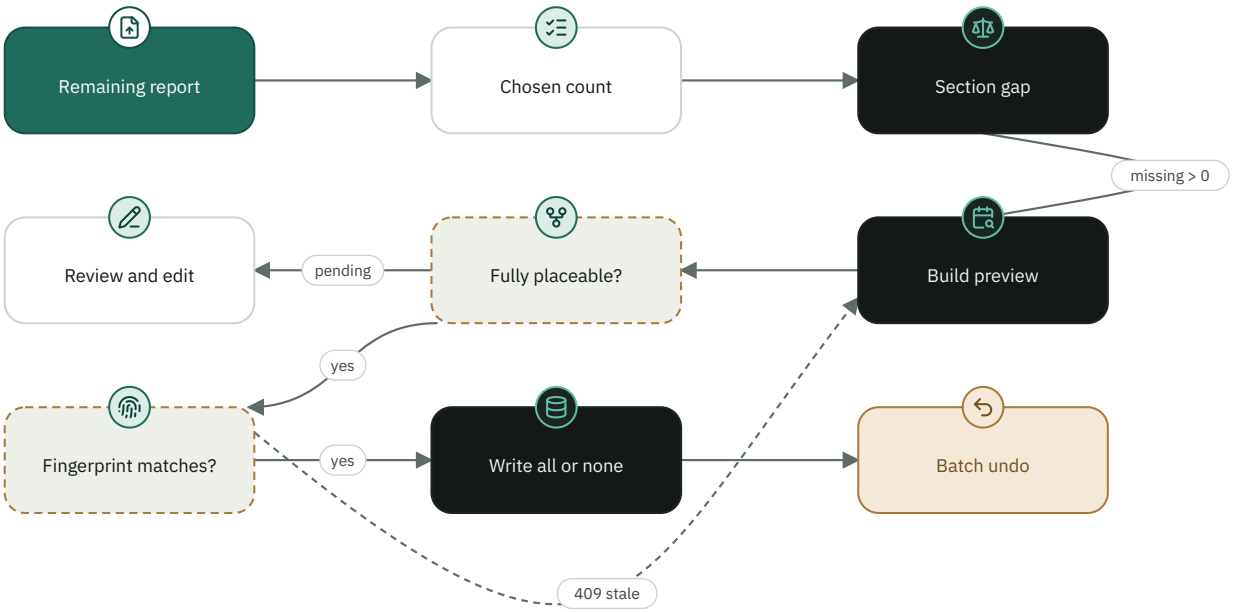
Department head — A study proposal for an instructor

1. Starts the proposal from the instructor's request card in the inbox and adds items (assign, new section, edit or replace).
2. Sees the difference on the schedule and the check result at once: blocker, note or information.
3. Sends the version and the instructor receives a personal link, with nothing reserved.
4. The instructor signs with the civil number to approve or requests a change.
5. The department reviews and commits in one transaction, and may undo if no appointment changed.

3.10.3 Diagrams

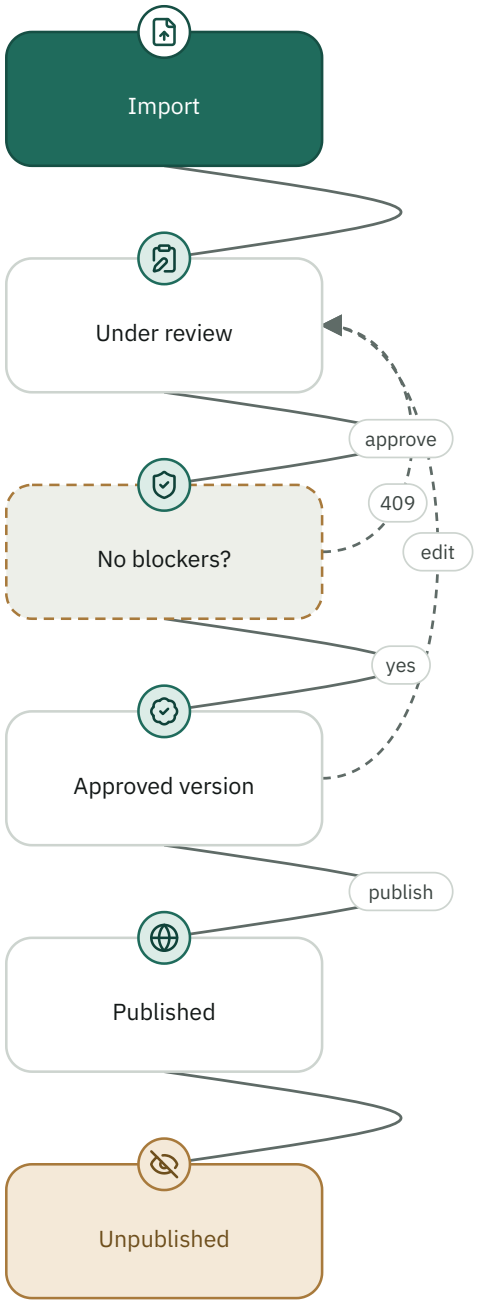
Section-build workflow

The decision in planning, the build on the server, and the write through the shared schedule path.



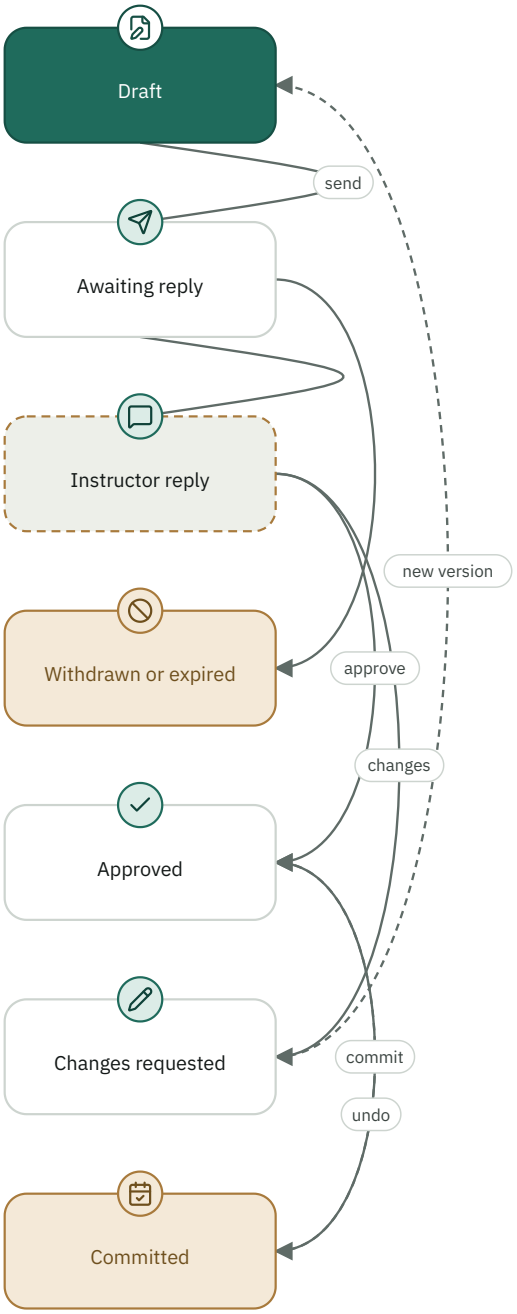
Plan sheet lifecycle

The approved version stays until the next edit is approved, and the public snapshot is independent.



Proposal states

Only the sent version accepts a response, and only the commit writes to the schedule.



Who can do what

	Section planning	Apply sections	Edit sheet rules	Approve sheet	Publish sheet	Plan upload and reference
Root administrator	✎	✎	✎	✎	✎	✎
Schedule committee chair	✎	✎	✎	✎	✎	—
Department head	✎	✎	✎	—	—	—
Department advisor	👁	—	—	—	—	—
Standard user	✎	✎	👁	—	—	—

✎ Write 👁 Read — No access

3.10.4 Screens

Section planning
Remaining-report upload, suggested range, chosen count, printed report.`src/components/SectionPlanning.tsx`

Apply sections
Gap, weekly preview, apply and undo.`src/components/SectionBuild.tsx`

Sections to add
A quiet badge in the approval bar opening a «5 to 7» list.`src/components/SectionGapNote.tsx`

Planning triage
Urgent, important and opportunity with done and ignored decisions.`src/components/PlanTriageView.tsx`

Course offering status
Four states, new and old sheet tabs and the reach verdict.`src/components/CourseOfferingPanel.tsx`

Curriculum plans
Plan creation, course linking, transitions and archiving.`src/components/CurriculumPlans.tsx`

Plan review
Level map, course rules and approval.`src/components/PlanReview.tsx`

Reference sheet
Comparison of the two reference sheets with the catalogue and their application.`src/components/CurriculumReferenceReview.tsx`

Study plan upload
Upload of the registration office's PDF.`src/components/PlanSheetUpload.tsx`

Sheet publishing
Link, QR and PDF for every approved sheet.`src/components/PlanSheetPublisher.tsx`

Study proposal
A schedule with visible differences and tabs for changes, vacancies and checks.`src/components/proposal/StudyProposalWorkspace.tsx`

3.10.5 Requirements

FR-PLN-001 Remaining-report import

The system shall read the registrar's remaining-students report, as a PDF or up to four page images, and show the reading as a preview before saving, without retaining the uploaded file.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A file above 24 MB is refused, and images whose total does not match the declared page sizes, or more than four images, are refused with status 400.
2. A portrait-oriented image is refused with status 422 and code orientation before any reading.
3. At most twenty document-reading requests per minute are accepted.

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/utils/remainingReport.ts` `src/components/SectionPlanning.tsx` `tests/section-planning-import-audit.ts` `tests/section-planning-audit.ts`

FR-PLN-002 Mismatched report refusal

The system shall refuse a report that belongs to a different department, branch or term than the one selected, import nothing from it, and state the refusal kind and the difference between what was read and what was chosen.

Implemented · Must · Functional · Department head, Schedule committee chair

RATIONALE

> Figures from another term would corrupt the plan.

ACCEPTANCE CRITERIA

1. The known refusal kinds are mismatch, wrongTerm, noCode, branchUnread, departmentUnread, noCourses, noColumn and unread, answered with status 422.
2. A report for a term unrelated to the chosen term is refused entirely with kind wrongTerm, not merely warned about.
3. Planning uses only the «not passed» column (notPassed) as its input.

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/utils/remainingReport.ts` `tests/section-planning-import-audit.ts`

FR-PLN-003 Registration statistics save

Inferred

The system shall store, per department and term, the chosen section counts, the remaining, unregistered and vacant figures after sanitising the values, with the user's name and the save time.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. No more than 2000 courses are accepted per map, and the remaining is clamped between 0 and 100000 with negatives stored as zero.
2. Vacant seats accept negatives down to -100000, and the chosen count accepts up to 500 sections.
3. A request outside the allowed departments is answered with status 403.

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/db/repository.ts#saveRegistrationStats` `tests/section-planning-audit.ts`

FR-PLN-004 Section count suggestion

The system shall suggest for each course a range of section counts anchored on the average sections opened in comparable terms (same season), tilted by the remaining figure and bounded by capacity.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The tilt stays within the baseline plus or minus 1, widening to 2 when the baseline is 4 or more.
2. A course never opened before starts at one section and is capped at 6 (NO_HISTORY_MAX) even if remaining divided by capacity is larger.
3. A zero or negative remaining suggests the usual historical count, neither «no sections» nor an increase.

EVIDENCE

`src/utils/sectionCountSuggestion.ts` `src/server/routes/planningRoutes.ts` `tests/section-planning-audit.ts`

FR-PLN-005 Same-season history and course lineage

Inferred

The system shall gather each course's history only from earlier terms of the same season, and let a renumbered or replaced course inherit its predecessor's sections and remaining figures when it was not itself opened.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A first term does not borrow history from a second term, nor the reverse.
2. A transition of kind removed inherits nothing.
3. A past remaining figure saved under a column other than «not passed» is excluded from comparison.

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/utils/sectionCountSuggestion.ts` `tests/section-planning-audit.ts`

FR-PLN-006 Section gap report

The system shall compare the chosen count in planning (otherwise the visible suggestion) with the number of distinct sections (SCode) in the term schedule for each course, and classify the row as done, missing, over or unset.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Required 6 with 5 existing gives remaining 1 and state missing.
2. A course with no chosen figure and no imported remaining has no target and stays unset.
3. The report is read-only and writes nothing to the schedule.

EVIDENCE

`src/utils/sectionGap.ts` `src/utils/sectionBuild.ts` `src/server/core/planning.ts` `src/server/routes/planningRoutes.ts` `src/components/SectionGapNote.tsx` `tests/section-gap-audit.ts` `tests/section-gap-live-audit.ts`

FR-PLN-007 Surplus and conflict review

The system shall present sections above the requirement as a review item, never a closure, and merge opposing opening and reduction estimates for one course into a single «conflicting estimates» item.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A surplus with no explicit choice, no comparable history and no positive remaining is not shown.
2. When the remaining exceeds required times capacity a capacity warning accompanies the surplus item.

EVIDENCE

`src/utils/sectionGap.ts` `tests/section-gap-audit.ts`

FR-PLN-008 Planning triage and decisions

The system shall sort planning items onto three shelves (urgent, important, opportunity) and store a «done» or «ignored» decision per course in the scope under the decider's name, lapsing the decision when the item's figures change.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A course with no section, or two or more missing, is urgent; one missing is important; a surplus is an opportunity.
2. An «ignored» decision without a reason is refused with status 400, and the reason is truncated at 140 characters.
3. Reading requires permission 7 or 14 and writing requires permission 7 only.

EVIDENCE

`src/utils/planTriage.ts` `src/server/routes/planningRoutes.ts` `src/components/PlanTriageView.tsx` `tests/decision-surfaces-audit.ts` `tests/section-gap-audit.ts`

FR-PLN-009 Course offering states

The system shall let the department set each course's offering state for the term: offered (default), closed, future sheet not yet reached, or limited sections with a maximum count.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A course with no entry is offered, and an unconfirmed suggestion changes nothing.
2. The maximum for limited sections lies between 0 and 99.
3. A limited course shows «remaining» = maximum minus existing, and becomes unselectable at zero.

EVIDENCE

`src/utils/courseOffering.ts` `src/server/routes/offeringRoutes.ts` `src/components/CourseOfferingPanel.tsx` `tests/owner-features-2026-10-28-audit.ts` `tests/instructor-offering-flow-audit.ts` `tests/prepublish-reach-audit.tsx`

FR-PLN-010 Offering effect on suggestions

The system shall not suggest new sections for a closed course or one from a sheet whose cohort has not reached its level, and shall trim a limited course to what remains of its ceiling.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. States closed and future yield zero new sections.
2. A limited course with ceiling 3 and 2 existing allows one new section.
3. Offering states are read through a 20-second cache that is cleared on every save.

EVIDENCE

`src/utils/courseOffering.ts` `src/server/offeringStates.ts` `src/server/routes/planningRoutes.ts` `tests/instructor-offering-flow-audit.ts` `tests/section-build-audit.ts`

FR-PLN-011 Not-yet-reached verdict

The system shall judge for each course, branch and sheet whether the student cohort has reached its level, from the sheet's rules and the branch's own history only, and shall not lock a course without evidence.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A course with no prerequisites opens when its level exceeds the highest level offered from the sheet in the branch by at most one level.
2. A branch without history, or a course with neither level nor prerequisites, stays «unknown» and shows without a tag.
3. An «open for offering» decision opens the course whatever the rule says.

EVIDENCE

`src/utils/courseReach.ts` `src/server/courseReachShared.ts` `src/components/CourseReach.tsx` `src/components/PrepublishReachSheet.tsx` `tests/course-reach-audit.ts` `tests/prepublish-reach-audit.tsx`

FR-PLN-012 Section-build preview

The system shall build a preview of the missing sections with the same opening engine: a conflict-free time, a documented hall of sufficient capacity, and an instructor who taught the course and has available load, hardest courses first, stating the reason for every section not built.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Courses of one level do not overlap within the batch.
2. The recorded reasons are: no documented hall large enough, no instructor who taught it, loads full or no recorded allowance, no conflict-free time.
3. A department with no appointment in the term counts as «not started» and no schedule is invented.
4. The chosen count is accepted between 0 and 40 per course in the build.

EVIDENCE

`src/utils/sectionBuild.ts` `src/utils/sectionOpening.ts` `src/server/routes/planningRoutes.ts` `src/components/SectionBuild.tsx` `tests/section-build-audit.ts` `tests/section-opening-feasibility-audit.ts` `tests/section-build-live-audit.ts`

FR-PLN-013 Pending-room sections from history

The system shall, when no hall and instructor are available, place the section at a time taken from the comparable term (first with first), with a «pending room» hall and the «faculty» placeholder instructor, without stopping the build.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Two pending sections of one course never fall at the same time.
2. The proposed hall comes from the department's documented halls, most used first.

EVIDENCE

`src/utils/sectionPending.ts` `src/server/routes/planningRoutes.ts` `tests/section-build-audit.ts`

FR-PLN-014 Confirmed section apply

The system shall recompute the build on the server at apply time so that no time is accepted from the browser, and write all picked sections or none through the shared schedule path.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The browser sends only «course:section» keys, up to 200 keys.
2. A differing week fingerprint yields 409 with code stale and nothing is written.
3. Picking a pending section yields 409 with code needs-review.
4. A failure of one section reverts those written before it, and the batch is recorded with an id beginning plan: and origin «section planning».

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/utils/sectionBuild.ts` `tests/section-build-live-audit.ts`

FR-PLN-015 Apply undo

The system shall allow undoing one apply batch by deleting only its rows, and refuse if any row changed after it was applied.

Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A batch id not starting with plan: is refused with status 400.
2. A row whose revision exceeds 1 yields 409 with code changed and nothing is deleted.
3. A batch with no rows in the schedule yields 404.

EVIDENCE

`src/server/routes/planningRoutes.ts` `tests/section-build-live-audit.ts`

FR-PLN-016 Plan sheet course rules

The system shall store, for each course in a plan sheet, its level (100 to 400), kind (compulsory, college, university, elective), prerequisites and corequisites, and for elective groups a minimum number of courses and units.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The legacy level values 1 to 4 are converted to 100 to 400.
2. A prerequisite accepts «all», «any of» and «minimum units» up to 200, and a group up to 20 courses and 60 units.
3. A course in two groups stays in the first when sanitised.

EVIDENCE

`src/utils/curriculumRules.ts` `src/utils/curriculumRulesImport.ts` `tests/plan-review-audit.ts`

FR-PLN-017 Plan sheet rule checks

The system shall check the plan sheet rules and classify findings as approval-blocking or warnings.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Blockers: a prerequisite cycle, an unknown reference, a course not in the sheet, a group that cannot meet its minimum courses or units, and a course in two groups.
2. Warnings: a prerequisite at a later level, an elective without a group, an empty group or one without a minimum.

EVIDENCE

`src/utils/curriculumRules.ts` `tests/plan-review-audit.ts`

FR-PLN-018 Draft editing with department guard

The system shall let a department edit only its own courses' rules in the plan sheet draft, and refuse on the server any edit of another department's course.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Editing another department's course yields 403 with code foreign-course.
2. The details editor accepts integer units 0 to 20, hours 0 to 40 in half-hour steps and capacity 0 to 200, and rejects any other field.
3. Every edit returns the draft to «under review» based on the approved version.

EVIDENCE

`src/server/planReviewRoutes.ts` `src/components/PlanReview.tsx` `tests/plan-review-audit.ts`

FR-PLN-019 Plan sheet approval

The system shall let the schedule committee chair (and system administration) approve a draft only on the version they saw, and move it to the approved store with a new version in one transaction.

Implemented · Must · Functional · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. A draft changed after viewing yields 409 with code stale.
2. A blocker yields 409 with code blockers and the list of blockers.
3. A non-chair yields 403 with code not-approver.
4. When the sheet is published its public snapshot is refreshed on approval.

EVIDENCE

`src/server/planReviewRoutes.ts` `tests/plan-review-audit.ts`

FR-PLN-020 Study plan PDF upload

The system shall read the admissions and registration office's study-plan PDF, apply only the owning department's courses by number, correct name, units and hours without touching capacity, for the main administrator only.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. The file limit is 2 MiB, and a non-PDF is refused with status 422.
2. A sheet combining several tracks is refused with kind combined, and a sheet of another department is refused before any write.
3. Re-uploading the same file changes nothing.
4. The course key is seven digits: «105 09 01» becomes 0109105.

EVIDENCE

`src/server/planReviewRoutes.ts` `src/utils/planPdfParser.ts` `src/server/planPdfExtract.ts` `src/utils/planSheetImport.ts` `src/server/sheetReconcile.ts` `src/components/PlanSheetUpload.tsx` `tests/plan-sheet-import-route-audit.ts` `tests/plan-pdf-audit.ts` `tests/plan-sheet-import-audit.ts`

FR-PLN-021 Reference sheets and their application

The system shall carry the two reference sheets of the educational technology department (old 130 units, new 125 units) and apply them to the catalogue with a default dry run before writing, for the main administrator only.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A request without dryRun=false previews and writes nothing.
2. A frozen name in transitional sheets changes only with confirmIdentity, while units and hours are always corrected.
3. The created rules are a draft never approved, and re-applying to a matching catalogue changes nothing.

EVIDENCE

`src/data/technologyEducationCurricula.ts` `src/data/marineMechanicalDiploma.ts` `src/utils/curriculumReference.ts` `src/utils/referenceApply.ts` `src/server/planReviewRoutes.ts` `src/components/CurriculumReferenceReview.tsx` `tests/reference-apply-audit.ts` `tests/curriculum-reference-pdf-audit.ts` `tests/curriculum-reference-audit.ts`

FR-PLN-022 Sheet membership and badges

The system shall derive each course's «sheet year» tag and its new and old tabs from one membership source, so tag and tab never disagree on any screen.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The tag appears only for a course in the new sheet alone, and the pair exists only for a department holding both sheets.
2. After «ended» all badges disappear and return when a newer sheet arrives.
3. Badges respect the account's scope, so no badge is returned for a course it cannot see.

EVIDENCE

`src/utils/sheetMembership.ts` `src/utils/sheetTag.ts` `src/utils/planBadges.ts` `src/server/planBadgeShared.ts` `src/server/planReviewRoutes.ts` `src/utils/courseTags.ts` `tests/sheet-membership-audit.ts` `tests/consistent-badges-audit.tsx`

FR-PLN-023 Department plans and transitions

The system shall allow creating a department plan, renaming it, linking its courses, recording transition relations between two plans, and checking archive readiness before archiving.

Implemented · Must · Functional · Department head, Root administrator

ACCEPTANCE CRITERIA

1. Transition kinds are renumbered, renamed, renumbered-renamed, replaced and removed, and all except removed require a counterpart course.
2. Removing a course from its last live plan while offered in the newest term is refused with status 409.
3. Archiving a plan still used in the newest term is refused with status 409 unless force=1 is sent.
4. Adding courses accepts up to 300 courses per request, all from the plan's department.

EVIDENCE

`src/server/routes/coursesRoutes.ts` `src/components/CurriculumPlans.tsx` `tests/curriculum-transition-audit.ts`

FR-PLN-024 Public sheet publishing

The system shall publish an approved sheet, for the schedule committee chair (and system administration) within their scope, as an independent snapshot, copy what is published for female students to its sub-sites, and allow unpublishing.

Implemented · Must · Functional · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. Publishing an unapproved sheet yields 409, and a non-chair yields 403.
2. Clearing the plan after approval does not remove the published page, while unpublishing removes it and its copies.
3. A copy carries mirrorOf, and a site with no matching sheet is reported and not guessed.
4. Publishing is limited to 30 operations per minute.

EVIDENCE

`src/server/planSheetRoutes.ts` `src/components/PlanSheetPublisher.tsx` `src/utils/collegeTrack.ts` `tests/plan-sheet-audit.ts`

IR-PLN-001 Public plan pages

The system shall serve published sheets without sign-in at /plans/, each department at /plans/s<department>/ and each sheet at /plans/s<department>/<sheet>/, with a QR code that always carries the official domain and a PDF download.

Implemented · Must · Interface · Public visitor, Student

ACCEPTANCE CRITERIA

1. Pages send X-Robots-Tag: noindex and are limited to 120 requests per minute.
2. A department with one sheet redirects with 302 to it, and an unpublished sheet yields 404.
3. The page shows the department's own courses only, with external requirements as notes rather than cards.
4. The /plans/ index does not repeat the sub-sites, and the static educational technology page remains until that department is published.

EVIDENCE

`src/server/planSheetRoutes.ts` `src/server/planSheetPage.ts` `src/server/planSheetReferenceLayout.ts` `public/plans/_template/sheet.html` `tests/plan-sheet-audit.ts`

FR-PLN-025 Study proposal drafting

The system shall let the department prepare a proposal of several items for an instructor who has a request in the inbox, with four operations: assign a section, new section, edit an appointment and replace, kept as a draft independent of the live schedule.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. At most 12 items per proposal, 60 proposals per instructor, a 1200-character message and a 600-character reply note.
2. A frozen term or locked schedule yields 409 with code term-frozen or schedule-locked.
3. A reference appointment cannot be used in two items, and a duplicated new section inside the proposal is rejected.

EVIDENCE

`src/server/studyProposalRoutes.ts` `src/utils/studyProposal.ts` `src/components/proposal/StudyProposalWorkspace.tsx` `tests/study-proposal-audit.ts` `tests/study-proposal-flow-audit.ts`

FR-PLN-026 Proposal check with the schedule checker

The system shall check a proposal by applying its operations to an in-memory copy of the term schedule and asking the same conflict checker that governs saving an appointment, classifying results as blockers, review notes and information.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Sending is refused with 409 and code not-ready while readiness.canSend is false.
2. An instructor appointment in a department outside the user's scope appears as occupancy only, with no course or section.
3. Sending reserves no section and no hall.

EVIDENCE

`src/server/studyProposalEngine.ts` `src/utils/studyProposal.ts` `src/server/studyProposalRoutes.ts` `tests/study-proposal-flow-audit.ts` `tests/study-proposal-audit.ts`

FR-PLN-027 Proposal versions and instructor response

The system shall number proposal versions and accept the instructor's response (approval or change request, independent per item or linked) on the current sent version only, signed with the civil number without storing it.

Implemented · Must · Functional · Faculty member, Department head

ACCEPTANCE CRITERIA

1. Every material change to what was sent raises the version and returns the status to draft.
2. A response to an older version yields 409 with code stale-version, and after expiry 409 with code closed.
3. The default validity is 7 days, between 1 and 30.
4. A change request needs a short reason or a note.
5. A duplicate response with identical content is not recorded twice.

EVIDENCE

`src/server/studyProposalRoutes.ts` `src/utils/studyProposal.ts` `src/server/studyProposalPage.ts` `tests/study-proposal-audit.ts` `tests/study-proposal-flow-audit.ts` `tests/study-proposal-public-page-audit.ts`

FR-PLN-028 Proposal commit and undo

The system shall commit a proposal to the schedule in one atomic transaction after the instructor's approval and re-check, and allow one complete undo unless an appointment changed after the commit.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A request without confirm=true is a dry run that states what would happen without writing.
2. The linked mode commits only with approval of the whole group.
3. A change to an appointment after commit yields 409 with code proposal-stale on undo and nothing is written.
4. Re-committing a committed proposal returns alreadyCommitted and does not write twice.

EVIDENCE

`src/server/studyProposalRoutes.ts` `src/utils/studyProposal.ts` `tests/study-proposal-flow-audit.ts`

FR-PLN-029 Vacancy ranking for the instructor

The system shall rank vacant sections (the «faculty» placeholder) of the department's own courses with a verdict per section, excluding those overlapping the instructor's lectures and then running the full proposal check with the assignment added.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Verdicts: ready, review, blocked, clash, used and unchecked.
2. The full check covers the nearest 24 sections per call and the rest are «unchecked».

EVIDENCE

`src/server/studyProposalEngine.ts` `src/utils/proposalVacancyScope.ts` `src/components/proposal/ProposalVacancies.tsx` `tests/study-proposal-studio-audit.ts`

IR-PLN-002 Instructor proposal page and alert

The system shall show the instructor, through the personal link, the proposal page (current and proposed schedule) and inject a «you have a study proposal» alert in the request page and card, without the page ever claiming the schedule changed before the department commits.

Implemented · Must · Interface · Faculty member

ACCEPTANCE CRITERIA

1. The alert reads up to 8 links and hides proposals whose gate is closed or that are committed.
2. A failure fetching the alert is silent and does not break the host page.

EVIDENCE

`src/server/studyProposalPage.ts` `src/server/studyProposalAlert.ts` `src/server/routes/publicRequestRoutes.ts#L1571` `tests/study-proposal-public-page-audit.ts`

NFR-PLN-001 Department scope and permission

The system shall restrict every planning route to the college and department scope allowed to the user and to the proper permission, and prevent read-only roles from writing.

Implemented · Must · Non-functional · Department head, Schedule committee chair, Department advisor, Standard user

ACCEPTANCE CRITERIA

1. A query outside the scope yields 403 «outside your allowed departments».
2. Writing in planning and build requires permission 7, while reading accepts 7 or 14.
3. The plan review is visible to the department head, committee chair and standard user within their scope, and always to system administration.

EVIDENCE

`src/server/routes/planningRoutes.ts` `src/server/planReviewRoutes.ts` `src/server/planSheetRoutes.ts` `src/server/routes/offeringRoutes.ts` `tests/plan-review-audit.ts` `tests/section-build-live-audit.ts`

NFR-PLN-002 Seven-digit course numbers and printed design

The system shall show course numbers in sheet and planning pages as seven digits (college 2, department 2, course 3) and correct values rather than the layout of printed reports.

Implemented · Must · Non-functional · Department head

ACCEPTANCE CRITERIA

1. A short code is expanded with the owning department code when matched to the reference sheet, becoming 0109105 not 105.
2. A two-digit number (99) is kept in both forms 013099 and 0130099 when reading the sheet.

EVIDENCE

`src/utils/curriculumReference.ts` `src/utils/planPdfParser.ts` `src/server/planSheetRoutes.ts` `tests/curriculum-reference-pdf-audit.ts` `tests/plan-pdf-audit.ts`

DR-PLN-001 Planning data stores

Inferred

The system shall persist planning data in separate stores: registration statistics per (college, department, term), offering states, triage decisions, rule drafts and versioned approved rules, published sheets, and study proposals with a numeric revision.

Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. The registration statistics key and the offering document key both take the form «college:department:term».
2. A concurrent write to a proposal with a stale revision yields 409.

EVIDENCE

`src/db/repository.ts` `src/utils/courseOffering.ts` `src/types.ts` `tests/study-proposal-flow-audit.ts`

3.10.6 Algorithms

ALG-PLN-01 · Section planning and curricula

Statistics from history

Section-count suggestion from same-season history

How many sections should the department open for each course this term, as a range anchored on what it actually opened in comparable terms?

1. History is gathered only from earlier terms of the same season (first with first, second with second, summer with summer); no season borrows another's history.
2. A renumbered or replaced course that has not itself been opened inherits its predecessors' sections and remaining figures; predecessors are found by a breadth-first walk over the recorded course transitions.
3. Baseline = plain average of the course's sections in those terms (each term weight 1), rounded, never below one.
4. Tilt: the current «not registered» figure is compared with its own historical average, the baseline is scaled by that ratio, and the change is clamped to ±1 (±2 when the baseline is 4 or more).
5. An upward tilt never exceeds ⌈remaining ÷ capacity⌉; if that fill is below the baseline, the lower end of the range drops to it and the reason says so.
6. A course with no history gets a range from 1 to min(⌈remaining ÷ capacity⌉, 6), starting at one section; a zero or negative remaining returns the usual historical count.

Inputs

Imported registrar «not registered» figures per course, course capacity (MaxStudent), timetables of same-season terms, and course transitions between plan sheets.

Output

Per course: a suggested value, a min–max range, its basis (history, pool, none, empty), a one-line reason and an optional backlog warning; plus a department warning when the total exceeds the department's historical peak.

Deterministic and read-only: nothing is written to the timetable, and a course with neither history nor a recorded capacity gets no invented number.

Tilt band ±1, widened to ±2 when the baseline is 4 or more

Cap for a course without history: NO_HISTORY_MAX = 6 sections

A difference under 5% from the usual is reported as «within the usual»

Backlog warning when the remaining figure rises over three consecutive comparable terms

`src/utils/sectionCountSuggestion.ts#suggestSectionCount` `src/utils/sectionCountSuggestion.ts#historicalBaseline` `src/utils/sectionCountSuggestion.ts#departmentLoadWarning` `src/server/routes/planningRoutes.ts#registerPlanningRoutes` `tests/section-planning-audit.ts`

FR-PLN-004 FR-PLN-005

ALG-PLN-02 · Section planning and curricula

Deterministic

Section gap and surplus review

Which courses lack sections against the plan, which exceed it, and where do the opening and reduction estimates contradict each other?

1. Target per course = the department's written chosen count when present, otherwise the visible suggestion from ALG-PLN-01.
2. Existing = number of distinct section codes (SCode) for the course in the term timetable; the state is unset, missing, over or done.
3. A surplus is listed only with evidence: a written choice, or comparable history with a positive remaining figure; it is flagged when remaining exceeds target × capacity.
4. A course with both an opening need and a surplus is merged into one neutral «conflicting estimates» review item.
5. Student wishes form a separate signal never mixed with the plan: ⌈students ÷ capacity⌉ – existing.

Inputs

Chosen counts, remaining figures and comparable-term history, the term's sections, course capacity, and student wishes.

Output

A list ordered by greatest need of missing, surplus and conflict items, each with its basis and the demand figure behind it.

Pure function that reads and writes nothing; a surplus is a review item and never closes or deletes a section.

Ordering: largest gap, then largest demand, then course number

`src/utils/sectionGap.ts#planTargets` `src/utils/sectionGap.ts#planSurpluses` `src/utils/sectionGap.ts#reconcilePlanAdjustments` `src/utils/sectionGap.ts#studentGaps` `src/utils/sectionBuild.ts#gapRows` `src/server/core/planning.ts#sectionGapData` `tests/section-gap-audit.ts` `tests/section-gap-live-audit.ts`

FR-PLN-006 FR-PLN-007

ALG-PLN-03 · Section planning and curricula

Search and ranking

Placement search for new sections

Where can each missing section be placed — day, start time, hall and instructor — without colliding with anything?

1. Shortfall = demand – open sections × capacity, and needed sections = ⌈shortfall ÷ capacity⌉; in the build preview demand is set to target × capacity so the needed count equals the gap exactly.
2. Courses are ordered hardest first: the largest same-level cohort, then the most sections remaining.
3. Candidates: each weekday × the department's learned start ladder × verified, confirmed halls the department has taught in (most used first, up to 24, large enough when capacity is known) × active instructors who taught the course and remain within their load cap.
4. A candidate is accepted only if physically free (no overlapping hall or instructor, with no exemption) and the conflict checker finds nothing, including overlaps with the same cohort's courses and the department's own turnaround minutes.
5. Each accepted placement joins the working week before the next search, so two new sections never share a hall-hour; the section number continues the course's own sequence within 501–999.
6. Lecture length is the most common length of the existing section, else the last term the course was taught, else the department rhythm; no length is invented.

Inputs

The department's week and the whole term week, courses, instructors and their loads, history, the demand reading with cohort pairs, the department rhythm and the verified hall registry.

Output

Per course: proposed placements with a plain sentence, the reason for any section not placed, and a list of courses with no recorded capacity.

Read-only: every placement remains a proposal until a person opens and saves it through the normal appointment path.

Section codes 501–999; a used code is never reissued

Up to 24 candidate halls per course

`src/utils/sectionOpening.ts#readSectionOpenings` `src/utils/sectionOpening.ts#nextSectionCodes` `src/utils/sectionBuild.ts#hardestFirst` `src/utils/sectionBuild.ts#demandFromTargets` `tests/section-opening-feasibility-audit.ts` `tests/section-build-audit.ts`

FR-PLN-012

ALG-PLN-04 · Section planning and curricula

Not-yet-reached course verdict

Deterministic

Has this branch's cohort on this plan sheet reached the course's level, so that offering it this term makes sense?

1. Only the branch's own history before the current term is read, and a new sheet from its start year; a branch with no history makes every course «unknown» and locks nothing.
2. Offered before → reached. Otherwise, with in-sheet prerequisites, it is reached when every «all» prerequisite and at least one of each «any-of» group was offered in an earlier term (summer counts).
3. Without prerequisites, it is reached when its level ≤ highest level offered + 100; a new sheet's first term opens only 100-level courses without prerequisites.
4. Co-requisites are mutual: a course whose twin is locked is locked too, propagated along the chain until nothing changes (fixed point).
5. The department's «open for offering» override wins; across several sheets, reached beats unknown, which beats locked.

Inputs

Sheet membership and rules (prerequisites, co-requisites, level), courses offered in each earlier term of the branch, and manual open overrides.

Output

Per course: reached / locked / unknown, the reason code, missing prerequisites, and a chain of at most four links when locked.

No course is locked without complete evidence and no radar data is used; the same rule serves the faculty card and the pre-publish sheet.

Level margin: highest offered level + 100 · Locked chain: at most four links

`src/utils/courseReach.ts#sheetReach` `src/utils/courseReach.ts#combineReach` `src/utils/courseReach.ts#classifySheets` `src/utils/prepublishReach.ts` `tests/course-reach-audit.ts` `tests/prepublish-reach-audit.tsx`

FR-PLN-011 · FR-REQ-007 · FR-REQ-008

ALG-PLN-05 · Section planning and curricula

Plan-sheet rule check and reconciliation

Deterministic

Are the sheet's rules fit for approval, and how are the department's courses reconciled with what the sheet file prints?

1. Each rule is checked: the course is on the sheet, its references are known, a prerequisite is not at the same or a later level, and an elective has a group.
2. Prerequisite cycles are detected by depth-first search with three-state marking over the prerequisite graph; any cycle blocks approval.
3. Elective groups: a course in two groups, or a minimum of courses or units above what the group offers, blocks; a shortfall caused entirely by courses not yet in the catalogue is only a warning.
4. Reconciliation takes the sheet as the source: a course follows the department its code points to (with fixed aliases) and is created there if absent, the record the department actually taught is preferred, and name, units and hours come from the sheet; a spelling-only difference does not rename.
5. Reconciliation can run as a dry run that counts writes without performing them, and uploaded rules remain a draft for the department to approve.

Inputs

Sheet rules and groups, sheet courses, the catalogue and its credits, what the college has taught historically, and the sheet parsed from its file.

Output

Findings classified as block or warn, and a reconciled catalogue and rule set (draft or approved).

Membership, groups and prerequisites exactly as printed, nothing added or removed; a number held by more than one record is left untouched.

Sheet levels 100, 200, 300 and 400

`src/utils/curriculumRules.ts#checkRuleSet` `src/utils/curriculumRules.ts#findPrereqCycle` `src/server/sheetReconcile.ts#createReconciler` `tests/plan-review-audit.ts` `tests/plan-sheet-import-route-audit.ts`

FR-PLN-017 · FR-IMP-034

ALG-PLN-06 · Section planning and curricula

Study proposal check, alternatives and vacancy ranking

Search and ranking

Is the instructor's proposal valid as it stands, what is the closest alternative time or hall for a conflicting item, and which vacant sections can be assigned to the instructor without conflict?

1. The proposal's operations are applied to an in-memory copy of the term timetable; an operation whose snapshot no longer matches the live timetable becomes a blocker.
2. Every appointment entering the instructor's week is judged by the same save gate as the add-appointment screen, and findings are classified blocker, review or info.
3. Alternatives for a conflicting item: starts every 10 minutes within the department's day window, day sets from the same pattern family, eligible halls; rank = |shift in minutes| + 90 for a day change + 14 for a hall change + 35 for a different hall type + 45 for an undersized hall.
4. Cheap filters run first (busy instructor or hall), then a full check of the whole proposal; an alternative that blocks any other item is rejected, and survivors are sorted by review notes, then by change in gaps.
5. Vacant «faculty» sections: those overlapping the instructor's lectures are excluded, the rest get a full check with the assignment added, and are ranked ready, review, unchecked, blocked, clash, used.

Inputs

Proposal operations, the whole term timetable and the person's records in every college, the hall registry and hall rights, and the department's scope and day window.

Output

Per-item findings and send readiness, up to four alternatives with reasons, and a ranked verdict per vacant section.

No second checker: the save gate decides; the search says plainly when it was partial, and writes nothing.

Start-time step: 10 minutes · At most 100 full checks and at most 4 alternatives · 24 vacant sections fully checked by default

`src/server/studyProposalEngine.ts#evaluateProposal` `src/server/studyProposalEngine.ts#suggestAlternatives` `src/server/studyProposalEngine.ts#rankVacancies` `tests/study-proposal-flow-audit.ts` `tests/study-proposal-studio-audit.ts`

FR-PLN-026 FR-PLN-029

3.10.7 Edge cases

⚠ Schedule changed between preview and apply

Scenario The department previews sections and a colleague adds an appointment before clicking apply.

Behaviour The week fingerprint differs, so the request is refused with 409 and code stale, nothing is written, and the preview is rerun.

⚠ Section without a confirmed hall or instructor

Scenario No documented hall of sufficient capacity and no instructor with allowance for a missing course.

Behaviour The system places a time from the comparable term with a «pending room»; direct apply refuses it with 409 and code needs-review and it is opened in «review and edit».

⚠ Negative remaining in the report

Scenario The report prints «not passed» as negative because enrolled students exceed those who have not passed.

Behaviour The remaining is stored as zero and does not mean «no sections»: the usual historical count is suggested, or zero with an explanation when no history exists.

⚠ Editing an approved sheet that is published

Scenario A department edits a course in an approved, published sheet.

Behaviour The draft returns to «under review» while the approved version stays; the public page keeps its snapshot until the edit is approved, then refreshes.

⚠ A course in two elective groups

Scenario A user places the same course in two groups.

Behaviour Single approval is blocked with 409 and code blockers, while approve-all keeps it in the first group and flags it as repaired.

⚠ Approval of an outdated proposal version

Scenario The department edits what was sent while the instructor opens the previous version and approves.

Behaviour The reply is refused with 409 and code stale-version and is not credited to the newer version, and the page tells the instructor to reload.

3.10.8 Standing owner decisions

- A department sees only its own courses in the sheet map and counters; another department's course is a note, not a card, and is not editable by other departments.
- A course number is always seven digits: college 2, department 2, course 3, such as 0109105.
- The design of the published sheet and printed reports is not changed; differences are fixed in data, not layout.
- Female students have one sheet copied as-is to the two sub-sites, male students have their own, and each college family keeps its independent sheet.
- Only what is approved is published, and the QR code always carries the official domain.

↑ Contents

3.11 · STU

Student and advising services

Masari for students, the registration sheet, section requests, seats and the registration rehearsal

36	100%	58	55
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.11.1 Overview

The module covers the permanent student door «Masari» (a profile with a personal code, wishes within their phase, request status), the registration sheet on which the department committee and then registration decide, section requests from advising to the schedule committee and then registration, the advisor seat balance, and seats (account chairs) with their teams. It includes the registration radar (rehearsal) that estimates registration-day crowding from student profiles and schedules. Everything the student sees is an estimate, not a reservation or an official registration, and student identity is sealed and shown only to those working on the case.

Student · Department advisor · Schedule committee chair · Department head · Registration officer · College registrar · University registrar · System administrator · Root administrator · College dean · Public visitor · The system (automatic)

3.11.2 User journeys

Student — **From civil ID to my suggested schedule**

1. Opens /me and enters the civil ID, learning that no profile exists.
2. Consents to storing the profile and chooses a 6-digit non-sequential code.
3. Chooses department and plan and marks what is done; the system drops what lost its prerequisite.
4. Reads the «suggestions only» card and presses «understood».
5. In the wishes phase writes units (12 up to the department cap), courses and time constraints.
6. After the freeze sees the ticket with three plans and window, without a time for a section whose schedule is unsigned.

Department advisor — A section request from advising to registration

1. Chooses course, campus and term and sees students in need from Masari, or pastes a list.
2. Sets the preferred period and count and sends the request to the course department's schedule committee.
3. The committee opens the section in the slot editor or links an existing one, writing its capacity into the seat balance.
4. The committee approves all or some students; advising receives «approved» and registration receives what awaits.
5. Registration marks who was registered, and advising sees «registered n of m».

Schedule committee chair — Committee then registration decision on a student request

1. Opens the registration sheet for the department and sees its students' cases and other departments' courses read-only.
2. Approves a course, choosing «approve + open section» when a new section is proved conflict-free.
3. Chooses for what it does not approve a reason from the defined set.
4. After the department schedule is signed, the registration officer sees what the committee handed over and says «registered» or «rejected» with a reason.
5. The student reads the decision with its reason in «my request status» under the case number.

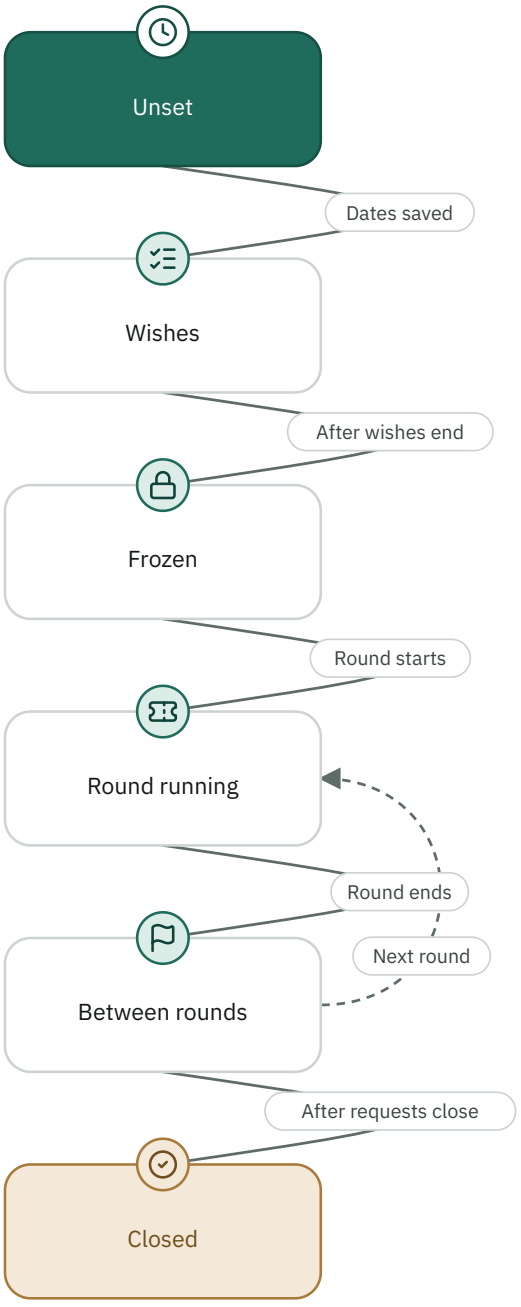
System administrator — Handing a seat to a new holder

1. Opens «Seats» and chooses the seat.
2. Records the new holder (from department faculty or manually) and hands over the seat.
3. The first password shows once, open devices are signed out and the team is frozen.
4. The new holder marks each member «continuing» or «ended».

3.11.3 Diagrams

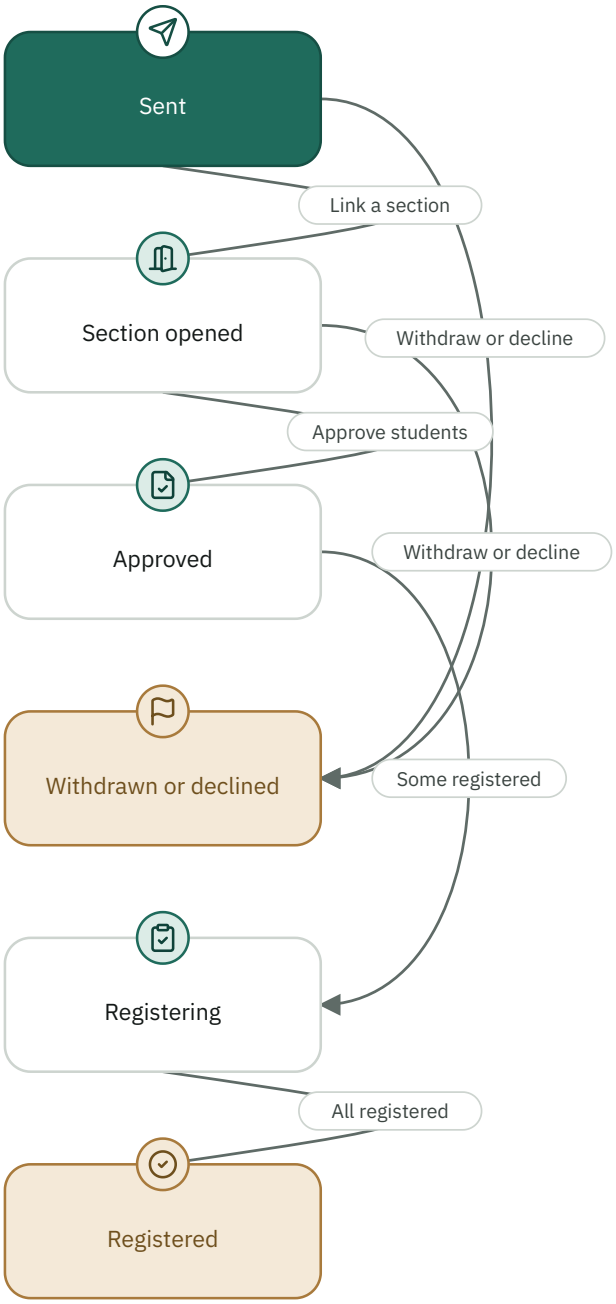
Student-journey phases in a term

The phase is read from today's date in Kuwait, each date inclusive of its whole day.



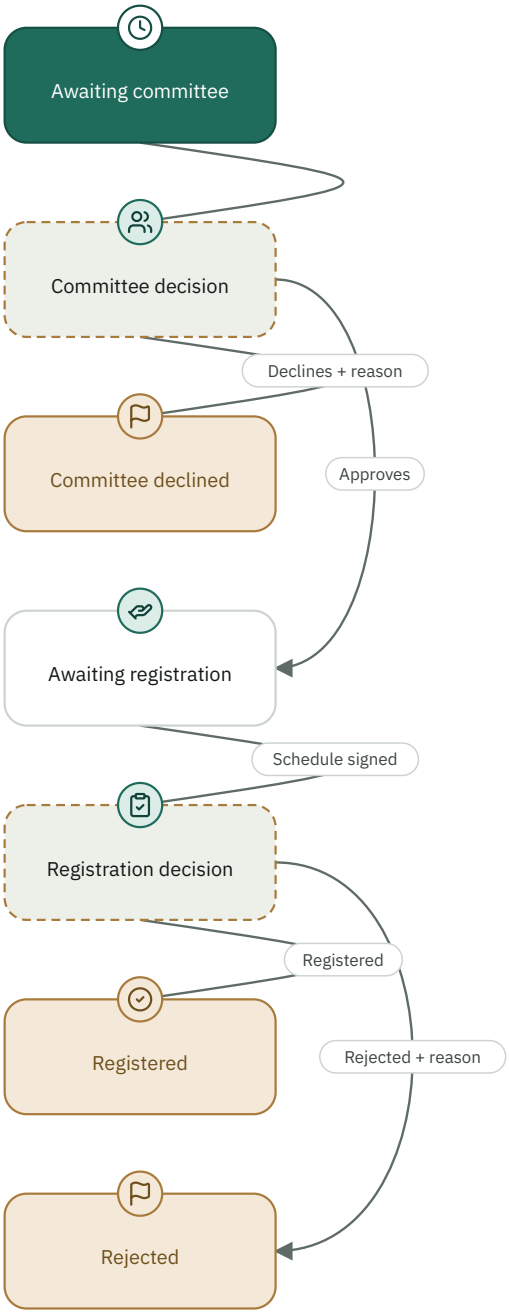
Section request stages

The stage is always derived from the document; approval closes linking, unlinking, withdrawal and decline.



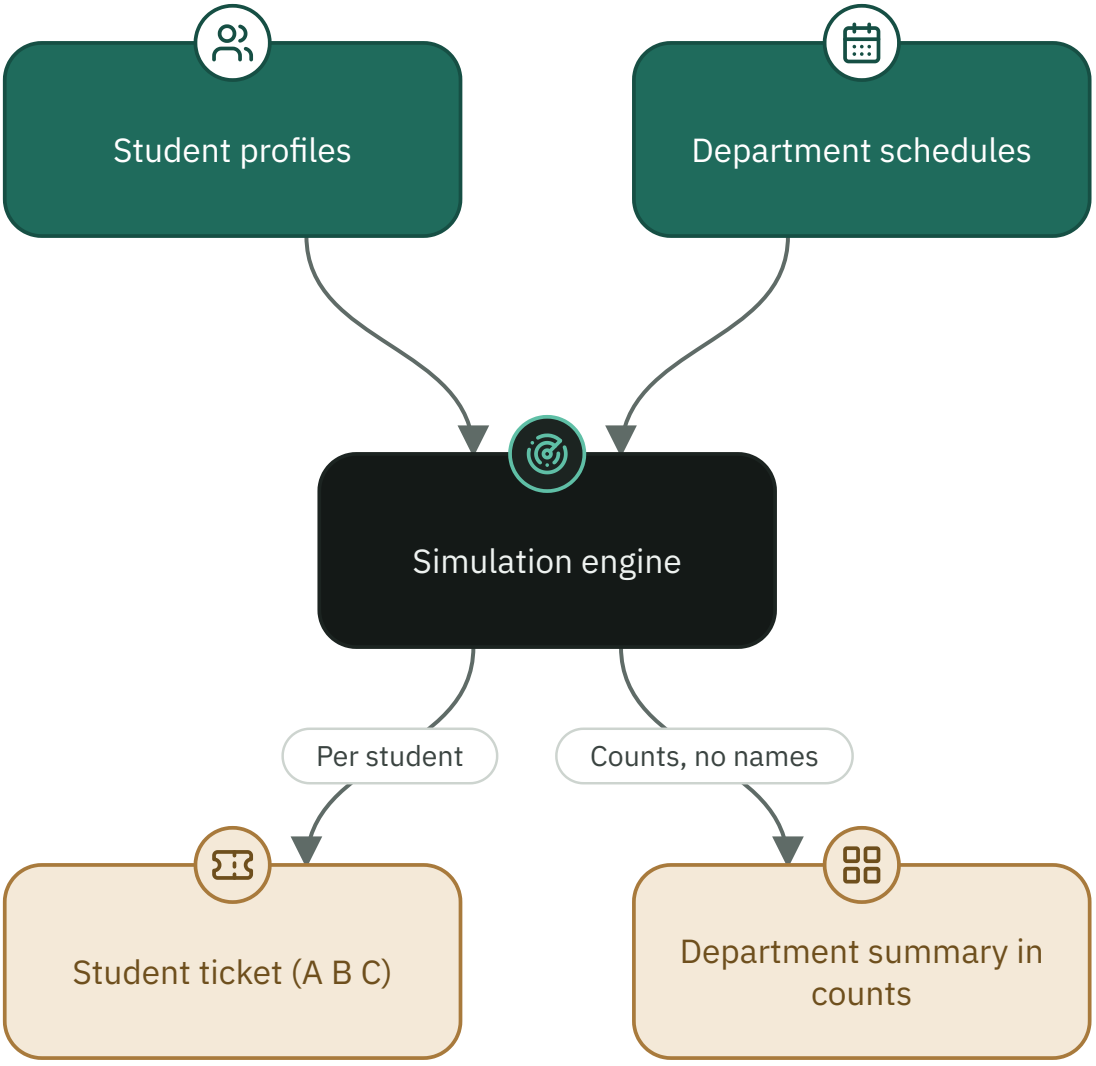
Case decision: committee then registration

Registration never writes on what the committee has not approved, and the committee never overturns registration.



Registration rehearsal

A deterministic estimate from profiles and schedules; no seat reservation and no official data.



Who does what

	Masari	Registration sheet (committee decision)	Registration sheet (registration decision)	Section request	Seat balance	Seats	Radar
Student	✎	—	—	—	—	—	👁
Department advisor	—	👁	—	✎	✎	—	👁
Schedule committee chair	—	✎	—	✎	👁	👁	👁
Department head	—	👁	—	👁	✎	👁	👁
Registration officer	—	—	✎	✎	—	👁	👁
College registrar	—	—	✎	✎	—	✎	👁
University registrar	—	—	✎	✎	—	✎	👁
College dean	—	👁	👁	👁	—	—	👁
System administrator	—	✎	✎	✎	✎	✎	✎

✎ Write 👁 Read — No access

3.11.4 Screens

Masari (/me)
The student door: entry, setup, wishes, «my plan», «did you register?», my cases.`src/server/studentHomePage.ts`

My map
The whole-plan board, computed identically to the server.`src/server/studentMapUi.ts`

My request status
By civil ID and case number.`src/server/pages/studentCaseStatusPage.ts`

Public student schedule
The approved department schedule or «provisional».`src/server/pages/studentSchedulePage.ts`

Registration sheet
One screen for the committee and registration.`src/components/StudentRegistration.tsx`

Section requests
For advising, the committee and registration.`src/components/sectionRequests/SectionRequestsPanel.tsx`

Advisor seat balance

Seats

A ring per section, atomic addition and exception request.`src/components/AdvisorSeatsPanel.tsx`

Managing seats, teams and access cards.`src/components/seats/SeatsAdmin.tsx`

Registration radar

Registration-day rehearsal in counts, not names.`src/components/RegistrationRehearsal.tsx`

Student journey

Dates for wishes, rounds and close.`src/components/StudentPhasesBlock.tsx`

3.11.5 Requirements

FR-STU-001 Masari entry by civil ID and personal code

The system shall let a student open or create a Masari profile from the permanent link /me using a 12-digit civil ID that passes the checksum and a 6-digit personal code the student chooses, shall create the profile only after explicit consent to store it, and shall reject a code made of one repeated digit or a run of consecutive digits.

● Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A 12-digit civil ID with a bad checksum yields 400 with code bad-civil.
2. Create without agree=true yields 400 no-consent; code 111111 or 123456 yields 400 weak-code.
3. Creating an existing profile yields 409 exists; a new profile yields 201 with a session cookie.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentProfile.ts` `src/server/studentDoor.ts` `tests/student-door-audit.ts` `tests/student-journey-audit.ts`

FR-STU-002 Profile lockout after wrong codes

The system shall lock a student profile for 15 minutes after 5 consecutive wrong codes, shall keep the lock even when a correct code arrives in parallel, and shall count against the network address only real guesses (a wrong code for an existing profile), not successful entries or wrong civil IDs.

● Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. The fifth failure yields 401 with locked; the next attempt yields 423 with retryInSeconds.
2. The address cap is 600 guesses per 10 minutes with a hard ceiling of five times that (3000); credit is returned when the request is not a guess.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentProfile.ts` `tests/student-door-audit.ts` `tests/student-door-shared-network-audit.ts`

FR-STU-003 Signed door session

The system shall issue after entry a signed (HMAC) sh_session cookie with HttpOnly and SameSite=Lax that carries the fingerprint, expiry and a code version but neither the name nor the civil ID, shall keep the session 12 hours (ending when the browser closes) or 180 days under «this is my device», and shall end every earlier session when the code changes.

● Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A cookie without remember has no Max-Age; with remember it has Max-Age = 180 × 86400 seconds.
2. After POST /api/public/me/code an old session returns 401 signed-out.
3. A tampered or expired signature is not accepted.

EVIDENCE

`src/server/studentDoor.ts` `src/server/studentDoorRoutes.ts` `tests/student-door-audit.ts`

DR-STU-001 Identity stored sealed

The system shall never store a student's name or civil ID in the clear: an HMAC fingerprint of the ID is the profile key, the name and ID are sealed with AES-256-GCM, the code is salted with scrypt, the civil ID is returned to the student only masked (last three digits), and sealed values open only for the session owner and authorised staff.

Implemented · Must · Data · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. The profile document holds no plain name or ID and holds nameCipher, civilCipher and codeHash.
2. The response says only hasIdentity and shows the ID masked to its last three digits.
3. A different case secret makes earlier sealed values unreadable, so the stored secret wins over the environment value on conflict.

EVIDENCE

`src/server/core/studentIdentity.ts` `src/server/studentCaseSecret.ts` `src/server/studentDoorRoutes.ts` `src/server/studentDoor.ts` `tests/student-door-audit.ts` `tests/masari-approval-integrity-audit.ts`

DR-STU-002 Student profile lifetime and deletion

The system shall compute for every profile a deletion date equal to the earlier of expected graduation + 365 days and last sign-in + 730 days (creation + 6 years when no plan is set), shall purge expired profiles at start-up and every 6 hours in batches of at most 400, and shall treat «forgot my code» as a deletion request executed after 30 days that the owner cancels by entering with the code.

Implemented · Must · Data · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. A profile with no plan created today gets deleteAt = last sign-in + 730 days (the nearer date).
2. POST /api/public/me/reset returns days = 30, and the owner's next entry clears resetRequestedAt.
3. There is no recovery of the code through phone or department; deletion is the only path.

EVIDENCE

`src/utils/studentProfile.ts` `src/server/studentDoorRoutes.ts` `tests/student-door-audit.ts`

FR-STU-004 Profile setup: department, plan, passed courses

The system shall save the student's department, plan (mandatory when two or more plans exist) and passed courses after closing them over prerequisites, dropping any course whose condition fails together with what depends on it except what a transfer student declared as equivalent, and shall keep the wishes while department and plan are unchanged, removing those that became passed or lost their prerequisite.

Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A department with two plans and no plan chosen yields 400 bad-plan.
2. Marking a course passed without its prerequisite drops it and returns droppedPassed ≥ 1.
3. Changing the department clears the wishes.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentProfile.ts` `src/utils/studentMap.ts` `tests/student-door-audit.ts` `tests/student-map-audit.ts`

FR-STU-005 Validation of the student-journey dates

The system shall enforce on term dates the order: schedule submission deadline, then wishes (open and close together), then rounds (up to 8, name up to 30 characters, no overlap), then request close after the last round, shall refuse dates spanning more than 400 days, shall name the offending field in every refusal, and shall not accept dates for an ended or closed term.

Implemented · Must · Functional · University registrar, System administrator

ACCEPTANCE CRITERIA

1. Wishes opening on or before the submission day is refused on wishesOpen.
2. Two overlapping rounds are refused on rounds.1; a round starting with the wishes close is refused on rounds.0.
3. POST /api/approvals/student-phases by anyone but the university registrar or an administrator yields 403, and for an ended term yields 409 term-closed.

EVIDENCE

`src/utils/termStudentPhases.ts` `src/server/routes/approvalsRoutes.ts` `src/components/StudentPhasesBlock.tsx` `tests/term-student-phases-audit.ts`

FR-STU-006 Phase reading and door-term choice

The system shall derive the student phase (waiting, wishes, frozen, round, between rounds, closed, or unset) from today's date in Kuwait with each date inclusive of its whole day, shall give the student door the newest unfinished term whose dates are not yet closed, otherwise the upcoming planning term, otherwise the current term, and shall let a term without dates behave as before, neither blocking nor opening anything.

Implemented · Must · Functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. Today = wishesClose gives mode wishes; the next day gives mode frozen.
2. The day after requestsClose gives mode closed and the door moves to the next term.
3. A term without dates gives key unset and mode none.

EVIDENCE

`src/utils/termStudentPhases.ts` `src/server/studentDoorRoutes.ts` `tests/term-student-phases-audit.ts` `tests/student-door-audit.ts`

FR-STU-007 Writing wishes during the wishes phase

The system shall accept wishes only during the wishes phase, after the profile is complete, with units from 12 up to the department cap, at most 12 courses taken from the remainder of the student's plan, shall exclude any course whose prerequisite is unmet, and shall store the three time constraints (no morning, no afternoon, a free day).

Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. PUT /api/public/me/wishes outside the phase yields 409 wishes-closed; without a department yields 409 no-setup.
2. targetUnits = 11 or above the cap yields 400 bad-units.
3. 15 submitted courses keep at most 12 and exclude any with an unmet prerequisite.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentNeedRequest.ts` `tests/student-door-audit.ts` `tests/student-journey-audit.ts`

FR-STU-008 Per-department student unit cap

The system shall let a department holding the schedule permission choose the student unit cap in Masari as 15, 18 or 21, shall apply it to the department and its campus family, shall default to 15 when nothing is chosen, and shall send the cap to the page in limits.wishUnitsMax.

Implemented · Should · Functional · Department head, Schedule committee chair, Student

ACCEPTANCE CRITERIA

1. PUT /api/section-student-settings with 16 yields 400 asking for 15, 18 or 21.
2. A department with no setting returns wishUnitsMax = 15.

EVIDENCE

`src/utils/studentUnitCap.ts` `src/server/routes/referenceRoutes.ts` `src/server/studentDoorRoutes.ts` `src/components/StudentUnitCapField.tsx` `tests/student-door-audit.ts`

FR-STU-009 Student advice and peer counts

The system shall suggest to the student the term's courses from the plan, its rules and what is actually offered, with units from 12 up to the department cap, ordered by what each course unlocks, giving the reason for every pick and an estimate of remaining terms (up to 14), and shall show «wanted by N students» only in a department with at least 10 profiles and for a course wanted by at least 5.

Implemented · Should · Functional · Student

ACCEPTANCE CRITERIA

1. A course lacking a prerequisite is not suggested and appears under waiting with what is missing.
2. A department with 9 profiles gives show=false and an empty byCourse; a course wanted by 4 is not sent.
3. No advice is returned without an open term (reason no-term) or without a plan (no-plan).

EVIDENCE

`src/utils/studentAdvice.ts` `src/utils/studentProfile.ts` `src/server/studentDoorRoutes.ts` `tests/student-advice-audit.ts` `tests/student-door-audit.ts`

FR-STU-010 The «suggestions only» acknowledgement

The system shall show on entering Masari a card stating «these are suggestions only, not a final registration», shall not let the student continue before acknowledging, shall store the acknowledgement in the profile on the server under a key of term, department and plan so it returns when the path changes, and shall keep a fixed «suggested · not final» ribbon on every screen afterwards.

Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. POST /api/public/me/ack with a key not matching masari-ack:... yields 400 bad-ack.
2. The acknowledgement is read from the profile on another device with no browser storage.

EVIDENCE

`src/server/masariDisclaimer.ts` `src/server/studentDoorRoutes.ts` `tests/student-journey-audit.ts` `tests/student-door-audit.ts`

NFR-STU-001 Aggregated home and no caching

The system shall serve the student home through one request GET /api/public/me/home that gathers nine parts (notices, cases, graduate, request-context, ticket, report, advice, offered, map) of which a slow one is dropped after 8 seconds without blocking the rest, and shall send every door response with Cache-Control: no-store.

Implemented · Should · Non-functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. A part exceeding 8 seconds returns parts[key] = {ok:false,status:504}.
2. Responses of /api/public/me/* carry no-store.

EVIDENCE

`src/server/studentDoorRoutes.ts` `tests/student-door-audit.ts` `tests/student-journey-audit.ts`

FR-STU-011 Student request to open a course or review a conflict

The system shall accept an individual student request (open a course, up to 12 courses, or a two-course conflict) only with the student's consent to send identity, only after the latest current simulation proves the student cannot reach 12 units, with a reason of 3 to 600 characters, shall send one case with a case number to each department owning a requested course, and shall accept a conflict only when published complete schedules prove it unresolvable.

● Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. Without consent yields 400 consent-needed; without a proven blocker yields 409 no-proven-blocker.
2. Details of length 2 yield 400 details-needed; a course outside the remainder or not blocked yields 400 course-outside-plan.
3. An unproven conflict yields 409 conflict-not-proven.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentNeedRequest.ts` `src/utils/studentCaseConflict.ts` `src/utils/studentNeedMerge.ts` `tests/masari-approval-integrity-audit.ts` `tests/registration-flow-audit.ts` `tests/student-needs-scale-audit.ts`

FR-STU-012 The «did you register?» question after a round

The system shall open the «did you register?» question only during a running round and after it until requests close, shall accept up to 40 answers (registered, or stuck with a reason: full, prerequisite, conflict, other) on courses of the student's own plan, shall tally them in 64 counter shards, and when a stuck student asks for department help shall create one case per department owning a stuck course after verifying the civil ID matches the profile fingerprint.

● Implemented · Must · Functional · Student, Department head

ACCEPTANCE CRITERIA

1. POST /api/public/me/report outside a round yields 409 report-closed; with no valid answer yields 400 empty.
2. A civil ID not matching the profile fingerprint yields 403 civil-mismatch.
3. Stuck courses of two departments create two cases, one per department.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/utils/studentNotices.ts` `src/utils/registrationCauses.ts` `tests/registration-flow-audit.ts` `tests/student-journey-audit.ts`

FR-STU-013 Graduate request and graduation-sheet proof

The system shall accept a graduate request (field-training kind: conflict or prerequisite) while requests are not closed even outside the wishes phase, provided the uploaded graduation sheet (PDF or image up to 14 MB, 12 attempts per 10 minutes per profile) is proved by matching the civil ID and programme to the student's department, with a signed proof token valid for 20 minutes, and shall fail closed when the department has no approved graduation rule.

● Implemented · Must · Functional · Student, Schedule committee chair

ACCEPTANCE CRITERIA

1. Uploading a sheet for a department without a saved graduation rule yields 422 without reading the document.
2. A civil ID on the sheet not matching the entered one yields 422; a programme not matching the department yields 422.
3. A request in the closed phase yields 409 requests-closed.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/server/core/studentIdentity.ts` `src/utils/studentCaseDecision.ts` `tests/masari-approval-integrity-audit.ts` `tests/student-journey-audit.ts`

FR-STU-014 Request status by case number

The system shall let a student read the status of a request from the link /m/:token using the civil ID and the case number together, shall give the same «no request» answer when either is wrong, shall limit attempts, and shall not promise registration: what the committee approved while the department's schedule is unsigned is reported as awaiting signatures.

Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A valid civil ID with a wrong case number returns found:false, identical to an ID with no request.
2. The case number is the first 8 uppercase characters of the record id.
3. Status approved for a department that has not signed returns awaiting-signatures.

EVIDENCE

`src/server/pages/studentCaseStatusPage.ts` `src/server/routes/surveyRoutes.ts` `src/utils/studentNeedMerge.ts` `src/components/StudentCaseStatus.tsx` `tests/student-case-status-audit.ts` `tests/masari-approval-integrity-audit.ts`

IR-STU-001 Public student schedule page

Inferred

The system shall serve the department schedule page to students from the link /t/:token of kind students, showing the approved schedule of the current term, and before approval the current schedule marked «provisional», with no sign-in, and shall refuse a link of another kind.

Implemented · Should · Interface · Student, Public visitor

ACCEPTANCE CRITERIA

1. The /api/public/students/:token response carries provisional:true when no approved rows exist and the schedule has rows.
2. A link of kind survey yields 404 stating it is not the students' schedule.

EVIDENCE

`src/server/pages/studentSchedulePage.ts` `src/server/routes/publicStaffRoutes.ts`

FR-STU-015 «Try as a student»: isolated demonstration

The system shall open at /me/demo a private demonstration sandbox for the visitor, empty of real data, with two cookies, a fictional student, wishes and a rehearsal run inside it, living 2 hours, at most 150 sandboxes, 30 creations per 10 minutes, and neither read from nor written to by real data.

Implemented · Could · Functional · Public visitor, Student

ACCEPTANCE CRITERIA

1. /me/demo returns 302 to /me?track=girls#now with cookies sh_demo_stu and sh_session.
2. An expired sandbox clears both cookies and requests return 401 against real data.

EVIDENCE

`src/server/studentShowcase.ts` `src/server/demoRadarSeed.ts` `tests/student-showcase-audit.ts`

FR-STU-016 Registration sheet: scope and signature gating

The system shall serve the registration sheet (GET /api/student-registration) to holders of the workshop (7) or department-report (14) permission within their scope, shall show the same requests the Analytics Center reads under the same ownership rule, shall withhold it from registration officers until the schedule committee chair and the department head have both signed the department's schedule, and shall hide identity from view-only roles.

Implemented · Must · Functional · Schedule committee chair, Registration officer, College registrar, College dean

ACCEPTANCE CRITERIA

1. A registration officer with an unsigned schedule gets 409 not-signed.
2. A department outside scope yields 403.
3. A dean (view-only) sees counts without name or civil ID.

EVIDENCE

`src/server/routes/studentRegistrationRoutes.ts` `src/utils/studentCaseScope.ts` `src/components/StudentRegistration.tsx` `src/components/StudentCasesTable.tsx` `tests/student-registration-audit.ts` `tests/masari-approval-integrity-audit.ts`

FR-STU-017 Committee first, then registration

The system shall order decisions on a student request as: the department committee approves or declines (with a reason from not eligible, not in plan, prerequisite, duplicate, other), then registration decides «registered» or «rejected» (with a reason from no seat, prerequisite, level, conflict, closed, other) only on what the committee handed over, shall not let the committee overturn a registration decision, and shall decide a graduate case with one decision for the whole case in the same order.

● Implemented · Must · Functional · Schedule committee chair, Department head, Registration officer, College registrar

ACCEPTANCE CRITERIA

1. The committee writing registered yields 403; registration writing committee-rejected yields 403.
2. Registration on a course the committee has not approved yields 409 with the committee-not-approved message.
3. A rejection without an allowed reasonCode yields 400.

EVIDENCE

`src/server/routes/studentRegistrationRoutes.ts` `src/utils/studentCaseDecision.ts` `tests/student-registration-audit.ts`

FR-STU-018 Stale-write refusal on decisions

The system shall refuse with 409 and code stale-write a decision built on a version that is no longer current (expectedAt for a course, expectedStamp for a case), inside the same transaction so two colleagues cannot overwrite each other on an old state, telling the writer to refresh and retry without changing anything.

● Implemented · Must · Functional · Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. A stamp not matching the current record yields 409 code=stale-write and the course state is unchanged.
2. A matching stamp yields 200 and a new at value.

EVIDENCE

`src/server/routes/studentRegistrationRoutes.ts` `src/utils/staleWrite.ts` `tests/student-registration-audit.ts` `tests/colleagues-advising-registration-audit.ts`

FR-STU-019 Committee approval with section opening

The system shall let a committee holding the schedule permission (7) use «approve + open section», opening the section through the same change path the workshop uses (scope, lock, conflict), so the approval is not recorded when the section is refused (stateSaved:false), the section is removed if saving the decision fails afterwards, and a section opened earlier is picked up through the reference need:<request>:course:<course> so it is never opened twice.

● Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. openSection without permission 7 yields 403 no-schedule-permission.
2. Two simultaneous presses create one section and opened.reused = true in the second.
3. Refusing the section for a conflict returns stateSaved:false and leaves the course state unchanged.

EVIDENCE

`src/server/routes/studentRegistrationRoutes.ts` `src/components/ApplyProposal.tsx` `src/server/core/scheduleWrites.ts` `tests/student-registration-audit.ts` `tests/masari-approval-integrity-audit.ts`

FR-STU-020 Advising sends a section request

The system shall let a department advisor send one section request for a course, campus and term with a preferred period (morning, evening or any), a count and a student list taken from Masari or pasted (identities sealed), with at most 400 students in the list, 1000 in the count and a note of up to 300 characters, only from the advisor's own department scope and in a term that is not frozen.

● Implemented · Must · Functional · Department advisor

ACCEPTANCE CRITERIA

1. A request without a valid period yields 400 asking for the preferred period.
2. A sender who is not an advisor in the department scope yields 403.
3. A count of 5000 is clipped to 1000; students duplicated by fingerprint are merged.

EVIDENCE

`src/server/sectionRequestsRoutes.ts` `src/utils/sectionRequests.ts` `src/components/sectionRequests/SectionRequestsPanel.tsx` `src/utils/sectionRequestHandoff.ts` `tests/section-requests-audit.ts`

FR-STU-021 Section request stages and transitions

The system shall always derive a section request's stage from its document (sent, section opened, approved, registering, registered, withdrawn, declined) rather than a stored field, and shall enforce the transitions: no approval before a section is linked, no linking, unlinking, withdrawal or decline after approval, registration only for approved students, no removal of approval from a registered student, and a decline carries a written reason that reaches advising.

● Implemented · Must · Functional · Department advisor, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. Approval with no linked section yields 409 asking to link a section first.
2. A decline without text yields 400; after the decline every write is refused with 409.
3. Registering an unapproved student yields 409 stating only approved students are registered.

EVIDENCE

`src/utils/sectionRequests.ts` `src/server/sectionRequestsRoutes.ts` `tests/section-requests-audit.ts`

FR-STU-022 Distributing request capacity across sections

The system shall distribute the request count over all its linked sections rounding up (count ÷ number of sections) and write each section's share as base capacity in the advisor seat balance up to 200 per section, shall redistribute on every link or unlink, and shall never touch a capacity an administrator wrote by hand.

● Implemented · Should · Functional · Schedule committee chair

ACCEPTANCE CRITERIA

1. 100 students and two sections give 50 each; after a third link, 34.
2. A hand-written capacity different from appliedCapacity stays as it is.
3. Unlinking a section the request wrote removes its capacity from the balance.

EVIDENCE

`src/utils/sectionRequests.ts` `src/server/sectionRequestsRoutes.ts` `src/utils/advisorSeats.ts` `tests/section-requests-audit.ts`

FR-STU-023 Party powers and notifications

The system shall decide what each party may do on a section request from role, scope and granted permissions rather than the role template (advising sends and withdraws, the course department's committee links, declines and approves, registration approves and records, a view-only role reads without names), and shall deliver to each party what awaits it through the bell and mobile alerts without waking the system administrator for every request.

Implemented · Must · Functional · Department advisor, Schedule committee chair, Registration officer, College dean

ACCEPTANCE CRITERIA

1. An advisor of another department requesting a request not from their department gets 404.
2. An account without permission 7 or 14 cannot read.
3. The bell shows up to 8 items derived at query time with nothing stored.

EVIDENCE

`src/server/sectionRequestsRoutes.ts` `src/utils/sectionRequests.ts` `tests/section-requests-audit.ts` `tests/colleagues-advising-registration-audit.ts`

FR-STU-024 Advisor seat balance

The system shall compute for each section a hard cap = the lesser of (base capacity + balance) and the hall capacity, shall let an advisor add seats (1 to 50) atomically, all or nothing, so two advisors acting together cannot exceed the cap, shall split the balance evenly under fair split, shall suggest alternative sections when full, and shall allow an exception request to the administrator (up to 50 seats and a 300-character note).

Implemented · Must · Functional · Department advisor, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Capacity 60 with balance 10 shows «60 + 0/10»; adding 11 is refused as full.
2. Adding 0 or 51 is invalid; an undefined base capacity yields 409.
3. A rev not matching the number of additions yields 409 stale-write.

EVIDENCE

`src/utils/advisorSeats.ts` `src/server/routes/offeringRoutes.ts` `src/components/AdvisorSeatsPanel.tsx` `tests/section-requests-audit.ts` `tests/colleagues-advising-registration-audit.ts`

FR-STU-025 Seat, holder and handover

The system shall treat an account as a fixed seat whose holder changes: one ledger per seat (holder, team, deputy, history up to 400 events); the holder or the system administrator hands it over, a first password is generated and shown once, all open devices are signed out and the team is frozen until the new holder marks each member «continuing» or «ended», and the session owner acts under their own name in the log and signature.

Implemented · Must · Functional · System administrator, Schedule committee chair, College registrar, University registrar

ACCEPTANCE CRITERIA

1. Handing over to the same current holder (same instructor number) yields 409.
2. Anyone other than the holder or the administrator yields 403.
3. After handover every member has status frozen.

EVIDENCE

`src/server/seatsRoutes.ts` `src/server/seats.ts` `src/utils/seats.ts` `src/components/seats/SeatsAdmin.tsx` `src/components/seats/SeatPanel.tsx` `src/components/seats/SeatDrawer.tsx` `tests/seats-audit.ts`

FR-STU-026 Team member link and deputy

The system shall let a team member enter through the personal link /k/<token> after confirming the last four digits of their civil ID, working under their own name and team role (a registration officer within their own scope), not with the seat password, and not signing; the link locks after 5 wrong attempts, a reset link expires after 24 hours if unopened, the team is limited to 40 members, and a deputy appointed from the team acts and signs until the end date.

● Implemented · Must · Functional · Schedule committee chair, Registration officer, College registrar, University registrar

ACCEPTANCE CRITERIA

1. Five wrong attempts yield 423, revoke the token and log link-locked.
2. A new member after 40 active members yields 409 stating the team is full.
3. A deputy end date before today yields 400.

EVIDENCE

`src/server/seatsRoutes.ts` `src/server/seats.ts` `src/utils/seats.ts` `src/server/seatTeamPage.ts` `src/components/seats/SeatLinkCard.tsx` `src/utils/seatAccessCard.ts` `tests/seats-audit.ts` `tests/seat-access-card-audit.tsx`

FR-STU-027 Registration rehearsal engine

The system shall deterministically simulate for each student with wishes the remaining courses under the plan's rules and build three plans: (A) closest to wishes and constraints, drawing on a virtual seat ledger in window order, (B) alternative sections, and (C) a safe minimum of 12 units, with units clamped between 12 and 18 (default 15), ordering students by greatest need first and splitting them evenly over rounds, and shall remain an estimate rather than a reservation.

● Implemented · Must · Functional · The system (automatic), System administrator

ACCEPTANCE CRITERIA

1. The same inputs produce the same outputs (determinism).
2. targetUnits = 21 is clamped to 18 in the simulation.
3. Plans B and C do not draw on the ledger.

EVIDENCE

`src/utils/registrationSim.ts` `src/server/simulationData.ts` `tests/registration-sim-audit.ts` `tests/registration-flow-audit.ts`

FR-STU-028 Running and refreshing the radar

The system shall restrict running the radar to system administration with an in-process lock and a database lock across servers, shall refresh it automatically every 24 hours (RADAR_AUTO_HOURS) during and after wishes until requests close and once after wishes close, checking every 5 minutes, stoppable with RADAR_AUTO=false and inactive in the demonstration, shall show departments their own courses as counts not names, and shall store for each student a ticket with no name or civil ID.

● Implemented · Must · Functional · System administrator, Root administrator, Department head, Student

ACCEPTANCE CRITERIA

1. POST /api/simulation/run by a non-administrator yields 403; during another run yields 409 busy.
2. A frozen term yields 409 term-frozen.
3. A ticket whose runId does not match the summary is ignored.

EVIDENCE

`src/server/simulationRoutes.ts` `src/server/simulationData.ts` `src/components/RegistrationRehearsal.tsx` `tests/registration-flow-audit.ts` `tests/rehearsal-audit.ts`

FR-STU-029 Why a student does not reach 12 units

The system shall explain a student's shortfall below 12 units by comparing scenarios: one constraint (capacity, conflict, student constraints, prerequisite) is lifted for everyone and the question is whether the student then reaches 12, so a single cause is named, more than one is «shared», and none is «course not offered or other», with the interface stating it is an estimate not proof.

● Implemented · Should · Functional · The system (automatic), Department head

ACCEPTANCE CRITERIA

1. Lifting capacity alone resolves the case so the cause is full.
2. Lifting two constraints resolves it so the cause is shared.
3. No constraint resolves it so the cause is other.

EVIDENCE

`src/utils/registrationCauses.ts` `src/utils/registrationSim.ts` `tests/registration-causes-audit.ts`

NFR-STU-002 Limits on a shared university network

The system shall withstand thousands of students entering from one network address under tuned limits: 24,000 identity queries and 600 code errors per address per 10 minutes, 60 writes per minute per student and a ceiling of 1200 writes per minute per address, adjustable through STUDENT_DOOR_* variables, and shall not count /api/public/me/* against the general public address counter.

● Implemented · Must · Non-functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. 600 students from one address within two minutes are not blocked by the identity limit.
2. The 61st write within a minute by one student yields 429 with the too-many-requests message.
3. The STUDENT_DOOR_WRITES_PER_MINUTE variable changes the limit.

EVIDENCE

`src/server/studentDoorRoutes.ts` `tests/student-door-shared-network-audit.ts` `tests/student-door-audit.ts`

IR-STU-002 Academic advice panel for departments

The system shall show a department a non-blocking academic reading from approved plans and wishes: course pairs needed by at least 3 eligible students whose times overlap with whether alternative sections exist, and a morning or afternoon preference for a course wanted by at least 3 (up to 160 pairs and 24 preferences), shall never change the save, sign or send verdict, and shall show neither identity nor counts below 3.

● Implemented · Should · Interface · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A single student wanting the pair gives shared = 0 and basis = curriculum.
2. A mandatory prerequisite is not simultaneous demand, so no pair forms between a course and its prerequisite.
3. The response carries nonBlocking:true.

EVIDENCE

`src/utils/academicAdvisory.ts` `src/server/academicAdvisoryData.ts` `src/server/academicAdvisoryVisual.ts` `src/components/AcademicAdvicePanel.tsx` `tests/academic-advice-audit.ts`

IR-STU-003 **Registration and radar tabs and the phase line**

The system shall join «registration sheet» and «registration-day radar» in two tabs atop the screen for someone holding both (someone holding one sees no bar), shall have the phase line say «numbers only» before registration and show the names of stuck students to the owning departments only once a round starts, shall carry a QR code button for the permanent /me link, and shall let the college and department picker narrow what the reader sees inside their scope, never widen it.

Implemented · Should · Interface · Department head, Registration officer, College registrar, System administrator

ACCEPTANCE CRITERIA

1. An account without radar or sheet permission gets no RegistrationTabs bar.
2. phase.namesVisible is false before a round and true during and after it.
3. ?collegeId&sectionId outside scope does not widen results.

EVIDENCE

`src/components/RegistrationTabs.tsx` `src/components/MasariQrButton.tsx` `src/components/AdvisingScopePicker.tsx` `src/server/simulationRoutes.ts` `src/utils/termStudentPhases.ts` `tests/registration-flow-audit.ts` `tests/rehearsal-audit.ts`

3.11.6 Algorithms

ALG-STU-01 · Student and advising services

Deterministic

Passed-course closure over prerequisites

Which of the courses a student marks as passed can be kept, given that every passed course requires its own prerequisites to be passed as well?

1. Start from the declared set and compute passed units from course credits.
2. Scan the set: the first course whose «all» prerequisites, one of each «any-of» group, or minimum units are not satisfied is dropped, and units are recomputed.
3. Repeat until a full pass drops nothing (fixed point), so everything built on a dropped course falls with it.
4. Courses a transfer student declared equivalent are exempt and still unlock their dependants.

Inputs

Courses declared passed, the sheet's prerequisite rules, course credits, and declared equivalents.

Output

Two lists: kept and dropped.

The same closure runs in the browser and on the server, and the server does not trust the page; an exemption never extends beyond the declared equivalent.

`src/utils/passedClosure.ts#closePassed` `src/utils/curriculumRules.ts#unmetPrereq` `src/server/studentDoorRoutes.ts` `tests/student-map-audit.ts`

FR-STU-004

ALG-STU-02 · Student and advising services

Graduate plan choice and graduation-sheet proof

Deterministic

Which plan (old or new) is this graduate measured against, and does the uploaded graduation sheet prove eligibility?

1. The sheet's «programme» row must match the selected department name; a missing programme row refuses the request.
2. One live plan → that plan. Otherwise the sheet's «required units» must equal exactly one live plan's degree total: the direct reading first, then nearby figures between 60 and 300, excluding the passed-units figure.
3. If the total is unreadable but every live plan has the same rule, the result is the same; otherwise the request is refused with a written reason: unreadable, unknown total, ambiguous, or rule not approved.
4. Passed units: the direct reading, otherwise bounded candidates between 60 and the plan's degree total, excluding the total itself, taking the largest remaining.
5. The result is compared with the plan rule's graduation threshold for a regular or summer term.

Inputs

Facts read from the graduation sheet (programme, required and passed units and their candidates), the department's live plans and their approved graduation rules, and the term name.

Output

The chosen plan, its rule and the evidence used, or a coded refusal with a reason; the passed units and eligibility.

No student is measured against another plan's rule and nothing is verified without a saved rule; a rule derived from the department name is only a seed suggestion.

Range for nearby total figures: 60 to 300

Seed suggestions (saved before use): 130 units → 107 regular, 109 summer; 132 → 109 and 111; 134 → 111 and 113

`src/utils/graduationPlan.ts#choosePlanForSheet` `src/utils/graduationPlan.ts#sheetTotalEvidence`
`src/utils/degreeRules.ts#suggestedDegreeRule` `src/server/core/studentIdentity.ts#graduationCandidatesForSection`
`tests/curriculum-graduation-audit.ts` `tests/student-journey-audit.ts`

FR-STU-013 FR-ENG-004

ALG-STU-03 · Student and advising services

Registration rehearsal and semester readiness

Search and ranking

If students register on the offered sections, who reaches 12 units or more, which sections fill up, and how ready is the semester?

1. Offerings are audited first: valid identity, a positive integer capacity, valid non-overlapping meetings, a unique id; flawed ones are excluded and their course marked uncertain.
2. Per student: remaining need from the approved plan; candidates = remaining compulsory courses, enough of each unfinished elective group, then wishes; priority = (1000 if required, else 400) − 20 × level + 8 × courses it unlocks + 150 if wished.
3. Ordering by need: students within 36 units of graduating first (2000 − remaining), then compulsory units, with a stable hash tie-break; students are then split evenly across registration windows.
4. Plan A: depth-first backtracking over course bundles (a course with its co-requisites) and their sections on a shared virtual seat ledger, maximising units up to the target; beyond 2,000 visited states the student becomes «unknown» rather than guessed.
5. Plans B (different sections) and C (a safe 12-unit minimum of required courses) are fall-backs that reserve no seat.
6. Each student falls into a band: 15+, 12–14, graduating below 12, below 12, unknown; readiness = 60% × share reaching 12+ + 40% × share of demand that found a seat.

Inputs

Student profiles (passed courses, wishes, constraints), offered sections with meetings and capacities, approved plan rules, course credits, and the number of windows.

Output

A ticket per student with three plans and a window, the seat ledger, unmet demand by reason, the band distribution, and the readiness score and tier.

No database, clock or randomness: the same inputs give the same tickets; seats are virtual and nothing is reserved.

Units: minimum 12, default 15, maximum 18 · A seat is «tight» when less than 10% of capacity remains · Search cap of 2,000 states per student · Readiness: «ready» from 85, «needs attention» from 65, and weak when «unknown» exceeds 20%

`src/utils/registrationSim.ts#simulateRegistration` `src/utils/registrationSim.ts#auditSimOfferings` `src/utils/semesterReadiness.ts#semesterReadiness` `src/server/simulationData.ts` `tests/registration-sim-audit.ts` `tests/registration-causes-audit.ts`

FR-STU-027 FR-DSH-014

ALG-STU-04 · Student and advising services

Shortfall cause attribution by scenario comparison

Deterministic

Why does this student not reach 12 units: full sections, time conflicts, the student's own constraints, or a prerequisite?

1. The registration rehearsal runs for students with wishes as the baseline, collecting those below 12 units.
2. It is re-run four times, each lifting one constraint for everyone: capacity, conflicts, student constraints, or prerequisites.
3. For each short student, record which lifts bring them to 12 units or more.
4. Exactly one → that cause; more than one → «shared»; none → «not offered or data»; totals add up to the short count.
5. Counts are grouped per college and department, offerings rejected by the audit are listed as incomplete data with a link to their row, and the result is cached for 10 minutes per source fingerprint.

Inputs

The same registration-rehearsal inputs for the term, and the reader's scope.

Output

Per scope: students, short students and the cause distribution; the dominant cause; and the list of offerings with incomplete data.

A scenario comparison shown as an estimate, not an absolute causal claim; counts, never names.

4 relaxation scenarios · Result cached 10 minutes; up to 5,000 data-quality items held

`src/utils/registrationCauses.ts#explainShortfalls` `src/utils/registrationCauses.ts#dominantCause` `src/server/radarInsight.ts#radarInsightFor` `tests/registration-causes-audit.ts` `tests/radar-insight-audit.ts`

FR-STU-029 FR-INT-020

ALG-STU-05 · Student and advising services

Advisor seat cap and section-request capacity

Deterministic

How many extra seats may advisors add in each section, and how is a section request's count spread over its linked sections?

1. Extra balance: the section's value, else the course's, else the global value (most specific wins), each clamped to 0–200.
2. Hard cap = min(base capacity + balance, the physical hall capacity when known).
3. Adding n seats (1–50) is all or nothing: refused when remaining < n or, under fair split, when the advisor's total would exceed ⌊extra ÷ advisors⌋; it runs inside a single atomic read-write transaction.
4. A section request's count is spread as ⌈count ÷ sections⌉ (up to 200) over all linked sections and recomputed on every link or unlink; a capacity written by an administrator is left untouched.
5. The request stage is derived, never stored: sent → opened → approved → registering → registered, or closed.

Inputs

Balance settings (global/course/section, fair split, advisor count), the section ledger and its adds, hall capacity, and the request's linked sections.

Output

Seat state (cap, remaining, fair share, ring tone), the add result with its message, written section capacities, and the request stage.

No student is added beyond the physical seats whatever the balance, two advisors acting together cannot exceed the cap, and the result does not depend on linking order.

One add: 1 to 50 seats | Red at zero remaining; yellow at 30% or less, or 2 seats or fewer

`src/utils/advisorSeats.ts#seatState` `src/utils/advisorSeats.ts#tryAddSeats` `src/utils/advisorSeats.ts#quotaFor` `src/utils/sectionRequests.ts#planCapacities` `src/utils/sectionRequests.ts#capacityShare` `src/utils/sectionRequests.ts#stageOf` `tests/section-requests-audit.ts`

FR-STU-021 FR-STU-022 FR-STU-024

3.11.7 Edge cases

⚠ Changing the code during an open session

Scenario A student changes the code on one device while another stays open.

Behaviour The code version in the cookie no longer matches the profile, so the other device receives 401 signed-out and must sign in again.

⚠ A lock raised while a correct code is being verified

Scenario Parallel requests: failures reach the fifth while a correct code arrives.

Behaviour The lock stays and is not erased by the later success, returning 423 locked.

⚠ A student removes a course after a decision

Scenario A student resubmits a request without a course registration already recorded.

Behaviour The request is updated, not replaced: the id and case number remain, and the decision stays marked «cancelled by the student after registration» so no held seat goes unnoticed.

⚠ Two people approving the same section list at once

Scenario Two committee members press approve on the same request at the same moment.

Behaviour The write is atomic with the rev stamp: one succeeds and the other gets 409 refresh-and-retry without any change.

⚠ A new term saved before the current term's requests close

Scenario Dates for a newer term are saved while the current term still has an open round.

Behaviour The door switches at once to the newer term as the newest unfinished one; the remedy is to move the current term's request close earlier or to set the new term after it closes.

⚠ A registration officer meeting an unsigned schedule

Scenario A decision is written on a course of a department whose schedule is unsigned while the other course in the request belongs to a signed one.

Behaviour Signature is checked for the requested course's own department, not the first department owning the request, so it returns 409 not-signed even if the other department has signed.

3.11.8 Standing owner decisions

- Course numbers are always seven digits in everything a student or staff member sees; the stored three-digit code and AdCourseId are never shown.
- The design of printed reports and sheets (the registration sheet and the plan sheet) is not changed; fixes go into data, not layout.
- Masari is always light and never follows the device's dark mode, and the word «radar» never appears in student-facing text.
- The seat is fixed and the person changes: no new account is created per person.
- What the student sees is an estimate, not a reservation or official registration, and every screen says so.

↑ Contents

3.12 · INT

Analytics Center and Guide

Deterministic schedule readings, next-term forecasts, a natural-language query layer that never executes, and an in-app Guide that runs in the browser.

34	100%	49	35
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.12.1 Overview

The Analytics Center is the screen where the department head, the schedule committee chair and the dean read the state of the schedule: its quality, approval blockers, room pressure, instructor loads and campus movement, and then try the effect before deciding. The module also holds the next-term forecast, the academic decision lab and the live schedule layer. In the natural-language query layer the Gemini service reads the sentence and proposes calls from five allowed functions only; the calls are then sanitised and executed on the server with deterministic logic inside the user's scope, so execution always passes through the deterministic validators and an explicit confirmation by the user. The Guide is an in-app assistant that understands questions in standard Arabic and the Kuwaiti dialect, shows the steps and walks the user through a tour on screen, and sends nothing out of the browser.

Department head · Schedule committee chair · College dean · Vice dean for academic affairs · System administrator · Root administrator · Standard user · Public visitor · The system (automatic) · External service

3.12.2 User journeys

Schedule committee chair — **From a sentence to a confirmed move**

1. The committee chair types in the query box: move 101 to Wednesday 11:00.
2. The server sends Gemini only the text, the scope and the names of the five allowed functions.
3. The proposed calls are sanitised; if the service returns nothing valid the server parses the sentence with its own rules.
4. The server runs the simulation, conflict check and regulations check against the real schedule without writing.
5. The screen shows a before/after preview, the schedule quality and the reason for refusal if a save blocker exists.
6. If the preview can be applied the user confirms explicitly, and the move goes through the usual move path and its validators.

Department head — Reading the schedule and trying a movement fix

1. The department head opens the Analytics Center and picks the department and term.
2. They read the quality out of 100, the blocker count, the largest gap, the day balance and the crowding map.
3. They open the movement reading and find an instructor moving between two buildings in less time than needed.
4. The system offers a room move or a room swap, and shows only what does not raise movement risk for any other instructor.
5. They turn the proposal into a draft for review; the published schedule does not change before approval.

Standard user — Asking about a task, then an on-screen tour

1. The user types in the Guide a Kuwaiti-dialect question about moving a lecture to another room.
2. The Guide normalises the text and matches it against the knowledge bank inside the browser with no network request.
3. It shows short standard-Arabic steps, four at most, and only what the user's role can reach.
4. The user presses the show-me action and the tour moves to the screen and highlights the element.
5. If the element does not appear the tour says so and stays alive; it closes only by the user's hand.

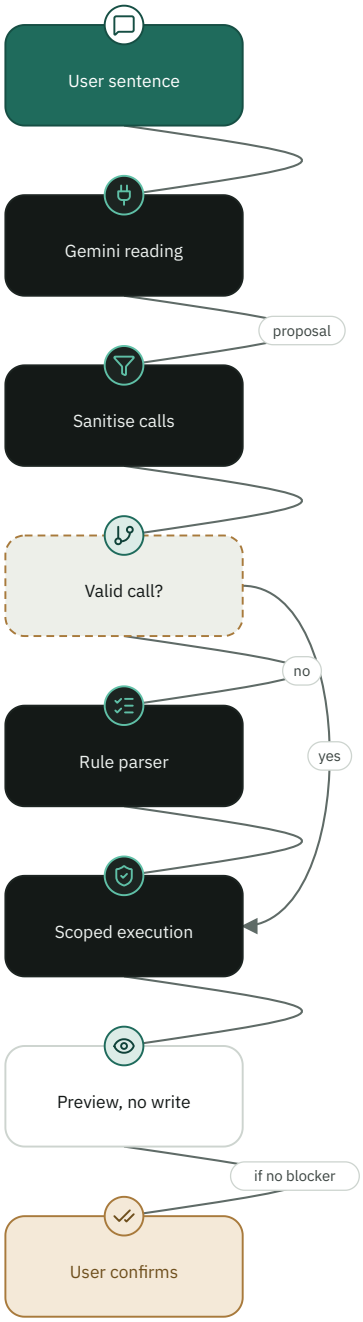
Root administrator — Reviewing where users struggle

1. The root administrator opens unanswered questions in the Guide to see the phrasings it did not understand.
2. They read the struggle points ranked by failure and help rate over the last 30 days against the previous 30.
3. If struggle rises after a new release they change the interface itself rather than the explanation.

3.12.3 Diagrams

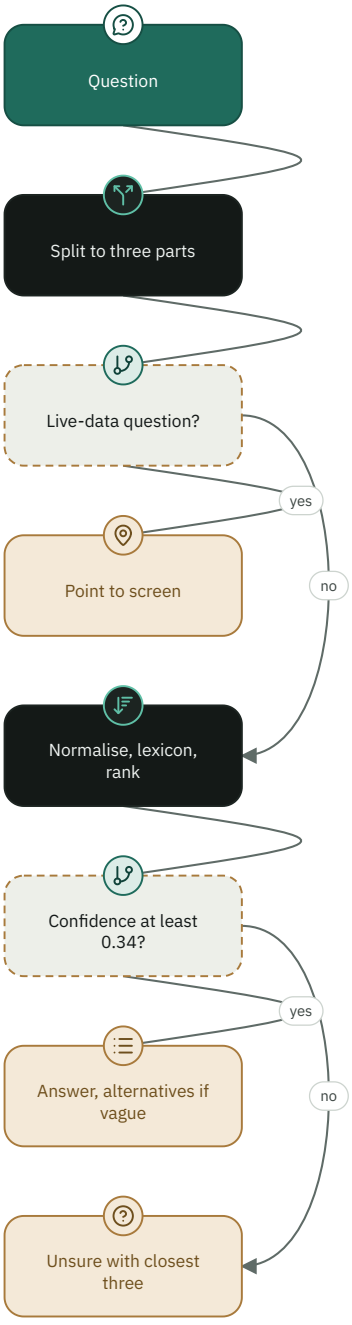
From sentence to safe preview

Gemini reads the sentence and only proposes; execution is deterministic and bound to the user's scope.



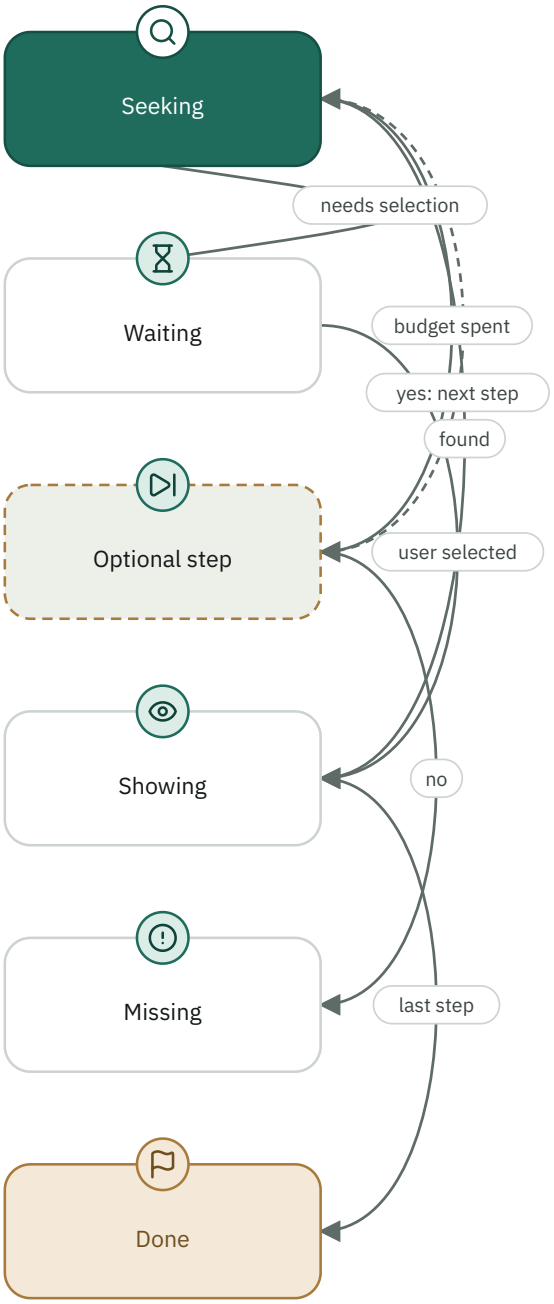
How the Guide understands a question

Every stage runs in the browser; permissions apply before ranking.



Guide tour states

A tour never disappears silently: element shown, waiting for an action, or an explicit not-found.



Forecast confidence gate

No one-click action unless confidence is medium or higher.



3.12.4 Screens

Analytics Center

Three scenes: understand, try and approve; tabs for quality, rooms, instructors, data health and forecast.`src/components/IntelligenceWorkspace.tsx`

Decision scope bar

Choose college, department and term; shows a read-only pill when offline.`src/components/IntelligenceContextBar.tsx`

Analysis scope switch
This college only, or every college of the department.`src/components/IntelligenceScopeSwitch.tsx`

Next-term forecast
Every figure carries its confidence and the number of terms behind it.`src/components/PredictiveInsightPanel.tsx`

Academic decision lab
Six read-only views: academic twin, before/after, domino, path rescue, balance and trust.`src/components/AcademicDecisionLabPanel.tsx`

Live schedule layer
Six scenes: overview, pressure map, health and fairness, one-minute brief, start of term and safety net.`src/components/LivingScheduleLayer.tsx`

Guide
Screens: ask, more, all, what is new, and unanswered questions for the root administrator.`src/components/guide/GuideAssistant.tsx`

Guide tour
A card that highlights the target element and waits for the user's action when needed.`src/components/guide/GuideTour.tsx`

Course impact
What precedes the course, what it opens, its plan sheets and the demand on it.`src/components/CourseImpact.tsx`

3.12.5 Requirements

FR-INT-001 **Analytical overview of the schedule**

The system shall compute, for a given department and term, the schedule quality out of 100, its readiness, approval blockers, instructor gaps, day balance and a crowding map in 30-minute slots within the configured day hours, and return them in one reading.

● Implemented · Must · Functional · Department head, Schedule committee chair, College dean

RATIONALE

> A decision maker needs one picture of where the problem is before opening any tool.

ACCEPTANCE CRITERIA

1. Quality = 100 minus penalties: booking safety up to 58, gaps up to 16, slots after 16:00 up to 10, day balance up to 10, data health up to 18; the result is clamped to 0–100.
2. Readiness is blocked when an approval blocker or an incomplete appointment exists; otherwise ready at 85 or more, review at 70 or more, else needs work.
3. A call with no department or term returns 400, and one for a department outside the user's permissions returns 403.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/overview` `src/utils/scheduleIntelligence.ts#analyzeSchedule` `tests/intelligence-scope-api.mjs`

FR-INT-002 Reading scope: one college or all colleges of a department

The system shall read with the «this college» scope by default, widen the reading on a «all colleges of the department» request to the matching departments in other colleges that the user is permitted to read, and treat any unknown scope value as the single-college scope.

Implemented · Must · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. A mathematics department in two colleges is read together on a department-scope request, and a department with a different name is not added.
2. A scope value such as «invalid» returns the single-college reading without error.
3. Term comparison and aggregation keep each row's source, so an identical course and section in two colleges are never merged.

EVIDENCE

`src/server/core/scope.ts#intelligenceReadScopes` `src/utils/intelligenceReadingScope.ts`
`src/components/IntelligenceScopeSwitch.tsx` `tests/intelligence-reading-scope-audit.ts` `tests/intelligence-scope-api.mjs`

FR-INT-003 Room insight

Inferred

The system shall show, for a room confirmed in the buildings and rooms registry and a given term, its weekly bookings, the free windows of at least 30 minutes and its usage split by department, hiding the department name of any booking outside the user's permissions.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A request with no term or no identified room returns 400, as does an unconfirmed or inactive room.
2. Another department's booking shows to a non-administrator as «booking from another department» with its time, and its usage is grouped under «other departments».

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/room`

FR-INT-004 Instructor insight

Inferred

The system shall show, for a given instructor and term, the weekly load, gaps, the appointments visible to the user and the count of commitments outside the user's permissions, and shall not show an instructor who appears in no department the non-administrator user may read.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A request with no instructor or term returns 400, and an instructor with no visible appointments returns 403 for a non-administrator.
2. Appointments in other departments are counted in externalCommitments and their rows are not shown.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/professor/:id`

FR-INT-005 Appointment story and context

Inferred

The system shall show, for a given appointment, its context: the instructor, course and room appointments visible to the user, conflicts with bookings outside the user's scope without revealing their details, the department's historical memory, the course life, the cost of a change, the reasons behind the room and instructor choice, and a timeline of how the appointment changed across versions, drafts, comments and the audit trail.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A missing appointment returns 404 and one outside the scope returns 403.
2. A conflict with another department's booking returns the other party's identifier as zero instead of revealing it.
3. A failure computing the why-reasons does not fail the response; it returns an empty reasons list.
4. The timeline reads up to 16 versions, 20 drafts and 30 audit entries for the appointment.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/context/:id`
`src/server/routes/intelligenceRoutes.ts#/api/intelligence/replay/:id` `src/utils/advancedIntelligence.ts`

FR-INT-006 Campus movement and safe room swaps

Inferred

The system shall detect an instructor moving between two buildings with a gap shorter than the campus movement profile requires, and propose moving the room or swapping two rooms without changing the time, instructor or days, and shall show a proposal only if it does not raise movement risk for any instructor in the term.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Margin = gap − travel time; below 0 is high risk, 0 to 10 minutes is a tight transfer, anything above is not listed.
2. Score = 100 − 18 per high risk − 6 per tight transfer, floored at 0.
3. The 12 worst high risks are examined, at most 8 proposals are returned, and a proposal is accepted only if term-wide high risks do not rise and the overall score does not fall.
4. Up to 40 risks are listed in the reading.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#spatialBurnoutAnalysis` `src/server/routes/intelligenceRoutes.ts#roomCastlingProposals`

FR-INT-007 Ripple forecast before a drop

Inferred

The system shall forecast the effect of moving an appointment to a proposed day and time, including conflicts, instructor effect and campus movement, before the user drops it, and shall write nothing.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A day outside Sunday–Thursday or a time not in HH:MM returns 400 «invalid drag position».
2. A time that pushes the appointment outside the configured day hours returns 400 naming the day bounds.
3. The response carries the movement score before and after and the difference.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/ripple/:id` `src/utils/scheduleInnovation.ts#forecastScheduleMove`

FR-INT-008 Rule-based analytical assistant

The system shall answer analytical-assistant questions about the schedule with explicit rules over the permitted department's data only, and return figures, bars and a before/after pair alongside the text when relevant, writing no change to the schedule.

Implemented · Should · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. A question shorter than two characters returns 400.
2. The assistant distinguishes: biggest problem, gaps (default threshold 3 hours, changed by «N hours»), a day-crowding investigation, «if I move…», best distribution, rooms, and a general reading when nothing matches.
3. An aggregate question such as who/how many/everyone is never read as an instructor name, and a named instructor with no appointment in the department is answered as having no data here, without mixing in another department.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/copilot` `tests/intelligence-scope-api.mjs`

FR-INT-009 Intent identification by Gemini within allowed functions

The system shall send the natural-language query to the Gemini service together with the scope identifiers and the names of five allowed functions only, and sanitise the proposed calls so that only allowed functions with bounded arguments are accepted, and no call is executed before sanitising.

Implemented · Must · Functional · Department head, Schedule committee chair

RATIONALE

> The Gemini service reads the text and proposes; execution is always through the deterministic validators and an explicit confirmation by the user.

ACCEPTANCE CRITERIA

1. Allowed functions: check_conflicts, find_rooms, check_instructors, check_regulations, simulate_schedule; any other name such as updateSchedule is dropped.
2. Up to 8 calls are accepted, with up to 24 arguments each, strings up to 240 characters, arrays up to 20 items and objects up to 2000 characters.
3. An argument whose name contains a write verb (save, publish, apply, delete…) or is commit or confirmed is removed, so {commit:true} never reaches execution.
4. The query text is cut to 800 characters, and fewer than two characters returns 400.

EVIDENCE

`src/utils/geminiScheduleLayer.ts#sanitizeGeminiScheduleCalls`
`src/server/routes/intelligenceInsightRoutes.ts#/api/intelligence/nl-schedule` `src/server/core/gemini.ts`
`tests/gemini-schedule-layer.ts`

FR-INT-010 Deterministic scoped execution of calls

Inferred

The system shall execute each accepted call on the server with its own deterministic logic over the permitted department and term schedule, without taking any execution result from the Gemini service.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. simulate_schedule returns the move preview; check_conflicts returns the critical blocker count and the first 6 alerts; check_regulations evaluates the department rules.
2. find_rooms examines up to 80 active confirmed rooms of the department and returns up to 12 that are free in a 30-minute window.
3. check_instructors matches the name and returns up to 12 instructors with their appointment counts.
4. A scope that is not permitted returns 403 before any call to the Gemini service.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts#executeGeminiScheduleCalls`

FR-INT-011 Fallback to the rule parser

When the Gemini service returns no valid call (no key, a refusal, a timeout or an invalid result), the system shall parse the sentence with a deterministic rule parser and produce the same kind of calls, and state their source in the response.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. «Move course 101 to Wednesday 11:00» yields simulate_schedule, check_conflicts, check_regulations in order, with day index 3 and time 11:00.
2. The source field is gemini-function-calls or deterministic-parser according to the origin.
3. A sentence from which nothing is extracted returns ok:false and a hint asking for the course, day, time or room.

EVIDENCE

`src/utils/geminiScheduleLayer.ts#deterministicSchedulingCalls` `src/utils/naturalQuery.ts#parseNaturalQuery`
`src/server/routes/intelligenceInsightRoutes.ts#/api/intelligence/nl-schedule` `tests/gemini-schedule-layer.ts`

FR-INT-012 Move preview without writing, separate confirmation

The system shall return, for a natural-language move request, a before/after preview with conflicts, regulations and schedule quality, declare that confirmation is required, and write nothing to the schedule; applying is a second explicit step through the usual move path.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The response carries commitRequired:true, canApply, and blockedReason when a save blocker exists.
2. A course code matching more than one appointment with no day given returns ambiguous with up to 6 options (section, days, time).
3. A code matching no course in the department and term returns a message rather than an HTTP error, and the lecture duration is preserved when the start time changes.
4. The response text states that the path is preview only and execution remains after the user's confirmation.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts#previewNaturalLanguageMove`
`src/server/routes/intelligenceInsightRoutes.ts#/api/intelligence/nl-move` `tests/gemini-schedule-layer.ts`

FR-INT-013 Academic decision lab

The system shall try, in the academic decision lab, moving the time of an existing section to alternative times and count the eligible students whose full wished bundle becomes arrangeable before and after, writing no row, and shall not show a result for fewer students than the privacy floor.

Implemented · Should · Functional · Department head, College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. The privacy floor is 3 students; fewer than 3 or more than 800 eligible students produces no scenarios.
2. Up to 12 courses with groups of 3 to 250 students are analysed, up to 3 sections per course, start times 08:00, 09:30, 11:00, 12:30, 14:00, 15:30 and 17:00 with a shift of at most 300 minutes.
3. A move that collides with an instructor or a room is dropped, and one that helps fewer than 3 is counted as below the floor and not shown.
4. Every result carries a caution that it is not applied and settles neither capacity, instructor consent nor regulations, and the diagnostics panel counts the funnel stages so an empty result explains itself.

EVIDENCE

`src/utils/academicDecisionLab.ts#buildAcademicDecisionLab` `src/server/academicAdvisoryData.ts`
`src/components/AcademicDecisionLabPanel.tsx` `tests/academic-advice-audit.ts`

FR-INT-014 Decision room, automatic optimiser and auto proposal

Inferred

The system shall offer alternatives for a department schedule through the decision room, the automatic optimiser and the auto proposal as scenarios to review, not apply, and shall propose a change only if it does not increase save blockers and does not lower schedule quality.

Implemented · Should · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The optimiser explores 240 attempts with a fixed-seed generator so the same input gives the same result, returns up to 3 options, and declares it does not publish, change days or rooms, and that the gate is human.
2. The auto proposal changes times only and keeps course, instructor, days and room; if the improvement is not safe it returns the schedule unchanged with zero changes.
3. A department with no appointments in the term receives a 200 response pointing to «start of term from the previous term» or import, not a 400 error.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/autopilot`
`src/server/routes/intelligenceRoutes.ts#/api/intelligence/auto-schedule`
`src/server/routes/intelligenceRoutes.ts#/api/intelligence/war-room` `src/utils/scheduleInnovation.ts#runScheduleAutopilot`

FR-INT-015 Custom scheduling rules

Inferred

The system shall let a power administrator create a scheduling rule for a department and term from six types, enable, disable and delete it, and evaluate the schedule and its scenarios against the enabled rules.

Implemented · Should · Functional · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Types: instructor latest end, instructor day off, department day off, course room, max instructor gap, room doorway; any other type returns 400.
2. Max gap is clamped between 30 and 480 minutes (default 120), and a doorway between 1 and 60 minutes.
3. No new rule is created for a retired or released instructor, and a course room must come from the official registry.
4. A user without power rights is refused on create, edit and delete.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/constraints`
`src/utils/scheduleInnovation.ts#evaluateScheduleConstraints`

FR-INT-016 Policy simulation

The system shall simulate the effect of a policy of four types on the term's appointments against the last ten academic years of history, and return the size of the effect on departments, instructors and rooms without changing any appointment or rule.

Implemented · Could · Functional · Department head, College dean, System administrator

ACCEPTANCE CRITERIA

1. Types: a day off, close a building, no classes after a time, growth by a percentage (default 10); any other type returns 400.
2. The university scope is accepted only from an administrator; others are held to their own scope.
3. The response carries a guardrail stating it is a simulation only.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/policy-simulate` `src/utils/advancedIntelligence.ts#simulatePolicy`
`tests/intelligence-scope-api.mjs`

FR-INT-017 Open decisions

Inferred

The system shall keep an open-decisions list per department and term to which the user adds a decision with a title, priority, owner and due date, merge into it decisions inferred from schedule alerts and logical anomalies, and close recorded ones by updating their status.

● Implemented · Could · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. A title under 3 characters returns 400; the title is cut at 140 characters, the detail at 500 and the owner at 100.
2. Priority is low, medium or high (default medium), and status is open or done.
3. Inferred items are up to 5 alerts and 4 non-low anomalies, and count towards the open total.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/open-decisions`

FR-INT-018 Next-term demand forecast

The system shall forecast, for each course, the expected number of sections for the next term as a value and a range from previous terms' sections, registration statistics for the last 6 terms and live radar wishes, together with prerequisite bottlenecks, crowded hours and graduation delay, within the reader's scope and without writing.

● Implemented · Should · Functional · Department head, College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. Each older term is weighted ×0.6, terms of the same season are preferred, and the result is deterministic with no randomness or clock.
2. The reader's scope is applied before computing, not after, and a department is not affected by another department's figures.
3. No student list or student identifier leaves the server; the distribution and reasons are counts only, and the graduation simulation runs up to 12 terms.
4. The backtest hides the measured term and later ones so the future does not leak, and each course's error is Σ|expected – actual| ÷ Σ actual.

EVIDENCE

`src/server/predictiveInsight.ts` `src/utils/predictiveIntelligence.ts`
`src/server/simulationRoutes.ts#/api/simulation/predictive` `src/components/PredictiveInsightPanel.tsx`
`tests/predictive-intelligence-audit.ts` `tests/predictive-backtest-audit.ts`

FR-INT-019 Forecast confidence and action gate

The system shall attach to every forecast figure an explicit confidence of four levels, enable the one-click open action only from medium confidence or higher, and show the reason for disabling it when confidence is weak.

● Implemented · Must · Functional · Department head, College dean

RATIONALE

> So that a guess is not presented as a fact.

ACCEPTANCE CRITERIA

1. The score is a weighted mean: number of terms ÷ 4 with weight 45, stability of the figures weight 25, agreement of history with wishes weight 20, input quality weight 10.
2. Levels: high at 0.70 or more, medium at 0.45 or more, else low; zero terms with no live signal is «history not enough».
3. One term or fewer caps the score at 0.34, no previous term of the same season multiplies it by 0.75, and the backtest error lowers it by at most half.
4. The surplus review action is enabled only for a surplus of 2 or more with enough confidence, and never closes anything automatically.

EVIDENCE

`src/utils/predictiveIntelligence.ts#confidenceFrom` `src/utils/predictiveIntelligence.ts#isActionable`
`src/components/PredictiveInsightPanel.tsx` `tests/predictive-intelligence-audit.ts` `tests/predictive-backtest-audit.ts`

FR-INT-020 Radar insight: shortfall causes and offering quality

The system shall explain, for students who did not reach the required hours, the shortfall cause in a classification that sums to their count, and list the offerings with incomplete data with a way to open their row, within the reader's scope.

Implemented · Should · Functional · Department head, College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. The sum of causes equals the number of students who did not reach 12 hours.
2. The computation is heavy, so it runs on request once per source fingerprint, is held for 10 minutes, and two simultaneous requests share one computation.
3. Up to 5000 quality items are kept in memory and 300 are shown after the reader's scope is applied, with a truncation flag.
4. A request without sign-in is refused (401 or 403).

EVIDENCE

`src/server/radarInsight.ts` `src/server/simulationRoutes.ts#/api/simulation/insight` `src/utils/registrationCauses.ts` `tests/radar-insight-audit.ts`

FR-INT-021 Live schedule layer

Inferred

The system shall give the schedule a live layer of six scenes: overview, pressure map, health and fairness, one-minute brief, start of term and safety net, from one reading that carries pulse, health, fairness, fragility, room resilience, conflict topology and decision memory.

Implemented · Should · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. The approval-blocker number in the layer is the approval bar's own number (same exemption for placeholder instructors and the same room identity).
2. The response is held for 120 seconds per scope (except in demo mode), so a later request returns it without recomputation.
3. The server yields to the main loop between the six readings so other requests are not frozen during computation.
4. The response states that the emergency capability has no screen yet (emergency:false) and that the one-minute brief says what changed since the last safety point.

EVIDENCE

`src/server/routes/intelligenceInsightRoutes.ts#/api/intelligence/living` `src/utils/livingSchedule.ts` `src/components/LivingScheduleLayer.tsx` `src/components/LivingScheduleScenes.tsx`

FR-INT-022 Course impact and appointment memory

The system shall show, for a course, what precedes it, what it opens, the live plan sheets that carry it, the demand on it and its seats, and show for an appointment its origin and the latest 12 recorded events, storing nothing new.

Implemented · Should · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. A missing course or appointment returns 404, outside permissions returns 403, and without sign-in 401 or higher.
2. The plan sheets shown exclude virtual and archived ones, and prerequisites and corequisites come from the sheet's rules.
3. The appointment memory carries a truncated flag when events exceed 12.

EVIDENCE

`src/server/insightRoutes.ts` `src/components/CourseImpact.tsx` `tests/radar-insight-audit.ts`

FR-INT-023 Guide knowledge bank

The system shall keep in the repository an authored knowledge bank covering every feature defined in the Guide, each feature having one to four short standard-Arabic steps and at least two question phrasings in standard Arabic, Kuwaiti dialect and English, with nothing generated at run time.

● Implemented · Must · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Every feature in the registry has a bank entry, its steps number 1 to 4, and the question phrasings are at least 2.
2. Step text is standard Arabic with no dialect words, and the team is called «فريق» not the legacy term.
3. Question phrasings are for understanding only and are never shown to the user.

EVIDENCE

`src/guide/bank.ts` `src/guide/smartGuide.ts#GUIDE_FEATURES` `tests/guide-brain-audit.ts`

FR-INT-024 Guide question understanding within role permissions

The Guide shall understand a question inside the browser with no network request, by splitting it into at most three parts and ranking features by weight, and shall not suggest to a user a feature that the user's role cannot reach, even if it is the closest.

● Implemented · Must · Functional · Standard user, Department head, Schedule committee chair, System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Weight = 0.62 for matching a phrasing + 0.30 coverage + 0.12 title, and the current screen adds 0.025 only when scores are close.
2. Confidence of 0.34 or more is an answer, below it is «unsure» with the closest three; confidence under 0.55, unknown words or a rival within 0.9 of the first adds a «did you mean» list.
3. A question asking how to move a course and how to print the report is split into two parts, and a report of instructors and departments stays one question.
4. A user with permission 7 only is not answered with the backup feature.
5. Accuracy on 160 questions written by two parties who had not seen the bank is at least 90%.

EVIDENCE

`src/guide/brain/engine.ts` `src/guide/smartGuide.ts#canAccessGuideFeature` `tests/guide-brain-audit.ts`

FR-INT-025 Kuwaiti-dialect understanding and spelling correction

The Guide shall normalise the question text and map Kuwaiti and English words to a single concept through an internal lexicon, correct spelling mistakes with an edit distance proportional to the word length, and show nothing of the lexicon to the user.

● Implemented · Should · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Normalisation unifies hamza forms, ta marbuta, alef maksura, elongation, diacritics and Arabic-Indic digits, so a decorated spelling of «the administrative hall 12» becomes its plain form.
2. Correction allows an edit distance of 1 below 8 letters and 2 from 8 letters, and a word under 5 letters is not corrected; a misspelled «clash» maps to its right word.
3. A word after the negators «mo/mob/mish/not» is not counted as requested.
4. A question outside the program's world (weather, salaries, restaurants…) is answered as outside what the Guide knows.

EVIDENCE

`src/guide/brain/normalize.ts` `src/guide/brain/lexicon.ts` `src/guide/brain/engine.ts#correct` `tests/guide-brain-audit.ts`

FR-INT-026 Guide on-screen tours

The Guide shall build a tour for each feature from its own definition and walk it on screen with a state machine that never disappears silently: the element appears, it waits for the user's action, or it states plainly that it did not find it.

Implemented · Should · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The engine looks for the element every 120 ms for 7000 ms, for an optional step 1200 ms, then reports «missing» and keeps looking every 800 ms without taking the card away.
2. The screen's safe command is resent every 8 polls during the search, and an absent optional step is skipped quietly.
3. A tour from another screen opens the menu, navigates and then shows a real step; a step waiting for a selection does not advance by itself.
4. Tour progress (feature and step number) is stored in the browser, so a tour left within a day resumes, a finished one is remembered for seven days, and 40 records are kept.
5. Every tour target and selector exists in the code.

EVIDENCE

`src/guide/tour/runner.ts` `src/guide/tour/tours.ts` `src/guide/awareness.ts#saveTourProgress` `src/components/guide/GuideTour.tsx` `tests/guide-tours-audit.ts` `tests/guide-targets-audit.ts`

FR-INT-027 Guide situational awareness and quiet hints

The Guide shall read the user's situation from the screen, the open dialog, the selected lecture, the import state and recent actions, offer at most three suggestions, and show a proactive hint only within cooldown limits.

Implemented · Could · Functional · Standard user, Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. An approval blocker puts «approval review» first even if its tour was finished before.
2. The situation produces an answer before ordinary suggestions: import stopped at notes, current schedule unreadable, pre-check failed, check complete, selected lecture in conflict.
3. The cooldown between hints is 90 seconds for high confidence, 6 minutes for medium and 12 minutes for low, a hint the user ignored does not recur within 24 hours, and in quiet mode only high-confidence hints appear.
4. The predicted next step needs at least 5 uses, 3 successes and a 0.6 success rate.
5. The session trail holds at most 8 actions in memory only, fixed identifiers with no names or rooms, and dies when the page closes.

EVIDENCE

`src/guide/awareness.ts` `src/guide/sessionTrail.ts` `src/guide/smartGuide.ts#canProactivelyHint` `src/guide/smartGuide.ts#predictedNextFeature` `tests/guide-tours-audit.ts`

FR-INT-028 Interface telemetry intake and anonymous Guide question log

The system shall accept interface telemetry batches from the signed-in user and record Guide events without the user's identity, and mask the digits and names in unanswered questions before they reach the server or are stored.

Implemented · Should · Functional · Standard user, Department head, Schedule committee chair, The system (automatic)

ACCEPTANCE CRITERIA

1. A batch holds up to 20 entries; an entry with a college and department outside the scope is dropped.
2. Kind is one of api, error, offline, sync, guide, otherwise error; name up to 140 characters, message up to 320, duration 0 to 120000 ms and breadcrumbs the last 12 steps.
3. A Guide record carries user id 0 and an empty name.
4. An unanswered question is cleaned in the browser by removing the name after a title, digits and emails with a 90-character limit, and digits are replaced on the server as well.

EVIDENCE

`src/server/routes/guideRoutes.ts` `src/guide/awareness.ts#anonymizeQuestion` `src/guide/brain/engine.ts#maskUnknownWords` `tests/guide-brain-audit.ts`

FR-INT-029 Guide friction points for the root administrator

Inferred

The system shall show the root administrator alone anonymous friction points over the last 30 days against the previous 30, ranked by failure and help rate, together with unanswered questions.

Implemented · Could · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A user other than the root administrator receives 403 «reserved for the main system administration».
2. Ranking is by (failed + abandoned) rate plus 0.35 of the help rate; at most 12 points, 8 items and 30 unanswered questions are shown.
3. Change versus the previous period is the difference in failure rate, and the response declares anonymous:true and windowDays:30.
4. The accompanying interpretation: rising friction after a new release may mean the interface needs redesign, not more explanation.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/guide-friction` `src/utils/clientTelemetry.ts#telemetryGuideJourney` `src/guide/smartGuide.ts#evaluateGuideFriction`

FR-INT-030 Experience health

Inferred

The system shall compute for the reading scope an experience-health score from response times, errors and offline events in the last 14 days, and show the slowest paths and error replays.

Implemented · Could · Functional · Department head, System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Score = 100 − 55 × failure share − 35 × slow share (≥ 1800 ms) − up to 10 for offline events, between 0 and 100.
2. Label: excellent at 90, good at 75, needs follow-up at 55, otherwise needs intervention.
3. p50 and p95 are computed from the last two hours' samples together with client measurements, showing the slowest 5 paths and up to 6 replays.

EVIDENCE

`src/server/routes/intelligenceRoutes.ts#/api/intelligence/experience-health`

NFR-INT-001 Read-only: analytics routes write nothing to the schedule

The simulation, forecast, optimiser, natural-language query and analytical-assistant routes shall not write any change to schedule appointments or rules; every application goes through the usual write path with its validators and an explicit confirmation.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Every response from the simulation, assistant and query routes carries text declaring it a preview or simulation only.
2. A what-if request changes no schedule row (the row fingerprint before and after is identical).
3. The natural-language query accepts neither a commit nor a confirmed argument nor a name implying a write.

EVIDENCE

`src/utils/geminiScheduleLayer.ts#sanitizeGeminiScheduleCalls` `src/server/routes/intelligenceRoutes.ts#/api/intelligence/policy-simulate` `tests/gemini-schedule-layer.ts` `tests/radar-insight-audit.ts`

NFR-INT-002 Privacy: no civil identity or student list leaves

No civil number or national identifier shall be sent to the Gemini service, no student list or student identifier shall leave the server in forecast and radar readings, the Guide shall store no name or room, and the service keys shall stay on the server.

Implemented · Must · Non-functional · The system (automatic), External service

ACCEPTANCE CRITERIA

1. The course and instructor catalogue sent to the service has the identifier and name only, and any key resembling a civil or national number is removed.
2. The student list is removed from the forecast reading before it is sent to the client.
3. The service key is read from a server environment variable and never reaches the client.
4. Tour progress and the Guide profile are kept in the browser with identifiers only.

EVIDENCE

`src/utils/geminiScheduleLayer.ts#buildSmartImportCatalogue` `src/server/predictiveInsight.ts` `src/server/core/gemini.ts` `src/guide/sessionTrail.ts` `tests/gemini-schedule-layer.ts` `tests/predictive-intelligence-audit.ts`

IR-INT-001 Analytics Center screen: scenes and scope

The Analytics Center screen shall show the decision scope bar (college, department, term) and the analysis scope switch, three scenes «understand», «try» and «approve», and tabs covering quality and readiness, before approval, attention, density, campus movement, rooms, instructors, data health, student requests, department fingerprint, plan sheet, schedule improvement opportunity and next-term forecast.

Implemented · Should · Interface · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. When offline, «read only» appears in the scope bar.
2. Dismissing the error notice clears it from the state, and the decision-lab tools include the rules board, decision room, automatic optimiser and policy lab.
3. An action's reading returns to its source and blocks stale answers when the term or scope changes.

EVIDENCE

`src/components/IntelligenceWorkspace.tsx` `src/components/IntelligenceContextBar.tsx` `src/components/IntelligenceScopeSwitch.tsx` `tests/intelligence-reading-scope-audit.ts`

IR-INT-002 Gemini service interface: request constraints and timeout

Inferred

The system shall call the Gemini service from the server only, with an instruction that confines it to understanding the request and forbids it from asking to save, publish, apply or delete, with function-calling mode restricted to the allowed names, and shall return the failure reason to the screen instead of staying silent.

Implemented · Must · Interface · The system (automatic), External service

ACCEPTANCE CRITERIA

1. With no key an empty value is returned and the deterministic fallback applies.
2. The function-call timeout is 45 seconds, the document-reading timeout 180 seconds, temperature 0.1 and a low reasoning level.
3. A service refusal (HTTP code) or a timeout returns a failed field and a reason cut to 300 characters, and is logged on the server.
4. The service reads and proposes; execution is always through the deterministic validators and an explicit confirmation by the user.

EVIDENCE

`src/server/core/gemini.ts#requestGeminiScheduleLayer`

3.12.6 Algorithms

ALG-INT-01 · Analytics Center and Guide

Next-term demand forecast with confidence

Statistics from history

How many sections will each course need next term, with what confidence, and is the confidence enough for a one-click action?

1. Demand value of each past term = distinct sections + ⌈(not registered − vacant seats) ÷ capacity⌉.
2. Same-season terms are used when there are at least two, otherwise all history; the series starts at the course's first appearance.
3. Recency-weighted mean (newest weight 1, ×0.6 per older term) plus a damped linear slope (only from 4+ points, ×0.5, clamped to ±1); range ± ⌈weighted standard deviation⌉ (±1 with fewer than 3 terms).
4. It is blended with live radar wishes ⌈wanted ÷ capacity⌉: weight 0.5 with 3+ terms, otherwise 0.6.
5. Confidence = weighted mean of coverage (45, terms ÷ 4), stability (25, 1 − coefficient of variation), agreement (20) and input quality (10); ×0.75 with no same-season term; capped at 0.34 with one term or fewer; reduced by up to half by the back-test error (predicting each past term from earlier terms only).
6. The one-click «open sections» action is enabled only at medium confidence or above; a surplus of two or more is review only. Prerequisite bottlenecks, crowded hours (60-minute slots) and graduation delay (up to 12 future terms at 15 units) are produced alongside with the same confidence rule.

Inputs

Past terms' sections, registration statistics for the last 6 terms, live radar wishes, student profiles and approved plan rules — within the reader's scope.

Output

Per course: expected value and range, planned count and difference, trend, a reason line, confidence with level and terms used, and an action enabled or disabled with its reason.

Deterministic with no randomness or clock, read-only, counts not names; no figure without earlier terms («not enough history»).

High confidence ≥ 0.70, medium ≥ 0.45, thin-history cap 0.34 · Recency decay 0.6; full coverage at four terms · Back-test penalty up to 50%; errors above 25% are stated

`src/utils/predictiveIntelligence.ts#predictDemand` `src/utils/predictiveIntelligence.ts#confidenceFrom` `src/utils/predictiveIntelligence.ts#backtestDemand` `src/server/predictiveInsight.ts#predictiveReadingFor` `tests/predictive-backtest-audit.ts` `tests/predictive-intelligence-audit.ts`

FR-INT-018 FR-INT-019

ALG-INT-02 · Analytics Center and Guide

Academic decision lab

Search and ranking

Which time shift of an existing section lets more students arrange their entire wished bundle?

1. Eligible people: profiles of the college and department with wishes for the term, on an approved sheet, with at least two offered eligible wished courses; the analysis runs only for 3 to 800 people.
2. Feasibility = can the whole wished bundle be arranged: backtracking over sections (fewest options first) with no-morning, no-afternoon and free-day constraints; counted infeasible beyond 20,000 steps.
3. The 12 most-wanted courses (3 to 250 interested) are taken, up to 3 sections each, and 7 standard starts (08:00 to 17:00) are tried with a shift of at most 300 minutes inside the day window; shifts colliding with an instructor or room are dropped.
4. Gain = feasible after − before; a gain below 3 is never shown, nor a comparison whose «before» count is 1 or 2 (privacy floor).
5. Score = 10 × gain + 3 × removed overlap pairs + min(4, downstream courses) − shift hours − 3 when resources are unverified; the best per section is kept, top 8, plus a combined chain.

Inputs

The college's term rows, student profiles with wishes and constraints, approved sheets, and course pairs that share students.

Output

Up to 8 scenarios with before, after, gain, reasons and cautions, and diagnostics that explain each stage when the result is empty.

Writes no row; a transparent score that claims no global optimum; seats and instructor consent are not decided here.

Privacy floor: 3 students · 7 standard starts: 08:00, 09:30, 11:00, 12:30, 14:00, 15:30, 17:00 · Backtracking cap: 20,000 steps per person

`src/utils/academicDecisionLab.ts#buildAcademicDecisionLab` `src/utils/academicDecisionLab.ts#feasibleAcademicBundle` `tests/academic-advice-audit.ts`

FR-INT-013

ALG-INT-03 · Analytics Center and Guide

Natural-language query layer

Assisted reading, confirmed

What does the user's sentence in the Analytics Center ask for, and which deterministic check should answer it?

1. The sentence is sent to the Gemini service with the scope identifiers and five read-only function names (check_conflicts, find_rooms, check_instructors, check_regulations, simulate_schedule) in a calling mode restricted to them.
2. Sanitising before any execution: at most 8 calls; a name that is not allowed or contains a write word (save, delete, publish, apply…) is dropped; at most 24 arguments, strings cut at 240 characters, and commit/confirmed keys removed.
3. The server executes each accepted call with its own deterministic logic over the permitted department and term timetable; no result is taken from the service.
4. With no key, a refusal, a timeout or invalid output, a rule parser extracts the intent, course number, day and time and produces the same kinds of calls; the response states the source.
5. A move request returns a before/after preview (conflicts, regulations, quality) and requires a separate explicit confirmation through the usual move path.

Inputs

The user's sentence, the user's scope (college, department, term), and the service key when configured.

Output

A sanitised list of calls with their source (service or rule parser), the deterministic check results, and a move preview with no write.

The service only identifies intent: it never decides, writes or confirms; any write happens only after a person confirms in a second step.

Function-call budget 45 seconds; temperature 0.1 · Up to 8 calls, 24 arguments, 240 characters per string

`src/utils/geminiScheduleLayer.ts#sanitizeGeminiScheduleCalls` `src/utils/geminiScheduleLayer.ts#deterministicSchedulingCalls` `src/server/core/gemini.ts#requestGeminiScheduleLayer` `src/utils/naturalQuery.ts#parseNaturalQuery` `tests/gemini-schedule-layer.ts`

FR-INT-009 FR-INT-010 FR-INT-011 FR-INT-012 IR-INT-002

ALG-INT-04 · Analytics Center and Guide

Guide question understanding by lexicon and weights

Deterministic

Which system feature is the user asking about in standard Arabic, Kuwaiti dialect or English, and is the match confident enough to answer?

1. Text is normalised, dialect phrases replaced, Kuwaiti, English and standard words mapped to one concept through the internal lexicon, the rest stemmed; stop-words and negated words («مو», «not») are dropped.
2. A compound question is split into at most three parts at punctuation and connectors.
3. An out-of-vocabulary word of 5+ letters is corrected to the nearest vocabulary word within edit distance 1 (2 for 8+ letters), weighted ×0.85.
4. Term weight = 1 + ln((N+1) ÷ number of features mentioning it) over the knowledge bank, so rare words weigh more.
5. Score of each permitted feature = 0.62 × best phrase overlap (Dice) + 0.30 × coverage + 0.12 × title hits, with small boosts for the current screen (+0.025), suggested (+0.05) and recent (+0.012).
6. An answer is given at a score of 0.34 or more; «did you mean?» alternatives are added below 0.55, with unknown words, or when a rival is within 90%; otherwise «unsure» with up to three closest.

Inputs

The question text, the knowledge bank, the features the user's role can reach, and the current screen.

Output

Per part: an answer with a feature, its confidence and alternatives, a live question, or «unsure» with the closest features.

Runs entirely in the browser with no network request; only features the role can reach are ranked; nothing of the lexicon is shown, and unknown words are masked in the anonymous log.

Answer threshold 0.34; weak-understanding threshold 0.55 · Edit distance 1 (5–7 letters) and 2 (8+ letters)

`src/guide/brain/engine.ts#understand` `src/guide/brain/engine.ts#rank` `src/guide/brain/engine.ts#maskUnknownWords` `src/guide/brain/normalize.ts` `src/guide/brain/lexicon.ts` `tests/guide-brain-audit.ts`

FR-INT-024 FR-INT-025 FR-INT-028

3.12.7 Edge cases

⚠ Gemini service absent or failing

Scenario No key is configured, the service refused the request, or the 45-second timeout elapsed during a natural-language query.

Behaviour The server parses the sentence with the rule parser and completes the same preview, declaring its source deterministic-parser; if nothing is extracted it returns a hint, not an error.

⚠ A proposal that asks to write

Scenario The service returns a call named updateSchedule, or an allowed call with a commit:true or save:true argument.

Behaviour The disallowed call is dropped entirely, the write argument is removed from the allowed one and the rest stays; nothing reaches a write.

⚠ A course code with several appointments

Scenario The user types «move 101 to 11:00» and the department has three sections of that course.

Behaviour The server picks no section; it returns ambiguous with up to 6 options (section, days, time) so the user chooses the section or day.

⚠ A department with no appointments in the term

Scenario The department head opens the decision room, the optimiser or the auto proposal for a term whose schedule is not yet built.

Behaviour A 200 response returns with empty:true and a message pointing to «start of term from the previous term» (a draft that changes nothing before publishing) or to schedule import, not an error.

⚠ Short history for the forecast

Scenario A course with only one term in history, or no term and no live wishes.

Behaviour A one-term confidence is capped at 0.34 so the one-click action stays disabled with its reason shown; zero terms with no live signal groups the course under «insufficient data» with no expected figure.

⚠ A Guide question about live data or outside its world

Scenario The user asks how many students are registered now, asks about today's weather, or names a person after a title.

Behaviour The Guide does not read data and points to the screen that holds the answer; an out-of-world question is answered honestly as outside what it knows; the name after a title is removed before logging.

⚠ A tour element does not appear

Scenario The tour navigates to a screen and the required element does not appear within 7 seconds.

Behaviour The tour reports «missing» and offers what can be done (retry or skip the step) while it keeps looking quietly; it closes only by the user's hand.

⚠ A named instructor outside the department scope

Scenario The user asks the assistant about the gaps of an instructor who is in the university directory but has appointments only in another department.

Behaviour It answers that the instructor has no data in this department and term, and does not mix another department's schedule into the answer.

3.12.8 Standing owner decisions

- Course numbers are 7 digits only: college (2) + department (2) + course (3), for example 0109105; the stored three-digit tail and AdCourseId are never shown on any screen of this module.
- The Gemini service reads the text and proposes; execution is always through the deterministic validators and an explicit confirmation by the user.
- Printed reports keep their design: fixes go in values and data, not layout.

↑ Contents

3.13 · RPT

Reports, queries, printing and export

One question, many lenses; A4 landscape sheets in their approved design; Excel files with seven-digit course numbers.

30	100%	10	15
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.13.1 Overview

This module covers the queries-and-reports screen, which has ten entry points (search and report for instructor, room, time and room-and-time, advanced search, and the department report). All of them open on one shared filter set and fifteen display lenses. Sheets print on A4 landscape without any change to their approved design, and Excel export uses the same filters. It also covers the report data sources on the server: department balance, room load, the visiting-instructor roster and history, the schedule-changes report, the Authority file comparison, and the plan-sheet badge bridge shared by several screens. Every course number shown is seven digits.

Department head · Schedule committee chair · College dean · Vice dean for academic affairs · Registration officer · College registrar · Dean of registration · University registrar · Department advisor · System administrator · Root administrator · The system (automatic)

3.13.2 User journeys

Department head — **Printing the comprehensive department report**

1. Opens the department report from the reports menu and picks college, department and term.
2. Presses the comprehensive-report print button; the sheet list offers this site, with the changes appendix, or all branches when the department has more than one site.
3. Selects a sheet with the arrow keys or a click, then presses Print or Enter.
4. The system opens the print dialog on A4 landscape; each page is measured and scaled to fill the sheet.
5. When printing ends the print attributes are removed and the screen returns to normal.

Schedule committee chair — **Finding a room at a given time and exporting the result**

1. Opens room-and-time from the queries menu.
2. Sets building, hall, days and a time window from a list in half-hour steps.
3. Results and filter chips update immediately from a single set of conditions.
4. Presses Export Excel and receives a file with the same filters and seven-digit course numbers.

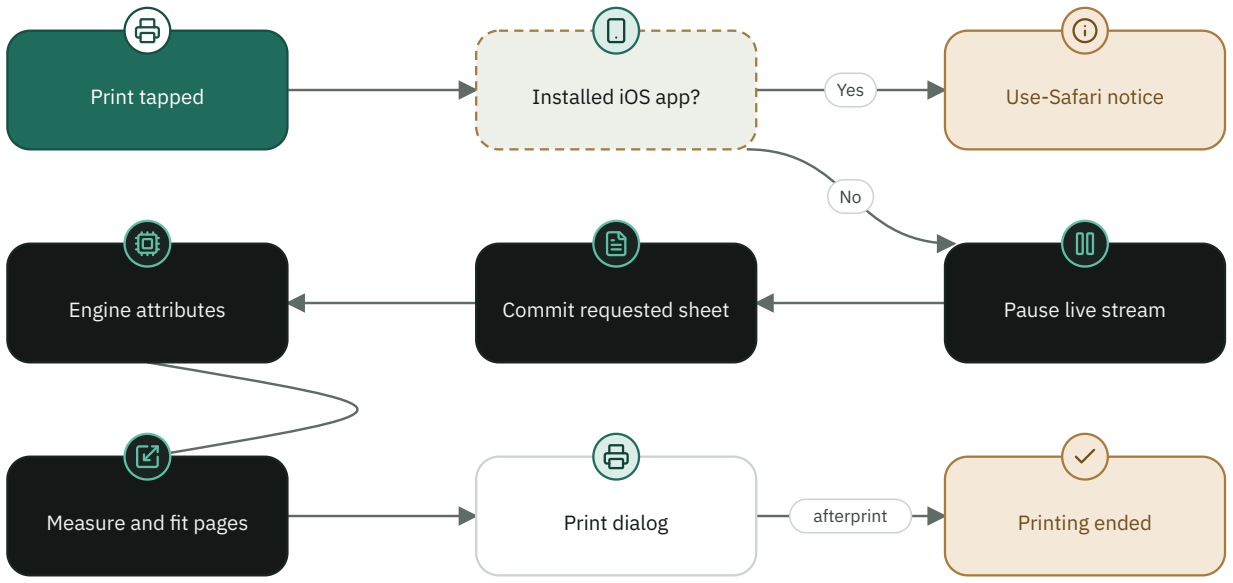
College dean — **Reading the department balance**

1. The dean opens the queries screen and it starts on the department balance lens.
2. Reads one line per department: appointments, instructors, halls, morning and evening shares, fairness and conflicts.
3. Only departments with approved schedules count, and only within the dean's college.
4. Prints the lens with the Print this view button.

3.13.3 Diagrams

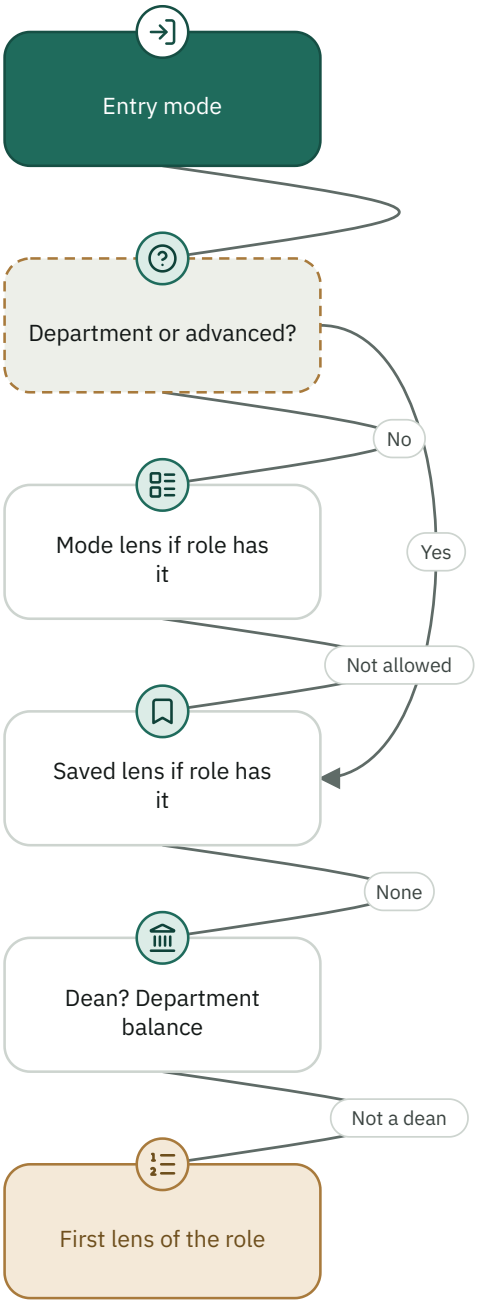
Print command flow

Every step before the print dialog runs synchronously in the same tap; print attributes are lifted only when printing has really ended.



Choosing the opening lens

The opened screen's lens outranks the saved one, except for the department report and advanced search where the saved one wins, and deans start on the department balance.



3.13.4 Screens

Queries and reports center

A single screen with ten entry modes (ReportMode), a lens bar, one filter bar, and the print and export buttons.`src/components/Reports.tsx`

Report sheet chooser

One row per sheet, a single selection, one print button, with keyboard navigation.`src/components/ReportChoicePanel.tsx`

Printed schedule-changes report
A document comparing the current schedule with the imported Authority file, using the same head as the comprehensive report.`src/components/AuthorityPdfReport.tsx`

Official document head
A head shared by the comprehensive report and the changes report.`src/components/PrintComprehensiveHead.tsx`

Occupancy rate grid
A day-by-hour heat map with five levels.`src/components/OccupancyHeatGrid.tsx`

3.13.5 Requirements

FR-RPT-001 Ten entry modes on one screen

The system shall provide ten entry modes for queries and reports, all opening on the same screen on the lens that fits the mode's question, and shall show only the modes the user holds permission for.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer

RATIONALE

> Each mode used to carry its own filters, layout and print path; unifying them keeps deep links and permissions unchanged.

ACCEPTANCE CRITERIA

1. The modes are searchInstructor, searchRoom, searchTime, searchRoomTime, searchAdvanced, reportDepartment, reportInstructor, reportRoom, reportTime and reportRoomTime.
2. Permissions: instructor 8, room 9, time 10, room-and-time 16, advanced search 17, department report 14.
3. The time and room-and-time modes open on the Times lens unless the role lacks it, in which case the Rooms by times lens is used.

EVIDENCE

`src/components/Reports.tsx` `src/App.tsx` `tests/report-print-menus-audit.ts`

FR-RPT-002 One filter set

The system shall apply the same filter set to every lens: college, department, term, instructor, civil number, instructor name, building, hall, course, course number, time window and days.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer, Department advisor

ACCEPTANCE CRITERIA

1. The time window matches any appointment that overlaps it by at least one minute; touching edges do not count.
2. Selecting several days keeps an appointment that falls on any of them.
3. An instructor query written in Arabic-Indic or Persian digits is read as ordinary digits and searches the civil number.
4. The course, instructor, building and hall lists narrow to the other chosen filters.

EVIDENCE

`src/components/Reports.tsx` `tests/course-number-seven-digit-audit.ts`

FR-RPT-003 24-hour time options

Inferred

The system shall list the start and end time options of the query screen in half-hour steps, from the start of the schedule day to its end.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. With the default window 08:00-20:00, 25 values are listed from 08:00 to 20:00 in 30-minute steps.
2. The empty value shows as --:-- and the filter stays inactive until both start and end are set.

EVIDENCE

`src/components/Reports.tsx` `src/utils/scheduleTime.ts`

FR-RPT-004 Fifteen display lenses

The system shall provide fifteen lenses, each answering one specific question from the same filtered results, and every lens shall have a print sheet with content.

Implemented · Must · Functional · Department head, Schedule committee chair, College dean, Registration officer

ACCEPTANCE CRITERIA

1. The lenses are: All, Week, Instructors, Courses by sites, Rooms, Occupancy rate, Rooms and instructors, Courses by rooms, Rooms by times, Times, Visiting instructors, All terms, Load fairness, Department balance, Decision board.
2. No lens prints a blank sheet titled undefined; every print kind has an entry in the title table.

EVIDENCE

`src/components/Reports.tsx` `tests/report-print-menus-audit.ts`

FR-RPT-005 Role-based lens lists

Inferred

The system shall shorten the visible lens list for the deans and registration roles to what concerns them, without the server blocking anyone, and shall open each on the first lens of its list.

Implemented · Should · Functional ·

College dean, Vice dean for academic affairs, Dean of registration, University registrar, College registrar, Registration officer

ACCEPTANCE CRITERIA

1. The dean sees 7 lenses starting with the department balance; the vice dean sees 13.
2. The dean of registration, university registrar and college registrar see 7 lenses; the registration officer sees 6 (no Rooms by times).
3. Everyone else sees all lenses except All terms (a second face of Visiting instructors), Department balance for non-administrators, and Decision board for non-eligible roles.
4. A saved lens the role does not hold falls back to the first available lens.

EVIDENCE

`src/components/Reports.tsx`

FR-RPT-006 Department balance

The server shall return one line per department in the reader's scope with appointments, instructors, halls, morning and evening shares, fairness, quality and conflicts, and departments outside the scope shall not enter the calculation or the totals.

Implemented · Must · Functional · College dean, Vice dean for academic affairs, Dean of registration, System administrator

ACCEPTANCE CRITERIA

1. GET /api/reports/department-balance requires permission 14.
2. Morning means starting before 14:00 (840 minutes), counted per appointment day; departments are sorted by appointment count, descending.
3. For the dean and vice dean only final schedules count, and a department with none is omitted.
4. The totals' scope label is the university, the college or the scope, depending on the reader.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx` `tests/dean-notify-audit.ts` `tests/role-scope-audit.ts`

FR-RPT-007 Room load and free windows

The server shall return the occupancy of the reader's halls computed from the whole term without revealing any course or instructor of another department, and shall count only verified official halls.

Implemented · Must · Functional · Schedule committee chair, Department head, College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. Each hall carries day, start and end in minutes and a mine flag, together with the day start and end.
2. A non-administrator with no college, or a department outside the scope, receives 403; a reader covering the whole college reads its halls without choosing a department.
3. A row pending room confirmation occupies nothing, while an unlinked text booking occupies the same hall.
4. Ordering is by building then hall, numerically: 7/31 before 8/22 and 9 before 10.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `tests/room-occupancy-audit.ts`

FR-RPT-008 Load fairness from one formula

The load fairness lens shall compute its index from the same fairness engine as the department balance, shall exclude records that are not persons, and shall show the average load in credit hours.

Implemented · Must · Functional · Department head, College dean

ACCEPTANCE CRITERIA

1. A faculty-placeholder record does not enter the index, and two instructors with equal load score at least 90.
2. The unit is written as average load (credit hours).

EVIDENCE

`src/components/Reports.tsx` `src/server/routes/reportsRoutes.ts` `tests/dean-notify-audit.ts`

FR-RPT-009 Occupancy rate

The occupancy rate lens shall show, for each day-by-hour cell, the number of lectures and occupied halls in five color levels, plus each day's hall occupancy rate, for the site or for the department family across all its colleges.

Implemented · Should · Functional · Schedule committee chair, College dean, Registration officer

ACCEPTANCE CRITERIA

1. Level 1 to 5 is ceil(lectures / grid maximum x 5); an empty cell is level 0.
2. Pressing a cell opens its lectures, and the lens prints 18 halls per page.

EVIDENCE

`src/utils/occupancyHeat.ts` `src/components/OccupancyHeatGrid.tsx` `src/components/Reports.tsx` `tests/occupancy-heat-audit.ts`

FR-RPT-010 Visiting-instructor query across department colleges

The system shall read the visiting instructors of a scientific department and their teaching across every college holding a matching department the reader may access, and shall mark the total incomplete and name what is not accessible instead of presenting a partial total as complete.

Implemented · Must · Functional · Department head, College dean, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. The visiting-roster, visiting-teaching and visiting-history routes accept permissions 7, 8, 9, 10, 14, 16 and 17.
2. visiting-teaching without college, department or term returns 400, and an unauthorised scope returns 403.
3. A section is (college, department, course, section code); a lecture and a lab appointment of one section count as one section and their hours add up.
4. Weekly hours = appointment duration x number of days, and a row arriving from two sources is not counted twice.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/server/visitingTeachingRead.ts` `src/utils/visitingTeaching.ts` `tests/visiting-teaching-audit.ts` `tests/visiting-family-audit.ts`

FR-RPT-011 Visiting history across terms

The system shall show the visiting-instructor history across terms as one year matrix shared by screen and sheet, and shall keep anyone delegated in a past term even if they have left the department directory since, with a mark.

Implemented · Should · Functional · Department head, College dean

ACCEPTANCE CRITERIA

1. The detailed year window defaults to 6; choices are 4, 6, 10 and all years, and a choice that changes nothing is not offered for a shorter archive.
2. A term with no visiting instructor does not become a column.
3. An instructor no longer in the directory carries listedNow = false.

EVIDENCE

`src/utils/visitingHistory.ts` `src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx` `tests/visiting-teaching-audit.ts`

FR-RPT-012 Courses by rooms

The courses-by-rooms lens shall list appointments whose registered room type differs from the course's room type, grouped by course.

Implemented · Should · Functional · Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. If either the course or the room has no type there is no comparison and no mismatch.
2. Sorting is by Arabic name then section code.

EVIDENCE

`src/components/Reports.tsx` `tests/course-room-type-sync-audit.ts`

FR-RPT-013 Read-only decision board

The decision board lens shall appear only to the department head, dean, vice dean and top administrator, read-only and within their scope.

Implemented · Should · Functional · Department head, College dean, Vice dean for academic affairs, Root administrator

ACCEPTANCE CRITERIA

1. The department head sees the chosen department only; the deans see the college departments their scope covers.
2. Anyone else does not see the lens at all, and the server still checks every request by scope.

EVIDENCE

`src/utils/decisionBoardScope.ts` `src/components/Reports.tsx` `tests/decision-surfaces-audit.ts`

FR-RPT-014 Seven-digit course number

The system shall show the course number in every printed report, on the query screen and in the Excel file as seven digits (college 2 + department 2 + course 3), built with authorityCourseCode and authorityDepartmentCode, and shall never show the stored three-digit code or AdCourseId.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer

RATIONALE

> Owner decision: the full Authority number is how the reader identifies a course.

ACCEPTANCE CRITERIA

1. Code 105 with department key 0109 is shown as 0109105, and code 5 as 0109005.
2. Arabic digits are read as digits; a code containing letters or of unusual length stays as stored and no number is invented.
3. All code chips on screen come from courseNumberOf, and the course-number filter matches both it and the stored code.

EVIDENCE

`src/components/Reports.tsx` `src/utils/authorityAcademicCodes.ts` `src/server/routes/reportsRoutes.ts` `src/components/AuthorityPdfReport.tsx` `tests/course-number-seven-digit-audit.ts`

FR-RPT-015 Excel export with screen filters

The system shall generate an Excel (xlsx) file from /api/reports/excel/:type using the same filters as the screen, with a seven-digit course number column.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer, College dean

ACCEPTANCE CRITERIA

1. Types are ListofTeacherCourseExcel (permission 14), TeacherWithCourseExcel (8) and ScheduleExcel (7); any other type returns 400 and a missing permission returns 403.
2. There are 13 columns in this order: college, department, course number, course name, section, units, hours, instructor, civil number, time, days, building, hall.
3. The sheet is named the schedule sheet, in a right-to-left workbook with an auto-filter, sorted by course name, section, then start time.
4. A visiting instructor's name carries the visiting suffix, and the default term is the newest.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx` `tests/dean-notify-audit.ts` `tests/course-number-seven-digit-audit.ts`

FR-RPT-016 Export through the request reader

The export route shall read rows through the same request reader as the screen, so deans receive final schedules only, and the civil number shall not be exported to view-only roles.

Implemented · Must · Functional · College dean, Vice dean for academic affairs, Registration officer

ACCEPTANCE CRITERIA

1. A dean's file contains no drafts that the screen does not show to the dean.
2. The civil-number column passes through instructorsForReader and is empty for readers who are not entitled.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/server/core/scheduleReads.ts` `tests/dean-notify-audit.ts`

FR-RPT-017 A titled sheet for every lens

The system shall print every lens on a sheet whose document title starts with SCHEDULE followed by the report name, and shall restore the original title after printing.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. The two comprehensive kinds have separate titles: the comprehensive schedule report and the same with all branches.
2. Printing one document clears the other documents (changes report and book) before the print command.

EVIDENCE

`src/components/Reports.tsx` `tests/report-print-menus-audit.ts`

FR-RPT-018 A4 landscape on every engine

The system shall print every report on A4 landscape with margins of 10 mm top, 8 mm sides and 14 mm bottom, shall paint the sheet rotated inside the portrait page of Safari and Chrome on iOS, and shall pin the geometry on desktop Chromium.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. The @page rule states A4 landscape explicitly rather than millimetres alone.
2. The root is stamped data-print-rotate for Safari and iOS Chrome, and data-print-chromium for desktop Chromium.
3. The attributes are removed on afterprint or when the reader returns to the page, never by a timer during preview.

EVIDENCE

`src/styles/08-print.css` `src/components/Reports.tsx` `tests/print-fit-audit.ts`

FR-RPT-019 Filling the sheet without clipping

The system shall measure every print page with the print rules before the print command, enlarging it up to 1.8 or shrinking it down to 0.6 of its size so that it fits a full A4 sheet.

Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Media are evaluated against the sheet, not the window: a (max-width:640px) rule does not apply to a 297 mm sheet.
2. A comprehensive page rotated in Safari is not enlarged a second time (ceiling 1).
3. Sheet fonts (weights 300-700 and Mono 400-600) are preloaded when reports open, before the first measurement.

EVIDENCE

`src/utils/printFit.ts` `tests/print-fit-audit.ts`

FR-RPT-020 Printed page row budgets

The system shall distribute each sheet's rows over pages with a capacity measured on A4 landscape so that no sliver spills onto a blank sheet, and shall keep a comprehensive-report row on the same page as on the registration sheet.

Implemented · Must · Functional · Department head, Registration officer

ACCEPTANCE CRITERIA

1. List 10 rows, instructors 7, room occupancy 18, room free windows 14, room directory 17, times 10, fairness 11, department balance 14, visiting 7, all terms 11.
2. A comprehensive page holds 28 rows, on the first page and every following one.

EVIDENCE

`src/components/Reports.tsx` `src/utils/printFit.ts` `tests/print-fit-audit.ts`

FR-RPT-021 Instructor report scope

The system shall print the instructor report for the chosen instructor, or, when none is chosen, for all instructors in the current view with their full load in the requested scope (the selected college or all colleges within permissions), and shall state the reason when nobody qualifies.

Implemented · Must · Functional · Department head, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. With no chosen instructor the view's instructors enter the report and a row with no instructor does not.
2. A name query matches across colleges; a civil-number query with no results in view gives a message, not an empty sheet.
3. The displayed count is unique instructors, not rows.

EVIDENCE

`src/utils/instructorReportScope.ts` `src/components/Reports.tsx` `tests/report-print-menus-audit.ts`

IR-RPT-001 Sheet chooser for the two composite reports

The interface of the comprehensive report and the schedule-changes report shall show a single list with one row per sheet, one selection and one print button, without losing any sheet that was available before unification.

Implemented · Should · Interface · Department head, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. Comprehensive: this site, with the changes appendix, and all branches when the department has more than one site.
2. Changes report: this site and all branches, each also with the civil number after the instructor name.
3. Arrows navigate (left is next in RTL), Home and End jump to first and last, Enter prints, Escape closes and restores focus.
4. The saved choice stays if still available, otherwise the first is chosen.

EVIDENCE

`src/components/ReportChoicePanel.tsx` `src/utils/reportChoices.ts` `src/styles/48-report-choice.css` `tests/report-choices-audit.ts`

FR-RPT-023 Changes report and Authority comparison sources

The server shall provide the changes report for one department and a comparison of the current schedule with the approved Authority file, within the reader's scope, and shall limit deans to an approved schedule.

Implemented · Must · Functional · Registration officer, College registrar, Department head, College dean

ACCEPTANCE CRITERIA

1. schedule-changes without term, college or department returns 400, outside scope returns 403, and is limited to 90 requests per minute.
2. Deans cannot read the report of an unapproved schedule: 403 with a message saying it is shown after approval.
3. authority-pdf-diff returns 404 when no copy exists, and with meta=1 returns a null draftId instead of 404.
4. A currently empty table is still compared and every original row is classified deleted, so the report is not suppressed.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/components/AuthorityPdfReport.tsx` `src/utils/authorityReportOrdering.ts` `tests/schedule-changes-audit.ts` `tests/changes-baseline-audit.ts` `tests/authority-net-diff-audit.ts`

FR-RPT-024 Authority report row ordering

The system shall order the rows of the Authority file report in the original PDF order, and shall place all true additions after all source rows while keeping the site sequence.

Implemented · Must · Functional · Registration officer, College registrar

ACCEPTANCE CRITERIA

1. A moved or renumbered original row stays in position by its original sourceOrder.
2. An invalid sourceOrder (negative, at least 1,000,000 or not a number) is treated as missing and sorted last.

EVIDENCE

`src/utils/authorityReportOrdering.ts` `tests/authority-net-diff-audit.ts`

FR-RPT-025 Plan-sheet badge bridge

The system shall provide a single server-side bridge whose source is set by plan-sheet review and read by the other screens, so every screen shows a course's sheet badge and requirement type from the same source, and the client shall make one shared fetch.

Implemented · Must · Functional · Department head, Schedule committee chair, Standard user

ACCEPTANCE CRITERIA

1. GET /api/plan-badges requires sign-in, returns only what falls inside the account's scope, and sets Cache-Control: no-store.
2. A badge does not cross department boundaries: the key is department:course, and the owner department's sheet is not overwritten by a service sheet.
3. A missing or failing source yields an empty result and does not break the screen.
4. Badges are refetched without cache on the schedule:plan-review-updated event.
5. A course side (new, old or shared) is read for several departments in one read, and returns null for a department with no new/old pairing so it stays one list.

EVIDENCE

`src/server/planBadgeShared.ts` `src/utils/planBadges.ts` `src/server/planReviewRoutes.ts` `tests/consistent-badges-audit.tsx` `tests/sheet-membership-audit.ts`

NFR-RPT-001 Printed sheet design is preserved

The system shall fix errors in printed reports by correcting values and data rather than layout, and shall reuse existing print classes when a layout change cannot be avoided.

Implemented · Must · Non-functional · Department head, Registration officer

RATIONALE

> A standing owner decision; his feedback on the instructor report (load, hours, course numbers) was resolved in the data, not the design.

ACCEPTANCE CRITERIA

1. Any unavoidable change is checked on a Chrome-printed A4 landscape PDF before merging.
2. The head is shared by the comprehensive report and the changes report through one component.

EVIDENCE

`src/styles/08-print.css` `src/components/PrintComprehensiveHead.tsx` `tests/print-fit-audit.ts`

NFR-RPT-002 Print command in the same tap

The system shall not place a network read or timer between the tap and the print command; it shall pause the live stream, commit the sheet and measure pages synchronously, and then request printing.

Implemented · Must · Non-functional · Department head, Schedule committee chair

RATIONALE

> Some browsers silently ignore the command if it comes later than the tap, and Safari may not print while a live stream is open.

ACCEPTANCE CRITERIA

1. Approval state is read with the scope before printing, not at print time, and a failed read prints the document without a stamp.
2. The appendix is fetched and awaited before printing is requested, and its failure prints the comprehensive report alone.
3. If the browser swallows the command, the stream resumes, the title returns and appendices are lifted without removing the sheet attributes.

EVIDENCE

`src/components/Reports.tsx` `tests/print-fit-audit.ts`

NFR-RPT-003 Printing from the installed iOS app

Inferred

The system shall detect the iOS app installed on the home screen, shall not attempt to print, and shall show a notice for 5 seconds asking the user to open the schedule in Safari or use a computer.

Implemented · Should · Non-functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. The notice has role=status and aria-live=polite, and any previous notice is removed before a new one is shown.
2. In any other environment the function returns false and printing continues.

EVIDENCE

`src/utils/pwaPrintHelp.ts`

DR-RPT-001 **Report data sources**

The query screen shall read its appointments from /api/schedules, which restricts them to the account's permissions, and shall read room load, department balance, visiting instructors and the changes report from /api/reports/* routes, so no reading falls outside the reader's scope.

Implemented · Must · Data · Department head, College dean, Registration officer, The system (automatic)

ACCEPTANCE CRITERIA

1. The routes are visiting-roster, visiting-teaching, visiting-history, schedule-changes, authority-pdf-diff, excel, department-balance and room-load.
2. College level: a reader covering the whole college reads the visiting instructors of all its departments without choosing one; a reader who does not cover it receives 403.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/components/Reports.tsx` `tests/role-scope-audit.ts`

IR-RPT-002 **Export Excel button**

The queries toolbar shall show an Export Excel button as a direct download link carrying the current view's filters, omitting empty, zero or false values.

Implemented · Should · Interface · Department head, Schedule committee chair, Registration officer

ACCEPTANCE CRITERIA

1. The link requests ListofTeacherCourseExcel and sends the time window only when both start and end are set.
2. Days are sent as sun=true, mon=true through thr=true.

EVIDENCE

`src/components/Reports.tsx` `tests/dean-notify-audit.ts`

3.13.6 Algorithms

ALG-RPT-01 · Reports, queries, printing and export

Load fairness index

Statistics from history

Is the teaching burden spread fairly among the scope's instructors, and who carries clearly more than average?

1. Records that are not persons («faculty» placeholder, rows without an instructor) are excluded.
2. Instructor burden = 8 × attendance days + 5 × gap hours + 2.5 × meetings starting before 09:00 + 4 × meetings starting at 16:00 or later + 1.2 × weekly teaching hours.
3. Fairness score = clamp(100 − 72 × coefficient of variation of burdens, 0, 100).
4. Labels: 90+ very fair, 78+ balanced, 62+ needs balancing, otherwise unfair; a warning (up to 5) for anyone above average by more than max(8, 28% of the average).
5. Department balance reuses the same engine per department, alongside the morning share (start before 14:00, per meeting day) and conflicts.

Inputs

The scope's appointments with days and times, and the instructor registry.

Output

The score and label, average burden and spread, instructor profiles ranked by distance from the average, and warnings.

One formula for the fairness lens and for department balance; departments outside the scope do not enter the calculation.

Coefficient-of-variation multiplier 72 · Label thresholds 90, 78 and 62 · Morning cut-off 14:00

`src/utils/livingSchedule.ts#buildFairnessEngine` `src/server/routes/reportsRoutes.ts#registerReportsRoutes` `tests/dean-notify-audit.ts`

FR-RPT-006 FR-RPT-008

ALG-RPT-02 · Reports, queries, printing and export

Occupancy rate grid

Deterministic

How many lectures fall in each hour of each day, and how many of the scope's halls are occupied?

1. Lectures with a valid day and time are kept; the hour axis runs from day start to day end, widened for any lecture outside it.
2. A lecture occupies every hour it intersects (09:30–10:45 falls in both the 9 and 10 o'clock hours).
3. Cell: number of lectures and distinct halls; a hall counts once per hour, and a lecture without a hall counts as a lecture but occupies none.
4. Day rate = occupied hall-hours ÷ (halls in scope × hours); the week rate likewise.
5. Colour level = ⌈count ÷ grid maximum × 5⌉, zero for an empty cell, and the peak is reported.

Inputs

Rows of the scope (the site or the department family across its colleges) and a hall key that distinguishes two halls with the same number at two sites.

Output

A day × hour grid with count, halls, rate and level, each day's and the week's rate, and the peak.

Pure function; no existing lecture is hidden after the day window is narrowed.

5 colour levels and a 60-minute cell

`src/utils/occupancyHeat.ts#buildOccupancyHeat` `tests/occupancy-heat-audit.ts`

FR-RPT-009

3.13.7 Edge cases

⚠ Instructor report with no instructor chosen

Scenario The user opens the screen and prints the instructor report before choosing anyone.

Behaviour A sheet is printed for all instructors in the current view; if there are none a message says there are no instructors in view, and the menu does not stay open silently.

⚠ Stored course code with letters or another department key

Scenario The catalogue holds a code such as ENG101, or 0102102 for a course of a department other than the one shown.

Behaviour A code with letters is shown as stored, a course of another department is not rebound to the displayed department, and with no department key the stored code stays and no number is invented.

⚠ Dean opens the screen before any schedule is approved

Scenario The dean's saved lens is All and nothing is approved yet, so results are empty.

Behaviour The department balance lens opens automatically once instead of an empty screen, and a dean who later returns to the list by choice is not moved again.

⚠ Hall not yet linked to the registry

Scenario An imported row carries building and hall as text only, or a row is pending room confirmation.

Behaviour Normalised text (spacing, case) occupies the same hall, whereas a pending row occupies nothing and does not inflate occupancy rates.

> ⚠ **Visiting instructors in a college the reader cannot access**
>
> **Scenario** The matching department has appointments in another college outside the reader's scope.
>
> **Behaviour** Only allowed colleges are counted, the inaccessible ones are named, and the total is marked incomplete (complete = false).

> ⚠ **Excel file for a dean**
>
> **Scenario** A dean exports the schedule of a department that has an unapproved draft.
>
> **Behaviour** The file contains final schedules only, like the screen, and the civil-number column is empty for readers who are not entitled.

3.13.8 Standing owner decisions

- Course numbers are seven digits only: college (2), department (2), course (3), for example 0109105, in every printed report, query screen and Excel file; the stored code and AdCourseId are never shown.
- Printed reports keep their design: fix values and data, not layout; an unavoidable change reuses the existing print classes and is checked on a Chrome-printed A4 landscape PDF before merging.
- The header decides the department in the Authority file (SWRSCHA): department and branch come from the page header even if a course number carries another department's code, and a three-digit tail alone never links a course.

↑ Contents

3.14 · NTF

Notifications, activity and live channel

A bell derived from state, a live channel that announces change, an immutable trail, and a calendar you subscribe to.

30	100%	20	36
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.14.1 Overview

The module gathers everything that tells a user what happened without being asked: the notification center and bell, push notifications (Web Push) for department coordinators and faculty cards, the live channel that broadcasts schedule changes across screens and instances, presence and team colleagues, stale-write protection through Refresh-and-retry, the immutable activity trail, row memory, and public subscription calendars (ICS). All of it relies on the server remaining the arbiter of scope and permission and on notification staying light: news without rows, and a short pulse instead of a stream held open all day.

Schedule committee chair · Department head · College dean · Vice dean for academic affairs · University registrar · College registrar · Registration officer · Department advisor · Faculty member · Student · Public visitor · Standard user · System administrator · The system (automatic)

3.14.2 User journeys

Schedule committee chair — **An instructor request reaches the bell at once**

1. The instructor sends a request from the card and the server emits a notify pulse.
2. The chair's open screen asks the pulse within seconds and finds notifiedAt has advanced.
3. The bell reads /api/notifications after 700 ms and a "just arrived" line floats for 9 seconds.
4. If phone alerts are active on the device the notice arrives there instead of a desktop notification.
5. Pressing the notice marks it read and opens the workspace on the department.

Faculty member — The department's decision reaches the instructor's phone

1. The instructor opens the card and presses the bell, grants permission, and the subscription is saved under the card.
2. The department fixes three items of the request within 10 minutes.
3. The server checks the text carries no room trace and sends one notice "3 appointments fixed".
4. The instructor taps the notice and the worker opens the same card.

Department advisor — A colleague got to the same request first

1. The advisor opens a section request and sees a colleague in the chip with a working-on-this mark.
2. The colleague saves a decision and the request's version number rises.
3. The advisor saves with the old stamp and the server answers 409 stale-write, touching nothing.
4. The bar "your colleague edited this before you" appears; the advisor presses Refresh, sees the decision and retries.

Faculty member — A calendar subscription that does not break

1. The instructor adds the /api/public/cal/... link from the card to the phone's calendar.
2. The phone re-reads the file every 6 hours and the server answers 304 when nothing changed.
3. The department publishes a new card and the same link stays valid because its key follows the instructor.
4. If no live card link remains an empty calendar with a stopped description arrives and lectures clear quietly.

College dean

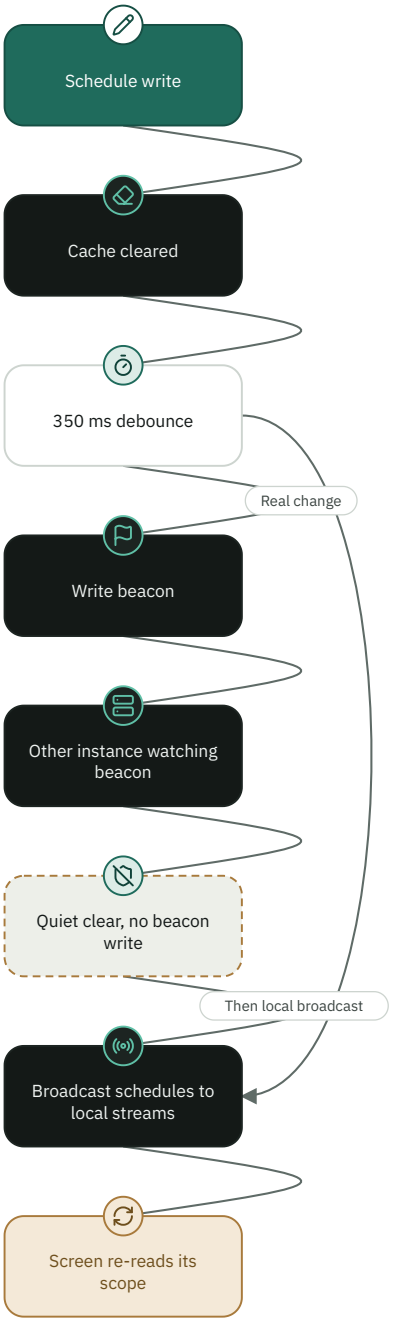
The dean reads the trail

1. The dean opens the trail sheet and the server fixes the audience: the vice dean and department heads of the college.
2. Their faces appear with a ring measuring the last 7 days and a timeline for the two terms.
3. Filtering by person or kind makes the server re-apply the verdict on each line.
4. Opening a line shows "before ← after" with no edit button.

3.14.3 Diagrams

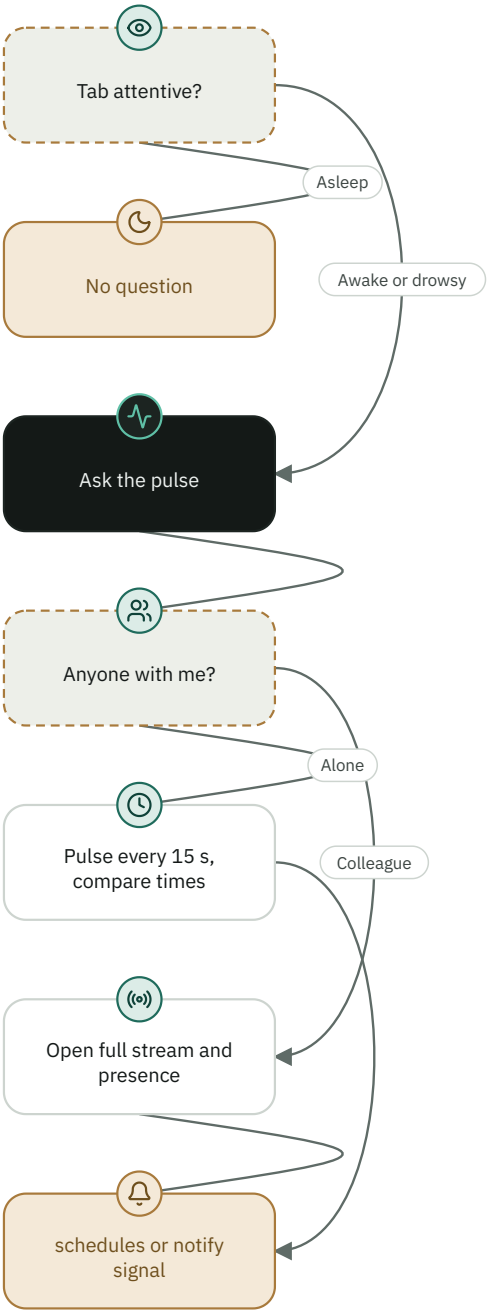
Path of a schedule-change announcement

The news carries no rows; each screen re-reads its own scope through the scoped road.



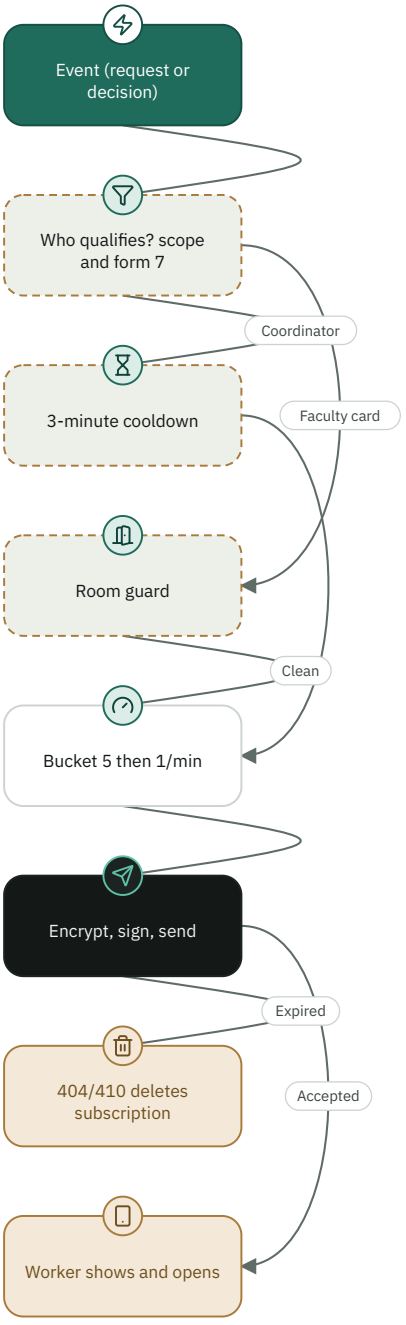
Pulse or stream?

A tab alone on its board holds no open stream.



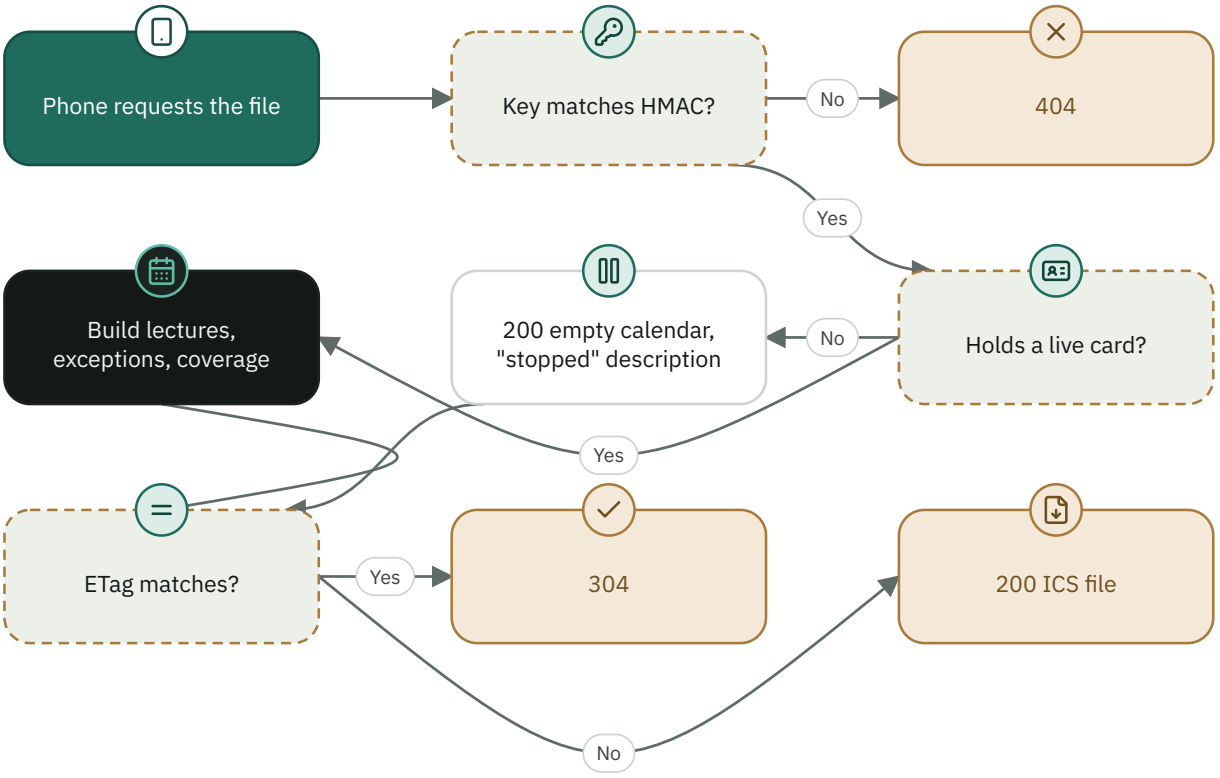
Life of a phone notice

A notice waits only for its own owner and never carries a room trace.



Stable calendar subscription

The key belongs to the instructor, not the link; no live card pauses without breaking.



3.14.4 Screens

Notification center (bell)

A bell, badge, three-group panel and floating notice, holding the "enable alerts on your phone" card.`src/components/NotificationCenter.tsx`

Phone alerts card

Enable, test and stop the subscription, plus a three-step iPhone guide.`src/components/PushCard.tsx`

Activity trail (side sheet)

Faces of who worked, a timeline, and person and kind filters; read-only.`src/components/trail/TrailSheet.tsx`

Last-edit stamp

"Last edit: person · yesterday 10:20" on hovering a row.`src/components/trail/TrailStamp.tsx`

Row "Why?"

The row's origin, its change history and its facts.`src/components/RowHistory.tsx`

Colleague faces

Up to 5 faces, "+N", and the list of those present.`src/components/ColleagueFaces.tsx`

Refresh-and-retry bar

A calm amber bar with a Refresh button.`src/componen ts/StaleWriteToast.tsx`

3.14.5 Requirements

FR-NTF-001 **Bell derived from current state**

The system shall derive bell notifications from the live state of schedules and requests at the moment GET /api/notifications is called, and shall not store any notification, so that no notice survives the removal of its cause.

Implemented · Must · Functional · Schedule committee chair, Department head, College registrar, College dean

ACCEPTANCE CRITERIA

1. After a department schedule is accepted and the bell is queried again, the pending-approval notice is gone and an accepted notice with the done tone appears.
2. A request with no current term and no explicit term returns { termId: 0, items: [] }.

EVIDENCE

`src/server/routes/notificationsRoutes.ts` `src/utils/notificationCenter.ts` `tests/notification-center-audit.ts`

FR-NTF-002 **Notification scope and tones**

The system shall build each role's notifications (committee chair, department head, registration, deans, dean of registration) inside that role's scope only, and shall order them by tone (alert, action, waiting, done) and then by most recent time.

Implemented · Must · Functional · Schedule committee chair, Department head, Registration officer, College dean

ACCEPTANCE CRITERIA

1. A user scoped to one department receives no notice about another department.
2. An alert-tone notice precedes an action-tone notice even if it is older.
3. A returned schedule idle for more than 3 days raises an alert-tone notice for the dean or vice dean.

EVIDENCE

`src/utils/notificationCenter.ts` `src/server/routes/notificationsRoutes.ts` `tests/notification-center-audit.ts` `tests/dean-notify-audit.ts`

FR-NTF-003 **Two-term bell**

The system shall read the bell for both the current term and the planning term (the newest unfinished term with activity), shall prefix planning-term item ids with term-<number>: and carry the term name, and shall answer for one explicit term alone when termId is passed.

Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. When the planning term differs from the current one, its item detail is prefixed with the term name.
2. With an explicit termId no planning term is read and no item carries the term- prefix.

EVIDENCE

`src/server/core/notifications.ts` `src/server/routes/notificationsRoutes.ts` `tests/notification-center-audit.ts`

FR-NTF-004 Feature notices and fault isolation

The system shall append to the bell notices for plan sheets under review (up to 4 groups, then a single "more" item), the registration radar, students who got stuck, Masari students' section needs (up to 4 departments out of the first 30), section requests, and advertising campaigns for the root administrator account outside the demo sandbox, and a failure in any one of them shall not drop the bell.

● Implemented · Should · Functional · Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. A department with two sheets of the same kind yields one notice, not two.
2. When the radar read fails the error is logged and the bell still returns the remaining items.

EVIDENCE

`src/utils/featureNotifications.ts` `src/server/routes/notificationsRoutes.ts` `tests/feature-notifications-audit.ts`

DR-NTF-005 Server-side read marks

The system shall store read marks per user on the server through POST /api/notifications/seen, keyed by term (seenKey), capped at 1000 ids with the newest kept and at most 200 characters per id, and shall permit the write for every role, read-only roles included.

● Implemented · Must · Data · Standard user, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. Sending 1050 ids keeps the last 1000 and drops the oldest.
2. A previous-term notice with the same id does not hide the new badge of the current term's notice.
3. Re-sending what is already stored writes nothing.

EVIDENCE

`src/utils/notificationSeen.ts` `src/server/routes/notificationsRoutes.ts` `src/server/roleGuard.ts` `tests/notification-center-audit.ts`

IR-NTF-006 Bell interface

Inferred

The system shall present one bell with a badge counting unread items that are required of the user (capped at 99+) and a panel with three groups ("Required of you", "Waiting on others", "Done"), shall float a newly arrived urgent notice for 9 seconds, and shall carry the count in the window title.

● Implemented · Should · Interface · Schedule committee chair, Department head, Registration officer

ACCEPTANCE CRITERIA

1. Three unread required items show 3 on the bell; after "mark all as read" the badge disappears.
2. The page title becomes "(3) ..." and returns to its original when nothing unread remains.

EVIDENCE

`src/components/NotificationCenter.tsx` `tests/notify-bell-yield-audit.ts` `tests/notification-center-audit.ts`

FR-NTF-007 Opening a notice on its department

When a notice is pressed the system shall mark it read and navigate to the screen it belongs to, carrying its college and department so the screen opens on it, and shall not route a role to a screen it does not hold.

● Implemented · Should · Functional · Department head, College dean, Schedule committee chair

ACCEPTANCE CRITERIA

1. A department head's notice about registrar notes opens Schedule Changes, not the schedule workspace.
2. Instructor-request items are not shown to a role without the schedule screen (form 7).

EVIDENCE

`src/utils/notificationCenter.ts` `src/utils/notifyFocus.ts` `src/components/NotificationCenter.tsx` `tests/dean-notify-audit.ts` `tests/notification-center-audit.ts`

NFR-NTF-008 Bell cost

The system shall memoise the student-queue aggregate and the blocking-conflict count for an accepted department for one minute (60,000 ms), shall clear them when a schedule changes or the queue is written, and shall coalesce notify pulses caused by registration-queue decisions into one pulse every 3 seconds.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Twenty consecutive queue decisions emit one notify pulse, not twenty.
2. A failed read is not cached, and a read begun before invalidation is not kept after it.

EVIDENCE

`src/server/notificationCache.ts` `src/server/core/notifications.ts` `src/server/core/liveChannel.ts` `tests/student-journey-audit.ts`

FR-NTF-009 Live schedule channel

The system shall broadcast a light event (schedules, with changedAt) to every open screen through GET /api/schedules/events after any schedule write with a 350 ms coalescing delay, carrying no rows, shall restrict the route to forms 7, 8, 9, 10, 14, 16 and 17, with a cap of 400 connections, a keep-alive ping every 25 seconds and a 4000 ms retry.

Implemented · Must · Functional · Schedule committee chair, Department head, Department advisor

ACCEPTANCE CRITERIA

1. Saving six cards in one drag reaches each subscriber as one schedules message.
2. The 401st connection is closed at once and its screen falls back to reading on demand.
3. The event payload contains no rows: each screen re-reads its own scope through the normal scoped road.

EVIDENCE

`src/server/routes/scheduleLiveRoutes.ts` `src/server/core/liveChannel.ts` `src/components/Schedules.tsx` `tests/live-pulse-http-audit.ts` `tests/live-pulse-audit.ts`

FR-NTF-010 Cross-instance change broadcast

The system shall write a beacon document on every real change and watch it in every server instance; when a change arrives from another instance it shall clear the schedule, queue and bell caches and rebroadcast locally without rewriting the beacon, to prevent an endless echo.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A write on a second instance reaches a screen held on the first instance as a schedules message.
2. The instance that wrote does not receive its own write back from the beacon.
3. A change made inside a demo sandbox does not write the beacon.

EVIDENCE

`src/server/core/liveChannel.ts` `src/db/beaconEcho.ts` `tests/live-pulse-audit.ts`

FR-NTF-011 Board presence

Inferred

The system shall keep each user's position on a board (cell under the pointer, card in hand, row open in the editor) in memory only and shall broadcast the roster to members of that same board only, as named fields with no password hash or login name; a mark is stale after 45 seconds and a silent connection is dropped after 90 seconds.

Implemented · Must · Functional · Schedule committee chair, Department advisor, Department head

ACCEPTANCE CRITERIA

1. A user from another college, an administrator included, does not see the names of those present on a board they are not scoped to.
2. A tab frozen for more than 45 seconds has its mark emptied and stays listed until 90 seconds.
3. POST /api/auth/presence only alters the calling user's own stream.

EVIDENCE

`src/server/core/liveChannel.ts` `src/server/routes/scheduleLiveRoutes.ts` `src/server/routes/authRoutes.ts` `src/components/schedulePresence.ts` `tests/live-pulse-http-audit.ts` `tests/colleague-guard-measure-audit.ts`

FR-NTF-012 Presence and change pulse

The system shall answer GET /api/schedules/pulse from memory with no database read, returning { company, changedAt, notifiedAt } and a no-store header, shall count in company the other people at their screens on the same board (not tabs), and shall return 403 to an account without permission so that the tab stops asking.

Implemented · Must · Functional · Schedule committee chair, Department advisor

ACCEPTANCE CRITERIA

1. A colleague with three tabs counts as one, and my own second tab counts as nobody.
2. A tab untouched for a while (active=0) or whose last pulse is older than 90 seconds is not company.
3. A pulse without conn, or for a scope not allowed, is answered with company = 0.

EVIDENCE

`src/server/routes/scheduleLiveRoutes.ts` `src/server/livePulse.ts` `tests/live-pulse-audit.ts` `tests/live-pulse-http-audit.ts`

NFR-NTF-013 Browser pulse cadence

The browser shall ask the pulse every 15 seconds while awake, every 8 seconds for a team that has company, every 60 seconds while drowsy, and never while asleep, shall open the full stream only when company exists, and shall tell the server it is leaving with sendBeacon.

Implemented · Should · Non-functional · Schedule committee chair, Department advisor

ACCEPTANCE CRITERIA

1. A sleeping tab sends no pulse, and the first look back asks at once.
2. An answer older than the latest the tab has seen does not make old news look new (the largest value is kept).

EVIDENCE

`src/utils/livePulse.ts` `src/utils/pageAwake.ts` `tests/live-pulse-audit.ts`

FR-NTF-014 Advising and registration team presence

The system shall tell members of the same team and scope (team + college + department + term) who is working now and on what (a section request, advisor seats, or a student case), with a "working on this" hold that expires 120 seconds after the last touch, at most 40 colleagues per answer, and each person listed once however many tabs they hold.

Implemented · Should · Functional · Department advisor, College registrar, Registration officer, Schedule committee chair

ACCEPTANCE CRITERIA

1. An advisor does not see the registration team and vice versa.
2. Two colleagues meeting on the same section request raises a notice and does not block saving.
3. The answer carries no account number or connection id: the peer id is derived and irreversible.

EVIDENCE

`src/server/teamPresence.ts` `src/utils/teamWork.ts` `src/server/routes/scheduleLiveRoutes.ts` `tests/colleagues-advising-registration-audit.ts`

IR-NTF-015 Colleague faces chip

Inferred

The system shall present presence as one chip for all teams: up to 5 faces with the name's initial and colour followed by "+N", where one tap opens the list of those present, most recently active first, with what each is working on, closed by Esc or a tap outside.

● Implemented · Should · Interface · Schedule committee chair, Department advisor, College registrar

ACCEPTANCE CRITERIA

1. Seven colleagues appear as five faces and "+2".
2. A colleague in a background tab is shown as away and ordered after those at their screens.

EVIDENCE

`src/components/ColleagueFaces.tsx` `src/components/ColleagueHoldNotice.tsx` `src/utils/colleagueFace.ts` `tests/colleagues-advising-registration-audit.ts` `tests/colleague-guard-measure-audit.ts`

FR-NTF-016 Stale-write protection

The system shall answer 409 with the code stale-write and the message "A colleague edited this before you. Refresh and retry." to any write on section requests, advisor seats or registration-queue decisions that carries a version stamp older than the current one, touching nothing, and the screen shall show a calm bar with a "Refresh" button that re-reads; a write without a stamp proceeds as before.

● Implemented · Must · Functional · Department advisor, College registrar, Registration officer, Schedule committee chair

ACCEPTANCE CRITERIA

1. A decision on a section request with rev + 7 returns 409 with stale-write and leaves the request unchanged.
2. An empty text stamp "" is valid and means nothing decided yet; 1.5, "1e3" or a negative number counts as not sent.
3. The Refresh button is disabled while reading and then folds the bar.

EVIDENCE

`src/utils/staleWrite.ts` `src/components/StaleWriteToast.tsx` `src/server/sectionRequestsRoutes.ts` `tests/colleagues-advising-registration-audit.ts`

FR-NTF-017 Push subscription

The system shall accept a Web Push subscription for an account (POST /api/push/subscribe) or for a faculty card by its token, after verifying that the endpoint is https on port 443 with no credentials and at most 2048 characters and that the p256dh key is an uncompressed P-256 point, shall bind the device to the login session, and shall delete a subscription only for its owner.

● Implemented · Must · Functional · Standard user, Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. An http endpoint, one on port 8443, or one with user and password is rejected with 400.
2. One device is one record: subscribing the same endpoint twice updates rather than duplicates.
3. A stopped or expired card answers 404 and accepts no subscription.

EVIDENCE

`src/server/pushRoutes.ts` `src/server/webPush.ts` `src/utils/pushClient.ts` `src/components/PushCard.tsx` `tests/web-push-audit.ts`

FR-NTF-018 Department coordinator push

The system shall push to the devices of users who hold the request inbox (form 7 or system administration) within the item's scope a notice when an instructor sends a request or reply, carrying the count present at send time, replacing the previous one by a department tag, with a 3-minute cooldown per department and kind stored in the database rather than in one instance's memory.

● Implemented · Should · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Three requests within 3 minutes for one department produce one notice.
2. An inactive, locked or form-7-less user receives no notice.
3. Two server instances share the same cooldown.

EVIDENCE

`src/server/pushRoutes.ts` `src/server/pushMessages.ts` `tests/web-push-audit.ts`

FR-NTF-019 Faculty card push

The system shall push to a faculty card a single notice that replaces the previous one (fixed, rejected, alternatives, department reply, training window opened) which gathers department decisions of the last 10 minutes and counts them ("3 appointments fixed"), shall not send any notice carrying a room trace in its title or body, and shall not push to a stopped card but erase its subscriptions.

● Implemented · Must · Functional · Faculty member

ACCEPTANCE CRITERIA

1. Three items fixed together arrive titled "3 appointments fixed" with the card-decisions tag.
2. A request naming room B-204 in any field sends no notice and the reason is logged.
3. A mixed-kind batch is summarised as icons with counts.

EVIDENCE

`src/server/pushRoutes.ts` `src/server/pushMessages.ts` `src/server/pushCard.ts` `tests/web-push-audit.ts`

NFR-NTF-020 Delivery limits

The system shall limit delivery with a bucket per subscription (5 consecutive notices, then one per minute), at most 50 devices per delivery, a 10-second call timeout and a 24-hour TTL, shall encrypt the payload (RFC 8291) to at most 3,993 plaintext bytes and sign it with VAPID, and the route shall wait for the hook at most 2,500 ms without throwing.

● Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Pressing Test six times in a row sends 5 and answers 429 "wait a minute".
2. A 404 or 410 from the push service deletes the subscription, while a 429 is a transient failure that does not.
3. A slow push service delays the route's reply by no more than 2.5 seconds.

EVIDENCE

`src/server/webPush.ts` `src/server/pushRoutes.ts` `tests/web-push-audit.ts`

FR-NTF-021 Subscription rotation and sign-out

The system shall accept a browser's subscription rotation (POST /api/public/push/rotate) by proof of ownership with the old endpoint and its secret in a constant-time comparison, moving the record to the new one under the same owner, and shall drop device subscriptions bound to a session on explicit sign-out but not on idle expiry.

● Implemented · Should · Functional · Standard user, Faculty member

ACCEPTANCE CRITERIA

1. A rotation with a wrong old secret answers 404 and changes nothing.
2. A rotation for a card that was stopped answers 410 and deletes the subscription.
3. An explicit sign-out removes only that session's devices.

EVIDENCE

`src/server/pushRoutes.ts` `public/push-sw.js` `tests/web-push-audit.ts`

IR-NTF-022 Push service worker

The push service worker (/push-sw.js) shall run with scope /push/ in the app and /r/<token> on a faculty card, shall clip the title to 120 characters, the body to 240 and the tag to 64, shall open only a same-origin link, shall focus an open tab instead of opening a new window, and shall cache no page nor answer from memory except the "updating" page on 500, 502, 503, 504 or a network failure.

● Implemented · Must · Interface · Faculty member, Standard user

ACCEPTANCE CRITERIA

1. A notice with a cross-origin link opens the worker's scope start, not that link.
2. Unregistering app workers during recovery keeps the push worker.
3. A 503 on a card navigation is replaced with the updating page, status 503 and header Retry-After: 5.

EVIDENCE

`public/push-sw.js` `src/utils/pushClient.ts` `src/server/pushCard.ts` `tests/web-push-audit.ts` `tests/updating-page-audit.ts`

FR-NTF-023 Activity trail recording

The system shall write one trail line after every successful write request (status below 300 and not a dry run) on a route in a closed list: who, which team, which scope, what, on what, and before ← after, with maximum lengths of 90 for target, 220 for detail, 90 for each of before/after, 80 for name and 120 for seat, and shall record no opens, views or calculate-only calls.

● Implemented · Must · Functional · The system (automatic), Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A conflict-check or preview POST writes no line; PUT /schedules/<id> writes an edit line carrying the row id.
2. A request refused with 409 records nothing.
3. A re-save with no difference (skip) is not written.

EVIDENCE

`src/server/activityTrail.ts` `src/utils/activityTrail.ts` `src/db/demoActivityTrail.ts` `tests/activity-trail-audit.ts`

FR-NTF-024 Who sees the trail

The system shall read the trail for the current and previous terms only and apply the owner's decision to the audience: the dean sees the vice dean and department heads, the vice dean nothing, a department head sees the department and its teams, a committee chair, advising head or college registrar sees their own team, the university registrar sees registration heads and their teams, a member sees only their own actions, and a system administrator sees all, with the verdict re-applied to each line before it is sent.

● Implemented · Must · Functional ·

College dean, Vice dean for academic affairs, Department head, Schedule committee chair, University registrar, System administrator

ACCEPTANCE CRITERIA

1. A vice dean receives audience = none and an empty list.
2. A deputy sees the team but not the head's own actions.
3. A response carries at most 160 lines (truncated = true beyond), a read budget of 240 spread over keys, and a person filter reads up to 200 lines for that key.

EVIDENCE

`src/utils/activityTrail.ts` `src/server/activityTrail.ts` `src/components/trail/TrailSheet.tsx` `tests/activity-trail-audit.ts`

FR-NTF-025 Read-only trail and last-edit stamp

The system shall answer 405 with Allow: GET to any non-GET method under /api/activity-trail, a system administrator included, and shall serve GET /api/activity-trail/stamp/<row> as a last-edit stamp to anyone whose scope contains the row, naming the person for those entitled to see them and otherwise the team.

● Implemented · Must · Functional · College dean, Department head, Schedule committee chair, System administrator

ACCEPTANCE CRITERIA

1. A DELETE or PUT on a trail line answers 405 and the line count is unchanged.
2. A stamp for a row outside scope returns 403, and for a missing row 404.
3. A dean sees the team name in the stamp, not a member's name they are not entitled to see.

EVIDENCE

`src/server/activityTrail.ts` `src/components/trail/TrailStamp.tsx` `src/components/trail/TrailButton.tsx` `tests/activity-trail-audit.ts`

FR-NTF-026 Row memory: why is it here

The system shall show on each row a "Why?" button that reads GET /api/schedules/<id>/history on open (up to 12 events from the audit log with a truncated flag), the row's origin badge if an approval door wrote it, and computed fact chips, shall show nothing when there is nothing to say, and shall refuse with 403 a row outside scope.

● Implemented · Should · Functional · Schedule committee chair, Department head

RATIONALE

> No new storage: it reads what was already written.

ACCEPTANCE CRITERIA

1. A row with no history, origin or facts renders nothing.
2. A row that came from the registration radar carries the Registration Radar badge with who applied it and when.
3. More than 12 events show "and older than that...".

EVIDENCE

`src/components/RowHistory.tsx` `src/server/insightRoutes.ts` `tests/row-reasons-audit.ts`

FR-NTF-027 Public subscription calendar (ICS)

The system shall serve a public schedule link as a calendar file (GET /api/public/ics/<token>) as text/calendar, inline, with a weak ETag computed without DTSTAMP and Cache-Control: no-cache so that unchanged content returns 304, as a weekly recurrence between the declared term bounds (otherwise 16 weeks with an "estimated" tag in the name), an optional reminder of 0 to 120 minutes, a suggested refresh every 6 hours, and shall remove a lecture cancelled on a specific date from that date only.

● Implemented · Must · Functional · Public visitor, Student, Faculty member

ACCEPTANCE CRITERIA

1. A second request with If-None-Match for an unchanged file returns 304 with no body.
2. ?alarm=500 is clamped to 120 minutes and ?alarm=-5 to 0.
3. A staff-card, survey, students or request link returns 404 and serves no calendar.
4. With no current term an instructor feed answers 409.

EVIDENCE

`src/server/routes/publicCalendarRoutes.ts` `src/utils/icalendar.ts` `tests/calendar-ics-audit.ts` `tests/term-calendar-audit.ts`

FR-NTF-028 Stable instructor feed and quiet pause

The system shall serve GET /api/public/cal/<token>/p<id>-<key>.ics with a key that follows the instructor (a 32-hex-character HMAC) so that republishing the card does not break it, shall drop dates the instructor is absent from (cancelled or handed to a colleague) and add dates they cover as single days, and when no live card link remains shall answer 200 with a valid empty calendar of the same name and a description saying the subscription stopped, resuming by itself when a new card is issued.

● Implemented · Must · Functional · Faculty member

RATIONALE

> Owner decision of 2026-10-29.

ACCEPTANCE CRITERIA

1. A key with or without the .ics suffix is accepted, and a key altered by one character returns 404.
2. After the last live link is revoked: 200 with no VEVENT and the REFRESH-INTERVAL kept.
3. The key reveals no civil ID and cannot be guessed without the secret.

EVIDENCE

`src/server/routes/publicCalendarRoutes.ts` `src/server/calendarSecret.ts` `src/server/core/shareLinks.ts` `tests/calendar-ics-audit.ts`

NFR-NTF-029 Calendar key secret

The system shall derive the calendar key from CALENDAR_SECRET when the operator sets it and otherwise from the stored shared secret under a calendar-specific label, lazily and once per instance, and shall not keep a failed read, so that no cold start generates a new secret and kills subscriptions on phones.

● Implemented · Must · Non-functional · The system (automatic), System administrator

ACCEPTANCE CRITERIA

1. With CALENDAR_SECRET set, existing subscriptions stay exactly as they are.
2. Two instances reading the same shared secret return the same key for the same card.
3. Rotating CALENDAR_SECRET invalidates every key.

EVIDENCE

`src/server/calendarSecret.ts` `src/server/core/shareLinks.ts` `tests/calendar-ics-audit.ts`

FR-NTF-030 One-day reminder file

Inferred

The system shall serve GET /api/public/ics-reminder as an inline text/calendar file that opens an "add to calendar" dialog, with a valid YYYY-MM-DD date d (otherwise 400), a title t of at most 120 characters and a note n of at most 300, a UID stable from date and title, two alarms (3 hours before and 8 hours after), and Cache-Control: no-store, with no name or number in the link.

● Implemented · Could · Functional · Student, Public visitor

ACCEPTANCE CRITERIA

1. d=2026-13-40 answers 400 "bad date".
2. Re-adding the same reminder updates the entry rather than duplicating it (stable UID).
3. ?dl=1 changes Content-Disposition to attachment.

EVIDENCE

`src/server/routes/publicCalendarRoutes.ts` `src/utils/icalendar.ts` `tests/calendar-ics-audit.ts`

3.14.6 Algorithms

ALG-NTF-01 · Notifications, activity and live channel

Live broadcast coalescing, foreign-write detection and stale-write refusal

Deterministic

When are open screens told of a change, and when is a write that did not see the latest edit refused?

1. Every schedule write is coalesced into a 350 ms window so each subscriber receives one row-less event.
2. The beacon document counts all writes and each instance's writes; foreign writes = (total delta) − (this instance's delta).
3. A counter going backwards counts as a foreign write; an instance never rewrites the beacon on receiving another's change, so there is no echo.
4. A write whose version stamp differs from the current one is refused with 409 and code stale-write, touching nothing; a write without a stamp proceeds.

Inputs

Write events, beacon counters, the sent stamp and the current version.

Output

One broadcast event per window, or a stale-write refusal.

A change is announced across all instances once; no write lands over an unseen colleague edit; the event carries no data.

350 ms coalescing, 400-connection cap, 25-second keep-alive.

`src/server/core/liveChannel.ts#listenForScheduleChangesAcrossInstances` `src/db/beaconEcho.ts#foreignWrites` `src/utils/staleWrite.ts#expectedRev` `src/server/sectionRequestsRoutes.ts` `tests/live-pulse-audit.ts` `tests/live-pulse-http-audit.ts`

FR-NTF-009 FR-NTF-010 FR-NTF-016

3.14.7 Edge cases

⚠ Stream cap reached

Scenario A 401st live-channel connection arrives while the instance holds 400 streams.

Behaviour The connection is closed at once without being registered; its screen falls back to reading on demand and to asking the pulse, and no error is shown.

⚠ Beacon echo

Scenario Two server instances watch the same beacon and one writes a change.

Behaviour The writing instance does not receive its own write and the other clears its caches quietly and broadcasts locally without writing the beacon, so the news does not bounce between instances forever.

⚠ Calendar with no live card

Scenario The instructor has no live card link left (stopped or expired) while the phone is still subscribed.

Behaviour The server answers 200 with a valid empty calendar of the same name and a description that the subscription stopped, so lectures clear quietly instead of a failure keeping them, and it resumes alone when a new card is issued.

⚠ Room name in notice text

Scenario An instructor-request item mentions a room (such as B-204) in the course name or alternatives.

Behaviour Room tokens are scrubbed from the course name and, if any trace remains in the final text, the notice is not sent and the reason is logged: the rule is never left to the text alone.

⚠ Stamp invalid or absent

Scenario An old tab or internal tool sends a write with no stamp, or a stamp like 1.5 or "1e3".

Behaviour An invalid stamp is treated as not sent so the write proceeds as before, whereas a valid but older stamp is refused with 409 stale-write and no change.

⚠ Deputy and the trail

Scenario A team head's deputy opens the trail sheet.

Behaviour The deputy sees the team's actions and their own but not the head's own actions (hideHolder), the sender's name is withheld in the send log, and the decision is made on the server, not in the interface.

3.14.8 Standing owner decisions

- The trail is read-only: no line is edited or deleted, not even by a system administrator, and who sees it is decided by role and scope under the owner's decision.
- The stable subscription pauses quietly (200 with an empty calendar) when no live card remains and resumes alone (owner decision of 2026-10-29).
- Nothing is pushed to a phone except what awaits its own owner, and a faculty notice never carries a room trace.
- Every course number a reader sees has seven digits (college 2 + department 2 + course 3, such as 0109105), and the printed reports keep their design.

↑ Contents

3.15 · ENG

Engagement and public presence

Student survey, partner advertising, audience dashboard and the public pages.

27	100%	40	40
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.15.1 Overview

This module covers everything that faces people who have no account: the student survey and the case-status page, advertising on public pages, partnership profiles and campaign reports for companies, the shared kit of public pages (colours, fonts, security policy, invalid-link page), the landing page and the presentation deck. For the main administrator it covers the Ads and Audience dashboard screens and the story mode. Measurement is anonymous by design: no name, civil ID or network address is stored, and the figures shown to companies are aggregated. No advertisement is ever shown on a page that asks for an identity.

Student · Public visitor · Partner organisation · Root administrator · The system (automatic)

3.15.2 User journeys

Student — **A student submits a course request and follows its status**

1. Opens the department link and enters the 12-digit civil ID; Arabic-Indic and Persian digits are accepted.
2. If a request already exists in this survey, the case number is requested; before it only the first letter of the name is revealed.
3. Chooses the request type (new course, conflict between two courses, or graduate) and then the courses from the list the department has actually taught.
4. Submits and receives a case number derived from the stored request itself.
5. Opens /m on the same link, enters the civil ID and the case number, and sees what arrived, when, and the committee and registration decisions if any.

Root administrator — Creating and approving an advertising campaign

1. Opens Ads and presses New campaign; the wizard opens on its first step.
2. Sets company, category, slot, dates and the destination link (public https).
3. Sets the target seasons, colleges or courses; the campaign is saved as a draft.
4. Uploads an image of the exact slot size; if rejected for quality he decides explicitly whether to override the warning.
5. Accepts the advertising rules and approves; if capacity or category conflicts, approval is refused with a written reason.
6. Sends the report link to the company, or rotates it so that the old link stops working.

Partner organisation — A company opens the partnership profile and asks for a campaign

1. Opens its private link; the first open is recorded for the administration without tracking any person.
2. Reads the aggregated audience figures, the ad slots with their capacity and the season calendar.
3. Picks a slot, season and number of weeks in the What will you get calculator and sees estimates drawn from the site record.
4. Presses New campaign and fills in name, mobile number and an optional email.
5. Returns with a success notice; the administration receives a bell notification until contact is recorded.

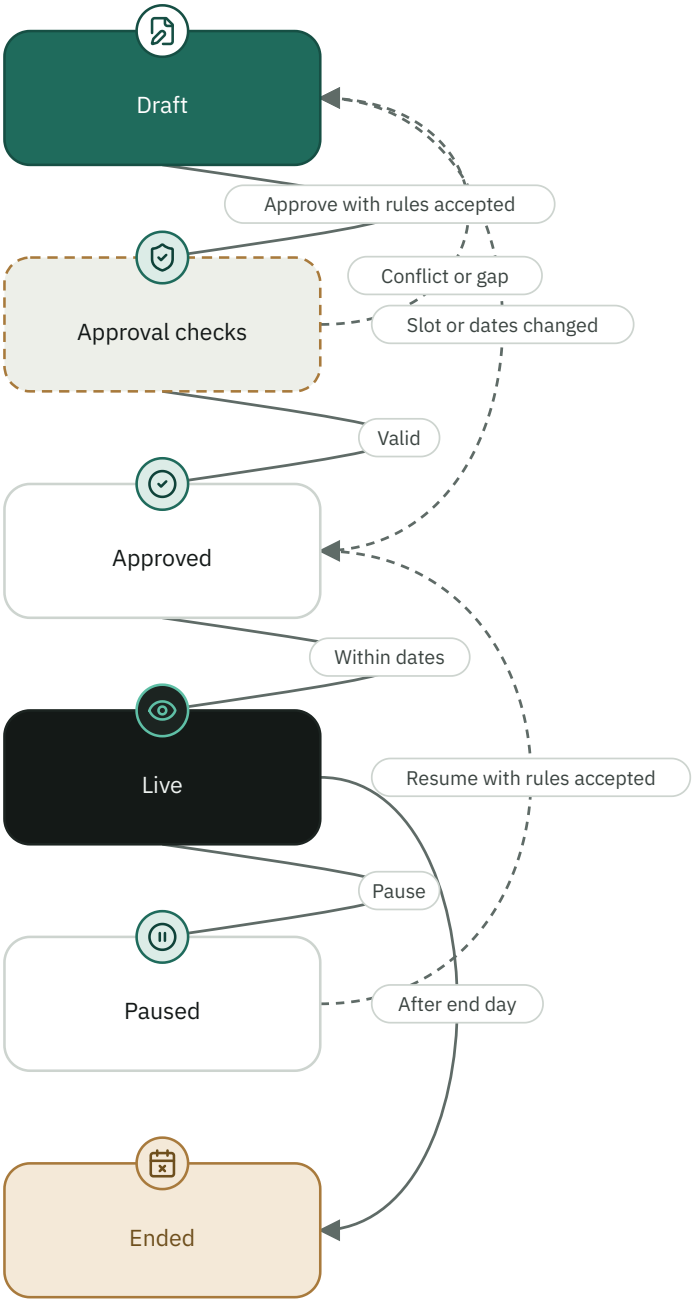
Root administrator — Presenting the audience in a company meeting

1. Opens the Audience dashboard and reviews the headline numbers, peak times and what students request.
2. Starts story mode; it enters full screen with five slides advancing every 8 seconds.
3. Pauses the autoplay with the space bar, or navigates with arrows or swipes.
4. Ends on Your next step (a free 7-day trial) and closes with Esc.

3.15.3 Diagrams

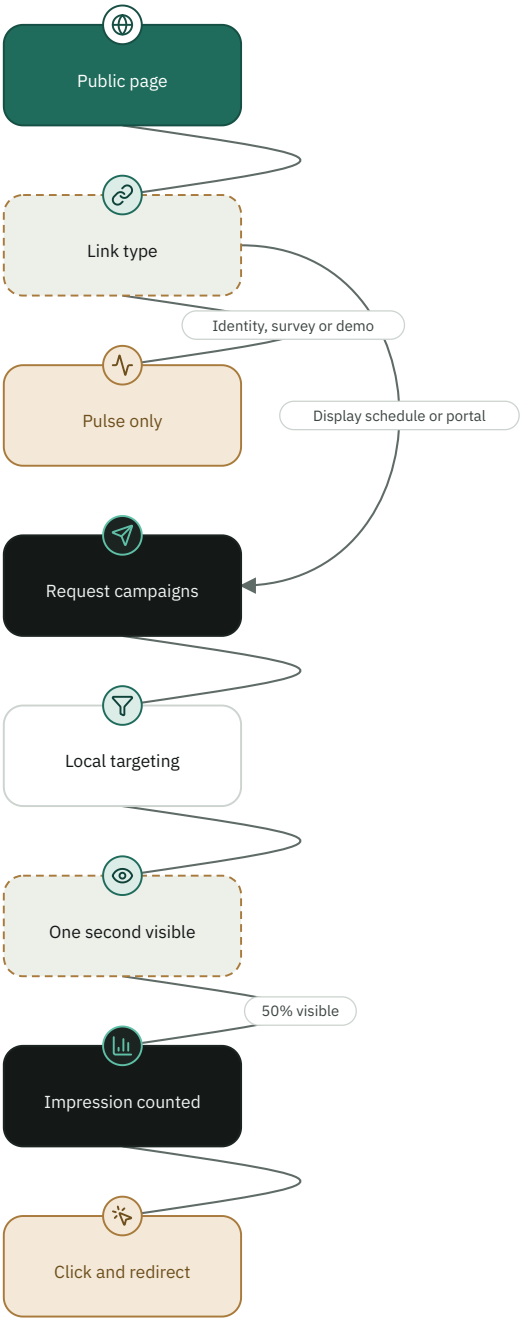
Campaign lifecycle

A campaign is shown only when approved and within its dates; changing its slot, category, dates, seasons or colleges after approval returns it to draft.



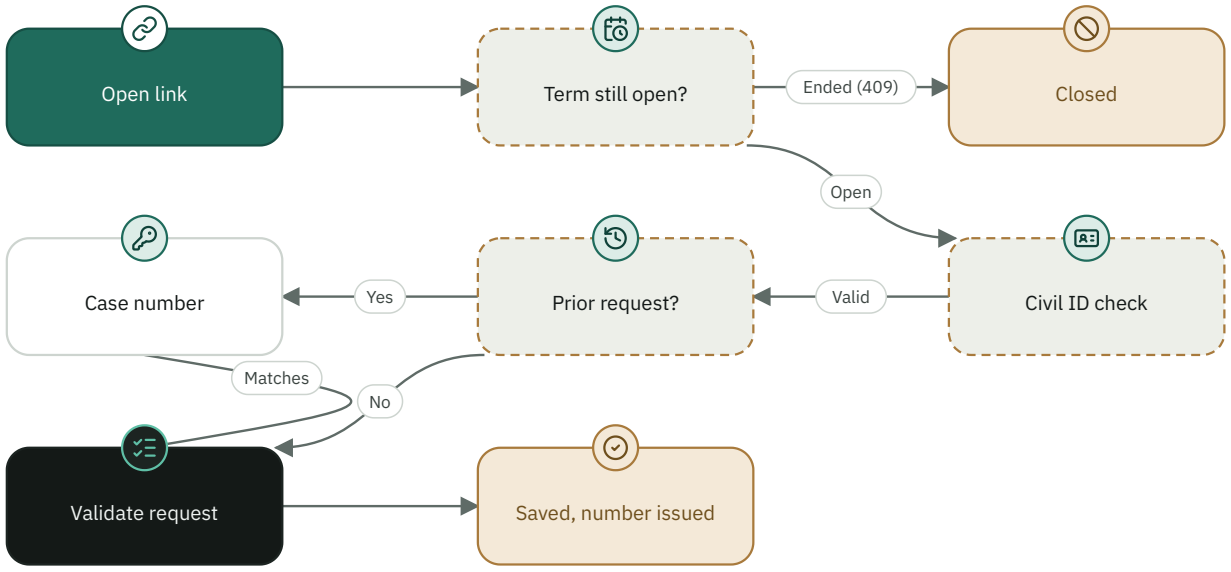
Serving and measuring an ad

The public page carries one script; the server sends eligible campaigns and personal targeting conditions are evaluated in the student's browser.



Survey submission

The civil ID alone never opens an existing request: replacing it also needs the case number.



Who reaches what

	Survey and case status	Ad display	Campaign management	Audience dashboard	Partnership profile and report
Student	✎	👁	—	—	—
Public visitor	—	👁	—	—	—
Partner organisation	—	—	—	—	✎
Root administrator	—	—	✎	👁	✎

✎ Write 👁 Read — No access

3.15.4 Screens

Ads manager

Four tabs: campaigns, requests, placements and seasons, partnership profile. Main administrator only and not available in the demo.`src/components/AdsManager.tsx`

Campaign wizard

Four steps: company and slot, audience, image, review and approval.`src/components/ads/CampaignWizard.tsx`

Audience dashboard

Aggregated anonymous figures refreshed every minute, with a print profile for companies.`src/compon`

Audience story mode

Five full-screen slides for a company meeting.`src/components/AudienceStory.tsx`

ents/AudienceDashboard.tsx

Staff dashboard ad slot
A hidden element that appears only when a campaign arrives; not mounted in the demo.`src/components/AdSlot.tsx`

Student survey page
Public page keyed by civil ID; in production the old link opens the student portal instead.`src/server/pages/surveyPages.ts`

Invalid-link page
One look for every public link: the SCHEDULE mark and the message only.`src/server/pages/publicInvalidLinkPage.ts`

Landing page
Served at /landing and on the home sub-domain.`public/landing/index.html`

Presentation deck
Full-screen slides with live numbers from two public endpoints.`public/schedule-presentation.html`

3.15.5 Requirements

FR-ENG-001 **Survey payload**

The system shall serve a survey link of kind survey only, and return to the page the link's department, college, term and the courses the department has actually taught and that are still operational, answering term-ended instead of the payload once the link's term has ended. The payload is cached for 10 minutes while term end is checked on every open outside that cache.

Implemented · Must · Functional · Student, The system (automatic)

RATIONALE

> A course nobody has taught for years is not something to ask a student about.

ACCEPTANCE CRITERIA

1. A link of another kind returns 404 with the message that the link is not a survey.
2. A link whose term has ended returns termEnded=true with the /m link and is sent with Cache-Control: no-store.
3. The cached payload expires after 600000 ms and the cache is cleared when it exceeds 200 entries.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `tests/student-case-status-audit.ts` `tests/student-journey-audit.ts`

FR-ENG-002 **Request validation on submit**

On survey submission the system shall accept a 12-digit civil ID with a valid checksum and a name of at least 3 characters (up to 60), and enforce by request type: at least one course for a new course, exactly two courses for a conflict, an approved reason and notes of 3 to 600 characters for a graduate, with no more than 12 courses stored.

Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A civil ID failing the checksum returns 400 with the invalid-civil-ID message.
2. A conflict request with one course returns 400 asking for exactly two conflicting courses.
3. A graduate request with notes shorter than 3 characters returns 400; longer notes are cut at 600 characters.
4. Success returns 201 with name, count, type and case number.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `tests/student-journey-audit.ts` `tests/student-needs-scale-audit.ts`

FR-ENG-003 Case number as second key and identity lock

The system shall read, edit or replace an existing request only with the civil ID together with the case number (normalised to at most 16 alphanumeric characters), reveal before that only that a request exists and the first letter of the name, and refuse changes to a previously verified name or home department.

● Implemented · Must · Functional · Student

RATIONALE

> A civil ID is known to colleagues and relatives, so it cannot be the only key.

ACCEPTANCE CRITERIA

1. Submitting for a civil ID with a prior request and no case number returns 409 with code case-ref-required.
2. The identity-status endpoint without a case number returns only exists and initial; with a wrong number it returns 403 with code case-ref-mismatch.
3. A name or department different from the verified one returns 409.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `tests/student-journey-audit.ts` `tests/student-case-status-audit.ts`

FR-ENG-004 Graduation-sheet proof

The system shall accept a graduation-sheet upload for a graduate request up to 14 MB, answer 413 with code file-too-large beyond it, reuse an earlier proof for the same student, department and term while re-applying the current academic rule every time, and accept no upload after the term has ended.

● Implemented · Must · Functional · Student

ACCEPTANCE CRITERIA

1. A file larger than 14×1024×1024 bytes is answered 413 with a message naming the size.
2. A prior proof is reused only for the holder of the case number, and not when the degree units in the rule changed or the passed units fall below the required threshold.
3. An upload after the term has ended is answered 409 with code term-ended.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `tests/student-case-status-audit.ts` `tests/student-journey-audit.ts`

FR-ENG-005 My request status

The system shall let a student read the status of his request from the same link (/m) with the civil ID and case number, keep reading open for 30 days after the later of link expiry and term end, answer someone with no request or a wrong number with one identical answer, and never claim a registration or a seat.

● Implemented · Must · Functional · Student

RATIONALE

> A difference between no request and wrong number would reveal to someone guessing who filled the survey.

ACCEPTANCE CRITERIA

1. A civil ID of a length other than 12 or with a wrong checksum returns 400.
2. A missing request or wrong case number returns found=false with only the term name.
3. After the 30-day grace the answer is 410 saying the follow-up period for the term has ended.
4. A committee approval while the department schedule is not fully signed appears as awaiting signatures, not as awaiting registration.
5. No name and no fingerprint appear in the payload.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `tests/student-case-status-audit.ts`

FR-ENG-006 Legacy survey link

Inferred

The system shall redirect the previously printed /q link to the student portal (/me) with a 302 for real tokens, keep the standalone survey page for demo tokens only, and answer 404 for a token that is not a survey.

Implemented · Should · Functional · Student, Public visitor

ACCEPTANCE CRITERIA

1. A real token of kind survey answers 302 to /me.
2. A demo token shows the survey page with the fictitious-number hint.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `src/server/core/publicPages.ts`

NFR-ENG-007 Attempt limit on survey endpoints

The system shall limit requests to the survey endpoints (submit, identity status, proof status, sheet upload, my request status) by a key made of the link and the network address, answering 429 when exceeded; the default is 10 attempts in a 10-minute window and is adjustable through environment settings.

Implemented · Must · Non-functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. The eleventh attempt within the window returns 429 with a wait message.
2. The key combines the link token and the network address.

EVIDENCE

`src/server/routes/surveyRoutes.ts` `src/server/core/shareLinks.ts` `tests/student-case-status-audit.ts`

NFR-ENG-008 Public page security policy

The system shall send with every public page a strict content security policy with a fresh nonce per response: default-src 'none', script-src by nonce only, connect-src 'self' and base-uri 'none', and forbid framing (frame-ancestors 'none') unless running inside an external applet host, and every visitor-supplied text displayed shall be escaped.

Implemented · Must · Non-functional · Public visitor, Student, Partner organisation

RATIONALE

> These pages are the only surface an unauthenticated stranger reaches, and they display text the system did not author.

ACCEPTANCE CRITERIA

1. The header contains a different nonce in each response and script-src does not include unsafe-inline.
2. A script without the nonce does not run.
3. The status page carries a robots meta tag with noindex,nofollow.

EVIDENCE

`src/server/core/publicPages.ts` `tests/public-pages-syntax-audit.ts` `tests/student-case-status-audit.ts`

IR-ENG-009 Public page kit and invalid link

Public pages shall use colours derived from the application's own tokens (light by default with a dark palette under the same names), the Plex font in both Arabic and Latin faces, a text time field with English digits that accepts 930 and writes 09:30, accepts Arabic-Indic and Persian digits and rejects times outside 08:00 to 20:00 by default, and one shared invalid-link page that shows only the message.

Implemented · Should · Interface · Public visitor, Student

ACCEPTANCE CRITERIA

1. The norm function turns "930" and "۱۵۳۰" into 09:30 and 15:30 and rejects "0700" and "2560".
2. The time field is of type text with ltr direction and no input of type time exists in a public page.
3. The invalid-link page carries the SCHEDULE mark and the message with no button or menu.

EVIDENCE

`src/server/publicKit.ts` `src/server/publicTheme.ts` `src/server/pages/publicInvalidLinkPage.ts` `tests/public-kit-audit.ts` `tests/public-pages-syntax-audit.ts`

FR-ENG-010 Where the script and ad are injected

The system shall inject the measurement script into successful HTML responses of public pages before </body> with the page's nonce, with an ad slot only for the public schedule (/s and /t), a pulse without an ad for /me, /m, /q and /r, and nothing for demo tokens, non-GET requests or non-200 responses.

Implemented · Must · Functional · The system (automatic), Student

ACCEPTANCE CRITERIA

1. injectPlanFor("/t/abc") returns pulse=schedule and slot=schedule; "/m/abc" returns pulse=status and an empty slot.
2. "/s/demo.abc", "/" and "/api/..." are not injected.
3. An injection failure does not prevent the original page from being sent.

EVIDENCE

`src/server/adsRoutes.ts` `tests/ads-audit.ts`

DR-ENG-011 Ad slots and image rules

The system shall define eight active slots with sizes and capacity: season sponsor 640×80 (exclusive, 1), student portal 640×200 (5), public schedule 640×160 (3), free time, after last lecture and between two lectures 480×160 (3 each), course sponsor 480×80 (8), and faculty and administrators 480×160 (3), with the order-status slot retired; and accept a PNG, JPEG or WebP image of exactly the slot size and at most 300 KB, up to two images per campaign.

Implemented · Must · Data · Root administrator

ACCEPTANCE CRITERIA

1. A 640×200 image for the public schedule slot is refused with a message naming the required 640×160.
2. A GIF or text file is refused as unsupported and a file over 300 KB is refused.
3. Adding a third image returns 409 with a maximum of two.
4. A new campaign on the order-status slot is refused because it is retired.

EVIDENCE

`src/utils/adsEngine.ts` `src/server/adsRoutes.ts` `tests/ads-audit.ts`

FR-ENG-012 Campaign validation and placement conflicts at approval

The system shall validate on save the company, category, start and end days (Kuwait time, end inclusive, one year at most), a public https link of up to 600 characters, and a free trial of at most 7 days once per company; and check at approval day by day: capacity per possible viewer (season, college), exclusivity of the season sponsor, a competitor of the same category in the same slot, and another sponsor of the same course.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. 30 February, an end before the start and a period over one year are refused.
2. A link using http, localhost, an IP address or embedded credentials is refused.
3. A sixth company in the student portal slot (capacity 5) in the same period is refused with 409 full.
4. The same category in another slot is allowed, and in the same slot with overlapping seasons or colleges is refused.
5. A course-sponsor campaign without a course, or targeted by near-graduation, is refused.

EVIDENCE

`src/utils/adsEngine.ts` `src/server/adsRoutes.ts` `tests/ads-audit.ts`

FR-ENG-013 Campaign state management and audit

The system shall allow only the main administrator to approve or resume a campaign on condition that the rules are accepted, an image exists and no conflict remains, to pause it immediately, return it to draft, and delete it unless it is live now; to return an approved campaign to draft when its slot, category, dates, seasons or colleges change; and to record every operation in the audit trail under entity ads with no person name or number.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Approval without rulesAccepted returns 400 and without an image returns 409.
2. Deleting a live campaign returns 409 asking to pause it first.
3. Changing the slot after images were uploaded returns 409; deleting the only image of an approved campaign returns 409.
4. Rotating the report token invalidates the old link.
5. A user other than the main administrator cannot reach /api/admin/ads routes.

EVIDENCE

`src/server/adsRoutes.ts` `src/components/AdsManager.tsx` `tests/ads-audit.ts` `tests/ads-preview-audit.ts`

FR-ENG-014 Image quality check

The system shall measure an uploaded image from its pixels and refuse it with 422 and written reasons if opacity is below 90%, one colour covers more than 97%, the WCAG contrast between the 5th and 95th percentiles is below 2, or clear edges are under 0.4%; the administrator may override the warning with an explicit header that marks the creative and is logged in the audit, and the check is skipped when the image-decoding library is absent.

Implemented · Should · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A single-colour image returns the nearly-empty reason and a transparent image returns the transparent reason.
2. An upload with X-Quality-Override: 1 is accepted and carries qualityOverride=true.
3. An image over 4,000,000 pixels is refused as having invalid dimensions.

EVIDENCE

`src/utils/adQuality.ts` `src/server/adQuality.ts` `src/server/adsRoutes.ts` `tests/ads-audit.ts`

FR-ENG-015 Choosing ads to serve

The system shall send for each serve request only approved, running campaigns matching the active season and the page's college, send no ad for an unknown link or a kind that asks for identity, shuffle them in every request with a rotation of 9000 ms, send the course and near-graduation targeting conditions (30 units or fewer) to be evaluated in the browser, and not serve draft images to the public.

Implemented · Must · Functional · The system (automatic), Student

ACCEPTANCE CRITERIA

1. A survey, staff-card or schedule-request token returns empty items.
2. A campaign targeted by college does not appear in the student portal for a visitor who did not send a college number, and only numbers from 1 to 99999 are accepted.
3. A draft campaign's image returns 404 from /_ads/img.
4. Campaigns are cached for 15 seconds so a pause takes effect within them.

EVIDENCE

`src/server/adsRoutes.ts` `src/utils/adsEngine.ts` `src/utils/adSeasons.ts` `tests/ads-audit.ts`

IR-ENG-016 Browser rendering and impression counting

The ad script shall render images with a fixed Ad label and rel=sponsored links, never use innerHTML for server text, count an impression after one second with at least half of the ad visible while the page is shown, stop automatic rotation for users who prefer reduced motion and provide a pause button, cap impressions at 60 per day per browser across all campaigns and 10 per campaign, and not repeat the sponsor bar in the ticket stations.

Implemented · Must · Interface · Student, Public visitor

ACCEPTANCE CRITERIA

1. After 60 impressions in a day no ad is requested until the next day (key adx:t:date).
2. An impression counts once per image per page load, and events are sent in batches of up to 20.
3. Under every ad a link How we measure leads to /measurement.
4. Ads are hidden when printing.

EVIDENCE

`src/server/adClientScript.ts` `src/components/AdSlot.tsx` `tests/ads-audit.ts`

FR-ENG-017 Clicks and filtering of automated and repeated

The system shall redirect an ad click to the company's destination with 302 and the utm_source=schedule, utm_medium=banner, utm_campaign and utm_content tags unless its link already carries any utm_*, and count the click as filtered rather than valid when it comes from an automated agent, repeats for the same campaign from the same address and agent within 4 seconds, or exceeds 120 clicks per minute per address.

Implemented · Must · Functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. A link https://a.example.com/x?y=1 gains utm tags while keeping y=1.
2. A link already containing the company's UTM_source stays unchanged.
3. An ended or unapproved campaign redirects to / and is not counted.

EVIDENCE

`src/server/adsRoutes.ts` `src/utils/adsEngine.ts` `tests/ads-audit.ts`

FR-ENG-018 Anonymous audience pulse

The system shall record one anonymous visit per day per page kind (student portal, schedule, order status, survey, staff card, app, install) per browser, count returning, mobile and peak hour for students only, accumulate counters in memory and write them every 30 seconds, on an administrator read and on shutdown, and store no name, civil ID or network address.

Implemented · Must · Functional · The system (automatic), Student

ACCEPTANCE CRITERIA

1. An unknown kind posted to /_pulse is not recorded.
2. A staff pulse does not add to returning, mobile or hours.
3. A failed write is merged with new pulses and not dropped.
4. Rate-limit keys (which contain the address) are swept every 30 seconds after their window ends.

EVIDENCE

`src/server/adsRoutes.ts` `src/utils/audienceStats.ts` `tests/ads-audit.ts`

FR-ENG-019 Unique coupon and store verification page

The system shall issue each device a unique coupon for a campaign that enabled it (an 8-character device key generated by the browser with a 6-character signature computed by the server) without any identity, the store page /verify/<id> shall check the signature and record the first use atomically, and the id shall be derived from the report token so that it does not open it.

Implemented · Could · Functional · Student, Partner organisation, The system (automatic)

ACCEPTANCE CRITERIA

1. The same device always receives the same coupon, and altering one character makes it invalid.
2. The first use shows done and the second shows already used.
3. Verification is limited to 60 requests a minute per address and issuing to 600.

EVIDENCE

`src/utils/adCoupons.ts` `src/server/adsRoutes.ts` `src/server/adsPages.ts` `tests/ads-audit.ts`

FR-ENG-020 Campaign report for the company

The system shall show at /ad-report/<token>, with no account, a single campaign's impressions, clicks, click-through rate, daily performance, week-over-week comparison, comparison of the two images, an A/B recommendation (at least 500 impressions per image, two-proportion z at 1.96) and a wrap-up of an ended campaign, allow daily reporting of code usage (unverified and excluded from any marketing total), a calendar subscription, and a script-free SVG badge for a campaign that has started.

Implemented · Should · Functional · Partner organisation, Root administrator

ACCEPTANCE CRITERIA

1. A token shorter than 20 characters or unknown returns a non-indexed 404 page.
2. A report dated outside the campaign, in the future, or with a count above 100,000 returns ?red=bad.
3. A difference under the significance threshold is shown as no significant difference and nothing is applied automatically.
4. The badge is not issued for a draft or a campaign that has not started, and its id does not reveal the report token.
5. No author name appears on the page.

EVIDENCE

`src/server/adsRoutes.ts` `src/server/adsPages.ts` `src/utils/adInsights.ts` `src/utils/adBadge.ts` `src/utils/adCalendar.ts` `tests/ads-audit.ts`

FR-ENG-021 Partnership profile and its links

The system shall let the main administrator create a partnership-profile link with a random token and an optional validity of 1 to 365 days and revoke it, and /partners/<token> shall show aggregated audience figures, ad slots with their capacity, the season calendar and an estimate calculator, compare the token in constant time, and record the first open excluding bots, administrator preview (?preview=1) and repeats within 30 seconds from the same browser.

Implemented · Should · Functional · Root administrator, Partner organisation

ACCEPTANCE CRITERIA

1. An expired, revoked or shorter-than-20-character token returns 404.
2. An unknown slot, season or weeks value in the calculator is ignored; accepted weeks are 1, 2 and 4.
3. The bell alerts the administration of a first open for one week.

EVIDENCE

`src/server/adsRoutes.ts` `src/server/adsPages.ts` `src/utils/partnerLinks.ts` `src/utils/adForecast.ts` `src/components/ads/PartnerLinks.tsx` `tests/ads-audit.ts`

FR-ENG-022 Campaign request from the profile

The system shall receive the New campaign form from a company as a plain script-free form and redirect with 303 to ?lead=ok, bad or rate, store it for the administration only after cleaning the fields (company up to 80 characters, name 60, mobile 8 to 15 digits, note 300), ignore the trap field website, limit to 5 requests per hour per address, create no second request for the same mobile and company within 24 hours, and cap new requests at 200.

Implemented · Should · Functional · Partner organisation, Root administrator

ACCEPTANCE CRITERIA

1. Filling the website field returns ?lead=ok without storing.
2. The sixth request in an hour returns ?lead=rate.
3. Indian digits in the mobile number are converted to Latin and symbols are removed.
4. The Contacted button removes the notification from the bell.

EVIDENCE

`src/server/adsRoutes.ts` `src/utils/adLeads.ts` `src/components/ads/LeadsPanel.tsx` `tests/ads-audit.ts`

FR-ENG-023 Optional email and administrator alerts

The system shall send no email unless AD_MAIL_API_KEY and AD_MAIL_FROM are set; once set it sends a weekly report on Sunday after 09:00 Kuwait time and a renewal reminder once in the last week (replacing the weekly one that Sunday) with an atomic claim before sending to prevent duplicates, and shall raise a bell notification for the main administrator for a campaign ending within 7 days or ended 3 days ago or less.

Implemented · Could · Functional · The system (automatic), Root administrator, External service

ACCEPTANCE CRITERIA

1. mailConfig without a key or sender returns null and an invalid email is rejected.
2. The week key starts on Sunday (2026-11-11 gives 2026-11-08).
3. A manual send bypasses an earlier claim, and the mail provider timeout is 10 seconds.
4. The company email appears in no public page and not in the audit trail.

EVIDENCE

`src/server/adMailer.ts` `src/server/adsRoutes.ts` `tests/ads-audit.ts`

FR-ENG-024 Audience dashboard

The system shall present to the main administrator only aggregated audience figures (daily visits, peak times, returning and mobile, survey growth, student continuity, most requested courses and the distribution over colleges under anonymous names by size), with a print profile and a refresh every minute, and add the administration-approved baseline (about 80,000 students, 5.3 million visits, 786 faculty members, 75 thousand opens) to what was measured after 2026-10-17, and show no course requested by fewer than two students.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A section with two students appears in demand; fewer is counted in hidden and not shown.
2. The baseline is shown once with the about label, and the faculty count is fixed and does not grow by measurement.
3. The term shown is the requested one, otherwise the operational current term, not the last term with answers.
4. The screen is unavailable in the demo and to anyone but the main administrator.

EVIDENCE

`src/utils/audienceStats.ts` `src/utils/audienceBaseline.ts` `src/components/AudienceDashboard.tsx` `src/server/adsRoutes.ts` `tests/ads-audit.ts`

FR-ENG-025 Audience story mode

The system shall provide a full-screen story mode of five slides (size, peak, what students request, slots, next step) advancing every 8000 ms without wrap-around, using the same figures as the dashboard, supporting pause with the space bar, navigation with arrows, PageUp/PageDown, Home/End and horizontal swipe (48 pixels or more), and starting paused for users who prefer reduced motion.

Implemented · Could · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Advancing from the last slide and going back from the first do not wrap.
2. Autoplay stops at the last slide and when paused.
3. Swiping right advances in the Arabic page, and the left arrow is next.
4. Leaving full screen closes the mode, and Esc closes it.

EVIDENCE

`src/utils/audienceStory.ts` `src/components/AudienceStory.tsx` `tests/ads-audit.ts`

FR-ENG-026 Landing page and presentation deck

Inferred

The system shall serve the landing page from memory at /landing and on the home sub-domain without touching system routes, and serve the presentation deck with live figures from /api/public/audience-headline and /api/journey, keeping the printed numbers when a read fails or returns zero, and the public figures endpoint shall be cached for 10 minutes and answer 503 if the computation fails.

Implemented · Could · Functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. /landing and any path under it that is not a static asset returns the landing page, and the page's assets are served without an index.
2. The public endpoint returns only since, asOf, students, visits, instructors and instructorOpens.
3. The deck's full screen: left arrow, space and PageDown for next, key f for full screen, n for notes and r for reading mode.

EVIDENCE

`src/server/adsRoutes.ts` `public/landing/index.html` `public/schedule-presentation.html` `server.ts`

FR-ENG-027 **Delivery of a landing-page inquiry**

The system shall deliver a «request a walkthrough» inquiry from the landing page to the system administration so it is never lost: it is validated, size- and rate-limited, stored in the landingInquiries collection, and shown in the notification bell to the system administrator only; and it shall show the visitor an honest result, saying «sent» only after an actual save.

● Implemented · Should · Functional · Public visitor, The system (automatic)

RATIONALE

> The request used to be written to the server log only and answered with success, so it was lost while the visitor was told it was sent.

ACCEPTANCE CRITERIA

1. A valid inquiry is stored (201) and appears in the system administrator’s bell with institution, name and email, and no academic role sees it.
2. If saving fails or the database is down the server answers 503 and the page says «your request was not sent» with the direct email, keeping the form.
3. An over-long field is refused (400), an oversized body (413), the sixth within ten minutes from one address (429), and a duplicate within an hour creates no second record.

EVIDENCE

`src/server/landingInquiry.ts` `server.ts#/api/landing/inquiry` `src/db/repository.ts#createLandingInquiry` `src/server/routes/notificationsRoutes.ts` `public/landing/index.html` `tests/landing-inquiry-audit.ts`

3.15.6 Algorithms

ALG-ENG-01 · Engagement and public presence

Ad image quality check from its pixels

Deterministic

Is the uploaded ad image fit for display: opaque, not empty, of readable contrast, and with visible detail?

1. The image is decoded into RGBA pixels and refused above 4,000,000 pixels.
2. Opacity = share of pixels with alpha ≥ 250.
3. Emptiness = share of the most common colour after quantising each channel to 16 levels.
4. Contrast = (95th-percentile luminance + 0.05) ÷ (5th-percentile luminance + 0.05), using WCAG relative luminance.
5. Detail = share of pixels whose brightness differs from the right or lower neighbour by more than 32.

Inputs

The uploaded image file.

Output

The four measurements and Arabic refusal reasons ordered by severity, or an empty list when the image passes.

Pure function; if the decoding library is unavailable the check is skipped and said so, rejecting no one for it; the administrator may override the warning.

Opacity ≥ 90% · Dominant colour ≤ 97% · Contrast ≥ 2 · Edges ≥ 0.4%

`src/utils/adQuality.ts#measureQuality` `src/utils/adQuality.ts#qualityProblems` `src/server/adQuality.ts#analyzeCreative` `tests/ads-audit.ts`

FR-ENG-014

ALG-ENG-02 · Engagement and public presence

Aggregated anonymous audience statistics

Statistics from history

What is the size and activity of the system's audience, in honest aggregate figures that identify no one?

1. A student = a unique survey fingerprint; a visit = one browser once per day per page kind; page opens are not counted.
2. Per term: answers and distinct students; funnel: created → setup done → wishes sent → returned after 24 hours or more; active within 7 and 30 days.
3. Continuity between consecutive terms is shown only when the earlier term has 30 or more students; requested courses appear only with two or more distinct students, top 8.
4. Colleges are anonymised as «college 1, college 2…» in order of size.
5. Headline totals = the administration-approved opening baseline up to 2026-10-17 + what was measured after it; instructor open rate = max(90%, measured).

Inputs

Survey answers and student profiles with fingerprints, daily visit counters, instructor requests, and ad campaigns with their statistics.

Output

A figures board: 30-day daily visits, peak times, returning and mobile shares, survey growth, continuity, most-requested courses, college distribution, and headline totals.

Counts only, no names or civil numbers; one function feeds both the dashboard and the partnership profile, so no figure differs between the two.

Per-course privacy floor: 2 students | Continuity floor: 30 students | Opening baseline: about 80,000 students, 5,300,000 visits and 75,000 card opens

`src/utils/audienceStats.ts#buildAudienceStats` `src/utils/audienceStats.ts#headlineTotals` `src/utils/audienceBaseline.ts#AUDIENCE_BASELINE` `tests/ads-audit.ts`

`FR-ENG-018` `FR-ENG-021` `FR-ENG-024`

3.15.7 Edge cases

⚠ A civil ID known to someone else

Scenario A colleague who knows the civil ID of a student with a submitted request tries to open the survey with it.

Behaviour Only the existence of a request and the first letter of the name are revealed; submission returns 409 case-ref-required and a wrong case number returns 403; ten attempts in a 10-minute window stop the guesser with 429.

⚠ Term ended while the link still circulates

Scenario A survey link's term ends or the link is revoked after the drop-and-add week, before registration decides.

Behaviour Submission and upload answer 409 term-ended, while reading My request status stays open for 30 days after the later of link end and term end and then answers 410.

⚠ Two competing campaigns in one slot

Scenario A Restaurants and cafés campaign is approved in the student portal and then another of the same category with overlapping dates is approved.

Behaviour The second approval is refused with 409 naming the rival company and its dates; the same category in the public schedule is allowed, and also in the same slot when the two campaigns' seasons or colleges do not overlap.

⚠ Editing an approved campaign that is live

Scenario The dates, slot, category, seasons or colleges of an approved campaign are changed.

Behaviour The campaign returns to draft and stops serving until a new approval that re-checks conflicts and re-accepts the rules; changing the slot after images were uploaded is refused outright because the image size is tied to it.

⚠ Two identical clicks from the university network

Scenario Two students behind one network address with an identical browser agent click an ad within 4 seconds.

Behaviour The second is counted as a filtered click, not a valid one; this is a deliberate conservatism because the network is one address for thousands of students.

3.15.8 Standing owner decisions

- No advertisement on a page that asks for an identity (survey, staff card, schedule request, order status), nor on approval, print and calendar pages.
- Measurement is anonymous: no name, civil ID or network address in any figure, and colleges are shown as College 1, 2… by size.
- A course number is always seven digits (college 2 + department 2 + course 3); the stored three-digit tail and the AdCourseId are never shown.
- Printed reports keep their design; the audience print profile is corrected through values, not by changing its layout.
- The numbers approved by the system administration (the baseline) are not modified and no unapproved numbers are added to them.

↑ Contents

3.16 · DSH

Dashboards and term awareness

The home dashboard, term story, semester readiness and the welcome tour.

23	100%	4	21
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Algorithms · 7. Edge cases · 8. Standing owner decisions

↑ Contents

3.16.1 Overview

This module gathers what a user sees first on entering: the home dashboard with its scope, displayed term, figures and day-and-week reading; the Term story card of the approval cycle; the Semester readiness panel of the rehearsal radar; the period-balance drawing; the mini rings and other helper drawings; the public Journey reading; the global search and sentence search; the welcome tour by role; and the safety net that redraws a screen after a rendering fault. Every number is read from the data at request time, and scope and term are decided on the server, not in the screen.

Schedule committee chair · Department head · College dean · Vice dean for academic affairs · University registrar · College registrar · Registration officer · Dean of registration · System administrator · Root administrator · Standard user · Faculty member · Public visitor

3.16.2 User journeys

Schedule committee chair — **Reading the dashboard in the morning**

1. Opens the home dashboard; three requests start together: the dashboard and two analytics readings.
2. Sees the operationally current term and a badge naming the upcoming term in preparation, if any.
3. The decision card states what needs his attention most (up to three items) or that the schedule is sound.
4. Reads today's lecture track and the current or next lecture, then the load by day, hour and hall.
5. Presses an item to go to the schedule screen and handle it.

College dean — A dean opens the dashboard while a new term is in preparation

1. Opens the dashboard; the current term is shown, not the newest in the list.
2. Sees his college's figures from the accepted version only.
3. A badge appears with the upcoming term's name; he presses it and the dashboard opens on it.
4. If nothing is approved in it the dashboard says the term's schedule has not been approved yet, not that the schedule is sound.
5. Returns to the original term through the default badge.

Department head — First login and the role tour

1. After login there is no seen-tour mark for this role, so the tour starts.
2. Sees the steps of his own work (schedule changes, notes, signing, additions) as a list, not a building stage.
3. Moves with the tabs or the Next button and closes with Esc at any moment.
4. On Start work the mark is saved for his role so the tour does not repeat.
5. Replays the tour whenever he wants from the menu; the mark is removed and it starts again.

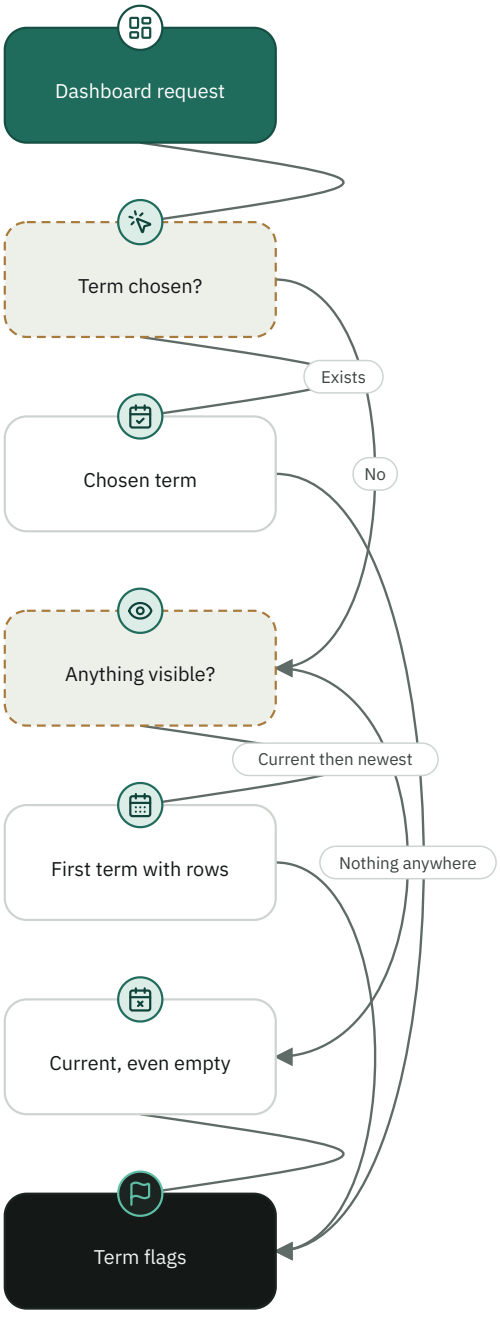
University registrar — Reading semester readiness in the radar

1. Opens the registration radar; the readiness ring, number and tier appear.
2. Reads the share reaching 12 units, the share of demand that found a seat and the count of full and not-offered courses.
3. Reads the biggest shortfall cause in the stacked bar.
4. If unknown exceeds 20% he reads the weak-estimate warning before basing a decision on the number.

3.16.3 Diagrams

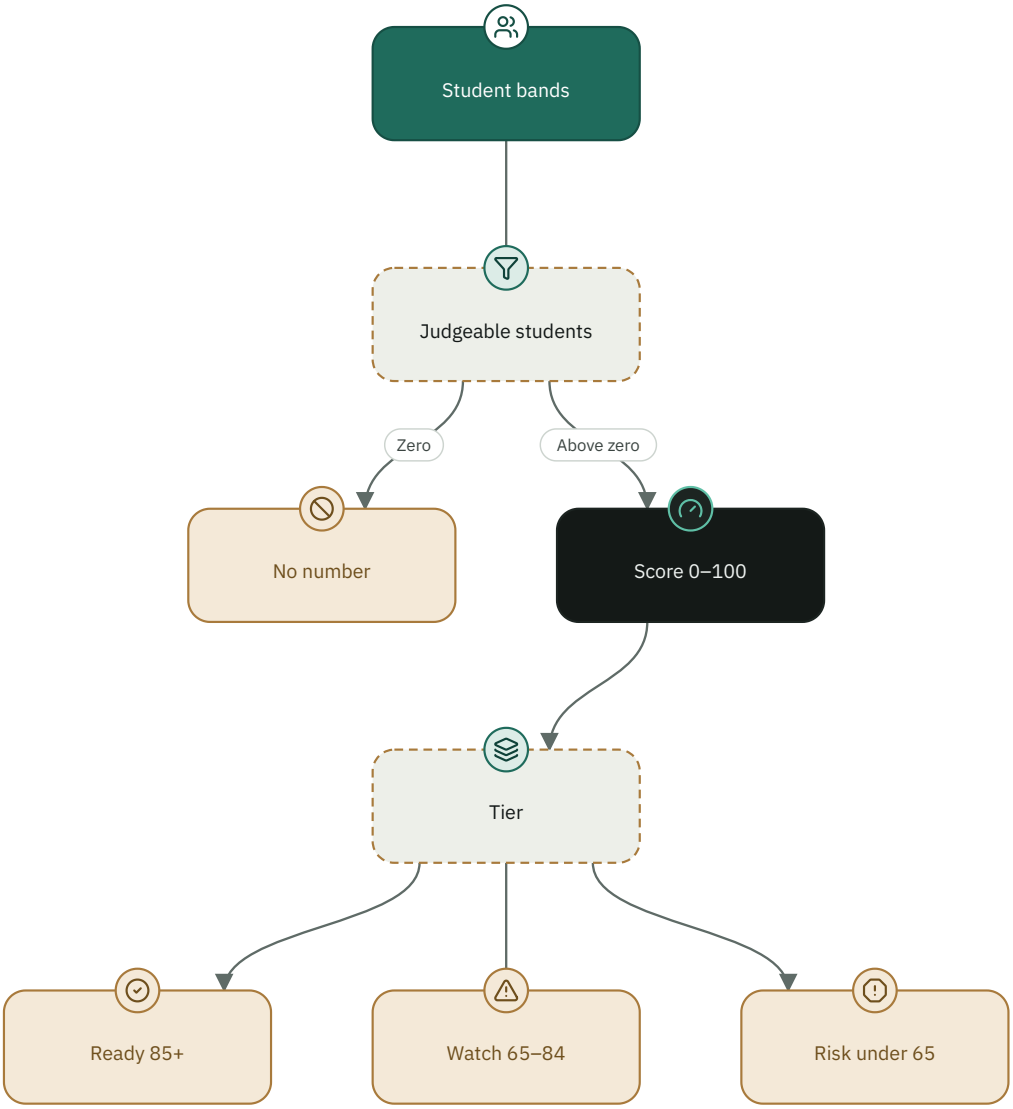
Choosing the dashboard term

The decision is on the server: the reader's explicit choice first, then the operationally current term, then the newest term with something visible, then the current term even if empty.



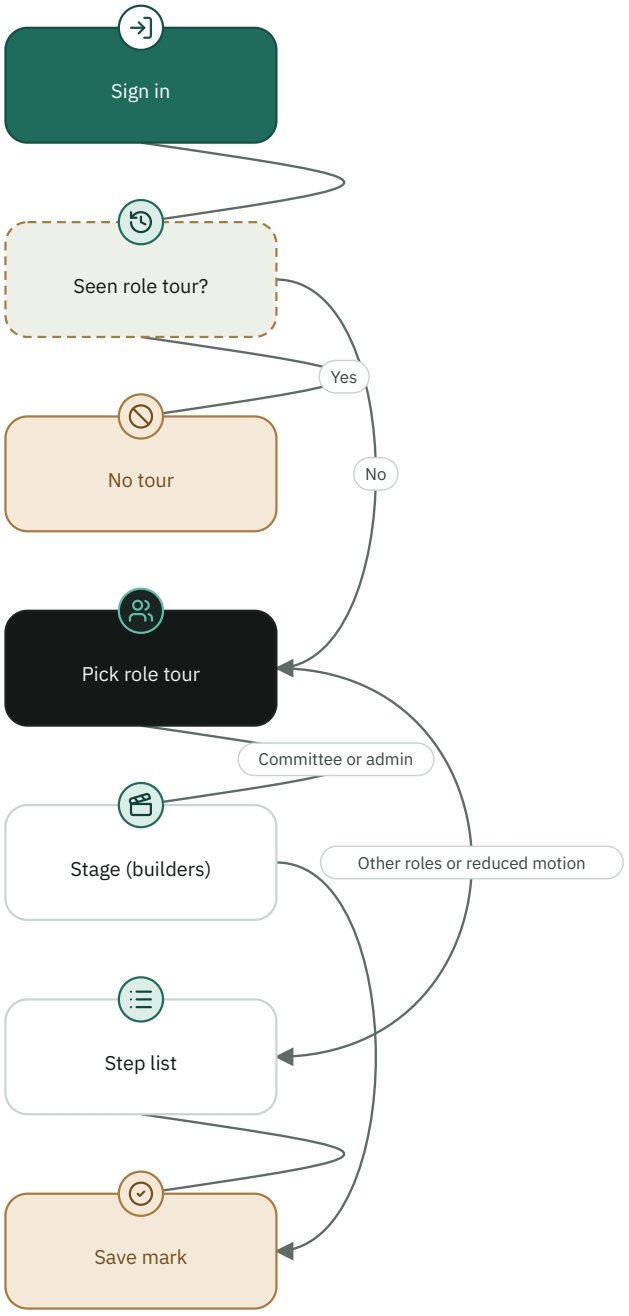
Computing semester readiness

Score = 60% share reaching 12 units + 40% share of demand that found a seat; unknown is left out of the denominator and only reported when large.



Welcome tour flow

The mark carries the user id and the role; whoever finished the old general tour is not shown it again unless his role now has its own tour.



What each role reads on the dashboard

	Scope figures	Personal figures	Readiness analytics	Public journey
Root administrator	👁	—	👁	👁
Schedule committee chair	👁	—	👁	👁
Department head	👁	—	—	👁
College dean	👁	—	—	👁
Faculty member	—	👁	—	👁
Public visitor	—	—	—	👁

✎ Write 👁 Read — No access

3.16.4 Screens

Home dashboard
Account and scope bar, decision card, readiness signals, today's lecture track, scope figures and the week-load reading.`src/components/Dashboard.tsx`

Term story
A card in the schedule-changes screen: appointments, what moved, rounds, notes and blockers, with a copy-as-sentence button for the dean.`src/components/TermStory.tsx`

Semester readiness
A ring and number at the top of the registration radar, three explanatory figures and a shortfall-cause bar.`src/components/SemesterReadiness.tsx`

Period balance
A strip of cells per rule 7/2 with a gauge for morning and one for evening.`src/components/PeriodBalanceInfographic.tsx`

Mini ring
A ring for a figure out of 100 with a colour tone.`src/components/MiniRing.tsx`

Visual bits
Building travel map, equation chips and a travel heat matrix.`src/components/VisualBits.tsx`

Welcome tour
A stage for those who build the schedule and a step list for other roles.`src/components/Onboarding.tsx`

Render recovery
Remounts the failed part in place, then shows one line and a Retry button.`src/components/ErrorBoundary.tsx`

3.16.5 Requirements

FR-DSH-001 Term shown on the dashboard

The system shall decide on the server the term the home dashboard opens on: the reader's explicit choice as long as the term exists, otherwise the operationally current term if the reader has something visible in it, otherwise the newest term with something visible within 6 earlier terms, otherwise the current term even if empty, and return the flags explicit, fallback, currentTermId and defaultTermId with it.

● Implemented · Must · Functional · Schedule committee chair, Department head, College dean, University registrar

RATIONALE

> A term just being prepared used to open an empty home for a dean.

ACCEPTANCE CRITERIA

1. The operationally current term is tried first even when a newer term exists in the list.
2. An explicit choice of an empty term is respected and the default is kept to return to.
3. A saved term that no longer exists falls back to the default with explicit=false.
4. Nothing for the reader in any term: the current term stays.

EVIDENCE

`src/utils/homeTerm.ts` `src/server/routes/dashboardRoutes.ts` `src/server/core/scheduleReads.ts` `tests/dashboard-home-term-audit.ts`

FR-DSH-002 Empty term state and upcoming term

The system shall return with the dashboard the flags empty, awaitingApproval and upcoming, and for a term with no visible row the screen shall show a card saying there are no appointments yet or the term's schedule is not yet approved instead of the schedule being sound, with no day or week figures built from zero, a badge opening the nearest upcoming term in preparation, and a badge to return to the default.

● Implemented · Must · Functional · College dean, Vice dean for academic affairs, Schedule committee chair

ACCEPTANCE CRITERIA

1. empty=true when no row is visible; awaitingApproval=true when the reader sees approved rows only and an unapproved row exists.
2. upcoming names a term later than the one shown and not the shown term itself, or null.
3. The decision card and attention list are emptied for an empty term.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/components/Dashboard.tsx` `src/utils/homeTerm.ts` `tests/dashboard-home-term-audit.ts`

FR-DSH-003 Scope of what the dashboard shows

The system shall show the whole university to the administrator, the personal rows to an account linked to a faculty member, and only the permitted department rows to anyone else, deans and the dean of registration reading the accepted version only, and return workspace.mode with one of admin, personal or scope.

● Implemented · Must · Functional · System administrator, Root administrator, Faculty member, College dean, Schedule committee chair

ACCEPTANCE CRITERIA

1. The administrator account sees the day's lectures from the whole university, not an empty list beside a large figure.
2. A dean does not see a second or summer term before it is accepted.
3. The administrator receives metrics with the total of schedules, terms and instructors, and others receive figures for their scope.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/server/core/scheduleReads.ts` `src/server/core/scope.ts` `tests/dashboard-home-term-audit.ts` `tests/role-scope-audit.ts`

FR-DSH-004 Today's lectures and the weekend

Inferred

The system shall return today's lectures sorted by start time for Sunday to Thursday, and on Friday and Saturday not show another day's lectures but return weekend=true and an empty list, and the screen shall say that Friday and Saturday are a holiday with no lectures today.

Implemented · Must · Functional · Schedule committee chair, Department head, Root administrator

ACCEPTANCE CRITERIA

1. On Friday or Saturday: today is empty and weekend=true with dayName Friday or Saturday.
2. Each row carries the instructor, course, start and end times and the hall.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/components/Dashboard.tsx`

FR-DSH-005 Scope load reading

The system shall compute for the scope's rows in the shown term: appointments per day from Sunday to Thursday, the 5 busiest start hours, the 5 most used halls, the unique hall count, and the room-occupancy peak as the percentage of the most halls occupied in any half-hour slot over all halls.

Implemented · Should · Functional · Schedule committee chair, Root administrator, Department head

ACCEPTANCE CRITERIA

1. weekdayLoad has five items keyed fsunday to fthursday.
2. busiestHours and busiestRooms are each at most 5, sorted by count descending.
3. Rows without a verified hall are excluded from hall occupancy.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `tests/dashboard-home-term-audit.ts`

FR-DSH-006 Trend over the last terms

The system shall return a trend line of up to 4 points in chronological term order (appointments, halls, instructors) ending with the shown term, examining up to 8 terms back, skipping a term in which the reader had nothing, and return previous (the preceding point); the screen shall show the figure's difference from the previous term with an arrow and show no mark when nothing changed.

Implemented · Should · Functional · Schedule committee chair, Department head, College dean

ACCEPTANCE CRITERIA

1. The order is chronological, not by id: an archive term with a larger id does not drop out of the line.
2. A point with zero appointments is dropped unless it is the shown term.
3. The line and the difference are not drawn for fewer than two points.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/components/Dashboard.tsx` `tests/dashboard-home-term-audit.ts`

FR-DSH-007 Dashboard loading and enrichment

The screen shall start the dashboard and the two analytics readings together for those who manage the schedule, an optional reading's failure shall not empty the dashboard, the previous term's data shall be cleared on a term switch before the answer arrives, and awaited items shall hold their space with a dash rather than a zero or a gap that pushes the page later.

Implemented · Should · Functional · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. A failure of the analytics overview reading does not prevent the figures and lectures from showing.
2. The two optional readings carry home=1 so they resolve the same term.
3. A pending indicator exposes aria-valuetext as reading in progress, not a zero value.

EVIDENCE

`src/components/Dashboard.tsx` `tests/dashboard-home-term-audit.ts`

FR-DSH-008 Decision card and signals

Inferred

The dashboard shall show a decision card with the most important alert (up to 3 items, in critical, warning or info tone), an attention list for the rest, and a signal strip: today's lectures, critical conflicts (hidden at zero), occupied halls, average gap, readiness as a ring, and a health ribbon with quality, resilience, fairness and occupancy; analytics signals are not shown to those who do not manage the schedule.

Implemented · Should · Functional · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. An item is critical when its severity is critical, danger, error, high or blocker.
2. The readiness ring shows quality as a percentage and turns good from 85 upward.
3. The handle button on an item navigates to the schedule screen.

EVIDENCE

`src/components/Dashboard.tsx`

IR-DSH-009 Today's lecture track

Inferred

The dashboard shall draw today's lectures over the working-hours axis, mark the current or next lecture with its time, give each hour column a readable description (lecture count or no lectures), and format its numbers as Latin digits in Kuwaiti Arabic locale (ar-KW-u-nu-latn).

Implemented · Should · Interface · Schedule committee chair, Department head, Faculty member

ACCEPTANCE CRITERIA

1. The current lecture is the one started and not ended at the current minute; otherwise the first later lecture.
2. The track's description states the day's pressure over its hours.

EVIDENCE

`src/components/Dashboard.tsx` `src/utils/scheduleTime.ts`

FR-DSH-010 Public journey reading

The system shall return at /api/journey aggregated figures computed at request time with no identity: counts of terms, appointments (counted on the server), courses, instructors, departments and colleges, the first academic year on record, and the newest term's unique figures, without authentication, and without inheriting a demo sandbox in a real deployment.

Implemented · Should · Functional · Public visitor, Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. The payload returns scoped=false, lifetime and current and carries no name or civil ID.
2. Adding a term raises terms by one on the next request without a code change.
3. since is the earliest start year among the term names.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/components/ScheduleJourney.tsx` `server.ts` `tests/journey-reveal-audit.ts`

FR-DSH-011 Global search

The system shall answer /api/search with a minimum of two characters about appointments (12), instructors (8), courses (8) and halls (8) in the newest term, for those holding any of screens 7, 8, 9, 10, 16 or 17, limit each category to its permission, scope and the dean's accepted version, and not expose instructor data the reader's scope does not allow.

Implemented · Should · Functional · Schedule committee chair, Department head, College dean, Root administrator

ACCEPTANCE CRITERIA

1. A one-character query returns empty lists.
2. Search reads by scope and gives a dean the final version only.
3. Instructor results pass through a reader-appropriate view (civil ID hidden from those not permitted).

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/server/core/scope.ts` `tests/dean-notify-audit.ts`

FR-DSH-012 Sentence search

The system shall interpret an Arabic sentence by fixed rules with no external service, answering free halls (up to 60), a hall's appointments, an instructor's gaps between lectures (30 minutes or more, up to 30), and an instructor's or a time's appointments (up to 40 rows), refuse with 403 a college or department outside the reader's scope, and return intent unknown with no rows for a sentence it cannot parse.

Implemented · Could · Functional · Schedule committee chair, Department head, College dean

RATIONALE

> The same sentence always resolves to the same rows, offline.

ACCEPTANCE CRITERIA

1. A free hall is one that no row occupies in the whole term at the stated day and time.
2. Rows are read with the request's own reader (scope and final version for a dean), not raw.
3. A gap under 30 minutes is not counted.

EVIDENCE

`src/server/routes/dashboardRoutes.ts` `src/utils/naturalQuery.ts` `tests/dean-notify-audit.ts` `tests/input-hardening-audit.ts`

FR-DSH-013 Term story

Inferred

The Term story card shall summarise a department's approval cycle in five figures (appointments, what moved or first review, rounds, notes handled out of the total, blockers) with a rounds strip showing its steps (sent, returned with its note count, accepted), and one ready sentence copyable for the dean with a button, and shall name the comparison basis: the approved authority version, the last registration review, or the last snapshot.

Implemented · Should · Functional · Department head, Schedule committee chair, College dean

ACCEPTANCE CRITERIA

1. A first review shows a dash and first review rather than moved 0.
2. The authority basis says it matches the approved authority version or that N differ from it.
3. Rounds count at least one, and zero blockers show no blockers.
4. After copying the button reads Copied for 2.2 seconds.

EVIDENCE

`src/components/TermStory.tsx` `src/components/ScheduleChanges.tsx`

FR-DSH-014 Semester readiness score

The system shall compute semester readiness = 60% × the share reaching 12 units or more (reaching 15, 12 to 14, or a graduate below 12) of those who can be judged + 40% × the share of demand that found a seat, classify it ready from 85, needs attention from 65, at risk below that, and invent no number when there are no students to measure (score=null).

Implemented · Must · Functional · University registrar, College registrar, Schedule committee chair

ACCEPTANCE CRITERIA

1. 80/10/10 with demand 100 and seats 95: score 92, tier ready.
2. 10/10/80 with demand 100 and seats 50: score 32, tier risk.
3. Unknown above 20% sets weak=true.
4. No radar or no students: no score and no tier.

EVIDENCE

`src/utils/semesterReadiness.ts` `tests/registration-causes-audit.ts`

IR-DSH-015 Semester readiness panel

The panel shall show the readiness ring with a readable description (semester readiness N out of 100: tier) and figures: the share reaching 12, the share of demand that found a seat, the counts of full and not-offered courses, a stacked bar of shortfall causes for those not reaching 12 with the largest named, a warning when precision is weak, a button leading to the list of courses needing a decision, and shall not draw when there is no score.

Implemented · Should · Interface · University registrar, College registrar, Schedule committee chair

ACCEPTANCE CRITERIA

1. The bar shows the six shortfall causes in proportion to their counts and states it is an estimate, not proof of one cause.
2. When the unknown share exceeds 20% a weak-estimate line appears.
3. A waiting skeleton shows while the analysis loads and not when it fails.

EVIDENCE

`src/components/SemesterReadiness.tsx` `src/components/RegistrationRehearsal.tsx` `src/utils/semesterReadiness.ts` `tests/radar-insight-audit.ts`

IR-DSH-016 Period-balance drawing

The drawing shall represent each appointment as a cell in one strip (morning 8 to 2 and evening 2 to 8), show two gauges for the limits of rule 7/2: morning 50% to 70% and evening 30% to 50%, compute the limit with whole numbers (upper = floor of 70%, lower = ceiling of 50%), colour the excess cells and write how many sections move and to which side.

Implemented · Should · Interface · Schedule committee chair, Department head, Registration officer

ACCEPTANCE CRITERIA

1. 90 appointments all in the morning: 27 move, not 28 (limit 63).
2. 35 with 26 in the morning: 2; 20 with eight: 2; 10 with six: 0 (within range).
3. The strip is tightened when appointments exceed 60.
4. The drawing has a role=img description stating both percentages and the move count.

EVIDENCE

`src/components/PeriodBalanceInfographic.tsx` `src/components/ScheduleReview.tsx` `src/components/ScheduleChanges.tsx` `tests/decision-1913-audit.ts` `tests/lecture-length-audit.ts`

IR-DSH-017 Mini ring

Inferred

The mini ring shall draw a figure out of 100 clamped between 0 and 100, with a good tone from 85, warning from 70, bad below, and neutral (empty) when the value is absent, be read to a screen reader as its name and N out of 100 unless decorative beside an already-read number, and turn text of the form NN/100 into a ring and anything else into nothing.

Implemented · Could · Interface · Schedule committee chair, Root administrator

ACCEPTANCE CRITERIA

1. miniRingTone(85) is good, (70) warning, (69) bad and null neutral.
2. ringFromScoreText("80/100") returns a ring and a non-matching text returns null.

EVIDENCE

`src/components/MiniRing.tsx` `src/components/Dashboard.tsx`

IR-DSH-018 Helper drawings

Inferred

The helper drawings (building travel map, equation chips a – b = c, travel heat matrix) shall draw figures the screen already holds, with no fetching, saving or new computation, make the travel line's thickness follow walking minutes and colour tiring hops in the danger colour.

● Implemented · Could · Interface · Schedule committee chair, College dean, Root administrator

ACCEPTANCE CRITERIA

1. The map places buildings on an ellipse and the line thickness grows from 2 in proportion to minutes against the largest.
2. The matrix is read-only and the input fields stay where they are.

EVIDENCE

`src/components/VisualBits.tsx` `src/components/Colleges.tsx` `src/components/IntelligenceWorkspace.tsx`

FR-DSH-019 Welcome tour by role

The system shall show each role its own tour: committee, administrator and a role-less account see the building stage with its six scenes, the department head four steps, the registration officer and college registrar four, the registrar head five (with the deadline), the dean and dean of registration five, and the vice dean six, and show in a role's tour nothing that role does not do, and store the seen-tour mark by account id and role.

● Implemented · Should · Functional ·

Schedule committee chair, Department head, University registrar, Registration officer, College registrar, College dean, Dean of registration, Vice dean for academic affairs

ACCEPTANCE CRITERIA

1. The key is schedule-onboarding-v5-<id>-<role>: dean number 7 gives schedule-onboarding-v5-7-dean.
2. Whoever finished the old v4 tour is not shown it again unless his role has its own tour.
3. The deadline and extension step is in the registrar head's tour only.
4. Replaying from the menu removes both the new and the old mark and starts from the first scene.

EVIDENCE

`src/components/Onboarding.tsx` `src/utils/onboardingKey.ts` `src/App.tsx` `tests/dean-notify-audit.ts`

IR-DSH-020 Tour playback and accessibility

The staged tour shall autoplay only through the third scene with a per-scene hold that depends on its text length between 3200 and 5000 ms, then leave the pace to the reader, scenes shall be shown as a motionless list for users who prefer reduced motion and for every role that does not build the schedule, Esc shall close it at any moment, and the narration shall announce itself politely while the stage is decorative to a screen reader.

● Implemented · Should · Interface · Schedule committee chair, Department head, College dean, University registrar

ACCEPTANCE CRITERIA

1. The hold is clamped between 3200 and 5000 for 900 + (word count ÷ 4) × 1000.
2. A manual press on a scene stops the autoplay.
3. Reading the reader's real week (/api/schedules/workspace) fails silently and the sample remains.

EVIDENCE

`src/components/Onboarding.tsx` `tests/dean-notify-audit.ts`

FR-DSH-021 Render fault recovery

When part of a screen fails to render the system shall remount that part in place silently for up to 3 attempts (the first at once and the others after 120 ms), with no navigation, reload or cache clearing, and if all fail shall show one line saying this part could not be shown with a Retry button that restarts the full cycle, and report each stage to telemetry without showing the reader anything.

● Implemented · Must · Functional · Schedule committee chair, Department head, College dean, Standard user

ACCEPTANCE CRITERIA

1. The quiet cover applies only while attempts are not exhausted.
2. The stuck state carries role=alert and the text that the data is saved and unaffected.
3. The retry button resets the counter, and no window.location.reload or assign is used.
4. The events are ui.render.recovery, ui.render.unrecovered and ui.render.manual-retry.

EVIDENCE

`src/components/ErrorBoundary.tsx` `tests/run-tests.ts`

IR-DSH-022 General dashboard presentation constraints

Inferred

The screens of this module shall present numbers as Latin digits in ar-KW format, give drawings that carry meaning a readable description (role=img with aria-label) and treat decoration as hidden from screen readers, announce live states (loading, data updated) with status roles, and never display a three-digit course number.

● Implemented · Should · Interface · Schedule committee chair, Department head, College dean, University registrar

ACCEPTANCE CRITERIA

1. Loading shows role=status with the text that the command dashboard is loading.
2. Any course number shown in side lists passes through the seven-digit function.

EVIDENCE

`src/components/Dashboard.tsx` `src/components/SemesterReadiness.tsx` `src/components/MiniRing.tsx`

FR-DSH-023 Demo limits

Inferred

The demo guide shall describe what the demo opens on: one college and three departments, with the scope preselected (College of Basic Education → Educational Technology → first term 2026/2027), roles the user switches from the side bar, and state that real administration (users, permissions, backup) cannot be modified from the demo and that the audience dashboard and ads are unavailable in it.

● Implemented · Could · Functional · Public visitor, System administrator

ACCEPTANCE CRITERIA

1. Administration (users, permissions, user-scopes, system-backup) answers 403 to writes in the demo.
2. The audience and ads screens return unauthorised for a demo account.

EVIDENCE

`docs/DEMO_GUIDE.md` `src/server/core/auth.ts` `src/App.tsx` `tests/demo-complete-audit.ts`

3.16.6 Algorithms

ALG-DSH-01 · Dashboards and term awareness

Term early warning

Deterministic

What are the three most important risks in the term timetable right now: missing sections, an instructor over load, or sections without a hall?

1. Demand = «not registered» from the imported report only; with no report there is no forecast.
2. When vacant seats are stored: covered = min(remaining, vacant) and shortage = ⌈(remaining − vacant) ÷ capacity⌉; otherwise required = ⌈remaining ÷ capacity⌉ and shortage = required − scheduled sections.
3. Demand coverage % = covered seats ÷ total remaining over courses with known capacity; an unknown capacity is reported on its own and never assumed.
4. Instructor overload = weekly credit hours above the recorded load allowance; a section without a hall = at least one row with no location.
5. Highlights (up to 3): the largest shortage, then instructor overload, then sections without a hall, then other shortages, then unknown capacity.

Inputs

The term's sections, course capacities, instructor load allowances, and the imported remaining and vacant figures.

Output

Status, coverage percentage and seats, number of short courses, up to three highlighted risks with tone, and the rest under «all details».

Fixed rules over stored data; anything without data is reported as unknown, never guessed, and no percentage is shown without seats.

Up to 3 highlighted risks

`src/utils/termForecast.ts#computeTermForecast` `src/server/routes/planningRoutes.ts#registerPlanningRoutes` `tests/term-forecast-audit.ts`

FR-DSH-008

3.16.7 Edge cases

⚠ A dean opens the dashboard when a new term has nothing approved

Scenario A new term in preparation has been created and registration has not accepted it; the dean sees approved rows only.

Behaviour The dashboard opens on the current term with its week and figures and names the new term in a badge; if the dean opens it, it says its schedule is not yet approved and never that the schedule is sound.

⚠ Friday and Saturday

Scenario A user opens the dashboard on a holiday.

Behaviour Today's list is empty with weekend=true and the screen writes that Friday and Saturday are a holiday, so Sunday's lectures are not stamped with another day's date.

⚠ Optional analytics read fails

Scenario One of the two companion analytics readings fails while /api/dashboard succeeds.

Behaviour The dashboard keeps its figures and lectures, and the awaited signals (gap, readiness, health) drop out once waiting ends instead of the screen being emptied by an error banner.

⚠ Readiness with no judgeable students

Scenario All student files are unknown or there are no students in the radar.

Behaviour No number and no ring is shown (score=null); if a little is judged with unknown above 20% the number is shown with the weak-estimate warning.

⚠ A render fault that does not heal

Scenario A part of the screen fails all three automatic attempts.

Behaviour No invisible cover over a blank screen: one line and a Retry button appear with an accessible announcement, and the user is not navigated away and the application is not reloaded.

3.16.8 Standing owner decisions

- The displayed term is decided by the server: the operationally current term unless the reader chooses another.
- Deans read the accepted version only on every screen.
- A course number is always seven digits (college 2 + department 2 + course 3).
- Printed reports keep their design; fixes are in values, not layout.
- Every tour shows only features that are switched on; a tour showing what its owner cannot find is worse than none.

↑ Contents

3.17 · OPS

Operations

From server boot to release: runtime, backups, the demo sandbox and the installable app.

30	100%	6	38
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Edge cases · 7. Standing owner decisions

↑ Contents

3.17.1 Overview

This module describes how the server is composed, booted and protected; how the database target is chosen and a wrong target refused; and how the whole system is exported and restored in resumable steps by the root administrator alone. It also covers the per-visitor demo sandbox, the installable app with its 'we are updating' page during a switch-over, and the build, check and release pipeline with its guards. Every other module depends on it, because a fault here is seen by every user.

Root administrator · System administrator · The system (automatic) · Public visitor

3.17.2 User journeys

Root administrator — **Export a full backup, then restore it**

1. The root account opens the system vault and starts an export job; the server answers 202 with the job.
2. The job advances in short steps (6500 ms budget each) and its chunks are staged in Firestore, so closing the browser loses nothing.
3. When ready, a gzip file is downloaded with its SHA-256 and document count in the response headers.
4. Later the administrator previews the file; the server checks format and digest and returns per-collection counts without writing anything.
5. Import starts with an explicit confirmation header; the server takes a safety export first, then replaces data at resumable checkpoints.
6. If the result is not what was wanted, the administrator returns to the safety restore point with 'undo'.

Public visitor — Trying the system in an isolated sandbox

1. The visitor presses 'try the system'; a sandbox is created in server memory and a session cookie starting with demo_ is set.
2. Sign-in opens at once on the preset department and term while the larger stories (64 student profiles and more) are seeded in the background, shown as 'preparing'.
3. He switches role from the bar (30 switches per minute) and sees the screens with that role's real scope and permissions.
4. Trying to edit an administrative screen (users, permissions, system vault) is refused with 403 and a note that it is display-only.
5. He resets with the reset button; the sandbox is replaced by a fresh one and the one-hour lifetime renews.

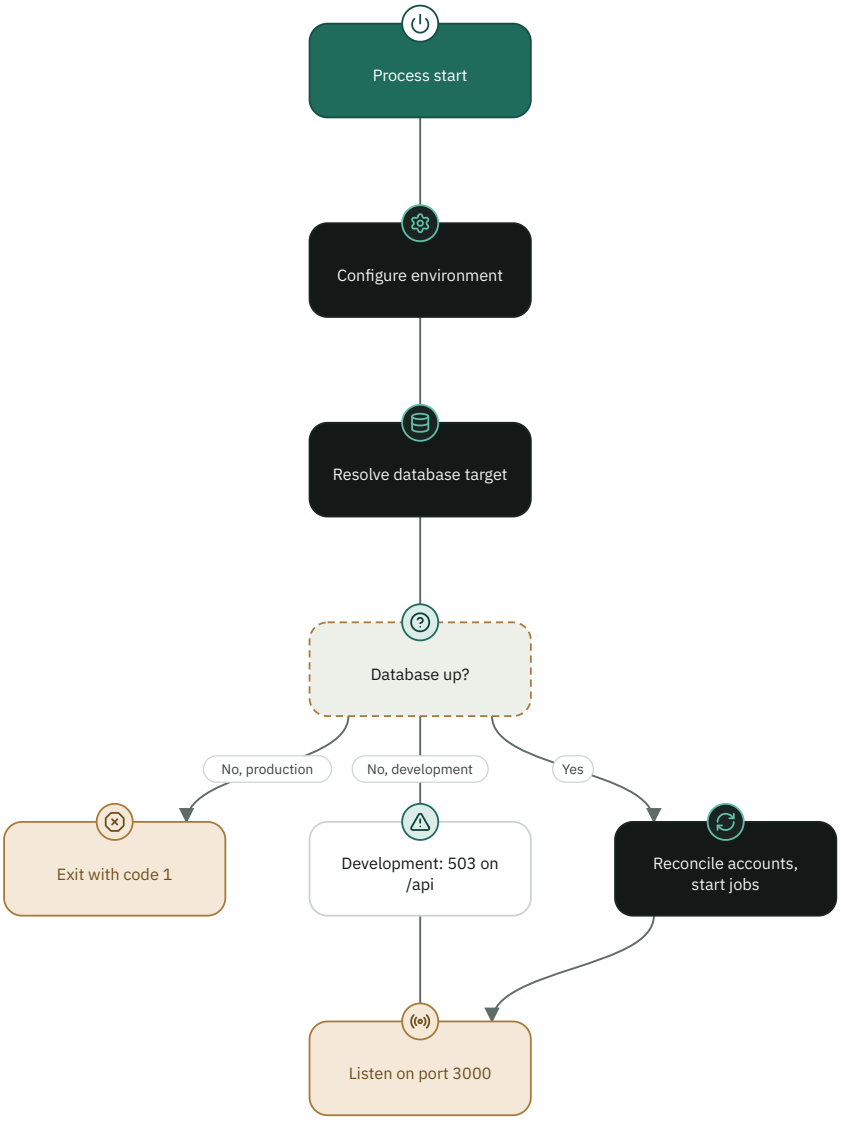
The system (automatic) — Releasing a new build to the official site

1. A change request runs the quick CI (skipping the seven slow tests) after the target and hygiene guards.
2. After the merge into main the full CI runs every test; only a commit that passed it can be released.
3. The release pipeline refuses any commit the full CI has not passed, and checks the kill switch and the owning repository.
4. One immutable image is built and pushed, then only the official service's image is changed, never its settings.
5. The pipeline compares the build id in sw.js on the official domain and on the service address, failing if they still differ after 20 tries.
6. Open pages notice the new build through /api/version and reload themselves.

3.17.3 Diagrams

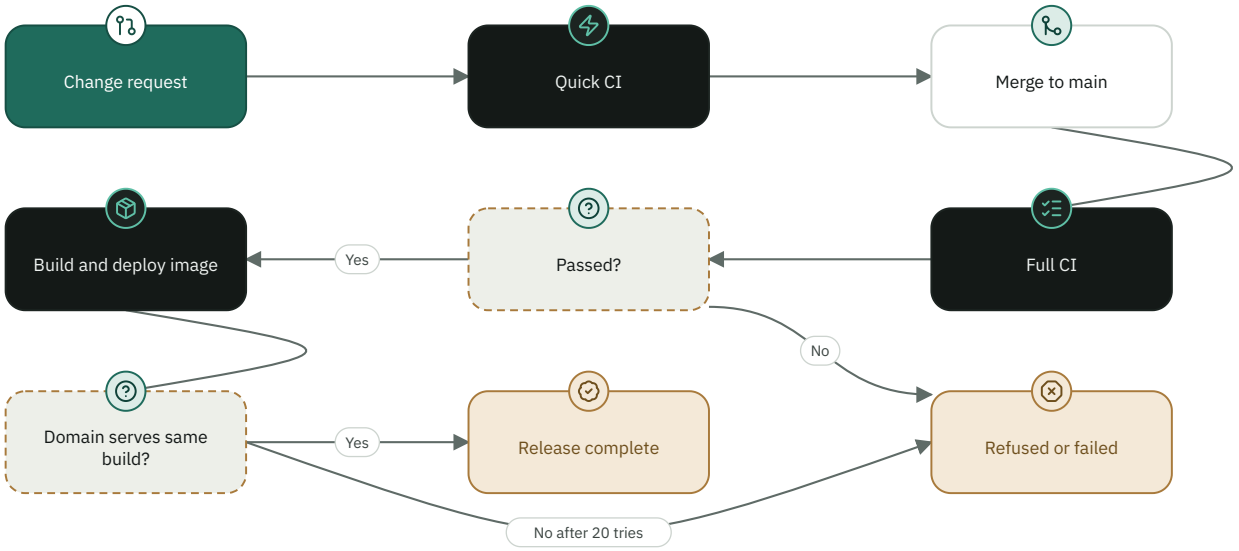
Server boot

A fixed order: environment first, then the database; in production the port is never opened without one.



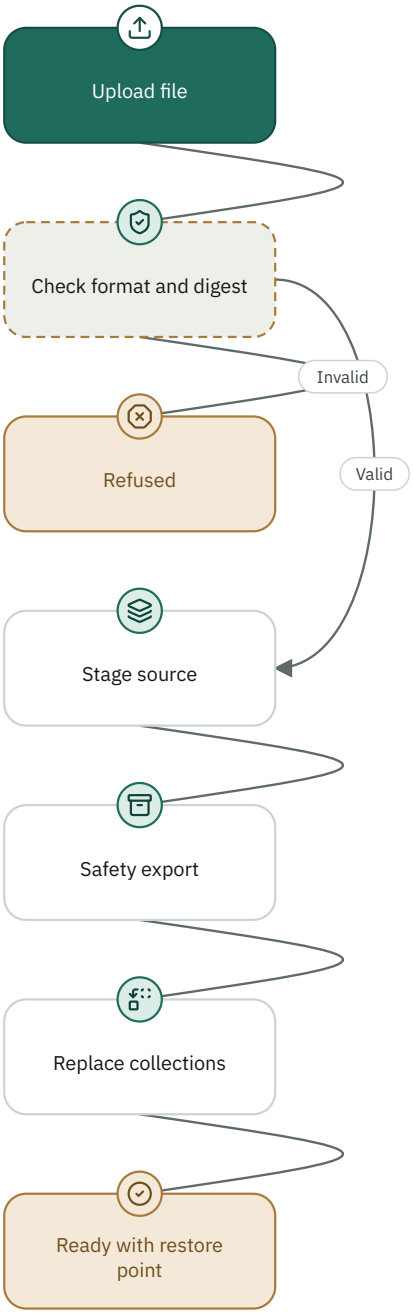
Release path

Only a commit that passed the full CI on main reaches the official site, and service settings are never touched.



Full backup import

Each step is a durable checkpoint; the safety export precedes any replacement.



3.17.4 Screens

System vault

Full export, import, reset and undo, for the root account only.`src/server/routes/adminRoutes.ts`

'We are updating' page

A standalone page with no external request that replaces the platform's error during a release.`public/updating.html`

Demo entry and role bar
Starts the visitor sandbox, switches role and resets it.`src/server/routes/authRoutes.ts`

3.17.5 Requirements

FR-OPS-001 **Boot order, private directory and environment file**

The system shall configure the runtime environment before any server module is loaded (the boot module remaining the first import in server.ts), read the environment file from a private directory outside the release folder (by default a schedule-private sibling, and under /tmp on Cloud Run), copy a local .env into it once with mode 600 without overwriting, and in production refuse a private directory inside the release folder.

● Implemented · Must · Functional · The system (automatic), System administrator

RATIONALE

> Many modules read variables at load time; loading them before configuration yields a silently different setup.

ACCEPTANCE CRITERIA

1. The first line of server.ts imports core/boot, which calls configureRuntimeEnvironment() after the original modules and before any core or routes module.
2. The route table and middleware order after the split equal the snapshot taken before the split (server parity).
3. On Cloud Run with SCHEDULE_PRIVATE_DIR set to a temporary folder, the file is read from there and nothing is written into the application folder.
4. In production the repository throws 'SCHEDULE_PRIVATE_DIR must be outside the application release directory in production' when it lies inside the release.

EVIDENCE

`server.ts` `src/server/core/boot.ts` `src/server/runtimeEnv.ts` `src/db/repository.ts` `scripts/protect-private-state.mjs` `docs/SAFE_DEPLOYMENT.md` `tests/server-route-parity-audit.ts` `tests/firestore-target-audit.ts`

FR-OPS-002 **Cloud Run production mode**

When the system detects it runs on Cloud Run (K_SERVICE, K_REVISION or K_CONFIGURATION) outside the preview environment, it shall force NODE_ENV=production, DATA_MODE=firestore and AUTO_IMPORT_LEGACY_DATA=false, with the sole exception of a dedicated demo service declaring SCHEDULE_DEMO_SERVICE=true, which runs on in-memory demo data.

● Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. With K_SERVICE=schedule, DATA_MODE becomes firestore even when the file says otherwise.
2. With SCHEDULE_DEMO_SERVICE=true, DATA_MODE becomes demo and the service never opens the business store.

EVIDENCE

`src/server/runtimeEnv.ts` `tests/firestore-target-audit.ts`

FR-OPS-003 Firestore target selection and refusal of an implicit target

The system shall resolve the Firestore project and database by a fixed precedence: on Cloud Run the application config file wins and the environment fills gaps, elsewhere the environment wins and the file fills gaps; it shall warn when an environment value contradicts the file and log the target actually used at boot, and shall refuse to start on Cloud Run when the project id or database id is missing or is '(default)', so it never connects to a database other than the one holding the university's records.

● Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. On Cloud Run a FIREBASE_DATABASE_ID that contradicts the file yields the file's value and exactly one warning; agreement yields none.
2. With FIREBASE_CONFIG_SOURCE=environment on Cloud Run both ids are required, and boot is refused if the file contradicts them.
3. An empty database id or '(default)' in any letter case throws 'Refusing to start on Cloud Run'.

EVIDENCE

`src/server/firestoreTarget.ts` `src/server/runtimeEnv.ts` `docs/DELIVERY_OPERATIONS.md` `src/db/repository.ts` `tests/firestore-target-audit.ts`

FR-OPS-004 No service without a database

When database initialisation fails, the system shall exit with code 1 in production so the previous revision keeps serving; in development it shall stay up and answer every /api request with 503 and an 8-hex-character reference without the driver's error text, and never fall back to packaged substitute data.

● Implemented · Must · Functional · The system (automatic), System administrator

RATIONALE

> A site that answers while showing a schedule that is not the university's is more dangerous than a site that is down.

ACCEPTANCE CRITERIA

1. With state.databaseDown set, any /api path (except /api/landing/inquiry and those registered ahead of the guard such as health and version) answers 503 JSON containing ref.
2. The body contains no hostnames or credential paths; the full text is in the server log under the same reference.

EVIDENCE

`server.ts` `src/server/core/state.ts` `tests/run-tests.ts`

FR-OPS-005 Account reconciliation and scheduled jobs at boot

After the database comes up the system shall migrate accounts without a role to the schedule committee chair role (except the root account), remove the workshop screen (7) from department-head accounts, and start the automatic next-term job, the expired-student-profile purge (every 6 hours, first run one minute after boot) and the automatic radar job; all are idempotent and a failure never blocks boot.

● Implemented · Should · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Restarting the server does not change an account that already has a role and writes nothing when there is nothing to migrate.
2. A migration failure is logged and boot continues.

EVIDENCE

`server.ts` `tests/role-guard-audit.ts` `tests/term-calendar-audit.ts`

IR-OPS-001 Health interface

The system shall provide an unauthenticated GET /api/health that needs no session lookup and returns JSON with ok, database (up or down) and, when down, a ref, and keeps answering even while the rest of /api answers 503.

Implemented · Must · Interface · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. With a healthy database: ok=true, database=up, no ref.
2. With state.databaseDown: ok=false, database=down and ref taken from state.databaseDownRef.

EVIDENCE

`src/server/routes/systemRoutes.ts` `src/server/core/state.ts` `tests/updating-page-audit.ts` `tests/instructor-offering-flow-audit.ts`

IR-OPS-002 Build identity and version interface

The system shall stamp every build with one identifier (timestamp plus random part) carried by the client bundle, the server bundle and the service worker, expose it through GET /api/version with a no-store header, and have an open page reload itself whenever the server's identifier differs from its own.

Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. After stamp-sw, dist/sw.js no longer contains the text __BUILD__ and holds the same identifier as the bundle.
2. If /api/version is answered by an error page (not ok), the page takes no decision and asks again later.
3. A tab that already reloaded for a build and still runs old code clears every cache and drops service workers (except the push worker), then reloads.

EVIDENCE

`src/server/routes/systemRoutes.ts` `scripts/stamp-build.mjs` `scripts/stamp-sw.mjs` `src/main.tsx` `tests/updating-page-audit.ts`

FR-OPS-006 Security headers on every response

Inferred

The system shall add to every response: X-Content-Type-Options=nosniff, Referrer-Policy=same-origin, a Permissions-Policy denying camera, microphone and geolocation, a CSP forbidding plugins (object-src 'none') and restricting base-uri and form-action to the same origin, X-Frame-Options=DENY with frame-ancestors 'none' outside the embedded preview environment, and Strict-Transport-Security for 31536000 seconds in production only; it shall disable the X-Powered-By header.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A request to any path returns the headers above, and HSTS appears only when NODE_ENV=production.
2. A full CSP is deliberately not imposed on the application shell; the server-built public pages carry a strict nonce policy (see NFR-NFR).

EVIDENCE

`server.ts` `tests/ads-audit.ts` `tests/room-door-audit.ts`

FR-OPS-007 Compression and request size limits

Inferred

The system shall compress text responses over 1024 bytes (never SSE or NDJSON streams), accept JSON up to 1 MB in general and 4 MB for the drafts path only, and accept a raw body up to 30 MB for the backup preview and import paths only.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A JSON request over 1 MB to an ordinary path is refused with 413 and code PAYLOAD_TOO_LARGE.
2. A text/event-stream or x-ndjson response leaves uncompressed.

EVIDENCE

`server.ts` `tests/student-journey-audit.ts`

FR-OPS-008 Refuse cross-site write requests

Inferred

The system shall refuse with 403 any POST, PUT, PATCH or DELETE to /api whose Sec-Fetch-Site is outside {same-origin, same-site, none} or whose Origin host differs from the request host or cannot be parsed; it enables CORS for no other origin.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A POST with Origin: https://other.example to a different host is refused with 403 and 'الطلب غير مسموح'.
2. GET requests and requests with neither Origin nor Sec-Fetch-Site (server tools) are not refused by this rule.

EVIDENCE

`server.ts`

FR-OPS-009 Public-surface and teacher-card rate limits

The system shall limit /api/public requests to 60 per window (60000 ms) per address, tunable by PUBLIC_RATE_LIMIT_MAX and PUBLIC_RATE_LIMIT_WINDOW_MS (0 disables it), answering 429 with a Retry-After in seconds when exceeded; and limit the teacher card path /api/public/request/:token to 240 requests per minute per link (keyed by a truncated sha256 fingerprint, never the token) with a ceiling of 1500 per address, tunable by PUBLIC_REQUEST_PER_LINK_PER_MINUTE and PUBLIC_REQUEST_PER_IP_PER_MINUTE, exempting the student journey path (/api/public/me), which has its own limits.

Implemented · Must · Functional · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. The 61st request from one address inside the window returns 429 with a Retry-After between 1 and the window length.
2. Expired windows are swept once per window so address rows do not accumulate.
3. 130 requests spread over four links from one address all pass.
4. A single link repeating without pause stops at its own limit with 429.

EVIDENCE

`src/server/core/rateLimits.ts` `server.ts` `tests/run-tests.ts` `tests/student-door-shared-network-audit.ts` `tests/teacher-card-flow-audit.ts`

FR-OPS-010 Anonymous-door guard

The system shall look up a session only for an id shaped sess_ or demo_ followed by 64 hex characters; remember dead ids for 30 minutes (capacity 20000); limit failed lookups per address to 120 per 10 minutes (hard ceiling 1200); and limit unmatched tokens on link pages (/s /q /r /m /t) to 40 per 10 minutes (hard ceiling 400), answering 429 with Retry-After=600; failures are counted, never successes, and each success earns credit.

Implemented · Must · Functional · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. Hundreds of valid links with the odd mistake from one address are never blocked.
2. An address that only fails stops at its limit and is read as signed out without asking Firestore.

EVIDENCE

`src/server/anonymousGuard.ts` `src/server/publicAttemptLimiter.ts` `server.ts` `tests/anonymous-guard-audit.ts`

FR-OPS-011 Sign-in attempt limit

Inferred

The system shall answer 429 with 'too many sign-in attempts' when one address reaches five attempts within a minute on sign-in, password change, demo entry and the /i/:username link.

Implemented · Must · Functional · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. The sixth attempt within 60 seconds from the same address is refused, and the window reopens after the minute passes.
2. The counter counts every attempt including successful ones and demo entry, and lives in instance memory.

EVIDENCE

`src/server/routes/authRoutes.ts`

FR-OPS-012 Document reading gate

The system shall allow two concurrent scanned-document reads (PDF or image), queue further ones in a queue of 8, answer 503 when it is full, and release the slot when the request finishes or the client closes.

Implemented · Must · Functional · The system (automatic)

RATIONALE

> Each page is rendered in process memory; four reads together took the server down for lack of memory.

ACCEPTANCE CRITERIA

1. The third read waits, and the eleventh (two active, eight waiting) is refused with 503.

EVIDENCE

`src/server/core/http.ts` `src/server/routes/pdfImportRoutes.ts` `tests/section-planning-audit.ts`

FR-OPS-013 Static serving and cache policy

Inferred

In production the system shall serve the dist folder with index.html and sw.js uncached, content-hashed js and css immutable for a year, other js and css always revalidated, and fonts and images cached for a year; it shall answer a plain 404 for any path under /assets/ or ending in a file extension and return the application shell only for navigation paths.

Implemented · Must · Functional · The system (automatic)

RATIONALE

> A missing chunk answered with HTML means a blank page with nothing wrong in the network log.

ACCEPTANCE CRITERIA

1. A request for a non-existent /assets file returns 404 with Cache-Control: no-store.
2. A file whose name matches -[A-Za-z0-9_-]{8,}.js is returned with public, max-age=31536000, immutable.

EVIDENCE

`server.ts` `vite.config.ts` `tests/updating-page-audit.ts`

FR-OPS-014 Error contract

The system shall catch every route error in one handler that logs a reference of the form E plus 8 characters with the method and URL (as an argument, never as a format string), answering 413 with PAYLOAD_TOO_LARGE for oversize bodies, 400 with BAD_JSON for invalid JSON, 409 for a request-reservation conflict, and otherwise 500 with INTERNAL and the reference without internal text; any unmatched /api path answers a JSON 404.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A request URL containing %s does not swallow the error text in the log.
2. The 500 body carries code=INTERNAL and ref and no collection name or document path.

EVIDENCE

`server.ts` `src/server/core/http.ts` `tests/input-hardening-audit.ts`

FR-OPS-015 Resumable full-system export

Inferred

The system shall let the root account alone export all durable documents through a job that advances in steps of at most 6500 ms over pages of 250 documents, with its gzip chunks staged in Firestore, excluding sessions, client telemetry, secrets, restore points and export/import jobs, and the output downloaded with X-Backup-SHA256 and X-Backup-Documents headers without walking the database.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. Any account other than the root receives 403 'this vault is for the root administration account only'.
2. Closing the browser mid-job and reopening the vault resumes the same job from its last cursor.
3. The download carries Cache-Control: private, no-store, and the old GET /api/system-backup/export answers 409.

EVIDENCE

`src/server/routes/adminRoutes.ts` `src/db/repository.ts` `src/server/core/auth.ts` `tests/role-guard-audit.ts`

FR-OPS-016 Backup preview, import, system reset and undo

Inferred

The system shall preview a backup before any write by checking its format (schedule-system-backup/1) and SHA-256 digest, decompress gzip with an 80 MB cap enforced during expansion rather than after, require the header x-schedule-confirm=FULL-SYSTEM-IMPORT for import and take a safety export before replacing data at durable checkpoints; reset business data only with the header FULL-SYSTEM-RESET and the Arabic confirmation phrase for 'reset the system' (keeping the root account and creating a restore point); and return to the safety point only with the header FULL-SYSTEM-UNDO.

Implemented · Must · Functional · Root administrator

ACCEPTANCE CRITERIA

1. A file with one altered byte is refused with 'backup integrity check failed (SHA-256)' and nothing is written.
2. An import request without the header answers 409 'full import needs explicit confirmation'.
3. The old POST /api/system-backup/import answers 409 and points to the stepped import.
4. A reset with the right header and the wrong phrase answers 409 'type the reset phrase to confirm'.
5. After undo the response carries restored and redoPoint.

EVIDENCE

`src/server/routes/adminRoutes.ts` `src/db/repository.ts` `tests/role-guard-audit.ts` `tests/safety-net-restore-audit.ts`

FR-OPS-017 An isolated demo sandbox per visitor

The system shall bind every request carrying a demo session cookie (demo_ + 64 hex characters) to a private in-memory data copy lasting 60 minutes, sweep expired ones at most once a minute, never let a demo request read or write Firestore or process-wide caches, build every cache key with a real or demo:<session> prefix, and answer 404 on demo entry when SCHEDULE_DEMO_ENABLED=false.

Implemented · Must · Functional · Public visitor, The system (automatic)

RATIONALE

> Marking the registration radar stale reached Firestore from a demo edit and changed real results.

ACCEPTANCE CRITERIA

1. Editing a slot inside a sandbox is invisible to every other visitor and to the real system.
2. Every repository branch that writes to Firestore checks for the absence of a sandbox context (firestoreDb && !demoSandboxContext).
3. Lifetime expiry or a server restart shows 'the demo session has ended' rather than 'sign in'.

EVIDENCE

`src/db/repository.ts` `src/server/dataContextCache.ts` `src/server/routes/authRoutes.ts` `src/server/core/auth.ts` `server.ts` `tests/demo-firestore-isolation-audit.ts` `tests/demo-complete-audit.ts` `tests/review-audit-2026-10-28.ts`

FR-OPS-018 Demo sandbox limits and management

The system shall refuse with 403 every write inside the demo sandbox to users, permissions, scopes, the system vault, teaching load and university day, expose GET /api/demo/config (enabled, 60 minutes, isolated, administration read-only), and allow role switching limited to 30 per minute and reset.

Implemented · Must · Functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. A POST to /api/system-backup/reset inside the sandbox answers 403 with a display-only notice.
2. The 31st switch within a minute answers 429.
3. Logout, heartbeat, reset and role are exempt from the write guard.

EVIDENCE

`server.ts` `src/server/core/auth.ts` `src/server/routes/authRoutes.ts` `src/server/routes/systemRoutes.ts` `tests/demo-roles-audit.ts` `tests/demo-screens-audit.ts`

FR-OPS-019 Demo seeding and role stories

The system shall prepare the core sandbox at sign-in and seed the larger stories in the background, serialised per session (role stories, other departments, 64 radar student profiles, student-journey demand, section requests, activity trail), expose the preparation state (preparing, ready or failed), never open a role on an empty screen, and keep every name, civil number and figure synthetic.

Implemented · Should · Functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. Sign-in does not wait for the large seeding, and /api/demo/seed-status returns the current state.
2. A reset during an older seed does not leave the new sandbox half-populated because jobs are serialised.
3. After preparation, every role in the switcher has non-empty lists on its main screens.

EVIDENCE

`src/server/routes/authRoutes.ts` `src/server/demoRadarSeed.ts` `src/server/demoPreviousTermSeed.ts` `src/server/demoSectionRequestsSeed.ts` `src/server/core/demo.ts` `src/db/demoSandbox.ts` `tests/demo-every-role-audit.ts` `tests/demo-cases-audit.ts`

FR-OPS-020 Service worker and offline behaviour

The service worker shall, at install, store 10 shell files plus the shell page and the updating page; serve navigation network-first and fall back to the stored shell when offline; cache GET responses for exactly fifteen named /api prefixes and replay them when offline; never cache /api/auth, /api/users or the two personal paths /api/schedules/demand and /api/student-registration; serve content-hashed files from cache without the network; and leave public link pages, the room door and advertisements to the network alone.

Implemented · Must · Functional · The system (automatic), System administrator

ACCEPTANCE CRITERIA

1. A cached read while offline returns the last successful copy; if none exists it returns 503 JSON with the X-Schedule-Offline header.
2. On activation every cache other than the current three is deleted, and the API cache is cleared when the user identity changes.
3. The first offline launch after install opens the stored shell, otherwise a 'no internet connection' page.

EVIDENCE

`public/sw.js` `src/App.tsx` `src/main.tsx` `tests/updating-page-audit.ts` `scripts/verify-pwa.mjs`

FR-OPS-021 The 'we are updating' page

The system shall serve /updating.html in place of any 500, 502, 503, 504 or dropped connection on a GET navigation (not /api, not non-GET, not a successful answer) with status 503 and Retry-After=5; the page asks /api/health every 3 seconds, slowing to 15, reopens the requested link when it answers, and after three stalls in two minutes slows to once every 30 seconds without looping.

● Implemented · Must · Functional · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. No 5xx answer is stored in any cache, and neither /api nor POST requests are answered with the page.
2. The page makes no external request, is RTL and in standard Arabic, and says there is no connection when navigator.onLine is false.

EVIDENCE

`public/updating.html` `public/sw.js` `src/utils/updatingNotice.ts` `tests/updating-page-audit.ts`

IR-OPS-003 Installable-app manifest

The manifest shall declare a standalone app (display=standalone) with scope /, language ar, direction rtl and start URL /?source=pwa, with 192 and 512 icons, an svg icon and maskable 192 and 512 icons, and the verify:pwa check shall guard these on every build.

● Implemented · Should · Interface · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. The listed icons exist with the right dimensions (the check reads the PNG header).
2. A verify:pwa failure stops npm run build before building.

EVIDENCE

`public/manifest.webmanifest` `scripts/verify-pwa.mjs` `index.html` `package.json` `scripts/verify-pwa.mjs`

FR-OPS-022 Continuous-integration pipeline

The CI pipeline shall run on every push to main and every change request against it with a 20-minute timeout, Node 22 and Python 3.12, in this order: deploy target guard, repository hygiene guard, dependency install, type check (tsc with no emit), tests (quick on change requests, full on main), production build, then start the built demo server and verify the Analytics Center scopes through its API; a run on main is never cancelled by a newer one.

● Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A change request runs npm test -- --quick skipping 7 slow tests; main runs them all.
2. A server left on port 3000 by an earlier test is killed before the final verification step.

EVIDENCE

`package.json` `tsconfig.json` `tests/repo-hygiene-guard.mjs`

FR-OPS-023 Deploy target guard and release pipeline

Deployment shall happen only from the owning repository with PRODUCTION_DEPLOY_ENABLED=true and only for a commit the full CI on main passed (the pipeline waits up to 60 checks 30 seconds apart); it shall change only the image of the official service in europe-west1, never the service account, variables, secrets or memory, never return to the retired europe-west2, and never touch the domain mapping; CI fails if the pipeline violates any of this, and after release the pipeline verifies the official domain and the service address serve the same build identifier from /sw.js.

● Implemented · Must · Functional · The system (automatic), Root administrator

RATIONALE

> One commit pointed every deploy at another project while CI stayed green, so the official site missed releases for a day; recreating the domain mapping took HTTPS down while a certificate was issued.

ACCEPTANCE CRITERIA

1. Any command touching the domain mapping in any yaml, js, ts, sh, json or py file fails the guard.
2. Adding --memory, --set-env-vars or --service-account to the deploy command fails the guard.
3. Verification failing after 20 tries 30 seconds apart fails the release with a message forbidding any change to the domain mapping.

EVIDENCE

`tests/deploy-target-guard.mjs` `docs/DELIVERY_OPERATIONS.md` `tests/deploy-target-guard.mjs`

FR-OPS-024 Repository hygiene guard

The guard shall scan text files and path names in the repository and fail on a reference to a banned development tool, an instruction or prompt file, or leftover scratch audit files, skipping the excluded folders (.git, node_modules, dist, coverage, build, database).

● Implemented · Should · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. The guard exits with code 1 listing the offending path, and with code 0 and a success message when there is none.

EVIDENCE

`tests/repo-hygiene-guard.mjs` `tests/repo-hygiene-guard.mjs`

FR-OPS-025 Test runner policy

npm test shall run every script listed in tests/suite.txt (252 scripts) even when some fail, count as failed any test that exits non-zero, skipped its live part because the port was busy, or left a server listening (killing that server), skip tests/slow.txt under --quick, reject a script that does not exist, is duplicated or is slow but unlisted, and fail a known failure that now passes until it is removed from known-failures.txt.

● Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Port 3000 busy before the start stops the run with a message to close whatever holds it.
2. No active known failure exists today in known-failures.txt (comments only).

EVIDENCE

`scripts/run-tests.mjs` `tests/suite.txt` `tests/slow.txt` `tests/known-failures.txt` `scripts/run-tests.mjs`

NFR-OPS-001 A small, fail-early runtime image

Inferred

The image shall be built in two stages from node:22-bookworm-slim so the runtime stage carries the package file, dist, only the dependencies the server bundle names, the Firestore target file and the two OCR language files; the build shall fail on the runner (never on the live site) if a package is missing (--smoke load check); it shall set MALLOC_ARENA_MAX=2 and run node dist/server.cjs on port 3000.

Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> Scan reading starts about 11 workers each with its own memory arena; on a 4 GiB server it was killed for lack of memory.

ACCEPTANCE CRITERIA

1. A package required by dist/server.cjs and missing fails the npm prune --omit=dev && node scripts/runtime-deps.mjs --prune --smoke step.
2. Procfile and CMD both run the same node dist/server.cjs.

EVIDENCE

`Dockerfile` `Procfile` `scripts/runtime-deps.mjs` `cloudbuild.yaml`

FR-OPS-026 Backup and recovery procedure

The system shall ship with a documented, executable procedure to protect the production database: enabling point-in-time recovery (7-day PITR), delete protection and scheduled exports to a storage bucket, with the steps to restore and verify, and the system itself shall protect its data in local mode and in the system vault.

Implemented · Should · Functional · Root administrator

RATIONALE

> Turning these options on is the hosting party’s decision on its own project because some carry a cost; the document commits to the ready procedure, and activation is listed under assumptions and dependencies.

ACCEPTANCE CRITERIA

1. The procedure gives the complete enable, restore and verify commands, and runs as written on any project.

EVIDENCE

`scripts/firestore-backup.md` `tests/firestore-backup-procedure-audit.ts`

3.17.6 Edge cases

⚠ Database unreachable at boot

Scenario A new revision starts on Cloud Run and Firestore initialisation fails (wrong target or outage).

Behaviour The process exits with code 1 so the previous revision keeps serving; the system never falls back to a packaged snapshot. In development the shell runs and /api answers 503 with a reference.

⚠ A release while people are using the system

Scenario The platform switches servers and its front end returns an English error page or the request drops.

Behaviour The service worker serves the 'we are updating' page and reopens the link when /api/health answers; inside the app a calm banner replaces an error banner, and no 5xx is stored in any cache.

⚠ A stale tab requests a removed chunk

Scenario A tab left open from the previous release requests a hashed file that no longer exists.

Behaviour The server answers a plain uncached 404 instead of an HTML shell with status 200, so the boot guard recovers the page by clearing caches and reloading.

⚠ A whole campus behind one address

Scenario Hundreds of students and teachers open their valid links from one IP address in the same minute.

Behaviour Failures are counted, not successes, each success earns credit, and teacher-card limits are per link rather than per address; so no valid-link holder is blocked, while a guesser stops at the hard ceiling.

⚠ A size-bomb backup

Scenario A compromised account uploads a small gzip file that expands to several gigabytes.

Behaviour Size is measured during expansion and stops at 80 MB, refusing the file before it fills memory or blocks the event loop; the path is in any case limited to the root account.

⚠ Demo session expiry or server restart

Scenario The visitor's sandbox expires after an hour or a new release wipes server memory.

Behaviour He is told the demo session ended and its data was erased and is asked to start a new trial, not 'sign in' to someone with no account.

3.17.7 Standing owner decisions

- The official domain mapping is never deleted, recreated or moved; certificate issuance interrupts the service.
- A normal release changes the service image only; no second service or region without the owner's decision.
- Printed reports keep their design; operational work never touches it and faults are fixed in values and data.

↑ Contents

3.18 · NFR

Non-functional requirements

Performance, security, privacy, accessibility, Arabic-first design and reliability, with measured figures.

39	100%	0	0
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Screens · 5. Requirements · 6. Edge cases · 7. Standing owner decisions

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3.18.1 Overview

This module gathers the qualities that govern the whole system rather than one screen: response under load, protection of accounts, links and data, the privacy policy, accessibility for every user, Arabic, text direction and numerals, phone layout and printing, write and recovery reliability, and maintainability. Each requirement is tied to an existing measurement, test or audit that fails the build if it regresses.

The system (automatic) · Root administrator · System administrator · Standard user · Faculty member · Student · Public visitor · External service

3.18.2 User journeys

Student — **A student opens the journey page on registration day from the campus network**

1. The student opens his link on a phone over the campus network, which the server sees as one address.
2. Only failures are counted against the address, so he is not blocked while entering his correct code.
3. His profile is created and his civil number stored sealed with encryption that logs cannot reveal.
4. The page renders in a self-hosted Arabic typeface, right to left, with no horizontal scroll.
5. Each of the eleven journey steps completes in under a third of a second on the server.

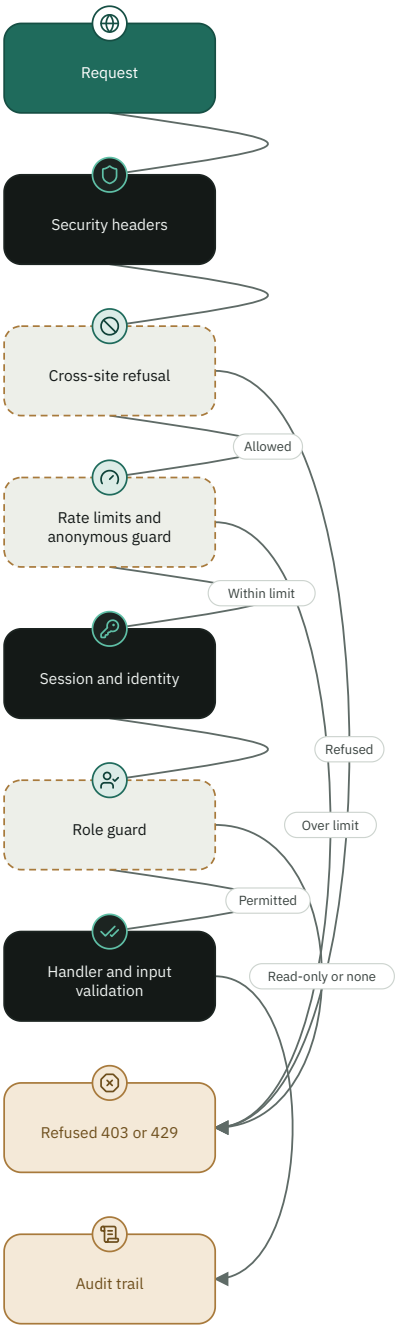
System administrator — **Auditing an operation after it happened**

1. The system administrator opens the audit log to learn who signed in and who tried and was refused.
2. He finds a line for every successful write: user, method, path, entity and status, without the request content.
3. He finds refused attempts with status 403 or 409 labelled 'outside permission' or 'conflict'.
4. He returns to the schedule version taken before the change and restores it if nobody else changed it afterwards.

3.18.3 Diagrams

Request defence layers

What an anonymous or signed-in request passes before reaching its handler; the server alone is the authority on permission.



3.18.4 Screens

Privacy policy

A public page that follows the device's light or dark setting.`public/privacy-policy.html`

3.18.5 Requirements

NFR-NFR-001 Student journey throughput

The system shall serve the student's first-visit journey (11 requests) at 15 students per second, i.e. 54 thousand per hour, without failure, with every step completing on the server in under a third of a second.

Implemented · Must · Non-functional · Student, The system (automatic)

RATIONALE

> A wishes or registration day brings twenty thousand students within hours.

ACCEPTANCE CRITERIA

1. Measured on 2026-10-24 on a production-built demo copy: 900 of 900 journeys succeeded from distinct addresses, the slowest step being sign-in at 148 ms and home at 107 ms.
2. The measurement shared processors with the emulator and the load generator, and the Firestore emulator is not the real service.

EVIDENCE

`scripts/loadtest/student-door.mjs` `src/server/studentDoorRoutes.ts` `docs/LOAD_TEST.md` `tests/student-door-shared-network-audit.ts`

NFR-NFR-002 Limit fairness on a shared network

Protection limits shall not block students from one address (a campus network or a mobile carrier behind NAT): writes 60 per minute per student rather than per address with a 1200 ceiling per address, identity questions 24000 per 10 minutes, and code errors 600 per 10 minutes with success credit and an absolute ceiling of 3000.

Implemented · Must · Non-functional · Student, The system (automatic)

RATIONALE

> Before the fix only 3.3% (20 of 600) of students behind one address completed their journey.

ACCEPTANCE CRITERIA

1. 600 students at 5 per second from one address: 600 of 600 after the fix.
2. A single student repeating writes stops at sixty per minute with 429.
3. The shrunken scenario (ceiling 60 instead of 600) blocked 199 valid-code holders on the old code and none on the new.

EVIDENCE

`src/server/studentDoorRoutes.ts` `src/server/publicAttemptLimiter.ts` `docs/LOAD_TEST.md` `tests/student-door-shared-network-audit.ts` `tests/student-door-audit.ts`

NFR-NFR-003 Opening the teacher card and department screen

The teacher card and the department screen shall be read with parallel Firestore reads (the link and its request together, one term context per term, college and department) rather than a chain of sequential trips, with responses identical before and after the change.

Implemented · Should · Non-functional · Faculty member, The system (automatic)

ACCEPTANCE CRITERIA

1. On a real database with 300 teachers and reads near 170 ms: opening the card from 1.9 s to 1.2 s, and the department screen (about 108 requests) from 3.5–5.4 s to 1.5 s.
2. Processor cost in the full run (one thousand teachers) was 381 CPU-seconds, under 0.4 s per teacher.

EVIDENCE

`src/server/core/instructorRequests.ts` `scripts/loadtest/teacher-card.mjs` `docs/LOAD_TEST.md` `tests/teacher-card-flow-audit.ts`

NFR-NFR-004 Guide performance budget

Understanding a guide question shall take no more than 8 ms, building its index no more than 300 ms, and the knowledge bank shall not exceed 70000 bytes, with the guide kept in a lazily loaded bundle; the production build fails if any is exceeded.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. The mean over 1000 questions after 200 warm-ups is under 8 ms per question.
2. Removing the lazy import of GuideAssistant from App.tsx fails the check.

EVIDENCE

`scripts/guide-performance-audit.mjs` `src/guide/bank.ts` `package.json` `scripts/guide-performance-audit.mjs`

NFR-NFR-005 Database read economy

A query on a large Firestore collection (Enterprise edition, which does not index automatically) shall be added only when its leading field heads an index in firestore.indexes.json, and the wishes counter shall read one department's student files, not a whole college's; the index audit fails when the condition is not met.

Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> Before indexes the fifty heaviest queries read about 590 million read units in a week, nearly all full scans; the wishes counter read about 300 thousand reads for a college with ten thousand files.

ACCEPTANCE CRITERIA

1. A new query on a counted large collection (such as schedules at about 16,000 rows) without an index fails tests/firestore-index-audit.ts.
2. College reads per counter window became the number of its files (about ten thousand), roughly 30 times fewer.

EVIDENCE

`firestore.indexes.json` `docs/LOAD_TEST.md` `docs/DELIVERY_OPERATIONS.md` `tests/firestore-index-audit.ts`

NFR-NFR-006 Caching layers and delivery weight

Inferred

The server shall hold session identity for 120 seconds (dropped at once on sign-out or account edit), reference data for 90 seconds and schedule rows for 300 seconds with invalidation on any write; text responses over 1024 bytes shall be compressed, hashed files cached for a year, the react and icon bundles and deferred CSS split from the main bundle, and two Arabic font weights preloaded.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Editing an account or its scope drops the cached entry so a revoked account cannot keep working from memory.
2. A static request (stylesheet or font) does no identity lookup; only /api is authenticated.

EVIDENCE

`src/server/core/auth.ts` `src/db/referenceCache.ts` `server.ts` `vite.config.ts` `index.html` `tests/demo-complete-audit.ts`

NFR-NFR-007 Permission enforced on the server

The server shall enforce permission at a single point at the entrance of all of /api: any non-read request from a read-only role (dean, vice dean, dean of registration) is refused with 403 whatever buttons the interface hides, and only approval-cycle paths are open to a reader according to the role; hiding an icon in the interface is not permission.

Implemented · Must · Non-functional · The system (automatic), Standard user

RATIONALE

> Checking in each route separately lets one omission open a write door nobody knows about.

ACCEPTANCE CRITERIA

1. Every combination of method, path and role is tested on the verdict, and the text of server.ts proves the guard is mounted in place.
2. A standard user without the required screen is refused on the server, not only in the interface.

EVIDENCE

`server.ts` `src/server/roleGuard.ts` `src/server/core/auth.ts` `docs/SECURITY_NOTES.md` `tests/role-guard-audit.ts` `tests/forbidden-fetch-guard-audit.ts`

NFR-NFR-008 Password protection

Passwords shall be stored hashed with scrypt and a per-password salt (64 derived bytes) and compared in constant time; sign-in with a legacy non-scrypt hash shall be blocked until reset; a new password shall be 8 to 128 characters; a temporary password shall be 14 characters valid for 24 hours for a single sign-in with no decryptable copy; and a first password (valid 7 days) shall force a change before any path outside /auth.

Implemented · Must · Non-functional · The system (automatic), Standard user, System administrator

ACCEPTANCE CRITERIA

1. A 7-character password is refused with an 'at least 8' message and a 129-character one as too long.
2. An account that must change its password reaches nothing but /auth until it does.
3. For the legacy screen the system keeps an AES-256-GCM decryptable copy of the password, protected by a separate key and permission for screen 11, and neither sign-in nor session endpoints return it.

EVIDENCE

`src/db/repository.ts` `src/server/accountPassword.ts` `src/server/routes/authRoutes.ts` `src/server/routes/usersRoutes.ts` `docs/SECURITY_NOTES.md` `tests/password-reset-audit.ts` `tests/first-login-password-audit.tsx`

NFR-NFR-009 Sessions and cookies

Inferred

The session id shall be 32 random bytes (prefix sess_ or demo_), its cookie HttpOnly, SameSite=Lax and Secure in production; a session shall end after 15 idle minutes on the server (renewed at most every 120 seconds), all sessions of an account shall be ended when it is handed over or edited, and abandoned session documents shall be swept by a Firestore TTL policy.

Implemented · Must · Non-functional · The system (automatic), Standard user

ACCEPTANCE CRITERIA

1. In production the cookie carries Secure, HttpOnly and SameSite=Lax; outside production without Secure.
2. An id not matching ^(sess|demo)_[0-9a-f]{64}$ is never looked up in Firestore.

EVIDENCE

`src/server/core/auth.ts` `src/server/routes/authRoutes.ts` `src/db/repository.ts` `src/server/anonymousGuard.ts` `tests/anonymous-guard-audit.ts`

NFR-NFR-010 Firestore rules deny every direct connection

Inferred

Firestore rules shall deny every read and write from the browser or client SDK (allow read, write: if false), so only the application server, using Admin SDK credentials that are not evaluated against rules, reaches the data; any later need for direct reads shall arrive as a narrow collection-scoped rule with a documented reason, never by widening this one.

Implemented · Must · Non-functional · The system (automatic)

RATIONALE

> The earlier 'authenticated only' rule would have let any identity the project mints read users, sessions and share links.

ACCEPTANCE CRITERIA

1. The file contains a single match /{document=**} rule with if false.

EVIDENCE

`firestore.rules` `firebase.json` `docs/SECURITY_NOTES.md`

NFR-NFR-011 Secrets handling

Secrets, vault keys and real snapshots shall not be kept in the repository or the image; external-service keys, the vault secret (PASSWORD_VAULT_KEY) and the calendar secret (CALENDAR_SECRET) shall come from the platform secret manager or private environment variables, and when the calendar secret is absent it shall be derived by HMAC from the stored shared secret under a dedicated label instead of a random key per boot.

Implemented · Must · Non-functional · The system (automatic), Root administrator

RATIONALE

> A random key per boot killed every calendar subscription on teachers' phones and gave two instances different keys.

ACCEPTANCE CRITERIA

1. An existing calendar subscription stays valid after a server restart and on a second instance.
2. The old administrator password that remained in git history was changed by the owner in the live system (2026-11-07), so it is no longer valid.

EVIDENCE

`src/server/calendarSecret.ts` `src/server/runtimeEnv.ts` `SECURITY-TODO.md` `docs/SECURITY_NOTES.md` `tests/credential-audit.ts` `tests/calendar-ics-audit.ts`

NFR-NFR-012 Public link tokens

A public link token shall be a long random string that expires (a 150-day fallback for term links, with reading until term end for certain kinds) and is revoked on withdrawal, and shall expose no civil number, phone, account or scope beyond its own; a calendar subscription shall carry a key derived by HMAC from the link and the instructor number, not the civil number; and limiter memory shall hold the token's fingerprint, never the token.

Implemented · Must · Non-functional · Faculty member, Student, Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. An expired link answers 410 and is not counted as a guess; a token matching nothing answers 404 and is counted.
2. A withdrawn link no longer yields its subscription key.

EVIDENCE

`src/server/core/shareLinks.ts` `src/utils/shareLinkLifetime.ts` `src/server/calendarSecret.ts` `src/server/anonymousGuard.ts` `src/server/core/rateLimits.ts` `tests/calendar-ics-audit.ts` `tests/staff-device-key-audit.ts` `tests/anonymous-guard-audit.ts`

NFR-NFR-013 Escaping and content policy for public pages

Every value inserted into server-built public pages (a label, an OCR'd name, a visitor's message) shall be escaped, and every such response shall carry a CSP with a fresh nonce per response: default-src 'none', script-src by nonce alone, connect-src, font-src, manifest-src and worker-src on the same origin, base-uri 'none' and frame-ancestors 'none' outside preview, so an injected script tag does not run.

Implemented · Must · Non-functional · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. Every script in the public pages is syntax-checked by the TypeScript parser; an injected script without the nonce does not execute.
2. style-src deliberately stays 'unsafe-inline' because inline styles cannot execute and a nonce cannot cover them.

EVIDENCE

`src/server/core/publicPages.ts` `src/server/publicKit.ts` `tests/public-pages-syntax-audit.ts` `tests/student-door-audit.ts`

NFR-NFR-014 Input validation and bounds

Reference-data inputs shall be accepted as trimmed non-empty text of at most 120 characters for a name and 40 for a code, and as whole numbers within bounds after Arabic-Indic and Persian digits are converted to Latin, with objects, arrays, booleans and NaN refused by one Arabic message naming the field; regular expressions shall not backtrack quadratically on hostile input.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. '٤٥' is accepted as 45; a 121-character name is refused.
2. Matching the transfer and whitespace-limit expressions on hostile input finishes within a generous time bound the test measures by the clock.

EVIDENCE

`src/server/core/input.ts` `src/utils/digits.ts` `tests/input-hardening-audit.ts`

NFR-NFR-015 Abuse resistance through rate limits

The server shall rate-limit requests in every instance: sign-in 5 per minute, public endpoints 60, and the teacher card 240 per link, with a clear message and a wait period when exceeded, without students on a shared network blocking one another.

Implemented · Should · Non-functional · The system (automatic), Root administrator

RATIONALE

> Per-instance limits stop guessing and flooding without any paid service. Unifying the counter at the network edge is a choice for the hosting party, listed under assumptions and dependencies.

ACCEPTANCE CRITERIA

1. A sixth sign-in attempt within a minute is refused with a message and a wait period.
2. Many students behind one network address open their cards without blocking one another.

EVIDENCE

`src/server/core/rateLimits.ts` `src/server/routes/authRoutes.ts` `SECURITY-TODO.md` `docs/LOAD_TEST.md` `tests/student-door-shared-network-audit.ts`

NFR-NFR-016 An audit trail without request content

Inferred

The system shall record every successful write through /api (user, method, path, action, entity, status and what changed where possible) and every sign-in, sign-out, refused sign-in attempt and 403 or 409 refusal, without storing the request body, excluding the heartbeat, the 'read' marker, conflict checks and telemetry so the log is not buried.

Implemented · Must · Non-functional · System administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. A password sent in a write request appears in no log line.
2. A wrong sign-in is recorded under the targeted account with the action 'refused sign-in attempt'.

EVIDENCE

`server.ts` `src/server/routes/adminRoutes.ts` `src/server/routes/authRoutes.ts` `tests/role-guard-audit.ts`

NFR-NFR-017 External-service boundary

The Gemini service shall only read text and propose; execution shall always go through the deterministic validators and with explicit confirmation from the user; the functions it may propose shall be limited to a fixed list with sanitised output, its key shall stay on the server from the platform secret and never reach the browser, only the data the request needs shall be sent to it, and any failure shall be answered with one uniform message that reveals neither the model nor the key.

Implemented · Must · Non-functional · External service, The system (automatic)

ACCEPTANCE CRITERIA

1. A call to a function not in GEMINI_SCHEDULE_FUNCTION_NAMES is dropped from the output.
2. A refusal or timeout is logged on the server with its status code and its text is not shown to the user.

EVIDENCE

`src/server/core/gemini.ts` `src/utils/geminiScheduleLayer.ts` `tests/gemini-schedule-layer.ts`

NFR-NFR-018 A public privacy policy

Inferred

The public page /privacy-policy.html shall publish a policy in Arabic with an English summary stating the data processed and its purpose, the parties processing it on the operator's behalf (Firestore, Cloud Run and the Gemini provider), protection by HTTPS encryption and role-based access, that data is neither sold nor used for advertising, and how to request access, correction or deletion by email.

Implemented · Must · Non-functional · Public visitor, Standard user

ACCEPTANCE CRITERIA

1. The page carries a last-updated date (5 November 2026), opens without an account and follows the device's light or dark setting.
2. The page needs no session and caches no personal data.

EVIDENCE

`public/privacy-policy.html` `public/robots.txt`

NFR-NFR-019 Personal data minimisation

A student's civil number shall be sealed with AES-256-GCM under a key derived from the shared secret and never kept as text in memory, log or URL; student profiles shall be deleted automatically when their lifetime ends (expected graduation plus a year, or inactivity), checked every 6 hours; the service worker shall not store on the device the personal case responses (/api/schedules/demand and /api/student-registration), /api/auth or /api/users, and the API cache shall be cleared when identity changes.

Implemented · Must · Non-functional · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. A calendar subscription carries a derived key rather than the civil number, and the number does not appear in the server log.
2. Sign-out sends CLEAR_API_CACHE and also deletes the API cache directly.

EVIDENCE

`src/server/core/studentIdentity.ts` `src/server/studentDoorRoutes.ts` `public/sw.js` `src/App.tsx` `tests/student-door-audit.ts`

NFR-NFR-020 44-pixel touch targets

Inferred

Every interactive element on phone and tablet shall have a touch target of at least 44 pixels (coarse pointer and width up to 760 pixels), with a small icon button enlarged by extending its hit area rather than changing its drawing.

Implemented · Should · Non-functional · Standard user, System administrator, Student

ACCEPTANCE CRITERIA

1. The src/styles/52-touch-targets.css layer gives buttons, tabs, selects, fields, button links and chips 44 pixels on phones and tablets, and small icon buttons a transparent touch layer reaching 44 without changing their drawing.
2. A sweep of 66 screens across all roles and public pages at 390 and 768 with touch: zero targets under 44×44 (was 322 and 881).

EVIDENCE

`src/styles/52-touch-targets.css` `src/index.css` `src/server/studentMapUi.ts` `src/server/staffCardVisual.ts` `public/landing/index.html` `tests/accessibility-audit.ts`

NFR-NFR-021 Colour contrast to WCAG AA

Normal text contrast shall be at least 4.5:1 under WCAG AA on every calibrated screen, with platform colours managed through variables rather than scattered values, except the course-number colours in plan review, which are the owner's rule: a number's colour is the colour of its arrows.

Implemented · Should · Non-functional · Standard user, Student, Public visitor

ACCEPTANCE CRITERIA

1. Every light-theme text colour variable reaches 4.5:1 on all five backgrounds and on its own soft tint, and no opacity dimming pushes text below it.
2. A sweep of 66 screens at 390 and 768: zero contrast failures (was 132 and 130).

EVIDENCE

`src/styles/01-foundation.css` `src/styles/06-intelligence.css` `src/styles/11-approval.css` `src/server/studentHomePage.ts` `tests/accessibility-audit.ts`

NFR-NFR-022 Controls with accessible names

Inferred

Every input, drop-down and icon button shall have a name a screen reader reads (label or aria-label), every copy or close button in public pages shall carry an explicit description, and a closed drawer shall not stay in the tab order.

Implemented · Should · Non-functional · Standard user, Student, Public visitor

ACCEPTANCE CRITERIA

1. Every field, select and icon button in the interface and in 29 server-built pages has a name a screen reader reads, with no exception list; test:accessibility fails if an unnamed element is added.
2. The closed drawer is visibility:hidden after 280 ms so it leaves the Tab order, and copy and close buttons on public pages carry an explicit description.
3. A sweep of 66 screens at 390 and 768: zero unnamed elements.

EVIDENCE

`src/styles/07-responsive.css` `src/components/AdminUsers.tsx` `src/components/ScheduleViewsMenu.tsx` `src/components/PlanReview.tsx` `src/server/pages/instructorRequestPage.ts` `src/server/pages/staffCardPage.ts` `tests/accessibility-audit.ts`

NFR-NFR-023 Reduced-motion respect

The whole motion layer shall run under prefers-reduced-motion: no-preference so no animation runs when the user asks for less, and motion shall be limited to transform and opacity at 120, 180 and 260 ms with no infinite loops in the student journey.

Implemented · Should · Non-functional · Standard user, Student

ACCEPTANCE CRITERIA

1. The number of animations running under reduced motion is zero on every screen at the four widths (320, 390, 768, 1280).
2. The student journey has no animation-iteration-count: infinite.

EVIDENCE

`src/styles/30-motion.css` `docs/QUALITY-SCORECARD.md` `docs/VISUAL-POLISH-AUDIT.md` `tests/journey-reveal-audit.ts`

NFR-NFR-024 Zoom and focus indication

Inferred

The viewport tag shall not disable pinch zoom (no maximum-scale and no user-scalable=no), and every focusable element shall draw a visible focus ring (2 pixels) under keyboard navigation, with form fields keeping the ring through border colour and shadow.

Implemented · Should · Non-functional · Standard user, Student, Public visitor

ACCEPTANCE CRITERIA

1. A real keyboard sweep (Tab ×40) on home, schedule, plan review and queries at 390 and 1280: every control draws the ring except text inputs (border colour) and the document body on wrap-around.

EVIDENCE

`index.html` `src/styles/02-primitives.css` `public/updating.html` `docs/QUALITY-SCORECARD.md`

NFR-NFR-025 Arabic-first, right-to-left

The application shell, public pages, updating page and manifest shall declare lang=ar and dir=rtl, styles shall use logical properties (inline-start and inline-end) so the interface does not break with physical offsets, and Latin numbers and codes inside Arabic text shall be isolated (bdi).

Implemented · Must · Non-functional · Standard user, Student, Faculty member, Public visitor

ACCEPTANCE CRITERIA

1. html in index.html and updating.html carries lang="ar" dir="rtl", and the manifest has dir=rtl and lang=ar.
2. The demo civil number is shown inside bdi dir=ltr so its digit order never flips.

EVIDENCE

`index.html` `public/updating.html` `public/manifest.webmanifest` `src/server/core/publicPages.ts` `tests/updating-page-audit.ts`

NFR-NFR-026 Self-hosted typefaces

Typefaces (IBM Plex Sans Arabic in weights 300, 400, 500, 600 and 700, and IBM Plex Mono) shall be served from the same server as woff2 with font-display: swap and unicode-range splits for Arabic and Latin, no page shall request a font from an external host, and the updating page shall keep working with a local font when none is stored.

Implemented · Should · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. public/fonts holds the listed weights, and the public-page CSP is font-src 'self'.
2. The updating page makes no external request.

EVIDENCE

`src/styles/01-foundation.css` `index.html` `public/updating.html` `public/sw.js` `tests/updating-page-audit.ts`

NFR-NFR-027 Numerals and Arabic counts

The system shall accept Arabic-Indic, Persian and Latin digits in any field and convert them at the edge to Latin for storage and comparison, always display the course number as seven digits (college 2, department 2, course 3, e.g. 0109105), form counts in correct Arabic grammar (singular, dual and plural), and never display the stored three digits or the internal course id.

Implemented · Must · Non-functional · Standard user, Student, Faculty member

ACCEPTANCE CRITERIA

1. A civil number typed in Arabic-Indic digits matches the same Latin typing at entry.
2. Every course number a reader sees in a printed report, query or Excel export has seven digits.

EVIDENCE

`src/utils/digits.ts` `src/utils/arabicCount.ts` `src/utils/authorityAcademicCodes.ts` `tests/arabic-count-audit.ts` `tests/course-number-seven-digit-audit.ts`

NFR-NFR-028 Standard Arabic and text integrity

Interface text shall be written in standard Arabic with no colloquial words from a list guarded by an automated check, Arabic presentation forms copied from PDFs shall be normalised to ordinary letters with NFKC on write, and name matching shall ignore differences in hamza, ta marbuta and ya.

Implemented · Should · Non-functional · Standard user, The system (automatic)

ACCEPTANCE CRITERIA

1. The check fails on a whole colloquial word in interface text and names the file and line.
2. The two spellings of 'Department of Social Studies' differing in hamza give one match key.

EVIDENCE

`src/utils/arabicText.ts` `tests/standard-arabic-ui-audit.ts` `tests/arabic-presentation-audit.ts`

NFR-NFR-029 Layouts that work from 320 pixels

No content shall overflow the screen width (no horizontal scroll) from 320 pixels up: iOS date and time fields by dropping their native appearance, grids with a minmax(0,1fr) column, a minimum for auto-fit grids written minmax(min(Npx,100%),1fr) when it is 160 pixels or more, and layouts recomposed rather than merely shrunk.

Implemented · Must · Non-functional · Standard user, Student, Faculty member

ACCEPTANCE CRITERIA

1. Third-round sweep: no horizontal overflow (H2 = 0) at 320, 390, 768 and 1280.
2. The live sweep covers every screen and default role at two widths (390 and 360) and exits with 1 on a violation.

EVIDENCE

`src/styles/46-mobile-overflow.css` `src/styles/07-responsive.css` `scripts/mobile-overflow-sweep.ts` `docs/QUALITY-SCORECARD.md` `tests/mobile-overflow-css-audit.ts` `tests/masari-responsive-audit.ts`

NFR-NFR-030 A4 landscape printing with a fixed design

Reports shall print on landscape A4 with zero margins governed by @page rules, every page shall be measured with the print rules themselves before the command and scaled between 0.6 and 1.8 to fill the sheet without clipping, and no work shall change the design of printed reports (faults are fixed in values and data).

Implemented · Must · Non-functional · Standard user, The system (automatic)

ACCEPTANCE CRITERIA

1. A teacher report with two lectures fills the sheet instead of a third of it, and seven lectures with long names are not clipped in Safari.
2. Every print path measures before the command, not after; @media rules are evaluated against the paper, not the window.
3. Any unavoidable layout change reuses the existing print classes and is checked on a Chrome-printed landscape A4 PDF before merging.

EVIDENCE

`src/utils/printFit.ts` `src/styles/08-print.css` `docs/EXHAUSTIVE_UI_AUDIT_2026-08-12.md` `tests/print-fit-audit.ts` `tests/print-attestation-audit.ts`

NFR-NFR-031 A deterministic decision engine

Conflict and approval-blocker verdicts shall be computed by deterministic code that returns the same result for the same input from every entry point (the count, the detailed list, the alert and the live board scan), and the fast indexed path shall stay identical to the original all-pairs sweep: no missing or extra conflict and no different reason, severity or order.

● Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. The test compares both paths over timetables with shared halls and lecturers, combined delivery, duplicates and whole random terms and fails on the first difference.
2. Instructor identity is compared as a number, not text, so a conflict whose id arrives as text is not missed.

EVIDENCE

`src/server/core/scheduleConflicts.ts` `src/utils/scheduleIntelligence.ts` `tests/conflict-parity.ts` `tests/blocker-oracle-audit.ts` `tests/id-type-conflict-audit.ts`

NFR-NFR-032 Stale-write protection

Every schedule row shall carry a revision number (rev) that increases with each write, a write carrying a revision that is no longer current shall be refused with ScheduleRevisionConflict so no half-write occurs, and an accepted proposal shall be applied atomically, refused with 409 'the slot changed' and nothing written if any slot has changed since it was built.

● Implemented · Must · Non-functional · The system (automatic), Standard user

ACCEPTANCE CRITERIA

1. An edit with an old revision answers 409 and leaves the row unchanged.
2. Applying a proposal containing changed slots answers proposal-stale and writes none of them.

EVIDENCE

`src/db/repository.ts` `src/server/core/scheduleWrites.ts` `src/server/studyProposalRoutes.ts` `tests/instructor-apply-atomic-audit.ts` `tests/schedule-apply-proposal-audit.ts`

NFR-NFR-033 Reversibility of change

A snapshot of the schedule scope (college, department, term) shall be captured before each change it protects (editing or moving a slot, publishing, importing, copying), its state after the change shall be recorded from the rows the request itself wrote, so a restore refuses to overwrite a colleague's edit made in between, and the snapshot shall be read from storage, not from the read cache.

● Implemented · Must · Non-functional · Standard user, System administrator

RATIONALE

> Incident: every snapshot lagged a step because the cache was read before being cleared and held the rows about to be deleted.

ACCEPTANCE CRITERIA

1. Publishing a file, undoing it, then undoing a deletion returns each time the state before that operation, not one a step older.
2. A list of scope versions is available and restore requires that nothing else changed.

EVIDENCE

`src/server/core/versions.ts` `src/server/core/scheduleWrites.ts` `src/server/timeMachineRoutes.ts` `tests/safety-net-restore-audit.ts` `tests/time-machine-audit.ts` `tests/schedule-versions-list-audit.ts`

NFR-NFR-034 Process survival on a single request failure

Inferred

A single error in an asynchronous handler or a rejected promise shall not bring the process down: all routing verbs are wrapped so a rejection reaches the error handler, any unhandled rejection or process-level exception is logged and the server continues, so in-flight university requests and the live schedule stream are not cut.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. A route handler whose promise rejects answers 500 with a reference code and does not end the process.
2. A four-argument handler (an error handler) is not wrapped and stays as written.

EVIDENCE

`src/server/core/http.ts` `server.ts` `tests/input-hardening-audit.ts`

NFR-NFR-035 Type checking and parity guards

npm run lint (tsc with no emit) and every test in tests/suite.txt shall pass before each merge, and parity tests shall guard what type checking cannot see: the route table, server middleware order and bytes of server-built pages against stored snapshots, the guide and brand audit (render-brand --check) on every build, and the syntax of public-page scripts.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Adding a route, reordering a middleware or changing a byte in a built page fails test:server-parity.
2. Type checking covers the whole repository (only node_modules and dist excluded) with eight strict options: alwaysStrict, noImplicitThis, useUnknownInCatchVariables, noFallthroughCasesInSwitch, strictBuiltinIteratorReturn, noImplicitOverride, strictFunctionTypes and strictBindCallApply; removing an option or adding an exclusion fails test:tsconfig-strictness.

EVIDENCE

`tsconfig.json` `package.json` `tests/suite.txt` `docs/SERVER_MAP.md` `tests/server-route-parity-audit.ts` `tests/public-pages-syntax-audit.ts` `tests/tsconfig-strictness-audit.ts`

NFR-NFR-036 Transactional identifier allocation

New identifiers in Firestore shall be reserved by a transaction on a shared counter (_meta/counters) that reserves a block of 32 identifiers per instance, so no two identifiers collide across server instances or during an import, and identifiers shall be compared numerically even when they arrive as text.

Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Reserving a new block reads and writes the counter in the same transaction.
2. A text instructor id '7' and a numeric 7 are treated as one identity in conflict detection.

EVIDENCE

`src/db/repository.ts` `tests/id-type-conflict-audit.ts`

NFR-NFR-037 Supported browsers

Inferred

The system shall work on modern desktop and phone browsers (Chromium and Safari on iOS) at the ES2022 level with service workers, handle Safari-specific behaviour (date and time fields, ignoring @page size) with dedicated rules, keep the overflow sweep runnable on WebKit, and present a single light interface (the theme is pinned to light).

Implemented · Should · Non-functional · Standard user, Student, Faculty member, Public visitor

ACCEPTANCE CRITERIA

1. The sweep runs on the webkit engine with ENGINE=webkit and an iPhone user agent.
2. No browserslist is declared; support is inferred from the TypeScript ES2022 target and Vite defaults.

EVIDENCE

`tsconfig.json` `vite.config.ts` `scripts/mobile-overflow-sweep.ts` `src/styles/46-mobile-overflow.css` `index.html` `tests/mobile-overflow-css-audit.ts`

NFR-NFR-038 **Observability** Inferred

Every server error a user sees shall be tied to a reference code that appears in the log with method, URL and error text, boot shall print the database target actually used, the server shall keep the last 5000 performance samples (path, method, duration, status, user and scope) excluding the heartbeat, and /api/telemetry/client shall receive interface errors from authenticated sessions.

● Implemented · Should · Non-functional · System administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. A 500 carries a ref of the form E plus 8 characters and the same value appears in the log line.
2. Sample number 5001 evicts the oldest so the store never exceeds 5000.

EVIDENCE

`server.ts` `src/server/core/http.ts` `src/server/routes/guideRoutes.ts` `src/utils/clientTelemetry.ts` `src/server/runtimeEnv.ts` `tests/input-hardening-audit.ts`

NFR-NFR-039 **A reversible, error-resistant release**

A normal release shall change only the official service's image so it can be reversed by routing traffic back to the previous revision with no data movement, only what the full CI passed shall be released, the pipeline shall verify the build identifier on the official domain after release, and production shall stay on the previous revision if the database fails at boot.

● Implemented · Must · Non-functional · The system (automatic), Root administrator

ACCEPTANCE CRITERIA

1. The rollback procedure is documented as update-traffic to the previous revision and moves, copies or renames no data at any step.

EVIDENCE

`docs/DELIVERY_OPERATIONS.md` `tests/deploy-target-guard.mjs` `tests/deploy-target-guard.mjs`

3.18.6 Edge cases

⚠ A whole campus network on registration day

Scenario Thousands of students open their journey page from one IP address in the same hour.

Behaviour Limits sit on the student, not the address, and failures are counted, not successes; so 600 of 600 journeys pass from one address while someone cycling through many numbers stops at the address ceiling.

⚠ A user returns to an old tab after signing out

Scenario A tab from a previous account stays on a shared device with responses left in the service worker cache.

Behaviour The API store is cleared on the way out and the way in, and personal cases and authentication responses are never stored; so the next account never sees its predecessor's data.

⚠ A screen reader facing an unnamed drop-down

Scenario A drop-down, field or icon button is added to the interface without a name a screen reader can read.

Behaviour The accessibility check (test:accessibility) fails at build time naming the file and line, so the element never reaches users without a name.

⚠ Printing on Safari, which ignores page orientation

Scenario A user prints a landscape report from Safari, which does not honour @page size.

Behaviour The page is measured with the print rules themselves and scaled between 0.6 and 1.8 to fit the sheet, so the report's bottom is not clipped even if the paper stays upright.

⚠ Two tabs writing the same slot

Scenario Two department members edit one slot and the second saves with a revision that has become stale.

Behaviour The second save is refused with a revision conflict and he is asked to refresh; nothing is written and his colleague's edit is not erased.

3.18.7 Standing owner decisions

- The course number is always seven digits: college 2, department 2, course 3; the stored code and the internal id are never displayed.
- Printed reports keep their design; fixes go in values and data, not in layout.
- No substitute populated data is added when the database fails; production stays on its previous revision.

↑ Contents

3.19 · DAT

Data model and integrity

What is stored, how it relates, and what keeps it sound

29	100%	0	5
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Requirements · 5. Algorithms · 6. Edge cases · 7. Standing owner decisions

↑ Contents

3.19.1 Overview

This module describes the entities persisted in Firestore (accounts, permissions, terms, colleges, departments, courses, instructors, class sections, versions, drafts and links), their identifiers, relationships, uniqueness and deletion rules. It also covers compatibility with the data imported from the legacy SQL Server backup, the snapshots that freeze an appointment's academic identity, short-lived read caches, the rule for converting an instructor between visiting and appointed, retention and indexes. The governing principle is that an old record stays valid as it is and every new field is optional.

Root administrator · System administrator · Department head · Schedule committee chair · The system (automatic)

3.19.2 User journeys

Root administrator — **Importing and verifying the legacy data**

1. The legacy backup is extracted to a JSON snapshot by the extraction script and the table counts are matched against the SQL Server catalogue.
2. When the server starts on an empty database with auto-import enabled, it writes the active tables under their original identifiers.
3. The 28 historical tables are kept under legacyArchive unchanged.
4. Counters are initialised from the highest legacy identifier of each table.
5. The server compares each collection's count with what was written; any mismatch fails startup and no login is accepted.
6. A _meta/legacyImport document records the counts and the backup fingerprint.

Department head — Trying to delete a course that is in use

1. The department head opens the courses screen and chooses a course to delete.
2. The server checks its membership in an academic plan sheet; if found, the delete is refused with 409 suggesting removal from the sheet first.
3. It then checks for appointments in any term that point to the course and refuses the delete in that case too.
4. If nothing refers to it, its document is copied to the deleted-records bin and then removed.

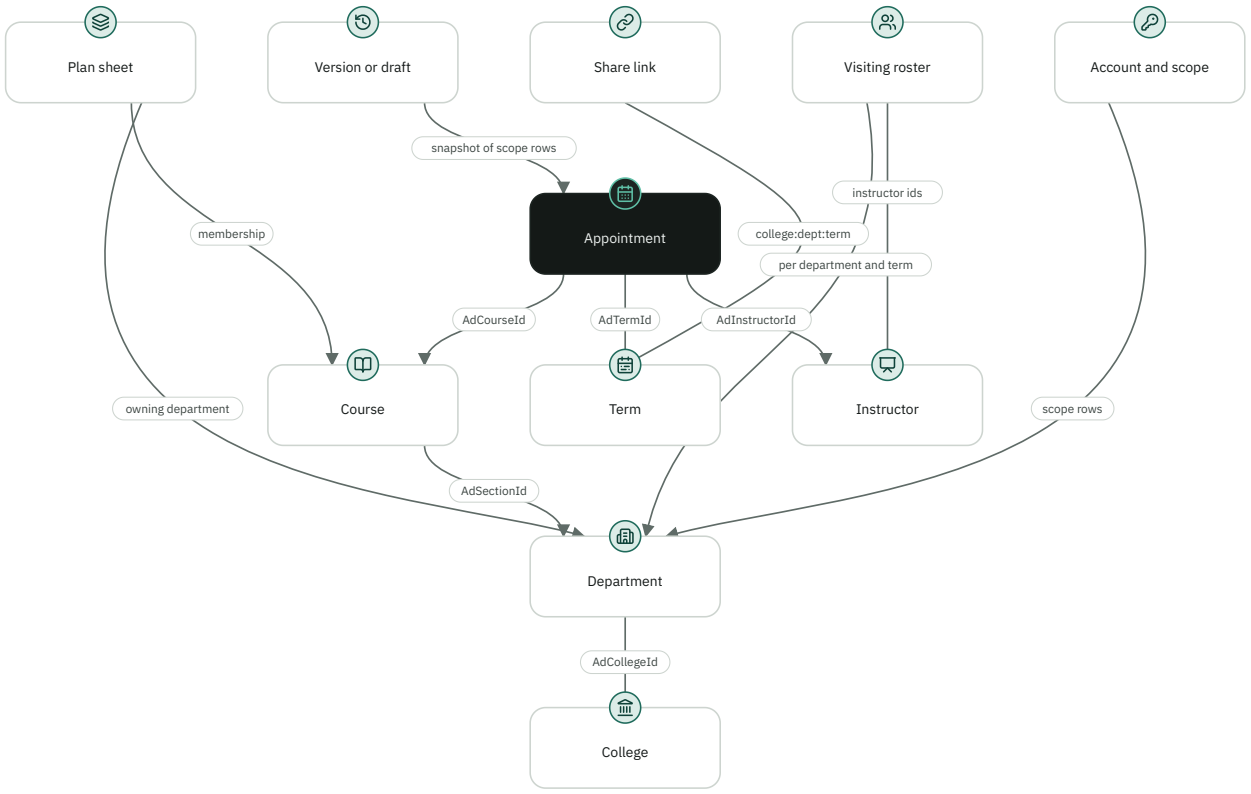
Department head — Converting a visiting instructor to appointed

1. The department head picks the instructor, asks for «appointed» and selects the department(s) the secondment is lifted from.
2. A pure plan of changes is computed: removal from the visiting directory and from the current and every future term roster.
3. A dry-run preview is shown without writing, and the whole plan is blocked if any department has an error.
4. On confirmation only those changes are written; no schedule and no past-term roster is touched.

3.19.3 Diagrams

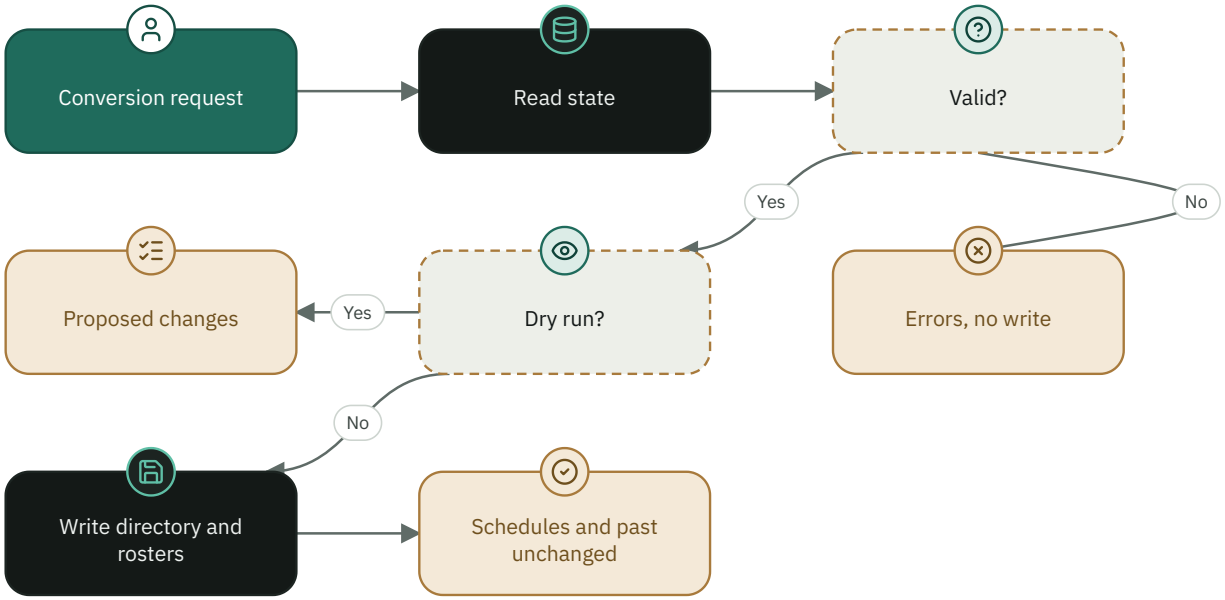
Entity relationships

Arrows mean «refers to» by numeric id; deletion is refused while an arrow still reaches the record.



Visiting ⇄ appointed conversion

The plan is a pure function that reads the state of the records and returns changes or errors; writing happens only afterwards.



3.19.4 Requirements

DR-DAT-001 **One store, closed to clients**

All durable data shall be kept in a single Firestore database reachable only by the server through the Admin SDK; the security rules shall deny every direct browser read and write, and the server shall never run on substitute data when the connection fails.

Implemented · Must · Data · The system (automatic), Root administrator

ACCEPTANCE CRITERIA

1. The rule match /{document=**} yields allow read, write: if false.
2. When the connection fails /api answers 503 and nothing is read from a packaged snapshot.

EVIDENCE

`firestore.rules` `src/db/repository.ts` `src/server/core/state.ts` `tests/demo-firestore-isolation-audit.ts`

DR-DAT-002 **Core entities and identifiers**

Each core entity shall have a numeric identifier and a document named prefix plus identifier: user_ for accounts, form_ for screens, permission_ and scope_ for permissions and scopes, and term_, college_, section_, instructor_, course_, schedule_ and room_.

Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. Course 418 is stored in course_418 and appointment 15430 in schedule_15430.
2. The department is stored under the field name AdSectionId and called «القسم» (Department) in the interface.

EVIDENCE

`src/types.ts` `src/db/repository.ts` `tests/parity_audit.py`

FR-DAT-003 **Block-based id allocation** Inferred

The system shall allocate new identifiers from a single counter in the _meta/counters document inside a transaction, in blocks of 32 identifiers per collection, and shall never reuse an identifier that was handed out even if it went unused.

Implemented · Must · Functional · The system (automatic)

ACCEPTANCE CRITERIA

1. Two concurrent course creations never receive the same identifier.
2. A creation that fails after allocation leaves a gap in the numbering and never a duplicate.

EVIDENCE

`src/db/repository.ts#reserveFirestoreIds`

DR-DAT-004 **Hierarchy: college, department, course**

A department shall refer to its college, a course to its department, and an appointment to its course, term and instructor by numeric identifiers, and the college of a course and the department and college of an appointment shall be derived from the course-to-department relation on every read rather than from a stale stored value.

Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. Moving a course to another department changes the department and college of its appointments on the next read with no write.
2. Every appointment row in the migrated snapshot points to an existing college, department, term, course and instructor.

EVIDENCE

`src/db/repository.ts#hydrateSchedules` `src/types.ts` `tests/parity_audit.py`

FR-DAT-005 Delete refused while referenced

The system shall refuse with 409 the deletion of a college that has departments, a department that has courses, a course that has appointments or plan-sheet membership, a term that has appointments, and an instructor that has appointments.

Implemented · Must · Functional · System administrator, Department head

ACCEPTANCE CRITERIA

1. Deleting a college with one department returns 409 and the document remains.
2. Deleting a course attached to a sheet returns 409 with a message suggesting its removal from the sheet first.

EVIDENCE

`src/server/routes/referenceRoutes.ts` `src/server/routes/coursesRoutes.ts` `src/server/routes/instructorsRoutes.ts` `tests/parity_audit.py`

DR-DAT-006 Course code unique within a department

Inferred

A course code shall be unique within its department; the code shall be claimed, in the same transaction that writes the course, through a document in courseCodeClaims, and a stale claim shall be taken over when the course that registered it no longer holds the code.

Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. Two identical create submissions: only one succeeds and the other is refused with «تم تسجيل رمز المقرر الدراسي هذا من قبل».
2. After a course is deleted a new course with the same code can be created.

EVIDENCE

`src/db/repository.ts#createCourse`

DR-DAT-007 Uniqueness of department code and term name

A department created from a file header shall exist once per code within a college (sectionCodeClaims), and an automatic term shall not be created twice for the same name or the same season and years (autoTermClaims keyed by a 32-character digest of the name).

Implemented · Should · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. Two concurrent requests for one department code produce one department and the second returns the existing one.
2. A manually entered term named with «2026/2027» prevents creating the automatic term matching it in season and years.

EVIDENCE

`src/db/repository.ts#createSectionOnce` `src/db/repository.ts#createTermIfAbsent` `tests/term-calendar-audit.ts`

DR-DAT-008 Instructor civil number

An instructor's civil number shall be stored in Latin digits only, 12 digits long, and shall pass the checksum inherited from the legacy system; a civil number shall not be registered twice, and an existing instructor is enrolled in the department instead of duplicating the record.

Implemented · Must · Data · System administrator, Department head

ACCEPTANCE CRITERIA

1. An 11-digit number is refused with 400 and the message «الرقم المدني 12 رقمًا بالضبط.».
2. Arabic and Persian digits and spaces are normalised to Latin digits before comparison and storage.

EVIDENCE

`src/utils/civilId.ts` `src/server/routes/instructorsRoutes.ts` `tests/legacy-data-audit.py`

DR-DAT-009 Core appointment fields

An appointment row shall carry weekdays as booleans (fsunday to fthursday, Sunday to Thursday), start and end times as text, the class-section code SCode, the building code AdRoomCode and hall AdRoomHall, plus optional references into the location registry (buildingId, roomId, companionRoomId and the locationStatus state).

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. locationStatus is one of four values: VERIFIED, PENDING_ROOM, LOCATION_REVIEW_REQUIRED and INVALID_HISTORICAL.
2. A legacy row without location references remains valid and is shown with its raw text.

EVIDENCE

`src/types.ts#FSchedule` `tests/parity_audit.py`

FR-DAT-010 Revision number and no overwriting of newer data

Each appointment shall carry a rev field that starts at 1 on creation and increases by one on every write, and a save shall be refused with a revision-conflict error when the stored number is newer than the one the request was based on; absence counts as zero.

● Implemented · Must · Functional · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Two saves based on revision 3: the first succeeds and the revision becomes 4, the second is refused as a conflict.
2. A legacy row without rev accepts a save expecting 0 and takes its first number.

EVIDENCE

`src/db/repository.ts` `src/types.ts#FSchedule` `tests/proposal-undo-emulator-audit.ts`

DR-DAT-011 Academic identity snapshot of an appointment

An appointment shall keep the course code, name and plan-sheet id as they stood when it was created (CourseCodeSnapshot, CourseNameSnapshot, CurriculumPlanIdSnapshot), a later rename of the course shall not rewrite an old term when read, and the displayed course number shall be built as seven digits (college 2 + department 2 + course 3), never from the stored three digits or from AdCourseId.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. A renamed course shows its then-name in a 2018 term and its new name in the current term.
2. A stored 105 with college 01 and department 09 is displayed as 0109105.

EVIDENCE

`src/db/repository.ts#hydrateSchedules` `src/utils/authorityAcademicCodes.ts` `tests/course-number-seven-digit-audit.ts` `tests/safety-net-restore-audit.ts`

DR-DAT-012 Scope key

The working scope of a timetable shall be identified by the key «college:department:term», under which versions, drafts, notes, open decisions, publications, share links and registration statistics are stored; the colons are replaced by underscores when the key becomes part of a document name.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. A version saved for college 6, department 7 and term 107 carries scopeKey = 6:7:107.
2. The visiting roster for the same scope is stored in a document named 6_7_107.

EVIDENCE

`src/db/repository.ts` `tests/schedule-versions-list-audit.ts`

DR-DAT-013 Timetable versions

Each version shall store a complete snapshot of the scope rows with its label, source (manual, draft, publish, undo, copy or import) and row count; the version list shall read metadata fields only, with no database-side ordering, a default limit of 60 and a maximum of 100; and full snapshots shall be fetched only for the newest (default 12, maximum 24).

Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. The version list uses no orderBy and does not read the rows field.
2. Requesting 500 versions returns at most 100.

EVIDENCE

`src/db/repository.ts#getScheduleVersions` `src/types.ts#ScheduleVersion` `tests/schedule-versions-list-audit.ts`

DR-DAT-014 Drafts

Inferred

A draft shall carry all its rows, its status (draft, published or archived), its source (what-if, auto, import or manual) and the fingerprint of the scope it was built on; a draft imported from PDF shall keep the scanned original rows unmodified (baselineRows) and the server-signed import receipt.

Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. Reading a scope's drafts returns at most 100.
2. Editing the draft rows does not change baselineRows.

EVIDENCE

`src/types.ts#ScheduleDraft` `src/db/repository.ts`

DR-DAT-015 Share links in storage

A share link shall be stored under a random 24-byte base64url identifier (32 characters) that is the token itself, carrying its scope, kind (department, staff, survey, request or students), expiry date, revoked flag and view count; a demo-sandbox token is prefixed «demo.».

Implemented · Must · Data · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A department timetable link is created for 30 days by default and at most 180; a students link for 365 days.
2. An instructor's personal card expires at the end of the term, or after 150 days when its dates are unknown.

EVIDENCE

`src/db/repository.ts#createShareLink` `src/server/routes/shareRoutes.ts` `src/utils/shareLinkLifetime.ts` `tests/calendar-ics-audit.ts`

DR-DAT-016 Accounts and passwords

An account shall store its password derived with scrypt and a random salt; the compatibility value for the legacy administration screen shall be stored AES-256-GCM encrypted in a separate field never sent to ordinary clients; plaintext passwords shall not be stored anywhere; the academic role is optional and its absence reads as «committee chair».

Implemented · Must · Data · Root administrator, System administrator

ACCEPTANCE CRITERIA

1. All 29 accounts in the original snapshot carry a scrypt-shaped hash.
2. An account created before the Role field existed works with committee-chair permissions without migration.

EVIDENCE

`src/types.ts#SystemUser` `src/db/repository.ts` `tests/legacy-data-audit.py`

DR-DAT-017 Permissions and scopes

A screen permission shall be stored as a row (user, screen) in formSecurity and a data scope as a row (user, college, department) in userScopes, each with a legacyId, and deleting an account shall delete its rows from both collections in one batch; department zero means «the whole college» for roles whose scope is a college only.

● Implemented · Must · Data · System administrator, Root administrator

ACCEPTANCE CRITERIA

1. Deleting an account with 4 permissions and 3 scopes leaves zero orphan rows for it.
2. Legacy orphan permission rows (28 in the original snapshot) stay unchanged and have no effect.

EVIDENCE

`src/db/repository.ts#deleteUser` `src/server/core/scope.ts` `tests/parity_audit.py` `tests/role-scope-audit.ts`

DR-DAT-018 Sessions

A session shall be stored in a document named by the session id (sess_ or demo_ followed by 64 hex characters) holding a numeric expiresAt and a timestamp twin expiresAtTtl for the automatic-deletion policy; it shall expire after 15 minutes idle (one hour for demo), an expired session shall be deleted on first read, and sessions shall never be part of any backup.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. An expired session is read, deleted and reported as not found.
2. A password change deletes the account's other sessions and keeps the current one.

EVIDENCE

`src/db/repository.ts#createSession` `src/server/core/auth.ts` `tests/anonymous-guard-audit.ts`

DR-DAT-019 Visiting-instructor model

Secondment shall not be an attribute of the instructor record; it is a name in the department's permanent visiting directory (departmentDelegates, per department and its family across colleges) and a name in the department's roster for a term (visitingRosters), so the instructor is not duplicated in the instructor register.

● Implemented · Must · Data · Department head

ACCEPTANCE CRITERIA

1. One instructor appears in the directories of two departments without two records.
2. A new term starts from a copy of the previous term's roster.

EVIDENCE

`src/types.ts#VisitingRoster` `src/db/repository.ts` `tests/visiting-conversion-audit.ts`

FR-DAT-020 Visiting ⇄ appointed conversion rule

When converting an instructor to «appointed» the system shall remove him from the visiting directory of the chosen departments and from the current and every future term roster of those departments; when converting to «visiting» it shall add him to the directory of exactly one department and to the roster of one current or future term; it shall never write a past-term roster, never touch any schedule, and shall offer a dry-run preview.

● Implemented · Must · Functional · Department head, Schedule committee chair

ACCEPTANCE CRITERIA

1. Converting to appointed with current term 5 and future term 6 removes him from the directory and from rosters 5 and 6 and leaves past roster 4 untouched.
2. Converting to visiting for a past or unknown term, or with no department, is refused and nothing is written.
3. Two departments of one family yield a single directory entry.

EVIDENCE

`src/utils/visitingConversion.ts` `src/db/visitingConversion.ts` `tests/visiting-conversion-audit.ts`

FR-DAT-021 Legacy import under original ids

The system shall import the legacy snapshot only when the users collection is empty and auto-import is not disabled, shall write each active table under its original identifiers (29 accounts, 18 screens, 219 permissions, 189 scopes, 31 terms, 13 colleges, 87 departments, 743 instructors, 1,404 courses, 15,430 schedule rows and 1 room), then match the counts before accepting any login and write _meta/legacyImport.

● Implemented · Must · Functional · Root administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. A count short of the expected in any collection ends startup with «فشل التحقق من نقل ...».
2. On Cloud Run with AUTO_IMPORT_LEGACY_DATA=false and an empty database the server refuses to start and writes nothing.

EVIDENCE

`src/db/repository.ts#seedFirestoreFromLocalSnapshotIfNeeded` `scripts/import-legacy-json.ts` `scripts/extract-legacy-bak.py` `tests/parity_audit.py` `tests/legacy-data-audit.py`

DR-DAT-022 Historical table archive

The 28 historical tables of the legacy backup shall be kept under legacyArchive/<table>/rows/row_000000001 onward without modification, with plaintext passwords as the only exclusion, and no screen shall read from it.

● Implemented · Should · Data · Root administrator

ACCEPTANCE CRITERIA

1. The document count of each archived table equals the recorded rowCount, otherwise verification fails.
2. Tables empty in the extraction report (such as Student and History) are recorded with zero rows.

EVIDENCE

`src/db/repository.ts#seedFirestoreFromLocalSnapshotIfNeeded` `docs/LEGACY_DATA_EXTRACTION_REPORT.json` `tests/parity_audit.py`

NFR-DAT-023 Short-lived read caches

The process shall keep reference collections (colleges, departments, terms, instructors, courses) for 90 seconds and schedule rows for 300 seconds, every write shall drop them at once, and the _meta/schedulesChanged document shall announce every write with a total counter and a per-writer counter so another instance counts the difference as foreign writes, and any counter that goes backwards counts as a foreign write.

● Implemented · Must · Non-functional · The system (automatic)

ACCEPTANCE CRITERIA

1. After a course edit the next read does not return the old value.
2. A snapshot with a total increase of 3 and 1 from this instance yields two foreign writes.

EVIDENCE

`src/db/referenceCache.ts` `src/db/beaconEcho.ts` `src/db/repository.ts` `tests/live-pulse-audit.ts`

DR-DAT-024 Demo world separated from the real one

The demo sandbox shall run on its own in-memory data per session without writing anything to Firestore, and the key of every cache that holds what the repository read shall start with «real|» for the real world or «demo:<session>|» for the demo.

● Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. Editing a sheet in the demo sandbox does not mark real radar runs stale.
2. What a demo request stores in a cache is never read by a real request.

EVIDENCE

`src/server/dataContextCache.ts` `src/db/demoSandbox.ts` `src/db/repository.ts` `tests/demo-firestore-isolation-audit.ts` `tests/demo-complete-audit.ts`

FR-DAT-025 Deleted-records bin

Inferred

Before deleting a college, department, term, instructor or course the system shall copy its complete document to deletedRecords under «<collection>__<document>» with the deletion time, a failed copy shall not block the requested deletion, and merged duplicate courses shall be archived the same way.

Implemented · Should · Functional · System administrator, Department head

ACCEPTANCE CRITERIA

1. Deleting course 418 produces a courses__course_418 document with the full payload.
2. No application path reads this collection; restoration is manual from the database.

EVIDENCE

`src/db/repository.ts#archiveBeforeDelete`

DR-DAT-026 Retention and automatic expiry

Inferred

Client diagnostics (clientTelemetry) shall expire after 30 days and be read for 14, sessions and these diagnostics shall declare an expiresAtTtl field, restore points shall keep only eight, and the local mode shall truncate lists to 4,000 diagnostics and 2,000 audit entries; the TTL policies are declared in the file but are not enabled in production, by the owner's decision.

Implemented · Should · Data · The system (automatic), Root administrator

ACCEPTANCE CRITERIA

1. The ninth restore point deletes the oldest.
2. firestore.indexes.json declares a TTL policy for sessions and clientTelemetry.

EVIDENCE

`src/db/repository.ts` `firestore.indexes.json`

DR-DAT-027 Indexes

firestore.indexes.json shall declare every index used in production (23 indexes: 5 on schedules, 4 on activityTrail, 2 each on scheduleComments, courses and instructorRequests, and one for the other large collections), every query on a large collection shall start with a field that leads an index, and the test shall fail on a new query without an index.

Implemented · Must · Data · The system (automatic)

ACCEPTANCE CRITERIA

1. A schedules query on AdTermId finds an index that starts with it.
2. The file is not deployed automatically: an index is created by hand in the console as the operations guide describes.

EVIDENCE

`firestore.indexes.json` `tests/firestore-index-audit.ts`

DR-DAT-028 Backup and restore

Inferred

The backup export (format schedule-system-backup/1, root administrator only) shall include everything durable and exclude sessions, clientTelemetry, systemSecrets, job collections and restore points, and a safety point shall be taken before any import or reset.

Implemented · Must · Data · Root administrator

ACCEPTANCE CRITERIA

1. Importing a backup does not drop the current login sessions.
2. The /api/system-backup/* routes refuse anyone who is not a root administrator.

EVIDENCE

`src/db/repository.ts` `src/server/routes/adminRoutes.ts`

DR-DAT-029 **Student data privacy** Inferred

A student's civil number shall not be stored in clear in a need record or profile; instead a keyed HMAC fingerprint serves as the uniqueness key, identity is held in encrypted fields, and the personal code is stored as a scrypt hash with a random salt, a failed-attempt counter and a temporary lock.

● Implemented · Must · Data · Student, The system (automatic)

ACCEPTANCE CRITERIA

1. One document per student in studentProfiles whose id is the fingerprint.
2. The service never stores the personal code itself.

EVIDENCE

`src/types.ts#StudentNeed` `src/types.ts#StudentProfile`

3.19.5 Algorithms

ALG-DAT-01 · Data model and integrity

Legacy migration with count parity

Deterministic

Has the legacy snapshot been moved completely, under its original identifiers, before the first login is accepted?

1. It starts only when the users collection is empty and auto-import is not disabled; on the cloud server with auto-import disabled and an empty database, the server refuses to start and writes nothing.
2. Each active table is written under its original identifier in batches of 400 records.
3. Every inactive table is archived row by row with its original name and row count.
4. Eleven collections are counted and compared with the snapshot; any difference ends startup with «فشل التحقق من نقل ...» before any login.

Inputs

The local snapshot extracted from the legacy backup, and the auto-import setting.

Output

A database whose counts match the snapshot plus a historical archive, or a halted startup naming the short collection.

No login is accepted on an incomplete migration, and the migration never runs over existing data.

Write batch: 400 records · 11 collections count-verified

`src/db/repository.ts#seedFirestoreFromLocalSnapshotIfNeeded` `scripts/import-legacy-json.ts` `tests/parity_audit.py` `tests/legacy-data-audit.py`

`FR-DAT-021`

3.19.6 Edge cases

⚠ Identifiers of two different types

Scenario An instructor identifier arrives as text from the database and as a number from the request.

Behaviour Identifiers are compared as numbers, so the instructor conflict is detected despite the type difference.

⚠ Reusing the code of a deleted course

Scenario A course was deleted and a new course with the same code is created in the same department while the old claim still exists.

Behaviour The claim is verified against the course that registered it; if it no longer holds the code the claim is taken over and the creation succeeds.

⚠ Converting to appointed while a past roster still holds his name

Scenario A visiting instructor is converted to appointed and a past-term roster still contains his name.

Behaviour The past roster stays unchanged, the «visiting» badge disappears from the present, and the roster history is kept.

⚠ The write-announcement counter goes backwards

Scenario The _meta/schedulesChanged document is recreated or the writers map is pruned.

Behaviour The regression counts as one foreign write and the instance drops its cache; one needless refresh is cheaper than a missed change.

⚠ Two simultaneous saves of the same appointment

Scenario Two coordinators open the same lecture and each saves a change.

Behaviour The first save succeeds and raises rev; the second is refused with a revision conflict and is shown the current row.

3.19.7 Standing owner decisions

- A displayed course number is always seven digits (college 2 + department 2 + course 3, e.g. 0109105); the stored CourseCode and AdCourseId are never shown.
- The design of printed reports is not touched: fixes go into values and data, not layout.
- No local data snapshot and no production credential in source control or delivery archives; they are configured in a private environment.

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3.20 · EXT

External interfaces

Everything the system connects to from outside, and under what contract

27	100%	0	1
reqs	done	routes	files

1. Overview · 2. User journeys · 3. Diagrams · 4. Requirements · 5. Edge cases · 6. Standing owner decisions

↑ Contents

3.20.1 Overview

This module fixes the contracts between the system and its surroundings: the Firestore database reached by the server alone, Google Cloud Run hosting and the domain, the Gemini service that only reads and proposes behind deterministic validators and explicit user confirmation, the Arabic and English OCR language packs, mobile notifications (Web Push) and outbound company e-mail, calendar subscription (ICS) and Excel export, the registration authority's timetable file (SWRSCHA) as an input, the browser contract (PWA and service worker), and the public link scheme with its token rules. Every number comes from the actual code.

External service · The system (automatic) · Root administrator · Schedule committee chair · Department head · Faculty member · Student · Public visitor · Partner organisation

3.20.2 User journeys

Faculty member — **Subscribing to my timetable in the phone calendar**

1. The instructor opens his card through his personal link (/r/<token>) and presses «add to calendar».
2. The server gives him a stable subscription link of the form /api/public/cal/<token>/p<number>-<key>.ics.
3. The calendar app reads the file and re-reads it every 6 hours; the server answers 304 when nothing changed.
4. A lecture cancelled on one date disappears from that date alone in every subscribed phone.
5. If the instructor has no live card left the subscription answers with an empty calendar explaining the pause, and returns on its own once a new card is issued.

Faculty member **Enabling mobile notifications for my card**

1. The instructor presses «enable notifications» in his card and the browser registers the push worker /push-sw.js scoped to the card.
2. The browser fetches the public VAPID key from /api/public/push/key and subscribes with the push service.
3. The server checks that the endpoint is https to a known push service and stores the record bound to the card token.
4. When his request is settled the server sends an aes128gcm-encrypted message with a 24-hour TTL and a topic that replaces earlier ones.
5. If the push service answers 404 or 410 the subscription is deleted.

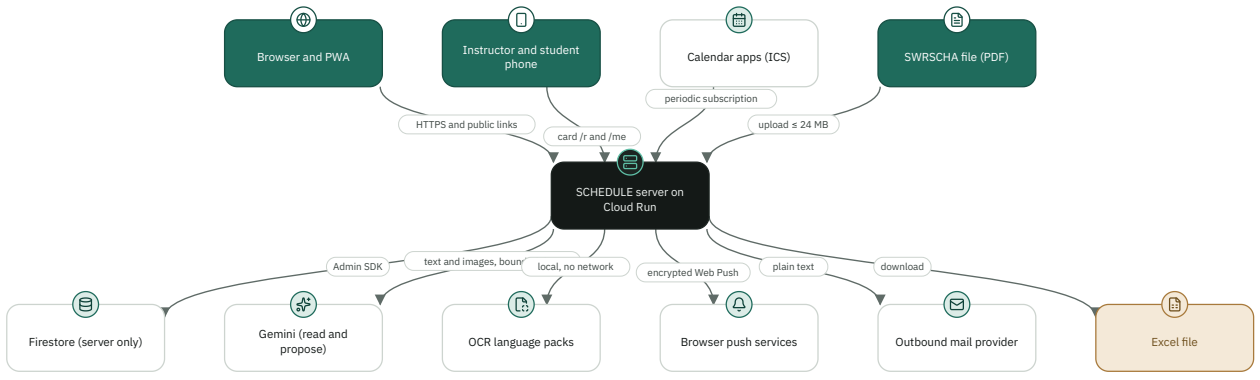
Root administrator **Releasing a version and verifying it**

1. The maintainer merges the change into the main branch and the full test run passes.
2. The «Deploy SCHEDULE» pipeline builds the image and replaces the image of the official schedule service only, leaving the domain and secrets untouched.
3. scripts/stamp-sw.mjs stamps the build identifier into sw.js so every open browser sees a real update.
4. The maintainer verifies that the domain and the platform's default service address return the same build identifier from /sw.js.

3.20.3 Diagrams

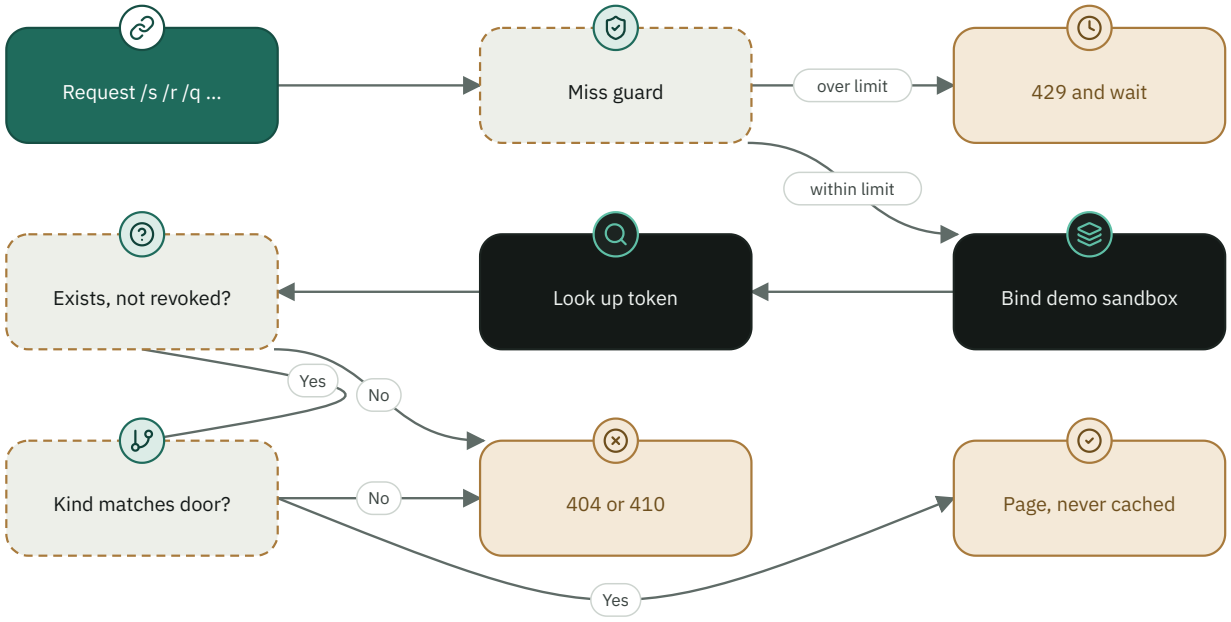
System context and external parties

The system sits in the middle; people reach it from the browser, and it reaches external services under defined contracts.



Opening a public link by token

Failure is counted, not success: someone guessing tokens is stopped after 40 misses, and a campus network opening a thousand valid links is never blocked.



3.20.4 Requirements

IR-EXT-001 Firestore through the server only

The system shall reach Firestore through the server alone, using the Admin SDK and the platform's application identity or a credential file kept outside the repository; it shall perform one real read at startup to verify the target and shall never run on substitute data if that fails: the server stays up and answers /api with 503 and a short reference number, while the full error text goes to the log, not the browser.

● Implemented · Must · Interface · The system (automatic), Root administrator

ACCEPTANCE CRITERIA

1. A wrong target or expired credential yields 503 on every /api route except /api/landing/inquiry, while the interface still loads.
2. No credential file exists in the repository or in the container image.

EVIDENCE

`src/db/repository.ts` `src/server/core/state.ts` `server.ts` `firestore.rules` `tests/firestore-target-audit.ts`

IR-EXT-002 Firestore project and database selection

The server shall select the Firestore project and database by one rule: on Cloud Run the application config file takes precedence over environment variables and a conflict is logged; elsewhere the variables win; with FIREBASE_CONFIG_SOURCE=environment the variables become the only source and startup is refused if the file contradicts them; and on Cloud Run it shall not start with a missing or «(default)» database.

● Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. On Cloud Run a FIREBASE_DATABASE_ID that differs from the file is ignored with an [firestore-target] warning.
2. An empty or «(default)» database throws «Refusing to start on Cloud Run».

EVIDENCE

`src/server/firestoreTarget.ts` `src/server/runtimeEnv.ts` `tests/firestore-target-audit.ts`

IR-EXT-003 Hosting on Cloud Run

The system shall be deployed as a single container (Node 22, port 3000, MALLOC_ARENA_MAX=2) on Cloud Run holding only the bundle, the interface build, the required packages, the config file and the two language packs, without sources or tests; on Cloud Run the runtime shall force production mode, the Firestore data mode and disabled auto-import, except for a dedicated demo service that is enabled explicitly and runs in memory.

Implemented · Must · Interface · Root administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. NODE_ENV becomes production, DATA_MODE=firestore and AUTO_IMPORT_LEGACY_DATA=false whenever K_SERVICE is present.
2. SCHEDULE_PRIVATE_DIR inside the release folder in production stops startup.

EVIDENCE

`Dockerfile` `src/server/runtimeEnv.ts` `src/db/snapshot.ts` `tests/firestore-target-audit.ts`

IR-EXT-004 Domain and official deployment

The «Deploy SCHEDULE» pipeline shall publish only the container image to the official schedule service in one region and shall not change the schedule-kw.com domain mapping, the service account, the secrets or the memory; after release the domain and the platform's default service address shall be verified to serve the same build identifier from /sw.js; cloudbuild.yaml is not the deploy path.

Implemented · Must · Interface · Root administrator

ACCEPTANCE CRITERIA

1. The deploy-target guard fails if the pipeline points to a project or region other than the official one or to a retired region.
2. The published sw.js carries a build identifier rather than the __BUILD__ placeholder.

EVIDENCE

`docs/DELIVERY_OPERATIONS.md` `scripts/stamp-sw.mjs` `tests/deploy-target-guard.mjs`

IR-EXT-005 Health endpoint

Inferred

GET /api/health shall answer from memory without reading Firestore with JSON holding ok, database («up» or «down») and a reference number on failure, so that it can keep the service warm during working hours.

Implemented · Should · Interface · The system (automatic), External service

ACCEPTANCE CRITERIA

1. A healthy database returns ok:true and database:up.
2. The warm-up scheduler asks every 5 minutes from Sunday to Thursday between 08:00 and 15:55 Kuwait time.

EVIDENCE

`src/server/routes/systemRoutes.ts` `docs/DELIVERY_OPERATIONS.md`

IR-EXT-006 Gemini call contract

The Gemini service shall be called with one HTTPS request per read, using a key from GEMINI_API_KEY or GOOGLE_GEMINI_API_KEY (nothing happens when absent) and a model from GEMINI_SCHEDULE_MODEL or GEMINI_MODEL, at temperature 0.1 and low thinking, with a 45-second budget for function calls and 180 seconds for a document read; it reads text and images and proposes alternatives or specific function calls, and never decides or writes.

Implemented · Must · Interface · External service, Schedule committee chair

ACCEPTANCE CRITERIA

1. Without a key the function returns null and the deterministic parser takes over.
2. Exceeding the budget returns failed with a «timeout» reason and does not hang the request.

EVIDENCE

`src/server/core/gemini.ts` `src/utils/geminiScheduleLayer.ts` `tests/gemini-schedule-layer.ts`

IR-EXT-007 Gemini output behind validators and confirmation

Nothing the Gemini service proposes shall be stored, published or applied unless it passes the server's deterministic validators and an explicit user confirmation: only five functions are accepted (check_conflicts, find_rooms, check_instructors, check_regulations and simulate_schedule), and any call or field carrying write, delete, publish or approve words is dropped.

● Implemented · Must · Interface · Schedule committee chair, External service

ACCEPTANCE CRITERIA

1. The nl-schedule route returns a preview only with commitRequired:true and writes nothing.
2. A call named delete_schedule is dropped and never reaches the executor.

EVIDENCE

`src/utils/geminiScheduleLayer.ts` `src/server/routes/intelligenceInsightRoutes.ts` `tests/gemini-schedule-layer.ts`

IR-EXT-008 When Gemini is unavailable

Inferred

When the Gemini service is unavailable (not configured, busy, timed out or refused), assisted timetable import shall return 503 with code GEMINI_SMART_IMPORT_UNAVAILABLE and a calm message stating that the timetable did not change, shall log rather than show the provider's English error text, and understanding of a text request shall fall back to the deterministic parser.

● Implemented · Must · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A 429 from the provider shows the user «the reading servers are busy», not the provider's text.
2. After any failure the schedule row count is unchanged.

EVIDENCE

`src/server/routes/smartImportRoutes.ts` `src/server/routes/intelligenceInsightRoutes.ts`

IR-EXT-009 OCR language packs

Scanned pages shall be read locally by Tesseract using the ara.traineddata and eng.traineddata packs located in the release root, searched in OCR_DATA_DIR, then the working directory, then the bundle folder, so they are never requested from the network, and a pool of workers shall serve one reading at a time.

● Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. Startup from the repository root works with no network connection.
2. The container image copies both files to the application root.

EVIDENCE

`src/utils/documentOcr.ts` `Dockerfile` `tests/pdf-import-hardening.ts`

IR-EXT-010 The authority's SWRSCHA file as an input

The system shall accept an authority timetable PDF (SWRSCHA) of up to 24 MB and 12 pages, reading the embedded text by its coordinates when present and otherwise the scanned page (landscape only), taking term, branch and department from each page header, and shall reject the whole file rather than part of it above 12 pages; scans never create a department or course.

● Implemented · Must · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. A 14-page file is rejected with a message stating the limit of 12 and no row is imported.
2. The department comes from the header even when the course number carries another department's code, and a three-digit tail alone never links.

EVIDENCE

`src/utils/documentOcr.ts` `src/server/routes/pdfImportRoutes.ts` `src/utils/authorityAcademicCodes.ts`
`tests/header-department-rule-audit.ts` `tests/authority-course-prefix-audit.ts`

IR-EXT-011 Web Push protocol

The server shall send mobile notifications per RFC 8030, RFC 8291 and RFC 8292: the payload is aes128gcm-encrypted to the browser's key (single record), the VAPID signature is ES256 valid for 12 hours with the subject https://schedule-kw.com, the TTL header is 86400 seconds, with an Urgency header and a 32-character base64url Topic that replaces a message not yet delivered; the VAPID keys come from VAPID_PUBLIC_KEY and VAPID_PRIVATE_KEY together, otherwise they are derived from the shared secret.

● Implemented · Must · Interface · The system (automatic), External service

ACCEPTANCE CRITERIA

1. Setting one key without the other, or two mismatched keys, throws an error.
2. A payload above the size limit is refused before sending.

EVIDENCE

`src/server/webPush.ts` `tests/web-push-audit.ts`

IR-EXT-012 Allowed push destinations and pruning

The server shall send only to https addresses on port 443 without credentials at known push services (FCM, legacy FCM links, Mozilla, Apple and Windows) to prevent internal requests, with a 10-second timeout, shall delete a subscription on a 404 or 410 answer, and shall refuse a subscription tied to a card that was stopped or expired.

● Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. An http address or an internal host is refused and no request is issued.
2. A 410 answer leaves the record deleted in the same response.

EVIDENCE

`src/server/webPush.ts#isAllowedPushEndpoint` `src/server/pushRoutes.ts` `tests/web-push-audit.ts`

IR-EXT-013 Per-device notification limit

Each subscription shall be limited by a bucket of five consecutive notifications and then one per minute, request-inbox notifications shall be cooled down for three minutes per department with the cooldown kept in the database, and the «try» button shall answer 429 when nothing was sent because of the limit.

● Implemented · Should · Interface · The system (automatic), Faculty member

ACCEPTANCE CRITERIA

1. The sixth within one minute is not sent and is counted as limited in the report.
2. After one minute one token returns to the bucket.

EVIDENCE

`src/server/webPush.ts#createPushRateLimiter` `src/server/pushRoutes.ts` `tests/web-push-audit.ts`

IR-EXT-014 Outbound company e-mail

Outbound e-mail shall stay disabled unless AD_MAIL_API_KEY and AD_MAIL_FROM are set; once set, a plain-text message is sent by a JSON POST with Authorization: Bearer to AD_MAIL_API_URL (default a Resend-compatible API) with a 10-second timeout, to the company address stored in its campaign or to the owner's AD_LEAD_NOTIFY_TO, with no personal numbers and no tracking, and a week key (starting Sunday) prevents sending the weekly summary twice.

● Implemented · Could · Interface · Partner organisation, The system (automatic)

ACCEPTANCE CRITERIA

1. Without configuration no mail leaves and nothing changes in production.
2. An invalid address or one longer than 120 characters is treated as absent.

EVIDENCE

`src/server/adMailer.ts` `tests/ads-audit.ts`

IR-EXT-015 ICS calendar file for a department timetable

GET /api/public/ics/:token shall serve an RFC 5545 calendar (CRLF endings, line folding at 75 bytes, a VTIMEZONE for Asia/Kuwait, weekly RRULE and EXDATE for cancelled dates) for public department timetables only, with an optional 0–120 minute alarm, a 6-hour refresh interval and a weak ETag answering 304 when unchanged, and a 16-week term length when its dates are not recorded, with an «estimated» label in the calendar name.

● Implemented · Must · Interface · Faculty member, Student, Public visitor

ACCEPTANCE CRITERIA

1. A staff-card, survey, students or request link answers 404 on this path.
2. A term without dates shows «(تواريخ الفصل غير مسجّلة)» in the name.

EVIDENCE

`src/server/routes/publicCalendarRoutes.ts` `src/utils/icalendar.ts` `tests/calendar-ics-audit.ts`

IR-EXT-016 The instructor's stable calendar subscription

The instructor's subscription key shall be of the form p<number>-<32-character HMAC> derived from CALENDAR_SECRET or from the shared secret and stable, not following the link token, so it does not die when the card is republished; when no live card remains it shall answer 200 with a valid empty calendar of the same name and a description stating the pause, and when the key matches nobody it shall answer 404.

● Implemented · Must · Interface · Faculty member

ACCEPTANCE CRITERIA

1. Two different tokens for two cards do not change the same instructor's key.
2. Rotating CALENDAR_SECRET invalidates every subscription.

EVIDENCE

`src/server/calendarSecret.ts` `src/server/routes/publicCalendarRoutes.ts` `tests/calendar-ics-audit.ts`

IR-EXT-017 One-day calendar reminder

GET /api/public/ics-reminder shall serve a one-day ICS file for the date d as YYYY-MM-DD with title t (120 characters) and note n (300 characters), with no-store, shall reject an invalid date with 400, and shall carry no name or personal number.

● Implemented · Could · Interface · Student

ACCEPTANCE CRITERIA

1. d=2026-13-40 returns 400 «bad date».
2. A 200-character title is cut to 120.

EVIDENCE

`src/server/routes/publicCalendarRoutes.ts` `tests/calendar-ics-audit.ts`

IR-EXT-018 Excel export

GET /api/reports/excel/:type shall serve an xlsx file for one of three types (ListofTeacherCourseExcel requiring permission 14, TeacherWithCourseExcel 8, ScheduleExcel 7), with a right-to-left sheet titled in Arabic «Schedule», 13 columns and an auto-filter, carrying all eleven screen filters, the seven-digit course number and the «visiting» suffix on a visiting instructor's name, under the file name schedule_report_<type>.xlsx.

● Implemented · Must · Interface · Schedule committee chair, Department head

ACCEPTANCE CRITERIA

1. An unknown type returns 400; a user without the permission returns 403.
2. Filtering by Sunday, course 321 and before noon produces the same rows as the screen.

EVIDENCE

`src/server/routes/reportsRoutes.ts` `src/utils/authorityAcademicCodes.ts` `tests/course-number-seven-digit-audit.ts`

IR-EXT-019 Service worker and PWA contract

sw.js shall be stamped with a fresh identifier on every build, cache the application shell and hashed assets (cache-first) and a fixed list of read APIs only when they succeed, never cache /api/auth, the user list and student cases (/api/schedules/demand and /api/student-registration), leave the /r, /s, /q, /m, /t, /me, /d, /b, /plans, /partners and /verify pages to the network without caching, and serve the «updating» page (503) on 500, 502, 503, 504 or a dropped navigation.

● Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. A navigation that returns 502 shows /updating.html with Retry-After: 5 and the failed answer is not cached.
2. An API read while offline returns the stored copy or a 503 with Arabic text.

EVIDENCE

`public/sw.js` `public/updating.html` `scripts/stamp-sw.mjs` `tests/updating-page-audit.ts`

IR-EXT-020 Push worker and manifests

The push worker /push-sw.js shall be independent of sw.js, stable between releases with no build stamp, shall cache no page and answer from no cache, scoped to /push/ in the application and to /r/<token> in the faculty card; the application has the manifest manifest.webmanifest (scope /) and Masari has me.webmanifest (scope /me).

● Implemented · Must · Interface · Faculty member, Student

ACCEPTANCE CRITERIA

1. Restoring the old version unregisters sw.js and keeps the push worker.
2. The application's start_url is «/?source=pwa» and the student manifest's is «/me?source=pwa».

EVIDENCE

`public/push-sw.js` `public/manifest.webmanifest` `public/me.webmanifest` `src/utils/pushClient.ts` `tests/web-push-audit.ts`

IR-EXT-021 Public link scheme

The server shall serve public pages at: /r/<token> faculty card, /s/<token> department or staff card, /q/<token> legacy survey (redirected to /me), /m/<token> my-request status, /t/<token> approved students timetable, /k/<token> chair team-member entry, /d/<token> room door, /b/<token> building board, /me «Masari», /verify/<token> signature or coupon verification, /partners/<token> partnership profile, /plans/<department>/<sheet>/ published plan sheets, and /landing the landing page; all are sent with no-store.

● Implemented · Must · Interface · Public visitor, Faculty member, Student, Partner organisation

ACCEPTANCE CRITERIA

1. An /s link of kind request is redirected to its own door and never opens the department timetable.
2. /plans/, /landing and /me carry no token.

EVIDENCE

`src/utils/demoLinkToken.ts` `src/server/routes/publicStaffRoutes.ts` `src/server/routes/surveyRoutes.ts` `src/server/routes/publicRequestRoutes.ts` `src/server/routes/verifyRoutes.ts` `src/server/pushCard.ts` `tests/public-pages-syntax-audit.ts`

IR-EXT-022 Token rules

The tokens of /s, /q, /m, /t and /r (share links) shall be random 24-byte base64url strings (32 characters), prefixed «demo.» in the demo sandbox, and the /r page accepts the form [A-Za-z0-9_-]{16,64}; the /d and /b token is a 16-character base64url string derived by HMAC from the stable room or building identifier and stays valid across releases; the /verify identifier is 32 lowercase hex characters for a coupon and a six-character code for a signature; no token carries a civil number.

Implemented · Must · Interface · The system (automatic), Public visitor

ACCEPTANCE CRITERIA

1. A token not matching the form returns the «link not found» page and is counted as a miss.
2. Moving a room between buildings or correcting its printed code does not change its door sticker token.

EVIDENCE

`src/server/roomDoorSecret.ts` `src/server/pushCard.ts` `src/utils/demoLinkToken.ts` `src/db/repository.ts#createShareLink` `tests/room-door-audit.ts` `tests/web-push-audit.ts`

IR-EXT-023 Anonymous door protection

The server shall count failure, not success, on anonymous doors: a link token matching nothing is counted against the address (limit 40 misses in 10 minutes, hard ceiling 400), a non-existent session is counted (limit 120), every success is credited to the address, and no success is ever limited; /api/public is limited to 60 requests per minute per address (configurable) except «Masari» and the faculty card, which have their own limits.

Implemented · Must · Interface · Public visitor, The system (automatic)

ACCEPTANCE CRITERIA

1. A campus on one address opens a thousand valid links without being blocked.
2. The forty-first miss within ten minutes answers with a Retry-After header.

EVIDENCE

`src/server/anonymousGuard.ts` `src/server/core/rateLimits.ts` `src/server/publicAttemptLimiter.ts` `tests/anonymous-guard-audit.ts`

IR-EXT-024 Browser contract: security and headers

Inferred

The server shall send on every response nosniff, Referrer-Policy: same-origin, a Permissions-Policy denying camera, microphone and location, a partial CSP (object-src 'none', base-uri 'self', form-action 'self', frame-ancestors 'none') and a one-year HSTS in production; it shall refuse POST, PUT, PATCH and DELETE requests to /api that come from another site (a foreign Sec-Fetch-Site or Origin) with 403.

Implemented · Must · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. A POST with an Origin of a different host returns 403 «الطلب غير مسموح».
2. HSTS is not sent outside production.

EVIDENCE

`server.ts`

IR-EXT-025 Body limits and asset caching

Inferred

JSON bodies shall be accepted up to 1 MB (4 MB for timetable drafts), PDF files up to 24 MB and system backups up to 30 MB; hashed assets shall be served with public, max-age=31536000, immutable, index.html and sw.js with no-store, and a missing file or a path under /assets/ shall answer 404 rather than the application shell.

Implemented · Should · Interface · The system (automatic)

ACCEPTANCE CRITERIA

1. A request for a missing /assets/old-chunk-abc12345.js returns 404, not HTML.
2. A 2 MB JSON body on an ordinary route is refused.

EVIDENCE

`server.ts` `src/server/routes/pdfImportRoutes.ts`

IR-EXT-026 Environment variables and secrets

Settings and secrets shall pass through the environment alone under fixed names: DATA_MODE, FIREBASE_PROJECT_ID, FIREBASE_DATABASE_ID, FIREBASE_CONFIG_SOURCE, GEMINI_API_KEY, GEMINI_SCHEDULE_MODEL, VAPID_PUBLIC_KEY, VAPID_PRIVATE_KEY, CALENDAR_SECRET, AD_MAIL_API_KEY, AD_MAIL_FROM, AD_MAIL_API_URL, AD_LEAD_NOTIFY_TO, PASSWORD_VAULT_KEY, SCHEDULE_PRIVATE_DIR, OCR_DATA_DIR and PUBLIC_RATE_LIMIT_MAX; no secret shall be written in the repository, documents or delivery archives.

● Implemented · Must · Interface · Root administrator, The system (automatic)

ACCEPTANCE CRITERIA

1. A missing CALENDAR_SECRET does not generate a new secret at each start but derives one from the stored shared secret.
2. PUBLIC_RATE_LIMIT_MAX=0 turns the public request limit off.

EVIDENCE

`src/server/runtimeEnv.ts` `src/server/calendarSecret.ts` `src/server/core/rateLimits.ts` `tests/firestore-target-audit.ts`

IR-EXT-027 Database target from environment variables

The server shall accept the Firestore project and database from environment variables alone (FIREBASE_CONFIG_SOURCE=environment with FIREBASE_PROJECT_ID and FIREBASE_DATABASE_ID), so the hosting party connects the system to its own database without editing source files, and it shall refuse to start when a variable is missing or contradicts the config file instead of choosing silently.

● Implemented · Could · Interface · Root administrator

RATIONALE

> It lets the receiving party connect the system to its database through runtime settings alone, with no data moved and no code edited.

ACCEPTANCE CRITERIA

1. With both variables and FIREBASE_CONFIG_SOURCE=environment the server connects to the named database and logs the source «environment».
2. A missing variable, or one contradicting the config file, stops start-up with an explicit message.

EVIDENCE

`src/server/firestoreTarget.ts` `src/server/runtimeEnv.ts` `docs/DELIVERY_OPERATIONS.md` `tests/firestore-target-audit.ts`

3.20.5 Edge cases

⚠ Gemini unavailable during a file import

Scenario A coordinator uploads a scanned file while the provider is busy or timed out.

Behaviour The server returns 503 with a calm message ending «جدولك لم يتغيّر»; no row is written and the provider text is logged for the operator only.

⚠ A push address chosen by someone else

Scenario A client submits a subscription with an http address or an internal host.

Behaviour The subscription is refused before storing, and if such a record reached sending, no request is issued and it is treated as gone.

⚠ Calendar subscription after the card ends

Scenario The instructor's card ended without a new card and the phone still reads the subscription.

Behaviour The server answers 200 with a valid empty calendar explaining the pause so lectures are cleared quietly, and the subscription returns on its own when a new card is issued.

⚠ A fourteen-page SWRSCHA file

Scenario A file above the 12-page limit is uploaded.

Behaviour The whole file is rejected with a message stating the limit and nothing is read from it, because an incomplete timetable is more dangerous than a refusal.

⚠ A browser holding an old build during a release

Scenario An open tab requests a hashed file that no longer exists, or a navigation returns 502 while the server is swapped.

Behaviour The missing file returns 404 rather than HTML so the boot guard clears the stale copy and reloads; the navigation shows the «updating» page which reopens the address when the server returns.

3.20.6 Standing owner decisions

- In a SWRSCHA file the header decides the department even when a course number carries another department's code; a missing department or course is created only after one confirmation, and scans never create anything.
- The course number in every exported file is seven digits (e.g. 0109105).
- A release never changes the domain mapping, the official service or its secrets; only the container image changes.
- The Gemini service only reads and proposes; execution always goes through the deterministic validators and an explicit user confirmation.

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ID	Title	Status	Priority	Code	Tests
FR-ACC-001	Sign-in with user name and password	Implemented	Must	**2**	**2**
FR-ACC-002	Rate limiting of anonymous requests	Implemented	Must	**2**	**1**
FR-ACC-003	Idle expiry and presence-based renewal	Implemented	Must	**3**	**0**
FR-ACC-004	Sign-out and session invalidation	Implemented	Must	**2**	**1**
NFR-ACC-005	Anonymous-door limits: count failures, never block success	Implemented	Must	**2**	**1**
DR-ACC-006	Password storage and non-disclosure	Implemented	Must	**3**	**1**
FR-ACC-007	Password-change gate	Implemented	Must	**2**	**1**
FR-ACC-008	New-password rules and change	Implemented	Must	**3**	**2**
FR-ACC-009	Current-password waiver at first sign-in	Implemented	Must	**2**	**1**
FR-ACC-010	Initial password, flagging and lifetime	Implemented	Must	**3**	**2**
FR-ACC-011	Temporary password for reset	Implemented	Must	**3**	**1**
DR-ACC-012	Catalogue of the ten roles	Implemented	Must	**2**	**2**
FR-ACC-013	Server guard for read-only roles	Implemented	Must	**2**	**1**
FR-ACC-014	Narrow doors for read-only roles	Implemented	Must	**2**	**2**

ID	Title	Status	Priority	Code	Tests
FR-ACC-015	Role template: screens and scope in one operation	Implemented	Must	**2**	**2**
FR-ACC-016	User administration	Implemented	Must	**3**	**1**
FR-ACC-017	Main-administration screens and the root account	Implemented	Must	**3**	**1**
FR-ACC-018	Permission and scope administration	Implemented	Must	**2**	**1**
DR-ACC-019	Screen and scope records	Implemented	Must	**3**	**1**
FR-ACC-020	The scope decision isScopeAllowed	Implemented	Must	**1**	**2**
FR-ACC-021	Scope filtering and directed reads	Implemented	Must	**2**	**1**
IR-ACC-022	Identity payload sent to the interface	Implemented	Must	**4**	**2**
FR-ACC-023	Hiding instructors' personal data from read-only roles	Implemented	Must	**1**	**1**
FR-ACC-024	Provisioning department-head and advisor accounts from a previewed plan	Implemented	Should	**2**	**1**
NFR-ACC-025	One identity resolution per request with a short cache	Implemented	Should	**1**	**0**
FR-ACC-026	Demo environment and role switching	Implemented	Should	**2**	**2**
FR-ACC-027	Audit trail for sign-in and refusals	Implemented	Should	**4**	**0**
FR-ACC-028	Backup vault for the root account	Implemented	Must	**2**	**0**
FR-ACC-029	Invitation link and access card without sending the password to the server	Implemented	Must	**4**	**2**
FR-ACC-030	Remember-this-device key for the faculty card	Implemented	Should	**1**	**1**
FR-ACD-001	Term create, update and delete	Implemented	Must	**4**	**1**
FR-ACD-002	Automatic creation of the next term	Implemented	Should	**3**	**1**
FR-ACD-003	Automatic closing of an ended term	Implemented	Must	**2**	**1**
FR-ACD-004	Scope-aware term list	Implemented	Should	**2**	**1**
FR-ACD-005	College CRUD and delete guard	Implemented	Must	**2**	**1**
FR-ACD-006	Department, faculty placeholder and delete guard	Implemented	Must	**3**	**1**
DR-ACD-007	Seven-digit authority course number	Implemented	Must	**2**	**2**
DR-ACD-008	Department key from college and department codes	Implemented	Must	**2**	**2**
FR-ACD-009	Linking a course to its printed number	Implemented	Must	**3**	**1**
FR-ACD-010	Course creation and numeric limits	Implemented	Must	**3**	**1**
FR-ACD-011	Freezing the identity of a course with history	Implemented	Must	**1**	**0**
FR-ACD-012	Block deleting a linked course	Implemented	Must	**1**	**0**
FR-ACD-013	Room type and teaching mode sync across copies	Implemented	Should	**2**	**1**
FR-ACD-014	Course tags in a fixed order	Implemented	Should	**2**	**1**
FR-ACD-015	Duplicate course detection and archived delete	Implemented	Should	**2**	**1**
FR-ACD-016	Spelling-twin merge with confirmation	Implemented	Should	**4**	**2**
DR-ACD-017	Faculty identity: civil number	Implemented	Must	**3**	**1**
FR-ACD-018	Quick instructor add and department enrolment	Implemented	Must	**3**	**2**
FR-ACD-019	Instructor search by name identity	Implemented	Must	**4**	**1**
NFR-ACD-020	Civil number and phone hidden from non-members	Implemented	Must	**2**	**1**
FR-ACD-021	Instructor status and category	Implemented	Must	**3**	**2**
FR-ACD-022	Rank and teaching load limits	Implemented	Should	**3**	**1**
FR-ACD-023	Load computation and over-load warning	Implemented	Should	**3**	**2**

ID	Title	Status	Priority	Code	Tests
FR-ACD-024	Visiting status through directory and term roster	Implemented	Must	**4**	**3**
FR-ACD-025	Visiting roster and copy between terms	Implemented	Should	**2**	**1**
FR-ACD-026	Converting an instructor between visiting and appointed	Implemented	Must	**4**	**1**
FR-ACD-027	Visiting instructor becomes faculty placeholder on import	Implemented	Must	**2**	**1**
FR-ACD-028	Course offering state per term	Implemented	Should	**5**	**2**
IR-ACD-029	Scoped course reads with sheet tags	Implemented	Should	**2**	**1**
NFR-ACD-030	Reference-data input validation	Implemented	Must	**3**	**1**
DR-LOC-001	Official building identity	Implemented	Must	**2**	**1**
DR-LOC-002	Hall record	Implemented	Must	**2**	**2**
DR-LOC-003	Registry confidence and seed	Implemented	Must	**3**	**1**
FR-LOC-004	Add a building	Implemented	Must	**2**	**1**
FR-LOC-005	Add halls in bulk	Implemented	Must	**1**	**1**
FR-LOC-006	Edit a building	Implemented	Must	**2**	**1**
FR-LOC-007	Deletion is deactivation	Implemented	Must	**1**	**1**
FR-LOC-008	Rename and move a hall	Implemented	Must	**2**	**1**
FR-LOC-009	The list is the permission	Implemented	Must	**3**	**2**
FR-LOC-010	Location check on every save	Implemented	Must	**2**	**2**
FR-LOC-011	Registry initialisation, migration and health	Implemented	Should	**3**	**1**
FR-LOC-012	Hall type differs by department	Implemented	Should	**4**	**2**
FR-LOC-013	Free-window board for barter	Implemented	Must	**3**	**2**
FR-LOC-014	Boys and girls campuses stay separate	Implemented	Must	**1**	**0**
FR-LOC-015	Requesting and approving a barter	Implemented	Must	**2**	**2**
FR-LOC-016	Cancelling and ending a barter	Implemented	Must	**1**	**2**
FR-LOC-017	Request expiry and inbox counter	Implemented	Should	**2**	**0**
FR-LOC-018	Enforcing loan windows on save	Implemented	Must	**3**	**2**
FR-LOC-019	Loan hours drawing	Implemented	Could	**2**	**1**
FR-LOC-020	Hall door page	Implemented	Must	**4**	**1**
FR-LOC-021	Building board and corridor screen	Implemented	Should	**4**	**1**
NFR-LOC-022	Sticker token is permanent and unguessable	Implemented	Must	**1**	**1**
NFR-LOC-023	Public page privacy and limits	Implemented	Must	**3**	**1**
FR-LOC-024	Door links and stickers	Implemented	Should	**3**	**1**
FR-LOC-025	Seven-digit course number on the door	Implemented	Must	**3**	**2**
FR-LOC-026	Occupancy rate	Implemented	Should	**4**	**3**
FR-LOC-027	Companion facility of a hall	Implemented	Should	**5**	**1**
FR-LOC-028	Travel policy between campuses	Implemented	Must	**6**	**1**
FR-LOC-029	Walking time between buildings on a campus	Implemented	Should	**2**	**0**
NFR-LOC-030	One identity for a hall	Implemented	Must	**3**	**2**
FR-SCH-001	Create an appointment	Implemented	Must	**2**	**2**
FR-SCH-002	Edit and delete with revision check	Implemented	Must	**3**	**1**

ID	Title	Status	Priority	Code	Tests
FR-SCH-003	Schedule lock after submission and term end	Implemented	Must	**3**	**2**
FR-SCH-004	Scope and course integrity	Implemented	Must	**2**	**1**
FR-SCH-005	Official place and companion facility	Implemented	Must	**2**	**1**
FR-SCH-006	Instructor, hall and duplicate-section conflict detection	Implemented	Must	**3**	**3**
FR-SCH-007	Same section in two places: warning, not a block	Implemented	Must	**3**	**2**
FR-SCH-008	A faculty placeholder is not a person who can be double-booked	Implemented	Must	**3**	**2**
FR-SCH-009	Turnaround gap and student clash: two soft notes	Implemented	Must	**3**	**2**
FR-SCH-010	What blocks a save and what does not	Implemented	Must	**3**	**3**
FR-SCH-011	What is disclosed about another department's appointment	Implemented	Must	**3**	**3**
FR-SCH-012	Other departments' halls and hall barter	Implemented	Must	**3**	**0**
FR-SCH-013	Instructor-request holds awaiting the department	Implemented	Must	**3**	**2**
FR-SCH-014	Two-minute advisory colleague hold	Implemented	Should	**3**	**1**
FR-SCH-015	University teaching day and department window	Implemented	Must	**6**	**4**
FR-SCH-016	Regulation days, lengths and start ladder	Implemented	Must	**4**	**2**
FR-SCH-017	Approved pattern library	Implemented	Must	**3**	**3**
FR-SCH-018	Weekly-hours ceiling per section	Implemented	Must	**3**	**2**
FR-SCH-019	Inter-campus travel time	Implemented	Should	**4**	**2**
FR-SCH-020	Department rhythm and memory	Implemented	Should	**3**	**2**
FR-SCH-021	Live appointment check before saving	Implemented	Must	**2**	**2**
FR-SCH-022	Atomic batch move	Implemented	Must	**2**	**0**
FR-SCH-023	Approving a proposal into the live schedule	Implemented	Must	**3**	**2**
FR-SCH-024	Assign, swap and replace a faculty-placeholder section	Implemented	Should	**3**	**2**
FR-SCH-025	Replace an instructor across a term	Implemented	Should	**1**	**0**
FR-SCH-026	Convert TA staff to the faculty placeholder	Implemented	Could	**2**	**1**
FR-SCH-027	Time and hall suggestions	Implemented	Should	**3**	**3**
FR-SCH-028	Common meeting time	Implemented	Could	**3**	**1**
FR-SCH-029	Cancel or cover one date, and rank substitutes	Implemented	Should	**3**	**2**
FR-SCH-030	Cell notes, department rebuttal and registration verdict	Implemented	Should	**3**	**3**
FR-SCH-031	Live push and colleague presence	Implemented	Should	**4**	**2**
FR-SCH-032	Snapshot before every write and no silent overwrite	Implemented	Must	**3**	**2**
FR-SCH-033	Wipe and restore a department schedule	Implemented	Must	**3**	**1**
FR-SCH-034	Copy a term's schedule into another term	Implemented	Should	**1**	**2**
FR-SCH-035	Read boundaries by role and scope	Implemented	Must	**2**	**2**
FR-SCH-036	Arabic ordering of terms, names and courses	Implemented	Should	**3**	**1**
FR-SCH-037	Teacher mini-week and week window	Implemented	Should	**4**	**3**
DR-SCH-001	Appointment (schedule row) data model	Implemented	Must	**3**	**1**
NFR-SCH-001	Indexed conflict detection with identical results	Implemented	Must	**4**	**1**
IR-SCH-001	Drag preview before the drop	Implemented	Should	**3**	**0**
IR-SCH-002	Saved views, undo and offline work	Implemented	Should	**4**	**1**

ID	Title	Status	Priority	Code	Tests
FR-DRF-001	Create a schedule draft	Implemented	Must	**1**	**3**
FR-DRF-002	Validate a draft before saving and cap its size	Implemented	Must	**1**	**0**
FR-DRF-003	Edit and delete draft rows	Implemented	Must	**1**	**1**
FR-DRF-004	Start-of-term draft from the previous term	Implemented	Must	**1**	**2**
FR-DRF-005	Appointment gate for start-of-term drafts	Implemented	Must	**2**	**1**
FR-DRF-006	Publish a draft with explicit confirmation, lock and deadline gates	Implemented	Must	**2**	**2**
FR-DRF-007	Never erase changes made after the draft's base	Implemented	Must	**4**	**1**
FR-DRF-008	Repair time collisions when publishing a start-of-term draft	Implemented	Should	**2**	**0**
FR-DRF-009	Publish every site or nothing	Implemented	Must	**1**	**1**
FR-DRF-010	Safety point before publish and approval-cycle notice	Implemented	Must	**2**	**2**
FR-DRF-011	Version list and two-version comparison	Implemented	Should	**2**	**1**
FR-DRF-012	Restore a previous version	Implemented	Must	**2**	**1**
FR-DRF-013	Compare two terms of a department	Implemented	Should	**2**	**2**
DR-DRF-014	Scenario record	Implemented	Must	**2**	**1**
FR-DRF-015	Who may read and simulate, and who may save and apply	Implemented	Must	**1**	**1**
FR-DRF-016	Mask the rest of the term outside the scope	Implemented	Must	**1**	**1**
FR-DRF-017	Apply a scenario atomically	Implemented	Must	**1**	**1**
FR-DRF-018	One-click restore point	Implemented	Must	**1**	**1**
FR-DRF-019	Green, yellow and red impact indicator	Implemented	Must	**2**	**1**
FR-DRF-020	Problem reading and three ranked solutions	Implemented	Must	**2**	**2**
FR-DRF-021	Student impact of a scenario from the radar only	Implemented	Must	**2**	**1**
FR-DRF-022	Notify only the affected instructors	Implemented	Should	**2**	**2**
FR-DRF-023	What-if on a proposal's impact before approval	Implemented	Should	**2**	**1**
FR-DRF-024	Refresh and read the registration radar within scope	Implemented	Should	**2**	**2**
FR-DRF-025	End the previous plan sheet for departments	Implemented	Should	**2**	**1**
IR-DRF-026	The «before apply» sheet	Implemented	Should	**4**	**1**
NFR-DRF-027	Seven-digit course number on time-machine screens	Implemented	Must	**5**	**2**
FR-VIW-001	Read-only department schedule	Implemented	Must	**1**	**1**
FR-VIW-002	Export the term to a file	Implemented	Must	**2**	**1**
FR-VIW-003	An ended term is read and export only	Implemented	Must	**2**	**0**
FR-VIW-004	Replace an instructor in a term in two steps	Implemented	Must	**2**	**2**
FR-VIW-005	Read what will carry over before copying a term	Implemented	Should	**2**	**1**
FR-VIW-006	Start-of-term screen: build, review, publish and undo	Implemented	Must	**2**	**3**
FR-VIW-007	The six living-schedule scenes	Implemented	Should	**3**	**2**
FR-VIW-008	One approval-blocker number everywhere	Implemented	Must	**1**	**1**
NFR-VIW-009	The living reading is computed once, not once per asker	Implemented	Should	**3**	**0**
FR-VIW-010	Last reading of the three indicators	Implemented	Could	**2**	**0**
FR-VIW-011	Previous-term ghost over the schedule	Implemented	Should	**1**	**0**
FR-VIW-012	The «decision now» panel	Implemented	Should	**2**	**0**

ID	Title	Status	Priority	Code	Tests
NFR-VIW-013	Analysis reads are deferred and quiet on failure	Implemented	Should	**1**	**0**
FR-VIW-014	Issue a share link	Implemented	Must	**2**	**3**
FR-VIW-015	One live department card per term	Implemented	Must	**2**	**2**
FR-VIW-016	Link lifetime: reading until the end of the term	Implemented	Must	**2**	**1**
FR-VIW-017	Opening and stopping a link	Implemented	Must	**1**	**1**
FR-VIW-018	Revoke a link	Implemented	Must	**1**	**1**
FR-VIW-019	Personal instructor links and send tracking	Implemented	Must	**1**	**2**
FR-VIW-020	The calendar subscription lives while a card link lives	Implemented	Should	**1**	**1**
FR-VIW-021	Saved views	Implemented	Could	**2**	**0**
FR-VIW-022	Warn when a saved view lost its scope	Implemented	Could	**1**	**0**
FR-VIW-023	The journey screen	Implemented	Could	**3**	**2**
NFR-VIW-024	Seven-digit course number on viewing and transfer screens	Implemented	Must	**6**	**1**
DR-APR-001	Approval record and its event log	Implemented	Must	**4**	**2**
DR-APR-002	Signature and captured version	Implemented	Must	**2**	**1**
FR-APR-001	Signing order and who may sign	Implemented	Must	**4**	**2**
FR-APR-002	Material conflict blocks signing, sending and acceptance	Implemented	Must	**2**	**3**
FR-APR-003	Other signing gates: capacity, requirements, regulation	Implemented	Must	**5**	**2**
FR-APR-004	Head signature, sending and its conditions	Implemented	Must	**2**	**2**
FR-APR-005	A new round for every send	Implemented	Must	**3**	**2**
FR-APR-006	Signature withdrawal and head return to committee	Implemented	Must	**4**	**1**
FR-APR-007	Sections added after the head's signature	Implemented	Must	**4**	**2**
FR-APR-008	Cell notes and their computed state	Implemented	Must	**4**	**3**
FR-APR-009	Department reply, registrar verdict and escalation	Implemented	Should	**2**	**2**
FR-APR-010	Returning a schedule with notes	Implemented	Must	**2**	**2**
FR-APR-011	Final acceptance by the registrar	Implemented	Must	**2**	**2**
FR-APR-012	Decision serialisation and binding to what was seen	Implemented	Must	**4**	**1**
FR-APR-013	Amendment round after acceptance	Implemented	Must	**3**	**2**
FR-APR-014	Schedule lock after sending	Implemented	Must	**2**	**3**
FR-APR-015	Ended term: the cycle is closed	Implemented	Must	**3**	**1**
FR-APR-016	Past terms frozen read-only	Implemented	Must	**3**	**2**
FR-APR-017	Term deadline and its effect after it passes	Implemented	Must	**7**	**2**
FR-APR-018	Bulk exception grants	Implemented	Must	**3**	**1**
FR-APR-019	Extension request from a department	Implemented	Should	**2**	**2**
FR-APR-020	Term student dates	Implemented	Could	**3**	**1**
FR-APR-021	Registrar inbox, lateness and counter	Implemented	Must	**4**	**4**
FR-APR-022	Term-wide status and department balance	Implemented	Should	**4**	**2**
FR-APR-023	Dean sees the final schedule only; council bulletin	Implemented	Should	**4**	**3**
FR-APR-024	Review readiness and detailed blockers	Implemented	Must	**3**	**2**
FR-APR-025	Approval score as an explicit formula	Implemented	Should	**2**	**1**

ID	Title	Status	Priority	Code	Tests
FR-APR-026	Approval dossier	Implemented	Should	**3**	**2**
FR-APR-027	Printed sheet stamp and its verification page	Implemented	Should	**6**	**1**
FR-APR-028	Instructor request signature check by civil id	Implemented	Should	**2**	**1**
NFR-APR-001	Scope on every route and inbox performance	Implemented	Must	**4**	**3**
NFR-APR-002	Kuwait-time dates with a single rule	Implemented	Must	**3**	**2**
NFR-APR-003	Privacy and attempt limits on public pages	Implemented	Must	**3**	**2**
IR-APR-001	Approval bar shows only what awaits the viewer	Implemented	Should	**3**	**2**
IR-APR-002	Submission deadlines panel	Implemented	Should	**3**	**1**
FR-IMP-001	Authority PDF intake	Implemented	Must	**3**	**2**
FR-IMP-002	Text layer or scanned reading	Implemented	Must	**2**	**2**
FR-IMP-003	Header proof before table reading	Implemented	Must	**2**	**2**
FR-IMP-004	Header decides the department	Implemented	Must	**2**	**1**
FR-IMP-005	Branch match by code	Implemented	Must	**2**	**2**
FR-IMP-006	Accepted source term rule	Implemented	Must	**2**	**1**
FR-IMP-007	Copy into an empty term and trial run	Implemented	Must	**4**	**2**
FR-IMP-008	Refusing unusable scans up front	Implemented	Must	**2**	**1**
FR-IMP-009	Scanned page verdict	Implemented	Must	**2**	**1**
NFR-IMP-010	One scan read at a time	Implemented	Must	**3**	**2**
FR-IMP-011	Reading progress and interruption	Implemented	Should	**3**	**2**
FR-IMP-012	Course identity and section number	Implemented	Must	**3**	**2**
FR-IMP-013	Instructor identity without guessing	Implemented	Must	**3**	**1**
FR-IMP-014	Building then room from the registry	Implemented	Must	**3**	**3**
FR-IMP-015	Registering branch sites the document proves	Implemented	Should	**1**	**1**
FR-IMP-016	Plausibility checks on scanned rows	Implemented	Must	**3**	**2**
DR-IMP-017	Per-cell evidence on every row	Implemented	Must	**2**	**1**
FR-IMP-018	Creating a missing department after one confirmation	Implemented	Must	**2**	**1**
FR-IMP-019	Creating missing courses after one confirmation	Implemented	Must	**3**	**1**
FR-IMP-020	A three-digit tail never links	Implemented	Must	**2**	**2**
DR-IMP-021	Signed header-check receipt	Implemented	Must	**3**	**0**
FR-IMP-022	Scanned page review confirmation	Implemented	Must	**3**	**1**
FR-IMP-023	Pre-publish check against the whole term	Implemented	Must	**3**	**1**
FR-IMP-024	Single import gate state	Implemented	Must	**3**	**1**
FR-IMP-025	Editable preview and name fill	Implemented	Should	**4**	**2**
FR-IMP-026	Comparing the schedule with the authority file	Implemented	Must	**5**	**4**
FR-IMP-027	Assisted reading through Gemini on explicit choice	Implemented	Should	**4**	**1**
FR-IMP-028	Personal-data redaction and proposal identity	Implemented	Must	**2**	**1**
FR-IMP-029	Assisted proposals never write on their own	Implemented	Must	**3**	**1**
IR-IMP-030	Gemini call interface and limits	Implemented	Should	**1**	**1**
FR-IMP-031	Spreadsheet import	Implemented	Must	**3**	**1**

ID	Title	Status	Priority	Code	Tests
FR-IMP-032	JSON schedule export and import	Implemented	Should	**2**	**0**
FR-IMP-033	Reading a study-plan sheet	Implemented	Should	**3**	**2**
FR-IMP-034	Reconciling courses with their sheet	Implemented	Should	**3**	**2**
FR-REQ-001	Personal request link per faculty member	Implemented	Must	**2**	**3**
FR-REQ-002	Request window and link lifetime	Implemented	Must	**4**	**2**
FR-REQ-003	Link mode: view, edit or training	Implemented	Must	**3**	**3**
FR-REQ-004	Publishing to selected faculty and the send log	Implemented	Must	**3**	**2**
FR-REQ-005	Reopening follows the new deadline	Implemented	Must	**2**	**2**
FR-REQ-006	Individual exception after closing	Implemented	Should	**2**	**1**
FR-REQ-007	Not-yet-reached course state	Implemented	Must	**4**	**2**
FR-REQ-008	Not-yet-reached sheet before sending	Implemented	Should	**2**	**2**
FR-REQ-009	Link resolution and error replies	Implemented	Must	**3**	**2**
FR-REQ-010	Guess limit on wrong links	Implemented	Must	**3**	**1**
NFR-REQ-011	Rate limits on public request APIs	Implemented	Must	**3**	**1**
FR-REQ-012	Civil-ID signature on requests	Implemented	Must	**3**	**2**
FR-REQ-013	Accepting a request and validating its items	Implemented	Must	**3**	**3**
FR-REQ-014	Engine verdict on a member's item	Implemented	Must	**3**	**3**
FR-REQ-015	Meeting length and pattern day completion	Implemented	Must	**3**	**3**
FR-REQ-016	Temporary hold and race settlement	Implemented	Must	**3**	**3**
FR-REQ-017	Time map and draft check	Implemented	Should	**2**	**2**
FR-REQ-018	Stick to my lecture	Implemented	Could	**3**	**3**
FR-REQ-019	Choosing a vacant section	Implemented	Must	**2**	**2**
FR-REQ-020	Confirming the schedule unchanged	Implemented	Must	**2**	**2**
FR-REQ-021	Replaying the previous term's request	Implemented	Could	**3**	**3**
NFR-REQ-022	What never reaches the member: rooms and internal history	Implemented	Must	**2**	**3**
IR-REQ-023	Live card stream	Implemented	Should	**2**	**1**
FR-REQ-024	Instructor inbox: decision list and its scope	Implemented	Must	**3**	**3**
FR-REQ-025	Department decision on an item: fix or reject	Implemented	Must	**3**	**3**
FR-REQ-026	Atomic approval of an item	Implemented	Must	**3**	**3**
FR-REQ-027	Item dialogue and accepting an alternative	Implemented	Must	**5**	**3**
FR-REQ-028	Cancelling a department-placed appointment and undoing a fix	Implemented	Should	**2**	**2**
NFR-REQ-029	Request revision stamp against overwriting a colleague	Implemented	Must	**2**	**2**
FR-REQ-030	Faculty card identity gate	Implemented	Must	**3**	**3**
FR-REQ-031	Remember this device	Implemented	Should	**2**	**1**
FR-REQ-032	Member notes from the faculty card	Implemented	Should	**1**	**2**
DR-REQ-033	Family card publication and schedule movement	Implemented	Must	**4**	**2**
IR-REQ-034	Location filter, congestion map and inbox question	Implemented	Could	**4**	**4**
IR-REQ-035	Save-blocker checks for the appointment form	Implemented	Should	**3**	**3**
IR-REQ-036	Faculty mini week in appointment forms	Implemented	Could	**2**	**3**

ID	Title	Status	Priority	Code	Tests
DR-REQ-037	Anonymous appointment-entry measure	Implemented	Could	**3**	**1**
FR-PLN-001	Remaining-report import	Implemented	Must	**3**	**2**
FR-PLN-002	Mismatched report refusal	Implemented	Must	**2**	**1**
FR-PLN-003	Registration statistics save	Implemented	Must	**2**	**1**
FR-PLN-004	Section count suggestion	Implemented	Must	**2**	**1**
FR-PLN-005	Same-season history and course lineage	Implemented	Must	**2**	**1**
FR-PLN-006	Section gap report	Implemented	Must	**5**	**2**
FR-PLN-007	Surplus and conflict review	Implemented	Must	**1**	**1**
FR-PLN-008	Planning triage and decisions	Implemented	Must	**3**	**2**
FR-PLN-009	Course offering states	Implemented	Must	**3**	**3**
FR-PLN-010	Offering effect on suggestions	Implemented	Must	**3**	**2**
FR-PLN-011	Not-yet-reached verdict	Implemented	Must	**4**	**2**
FR-PLN-012	Section-build preview	Implemented	Must	**4**	**3**
FR-PLN-013	Pending-room sections from history	Implemented	Must	**2**	**1**
FR-PLN-014	Confirmed section apply	Implemented	Must	**2**	**1**
FR-PLN-015	Apply undo	Implemented	Must	**1**	**1**
FR-PLN-016	Plan sheet course rules	Implemented	Must	**2**	**1**
FR-PLN-017	Plan sheet rule checks	Implemented	Must	**1**	**1**
FR-PLN-018	Draft editing with department guard	Implemented	Must	**2**	**1**
FR-PLN-019	Plan sheet approval	Implemented	Must	**1**	**1**
FR-PLN-020	Study plan PDF upload	Implemented	Must	**6**	**3**
FR-PLN-021	Reference sheets and their application	Implemented	Must	**6**	**3**
FR-PLN-022	Sheet membership and badges	Implemented	Must	**6**	**2**
FR-PLN-023	Department plans and transitions	Implemented	Must	**2**	**1**
FR-PLN-024	Public sheet publishing	Implemented	Must	**3**	**1**
IR-PLN-001	Public plan pages	Implemented	Must	**4**	**1**
FR-PLN-025	Study proposal drafting	Implemented	Must	**3**	**2**
FR-PLN-026	Proposal check with the schedule checker	Implemented	Must	**3**	**2**
FR-PLN-027	Proposal versions and instructor response	Implemented	Must	**3**	**3**
FR-PLN-028	Proposal commit and undo	Implemented	Must	**2**	**1**
FR-PLN-029	Vacancy ranking for the instructor	Implemented	Should	**3**	**1**
IR-PLN-002	Instructor proposal page and alert	Implemented	Must	**3**	**1**
NFR-PLN-001	Department scope and permission	Implemented	Must	**4**	**2**
NFR-PLN-002	Seven-digit course numbers and printed design	Implemented	Must	**3**	**2**
DR-PLN-001	Planning data stores	Implemented	Must	**3**	**1**
FR-STU-001	Masari entry by civil ID and personal code	Implemented	Must	**3**	**2**
FR-STU-002	Profile lockout after wrong codes	Implemented	Must	**2**	**2**
FR-STU-003	Signed door session	Implemented	Must	**2**	**1**
DR-STU-001	Identity stored sealed	Implemented	Must	**4**	**2**

ID	Title	Status	Priority	Code	Tests
DR-STU-002	Student profile lifetime and deletion	Implemented	Must	**2**	**1**
FR-STU-004	Profile setup: department, plan, passed courses	Implemented	Must	**3**	**2**
FR-STU-005	Validation of the student-journey dates	Implemented	Must	**3**	**1**
FR-STU-006	Phase reading and door-term choice	Implemented	Must	**2**	**2**
FR-STU-007	Writing wishes during the wishes phase	Implemented	Must	**2**	**2**
FR-STU-008	Per-department student unit cap	Implemented	Should	**4**	**1**
FR-STU-009	Student advice and peer counts	Implemented	Should	**3**	**2**
FR-STU-010	The «suggestions only» acknowledgement	Implemented	Must	**2**	**2**
NFR-STU-001	Aggregated home and no caching	Implemented	Should	**1**	**2**
FR-STU-011	Student request to open a course or review a conflict	Implemented	Must	**4**	**3**
FR-STU-012	The «did you register?» question after a round	Implemented	Must	**3**	**2**
FR-STU-013	Graduate request and graduation-sheet proof	Implemented	Must	**3**	**2**
FR-STU-014	Request status by case number	Implemented	Must	**4**	**2**
IR-STU-001	Public student schedule page	Implemented	Should	**2**	**0**
FR-STU-015	«Try as a student»: isolated demonstration	Implemented	Could	**2**	**1**
FR-STU-016	Registration sheet: scope and signature gating	Implemented	Must	**4**	**2**
FR-STU-017	Committee first, then registration	Implemented	Must	**2**	**1**
FR-STU-018	Stale-write refusal on decisions	Implemented	Must	**2**	**2**
FR-STU-019	Committee approval with section opening	Implemented	Should	**3**	**2**
FR-STU-020	Advising sends a section request	Implemented	Must	**4**	**1**
FR-STU-021	Section request stages and transitions	Implemented	Must	**2**	**1**
FR-STU-022	Distributing request capacity across sections	Implemented	Should	**3**	**1**
FR-STU-023	Party powers and notifications	Implemented	Must	**2**	**2**
FR-STU-024	Advisor seat balance	Implemented	Must	**3**	**2**
FR-STU-025	Seat, holder and handover	Implemented	Must	**6**	**1**
FR-STU-026	Team member link and deputy	Implemented	Must	**6**	**2**
FR-STU-027	Registration rehearsal engine	Implemented	Must	**2**	**2**
FR-STU-028	Running and refreshing the radar	Implemented	Must	**3**	**2**
FR-STU-029	Why a student does not reach 12 units	Implemented	Should	**2**	**1**
NFR-STU-002	Limits on a shared university network	Implemented	Must	**1**	**2**
IR-STU-002	Academic advice panel for departments	Implemented	Should	**4**	**1**
IR-STU-003	Registration and radar tabs and the phase line	Implemented	Should	**5**	**2**
FR-INT-001	Analytical overview of the schedule	Implemented	Must	**2**	**1**
FR-INT-002	Reading scope: one college or all colleges of a department	Implemented	Must	**3**	**2**
FR-INT-003	Room insight	Implemented	Should	**1**	**0**
FR-INT-004	Instructor insight	Implemented	Should	**1**	**0**
FR-INT-005	Appointment story and context	Implemented	Should	**3**	**0**
FR-INT-006	Campus movement and safe room swaps	Implemented	Should	**2**	**0**
FR-INT-007	Ripple forecast before a drop	Implemented	Should	**2**	**0**

ID	Title	Status	Priority	Code	Tests
FR-INT-008	Rule-based analytical assistant	Implemented	Should	**1**	**1**
FR-INT-009	Intent identification by Gemini within allowed functions	Implemented	Must	**3**	**1**
FR-INT-010	Deterministic scoped execution of calls	Implemented	Must	**1**	**0**
FR-INT-011	Fallback to the rule parser	Implemented	Must	**3**	**1**
FR-INT-012	Move preview without writing, separate confirmation	Implemented	Must	**2**	**1**
FR-INT-013	Academic decision lab	Implemented	Should	**3**	**1**
FR-INT-014	Decision room, automatic optimiser and auto proposal	Implemented	Should	**4**	**0**
FR-INT-015	Custom scheduling rules	Implemented	Should	**2**	**0**
FR-INT-016	Policy simulation	Implemented	Could	**2**	**1**
FR-INT-017	Open decisions	Implemented	Could	**1**	**0**
FR-INT-018	Next-term demand forecast	Implemented	Should	**4**	**2**
FR-INT-019	Forecast confidence and action gate	Implemented	Must	**3**	**2**
FR-INT-020	Radar insight: shortfall causes and offering quality	Implemented	Should	**3**	**1**
FR-INT-021	Live schedule layer	Implemented	Should	**4**	**0**
FR-INT-022	Course impact and appointment memory	Implemented	Should	**2**	**1**
FR-INT-023	Guide knowledge bank	Implemented	Must	**2**	**1**
FR-INT-024	Guide question understanding within role permissions	Implemented	Must	**2**	**1**
FR-INT-025	Kuwaiti-dialect understanding and spelling correction	Implemented	Should	**3**	**1**
FR-INT-026	Guide on-screen tours	Implemented	Should	**4**	**2**
FR-INT-027	Guide situational awareness and quiet hints	Implemented	Could	**4**	**1**
FR-INT-028	Interface telemetry intake and anonymous Guide question log	Implemented	Should	**3**	**1**
FR-INT-029	Guide friction points for the root administrator	Implemented	Could	**3**	**0**
FR-INT-030	Experience health	Implemented	Could	**1**	**0**
NFR-INT-001	Read-only: analytics routes write nothing to the schedule	Implemented	Must	**2**	**2**
NFR-INT-002	Privacy: no civil identity or student list leaves	Implemented	Must	**4**	**2**
IR-INT-001	Analytics Center screen: scenes and scope	Implemented	Should	**3**	**1**
IR-INT-002	Gemini service interface: request constraints and timeout	Implemented	Must	**1**	**0**
FR-RPT-001	Ten entry modes on one screen	Implemented	Must	**2**	**1**
FR-RPT-002	One filter set	Implemented	Must	**1**	**1**
FR-RPT-003	24-hour time options	Implemented	Should	**2**	**0**
FR-RPT-004	Fifteen display lenses	Implemented	Must	**1**	**1**
FR-RPT-005	Role-based lens lists	Implemented	Should	**1**	**0**
FR-RPT-006	Department balance	Implemented	Must	**2**	**2**
FR-RPT-007	Room load and free windows	Implemented	Must	**1**	**1**
FR-RPT-008	Load fairness from one formula	Implemented	Must	**2**	**1**
FR-RPT-009	Occupancy rate	Implemented	Should	**3**	**1**
FR-RPT-010	Visiting-instructor query across department colleges	Implemented	Must	**3**	**2**
FR-RPT-011	Visiting history across terms	Implemented	Should	**3**	**1**
FR-RPT-012	Courses by rooms	Implemented	Should	**1**	**1**

ID	Title	Status	Priority	Code	Tests
FR-RPT-013	Read-only decision board	Implemented	Should	**2**	**1**
FR-RPT-014	Seven-digit course number	Implemented	Must	**4**	**1**
FR-RPT-015	Excel export with screen filters	Implemented	Must	**2**	**2**
FR-RPT-016	Export through the request reader	Implemented	Must	**2**	**1**
FR-RPT-017	A titled sheet for every lens	Implemented	Must	**1**	**1**
FR-RPT-018	A4 landscape on every engine	Implemented	Must	**2**	**1**
FR-RPT-019	Filling the sheet without clipping	Implemented	Must	**1**	**1**
FR-RPT-020	Printed page row budgets	Implemented	Must	**2**	**1**
FR-RPT-021	Instructor report scope	Implemented	Must	**2**	**1**
IR-RPT-001	Sheet chooser for the two composite reports	Implemented	Should	**3**	**1**
FR-RPT-023	Changes report and Authority comparison sources	Implemented	Must	**3**	**3**
FR-RPT-024	Authority report row ordering	Implemented	Must	**1**	**1**
FR-RPT-025	Plan-sheet badge bridge	Implemented	Must	**3**	**2**
NFR-RPT-001	Printed sheet design is preserved	Implemented	Must	**2**	**1**
NFR-RPT-002	Print command in the same tap	Implemented	Must	**1**	**1**
NFR-RPT-003	Printing from the installed iOS app	Implemented	Should	**1**	**0**
DR-RPT-001	Report data sources	Implemented	Must	**2**	**1**
IR-RPT-002	Export Excel button	Implemented	Should	**1**	**1**
FR-NTF-001	Bell derived from current state	Implemented	Must	**2**	**1**
FR-NTF-002	Notification scope and tones	Implemented	Must	**2**	**2**
FR-NTF-003	Two-term bell	Implemented	Should	**2**	**1**
FR-NTF-004	Feature notices and fault isolation	Implemented	Should	**2**	**1**
DR-NTF-005	Server-side read marks	Implemented	Must	**3**	**1**
IR-NTF-006	Bell interface	Implemented	Should	**1**	**2**
FR-NTF-007	Opening a notice on its department	Implemented	Should	**3**	**2**
NFR-NTF-008	Bell cost	Implemented	Should	**3**	**1**
FR-NTF-009	Live schedule channel	Implemented	Must	**3**	**2**
FR-NTF-010	Cross-instance change broadcast	Implemented	Must	**2**	**1**
FR-NTF-011	Board presence	Implemented	Must	**4**	**2**
FR-NTF-012	Presence and change pulse	Implemented	Must	**2**	**2**
NFR-NTF-013	Browser pulse cadence	Implemented	Should	**2**	**1**
FR-NTF-014	Advising and registration team presence	Implemented	Should	**3**	**1**
IR-NTF-015	Colleague faces chip	Implemented	Should	**3**	**2**
FR-NTF-016	Stale-write protection	Implemented	Must	**3**	**1**
FR-NTF-017	Push subscription	Implemented	Must	**4**	**1**
FR-NTF-018	Department coordinator push	Implemented	Should	**2**	**1**
FR-NTF-019	Faculty card push	Implemented	Must	**3**	**1**
NFR-NTF-020	Delivery limits	Implemented	Must	**2**	**1**
FR-NTF-021	Subscription rotation and sign-out	Implemented	Should	**2**	**1**

ID	Title	Status	Priority	Code	Tests
IR-NTF-022	Push service worker	Implemented	Must	**3**	**2**
FR-NTF-023	Activity trail recording	Implemented	Must	**3**	**1**
FR-NTF-024	Who sees the trail	Implemented	Must	**3**	**1**
FR-NTF-025	Read-only trail and last-edit stamp	Implemented	Must	**3**	**1**
FR-NTF-026	Row memory: why is it here	Implemented	Should	**2**	**1**
FR-NTF-027	Public subscription calendar (ICS)	Implemented	Must	**2**	**2**
FR-NTF-028	Stable instructor feed and quiet pause	Implemented	Must	**3**	**1**
NFR-NTF-029	Calendar key secret	Implemented	Must	**2**	**1**
FR-NTF-030	One-day reminder file	Implemented	Could	**2**	**1**
FR-ENG-001	Survey payload	Implemented	Must	**1**	**2**
FR-ENG-002	Request validation on submit	Implemented	Must	**1**	**2**
FR-ENG-003	Case number as second key and identity lock	Implemented	Must	**1**	**2**
FR-ENG-004	Graduation-sheet proof	Implemented	Must	**1**	**2**
FR-ENG-005	My request status	Implemented	Must	**1**	**1**
FR-ENG-006	Legacy survey link	Implemented	Should	**2**	**0**
NFR-ENG-007	Attempt limit on survey endpoints	Implemented	Must	**2**	**1**
NFR-ENG-008	Public page security policy	Implemented	Must	**1**	**2**
IR-ENG-009	Public page kit and invalid link	Implemented	Should	**3**	**2**
FR-ENG-010	Where the script and ad are injected	Implemented	Must	**1**	**1**
DR-ENG-011	Ad slots and image rules	Implemented	Must	**2**	**1**
FR-ENG-012	Campaign validation and placement conflicts at approval	Implemented	Must	**2**	**1**
FR-ENG-013	Campaign state management and audit	Implemented	Must	**2**	**2**
FR-ENG-014	Image quality check	Implemented	Should	**3**	**1**
FR-ENG-015	Choosing ads to serve	Implemented	Must	**3**	**1**
IR-ENG-016	Browser rendering and impression counting	Implemented	Must	**2**	**1**
FR-ENG-017	Clicks and filtering of automated and repeated	Implemented	Must	**2**	**1**
FR-ENG-018	Anonymous audience pulse	Implemented	Must	**2**	**1**
FR-ENG-019	Unique coupon and store verification page	Implemented	Could	**3**	**1**
FR-ENG-020	Campaign report for the company	Implemented	Should	**5**	**1**
FR-ENG-021	Partnership profile and its links	Implemented	Should	**5**	**1**
FR-ENG-022	Campaign request from the profile	Implemented	Should	**3**	**1**
FR-ENG-023	Optional email and administrator alerts	Implemented	Could	**2**	**1**
FR-ENG-024	Audience dashboard	Implemented	Must	**4**	**1**
FR-ENG-025	Audience story mode	Implemented	Could	**2**	**1**
FR-ENG-026	Landing page and presentation deck	Implemented	Could	**4**	**0**
FR-ENG-027	Delivery of a landing-page inquiry	Implemented	Should	**5**	**1**
FR-DSH-001	Term shown on the dashboard	Implemented	Must	**3**	**1**
FR-DSH-002	Empty term state and upcoming term	Implemented	Must	**3**	**1**
FR-DSH-003	Scope of what the dashboard shows	Implemented	Must	**3**	**2**

ID	Title	Status	Priority	Code	Tests
FR-DSH-004	Today's lectures and the weekend	Implemented	Must	**2**	**0**
FR-DSH-005	Scope load reading	Implemented	Should	**1**	**1**
FR-DSH-006	Trend over the last terms	Implemented	Should	**2**	**1**
FR-DSH-007	Dashboard loading and enrichment	Implemented	Should	**1**	**1**
FR-DSH-008	Decision card and signals	Implemented	Should	**1**	**0**
IR-DSH-009	Today's lecture track	Implemented	Should	**2**	**0**
FR-DSH-010	Public journey reading	Implemented	Should	**3**	**1**
FR-DSH-011	Global search	Implemented	Should	**2**	**1**
FR-DSH-012	Sentence search	Implemented	Could	**2**	**2**
FR-DSH-013	Term story	Implemented	Should	**2**	**0**
FR-DSH-014	Semester readiness score	Implemented	Must	**1**	**1**
IR-DSH-015	Semester readiness panel	Implemented	Should	**3**	**1**
IR-DSH-016	Period-balance drawing	Implemented	Should	**3**	**2**
IR-DSH-017	Mini ring	Implemented	Could	**2**	**0**
IR-DSH-018	Helper drawings	Implemented	Could	**3**	**0**
FR-DSH-019	Welcome tour by role	Implemented	Should	**3**	**1**
IR-DSH-020	Tour playback and accessibility	Implemented	Should	**1**	**1**
FR-DSH-021	Render fault recovery	Implemented	Must	**1**	**1**
IR-DSH-022	General dashboard presentation constraints	Implemented	Should	**3**	**0**
FR-DSH-023	Demo limits	Implemented	Could	**3**	**1**
FR-OPS-001	Boot order, private directory and environment file	Implemented	Must	**6**	**2**
FR-OPS-002	Cloud Run production mode	Implemented	Must	**1**	**1**
FR-OPS-003	Firestore target selection and refusal of an implicit target	Implemented	Must	**4**	**1**
FR-OPS-004	No service without a database	Implemented	Must	**2**	**1**
FR-OPS-005	Account reconciliation and scheduled jobs at boot	Implemented	Should	**1**	**2**
IR-OPS-001	Health interface	Implemented	Must	**2**	**2**
IR-OPS-002	Build identity and version interface	Implemented	Must	**4**	**1**
FR-OPS-006	Security headers on every response	Implemented	Must	**1**	**2**
FR-OPS-007	Compression and request size limits	Implemented	Must	**1**	**1**
FR-OPS-008	Refuse cross-site write requests	Implemented	Must	**1**	**0**
FR-OPS-009	Public-surface and teacher-card rate limits	Implemented	Must	**2**	**3**
FR-OPS-010	Anonymous-door guard	Implemented	Must	**3**	**1**
FR-OPS-011	Sign-in attempt limit	Implemented	Must	**1**	**0**
FR-OPS-012	Document reading gate	Implemented	Must	**2**	**1**
FR-OPS-013	Static serving and cache policy	Implemented	Must	**2**	**1**
FR-OPS-014	Error contract	Implemented	Must	**2**	**1**
FR-OPS-015	Resumable full-system export	Implemented	Must	**3**	**1**
FR-OPS-016	Backup preview, import, system reset and undo	Implemented	Must	**2**	**2**
FR-OPS-017	An isolated demo sandbox per visitor	Implemented	Must	**5**	**3**

ID	Title	Status	Priority	Code	Tests
FR-OPS-018	Demo sandbox limits and management	Implemented	Must	**4**	**2**
FR-OPS-019	Demo seeding and role stories	Implemented	Should	**6**	**2**
FR-OPS-020	Service worker and offline behaviour	Implemented	Must	**3**	**2**
FR-OPS-021	The 'we are updating' page	Implemented	Must	**3**	**1**
IR-OPS-003	Installable-app manifest	Implemented	Should	**4**	**1**
FR-OPS-022	Continuous-integration pipeline	Implemented	Must	**2**	**1**
FR-OPS-023	Deploy target guard and release pipeline	Implemented	Must	**2**	**1**
FR-OPS-024	Repository hygiene guard	Implemented	Should	**1**	**1**
FR-OPS-025	Test runner policy	Implemented	Must	**4**	**1**
NFR-OPS-001	A small, fail-early runtime image	Implemented	Must	**4**	**0**
FR-OPS-026	Backup and recovery procedure	Implemented	Should	**1**	**1**
NFR-NFR-001	Student journey throughput	Implemented	Must	**3**	**1**
NFR-NFR-002	Limit fairness on a shared network	Implemented	Must	**3**	**2**
NFR-NFR-003	Opening the teacher card and department screen	Implemented	Should	**3**	**1**
NFR-NFR-004	Guide performance budget	Implemented	Should	**3**	**1**
NFR-NFR-005	Database read economy	Implemented	Must	**3**	**1**
NFR-NFR-006	Caching layers and delivery weight	Implemented	Should	**5**	**1**
NFR-NFR-007	Permission enforced on the server	Implemented	Must	**4**	**2**
NFR-NFR-008	Password protection	Implemented	Must	**5**	**2**
NFR-NFR-009	Sessions and cookies	Implemented	Must	**4**	**1**
NFR-NFR-010	Firestore rules deny every direct connection	Implemented	Must	**3**	**0**
NFR-NFR-011	Secrets handling	Implemented	Must	**4**	**2**
NFR-NFR-012	Public link tokens	Implemented	Must	**5**	**3**
NFR-NFR-013	Escaping and content policy for public pages	Implemented	Must	**2**	**2**
NFR-NFR-014	Input validation and bounds	Implemented	Must	**2**	**1**
NFR-NFR-015	Abuse resistance through rate limits	Implemented	Should	**4**	**1**
NFR-NFR-016	An audit trail without request content	Implemented	Must	**3**	**1**
NFR-NFR-017	External-service boundary	Implemented	Must	**2**	**1**
NFR-NFR-018	A public privacy policy	Implemented	Must	**2**	**0**
NFR-NFR-019	Personal data minimisation	Implemented	Must	**4**	**1**
NFR-NFR-020	44-pixel touch targets	Implemented	Should	**5**	**1**
NFR-NFR-021	Colour contrast to WCAG AA	Implemented	Should	**4**	**1**
NFR-NFR-022	Controls with accessible names	Implemented	Should	**6**	**1**
NFR-NFR-023	Reduced-motion respect	Implemented	Should	**3**	**1**
NFR-NFR-024	Zoom and focus indication	Implemented	Should	**4**	**0**
NFR-NFR-025	Arabic-first, right-to-left	Implemented	Must	**4**	**1**
NFR-NFR-026	Self-hosted typefaces	Implemented	Should	**4**	**1**
NFR-NFR-027	Numerals and Arabic counts	Implemented	Must	**3**	**2**
NFR-NFR-028	Standard Arabic and text integrity	Implemented	Should	**1**	**2**

ID	Title	Status	Priority	Code	Tests
NFR-NFR-029	Layouts that work from 320 pixels	Implemented	Must	**4**	**2**
NFR-NFR-030	A4 landscape printing with a fixed design	Implemented	Must	**3**	**2**
NFR-NFR-031	A deterministic decision engine	Implemented	Must	**2**	**3**
NFR-NFR-032	Stale-write protection	Implemented	Must	**3**	**2**
NFR-NFR-033	Reversibility of change	Implemented	Must	**3**	**3**
NFR-NFR-034	Process survival on a single request failure	Implemented	Must	**2**	**1**
NFR-NFR-035	Type checking and parity guards	Implemented	Must	**4**	**3**
NFR-NFR-036	Transactional identifier allocation	Implemented	Must	**1**	**1**
NFR-NFR-037	Supported browsers	Implemented	Should	**5**	**1**
NFR-NFR-038	Observability	Implemented	Should	**5**	**1**
NFR-NFR-039	A reversible, error-resistant release	Implemented	Must	**2**	**1**
DR-DAT-001	One store, closed to clients	Implemented	Must	**3**	**1**
DR-DAT-002	Core entities and identifiers	Implemented	Must	**2**	**1**
FR-DAT-003	Block-based id allocation	Implemented	Must	**1**	**0**
DR-DAT-004	Hierarchy: college, department, course	Implemented	Must	**2**	**1**
FR-DAT-005	Delete refused while referenced	Implemented	Must	**3**	**1**
DR-DAT-006	Course code unique within a department	Implemented	Must	**1**	**0**
DR-DAT-007	Uniqueness of department code and term name	Implemented	Should	**2**	**1**
DR-DAT-008	Instructor civil number	Implemented	Must	**2**	**1**
DR-DAT-009	Core appointment fields	Implemented	Must	**1**	**1**
FR-DAT-010	Revision number and no overwriting of newer data	Implemented	Must	**2**	**1**
DR-DAT-011	Academic identity snapshot of an appointment	Implemented	Must	**2**	**2**
DR-DAT-012	Scope key	Implemented	Must	**1**	**1**
DR-DAT-013	Timetable versions	Implemented	Must	**2**	**1**
DR-DAT-014	Drafts	Implemented	Must	**2**	**0**
DR-DAT-015	Share links in storage	Implemented	Must	**3**	**1**
DR-DAT-016	Accounts and passwords	Implemented	Must	**2**	**1**
DR-DAT-017	Permissions and scopes	Implemented	Must	**2**	**2**
DR-DAT-018	Sessions	Implemented	Must	**2**	**1**
DR-DAT-019	Visiting-instructor model	Implemented	Must	**2**	**1**
FR-DAT-020	Visiting ⇄ appointed conversion rule	Implemented	Must	**2**	**1**
FR-DAT-021	Legacy import under original ids	Implemented	Must	**3**	**2**
DR-DAT-022	Historical table archive	Implemented	Should	**2**	**1**
NFR-DAT-023	Short-lived read caches	Implemented	Must	**3**	**1**
DR-DAT-024	Demo world separated from the real one	Implemented	Must	**3**	**2**
FR-DAT-025	Deleted-records bin	Implemented	Should	**1**	**0**
DR-DAT-026	Retention and automatic expiry	Implemented	Should	**2**	**0**
DR-DAT-027	Indexes	Implemented	Must	**1**	**1**
DR-DAT-028	Backup and restore	Implemented	Must	**2**	**0**

ID	Title	Status	Priority	Code	Tests
DR-DAT-029	Student data privacy	Implemented	Must	**2**	**0**
IR-EXT-001	Firestore through the server only	Implemented	Must	**4**	**1**
IR-EXT-002	Firestore project and database selection	Implemented	Must	**2**	**1**
IR-EXT-003	Hosting on Cloud Run	Implemented	Must	**3**	**1**
IR-EXT-004	Domain and official deployment	Implemented	Must	**2**	**1**
IR-EXT-005	Health endpoint	Implemented	Should	**2**	**0**
IR-EXT-006	Gemini call contract	Implemented	Must	**2**	**1**
IR-EXT-007	Gemini output behind validators and confirmation	Implemented	Must	**2**	**1**
IR-EXT-008	When Gemini is unavailable	Implemented	Must	**2**	**0**
IR-EXT-009	OCR language packs	Implemented	Must	**2**	**1**
IR-EXT-010	The authority's SWRSCHA file as an input	Implemented	Must	**3**	**2**
IR-EXT-011	Web Push protocol	Implemented	Must	**1**	**1**
IR-EXT-012	Allowed push destinations and pruning	Implemented	Must	**2**	**1**
IR-EXT-013	Per-device notification limit	Implemented	Should	**2**	**1**
IR-EXT-014	Outbound company e-mail	Implemented	Could	**1**	**1**
IR-EXT-015	ICS calendar file for a department timetable	Implemented	Must	**2**	**1**
IR-EXT-016	The instructor's stable calendar subscription	Implemented	Must	**2**	**1**
IR-EXT-017	One-day calendar reminder	Implemented	Could	**1**	**1**
IR-EXT-018	Excel export	Implemented	Must	**2**	**1**
IR-EXT-019	Service worker and PWA contract	Implemented	Must	**3**	**1**
IR-EXT-020	Push worker and manifests	Implemented	Must	**4**	**1**
IR-EXT-021	Public link scheme	Implemented	Must	**6**	**1**
IR-EXT-022	Token rules	Implemented	Must	**4**	**2**
IR-EXT-023	Anonymous door protection	Implemented	Must	**3**	**1**
IR-EXT-024	Browser contract: security and headers	Implemented	Must	**1**	**0**
IR-EXT-025	Body limits and asset caching	Implemented	Should	**2**	**0**
IR-EXT-026	Environment variables and secrets	Implemented	Must	**3**	**1**
IR-EXT-027	Database target from environment variables	Implemented	Could	**3**	**1**

B Inferred requirements for confirmation

↑ Contents

These requirements were derived from the behaviour of the code and no direct test proves them. They need owner confirmation before final approval.

ID	Title	Module
FR-ACC-003	Idle expiry and presence-based renewal	ACC
FR-ACC-017	Main-administration screens and the root account	ACC
FR-ACC-018	Permission and scope administration	ACC
NFR-ACC-025	One identity resolution per request with a short cache	ACC
FR-ACC-027	Audit trail for sign-in and refusals	ACC
FR-ACC-028	Backup vault for the root account	ACC
FR-ACD-011	Freezing the identity of a course with history	ACD
FR-ACD-012	Block deleting a linked course	ACD
DR-ACD-017	Faculty identity: civil number	ACD
NFR-ACD-030	Reference-data input validation	ACD
FR-LOC-014	Boys and girls campuses stay separate	LOC
FR-LOC-017	Request expiry and inbox counter	LOC
FR-LOC-029	Walking time between buildings on a campus	LOC
FR-SCH-012	Other departments' halls and hall barter	SCH
FR-SCH-022	Atomic batch move	SCH
FR-SCH-025	Replace an instructor across a term	SCH
IR-SCH-001	Drag preview before the drop	SCH
IR-SCH-002	Saved views, undo and offline work	SCH
FR-DRF-002	Validate a draft before saving and cap its size	DRF
FR-DRF-008	Repair time collisions when publishing a start-of-term draft	DRF
FR-DRF-011	Version list and two-version comparison	DRF
FR-VIW-003	An ended term is read and export only	VIW
NFR-VIW-009	The living reading is computed once, not once per asker	VIW
FR-VIW-010	Last reading of the three indicators	VIW
FR-VIW-011	Previous-term ghost over the schedule	VIW
FR-VIW-012	The «decision now» panel	VIW
NFR-VIW-013	Analysis reads are deferred and quiet on failure	VIW
FR-VIW-017	Opening and stopping a link	VIW
FR-VIW-021	Saved views	VIW
FR-VIW-022	Warn when a saved view lost its scope	VIW
DR-IMP-021	Signed header-check receipt	IMP
IR-IMP-030	Gemini call interface and limits	IMP
FR-IMP-032	JSON schedule export and import	IMP
FR-REQ-028	Cancelling a department-placed appointment and undoing a fix	REQ
DR-REQ-037	Anonymous appointment-entry measure	REQ

ID	Title	Module
FR-PLN-003	Registration statistics save	PLN
FR-PLN-005	Same-season history and course lineage	PLN
DR-PLN-001	Planning data stores	PLN
IR-STU-001	Public student schedule page	STU
FR-INT-003	Room insight	INT
FR-INT-004	Instructor insight	INT
FR-INT-005	Appointment story and context	INT
FR-INT-006	Campus movement and safe room swaps	INT
FR-INT-007	Ripple forecast before a drop	INT
FR-INT-010	Deterministic scoped execution of calls	INT
FR-INT-014	Decision room, automatic optimiser and auto proposal	INT
FR-INT-015	Custom scheduling rules	INT
FR-INT-017	Open decisions	INT
FR-INT-021	Live schedule layer	INT
FR-INT-029	Guide friction points for the root administrator	INT
FR-INT-030	Experience health	INT
IR-INT-002	Gemini service interface: request constraints and timeout	INT
FR-RPT-003	24-hour time options	RPT
FR-RPT-005	Role-based lens lists	RPT
NFR-RPT-003	Printing from the installed iOS app	RPT
IR-NTF-006	Bell interface	NTF
FR-NTF-011	Board presence	NTF
IR-NTF-015	Colleague faces chip	NTF
FR-NTF-030	One-day reminder file	NTF
FR-ENG-006	Legacy survey link	ENG
FR-ENG-026	Landing page and presentation deck	ENG
FR-DSH-004	Today's lectures and the weekend	DSH
FR-DSH-008	Decision card and signals	DSH
IR-DSH-009	Today's lecture track	DSH
FR-DSH-013	Term story	DSH
IR-DSH-017	Mini ring	DSH
IR-DSH-018	Helper drawings	DSH
IR-DSH-022	General dashboard presentation constraints	DSH
FR-DSH-023	Demo limits	DSH
FR-OPS-006	Security headers on every response	OPS
FR-OPS-007	Compression and request size limits	OPS
FR-OPS-008	Refuse cross-site write requests	OPS
FR-OPS-011	Sign-in attempt limit	OPS
FR-OPS-013	Static serving and cache policy	OPS

ID	Title	Module
FR-OPS-015	Resumable full-system export	OPS
FR-OPS-016	Backup preview, import, system reset and undo	OPS
NFR-OPS-001	A small, fail-early runtime image	OPS
NFR-NFR-006	Caching layers and delivery weight	NFR
NFR-NFR-009	Sessions and cookies	NFR
NFR-NFR-010	Firestore rules deny every direct connection	NFR
NFR-NFR-016	An audit trail without request content	NFR
NFR-NFR-018	A public privacy policy	NFR
NFR-NFR-020	44-pixel touch targets	NFR
NFR-NFR-022	Controls with accessible names	NFR
NFR-NFR-024	Zoom and focus indication	NFR
NFR-NFR-034	Process survival on a single request failure	NFR
NFR-NFR-037	Supported browsers	NFR
NFR-NFR-038	Observability	NFR
FR-DAT-003	Block-based id allocation	DAT
DR-DAT-006	Course code unique within a department	DAT
DR-DAT-014	Drafts	DAT
FR-DAT-025	Deleted-records bin	DAT
DR-DAT-026	Retention and automatic expiry	DAT
DR-DAT-028	Backup and restore	DAT
DR-DAT-029	Student data privacy	DAT
IR-EXT-005	Health endpoint	EXT
IR-EXT-008	When Gemini is unavailable	EXT
IR-EXT-024	Browser contract: security and headers	EXT
IR-EXT-025	Body limits and asset caching	EXT

C Glossary

↑ Contents

Academic decision lab
مختبر القرار الأكاديمي
Bounded read-only what-if calculations over approved plan sheets and student wishes.

Access card
بطاقة الدخول
A card with a QR code and personal link sent to someone invited to the system.

Access scope
النطاق
The colleges and departments whose data an account may see; a row with department 0 means the whole college for college and all-colleges roles.

Activity trail
الأثر
A read-only record of everything that changed something: who did what, and to what.

Ad slot
الموقع الإعلاني
A place of fixed size and capacity in a page, such as the student portal or the public schedule.

Advisor seat balance
رصيد مقاعد المرشد
Extra seats the administrator grants advisors above a section's base capacity.

Analytics Center
مركز التحليلات
The screen from which departments read the same student-case register that the sheet reads.

Anonymous guard
الحارس المجهول
A limit that counts failures from unauthenticated addresses and never blocks success.

Appointed instructor
معيّن
An instructor who is not visiting a department; conversion to it removes him from the directory and rosters.

Appointment
موعد
One schedule row: a class section with its days, time, hall and instructor (FSchedule).

Approval
الاعتماد
The review cycle of a department's schedule in a term, from signature to registrar acceptance.

Approval blocker
مانع اعتماد
A physical conflict that prevents approving the schedule.

Approval dossier
ملف الاعتماد
An A4 sheet gathering what changed, blockers, signatures and department status.

Assisted reading
القراءة الأدق
An optional reading through the Gemini service for pages with unread cells; it proposes and never writes.

Authority document (SWRSCHA)
وثيقة الجهة المعتمدة
The official timetable file issued by the registration authority for a department, branch and term, whose header carries the term, branch and department.

Baseline
خط الأساس
The first reading of the document kept unchanged to compare with the live schedule.

Baseline
الرصيد
The audience figures since 2016 approved by the system administration, to which what is measured after 2026-10-17 is added.

Bell
الجرس
The notification center icon carrying the number of unread items required of the user.

Build identifier
معرّف البناء
The stamp a release carries in the client, server and service worker to tell which version is running.

Building
المبنى
A unit of the official registry with a code made of the site prefix and two digits.

Building board / Corridor screen
لوحة المبنى / شاشة الممر
A page with a tile per hall of the building, and its full-screen mode for a TV.

Calendar subscription
اشتراك التقويم
An ICS file one subscribes to so that it updates itself.

Campaign
الحملة
A company's advertisement in one slot with dates and one or two images; it is a draft, approved or paused.

Campus movement
الحركة المكانية
An instructor's move between two buildings between lectures, measured as a margin in minutes against the travel time.

Case number
رقم الحالة
The first 8 uppercase characters of the record id, held by the student.

Chosen count
المختار
The section count a department decides for a course in planning, taking precedence over the suggestion.

Class section
الشعبة
A class of a course in a given term, identified by its section code (501 onward in authority schedules).

Class section
شعبة
A group of students with a time, hall and instructor for a course in a term; counted in the schedule by its distinct section code (SCode).

College
الكلية
A top academic unit with a code and a name; the code may hold the branch code, so its first two digits give the official college identity.

Companion facility
المرفق المرافق
A studio or lab booked together with the main hall for the same appointment.

Comprehensive report
التقرير الشامل
The official main schedule document, read page by page beside the registration sheet.

Confidence
الثقة
An explicit four-level grade attached to every forecast figure that governs whether the action is enabled.

Course
المقرر
A subject with credits, hours and capacity; its official number has seven digits: college (2), department (2), course (3).

Deadline exception
الاستثناء
A department-specific submission date that takes precedence over the term deadline when later, granted with a written reason.

Decision board
لوحة القرار
A read-only planning lens for the department head, dean and above.

Demo sandbox
البيئة التجريبية
A per-visitor in-memory data copy lasting one hour that never touches real data.

Department
القسم
The scientific department (sectionId in code) to which courses belong; it does not mean a class section.

Department
قسم
The scientific department; stored in code as AdSectionId and belonging to a college.

Department balance
ميزان الأقسام
A table with one row per department in the term: appointments, instructors, rooms, fairness, quality, blockers and approval status.

Department family
عائلة القسم
A group of departments with the same academic code across colleges and sites (boys, girls, branches).

Department window
نافذة القسم
An exception narrower than the university day for one department; it never widens it.

Draft
مسودة
A proposed schedule for a department and term that is reviewed and edited and does not change the live schedule until published.

Draft
المسودة
The plan sheet rule set under review before approval.

Drift
انحراف
A row the scenario touches that was deleted or whose revision changed after the scenario was built.

Ended term
الفصل المنتهي
An archive term that is not written to and is only read and exported.

Faculty card
بطاقة الأستاذ
A public page where an instructor opens their schedule and asks for a change, through the department link or a personal link.

Faculty member
عضو هيئة التدريس
The instructor in the instructor register with identity (civil number and name), status (active, retired or on sabbatical) and category (instructor or TA).

Faculty placeholder
هيئة تدريسية
A record each department owns that is not a person: it receives sections with no fixed instructor and is excluded from double-booking and fairness checks.

Guide
المرشد
An in-app guide that answers usage questions and walks the user through a tour on screen, running entirely in the browser.

Hall
القاعة
A teaching room or lab inside a building, identified by its code there.

Hall barter
استعارة القاعات
A digital reservation of a free window in another department's hall for one term, with the host's approval.

Hall door page
باب القاعة
A public page reached by QR code stating the hall's state now.

Header
الترويسة
The page-1 header that decides term, branch and department before rows are read.

Historical form
الصيغة التاريخية
An old hall or building code that stays recognised after its code changes.

Hold

حجز

Either an instructor-request hold awaiting the department (blocks, and is asked about) or an advisory two-minute colleague hold (never blocks).

Home dashboard

لوحة البداية

The first screen after sign-in: scope, term, its figures and the reading of the day and week.

Individual exception

الاستثناء الفردي

An edit window for one member with a written reason and a dedicated closing time.

Initial password

الكلمة الأولى

A password an administrator sets for the account owner, changed at first sign-in; the default one lapses after 7 days.

Instructor inbox

وارد الأساتذة

The department screen where faculty requests are reviewed and decided item by item.

Instructor replacement

استبدال أستاذ

Handing all of an instructor's appointments in a term to a replacement or leaving them without an instructor.

Instructor request card

بطاقة طلب الأستاذ

The public page (/r/) a faculty member opens with the personal link to request changes to their schedule.

Journey

الرحلة

A narrative screen of aggregate counts about the system's memory.

Lender / Borrower

المُعير / المستعير

The host department owning the hall, and the department requesting the window.

Lens

العدسة

A way of grouping the same filtered results to answer one specific question.

Live channel

القناة الحيّة

A light server-to-screen broadcast announcing that something changed without carrying data.

Live schedule layer

الطبقة الحية

Six live scenes about the schedule's state: overview, pressure, health, brief, start of term and safety net.

Living layer

الطبقة الحيّة

The living-schedule window with its six scenes.

Load fairness

عدالة الحمل

An index of the spread of weekly teaching load between instructors.

Main administration account

حساب الإدارة الرئيسي

An IsAdminUser account or the root identifier; it assigns roles and sits above them.

Masari

مساري

The permanent student door /me: a profile with a personal code, wishes, a plan and request status.

Material conflict

التعارض المادّي

An instructor or room double booking in overlapping appointments, or an indistinguishable duplicate; the general rule's only blocker for signing and acceptance.

Natural-language query

الاستعلام بلغة طبيعية

A sentence the user writes that is turned into allowed calls executed deterministically and ending in a preview, not a write.

Nonce

نونس

A random value created per response that only the page's own scripts may carry.

Not yet reached

لم نصل بعد

A judgement that a course in a branch is beyond the level or prerequisites the branch's students have reached.

Notification center
مركز الإشعارات
A panel showing what remains for a role holder, derived from state and not stored.

Occupancy rate
نسبة الإشغال
The share of the scope's halls occupied at each day and hour.

Offering state
حالة الطرح
A per-course entry in a term and department: offered, closed, future or limited sections.

Official target
الوجهة الرسمية
The Cloud Run service and Firestore database that serve schedule-kw.com and nothing else.

Operationally current term
الفصل الجاري تشغيلياً
The term in which the institution is working now, the same arbiter used by the staff card and the schedule.

Partnership profile
ملف الشراكة
A public page with a private token that shows a prospective company the audience figures, slots and terms.

Pattern library
مكتبة الأنماط
The approved distributions of a course's days and lengths by its hours (17 patterns).

Pending hall confirmation
بانتظار تثبيت القاعة
The state of an appointment saved before an official hall is chosen (PENDING_ROOM).

Personal link
رابط شخصي
A card opened only for one instructor, living until the end of the term.

Plan sheet
الصحيفة
The study-plan PDF whose text layer is read and against which a department's courses are reconciled.

Plan-sheet badge
شارة الصحيفة
A tag beside the course name showing its plan sheet and requirement type.

Presence
الحضور
A colleague's current position on a board or screen; it lives seconds and is never stored.

Preview
المعاينة
The editable review table showing read rows and their notes before saving.

Previous term
الفصل السابق
The one earlier in time than the current term by term chronology, not by identifier order.

Pulse
النبضة
A short question answered from memory: who is with me, did anything change, does the bell have news.

Pulse
النبض
An anonymous signal counting one visit per day per page kind per browser.

Push notifications
تنبيهات الجوال
Encrypted Web Push notices to a device subscribed by account or by card.

Queries
الاستعلامات
The entry modes starting with search; their counterpart, Reports, are the modes starting with report.

Receipt
الإيصال
An HMAC signature proving the scope of a reading or the gaps and preventing the screen from bypassing checks.

Refresh and retry
حدّث ثم أعد
The refusal of a write with an older stamp because a colleague saved first.

Registrar note
ملاحظة التسجيل
A one-click note on a cell of a row, carrying the row, field, value at writing time and the round.

Registration radar
رادار التسجيل
A simulation of student registration over the offered sections, with counts not names, each figure carrying its basis.

Registration sheet
كشف التسجيل
The one screen where the department committee and then registration decide on student requests.

Release path
مسار الإصدار
The chain from change request to release and verification on the official domain.

Remaining
المتبقي
The number of students who have not yet registered the course, from the «not passed» column of the registration report.

Request item
بند الطلب
One line in a request: keep, change, delete or add, with its decision and dialogue.

Request window
نافذة الطلبات
The period in which the server accepts submission, governed by the closing time in Kuwait time or by an individual exception.

Restore point
نقطة استعادة
Row images before applying and their revisions after, written with the apply and used for the restore.

Restore point
نقطة التراجع
A snapshot the server takes before a large system operation so the administrator can return to it.

Revision number
رقم المراجعة
A counter that grows with each write to a slot so a write built on an old copy is detected.

Role
الصفة
States who the account holder is in the college and determines read or write access, screens and scope.

Role template
القالب
The screens and scope a role writes automatically when it is chosen.

Round
الجولة
A numbered stage that starts at each send to the registrar or at the first edit after acceptance (amendment round).

Safety net
شبكة الأمان
The points captured before every large change and used to go back.

Safety point
نقطة أمان
A saved copy of the schedule before a major change that can be restored, with the restore effect stated.

Saved view
عرض محفوظ
A personal way of looking at the schedule saved under a name, not a copy of it.

Scenario
سيناريو
An ordered layer of changes (move, assign, remove, add) over the live schedule that is not written into it before approval.

Schedule
الجدول
All appointments of a department in a term.

Schedule pulse
نبض الجدول
A sentence and a ranking of items worth the coordinator's decision.

Scope
نطاق
The «college:department:term» combination under which versions, drafts and links are stored.

Screen (permission)
الشاشة (الصلاحية)
A number from 1 to 17 in FormSecurity that grants its holder what the screen contains.

Seat (account chair)
الكرسي
An account fixed by role and scope whose holder changes while its ledger stays.

Section request
طلب شعبة
A request from advising to open a section for a number of students, passing through the committee and then registration.

Section simulator
محاكي الشعب
A tab that tries removing, adding, merging or splitting a section with an immediate effect.

Semester readiness
جاهزية الفصل
A score out of 100 combining students reaching 12 units and demand finding a seat; an estimate, not a guarantee.

Service worker
عامل الخدمة
The script that provides offline operation and serves the updating page (public/sw.js).

Session
الجلسة
A record bound to the session_id cookie that establishes identity and ends after 15 minutes of inactivity.

Share link
رابط نشر
A long random token of a kind (department, faculty cards, survey, students) that opens a limited reading without an account.

Share link
رابط مشاركة
A record with a token, scope, kind and expiry that opens a defined read without an account.

Shared hall
القاعة المشتركة
A hall linked to more than one department.

Sheet stamp
ختم الورقة
A QR code and eight-character code in the printed report footer used to compare against the live schedule.

Snapshot
لقطة
A frozen copy of an appointment's academic identity or of a whole scope's rows.

Spelling twin
توأم الإملاء
Two courses in one department with the same name written with different spellings (hamza, ta marbuta or spacing).

Start ladder
سُلّم البدايات
The regulation start times: every hour from 08:00 for Sun/Tue/Thu and every 90 minutes for Mon/Wed.

Start of term
بداية الفصل
A draft built from a department's previous-term schedule, whose name begins with «بداية الفصل».

Stick to my lecture
لاصق بمحاضرتي
A suggested time just before or after an existing lecture, separated by the regulation gap.

Student survey
الاستبيان
An account-free public form completed with the civil ID to request a course, report a conflict or submit a graduate request.

Student ticket
التذكرة
What the simulation estimates for one student: three plans and a window, with no name.

Study proposal
المقترح الدراسي
A draft a department prepares for an instructor, separate from the live schedule until the instructor approves and the department commits.

SWRSCHA
SWRSCHA
The code of the registration authority's timetable report that the timetable is imported from.

System vault
خزنة النظام
The root administration screen for full export, import, reset and undo.

Teaching load
النصاب
An instructor's maximum term hours by rank or exception; overtime is added to form the ceiling.

Team
الفريق
One account with its head, members and deputy; the unit of visibility in the trail and presence.

Temporary hold
الحجز المؤقت
A hold that hides the chosen time from colleagues for ten minutes while editing, becoming timerless after submission.

Temporary password
الكلمة المؤقتة
A random 14-character reset password valid for 24 hours and one sign-in.

Term
الفصل
An academic term with a name such as first term 2026/2027, a start date, a week count and an ended flag.

Term freeze
التجميد
Making a past term read-only: it can be read and not written, announced by the 'Past term · view only' ribbon.

Term roster
الروستر
The list of visiting instructors of one department for one term.

Term story
قصة الفصل
A card summarising a department schedule's approval cycle in figures, rounds and a copyable sentence.

Time machine
آلة الزمن
The scenarios screen: emergency, lab and goal.

Token
رمز
A random string in the link that is the permission itself; it carries no personal data.

Touch target
هدف اللمس
The pressable area of an interactive element, at least 44 pixels.

Trial run (no save)
تجربة بلا حفظ
A full reading with the same checks for a term that has a schedule, drawing the difference without writing.

Turnaround gap
الفاصل
The minutes between the end of one lecture and the start of the next in the same hall (10 minutes).

University teaching day
الدوام الشامل
A single limit for the start and end of the teaching day for the whole programme (08:00–20:00 by default).

Vacant section
شعبة شاغرة
An existing section assigned to the placeholder faculty that a member can request instead of creating a new appointment.

View-only (read-only)
الاطّلاع (قراءة فقط)
A role from which the server refuses every write except the narrow approval-cycle doors.

Viewable impression
الظهور المرئي
An impression counted after half of the ad stayed visible for a full second while the page was open.

Visiting instructor

منتدب

An instructor named in a department's visiting directory and in a term roster; the status belongs to the department, not to the person.

Visiting instructor

المنتدب

An instructor listed in the department's visiting-instructor directory for the term.

Welcome tour

الجولة التعريفية

A role-specific welcome shown once per account and role.

Wishes

الرغبات

A student's units, courses and time constraints during the wishes phase.

D Revision history

↑ Contents

Version	Date	Changes
1.0	2026-11-07	First release: the baseline document with all its modules, requirement-to-code-and-test traceability, and the interactive page.